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Exercise - I (Conceptual Questions) Build Up Your Understanding

TRIGONOMETRY

1. As θ increases from 0° to 90°, the value of sinθ :-
(1) Increases
(2) Decreases
(3) Remains constant
(4) First decreases then increases.

PBM001

2. If $\sin\theta = \frac{\sqrt{2}}{\sqrt{3}}$ and θ lies in the first quadrant, the value of tan θ is :

(1) $\sqrt{2}$ (2) $\frac{1}{\sqrt{2}}$

(3) $\frac{\sqrt{3}}{\sqrt{2}}$ (4) $\frac{1}{2}$

PBM002

3. Find θ for which sinθ = cosθ, if 180° < θ < 360°
(1) 135° (2) 315°
(3) 225° (4) 150°

PBM003

4. If $\sin\theta_1 + \sin\theta_2 + \sin\theta_3 = 3$
then value of $\cos\theta_1 + \cos\theta_2 + \cos\theta_3$ is :-
(1) 3 (2) 0 (3) –3 (4) 1

PBM004

5. The greatest value of the function 7 sinθ – 24 cosθ is
(1) 12 (2) 13 (3) 25 (4) 17

PBM010

6. The length of hypotenuse of a right angle triangle exceeds the length of its base by 2 cm and exceeds twice the length of altitude by 1 cm. Find length of each side of the triangle.
(1) 6, 8, 10 (2) 7, 24, 25
(3) 8, 15, 17 (4) 7, 40, 41

PBM011

CALCULUS

7. If $x = at^4$ and $y = bt^3$. Find $\frac{dy}{dx}$

(1) $\frac{4a}{3bt}$ (2) $\frac{3b}{4at}$

(3) $\frac{3a}{4bt}$ (4) $\frac{4b}{3at}$

PBM012

8. A metallic disc is being heated. Its area at any time t is given by $A = 5t^2 + 4t + 8$. Calculate rate of increase in area at t = 3s.
(1) 30 m^2/s (2) 34 m^2/s
(3) 28 m^2/s (4) 20 m^2/s

PBM013

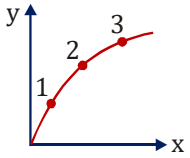
9. The side of a square is increasing at the rate of 0.1 cm/s. The rate of increase of perimeter w.r.t. time is :
(1) 0.2 cm/s (2) 0.4 cm/s
(3) 0.6 cm/s (4) 0.8 cm/s

PBM014

10. A particle moves along the straight line 3y =x+5. Which coordinate changes at a faster rate ?
(1) x–coordinate
(2) y–coordinate
(3) Both x and y coordinates
(4) Data insufficient.

PBM015

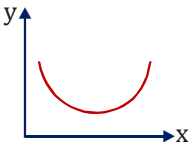
11. The slope of graph as shown in figure at points 1, 2 and 3 is m_1, m_2 and m_3 respectively then



(1) $m_1 > m_2 > m_3$ (2) $m_1 < m_2 < m_3$
(3) $m_1 = m_2 = m_3$ (4) $m_1 = m_3 > m_2$

PBM016

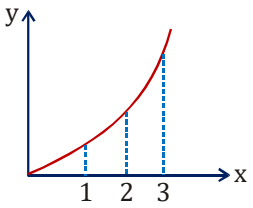
12. Magnitude of slope of the shown graph.



(1) First increases then decreases
(2) First decreases then increases
(3) Increases
(4) Decreases

PBM017

13. Calculate the area enclosed under the curve $f(x) = x^2$ between the limits x = 2 and x = 3



(1) 5 (2) $\frac{19}{3}$ (3) $\frac{17}{3}$ (4) 8

PBM018

14. Find the value of $\int_0^4 |(1-x)|.dx$

(1) zero (2) 1 (3) 4 (4) 5

PBM019

GEOMETRY & GRAPHS

15. The equation of a curve is given as $y = x^2+2-3x$. The curve intersects the y-axis at

(1) (1, 0) (2) (2, 0)

(3) (0, 2) (4) No where

PBM020

16. Two particles A and B are moving in XY-plane. Their positions vary with time t according to relation :

$x_A(t) = 3t$, $x_B(t) = 6$

$y_A(t) = t$, $y_B(t) = 2 + 3t^2$

Distance between two particles at t = 2 is :

(1) 12 (2) 13 (3) 5 (4) $\sqrt{12}$

PBM021

17. The distance between points (a + b, b + c) and (a – b, c – b) is :-

(1) $2\sqrt{a^2+b^2}$ (2) $2\sqrt{b^2+c^2}$

(3) $2\sqrt{2}b$ (4) $\sqrt{a^2-c^2}$

PBM022

18. A dog is at point A(0, 3, 4)m and cat is at B(5,3,–8)m The dog is free to move but cat is fixed. The minimum distance travelled by dog to catch the cat is :-

(1) 25m (2) 12m (3) 13m (4) 20m

PBM023

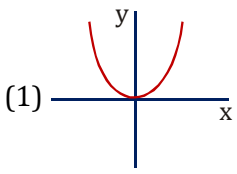
19. A particular straight line passes through origin and a point whose abscissa is equal to ordinate. The equation of such straight line is :

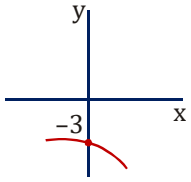
(1) y = x (2) y = 2x

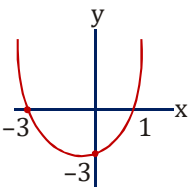
(3) y = –4x (4) $y = -\frac{x}{4}$

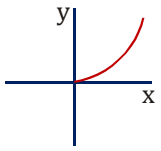
PBM024

20. If $y = x^2 + 2x - 3$, then y-x graph is :-

(1) 

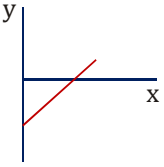
(2) 

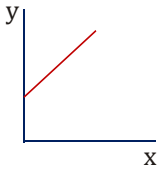
(3) 

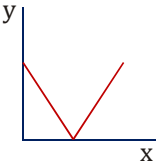
(4) 

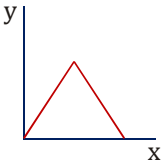
PBM025

21. If y =|x–1|, then y-x graph is :-

(1) 

(2) 

(3) 

(4) 

PBM026

22. Frequency f of a simple pendulum depends on its length ℓ and acceleration g due to gravity according to the following equation $f = \frac{1}{2\pi}\sqrt{\frac{g}{\ell}}$. Graph between which of the following quantities is a parabola ?

(1) f on the ordinate and $1/\ell$ on the abscissa

(2) f on the ordinate and $\sqrt{\ell}$ on the abscissa

(3) f^2 on the ordinate and ℓ on the abscissa

(4) f^2 on the ordinate and $1/\ell$ on the abscissa

PBM028

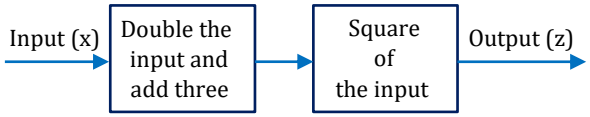
ALGEBRA

23. The sum of the series $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + ...\infty$ is :-

(1) $\frac{8}{7}$ (2) $\frac{6}{5}$ (3) $\frac{2}{3}$ (4) $\frac{3}{2}$

PBM029

24. In the given figure, each box represents a function machine. A function machine illustrates what it does with the input.



Which of the following statements is correct ?

(1) $z = (2x+3)^2$ (2) $z = 2(x+3)$

(3) $z=\sqrt{2x+3}$ (4) $z=\sqrt{2(x+3)}$

PBM030

DEFINITION & TYPES OF VECTOR

25. Which of the following statements is false :

(1) Mass, speed and energy are scalars

(2) Momentum, force and torque are vectors

(3) Distance is a vector while displacement is a scalar

(4) A scalar has only magnitude whereas as a vector has both magnitude and direction

PBM031

26. A physical quantity which has a direction and obeys triangle law of vector addition :

(1) must be a vector (2) may be a vector

(3) must be a scalar (4) none of the above

PBM032

27. The forces, which meet at one point and their lines of action lie in one plane, are called :

(1) non-concurrent and non-coplanar forces

(2) non-concurrent and coplanar forces

(3) concurrent and non-coplanar forces

(4) concurrent and coplanar forces

PBM033

28. Which of the following physical quantities is not an axial vector ?

(1) angular velocity

(2) angular momentum

(3) velocity

(4) torque

PBM034

29. The direction of the angular velocity vector is along :

(1) Along the tangent of circular path

(2) Along the direction of radius vector

(3) Opposite to the direction of radius vector

(4) Along the axis of rotation

PBM035

30. If $\hat{n}$ is a unit vector in the direction of the vector $\vec{A}$, then :-

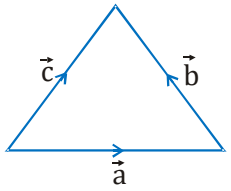
(1) $\hat{n}=\frac{\vec{A}}{|\vec{A}|}$ (2) $\hat{n}=\vec{A}|\vec{A}|$

(3) $\hat{n}=\frac{|\vec{A}|}{\vec{A}}$ (4) None of the above

PBM036

ADDITION & SUBTRACTION, MULTIPLICATION & DIVISION OF A VECTOR BY A SCALAR

31. In the given figure, $\vec{a}+\vec{b}+\vec{c}$ is



(1) $2\vec{a}$ (2) $2\vec{b}$ (3) $2\vec{c}$ (4) $\vec{a}+\vec{b}$

PBM037

32. Vector sum of two forces of 10N and 5N can be :

(1) 4N (2) 18N (3) 12N (4) 2N

PBM038

33. The vector sum of the forces of 10 newton and 8 newton cannot be :

(1) 2N (2) 8N

(3) 18N (4) 20N

PBM039

34. Which of the following pair of forces can give a resultant force of 2 N ?

(1) 1 N and 2 N (2) 1 N and 5 N

(3) 2 N and 5 N (4) 1 N and 4 N

PBM040

35. Two vectors of equal magnitude have a resultant equal to either of them in magnitude. The angle between the vector is :

(1) 60° (2) 90°

(3) 105° (4) 120°

PBM041

36. $\vec{R}=\vec{a}+\vec{b}$, where $\vec{R},\vec{a}$ and $\vec{b}$ are non-zero vector.

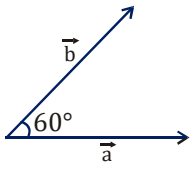
If R = a = b, the angle between $\vec{R}$ and $\vec{a}$ is :-

(1) $\frac{\pi}{3}$ (2) $\frac{2\pi}{3}$

(3) $\frac{\pi}{4}$ (4) $\frac{\pi}{6}$

PBM042

37. If resultant of two vectors $\vec{a}$ and $\vec{b}$ shown in the figure is $\sqrt{7}\,b$, then value of $\frac{b}{a}$ is :-



(1) $\frac{1}{2}$ (2) 1 (3) 2 (4) 3

PBM043

38. If $\vec{A}+\vec{B}=\vec{C}$ and A + B = C, then the angle between $\vec{A}$ and $\vec{C}$ is :

(1) 0 (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{2}$ (4) π

PBM044

39. Two vectors $\vec{A}$ and $\vec{B}$ are such that $\vec{A}+\vec{B}=\vec{C}$ and A – B = C. Which of the following statements, is correct ?

(1) $\vec{A}$ is parallel to $\vec{B}$

(2) $\vec{A}$ is anti-parallel to $\vec{B}$

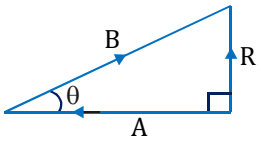
(3) $\vec{A}$ is perpendicular to $\vec{B}$

(4) $\vec{A}$ and $\vec{B}$ are equal in magnitude

PBM045

40. In vector diagram shown in figure where $(\vec{R})$ is the resultant of vectors ($\vec{A}$) and ($\vec{B}$).

If R = $\frac{B}{2}$, then value of angle θ is :



(1) 30° (2) 45°

(3) 60° (4) 75°

PBM046

41. The resultant of $\vec{A}$ and $\vec{B}$ makes an angle α with $\vec{A}$ and β with $\vec{B}$, then :

(1) α < β

(2) α < β if A < B

(3) α < β if A > B

(4) α < β if A = B

PBM047

42. If two forces of equal magnitude acting on a particle. The angle between forces is 60°. The resultant force on the particle is :-

(1) at an angle of 30° with each force.

(2) at an angle of 45° with one of the force

(3) at an angle of 15° with one of the force

(4) at an angle lying between 45° to 60°

PBM048

43. The resultant of $\vec{A}$ and $\vec{B}$ is perpendicular to $\vec{B}$. What is the angle between $\vec{A}$ and $\vec{B}$?

(1) $\cos^{-1}\left(\frac{A}{B}\right)$ (2) $\cos^{-1}\left(-\frac{B}{A}\right)$

(3) $\sin^{-1}\left(\frac{A}{B}\right)$ (4) $\sin^{-1}\left(-\frac{A}{B}\right)$

PBM049

44. The resultant of two forces make 30° and 60° angles with them and has magnitude 40 N. The magnitudes of two vectors are :-

(1) $20\sqrt{3}\,N, 20N$ (2) $30N, 20\sqrt{3}\,N$

(3) 40N, 30N (4) $3\sqrt{3}\,N, 40\sqrt{3}\,N$

PBM050

45. Given that $\vec{P}+\vec{Q}=\vec{P}-\vec{Q}$. This can be true when:

(1) $\vec{P}=\vec{Q}$

(2) $\vec{Q}=\vec{0}$

(3) Neither $\vec{P}$ nor $\vec{Q}$ is a null vector

(4) Data Insufficient

PBM051

46. The resultant of $\vec{A}$ & $\vec{B}$ is $\vec{R}_1$. On reversing the vector $\vec{B}$, the resultant becomes $\vec{R}_2$. What is the value of $2(R_1^2+R_2^2)$?

(1) $4(A^2 + B^2)$ (2) $A^2 - B^2$

(3) $2(A^2 + B^2)$ (4) $2(A^2 - B^2)$

PBM052

47. If vectors $\vec{A}$ and $\vec{B}$ are such that $|\vec{A}+\vec{B}| = |\vec{A}|=|\vec{B}|$, then $|\vec{A}-\vec{B}|$ may be equated to

(1) $\frac{\sqrt{3}}{2}|\vec{A}|$ (2) $|\vec{A}|$

(3) $\sqrt{2}|\vec{A}|$ (4) $\sqrt{3}|\vec{A}|$

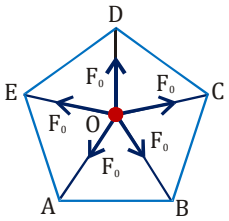
PBM053

48. If the difference of two unit vectors is a unit vector, then the magnitude of their sum is :

(1) $\sqrt{2}$ (2) $\sqrt{3}$ (3) $\frac{1}{\sqrt{2}}$ (4) $\sqrt{5}$

PBM054

49. In the given figure, O is the centre of the regular pentagon ABCDE. Five forces each of magnitude F_0 are acted as shown in the figure. The resultant force is :-



(1) 5 F_0 (2) 5 F_0 cos72°

(3) 5 F_0 sin72° (4) zero

PBM055

50. Two vectors $\vec{A}$ and $\vec{B}$ lie in a plane, another vector $\vec{C}$ lies in the same plane, then the resultant of these three vectors i.e. $\vec{A}+\vec{B}+\vec{C}$:

(1) may be zero

(2) cannot be zero

(3) lies in the plane containing $\vec{A}$ & $\vec{B}$

(4) Both 1 & 3

PBM056

51. Which of the following sets of concurrent forces may be in equilibrium ?

(1) F_1 = 3N, F_2 = 5N, F_3 = 1N, F_4 = 10N

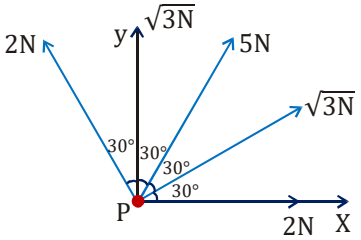
(2) F_1 = 3N, F_2 = 5N, F_3 = 9N, F_4 = 4N

(3) F_1 = 3N, F_2 = 5N, F_3 = 6N, F_4 = 15N

(4) F_1 = 3N, F_2 = 5N, F_3 =15N, F_4 = 5N

PBM057

52. Five forces 2N, $\sqrt{3}$N, 5N, $\sqrt{3}$N and 2N respectively act at a particle P as shown in the figure.



The resultant force on the particle P is.

(1) 10 N making angle 60° with x-axis.

(2) 10 N making angle 60° with y-axis.

(3) 20 N along y-axis

(4) None of these

PBM058

53. If $\vec{a}+\vec{b}+\vec{c}=0$. The angle between $\vec{a}$ and $\vec{b}$, $\vec{b}$ and $\vec{c}$ are 150° and 120°, respectively. Then, the magnitude of vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are in ratio of :-

(1) 1 : 2 : 3 (2) 1 : 2 : $\sqrt{3}$

(3) $\sqrt{3}$: 2 : 1 (4) 2 : $\sqrt{3}$: 1

PBM059

54. The magnitudes of vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$ are respectively 7, 24 and 25 units and $\vec{A}+\vec{B}=\vec{C}$, then the angle between $\vec{A}$ and $\vec{B}$ is :

(1) 0 (2) $\pi/3$

(3) $\pi/2$ (4) $\pi/4$

PBM060

55. If vectors $\vec{P}$, $\vec{Q}$ and $\vec{R}$ have magnitudes 5, 12 and 13 units and $\vec{P}+\vec{Q}=\vec{R}$, the angle between $\vec{P}$ and $\vec{R}$ is :

(1) $\cos^{-1}\left(\frac{5}{12}\right)$ (2) $\cos^{-1}\left(\frac{5}{13}\right)$

(3) $\cos^{-1}\left(\frac{12}{13}\right)$ (4) $\cos^{-1}\left(\frac{2}{13}\right)$

PBM061

56. If the vectors $\left(\hat{i}+\hat{j}+\hat{k}\right)$ and $3\hat{i}$ form two sides of a triangle, then 3rd side of the triangle can be :

(1) $\sqrt{18}$ unit (2) $\sqrt{6}$ unit

(3) $\frac{3}{\sqrt{2}}$ unit (4) both (1) & (2)

PBM062

57. The minimum number of vectors of equal magnitude required to produce a zero resultant is:

(1) 2 (2) 3

(3) 4 (4) more than 4

PBM063

58. How many minimum number of coplanar vectors having different magnitudes can be added to give zero resultant ?
(1) 2 (2) 3 (3) 4 (4) 5
PBM064

59. How many minimum number of vectors in different planes can be added to give zero resultant ?
(1) 2 (2) 3 (3) 4 (4) 5
PBM065

60. What happens, when we multiply a vector by 2?
(1) direction reverses and unit changes
(2) direction remains same and magnitude is doubled
(3) direction remains unchanged and unit changes
(4) none of these
PBM066

RESOLUTION OF VECTOR

61. What is the maximum number of components into which a vector can be split ?
(1) 2 (2) 3
(3) 4 (4) Infinite
PBM067

62. What is the maximum number of rectangular components into which a vector can be split in its own plane ?
(1) 2 (2) 3
(3) 4 (4) Infinite
PBM068

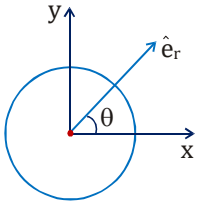
63. What is the maximum number of rectangular components into which a vector can be split in space ?
(1) 2 (2) 3 (3) 4 (4) ∞
PBM069

64. The unit vector along $\hat{i}-2\hat{j}$ is :
(1) $\frac{\hat{i}-2\hat{j}}{\sqrt{5}}$ (2) $\hat{i}+\hat{j}$
(3) $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$ (4) $\frac{\hat{i}-\hat{j}}{\sqrt{5}}$
PBM070

65. If $\vec{A}+\vec{B}$ is a unit vector along y-axis and $\vec{A}=\hat{i}-\hat{j}+\hat{k}$, then what is $\vec{B}$?
(1) $\hat{j}+\hat{k}$ (2) $\hat{j}-\hat{k}$
(3) $\hat{i}+\hat{j}+\hat{k}$ (4) $2\hat{j}-\hat{i}-\hat{k}$
PBM071

66. If a unit vector is represented by $0.3\hat{i}-0.4\hat{j}+c\hat{k}$, then the value of 'c' is :
(1) $\sqrt{0.75}$ (2) $\sqrt{0.25}$
(3) $\sqrt{0.01}$ (4) $\sqrt{0.39}$
PBM072

67. $\hat{e}_r$ is unit vector along radius of a circle shown in figure. $\hat{e}_r$ can be represented as :-



(1) $\cos\theta\hat{i}+\sin\theta\hat{j}$ (2) $\sin\theta\hat{i}+\cos\theta\hat{j}$
(3) $\cos\theta\hat{i}-\sin\theta\hat{j}$ (4) $-\cos\theta\hat{i}+\sin\theta\hat{j}$
PBM073

68. Forces 7N, 24N, 25N act at a point in mutually perpendicular directions. The magnitude of the resultant force is :
(1) 19 N (2) 13 N
(3) 26 N (4) $25\sqrt{2}$ N
PBM074

69. The angle that the vector $\vec{A}=2\hat{i}+3\hat{j}$ makes with x-axis is :
(1) $\tan^{-1}(3/2)$ (2) $\tan^{-1}(2/3)$
(3) $\sin^{-1}(2/3)$ (4) $\cos^{-1}(3/2)$
PBM075

70. What vector must be added to the other vectors $\hat{i}-2\hat{j}+2\hat{k}$ and $2\hat{i}+\hat{j}-\hat{k}$, so that the resultant may be a unit vector along y-axis?
(1) $2\hat{i}+\hat{j}-\hat{k}$ (2) $-3\hat{i}+2\hat{j}-\hat{k}$
(3) $-2\hat{i}-\hat{j}-\hat{k}$ (4) $3\hat{i}-2\hat{j}-\hat{k}$
PBM076

71. The unit vector parallel to the resultant of the vectors $\vec{A}=4\hat{i}+3\hat{j}+6\hat{k}$ and $\vec{B}=2\hat{i}-3\hat{j}+2\hat{k}$ is :
(1) $\frac{1}{10}\left[8\hat{i}+6\hat{k}\right]$ (2) $\frac{1}{10}\left[6\hat{i}+8\hat{k}\right]$
(3) $\frac{1}{10}\left[6\hat{i}+6\hat{k}+6\hat{j}\right]$ (4) $\frac{1}{10}\left[6\hat{j}+8\hat{k}\right]$
PBM077

72. If $\vec{a}=2\hat{i}+2\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}+\hat{k}$. Find a vector $\vec{c}$ which is parallel to $\vec{a}$ but has magnitude three times that of $\vec{b}$.

(1) $\vec{c}=3\hat{i}+3\hat{j}+3\hat{k}$

(2) $\vec{c}=2\sqrt{3}\hat{i}+2\sqrt{3}\hat{j}-\sqrt{3}\hat{k}$

(3) $\vec{c}=6\hat{i}+6\hat{j}-3\hat{k}$

(4) $\vec{c}=3\hat{i}+6\hat{j}-6\hat{k}$

PBM078

73. The velocity of a particle is $\vec{v}=(\hat{i}+\hat{j}-\hat{k})$m/s. A force of $10\sqrt{3}$N parallel to velocity in vector form is :-

(1) $(10\hat{i}+10\hat{j}-10\hat{k})$N

(2) $(\hat{i}+\hat{j}-\hat{k})10\sqrt{3}$N

(3) $(\hat{i}+\hat{j}-\hat{k})$N

(4) None of these

PBM079

74. The direction cosines of a vector $\sqrt{2}\hat{i}+\sqrt{2}\hat{j}+\hat{k}$ are:-

(1) $\frac{\sqrt{2}}{\sqrt{5}},\frac{\sqrt{2}}{\sqrt{5}},\frac{1}{\sqrt{5}}$ (2) $\frac{1}{\sqrt{5}},\frac{1}{\sqrt{5}},\frac{1}{5}$

(3) $\frac{1}{5},\frac{1}{5},\frac{1}{\sqrt{5}}$ (4) $\frac{1}{\sqrt{5}},\frac{1}{\sqrt{5}},\frac{1}{\sqrt{5}}$

PBM080

75. Vector $\vec{P}$ makes angles α, β & γ with the X, Y and Z axes respectively, then $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=$

(1) 0 (2) 1 (3) 2 (4) 3

PBM081

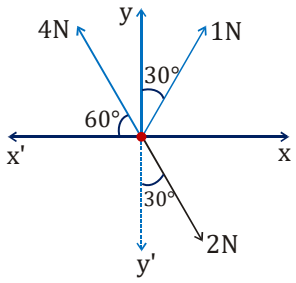
76. Find the direction cosines of vector $(\vec{a}-\vec{b})$, if $\vec{a}=2\hat{i}+3\hat{j}+\hat{k}$ and $\vec{b}=2\hat{i}+2\hat{j}+3\hat{k}$

(1) $0,\frac{1}{\sqrt{5}},\frac{-2}{\sqrt{5}}$ (2) 0, 0, 0

(3) $\frac{1}{\sqrt{5}},\frac{-2}{\sqrt{5}},\frac{2}{\sqrt{5}}$ (4) 1, 1, –1

PBM082

77. Three forces are acting on a particles as shown in the figure. To have the resultant forces only along the y-direction, the magnitude of the minimum additional force needed is :-



(1) 0.866 N (2) 1.732 N

(3) 0.5 N (4) 4 N

PBM083

78. The force $\vec{F}=(\hat{i}+b\hat{j}+3\hat{k})$N is rotated through an angle α, then it becomes $\{2\hat{i}+(2b-1)\hat{j}+\hat{k}\}$N. The value of b is :-

(1) 2 (2) $-\frac{2}{3}$

(3) $\frac{2}{3}$ (4) Both (1) and (2)

PBM084

DOT PRODUCT

79. If $\vec{P}.\vec{Q}=-PQ$, then angle between $\vec{P}$ and $\vec{Q}$ is :

(1) 0° (2) 180° (3) 45° (4) 60°

PBM085

80. A force $\vec{F}=(2\hat{i}+3\hat{j})$N acts on a body and displaces it by $\vec{S}=(3\hat{i}+4\hat{j})$m. The work done $(W=\vec{F}\cdot\vec{S})$ by the force is :

(1) 10J (2) 12J (3) 18J (4) 25J

PBM086

81. A force $(2\hat{i}+2\hat{j})$N displaces an object through a distance $(2\hat{i}-3\hat{j})$m. The work $(W=\vec{F}\cdot\vec{S})$ done is:

(1) –2J (2) 12J (3) 5J (4) 13J

PBM087

82. A force of 14 N acts on a particle along the vector $(3\hat{i}+2\hat{j}-6\hat{k})$. If the particle displaces from (0, 0, 0) to (2, 4, –2), the work $(W=\vec{F}\cdot\vec{S})$ done by force on the particle is :-

(1) 10J (2) 52J (3) –48J (4)14 J

PBM088

83. If a vector $(2\hat{i}+3\hat{j}+8\hat{k})$ is perpendicular to the vector $(4\hat{i}-4\alpha\hat{j}+\alpha\hat{k})$, then the value of α is :

(1) –1 (2) 1/2 (3) –1/2 (4) 2

PBM089

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84. The vector $\vec{P}=a\hat{i}+a\hat{j}+3\hat{k}$ and $\vec{Q}=a\hat{i}-2\hat{j}-\hat{k}$ are perpendicular to each other. The positive value of a is :
(1) 3 (2) 2 (3) 1 (4) zero
PBM090

85. A vector perpendicular to $\left(4\hat{i}+3\hat{j}\right)$ may be :
(1) $4\hat{i}+3\hat{j}$ (2) $7\hat{k}$
(3) $6\hat{i}$ (4) $3\hat{i}+4\hat{j}$
PBM091

86. Let $\vec{A}=\hat{i}A\cos\theta-\hat{j}A\sin\theta$, be any vector. Another vector $\vec{B}$ which is normal to $\vec{A}$ is :
(1) $\hat{i}B\cos\theta+\hat{j}B\sin\theta$
(2) $\hat{i}B\sin\theta+\hat{j}B\cos\theta$
(3) $\hat{i}B\sin\theta-\hat{j}B\cos\theta$
(4) $\hat{i}A\cos\theta-\hat{j}A\sin\theta$
PBM092

87. If $\hat{n}=a\hat{i}+b\hat{j}$ is perpendicular to the vector $\left(\hat{i}-\hat{j}\right)$, then the value of a and b may be :
(1) 1, 0 (2) –2, 0
(3) 3, 0 (4) $\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}$
PBM094

88. The vector sum of two forces is perpendicular to their vector difference. In that case, the force :
(1) Are equal to each other.
(2) Are equal to each other in magnitude.
(3) Are not equal to each other in magnitude.
(4) Cannot be predicted.
PBM095

89. Given that A = B and $\vec{A}\perp\vec{B}$. What is the angle between $(\vec{A}+\vec{B})$ and $(\vec{A}-\vec{B})$?
(1) 30^0 (2) 60^0 (3) 90^0 (4) 180^0
PBM096

90. A parallelogram is formed with $\vec{a}$ and $\vec{b}$ as the sides. Let $\vec{d}_1$ and $\vec{d}_2$ be the diagonals of the parallelogram. The value of $a^2 + b^2$ is :-
(1) $d_1^2+d_2^2$ (2) $\frac{d_1^2+d_2^2}{2}$
(3) $(d_1+d_2)^2$ (4) $d_1^2+d_2^2+d_1d_2$
PBM097

91. The angle between the two vectors $\vec{A}=3\hat{i}+4\hat{j}+5\hat{k}$ and $\vec{B}=6\hat{i}+8\hat{j}+10\hat{k}$ will be :
(1) 0° (2) 180° (3) 90° (4) 45°
PBM098

92. What is the angle between $\vec{A}$ and the resultant of $\left(\vec{A}+2\hat{B}\right)$ and $\left(2\vec{A}-2\hat{B}\right)$?
(1) 0° (2) $\tan^{-1}\left(\frac{A}{B}\right)$
(3) $\tan^{-1}\left(\frac{B}{A}\right)$ (4) $\tan^{-1}\left(\frac{A-B}{A+B}\right)$
PBM099

93. The angle between vectors $\left(\hat{i}+\hat{j}\right)$ and $\left(\hat{i}+\hat{k}\right)$ is:
(1) 90° (2) 180° (3) 0° (4) 60°
PBM100

94. The angle between $2\hat{i}+\hat{j}+2\hat{k}$ and $\hat{i}-\hat{j}+\hat{k}$ is :-
(1) 30° (2) 60°
(3) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (4) $\cos^{-1}\left(\frac{2}{3}\right)$
PBM101

95. What is the projection of $3\hat{i}+4\hat{k}$ on the z-axis ?
(1) 3 (2) 4
(3) 5 (4) zero
PBM102

96. What is the projection of $\vec{B}$ on $\vec{A}$?
(1) $\vec{A}.\vec{B}$ (2) $\vec{A}.\hat{B}$ (3) $\vec{B}.\hat{A}$ (4) $\hat{A}.\hat{B}$
PBM103

CROSS PRODUCT

97. If $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along X, Y & Z axis respectively, then tick the wrong statement :
(1) $\hat{i}\times\hat{i}=\vec{0}$ (2) $\hat{i}\times\hat{j}=\hat{k}$
(3) $\hat{i}.\hat{j}=1$ (4) $\hat{i}\times\hat{k}=-\hat{j}$
PBM105

98. The magnitude of the vector product of two vectors $\vec{A}$ and $\vec{B}$ may be :
(1) Equal to 0 (2) Less than AB
(3) Equal to AB (4) All of above
PBM106

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99. If $\vec{P}\times\vec{Q}=\vec{R}$, then which of the following statements is true ?

(1) $\vec{R}\perp\vec{P}$ (2) $\vec{R}\perp\vec{Q}$

(3) $\vec{R}\perp(\vec{P}+\vec{Q})$ (4) All of above

PBM107

100. Two vectors $\vec{P}$ and $\vec{Q}$ are inclined to each other at angle θ. Which of the following is the unit vector perpendicular to $\vec{P}$ and $\vec{Q}$?

(1) $\frac{\vec{P}\times\vec{Q}}{P.Q}$ (2) $\frac{\hat{P}\times\hat{Q}}{\sin\theta}$

(3) $\frac{\vec{P}\times\vec{Q}}{|\vec{P}\times\vec{Q}|}$ (4) Both 2 and 3

PBM108

101. Which of the following vector identities is false ?

(1) $\vec{P}+\vec{Q}=\vec{Q}+\vec{P}$ (2) $\vec{P}\times\vec{Q}=\vec{Q}\times\vec{P}$

(3) $\vec{P}.\vec{Q}=\vec{Q}.\vec{P}$ (4) $\vec{P}\times\vec{Q}=-(\vec{Q}\times\vec{P})$

PBM109

102. What is the value of $(\vec{A}+\vec{B})\times(\vec{A}-\vec{B})$?

(1) $-2(\vec{A}\times\vec{B})$ (2) $2(\vec{B}\times\vec{A})$

(3) $\vec{0}$ (4) both 1 and 2

PBM110

103. The angle between vectors $(\vec{A}\times\vec{B})$ and $-(\vec{B}\times\vec{A})$ is:

(1) π rad (2) $\frac{\pi}{2}$ rad

(3) $\frac{\pi}{4}$ rad (4) zero

PBM111

104. A vector $\vec{A}$ points vertically upward and $\vec{B}$ points towards south. The vector product $\vec{A}\times\vec{B}$ is

(1) zero

(2) along west

(3) along east

(4) vertically downward

PBM112

105. If $\frac{3}{4}|\vec{A}\times\vec{B}|=|\vec{A}.\vec{B}|$, then the angle between $\vec{A}$ and $\vec{B}$ will be :

(1) 30° (2) 45° (3) 60° (4) 53°

PBM113

106. If $\vec{A}=3\hat{i}+4\hat{j}$ and $\vec{B}=6\hat{i}+8\hat{j}$ and A and B are the magnitudes of $\vec{A}$ and $\vec{B}$, then which of the following is true ?

(1) $\vec{A}\times\vec{B}=\vec{0}$ (2) $\frac{A}{B}=2$

(3) $\vec{A}.\vec{B}=48$ (4) A = 10

PBM114

107. Two non zero vectors $\vec{A}$ and $\vec{B}$ are such that $|\vec{A}+\vec{B}|=|\vec{A}-\vec{B}|$. Then select correct alternative

(1) $\vec{A}.\vec{B}=0$ (2) $\vec{A}\times\vec{B}=\vec{0}$

(3) $\vec{A}=\vec{0}$ (4) $\vec{B}=\vec{0}$

PBM115

108. A vector $\vec{F}_1$ is along the positive Y-axis. If its vector product with another vector $\vec{F}_2$ is zero then $\vec{F}_2$ may be :-

(1) $4\hat{j}$ (2) $-(\hat{i}+\hat{j})$

(3) $(\hat{i}+\hat{k})$ (4) $-4\hat{i}$

PBM116

109. If $\vec{A}\times\vec{B}=\vec{0}$ and $\vec{B}\times\vec{C}=\vec{0}$, then the angle between $\vec{A}$ and $\vec{C}$ may be :

(1) π (2) $\frac{\pi}{4}$

(3) $\frac{\pi}{2}$ (4) None

PBM117

110. The vector $\vec{B}=6\hat{i}+2\hat{j}+S\hat{k}$ is parallel to the vector $\vec{A}=3\hat{i}+\hat{j}+2\hat{k}$ if S =

(1) 1 (2) 4 (3) 6 (4) 8

PBM118

111. If three vectors satisfy the relation $\vec{A}.\vec{B}=0$ and $\vec{A}.\vec{C}=0$, then $\vec{A}$ can be parallel to

(1) $\vec{C}$ (2) $\vec{B}$

(3) $\vec{B}\times\vec{C}$ (4) $\vec{B}.\vec{C}$

PBM119

112. For a body, angular velocity $\vec{\omega}=\hat{i}-\hat{j}+\hat{k}$ and radius vector $\vec{r}=\hat{i}+\hat{j}+\hat{k}$, then its velocity $(\vec{v}=\vec{\omega}\times\vec{r})$ is:

(1) $-5\hat{i}+2\hat{j}+3\hat{k}$ (2) $-5\hat{i}+2\hat{j}+3\hat{k}$

(3) $-2\hat{i}+2\hat{k}$ (4) $-5\hat{i}-2\hat{j}-3\hat{k}$

PBM120

113. Find unit vector perpendicular to the plane of $\vec{a}=2\hat{i}-2\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}+2\hat{k}$

(1) $\frac{2}{3}\hat{i}+\frac{1}{3}\hat{j}+\frac{2}{3}\hat{k}$ (2) $-\frac{2}{3}\hat{i}-\frac{1}{3}\hat{j}-\frac{2}{3}\hat{k}$

(3) $-\frac{2}{3}\hat{i}-\frac{1}{3}\hat{j}+\frac{2}{3}\hat{k}$ (4) $\frac{2}{3}\hat{i}+\frac{2}{3}\hat{j}-\frac{1}{3}\hat{k}$

PBM121

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	3	2	3	3	2	2	2	1	1	2	2	4	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	3	3	1	3	3	1	4	1	3	1	4	3	4	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	3	4	1	4	1	1	1	2	1	3	1	2	1	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	4	2	4	4	2	1	3	3	2	4	1	2	3	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	4	1	2	1	4	1	1	4	1	2	2	2	1	1	2
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	1	3	4	2	3	1	2	4	1	2	2	4	2	3	2
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	1	1	4	3	2	3	3	4	4	4	2	4	4	3	4
Question	106	107	108	109	110	111	112	113							
Answer	1	1	1	1	2	3	3	3							

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Exercise - II (Previous Year Questions) AIPMT/NEET

Re-AIPMT 2015

1. If vectors $\vec{A}=\cos\omega t\hat{i}+\sin\omega t\hat{j}$ and

$\vec{B}=\cos\frac{\omega t}{2}\hat{i}+\sin\frac{\omega t}{2}\hat{j}$

are functions of time, then the value of t at which they are orthogonal to each other is :

(1) t = 0 (2) $t=\frac{\pi}{4\omega}$

(3) $t=\frac{\pi}{2\omega}$ (4) $t=\frac{\pi}{\omega}$

PBM131

NEET-I 2016

2. If the magnitude of sum of two vectors is equal to the magnitude of difference of the two vectors, the angle between these vectors is :-

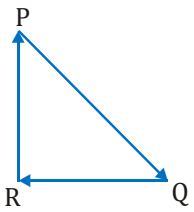
(1) 0° (2) 90°

(3) 45° (4) 180°

PBM132

NEET(UG) 2019

3. A particle moving with velocity $\vec{V}$ is acted by three forces shown by the vector triangle PQR. The velocity of the particle will:



(1) increase

(2) decrease

(3) remain constant

(4) change according to the smallest force $\overrightarrow{QR}$

PBM133

RE-NEET(UG) 2022

4. If $\vec{F}=2\hat{i}+\hat{j}-\hat{k}$ and $\vec{r}=3\hat{i}+2\hat{j}-2\hat{k}$, then the scalar and vector products of $\vec{F}$ and $\vec{r}$ have the magnitudes respectively as :

(1) 5, $\sqrt{3}$ (2) 4, $\sqrt{5}$

(3) 10, $\sqrt{2}$ (4) 10, 2

PBM134

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4
Answer	4	2	3	3

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Exercise - III (Analytical Questions) Master Your Understanding

1. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Current has magnitude as well as direction but still not considered as vector.
Reason (R) : Current do not follow vector algebra.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PBM135

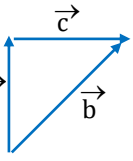
2. Two vectors $\vec{A}$ & $\vec{B}$ have equal magnitude equal to Z. If angle between $\vec{A}$ & $\vec{B}$ is 60° then match the following :

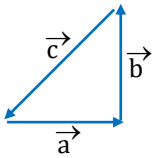
(A) $\lvert\vec{A}+\vec{B}\rvert$	(P) $\frac{\sqrt{3}}{2}Z^2$
(B) $\lvert\vec{A}-\vec{B}\rvert$	(Q) Z
(C) $\vec{A}\cdot\vec{B}$	(R) $\sqrt{3}Z$
(D) $\lvert\vec{A}\times\vec{B}\rvert$	(S) None

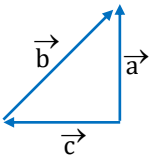
(1) (A)–(R), (B)–(S), (C)–(Q), (D)–(P)
(2) (A)–(R), (B)–(Q), (C)–(S), (D)–(P)
(3) (A)–(P), (B)–(Q), (C)–(R), (D)–(S)
(4) (A)–(Q), (B)–(P), (C)–(S), (D)–(P)

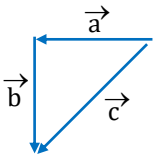
PBM137

3. Match the following

(A) $\vec{a}+\vec{b}=\vec{c}$ (P) 

(B) $\vec{a}-\vec{c}=\vec{b}$ (Q) 

(C) $\vec{b}-\vec{a}=\vec{c}$ (R) 

(D) $\vec{a}+\vec{b}+\vec{c}=\vec{0}$ (S) 

(1) (A)–(S), (B)–(R), (C)–(P), (D)–(Q)
(2) (A)–(R), (B)–(S), (C)–(Q), (D)–(P)
(3) (A)–(P), (B)–(Q), (C)–(R), (D)–(S)
(4) (A)–(S), (B)–(R), (C)–(Q), (D)–(P)

PBM138

4. Given below are two statements :
Statement I : Resultant of 2 forces of magnitude 4N and 5N can be 2N in magnitude.
Statement II : $\left|\lvert\vec{a}\rvert-\lvert\vec{b}\rvert\right| \le \lvert\vec{a}+\vec{b}\rvert \le \lvert\vec{a}\rvert+\lvert\vec{b}\rvert$
In the light of the above statements, choose the most appropriate answer from the options given below :
(1) Both statement I and statement II are correct.
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statements I and statements II are incorrect.

PBM139

5. Given below are two statements :
Statement I : Two null vector have same direction.
Statement II : $\vec{A}\times\vec{B}$ lies in the plane of $\vec{A}+\vec{B}$
In the light of the above statements, choose the most appropriate answer from the options given below :
(1) Both statement I and statement II are correct.
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statements I and statements II are incorrect.

PBM140

6. Which of the following is correct :

(i) $\vec{A}\cdot\vec{B}$ is a vector quantity

(ii) $\vec{A}\times\vec{B}$ is perpendicular to plane of $\vec{A}+\vec{B}$

(iii) For two orthogonal vectors $\vec{A}\cdot\vec{B}=0$

(iv) If vectors are parallel or antiparallel, then $\vec{A}\times\vec{B}=\vec{0}$

(1) (i) only (2) (i) & (ii)

(2) (iii) & (iv) only (4) (ii), (iii) & (iv)

PBM141

7. Given below are two statements :

Statement-I : For every small angle θ, we may use approximation $\sin\theta \approx \theta \approx \tan\theta$.

Statement-II : For very small angle θ, the hypotenuse and the base become approximately of the same length.

(1) Statement-I is true, Statement-II is true; Statement-II is a correct explanation for Statement-I.

(2) Statement-I is true, Statement-II is true; Statement-II is not a correct explanation for Statement-I.

(3) Statement-I is true, Statement-II is false.

(4) Statement-I is false, Statement-II is true.

PBM143

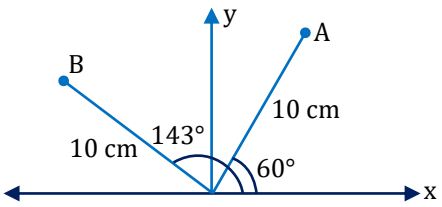
8. Refer the given figure and identify correct statement(s)

(A) Distance of A from x-axis is $5\sqrt{3}$ cm.

(B) Distance of B from x-axis is 6 cm.

(C) Distance of A from y-axis is 5 cm.

(D) Distance of B from y-axis is 8 cm.



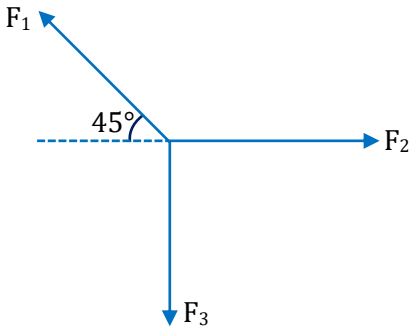
Options :-

(1) A, C (2) A, D

(3) A, B, C (4) A, B, C, D

PBM145

9. Three forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$ are represented as shown. Each of them is of equal magnitude.



Column I (Combination)	Column II (Approximate Direction)
(A) $\vec{F}_1+\vec{F}_2+\vec{F}_3$	(P)
(B) $\vec{F}_1-\vec{F}_2+\vec{F}_3$	(Q)
(C) $\vec{F}_1-\vec{F}_2-\vec{F}_3$	(R)
(D) $\vec{F}_2-\vec{F}_1-\vec{F}_3$	(S)

Options :-

(1) A → R, B → Q, C → P, D → S

(2) A → Q, B → P, C → R, D → S

(3) A → Q, B → R, C → P, D → S

(4) A → S, B → P, C → R, D → Q

PBM147

10. Which of the following statement is/are true ?

(a) Two vectors of unequal magnitude can add up to zero.

(b) Three vectors of unequal magnitude can add up to zero, if they lie in a plane.

(c) Three vectors of unequal magnitude can added upto zero, if they do not lie in same plane.

Options :-

(1) Only a (2) a, b and c

(3) only b (4) only c

PBM148

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	1	2	1	1	4	4	1	4	3	3

Exercise - I (Conceptual Questions) Build Up Your Understanding

TRIGONOMETRY

1. As θ increases from 0° to 90°, the value of sinθ :-
(1) Increases
(2) Decreases
(3) Remains constant
(4) First decreases then increases.

PBM001

2. If $\sin\theta = \frac{\sqrt{2}}{\sqrt{3}}$ and θ lies in the first quadrant, the value of tan θ is :

(1) $\sqrt{2}$ (2) $\frac{1}{\sqrt{2}}$

(3) $\frac{\sqrt{3}}{\sqrt{2}}$ (4) $\frac{1}{2}$

PBM002

3. Find θ for which sinθ = cosθ, if 180° < θ < 360°
(1) 135° (2) 315°
(3) 225° (4) 150°

PBM003

4. If $\sin\theta_1 + \sin\theta_2 + \sin\theta_3 = 3$
then value of $\cos\theta_1 + \cos\theta_2 + \cos\theta_3$ is :-
(1) 3 (2) 0 (3) –3 (4) 1

PBM004

5. The greatest value of the function 7 sinθ – 24 cosθ is
(1) 12 (2) 13 (3) 25 (4) 17

PBM010

6. The length of hypotenuse of a right angle triangle exceeds the length of its base by 2 cm and exceeds twice the length of altitude by 1 cm. Find length of each side of the triangle.
(1) 6, 8, 10 (2) 7, 24, 25
(3) 8, 15, 17 (4) 7, 40, 41

PBM011

CALCULUS

7. If $x = at^4$ and $y = bt^3$. Find $\frac{dy}{dx}$

(1) $\frac{4a}{3bt}$ (2) $\frac{3b}{4at}$

(3) $\frac{3a}{4bt}$ (4) $\frac{4b}{3at}$

PBM012

8. A metallic disc is being heated. Its area at any time t is given by $A = 5t^2 + 4t + 8$. Calculate rate of increase in area at t = 3s.
(1) 30 m^2/s (2) 34 m^2/s
(3) 28 m^2/s (4) 20 m^2/s

PBM013

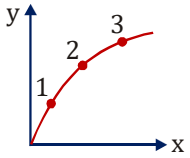
9. The side of a square is increasing at the rate of 0.1 cm/s. The rate of increase of perimeter w.r.t. time is :
(1) 0.2 cm/s (2) 0.4 cm/s
(3) 0.6 cm/s (4) 0.8 cm/s

PBM014

10. A particle moves along the straight line 3y =x+5. Which coordinate changes at a faster rate ?
(1) x–coordinate
(2) y–coordinate
(3) Both x and y coordinates
(4) Data insufficient.

PBM015

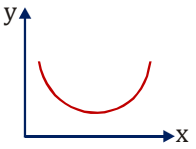
11. The slope of graph as shown in figure at points 1, 2 and 3 is m_1, m_2 and m_3 respectively then



(1) $m_1 > m_2 > m_3$ (2) $m_1 < m_2 < m_3$
(3) $m_1 = m_2 = m_3$ (4) $m_1 = m_3 > m_2$

PBM016

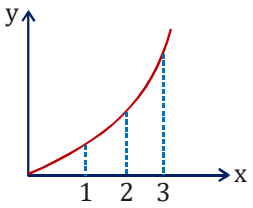
12. Magnitude of slope of the shown graph.



(1) First increases then decreases
(2) First decreases then increases
(3) Increases
(4) Decreases

PBM017

13. Calculate the area enclosed under the curve $f(x) = x^2$ between the limits x = 2 and x = 3



(1) 5 (2) $\frac{19}{3}$ (3) $\frac{17}{3}$ (4) 8

PBM018

14. Find the value of $\int_0^4 |(1-x)|.dx$

(1) zero (2) 1 (3) 4 (4) 5

PBM019

GEOMETRY & GRAPHS

15. The equation of a curve is given as $y = x^2+2-3x$. The curve intersects the y-axis at

(1) (1, 0) (2) (2, 0)

(3) (0, 2) (4) No where

PBM020

16. Two particles A and B are moving in XY-plane. Their positions vary with time t according to relation :

$x_A(t) = 3t$, $x_B(t) = 6$

$y_A(t) = t$, $y_B(t) = 2 + 3t^2$

Distance between two particles at t = 2 is :

(1) 12 (2) 13 (3) 5 (4) $\sqrt{12}$

PBM021

17. The distance between points (a + b, b + c) and (a – b, c – b) is :-

(1) $2\sqrt{a^2+b^2}$ (2) $2\sqrt{b^2+c^2}$

(3) $2\sqrt{2}b$ (4) $\sqrt{a^2-c^2}$

PBM022

18. A dog is at point A(0, 3, 4)m and cat is at B(5,3,–8)m The dog is free to move but cat is fixed. The minimum distance travelled by dog to catch the cat is :-

(1) 25m (2) 12m (3) 13m (4) 20m

PBM023

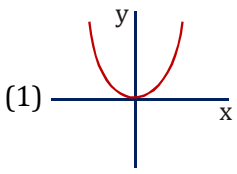
19. A particular straight line passes through origin and a point whose abscissa is equal to ordinate. The equation of such straight line is :

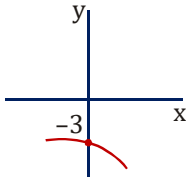
(1) y = x (2) y = 2x

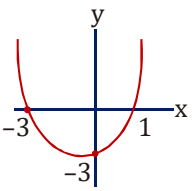
(3) y = –4x (4) $y = -\frac{x}{4}$

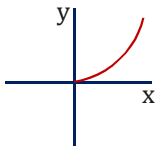
PBM024

20. If $y = x^2 + 2x - 3$, then y-x graph is :-

(1) 

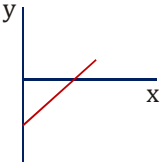
(2) 

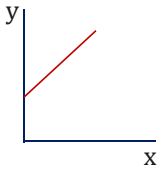
(3) 

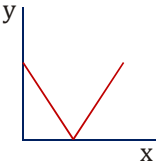
(4) 

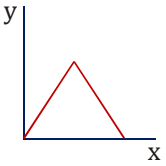
PBM025

21. If y =|x–1|, then y-x graph is :-

(1) 

(2) 

(3) 

(4) 

PBM026

22. Frequency f of a simple pendulum depends on its length ℓ and acceleration g due to gravity according to the following equation $f = \frac{1}{2\pi}\sqrt{\frac{g}{\ell}}$. Graph between which of the following quantities is a parabola ?

(1) f on the ordinate and $1/\ell$ on the abscissa

(2) f on the ordinate and $\sqrt{\ell}$ on the abscissa

(3) f^2 on the ordinate and ℓ on the abscissa

(4) f^2 on the ordinate and $1/\ell$ on the abscissa

PBM028

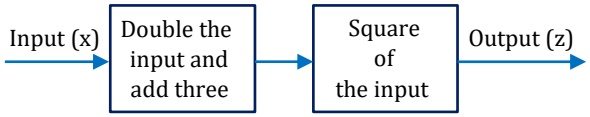
ALGEBRA

23. The sum of the series $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + ...\infty$ is :-

(1) $\frac{8}{7}$ (2) $\frac{6}{5}$ (3) $\frac{2}{3}$ (4) $\frac{3}{2}$

PBM029

24. In the given figure, each box represents a function machine. A function machine illustrates what it does with the input.



Which of the following statements is correct ?

(1) $z = (2x+3)^2$ (2) $z = 2(x+3)$

(3) $z=\sqrt{2x+3}$ (4) $z=\sqrt{2(x+3)}$

PBM030

DEFINITION & TYPES OF VECTOR

25. Which of the following statements is false :

(1) Mass, speed and energy are scalars

(2) Momentum, force and torque are vectors

(3) Distance is a vector while displacement is a scalar

(4) A scalar has only magnitude whereas as a vector has both magnitude and direction

PBM031

26. A physical quantity which has a direction and obeys triangle law of vector addition :

(1) must be a vector (2) may be a vector

(3) must be a scalar (4) none of the above

PBM032

27. The forces, which meet at one point and their lines of action lie in one plane, are called :

(1) non-concurrent and non-coplanar forces

(2) non-concurrent and coplanar forces

(3) concurrent and non-coplanar forces

(4) concurrent and coplanar forces

PBM033

28. Which of the following physical quantities is not an axial vector ?

(1) angular velocity

(2) angular momentum

(3) velocity

(4) torque

PBM034

29. The direction of the angular velocity vector is along :

(1) Along the tangent of circular path

(2) Along the direction of radius vector

(3) Opposite to the direction of radius vector

(4) Along the axis of rotation

PBM035

30. If $\hat{n}$ is a unit vector in the direction of the vector $\vec{A}$, then :-

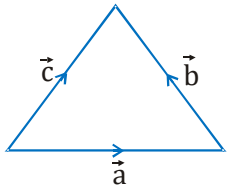
(1) $\hat{n}=\frac{\vec{A}}{|\vec{A}|}$ (2) $\hat{n}=\vec{A}|\vec{A}|$

(3) $\hat{n}=\frac{|\vec{A}|}{\vec{A}}$ (4) None of the above

PBM036

ADDITION & SUBTRACTION, MULTIPLICATION & DIVISION OF A VECTOR BY A SCALAR

31. In the given figure, $\vec{a}+\vec{b}+\vec{c}$ is



(1) $2\vec{a}$ (2) $2\vec{b}$ (3) $2\vec{c}$ (4) $\vec{a}+\vec{b}$

PBM037

32. Vector sum of two forces of 10N and 5N can be :

(1) 4N (2) 18N (3) 12N (4) 2N

PBM038

33. The vector sum of the forces of 10 newton and 8 newton cannot be :

(1) 2N (2) 8N

(3) 18N (4) 20N

PBM039

34. Which of the following pair of forces can give a resultant force of 2 N ?

(1) 1 N and 2 N (2) 1 N and 5 N

(3) 2 N and 5 N (4) 1 N and 4 N

PBM040

35. Two vectors of equal magnitude have a resultant equal to either of them in magnitude. The angle between the vector is :

(1) 60° (2) 90°

(3) 105° (4) 120°

PBM041

36. $\vec{R}=\vec{a}+\vec{b}$, where $\vec{R},\vec{a}$ and $\vec{b}$ are non-zero vector.

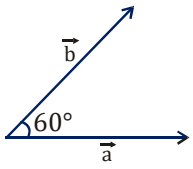
If R = a = b, the angle between $\vec{R}$ and $\vec{a}$ is :-

(1) $\frac{\pi}{3}$ (2) $\frac{2\pi}{3}$

(3) $\frac{\pi}{4}$ (4) $\frac{\pi}{6}$

PBM042

37. If resultant of two vectors $\vec{a}$ and $\vec{b}$ shown in the figure is $\sqrt{7}\,b$, then value of $\frac{b}{a}$ is :-



(1) $\frac{1}{2}$ (2) 1 (3) 2 (4) 3

PBM043

38. If $\vec{A}+\vec{B}=\vec{C}$ and A + B = C, then the angle between $\vec{A}$ and $\vec{C}$ is :

(1) 0 (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{2}$ (4) π

PBM044

39. Two vectors $\vec{A}$ and $\vec{B}$ are such that $\vec{A}+\vec{B}=\vec{C}$ and A – B = C. Which of the following statements, is correct ?

(1) $\vec{A}$ is parallel to $\vec{B}$

(2) $\vec{A}$ is anti-parallel to $\vec{B}$

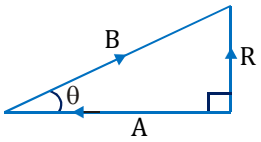
(3) $\vec{A}$ is perpendicular to $\vec{B}$

(4) $\vec{A}$ and $\vec{B}$ are equal in magnitude

PBM045

40. In vector diagram shown in figure where $(\vec{R})$ is the resultant of vectors ($\vec{A}$) and ($\vec{B}$).

If R = $\frac{B}{2}$, then value of angle θ is :



(1) 30° (2) 45°

(3) 60° (4) 75°

PBM046

41. The resultant of $\vec{A}$ and $\vec{B}$ makes an angle α with $\vec{A}$ and β with $\vec{B}$, then :

(1) α < β

(2) α < β if A < B

(3) α < β if A > B

(4) α < β if A = B

PBM047

42. If two forces of equal magnitude acting on a particle. The angle between forces is 60°. The resultant force on the particle is :-

(1) at an angle of 30° with each force.

(2) at an angle of 45° with one of the force

(3) at an angle of 15° with one of the force

(4) at an angle lying between 45° to 60°

PBM048

43. The resultant of $\vec{A}$ and $\vec{B}$ is perpendicular to $\vec{B}$. What is the angle between $\vec{A}$ and $\vec{B}$?

(1) $\cos^{-1}\left(\frac{A}{B}\right)$ (2) $\cos^{-1}\left(-\frac{B}{A}\right)$

(3) $\sin^{-1}\left(\frac{A}{B}\right)$ (4) $\sin^{-1}\left(-\frac{A}{B}\right)$

PBM049

44. The resultant of two forces make 30° and 60° angles with them and has magnitude 40 N. The magnitudes of two vectors are :-

(1) $20\sqrt{3}\,N, 20N$ (2) $30N, 20\sqrt{3}\,N$

(3) 40N, 30N (4) $3\sqrt{3}\,N, 40\sqrt{3}\,N$

PBM050

45. Given that $\vec{P}+\vec{Q}=\vec{P}-\vec{Q}$. This can be true when:

(1) $\vec{P}=\vec{Q}$

(2) $\vec{Q}=\vec{0}$

(3) Neither $\vec{P}$ nor $\vec{Q}$ is a null vector

(4) Data Insufficient

PBM051

46. The resultant of $\vec{A}$ & $\vec{B}$ is $\vec{R}_1$. On reversing the vector $\vec{B}$, the resultant becomes $\vec{R}_2$. What is the value of $2(R_1^2+R_2^2)$?

(1) $4(A^2 + B^2)$ (2) $A^2 - B^2$

(3) $2(A^2 + B^2)$ (4) $2(A^2 - B^2)$

PBM052

47. If vectors $\vec{A}$ and $\vec{B}$ are such that $|\vec{A}+\vec{B}| = |\vec{A}|=|\vec{B}|$, then $|\vec{A}-\vec{B}|$ may be equated to

(1) $\frac{\sqrt{3}}{2}|\vec{A}|$ (2) $|\vec{A}|$

(3) $\sqrt{2}|\vec{A}|$ (4) $\sqrt{3}|\vec{A}|$

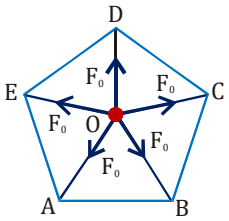
PBM053

48. If the difference of two unit vectors is a unit vector, then the magnitude of their sum is :

(1) $\sqrt{2}$ (2) $\sqrt{3}$ (3) $\frac{1}{\sqrt{2}}$ (4) $\sqrt{5}$

PBM054

49. In the given figure, O is the centre of the regular pentagon ABCDE. Five forces each of magnitude F_0 are acted as shown in the figure. The resultant force is :-



(1) 5 F_0 (2) 5 F_0 cos72°

(3) 5 F_0 sin72° (4) zero

PBM055

50. Two vectors $\vec{A}$ and $\vec{B}$ lie in a plane, another vector $\vec{C}$ lies in the same plane, then the resultant of these three vectors i.e. $\vec{A}+\vec{B}+\vec{C}$:

(1) may be zero

(2) cannot be zero

(3) lies in the plane containing $\vec{A}$ & $\vec{B}$

(4) Both 1 & 3

PBM056

51. Which of the following sets of concurrent forces may be in equilibrium ?

(1) F_1 = 3N, F_2 = 5N, F_3 = 1N, F_4 = 10N

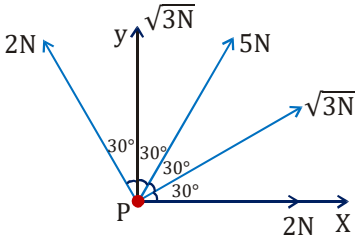
(2) F_1 = 3N, F_2 = 5N, F_3 = 9N, F_4 = 4N

(3) F_1 = 3N, F_2 = 5N, F_3 = 6N, F_4 = 15N

(4) F_1 = 3N, F_2 = 5N, F_3 =15N, F_4 = 5N

PBM057

52. Five forces 2N, $\sqrt{3}$N, 5N, $\sqrt{3}$N and 2N respectively act at a particle P as shown in the figure.



The resultant force on the particle P is.

(1) 10 N making angle 60° with x-axis.

(2) 10 N making angle 60° with y-axis.

(3) 20 N along y-axis

(4) None of these

PBM058

53. If $\vec{a}+\vec{b}+\vec{c}=0$. The angle between $\vec{a}$ and $\vec{b}$, $\vec{b}$ and $\vec{c}$ are 150° and 120°, respectively. Then, the magnitude of vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are in ratio of :-

(1) 1 : 2 : 3 (2) 1 : 2 : $\sqrt{3}$

(3) $\sqrt{3}$: 2 : 1 (4) 2 : $\sqrt{3}$: 1

PBM059

54. The magnitudes of vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$ are respectively 7, 24 and 25 units and $\vec{A}+\vec{B}=\vec{C}$, then the angle between $\vec{A}$ and $\vec{B}$ is :

(1) 0 (2) $\pi/3$

(3) $\pi/2$ (4) $\pi/4$

PBM060

55. If vectors $\vec{P}$, $\vec{Q}$ and $\vec{R}$ have magnitudes 5, 12 and 13 units and $\vec{P}+\vec{Q}=\vec{R}$, the angle between $\vec{P}$ and $\vec{R}$ is :

(1) $\cos^{-1}\left(\frac{5}{12}\right)$ (2) $\cos^{-1}\left(\frac{5}{13}\right)$

(3) $\cos^{-1}\left(\frac{12}{13}\right)$ (4) $\cos^{-1}\left(\frac{2}{13}\right)$

PBM061

56. If the vectors $\left(\hat{i}+\hat{j}+\hat{k}\right)$ and $3\hat{i}$ form two sides of a triangle, then 3rd side of the triangle can be :

(1) $\sqrt{18}$ unit (2) $\sqrt{6}$ unit

(3) $\frac{3}{\sqrt{2}}$ unit (4) both (1) & (2)

PBM062

57. The minimum number of vectors of equal magnitude required to produce a zero resultant is:

(1) 2 (2) 3

(3) 4 (4) more than 4

PBM063

58. How many minimum number of coplanar vectors having different magnitudes can be added to give zero resultant ?
(1) 2 (2) 3 (3) 4 (4) 5
PBM064

59. How many minimum number of vectors in different planes can be added to give zero resultant ?
(1) 2 (2) 3 (3) 4 (4) 5
PBM065

60. What happens, when we multiply a vector by 2?
(1) direction reverses and unit changes
(2) direction remains same and magnitude is doubled
(3) direction remains unchanged and unit changes
(4) none of these
PBM066

RESOLUTION OF VECTOR

61. What is the maximum number of components into which a vector can be split ?
(1) 2 (2) 3
(3) 4 (4) Infinite
PBM067

62. What is the maximum number of rectangular components into which a vector can be split in its own plane ?
(1) 2 (2) 3
(3) 4 (4) Infinite
PBM068

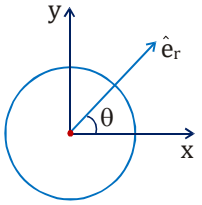
63. What is the maximum number of rectangular components into which a vector can be split in space ?
(1) 2 (2) 3 (3) 4 (4) ∞
PBM069

64. The unit vector along $\hat{i}-2\hat{j}$ is :
(1) $\frac{\hat{i}-2\hat{j}}{\sqrt{5}}$ (2) $\hat{i}+\hat{j}$
(3) $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$ (4) $\frac{\hat{i}-\hat{j}}{\sqrt{5}}$
PBM070

65. If $\vec{A}+\vec{B}$ is a unit vector along y-axis and $\vec{A}=\hat{i}-\hat{j}+\hat{k}$, then what is $\vec{B}$?
(1) $\hat{j}+\hat{k}$ (2) $\hat{j}-\hat{k}$
(3) $\hat{i}+\hat{j}+\hat{k}$ (4) $2\hat{j}-\hat{i}-\hat{k}$
PBM071

66. If a unit vector is represented by $0.3\hat{i}-0.4\hat{j}+c\hat{k}$, then the value of 'c' is :
(1) $\sqrt{0.75}$ (2) $\sqrt{0.25}$
(3) $\sqrt{0.01}$ (4) $\sqrt{0.39}$
PBM072

67. $\hat{e}_r$ is unit vector along radius of a circle shown in figure. $\hat{e}_r$ can be represented as :-



(1) $\cos\theta\hat{i}+\sin\theta\hat{j}$ (2) $\sin\theta\hat{i}+\cos\theta\hat{j}$
(3) $\cos\theta\hat{i}-\sin\theta\hat{j}$ (4) $-\cos\theta\hat{i}+\sin\theta\hat{j}$
PBM073

68. Forces 7N, 24N, 25N act at a point in mutually perpendicular directions. The magnitude of the resultant force is :
(1) 19 N (2) 13 N
(3) 26 N (4) $25\sqrt{2}$ N
PBM074

69. The angle that the vector $\vec{A}=2\hat{i}+3\hat{j}$ makes with x-axis is :
(1) $\tan^{-1}(3/2)$ (2) $\tan^{-1}(2/3)$
(3) $\sin^{-1}(2/3)$ (4) $\cos^{-1}(3/2)$
PBM075

70. What vector must be added to the other vectors $\hat{i}-2\hat{j}+2\hat{k}$ and $2\hat{i}+\hat{j}-\hat{k}$, so that the resultant may be a unit vector along y-axis?
(1) $2\hat{i}+\hat{j}-\hat{k}$ (2) $-3\hat{i}+2\hat{j}-\hat{k}$
(3) $-2\hat{i}-\hat{j}-\hat{k}$ (4) $3\hat{i}-2\hat{j}-\hat{k}$
PBM076

71. The unit vector parallel to the resultant of the vectors $\vec{A}=4\hat{i}+3\hat{j}+6\hat{k}$ and $\vec{B}=2\hat{i}-3\hat{j}+2\hat{k}$ is :
(1) $\frac{1}{10}\left[8\hat{i}+6\hat{k}\right]$ (2) $\frac{1}{10}\left[6\hat{i}+8\hat{k}\right]$
(3) $\frac{1}{10}\left[6\hat{i}+6\hat{k}+6\hat{j}\right]$ (4) $\frac{1}{10}\left[6\hat{j}+8\hat{k}\right]$
PBM077

72. If $\vec{a}=2\hat{i}+2\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}+\hat{k}$. Find a vector $\vec{c}$ which is parallel to $\vec{a}$ but has magnitude three times that of $\vec{b}$.

(1) $\vec{c}=3\hat{i}+3\hat{j}+3\hat{k}$

(2) $\vec{c}=2\sqrt{3}\hat{i}+2\sqrt{3}\hat{j}-\sqrt{3}\hat{k}$

(3) $\vec{c}=6\hat{i}+6\hat{j}-3\hat{k}$

(4) $\vec{c}=3\hat{i}+6\hat{j}-6\hat{k}$

PBM078

73. The velocity of a particle is $\vec{v}=(\hat{i}+\hat{j}-\hat{k})$m/s. A force of $10\sqrt{3}$N parallel to velocity in vector form is :-

(1) $(10\hat{i}+10\hat{j}-10\hat{k})$N

(2) $(\hat{i}+\hat{j}-\hat{k})10\sqrt{3}$N

(3) $(\hat{i}+\hat{j}-\hat{k})$N

(4) None of these

PBM079

74. The direction cosines of a vector $\sqrt{2}\hat{i}+\sqrt{2}\hat{j}+\hat{k}$ are:-

(1) $\frac{\sqrt{2}}{\sqrt{5}},\frac{\sqrt{2}}{\sqrt{5}},\frac{1}{\sqrt{5}}$ (2) $\frac{1}{\sqrt{5}},\frac{1}{\sqrt{5}},\frac{1}{5}$

(3) $\frac{1}{5},\frac{1}{5},\frac{1}{\sqrt{5}}$ (4) $\frac{1}{\sqrt{5}},\frac{1}{\sqrt{5}},\frac{1}{\sqrt{5}}$

PBM080

75. Vector $\vec{P}$ makes angles α, β & γ with the X, Y and Z axes respectively, then $\cos^2\alpha+\cos^2\beta+\cos^2\gamma=$

(1) 0 (2) 1 (3) 2 (4) 3

PBM081

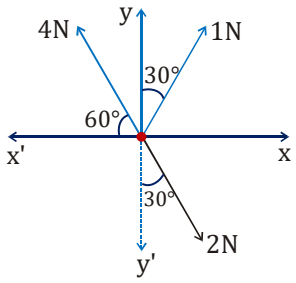
76. Find the direction cosines of vector $(\vec{a}-\vec{b})$, if $\vec{a}=2\hat{i}+3\hat{j}+\hat{k}$ and $\vec{b}=2\hat{i}+2\hat{j}+3\hat{k}$

(1) $0,\frac{1}{\sqrt{5}},\frac{-2}{\sqrt{5}}$ (2) 0, 0, 0

(3) $\frac{1}{\sqrt{5}},\frac{-2}{\sqrt{5}},\frac{2}{\sqrt{5}}$ (4) 1, 1, –1

PBM082

77. Three forces are acting on a particles as shown in the figure. To have the resultant forces only along the y-direction, the magnitude of the minimum additional force needed is :-



(1) 0.866 N (2) 1.732 N

(3) 0.5 N (4) 4 N

PBM083

78. The force $\vec{F}=(\hat{i}+b\hat{j}+3\hat{k})$N is rotated through an angle α, then it becomes $\{2\hat{i}+(2b-1)\hat{j}+\hat{k}\}$N. The value of b is :-

(1) 2 (2) $-\frac{2}{3}$

(3) $\frac{2}{3}$ (4) Both (1) and (2)

PBM084

DOT PRODUCT

79. If $\vec{P}.\vec{Q}=-PQ$, then angle between $\vec{P}$ and $\vec{Q}$ is :

(1) 0° (2) 180° (3) 45° (4) 60°

PBM085

80. A force $\vec{F}=(2\hat{i}+3\hat{j})$N acts on a body and displaces it by $\vec{S}=(3\hat{i}+4\hat{j})$m. The work done $(W=\vec{F}\cdot\vec{S})$ by the force is :

(1) 10J (2) 12J (3) 18J (4) 25J

PBM086

81. A force $(2\hat{i}+2\hat{j})$N displaces an object through a distance $(2\hat{i}-3\hat{j})$m. The work $(W=\vec{F}\cdot\vec{S})$ done is:

(1) –2J (2) 12J (3) 5J (4) 13J

PBM087

82. A force of 14 N acts on a particle along the vector $(3\hat{i}+2\hat{j}-6\hat{k})$. If the particle displaces from (0, 0, 0) to (2, 4, –2), the work $(W=\vec{F}\cdot\vec{S})$ done by force on the particle is :-

(1) 10J (2) 52J (3) –48J (4)14 J

PBM088

83. If a vector $(2\hat{i}+3\hat{j}+8\hat{k})$ is perpendicular to the vector $(4\hat{i}-4\alpha\hat{j}+\alpha\hat{k})$, then the value of α is :

(1) –1 (2) 1/2 (3) –1/2 (4) 2

PBM089

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84. The vector $\vec{P}=a\hat{i}+a\hat{j}+3\hat{k}$ and $\vec{Q}=a\hat{i}-2\hat{j}-\hat{k}$ are perpendicular to each other. The positive value of a is :
(1) 3 (2) 2 (3) 1 (4) zero
PBM090

85. A vector perpendicular to $\left(4\hat{i}+3\hat{j}\right)$ may be :
(1) $4\hat{i}+3\hat{j}$ (2) $7\hat{k}$
(3) $6\hat{i}$ (4) $3\hat{i}+4\hat{j}$
PBM091

86. Let $\vec{A}=\hat{i}A\cos\theta-\hat{j}A\sin\theta$, be any vector. Another vector $\vec{B}$ which is normal to $\vec{A}$ is :
(1) $\hat{i}B\cos\theta+\hat{j}B\sin\theta$
(2) $\hat{i}B\sin\theta+\hat{j}B\cos\theta$
(3) $\hat{i}B\sin\theta-\hat{j}B\cos\theta$
(4) $\hat{i}A\cos\theta-\hat{j}A\sin\theta$
PBM092

87. If $\hat{n}=a\hat{i}+b\hat{j}$ is perpendicular to the vector $\left(\hat{i}-\hat{j}\right)$, then the value of a and b may be :
(1) 1, 0 (2) –2, 0
(3) 3, 0 (4) $\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}$
PBM094

88. The vector sum of two forces is perpendicular to their vector difference. In that case, the force :
(1) Are equal to each other.
(2) Are equal to each other in magnitude.
(3) Are not equal to each other in magnitude.
(4) Cannot be predicted.
PBM095

89. Given that A = B and $\vec{A}\perp\vec{B}$. What is the angle between $(\vec{A}+\vec{B})$ and $(\vec{A}-\vec{B})$?
(1) 30^0 (2) 60^0 (3) 90^0 (4) 180^0
PBM096

90. A parallelogram is formed with $\vec{a}$ and $\vec{b}$ as the sides. Let $\vec{d}_1$ and $\vec{d}_2$ be the diagonals of the parallelogram. The value of $a^2 + b^2$ is :-
(1) $d_1^2+d_2^2$ (2) $\frac{d_1^2+d_2^2}{2}$
(3) $(d_1+d_2)^2$ (4) $d_1^2+d_2^2+d_1d_2$
PBM097

91. The angle between the two vectors $\vec{A}=3\hat{i}+4\hat{j}+5\hat{k}$ and $\vec{B}=6\hat{i}+8\hat{j}+10\hat{k}$ will be :
(1) 0° (2) 180° (3) 90° (4) 45°
PBM098

92. What is the angle between $\vec{A}$ and the resultant of $\left(\vec{A}+2\hat{B}\right)$ and $\left(2\vec{A}-2\hat{B}\right)$?
(1) 0° (2) $\tan^{-1}\left(\frac{A}{B}\right)$
(3) $\tan^{-1}\left(\frac{B}{A}\right)$ (4) $\tan^{-1}\left(\frac{A-B}{A+B}\right)$
PBM099

93. The angle between vectors $\left(\hat{i}+\hat{j}\right)$ and $\left(\hat{i}+\hat{k}\right)$ is:
(1) 90° (2) 180° (3) 0° (4) 60°
PBM100

94. The angle between $2\hat{i}+\hat{j}+2\hat{k}$ and $\hat{i}-\hat{j}+\hat{k}$ is :-
(1) 30° (2) 60°
(3) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ (4) $\cos^{-1}\left(\frac{2}{3}\right)$
PBM101

95. What is the projection of $3\hat{i}+4\hat{k}$ on the z-axis ?
(1) 3 (2) 4
(3) 5 (4) zero
PBM102

96. What is the projection of $\vec{B}$ on $\vec{A}$?
(1) $\vec{A}.\vec{B}$ (2) $\vec{A}.\hat{B}$ (3) $\vec{B}.\hat{A}$ (4) $\hat{A}.\hat{B}$
PBM103

CROSS PRODUCT

97. If $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along X, Y & Z axis respectively, then tick the wrong statement :
(1) $\hat{i}\times\hat{i}=\vec{0}$ (2) $\hat{i}\times\hat{j}=\hat{k}$
(3) $\hat{i}.\hat{j}=1$ (4) $\hat{i}\times\hat{k}=-\hat{j}$
PBM105

98. The magnitude of the vector product of two vectors $\vec{A}$ and $\vec{B}$ may be :
(1) Equal to 0 (2) Less than AB
(3) Equal to AB (4) All of above
PBM106

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99. If $\vec{P}\times\vec{Q}=\vec{R}$, then which of the following statements is true ?

(1) $\vec{R}\perp\vec{P}$ (2) $\vec{R}\perp\vec{Q}$

(3) $\vec{R}\perp(\vec{P}+\vec{Q})$ (4) All of above

PBM107

100. Two vectors $\vec{P}$ and $\vec{Q}$ are inclined to each other at angle θ. Which of the following is the unit vector perpendicular to $\vec{P}$ and $\vec{Q}$?

(1) $\frac{\vec{P}\times\vec{Q}}{P.Q}$ (2) $\frac{\hat{P}\times\hat{Q}}{\sin\theta}$

(3) $\frac{\vec{P}\times\vec{Q}}{|\vec{P}\times\vec{Q}|}$ (4) Both 2 and 3

PBM108

101. Which of the following vector identities is false ?

(1) $\vec{P}+\vec{Q}=\vec{Q}+\vec{P}$ (2) $\vec{P}\times\vec{Q}=\vec{Q}\times\vec{P}$

(3) $\vec{P}.\vec{Q}=\vec{Q}.\vec{P}$ (4) $\vec{P}\times\vec{Q}=-(\vec{Q}\times\vec{P})$

PBM109

102. What is the value of $(\vec{A}+\vec{B})\times(\vec{A}-\vec{B})$?

(1) $-2(\vec{A}\times\vec{B})$ (2) $2(\vec{B}\times\vec{A})$

(3) $\vec{0}$ (4) both 1 and 2

PBM110

103. The angle between vectors $(\vec{A}\times\vec{B})$ and $-(\vec{B}\times\vec{A})$ is:

(1) π rad (2) $\frac{\pi}{2}$ rad

(3) $\frac{\pi}{4}$ rad (4) zero

PBM111

104. A vector $\vec{A}$ points vertically upward and $\vec{B}$ points towards south. The vector product $\vec{A}\times\vec{B}$ is

(1) zero

(2) along west

(3) along east

(4) vertically downward

PBM112

105. If $\frac{3}{4}|\vec{A}\times\vec{B}|=|\vec{A}.\vec{B}|$, then the angle between $\vec{A}$ and $\vec{B}$ will be :

(1) 30° (2) 45° (3) 60° (4) 53°

PBM113

106. If $\vec{A}=3\hat{i}+4\hat{j}$ and $\vec{B}=6\hat{i}+8\hat{j}$ and A and B are the magnitudes of $\vec{A}$ and $\vec{B}$, then which of the following is true ?

(1) $\vec{A}\times\vec{B}=\vec{0}$ (2) $\frac{A}{B}=2$

(3) $\vec{A}.\vec{B}=48$ (4) A = 10

PBM114

107. Two non zero vectors $\vec{A}$ and $\vec{B}$ are such that $|\vec{A}+\vec{B}|=|\vec{A}-\vec{B}|$. Then select correct alternative

(1) $\vec{A}.\vec{B}=0$ (2) $\vec{A}\times\vec{B}=\vec{0}$

(3) $\vec{A}=\vec{0}$ (4) $\vec{B}=\vec{0}$

PBM115

108. A vector $\vec{F}_1$ is along the positive Y-axis. If its vector product with another vector $\vec{F}_2$ is zero then $\vec{F}_2$ may be :-

(1) $4\hat{j}$ (2) $-(\hat{i}+\hat{j})$

(3) $(\hat{i}+\hat{k})$ (4) $-4\hat{i}$

PBM116

109. If $\vec{A}\times\vec{B}=\vec{0}$ and $\vec{B}\times\vec{C}=\vec{0}$, then the angle between $\vec{A}$ and $\vec{C}$ may be :

(1) π (2) $\frac{\pi}{4}$

(3) $\frac{\pi}{2}$ (4) None

PBM117

110. The vector $\vec{B}=6\hat{i}+2\hat{j}+S\hat{k}$ is parallel to the vector $\vec{A}=3\hat{i}+\hat{j}+2\hat{k}$ if S =

(1) 1 (2) 4 (3) 6 (4) 8

PBM118

111. If three vectors satisfy the relation $\vec{A}.\vec{B}=0$ and $\vec{A}.\vec{C}=0$, then $\vec{A}$ can be parallel to

(1) $\vec{C}$ (2) $\vec{B}$

(3) $\vec{B}\times\vec{C}$ (4) $\vec{B}.\vec{C}$

PBM119

112. For a body, angular velocity $\vec{\omega}=\hat{i}-\hat{j}+\hat{k}$ and radius vector $\vec{r}=\hat{i}+\hat{j}+\hat{k}$, then its velocity $(\vec{v}=\vec{\omega}\times\vec{r})$ is:

(1) $-5\hat{i}+2\hat{j}+3\hat{k}$ (2) $-5\hat{i}+2\hat{j}+3\hat{k}$

(3) $-2\hat{i}+2\hat{k}$ (4) $-5\hat{i}-2\hat{j}-3\hat{k}$

PBM120

113. Find unit vector perpendicular to the plane of $\vec{a}=2\hat{i}-2\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}+2\hat{k}$

(1) $\frac{2}{3}\hat{i}+\frac{1}{3}\hat{j}+\frac{2}{3}\hat{k}$ (2) $-\frac{2}{3}\hat{i}-\frac{1}{3}\hat{j}-\frac{2}{3}\hat{k}$

(3) $-\frac{2}{3}\hat{i}-\frac{1}{3}\hat{j}+\frac{2}{3}\hat{k}$ (4) $\frac{2}{3}\hat{i}+\frac{2}{3}\hat{j}-\frac{1}{3}\hat{k}$

PBM121

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	3	2	3	3	2	2	2	1	1	2	2	4	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	3	3	1	3	3	1	4	1	3	1	4	3	4	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	3	4	1	4	1	1	1	2	1	3	1	2	1	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	4	2	4	4	2	1	3	3	2	4	1	2	3	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	4	1	2	1	4	1	1	4	1	2	2	2	1	1	2
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	1	3	4	2	3	1	2	4	1	2	2	4	2	3	2
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	1	1	4	3	2	3	3	4	4	4	2	4	4	3	4
Question	106	107	108	109	110	111	112	113							
Answer	1	1	1	1	2	3	3	3							

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Exercise - II (Previous Year Questions) AIPMT/NEET

Re-AIPMT 2015

1. If vectors $\vec{A}=\cos\omega t\hat{i}+\sin\omega t\hat{j}$ and

$\vec{B}=\cos\frac{\omega t}{2}\hat{i}+\sin\frac{\omega t}{2}\hat{j}$

are functions of time, then the value of t at which they are orthogonal to each other is :

(1) t = 0 (2) $t=\frac{\pi}{4\omega}$

(3) $t=\frac{\pi}{2\omega}$ (4) $t=\frac{\pi}{\omega}$

PBM131

NEET-I 2016

2. If the magnitude of sum of two vectors is equal to the magnitude of difference of the two vectors, the angle between these vectors is :-

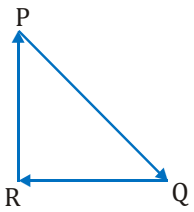
(1) 0° (2) 90°

(3) 45° (4) 180°

PBM132

NEET(UG) 2019

3. A particle moving with velocity $\vec{V}$ is acted by three forces shown by the vector triangle PQR. The velocity of the particle will:



(1) increase

(2) decrease

(3) remain constant

(4) change according to the smallest force $\overrightarrow{QR}$

PBM133

RE-NEET(UG) 2022

4. If $\vec{F}=2\hat{i}+\hat{j}-\hat{k}$ and $\vec{r}=3\hat{i}+2\hat{j}-2\hat{k}$, then the scalar and vector products of $\vec{F}$ and $\vec{r}$ have the magnitudes respectively as :

(1) 5, $\sqrt{3}$ (2) 4, $\sqrt{5}$

(3) 10, $\sqrt{2}$ (4) 10, 2

PBM134

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4
Answer	4	2	3	3

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Exercise - III (Analytical Questions) Master Your Understanding

1. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Current has magnitude as well as direction but still not considered as vector.
Reason (R) : Current do not follow vector algebra.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PBM135

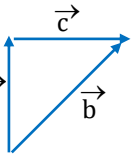
2. Two vectors $\vec{A}$ & $\vec{B}$ have equal magnitude equal to Z. If angle between $\vec{A}$ & $\vec{B}$ is 60° then match the following :

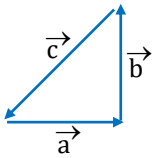
(A) $\lvert\vec{A}+\vec{B}\rvert$	(P) $\frac{\sqrt{3}}{2}Z^2$
(B) $\lvert\vec{A}-\vec{B}\rvert$	(Q) Z
(C) $\vec{A}\cdot\vec{B}$	(R) $\sqrt{3}Z$
(D) $\lvert\vec{A}\times\vec{B}\rvert$	(S) None

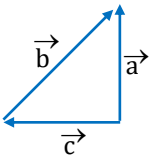
(1) (A)–(R), (B)–(S), (C)–(Q), (D)–(P)
(2) (A)–(R), (B)–(Q), (C)–(S), (D)–(P)
(3) (A)–(P), (B)–(Q), (C)–(R), (D)–(S)
(4) (A)–(Q), (B)–(P), (C)–(S), (D)–(P)

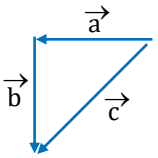
PBM137

3. Match the following

(A) $\vec{a}+\vec{b}=\vec{c}$ (P) 

(B) $\vec{a}-\vec{c}=\vec{b}$ (Q) 

(C) $\vec{b}-\vec{a}=\vec{c}$ (R) 

(D) $\vec{a}+\vec{b}+\vec{c}=\vec{0}$ (S) 

(1) (A)–(S), (B)–(R), (C)–(P), (D)–(Q)
(2) (A)–(R), (B)–(S), (C)–(Q), (D)–(P)
(3) (A)–(P), (B)–(Q), (C)–(R), (D)–(S)
(4) (A)–(S), (B)–(R), (C)–(Q), (D)–(P)

PBM138

4. Given below are two statements :
Statement I : Resultant of 2 forces of magnitude 4N and 5N can be 2N in magnitude.
Statement II : $\left|\lvert\vec{a}\rvert-\lvert\vec{b}\rvert\right| \le \lvert\vec{a}+\vec{b}\rvert \le \lvert\vec{a}\rvert+\lvert\vec{b}\rvert$
In the light of the above statements, choose the most appropriate answer from the options given below :
(1) Both statement I and statement II are correct.
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statements I and statements II are incorrect.

PBM139

5. Given below are two statements :
Statement I : Two null vector have same direction.
Statement II : $\vec{A}\times\vec{B}$ lies in the plane of $\vec{A}+\vec{B}$
In the light of the above statements, choose the most appropriate answer from the options given below :
(1) Both statement I and statement II are correct.
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statements I and statements II are incorrect.

PBM140

6. Which of the following is correct :

(i) $\vec{A}\cdot\vec{B}$ is a vector quantity

(ii) $\vec{A}\times\vec{B}$ is perpendicular to plane of $\vec{A}+\vec{B}$

(iii) For two orthogonal vectors $\vec{A}\cdot\vec{B}=0$

(iv) If vectors are parallel or antiparallel, then $\vec{A}\times\vec{B}=\vec{0}$

(1) (i) only (2) (i) & (ii)

(2) (iii) & (iv) only (4) (ii), (iii) & (iv)

PBM141

7. Given below are two statements :

Statement-I : For every small angle θ, we may use approximation $\sin\theta \approx \theta \approx \tan\theta$.

Statement-II : For very small angle θ, the hypotenuse and the base become approximately of the same length.

(1) Statement-I is true, Statement-II is true; Statement-II is a correct explanation for Statement-I.

(2) Statement-I is true, Statement-II is true; Statement-II is not a correct explanation for Statement-I.

(3) Statement-I is true, Statement-II is false.

(4) Statement-I is false, Statement-II is true.

PBM143

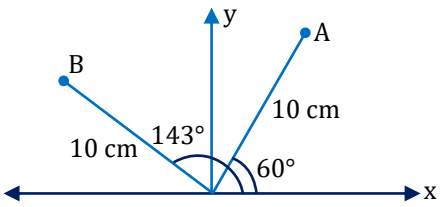
8. Refer the given figure and identify correct statement(s)

(A) Distance of A from x-axis is $5\sqrt{3}$ cm.

(B) Distance of B from x-axis is 6 cm.

(C) Distance of A from y-axis is 5 cm.

(D) Distance of B from y-axis is 8 cm.



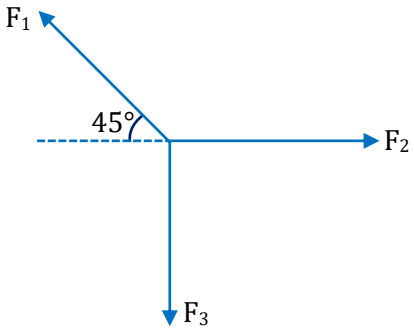
Options :-

(1) A, C (2) A, D

(3) A, B, C (4) A, B, C, D

PBM145

9. Three forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$ are represented as shown. Each of them is of equal magnitude.



Column I (Combination)	Column II (Approximate Direction)
(A) $\vec{F}_1+\vec{F}_2+\vec{F}_3$	(P)
(B) $\vec{F}_1-\vec{F}_2+\vec{F}_3$	(Q)
(C) $\vec{F}_1-\vec{F}_2-\vec{F}_3$	(R)
(D) $\vec{F}_2-\vec{F}_1-\vec{F}_3$	(S)

Options :-

(1) A → R, B → Q, C → P, D → S

(2) A → Q, B → P, C → R, D → S

(3) A → Q, B → R, C → P, D → S

(4) A → S, B → P, C → R, D → Q

PBM147

10. Which of the following statement is/are true ?

(a) Two vectors of unequal magnitude can add up to zero.

(b) Three vectors of unequal magnitude can add up to zero, if they lie in a plane.

(c) Three vectors of unequal magnitude can added upto zero, if they do not lie in same plane.

Options :-

(1) Only a (2) a, b and c

(3) only b (4) only c

PBM148

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	1	2	1	1	4	4	1	4	3	3

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Exercise - I (Conceptual Questions) Build Up Your Understanding

BASIC QUESTION RELATED TO CONCEPT OF FORCE AND NEWTON'S LAWS OF MOTION

1. A particle is in a straight line motion with uniform velocity. A force is not required :-
(1) To increase the speed
(2) To decrease the speed
(3) To maintain the same speed
(4) To change the direction

PLM001

2. Essential characteristic of equilibrium is :-
(1) Momentum equals zero
(2) Acceleration equals zero
(3) K.E. equals zero
(4) Velocity equals zero

PLM002

3. When a constant force is applied to a body, it moves with uniform :-
(1) Acceleration (2) Velocity
(3) Speed (4) Momentum

PLM003

4. A 10 kg mass is accelerated from rest to 30 m/s in 2 s. What force acts on it : -
(1) 300 N (2) 150 N
(3) 75 N (4) 600 N

PLM004

5. A body of mass 80 g is moving with a constant velocity of 4 cm/s on a horizontal frictionless table. The force on the body (in dynes) is :-
(1) Zero (2) 39200
(3) 78400 (4) 8000

PLM005

6. A body of mass 6 kg moving on a horizontal surface with an initial velocity of 8 m/s comes to rest after 3 s. If one wants to keep this body moving on the same surface with a velocity of 8 m/s, the force required is :-
(1) 8 N (2) 16 N (3) Zero (4) 6 N

PLM006

7. An object with a mass 5 kg moves at a constant velocity of 8 m/s. A constant force then acts for 6 seconds on the object giving it a speed of 4 m/s in opposite direction. The acceleration produced is :-
(1) 1.5 m/s^2 (2) –2 m/s^2
(3) 0.67 m/s^2 (4) –0.5 m/s^2

PLM007

8. The velocity acquired by a mass m in travelling a certain distance d starting from rest under the action of a constant force is directly proportional to :-
(1) $\sqrt{m}$ (2) m^0 (3) $\frac{1}{\sqrt{m}}$ (4) m

PLM008

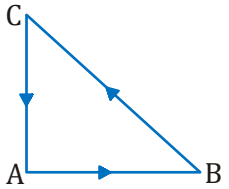
9. A ship of mass 5×10^7 kg initially at rest is pulled by a force of 25×10^4 N through a distance of 100m. Neglecting friction, the speed of the ship at this moment is :
(1) 3.0 m/s (2) 1.5 m/s
(3) 1 m/s (4) 2 m/s

PLM009

10. A body of mass 40 gm is moving with a constant velocity of 2cm/sec on a horizontal frictionless table. The force on the table by the body is :-
(1) 39200 dyne (2) 160 dyne
(3) 80 dyne (4) zero

PLM010

11. Three forces start acting simultaneously on a particle moving with velocity $\vec{v}$. These forces are represented in magnitude and direction by the three sides of a triangle ABC (as shown). The particle will now move with velocity



(1) less than $\vec{v}$
(2) greater than $\vec{v}$
(3) $\vec{v}$, in the direction of largest force BC
(4) $\vec{v}$ remains unchanged

PLM011

12. A body with mass 5 kg is acted upon by a force $\vec{F}=(-3\hat{i}+4\hat{j})N$. If its initial velocity at t = 0 is $\vec{u}=(6\hat{i}-12\hat{j})ms^{-1}$, the time at which it will just have a velocity along the Y-axis is :-
(1) never (2) 10 s (3) 2 s (4) 15 s

PLM012

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13. When forces F_1 , F_2 , F_3 are acting on a particle of mass m such that F_2 and F_3 are mutually perpendicular then the particle remains stationary. If the force F_1 is now removed then the acceleration of the particle is :-
(1) F_1/m (2) F_2F_3/mF_1
(3) $(F_2 - F_3)/m$ (4) F_2/m
PLM013

14. A stone of mass 1 kg is lying on the floor of a train which is accelerating with 1 ms^{-2}. The net force acting on the stone is :-
(1) zero (2) 1 N (3) 5 N (4) 10 N
PLM014

15. A car accelerates on a horizontal road due to the force exerted by :-
(1) the engine of the car
(2) the driver of the car
(3) the car on earth
(4) the road on the car
PLM015

MOMENTUM, IMPULSE, RATE OF CHANGE OF MOMENTUM AND AVERAGE FORCE RELATED PROBLEMS

16. Newton's II law of motion connects :
(1) Momentum and acceleration
(2) Change of momentum and velocity
(3) Rate of change of momentum and external force
PLM016
(4) Rate of change of force and momentum

17. When we kick a stone, we get hurt. Due to which of the following properties of stone does it happens ?
(1) Inertia (2) Velocity
(3) Reaction (4) Momentum
PLM017

18. A force of 20 N acts on a body of mass 10 kg for 20 seconds. The change in its momentum is
(1) 40 kg-m/s (2) 100 kg-m/s
(3) 400 kg-m/s (4) 200 kg-m/s
PLM018

19. A player catches a ball of 100 g moving with a speed of 5 m/s. If the time taken to complete the catch is 0.25 s, the force exerted on the player's hand is :-
(1) 8 N (2) 4 N (3) 2 N (4) 0 N
PLM019

20. A football of mass 0.5 kg coming towards player with a velocity of 10 m/s. Player hit straight back the football with velocity 15 m/s. If the contact time is 0.02 second then the average force involved is:-
(1) 250 N (2) 1250 N
(3) 500 N (4) 625 N
PLM020

21. A 100 g tennis ball coming at a speed of 20 m/s is hit straight back by a bat to speed of 40 m/s. The magnitude of the average force F on the ball, when it is in contact for 5 ms with the bat is:-
(1) 1200 N (2) 6000 N
(3) 2400 N (4) 1000 N
PLM021

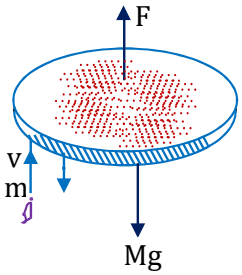
22. A ball weighing 50 g hits a hard surface vertically with a speed of 25 m/s and rebounds with the same speed. The ball remains in contact with the surface for 0.05 s. The average force exerted by the surface on the ball is :-
(1) 100 N (2) 25 N (3) 50 N (4) 5 N
PLM022

23. A tennis ball is dropped on the floor from a height of 10 m. It rebounds to a height of 5 m. The ball was in contact with the floor for 0.01 s. What was its average acceleration during the contact ? ($g=10m/s^2$)
(1) 800 m/s^2 (2) 200 m/s^2
(3) 1600 m/s^2 (4) 2400 m/s^2
PLM023

24. A 150 g ball, in horizontal flight with a speed of 40 m/s, is struck by a bat. After leaving the bat, the ball travels in the opposite direction with speed 40 m/s. If the impact time for the ball-bat collision is 1.20 ms, what average force acts on the ball ?
(1) 1500 N (2) 1000 N
(3) 10000 N (4) 9800 N
PLM024

25. n bullet strike per second elastically on a wall with speed v and rebound. What will be the force exerted on the wall by bullets if mass of each bullet is m :-
(1) mnv (2) 4mnv
(3) 2mnv (4) $\frac{mnv}{2}$
PLM025

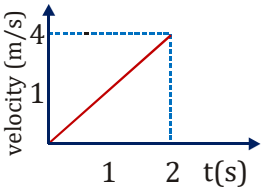
26. A disc of mass 2 kg is kept floating horizontally in air by firing bullets of mass 0.1 kg each vertically at it, at the rate of 5 per second. If the bullets rebound with the same speed, the speed with which these are fired will be–



(1) 1·96 m/s (2) 0·98 m/s
(3) 9·8 m/s (4) 19.6 m/s

PLM026

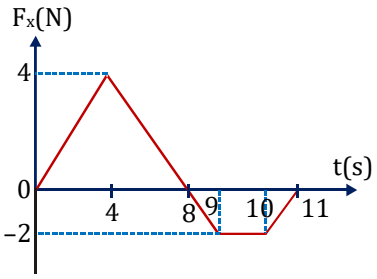
27. For a body of 25 kg mass, the velocity-time graph is shown in figure. The force acting on the body is :



(1) 25 N (2) 50 N
(3) 12.5 N (4) 100 N

PLM027

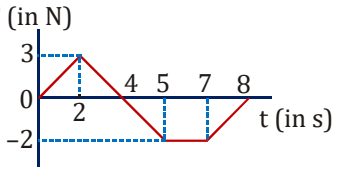
28. A 2 kg toy car starts from rest (i.e. at t = 0, v = 0) can move along x-axis. Graph shows force F_x, acting on the car. The velocity of the car at t = 10 s is :-



(1) 2 m/s (2) 6 m/s
(3) 6.5 m/s (4) 5 m/s

PLM028

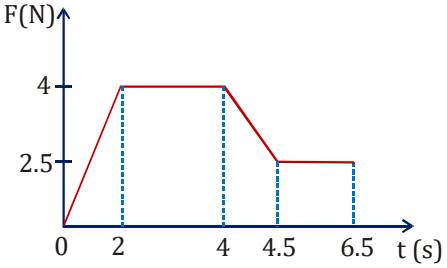
29. Force-time graph for the motion of a body is shown in fig. Change in linear momentum between 0 to 8 s is :-



(1) Zero (2) 4 N-s (3) 8 N-s (4) 6 N-s

PLM029

30. A body of mass 2 kg has an initial speed 5 ms^{-1}. A force acts on it for some time in the direction of motion. The force-time graph is shown in figure. The final speed of the body is :-



(1) 9.25 ms^{-1} (2) 5 ms^{-1}
(3) 14.3 ms^{-1} (4) 4.25 ms^{-1}

PLM030

31. Which one of the following statements is not true?

(1) The same force for the same time causes the same change in momentum for different bodies.
(2) The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction in which the force acts.
(3) A greater opposing force is needed to stop a heavy body than a light body in the same time, if they are moving with the same speed.
(4) The greater the change in the momentum in a given time, the lesser is the force that needs to be applied.

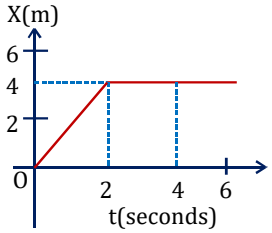
PLM031

32. If force F = 50 – 20t, then impulse in interval [0, t] :-

(1) $50t - 10t^2$ (2) 50t – 10
(3) $50 - t^2$ (4) $25t^2$

PLM032

33. In the figure given below, the position–time graph of a particle of mass 0·1 kg is shown. The impulse at t=2 sec is –



(1) 0·2 kg-m/s (2) –0·2 kg-m/s
(3) 0·1 kg-m/s (4) –0·4 kg-m/s

PLM033

34. Gravel is dropped onto a conveyer belt at a rate of 1.5 kg/s. The extra force required in newton to keep the belt moving at 3 m/s is :-
(1) 1.5 N (2) 2 N (3) 4 N (4) 4.5 N

PLM034

35. A block of metal weighing 1.5 kg is resting on a frictionless plane. It is struck by a jet releasing water at a rate of 3 kg/s with a speed of 10 m/s. The initial acceleration of the block will be :-
(1) 20 m/s^2 (2) 5 m/s^2
(3) 10 m/s^2 (4) 15 m/s^2

PLM035

36. A block of metal weighing 2 kg is resting on a frictionless plane. It is struck by a jet releasing water at a rate of 1 kg/s and a speed of 5 m/s. The initial acceleration of the block will be :-
(1) 2.5 m/s^2 (2) 5 m/s^2
(3) 10 m/s^2 (4) 20 m/s^2

PLM036

37. If the force on a rocket moving in force free space with an exhaust velocity of gases 400 m/sec is 300N, then the rate of combustion of the fuel, is :-
(1) 1.3 kg/s (2) 0.75 kg/s
(3) 2.7 kg/s (4) 0.3 kg/s

PLM037

38. A satellite in force free space sweeps stationary interplanetary dust at a rate (dM/dt) = + αv^2. Here v is the velocity. The acceleration of satellite of mass M is :-
(1) – 2 $\alpha v^2/M$ (2) – 3 $\alpha v^2/M$
(3) – $\alpha v^3/M$ (4) – αv^2

PLM038

39. A rocket of mass 60 kg. is fired in a gravity free space is ejecting gases with velocity 300 m/s at the rate of 2 kg/s. What will be the initial acceleration of the rocket ?
(1) 30 m/s^2 (2) 5 m/s^2
(3) 10 m/s^2 (4) 15 m/s^2

PLM039

40. For a Rocket propulsion velocity of exhaust gases relative to rocket is 1.6 km/s. If mass of rocket system is 800 kg, then the rate of fuel consumption for a rocket to rise up with an acceleration 4.2 m/s^2 will be :-
(1) 12.25 kg/s (2) 17.5 kg/s
(3) 7 kg/s (4) 5.2 kg/s

PLM040

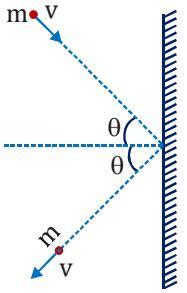
41. The force on a rocket, moving in gravity free space is 210 N. If the gases ejects from the rocket with speed 300 m/s (w.r.t. rocket) then the rate of combustion of the fuel is :-
(1) 0.07 kg s^{-1} (2) 1.4 kg s^{-1}
(3) 0.7 kg s^{-1} (4) 10.7 kg s^{-1}

PLM041

42. A rocket of mass 800 kg is to be projected vertically upwards. The gases are exhausted vertically downwards with velocity 40 m/s with respect to the rocket. What is the minimum rate of burning fuel, so as to just lift the rocket upwards against the gravitational attraction ? (Take g = 10 m/s^2)
(1) 80 kg/s (2) 400 kg/s
(3) 200 kg/s (4) 800 kg/s

PLM042

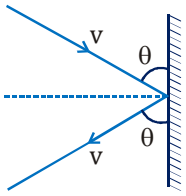
43. A ball is moving towards the wall as shown in diagram then the momentum of the ball is conserved



(1) along the wall
(2) along the normal of the wall
(3) along any direction
(4) Both (1) and (2)

PLM043

44. A water jet, whose cross sectional area is 'a' strikes a wall making an angle 'θ' with the wall and rebounds elastically. The velocity of water of density 'd' is v. Force exerted on wall is :-

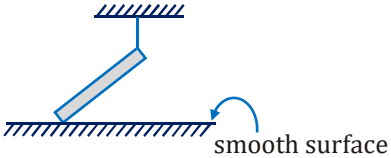


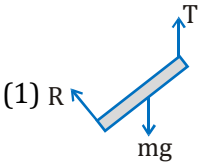
(1) $2av^2d\cos\theta$ (2) $2av^2d\sin\theta$
(3) $2avd\cos\theta$ (4) $avd\cos\theta$

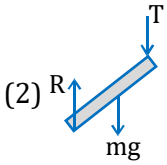
PLM044

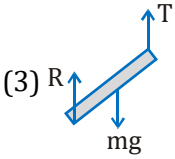
FREE BODY DIAGRAM, NEWTON'S THIRD LAW

45. Which figure represents the correct F.B.D. of rod of mass m as shown in figure :-



(1) 

(2) 

(3) 

(4) None of these

PLM045

46. Working of rocket or jet is based on :-
(1) Newton's I law (2) Newton's II law
(3) Newton's III law (4) All the three laws

PLM046

47. When a horse pulls a wagon, the force that causes the horse to move forward is the force
(1) He exerts on the wagon
(2) The wagon exerts on him
(3) The ground exerts on him
(4) He exerts on the ground

PLM047

48. A man is at rest in the middle of a pond on perfectly smooth ice. He can get himself to the shore by making use of Newton's :-
(1) First law
(2) Second law
(3) Third law
(4) Law of gravitation

PLM048

49. A material body A of mass m exerts a force on another material body B of mass 2m. If the acceleration of B be a, the magnitude of the acceleration of A is :-
(1) Zero (2) a
(3) 2a (4) a/2

PLM049

50. If the force of gravity suddenly increase, then :-
(1) The mass of all bodies will increase
(2) The weight of all bodies will increase
(3) Both mass and weight of all bodies will increase
(4) Neither mass nor weight of all bodies will increase

PLM050

51. Two bodies of masses 5 kg and 8 kg are acted upon by the same force. If the acceleration of lighter body is 2 m/s^2, then the acceleration of the heavier body is :-
(1) 4 m/s^2 (2) 2.5 m/s^2
(3) 6.25 m/s^2 (4) 1.25 m/s^2

PLM051

52. Action and reaction :- (For a given system)
(a) Act on the same object
(b) Have same direction
(c) Have different magnitudes
(d) Have non-zero resultant
(1) a, b, c
(2) b, c, d
(3) All of the above
(4) None of the above

PLM052

53. Weight is defined as :-
(1) Force of attraction exerted by the earth
(2) Mass of a body
(3) Nature of a body
(4) None of these

PLM053

54. A rider on horse falls back when horse starts running, all of a sudden because :-
(1) rider is taken back
(2) rider is suddenly afraid of falling
(3) inertia of rest keeps the upper part of body at rest while lower part of the body moves forward with the horse.
(4) none of the these

PLM054

55. The force exerted by the floor of an elevator on the foot for a person standing there, is more than the weight of the person, if the elevator is
A. Going up with increasing speed
B. Going down with decreasing speed
(1) A is true but B is false
(2) A is false but B is true
(3) Both A and B are true
(4) Both A and B are false

PLM055

56. A cork of mass 10 g is floating on water. The net force acting on the cork is :-
(1) 10 N (2) 10^{-3} N (3) 10^{-2} N (4) zero

PLM056

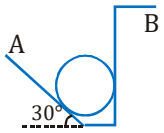
57. Ten one-rupee coins are put on top of each other on a table. Each coin has a mass m. Which of the following statements is true ? (counted from the bottom)
(1) The force on the 3rd coin due to all the coins on its top is equal to 3mg (downwards).
(2) The force on the 6th coin due to the 7th coin is 4mg (downwards)
(3) The reaction of the 6th coin on the 7th coin is 3mg (upwards).
(4) The total force on the 10th coin is 9mg (downwards)

PLM057

58. Ten one rupees coins are put on top of each other on a table. Each coin has a mass 'm' kg., then the force on the 3rd coin (counted from the bottom) due to all the coins on its top :-
(1) 3 mg (2) 7 mg (3) 2 mg (4) 5 mg

PLM058

59. The 50 kg homogeneous smooth sphere rests on the 30° incline A and bears against the smooth vertical wall B. The contact forces at A and B are:-



(1) $N_A = \frac{1000}{\sqrt{3}}N, N_B = \frac{500}{\sqrt{3}}N$

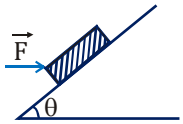
(2) $N_A = \frac{1000}{\sqrt{3}}N, N_B = \frac{1000}{\sqrt{3}}N$

(3) $N_A = \frac{500}{\sqrt{3}}N, N_B = \frac{500}{\sqrt{3}}N$

(4) $N_A = \frac{500}{\sqrt{3}}N, N_B = \frac{400}{\sqrt{3}}N$

PLM059

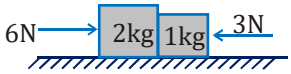
60. The figure shows a horizontal force $\vec{F}$ acting on a block of mass m placed on an inclined plane (angle θ). What is the normal reaction on the block?



(1) mgsinθ + Fcosθ (2) mgsinθ – Fcosθ
(3) mgcosθ – Fsinθ (4) mgcosθ + Fsinθ

PLM060

61. Two forces of 6N and 3N are acting on the two blocks of 2kg and 1kg kept on frictionless floor. What is the force exerted on 2kg block by 1kg block?



(1) 1N (2) 2N (3) 4N (4) 5N

PLM061

62. Two blocks of masses 2 kg and 1 kg are in contact with each other on a frictionless table. When a horizontal force of 3.0 N is applied to the block of mass 1 kg the value of the force of contact between the two blocks is -
(1) 2 N (2) 3 N (3) 5 N (4) 1 N

PLM062

63. A person is standing in an elevator. In which situation he finds his weight more ?
(1) when the elevator moves downward with constant velocity
(2) when the elevator moves downward with constant acceleration
(3) when the elevator moves upward with uniform velocity
(4) when the elevator moves downward with decreasing velocity

PLM063

64. A man weighs 80 kg. He stands on a weighing scale in a lift which is moving downwards with a uniform acceleration of 5m/s². What would be the reading on the scale ? (g = 10 m/s²)
(1) Zero (2) 400 N
(3) 800 N (4)1200 N

PLM064

65. A man, of mass 60 kg, is riding in a lift. The ratio of the apparent weights of the man when the lift is accelerating upwards and downwards at 2 m/s² is:-
(Taking g = 10 m/s²)
(1) 2 : 3 (2) 1 : 1
(3) 3 : 2 (4) none of these

PLM065

EQUILIBRIUM FORCES OR CONSTANT FORCES – LAMI THEOREM

66. Two persons hold a rope of negligible weight tightly at its ends so that it is horizontal. A 15 kg weight is attached to the rope at the mid point which is now no longer remains horizontal. The minimum tension required to completely straighten the top is

(1) 15 kg (2) 15/2 kg
(3) 5 kg (4) Infinitely large

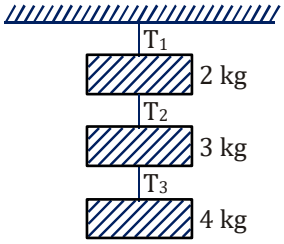
PLM066

67. A boy of mass 40 kg is hanging from a horizontal branch of a tree. The tension in his arms is minimum when the angle between the arms is :-

(1) 0° (2) 90° (3) 120° (4) 180°

PLM067

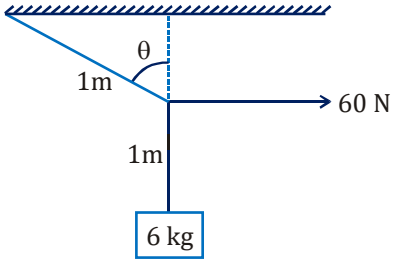
68. Find the ratio T_1/T_3 for the system shown in fig.



(1) 2 : 1 (2) 1 : 4 (3) 9 : 4 (4) 3 : 1

PLM068

69. A mass of 6 kg is suspended by a rope of length 2 m from a ceiling. A force of 60 N is applied in horizontal direction at the mid point of the rope. What is the angle between the rope and the vertical in equilibrium :-

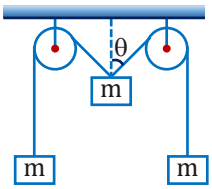


(1) $\tan^{-1}\left(\frac{4}{5}\right)$ (2) $\tan^{-1}\left(\frac{5}{4}\right)$

(3) $\tan^{-1}\left(\frac{5}{6}\right)$ (4) 45°

PLM069

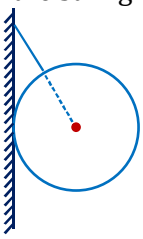
70. The pulleys and strings shown in the fig. are smooth and are of negligible mass. For the system to remain in equilibrium, the angle θ should be



(1) 0° (2) 30° (3) 45° (4) 60°

PLM070

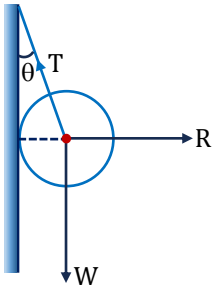
71. A uniform sphere of weight W and radius 3 m is being held by a string of length 2m attached to a frictionless wall as shown in the figure. The tension in the string will be :-



(1) 5 W/4 (2) 15 W/4
(3) 15 W/16 (4) none of these

PLM071

72. A metal sphere is hung by a string fixed to a wall. The forces acting on the sphere are shown in fig. Which of the following statements is not correct ?

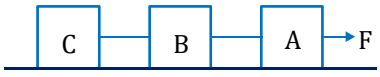


(a) $\vec{R}+\vec{T}+\vec{W}=0$ (b) $T^2 = R^2 + W^2$
(c) T = R + W (d) R = W tan θ
(1) a (2) b (3) c (4) d

PLM072

MOTION OF BODIES IN CONTACT OR CONNECTED BY STRINGS, PULLEY SYSTEM

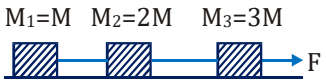
73. Three identical blocks of mass m = 2kg are drawn by a force of 10.2 N on frictionless surface. The tension (in N) in the string between the blocks B and C is :-



(1) 9.2 (2) 8 (3) 3.4 (4) 9.8

PLM073

74. Three masses M_1, M_2 and M_3 are lying on a frictionless table. The masses are connected by massless threads as shown. The mass M_3 is pulled by a constant force F as shown. The tension in the thread between masses M_2 and M_3 is



(1) $\frac{F}{2}$ (2) $\frac{F}{3}$ (3) $\frac{F}{4}$ (4) $\frac{F}{6}$

PLM074

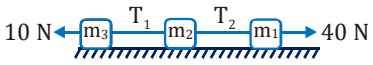
75. Three blocks are connected as shown in fig. on a horizontal frictionless table. If M_1 = 1 kg, M_2 = 8 kg, M_3 = 27 kg and F = 36N, Then T' will be :-



(1) 18 N (2) 8 N
(3) 35 N (4) 27 N

PLM075

76. Three blocks of masses m_1, m_2 and m_3 are connected by massless strings as shown in the figure on a frictionless table. Forces of 10 N and 40 N are applied on them as shown in diagram. If m_1 = 10kg, m_2 = 6 kg and m_3 = 4 kg, then tension T_2 will be:-



(1) 10 N (2) 20 N
(3) 25 N (4) 35 N

PLM076

77. Two bodies A and B of masses 10 kg and 15 kg respectively kept on a smooth, horizontal surface are tied to the ends of a light string. If T represents the tension in the string when a horizontal force F = 500 N is applied to A (as shown in figure 1) and T' be the tension when F = 750 N is applied to B (figure2), then which of the following is true?

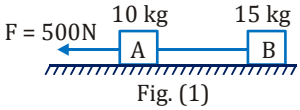


Fig. (1)

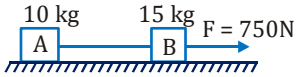
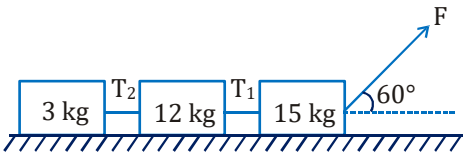


Fig. (2)

(1) T = T' = 500 N
(2) T = T' = 250 N
(3) T = 300 N, T' = 200 N
(4) T = 300 N, T' = 300 N

PLM077

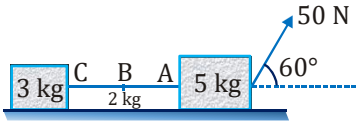
78. The surface shown in diagram is frictionless and T_1 is 30 N, then the value of F is :-



(1) 30 N (2) 60 N
(3) 120 N (4) 69.36 N

PLM078

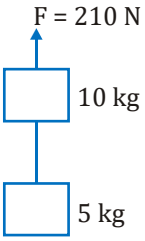
79. Two blocks of masses 5 kg and 3 kg connected by a rope of mass 2 kg are resting on a frictionless floor as shown in the following figure. If a constant force of 50 N is applied to 5 kg block, then the tension in the rope at points A, B and C are respectively given by:



(1) 7.5N, 10N, 12.5N (2) 12.5N, 10N, 7.5N
(3) 20N, 20N, 20N (4) 50N, 50N, 50N

PLM079

80. Two blocks of masses 10 kg and 5 kg are connected from a massless string as shown in figure. The system is pulled up with a force of 210 N, then the acceleration of the system is :-



(1) 14 m/s² (2) 11 m/s²
(3) 4 m/s² (4) zero

PLM080

81. A block of weight 40 N is supported by two ropes. One rope is horizontal and the other makes an angle of 30° with the ceiling. The tension in the rope attached to the ceiling is approximately.

(1) 80 N (2) 40 N

(3) $40\sqrt{3}$ N (4) $\frac{40}{\sqrt{3}}$ N

PLM081

82. A monkey of mass 15 kg is holding a vertical rope. The rope will not break when a mass of 30 kg is suspended from it but will break if the mass exceeds 30 kg what is the maximum acceleration with which the monkey can climb up along the rope ? (g = 10 m/s²)

(1) 15 m/s² (2) 10 m/s²

(3) 25 m/s² (4) 20 m/s²

PLM082

83. A uniform rope of length L resting on a frictionless horizontal surface is pulled at one end by a force F. What is the tension in the rope at a distance ℓ from the end where the force is applied.

(1) F (2) F(1+ℓ/L)

(3) F/2 (4) F(1–ℓ/L)

PLM083

84. A string of length L and mass M is lying on a horizontal table. A force F is applied at one of its ends. Tension in the string at a distance x from the end other than at which force is applied is

(1) Zero (2) Fx/L

(3) F(L – x)/L (4) F(L – x)/M

PLM084

85. A block of mass m is pulled along a horizontal frictionless surface by a rope of mass m. If a force F is applied at one end of the rope, the tension at the mid-point of the rope is :-

(1) $\frac{F}{2}$ (2) $\frac{3F}{4}$ (3) $\frac{F}{4}$ (4) $\frac{F}{3}$

PLM085

86. Two particles of masses m and 2M (M > m) are connected by a cord that passes over a massless and frictionless pulley. The tension T in the string and the acceleration a of the particles is :-

(1) $T=\frac{2mM}{(M-m)}g;\ a=\left(\frac{M-m}{M+m}\right)g$

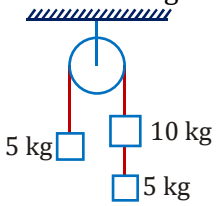
(2) $T=\frac{4mM}{(2M+m)}g;a=\left(\frac{2M-m}{2M+m}\right)g$

(3) $T=\left(\frac{2M-m}{2M+m}\right)g;\ a=\left(\frac{4mM}{M+m}\right)g$

(4) $T=\left(\frac{2Mm}{2M+m}\right)g;\ a=\left(\frac{4mM}{2M+m}\right)g$

PLM086

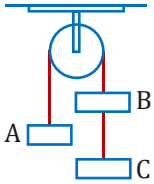
87. What will be the acceleration of the blocks if they are set free to move?
(g = acceleration due to gravity)



(1) g (2) g/2 (3) g/3 (4) g/4

PLM087

88. Three blocks each of mass 5 kg are hanging on a string passing over a fixed pulley as shown in fig. What is the tension in the string connecting blocks A and B.

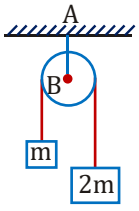


(1) Zero (2) 33.3 N

(3) 66.67 N (4) 50 N

PLM088

89. In the arrangement shown in the figure, the pulley has a mass 3m. Neglecting friction on the contact surface, the force exerted by the supporting rope AB on the ceiling is :-

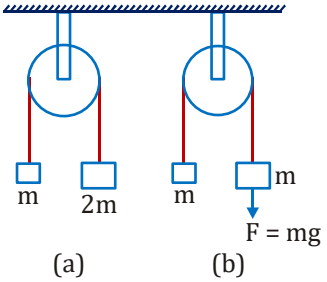


(1) 5.67mg (2) 3mg

(3) 4mg (4) 2.67mg

PLM089

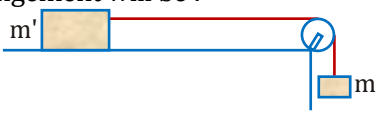
90. The pulley arrangements shown in the figure are ideal, the mass of the rope being negligible. In case (a) mass m is lifted by attaching a mass of 2m to the other end of the rope. In case (b) the mass m is connected with an another mass m which is pulling by a rope with a constant downward force F = mg, where g is the acceleration due to gravity. The acceleration of mass m in case (a) is :-



(1) Zero
(2) More than that in case (b)
(3) Less than that in case (b)
(4) Equal to that in case (b)

PLM090

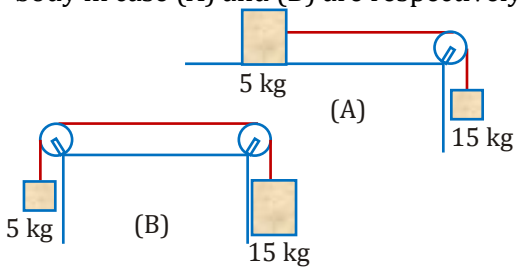
91. In the fig. given below masses m and m' are tied with a thread passing over a pulley, m' is on a frictionless horizontal surface. If acceleration due to gravity is g, the acceleration of m in this arrangement will be :-



(1) g
(2) m'g/(m + m')
(3) mg/(m + m')
(4) mg/(m – m')

PLM091

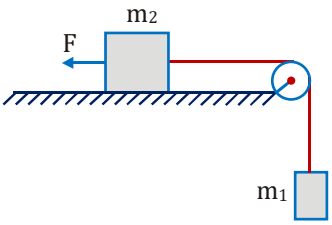
92. Two bodies of masses 15 kg and 5 kg are arranged in two different ways as shown in fig. (A) and (B). If the pulleys and the table are perfectly smooth, the acceleration of the 5 kg body in case (A) and (B) are respectively :-



(1) 3g/4 and g/2
(2) g/5 and g/2
(3) g/2 and 3g/4
(4) 15g/2 and g/5

PLM092

93. A constant force F = $m_1g/2$ is applied on the block of mass m_2 as shown in figure. The string and the pulley are light and the surface of the table is smooth. Find the acceleration of m_2



(1) $\frac{m_2g}{m_1+m_2}$
(2) $\frac{m_1g}{m_1+m_2}$
(3) $\frac{m_1g}{2(m_1+m_2)}$
(4) $\frac{m_2g}{2(m_1+m_2)}$

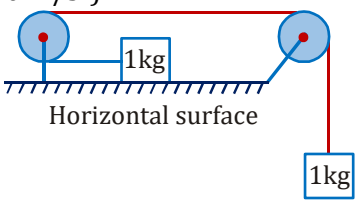
PLM093

94. Two bodies A (20 kg) and B (30 kg) tied with a light string are placed on a frictionless table. A force F acting at B pulls this system with an acceleration of $3ms^{-2}$. The tension in the string is:

(1) 60 N
(2) 120 N
(3) 90 N
(4) 140 N

PLM094

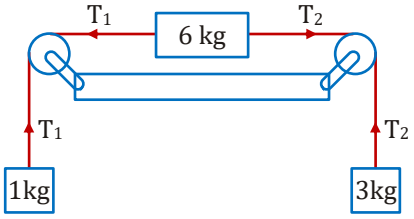
95. Consider the system as shown in the figure. The pulley and the string are light and all the surfaces are frictionless. The tension in the string is
($g = 10\ m/s^2$)



(1) zero
(2) 1N
(3) 2N
(4) 5N

PLM095

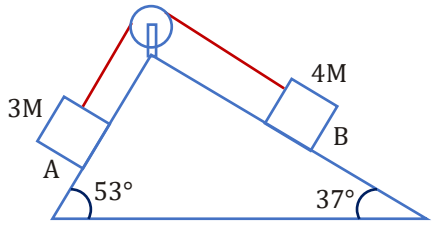
96. Three masses of 1 kg, 6 kg and 3 kg are connected to each other with threads and are placed on a table as shown in figure. The ratio $\frac{T_1}{T_2}$ is :- (Take $g = 10\ m/s^2$)



(1) 1 : 1
(2) 2 : 1
(3) 1 : 2
(4) 3 : 1

PLM096

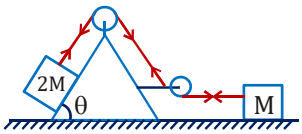
97. Two blocks of mass 3M and 4M are resting on a frictionless inclined planes as shown in fig. then



(1) The block A moves down the plane
(2) The block B moves down the plane
(3) Both the blocks remain at rest
(4) Both the blocks move down the plane.

PLM097

98. The acceleration of the block shown in figure :-

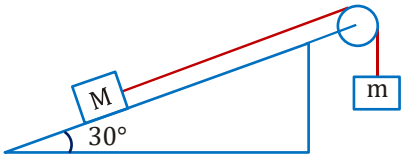


(1) 0 (2) $\frac{g\sin\theta}{2}$

(3) $\frac{2g\sin\theta}{3}$ (4) $2g\sin\theta$

PLM098

99. In the fig. mass m = 10 kg then in order to keep it at rest, the value of mass M will be:



(1) 10 kg (2) 5 kg
(3) 20 kg (4) 40 kg

PLM099

100. A block of mass 4 kg is suspended through two light spring balances A and B in parallel. Then A and B will read respectively.

(1) 4 kg and zero kg (2) zero kg and 4 kg
(3) 4 kg and 4 kg (4) 2 kg and 2 kg

PLM100

101. Two masses m_1 and m_2 are joined by a spring as shown. The system is dropped to the ground from a certain height. The spring will be :-



(1) Stretched when $m_2 > m_1$
(2) compressed when $m_2 < m_1$
(3) neither compressed nor stretched only when $m_1 = m_2$
(4) neither compressed nor stretched regardless of the values of m_1 and m_2.

PLM101

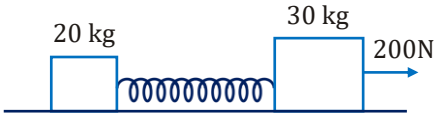
102. Two masses of 1 kg and 2 kg respectively are connected by a massless spring as shown in figure. A force of 20 N acts on the 2 kg mass at the instant when the 1 kg mass has an acceleration of 10 ms^{-2} towards right, the acceleration of the 2 kg mass is:-



(1) 2 ms^{-2} (2) 5 ms^{-2}
(3) 10 ms^{-2} (4) 20 ms^{-2}

PLM102

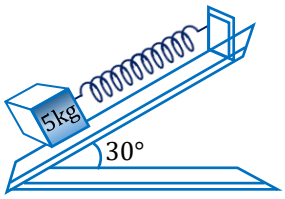
103. Two masses of 20 kg and 30 kg respectively are connected by a massless spring as shown in the figure. A force of 200 N acts on the 30 kg mass. At the instant shown the 20 kg mass has an acceleration 2 m/s^2 rightwards. What is the acceleration of 30 kg mass ?



(1) Zero (2) 5.3 m/s^2
(3) 4 m/s^2 (4) 8 m/s^2

PLM103

104. A body of mass 5kg is suspended by a spring balance on an inclined plane as shown in figure. The spring balance measure



(1) 50 N (2) 25 N (3) 500 N (4) 10 N

PLM104

105. A man is standing at a spring platform. Reading of spring balance is 60 kg-wt. If the man jumps off from the platform, then reading of spring balance

(1) First increases then decreases to zero
(2) Decreases
(3) Increases
(4) Remains same

PLM105

FRAME OF REFERENCE - INTERTIAL OR NON INERTIAL FRAME, PSEUDO FORCE, MECHANICAL ADVANTAGE

106. In Newton's second Law $\vec{F}=m\vec{a}$ (for constant mass m), $\vec{a}$ is the acceleration of the mass m with respect to
(1) any observer
(2) any inertial observer
(3) an observer at rest only
(4) an observer moving with uniform speed only

PLM106

107. A frame will be inertial, if it moves with respect to another inertial frame with a constant :-
(1) Linear velocity
(2) Angular velocity
(3) Linear acceleration
(4) All of the above

PLM107

108. A small sphere of mass m is suspended by a string from the ceiling of a car. If the car begins to move with a constant acceleration $\frac{g}{2}$, then the tension in the string is :-

(1) mg (2) $\frac{3}{2}$mg
(3) $\frac{\sqrt{5}}{2}$mg (4) $\frac{mg}{2}$

PLM108

109. A boy sitting on the upper berth in the compartment of an accelerated train, which is just left the railway station, drops an apple aiming at the open hand of his brother vertically below his hands at a distance of about 2 m. The apple will fall :-
(1) In the hand of his brother
(2) Slightly away from the hands of his brother in the direction of motion of the train
(3) Slightly away from the hands of his brother in the direction opposite to the direction of motion of the train
(4) None of the above

PLM109

110. The force exerted by a person on the floor of an elevator is less than the weight of the person if the elevator is :-
(a) Going up and slowing down
(b) Going up and speeding up
(c) Going down and slowing down
(d) Going down and speeding up
(1) a, c (2) b, c (3) a, d (4) b, d

PLM110

111. Two equal masses are kept on the pans of a simple balance in a lift accelerating upward. Then
(1) Pans will remain at the same level.
(2) Nothing can be said as data is incomplete
(3) Left side pan will lower down.
(4) Right side pan will lower down.

PLM111

112. The ratio of weights of a man in a lift moving with acceleration 'a' in upward direction and moving with acceleration 'a' in downward direction is 3 : 1. Then the value of a is :-

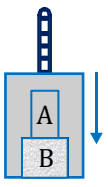
(1) $\frac{g}{3}$ (2) $\frac{g}{2}$ (3) $\frac{g}{5}$ (4) $\frac{4}{3}$g

PLM112

113. A man weighing 100 kg carries a load of 10 kg on his head. He jumps from a tower with the load on his head. What will be the weight of the load as experienced by the man ?
(1) zero
(2) 10 kg
(3) slightly more than 10 kg
(4) 110 kg

PLM113

114. The elevator shown in figure is descending, with a retardation of 2 m/s^2. The mass of the block A is 0.5 kg. The force exerted by the block A on the block B is :



(1) 2 N (2) 4 N
(3) 6 N (4) 8 N

PLM114

115. Drums of oil are carried in a moving truck. Suddenly breaks are applied on the truck then the surface of the oil in the drum will -
(1) Remain unaffected
(2) Rise in the forward direction
(3) Rise in the backward direction
(4) Nothing is certain

PLM115

116. A block of mass m is placed on a smooth wedge of inclination θ. The whole system is accelerated horizontally so that the block does not slip on the wedge. The force exerted by the wedge on the block (g is acceleration due to gravity) will be :–
(1) mg sinθ (2) mg
(3) mg/cosθ (4) mg cosθ

PLM116

117. A body kept on a smooth inclined plane of inclination 1 in x will remain stationary relative to the inclined plane if the plane is given a horizontal acceleration equal to :-

(1) $\sqrt{x^2-1}g$ (2) $\frac{\sqrt{x^2-1}}{x}g$

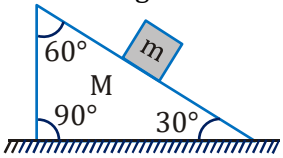
(3) $\frac{gx}{\sqrt{x^2-1}}$ (4) $\frac{g}{\sqrt{x^2-1}}$

PLM117

118. An object of mass 2 kg moving with constant velocity $10\hat{i}$ m/s is seen in a frame moving with constant velocity $10\hat{i}$ m/s. The value of 'pseudo force' acting on object in this frame will be :-
(1) 20 N (2) 0 N (3) 10 N (4) 2 N

PLM118

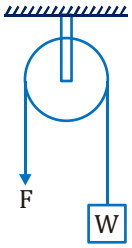
119. A triangular wedge of mass M rests on a smooth surface as shown in figure. A cubical block of mass m rests on the inclined surface of wedge. If all surfaces are frictionless, the horizontal force that must be applied to the wedge so as to keep the block stationary relative to the wedge is :-



(1) Mg tan30° (2) mg tan30°
(3) (M+m)g tan30° (4) (M+m)g cos30°

PLM119

120. What is the mechanical advantage of single fixed pulley ?



(1) 1 (2) 2 (3) 0.5 (4) 4

PLM120

121. The force required to lift a 400 kg wt is 80 kg wt then the mechanical advantage is :-
(1) 8 (2) 10 (3) 5 (4) 4

PLM121

FRICTION

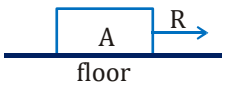
122. The coefficient of static friction between two surfaces depend on
(1) the normal reaction on the surface
(2) the shape of the surface in contact
(3) the area of contact
(4) None of the above

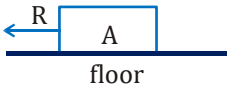
PLM122

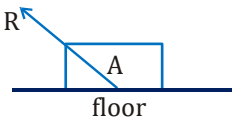
123. A girl horizontally press her physics text book against a rough vertical wall with her hand. The direction of the frictional force on the book exerted by the wall is :-
(1) downwards (2) upwards
(3) out from the wall (4) into the wall

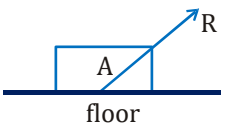
PLM123

124. A box 'A' is lying on the horizontal floor of the compartment of a train running along horizontal rails from left to right. At time 't', it retards. Then the reaction R by the floor on the box is given best by:-

(1) 

(2) 

(3) 

(4) 

PLM124

125. A horizontal force F acts on the block of mass m and the block remains stationary, the value of friction force on the block is :-

(1) μmg (2) $\mu mg - F$

(3) F (4) zero

PLM125

126. A block of mass 4 kg is placed on the floor. The coefficient of static friction is 0.4. Force of 8 N is applied on the block. The force of friction between the block and the floor is

(1) 16 N (2) 8.0 N (3) 2.0 N (4) zero

PLM126

127. A block of mass 3 kg is placed on a rough horizontal surface (μ_s = 0.4). A horizontal force of 8.7 N is applied on the block. The force of friction between the block and floor is :-

(1) 8.7 N (2) 12N (3) 10N (4) zero

PLM127

128. The frictional force of the air on a body of mass 0.5 kg, falling with an acceleration of 9.0 m/s^2, will be (g = 10 m/sec^2) :

(1) 4.9 N (2) 0.5 N

(3) 4.5 N (4) 5 N

PLM128

129. Consider a car moving on a straight road with a speed of 100 m/s. The distance at which car can be stopped is [μ_k = 0.5]

(1) 1000 m (2) 800 m

(3) 400 m (4) 100 m

PLM129

130. A block of mass 10 kg is placed on a long trolley. The coefficient of friction between the block and trolley is 0.2. The trolley accelerates from rest with 0.5 m/s^2 for 20 s. then what is the friction force ?

(1) 20 N (2) 100 N (3) 5 N (4) 10 N

PLM130

131. A 20 kg block is initially at rest. A 75 N force is required to set the block in motion. After the motion start, a force of 60 N is applied to keep the block moving with constant speed. The coefficient of static friction is :-

(1) 0.6 (2) 0.52

(3) 0.44 (4) 0.375

PLM131

132. A block of mass 1 kg lies on a horizontal surface in a truck. The coefficient of static friction between the block and the surface is 0.6. If the acceleration of the truck is 5 m/s^2, the frictional force acting on the block is :-

(1) 5N (2) 6N (3) 10N (4) 15N

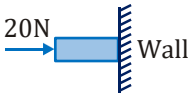
PLM132

133. A body of mass 10 kg lies on a rough horizontal surface. When a horizontal force of F newton acts on it, it gets an acceleration of 5m/s^2, and when the horizontal force is doubled, it gets an acceleration of 18m/s^2. The coefficient of friction between the body and the horizontal surface is :-

(1) 0.2 (2) 0.4 (3) 0.6 (4) 0.8

PLM133

134. A horizontal force of 20 N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and wall is 0.15. The weight of the block is :-

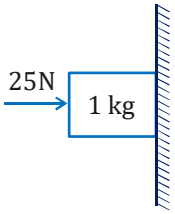


(1) 20 N (2) 3 N

(3) 30 N (4) 0.3 N

PLM134

135. A block of mass 1 kg. is pressed against a wall with a horizontal force of 25 N as shown in the figure. If the coefficient of friction between the wall and the block is 0.5 then the frictional force acting on the block will be (g = 9.8 m/s^2) :-

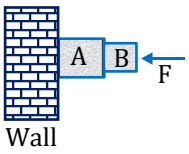


(1) 9.8 N (2) 2.5 N

(3) 12.5 N (4) 4.9 N

PLM135

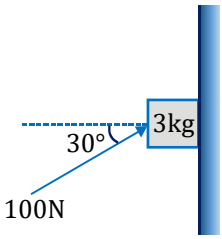
136. Adjoining figure shows two blocks A and B pushed against the wall with a force F. The wall is smooth but the surfaces in contact of A and B are rough. Which of the following is true for the system of blocks to be at rest against the wall ?



(1) F should be more than the weight of A and B
(2) F should be equal to the weight of A and B
(3) F should be less than the weight of A and B
(4) system cannot be in equilibrium

PLM136

137. A force of 100N is applied on a block of mass 3 kg as shown in the figure. The coefficient of friction between the surface and the block is $\mu = \frac{1}{\sqrt{3}}$. The frictional force acting on the block is :-



(1) 15 N downwards
(2) 25 N upwards
(3) 20 N downwards
(4) 30 N upwards

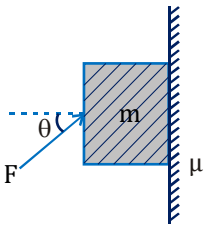
PLM137

138. In previous problem the acceleration of block is :-

(1) zero
(2) 10 m/s² upwards
(3) 10 m/s² downwards
(4) none of these

PLM138

139. The minimum value of force F required to keep the block stationary is :-



(1) $\frac{mg}{\mu\cos\theta}$ (2) $\frac{mg}{\sin\theta+\mu\cos\theta}$

(3) $\frac{mg}{\sin\theta-\mu\cos\theta}$ (4) $\frac{mg}{\mu\tan\theta}$

PLM139

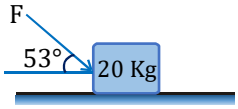
140. In previous question the value of F for which friction force between the block and the wall is zero.

(1) mg (2) $\frac{mg}{\sin\theta}$

(3) $\frac{mg}{\cos\theta}$ (4) $\frac{mg}{\tan\theta}$

PLM140

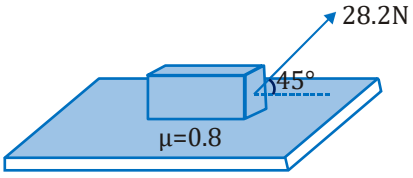
141. A block of mass 20 kg is acted upon by a force F = 30 N at an angle 53° with the horizontal in downward direction as shown in diagram. The coefficient of friction between the block and the horizontal surface is 0.2. The friction force acting on the block by the ground is ($g = 10\ m/s^2$)



(1) 40.0 N (2) 30.0 N
(3) 18.0 N (4) 44.8 N

PLM141

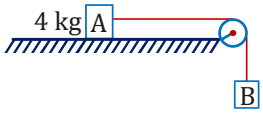
142. A body of weight 50 N is placed on a horizontal surface. A force of 28.2 N is applied on it as shown in diagram. The frictional force and normal reaction on the body are



(1) 2N, 3N (2) 5N, 6N
(3) 10N, 15N (4) 20N, 30N

PLM142

143. The coefficient of static friction, μ_s, between block A of mass 4 kg and the table as shown in the figure is 0.3. What would be the maximum mass value of block B so that the two blocks do not move ? The string and the pulley are assumed to be smooth and massless.
($g = 10\ m/s^2$)



(1) 12 kg (2) 0.2 kg
(3) 0.12 kg (4) 1.2 kg

PLM143

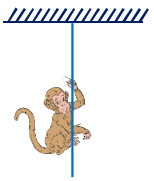
144. A rope lies on a table such that a part of it hangs down the table. When the length of hanging part is 1/5 of entire length the rope just begins to slide. The coefficient of friction between the rope and the table is :-
(1) 1/4 (2) 1/6 (3) 1/5 (4) 3/4

PLM144

145. A block of mass m = 2 kg is placed at rest with respect to moving plank accelerating with acceleration a = 1 m/s^2. If coefficient of friction between plank and block is $\mu = 0.2$. The frictional force acting on the block is :-
(1) 2 N (2) 4 N
(3) 3 N (4) none of these

PLM145

146. In the given figure a monkey of mass m is climbing up a rope hanging from the roof with acceleration a. The coefficient of static friction between the body of the monkey and the rope is μ. value of friction force on the monkey is :-



(1) mg (2) m(g + a)
(3) μ mg (4) m(g – a)

PLM146

147. A block has been placed on an inclined plane with the slope angle θ, the block slides down the plane at constant speed. The coefficient of kinetic friction is equal to :-
(1) $\sin\theta$ (2) $\cos\theta$ (3) g (4) $\tan\theta$

PLM147

148. A car is moving along a straight horizontal road with a speed $2v_0$. If the coefficient of friction between tyres and the road is μ. The shortest distance in which the car can be stopped is :

(1) $\frac{v_0^2}{2\mu g}$ (2) $\frac{v_0^2}{\mu g}$
(3) $\left(\frac{v_0}{\mu g}\right)^2$ (4) $\frac{2v_0^2}{\mu g}$

PLM148

149. A body of mass 2 kg is placed on a inclined plane with an inclination of 30^0. What is the frictional force experienced, if coefficient of friction is 0.8 ? (Take $g = 10\ m/s^2$)
(1) 8.5 N (2) 13.6 N (3) 10 N (4) 8 N

PLM149

150. The force required to just move a body up an inclined plane is 3 times the force required to just prevent the body from sliding down the plane. The coefficient of friction is μ. The inclination θ of the plane is :-
(1) $\tan^{-1}(\mu)$ (2) $\tan^{-1}(\mu/2)$
(3) $\tan^{-1}(2\mu)$ (4) $\tan^{-1}(3\mu)$

PLM150

151. A body is sliding down an inclined plane (angle of inclination 45°). If the coefficient of friction is 0.25 and $g = 10\ m/s^2$. then the approximate downward acceleration of the body in m/s^2 is :-
(1) 7 (2) 14 (3) 3.5 (4) 5.4

PLM151

152. A block of mass m is lying on an inclined plane. The coefficient of friction between the plane and the block is μ. The force (F_1) required to move the block up the inclined plane will be:-
(1) $mg\sin\theta + \mu mg\cos\theta$
(2) $mg\cos\theta - \mu mg\sin\theta$
(3) $mg\sin\theta - \mu mg\cos\theta$
(4) $mg\cos\theta + \mu mg\sin\theta$

PLM152

AIM: FREE EDUCATION TO ALL APUL

153. A block slides with constant velocity on a plane inclined at an angle θ. The same block is projected up the plane with an initial velocity v_0. The distance covered by the block before coming to rest is :-

(1) $\frac{v_0^2}{2g\sin\theta}$ (2) $\frac{v_0^2}{4g\sin\theta}$

(3) $\frac{v_0^2\sin^2\theta}{2g}$ (4) $\frac{v_0^2\sin^2\theta}{4g}$

PLM153

154. A given object takes n times as much time to slide down a 45^0 rough incline as it takes to slide down a perfectly smooth 45^0 incline. The coefficient of kinetic friction between the object and the incline is given by :

(1) $\left(1-\frac{1}{n^2}\right)$ (2) $\left(\frac{1}{1-n^2}\right)$

(3) $\sqrt{\left(1-\frac{1}{n^2}\right)}$ (4) $\sqrt{\left(\frac{1}{1-n^2}\right)}$

PLM154

155. A body is placed on an inclined plane and has to be pushed down in order to make it move. The angle made by the normal reaction with the vertical will be :-

(1) Equal to angle of repose
(2) Equal to the angle of friction
(3) Less than the angle of repose
(4) More than the angle of friction

PLM155

156. A block 'A' kept on an inclined surface just begins to slide if the inclination is 30° with the horizontal. The block is replaced by another block 'B' and it is found that it just begins to slide if the inclination is 40° with the horizontal :-

(1) mass of A > mass of B
(2) mass of A < mass of B
(3) mass of A = mass of B
(4) all the three are possible

PLM156

157. If the coefficient of friction of a surface is $\sqrt{3}$, then the angle of inclination of this surface from the horizontal to make a body on it just to slide, is :-

(1) 30° (2) 45° (3) 60° (4) 75°

PLM157

158. A block is placed on a rough floor and a horizontal force F is applied on it. The force of friction f by the floor on the block is measured for different values of F and a graph is plotted between them ($\mu_k < \mu_s$):-

(a) The graph is a straight line of slope 45°
(b) The graph is straight line parallel to the F-axis
(c) The graph is a straight line of slope 45° for small F and a straight line parallel to the F-axis for large F.
(d) There is small kink on the graph.

(1) c, d (2) a, d (3) a, b (4) a, c

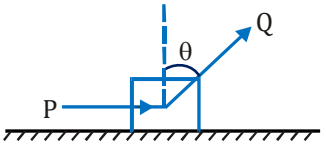
PLM158

159. A block of mass 4 kg rests on an inclined plane. The inclination of the plane is gradually increased. it is found that when the inclination is 3 in 5, the block just begins to slide down the plane. The coefficient of friction between the block and the plane is :-

(1) 0.4 (2) 0.6 (3) 0.8 (4) 0.75

PLM159

160. A block of mass m lying on a rough horizontal plane is acted upon by a horizontal force P and another force Q inclined at an angle θ to the vertical. The block will remain in equilibrium if the coefficient of friction between it and the surface is :-

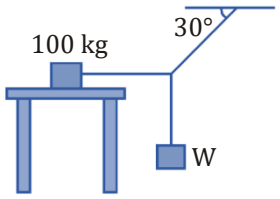


(1) $\frac{P+Q\sin\theta}{mg+Q\cos\theta}$ (2) $\frac{P\cos\theta+Q}{mg-Q\sin\theta}$

(3) $\frac{P+Q\cos\theta}{mg+Q\sin\theta}$ (4) $\frac{P+Q\sin\theta}{mg-Q\cos\theta}$

PLM160

161. The system shown in the figure is in equilibrium. The maximum value of W, so that the maximum value of static frictional force on 100 kg body is 346 N, will be :-



(1) 100 N (2) 250 N
(3) 200 N (4) 1000 N

PLM161

TWO BLOCK SYSTEMS

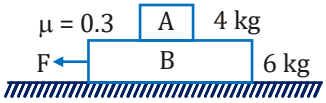
162. A 8 kg block (A) is placed on 2 kg block (B) which rests on a table. Coefficient of friction between (A) and (B) is 0.5 and between (B) and table is 0.2. A 60 N horizontal force is applied on the block (A), then the friction force between the blocks (A) and (B) is :-

(1) Zero (2) 2.5 N

(3) 28 N (4) 40 N

PLM162

163. Two blocks (A) 4 kg and (B) 6 kg rest one over the other on a smooth horizontal plane. The coefficient of static and dynamic friction between (A) and (B) is the same and equal to 0.3. The maximum horizontal force F that can be applied to (B) in order that both (A) and (B) do not have any relative motion is :

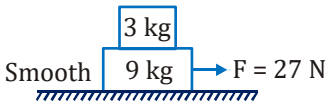


(1) 30 N (2) 30 kgf

(3) 12 kgf (4) 12 N

PLM163

164. In the arrangement coefficient of friction between the two blocks is $\mu = \frac{1}{3}$. The force of friction acting between the two blocks is :-



(1) 10 N (2) 6.75 N

(3) 40 N (4) 13.5 N

PLM164

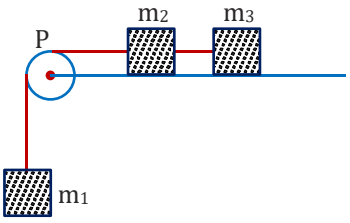
EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	1	2	1	2	2	3	3	1	4	2	1	2	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	1	3	3	4	1	3	4	3	3	4	2	3	1	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	1	2	4	1	1	2	3	3	3	3	3	1	2	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	3	3	3	2	4	4	1	3	3	4	2	2	1	4
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	3	1	4	2	3	4	1	3	4	4	1	3	3	1	3
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	3	4	3	2	3	1	2	4	2	2	2	2	3	1	3
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	3	1	3	1	4	3	3	3	3	4	4	2	2	2	1
Question	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	2	1	3	3	3	1	2	1	3	2	3	4	2	3	1
Question	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Answer	3	4	2	3	3	2	1	2	1	3	4	1	4	2	1
Question	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Answer	4	3	1	2	2	3	4	4	1	1	2	4	4	3	3
Question	151	152	153	154	155	156	157	158	159	160	161	162	163	164	
Answer	4	1	2	1	3	4	3	1	4	4	3	3	1	2	

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. A system consists of three masses m_1, m_2 and m_3 connected by a string passing over a pulley P. The mass m_1 hangs freely m_2 and m_3 are on a rough horizontal table (the coefficient of friction = μ). The pulley is frictionless and is of negligible mass. The downward acceleration of mass m_1 is:
(Assume $m_1 = m_2 = m_3 = m$)

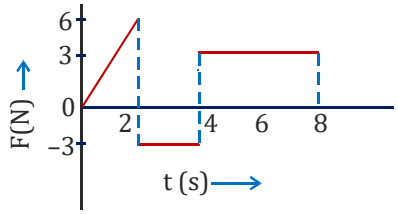


(1) $\frac{g(1-2\mu)}{9}$ (2) $\frac{2g\mu}{3}$

(3) $\frac{g(1-2\mu)}{3}$ (4) $\frac{g(1-2\mu)}{2}$

PLM171

2. The force 'F' acting on a particle of mass 'm' is indicated by the force-time graph shown below. The change in momentum of the particle over the time interval from zero to 8 s is :-



(1) 24 Ns (2) 20 Ns (3) 12 Ns (4) 6 Ns

PLM172

3. A balloon with mass 'm' is descending down with an acceleration 'a' (where a < g). How much mass should be removed from it so that it starts moving up with an acceleration 'a' ? (Assume that it's volume does not change)

(1) $\frac{2ma}{g+a}$ (2) $\frac{2ma}{g-a}$ (3) $\frac{ma}{g+a}$ (4) $\frac{ma}{g-a}$

PLM173

AIPMT 2015

4. A block A of mass m_1 rests on a horizontal table. A light string connected to it passes over a frictionless pulley at the edge of table and from its other end another block B of mass m_2 is suspended. The coefficient of kinetic friction between the block and the table is μ_k. When the block A is sliding on the table, the tension in the string is :-

(1) $\frac{(m_2-\mu_k m_1)g}{(m_1+m_2)}$ (2) $\frac{m_1m_2(1+\mu_k)g}{(m_1+m_2)}$

(3) $\frac{m_1m_2(1-\mu_k)g}{(m_1+m_2)}$ (4) $\frac{(m_2+\mu_k m_1)g}{(m_1+m_2)}$

PLM174

5. Three blocks A, B and C of masses 4 kg, 2 kg and 1 kg respectively, are in contact on a frictionless surface, as shown. If a force of 14 N is applied on the 4 kg block, then the contact force between A and B is :

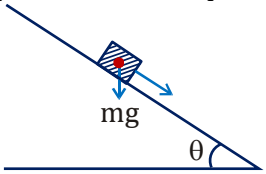


(1) 6 N (2) 8 N (3) 18 N (4) 2 N

PLM175

Re-AIPMT 2015

6. A plank with a box on it at one end is gradually raised about the other end. As the angle of inclination with the horizontal reaches 30°, the box starts to slip and slides 4.0 m down the plank in 4.0s. The coefficients of static and kinetic friction between the box and the plank will be, respectively :



(1) 0.4 and 0.3 (2) 0.6 and 0.6
(3) 0.6 and 0.5 (4) 0.5 and 0.6

PLM176

NEET-II 2016

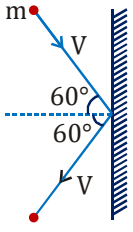
7. A rigid ball of mass m strikes a rigid wall at 60° and gets reflected without loss of speed as shown in the figure below. The value of impulse imparted by the wall on the ball will be :-

(1) $\frac{mV}{2}$

(2) $\frac{mV}{3}$

(3) mV

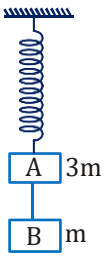
(4) 2mV



PLM177

NEET(UG) 2017

8. Two blocks A and B of masses 3 m and m respectively are connected by a massless and inextensible string. The whole system is suspended by a massless spring as shown in figure. The magnitudes of acceleration of A and B immediately after the string is cut, are respectively :-



(1) $\frac{g}{3}$, g (2) g, g (3) $\frac{g}{3}$, $\frac{g}{3}$ (4) g, $\frac{g}{3}$

PLM178

NEET(UG) 2018

9. Which one of the following statements is **incorrect** ?

(1) Rolling friction is smaller than sliding friction

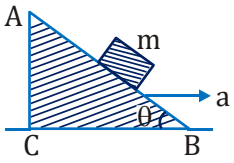
(2) Limiting value of static friction is directly proportional to normal reactions

(3) Frictional force opposes the relative motion

(4) Coefficient of sliding friction has dimensions of length

PLM179

10. A block of mass m is placed on a smooth inclined wedge ABC of inclination θ as shown in the figure. The wedge is given an acceleration 'a' towards the right. The relation between a and θ for the block to remain stationary on the wedge is :-



(1) $a = \frac{g}{\text{cosec}\,\theta}$ (2) $a = \frac{g}{\sin\theta}$

(3) a = g cos θ (4) a = g tan θ

PLM180

NEET(UG) 2019 (Odisha)

11. A truck is stationary and has a bob suspended by a light string, in a frame attached to the truck. The truck, suddenly moves to the right with an acceleration of a. The pendulum will tilt :-

(1) to the left and angle of inclination of the pendulum with the vertical is $\sin^{-1}(g/a)$

(2) to the left and angle of inclination of the pendulum with the vertical is $\tan^{-1}(a/g)$

(3) to the left and angle of inclination of the pendulum with the vertical is $\sin^{-1}(a/g)$

(4) to the left and angle of inclination of the pendulum with the vertical is $\tan^{-1}(g/a)$

PLM181

12. A body of mass m is kept on a rough horizontal surface (coefficient of friction = μ) A horizontal force is applied on the body, but it does not move. The resultant of normal reaction and the frictional force acting on the object is given by F, where F is :

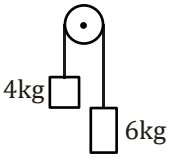
(1) $|\vec{F}| = mg + \mu mg$ (2) $|\vec{F}| = \mu mg$

(3) $|\vec{F}| \le mg\sqrt{1+\mu^2}$ (4) $|\vec{F}| = mg$

PLM182

NEET(UG) 2020

13. Two bodies of mass 4kg and 6kg are tied to the ends of a massless string. The string passes over a pulley which is frictionless (see figure). The acceleration of the system in terms of acceleration due to gravity (g) is :

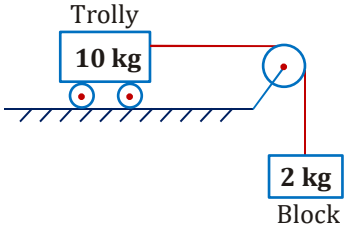


(1) g/10 (2) g (3) g/2 (4) g/5

PLM183

NEET(UG) 2020 (COVID-19)

14. Calculate the acceleration of the block and trolly system shown in the figure. The coefficient of kinetic friction between the trolly and the surface is 0.05. (g = 10 m/s², mass of the string is negligible and no other friction exists).



(1) 1.25 m/s² (2) 1.50 m/s²

(3) 1.66 m/s² (4) 1.00 m/s²

PLM184

NEET(UG) 2021

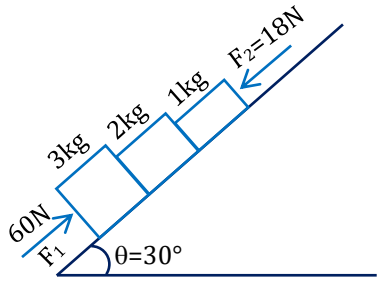
15. A ball of mass 0.15 kg is dropped from a height 10 m, strikes the ground and rebounds to the same height. The magnitude of impulse imparted to the ball is ($g = 10\ m/s^2$) nearly :

(1) 0 kg m/s (2) 4.2 kg m/s
(3) 2.1 kg m/s (4) 1.4 kg m/s

PLM185

RE-NEET(UG) 2022

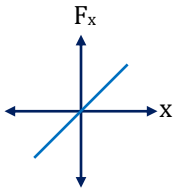
16. 22. In the diagram shown, the normal reaction force between 2 kg and 1 kg is (Consider the surface, to be smooth):
Given $g = 10\ ms^{-2}$

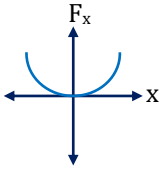


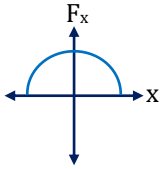
(1) 25 N (2) 39 N (3) 6 N (4) 10 N

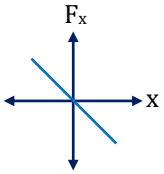
PLM186

17. The restoring force of a spring with a block attached to the free end of the spring is represented by :

(1) 

(2) 

(3) 

(4) 

PLM187

18. The distance covered by a body of mass 5 g having linear momentum 0.3 kg m/s in 5 s is:

(1) 300 m (2) 30 m (3) 3 m (4)0.3 m

PLM188

NEET(UG) 2023

19. A football player is moving southward and suddenly turns eastward with the same speed to avoid an opponent. The force that acts on the player while turning is :

(1) along northward (2) along north-east
(3) along south-west (4) along eastward

PLM202

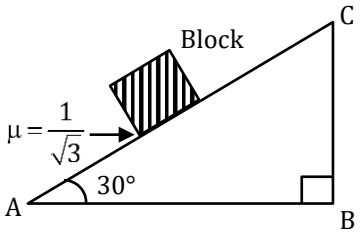
20. Calculate the maximum acceleration of a moving car so that a body lying on the floor of the car remains stationary. The coefficient of static friction between the body and the floor is 0.15 ($g = 10\ m\ s^{-2}$).

(1) 150 m s^{-2} (2) 1.5 m s^{-2}
(3) 50 m s^{-2} (4) 1.2 m s^{-2}

PLM203

RE-NEET(UG) 2023

21. A block of mass 2 kg is placed on inclined rough surface AC (as shown in figure) of coefficient of friction μ. If $g = 10\ m\ s^{-2}$, the net force (in N) on the block will be :



(1) $10\sqrt{3}$ (2) zero
(3) 10 (4) 20

PLM204

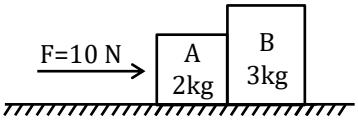
22. A 1 kg object strikes a wall with velocity 1 m s^{-1} at an angle of 60° with the wall and reflects at the same angle. If it remains in contact with wall for 0.1 s, then the force exerted on the wall is :-

(1) $30\sqrt{3}$ N (2) Zero
(3) $10\sqrt{3}$ N (4) $20\sqrt{3}$ N

PLM205

NEET(UG) 2024

23. A horizontal force 10 N is applied to a block A as shown in figure. The mass of blocks A and B are 2 kg and 3 kg, respectively. The blocks slide over a frictionless surface. The force exerted by block A on block B is :



(1) zero (2) 4 N
(3) 6 N (4) 10 N

PLM206

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RE-NEET(UG) 2024

24. A box of mass 5 kg is pulled by a cord, up along a frictionless plane inclined at 30° with the horizontal. The tension in the cord is 30 N. The acceleration of the box is
(Take g = 10 m s^{-2})
(1) 2 m s^{-2} (2) zero
(3) 0.1 m s^{-2} (4) 1 m s^{-2}

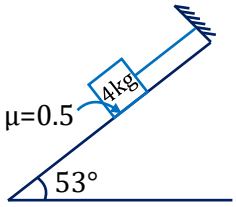
PLM207

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	1	2	1	3	3	1	4	4	2	3	4	1	2
Question	16	17	18	19	20	21	22	23	24						
Answer	1	4	1	2	2	2	3	3	4						

Exercise - III (Analytical Questions) Master Your Understanding

1. For the given figure, match column-I to column-II

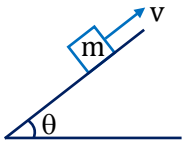


Column-I		Column-II	
(a)	Tension in the string	(P)	24 N
(b)	Normal on the block	(Q)	20 N
(c)	Contact force between block and inclined plane	(R)	12 N
(d)	Friction on the block	(S)	27 N

(1) (a) – Q, (b) – P, (c) – S, (d) – R
(2) (a) – P, (b) – Q, (c) – S, (d) – R
(3) (a) – P, (b) – Q, (c) – R, (d) – S
(4) (a) – Q, (b) – P, (c) – R, (d) – S

PLM190

2. A block of mass m is thrown upwards with some initial velocity as shown. On the block



Column-I		Column-II	
(a)	Net force in horizontal direction	(P)	zero
(b)	Net force in vertical direction	(Q)	$m(g\sin\theta+\mu g\cos\theta)$
(c)	Net force along the plane	(R)	$m(g\sin\theta\cos\theta+\mu g\cos^2\theta)$
(d)	Net force perpendicular to plane	(S)	$m(g\sin^2\theta+\mu g\sin\theta\cos\theta)$
		(T)	None

(1) (a) – P (b) – R (c) – Q (d) – P
(2) (a) – R (b) – S (c) – Q (d) – P
(3) (a) – P (b) – S (c) – Q (d) – P
(4) (a) – Q (b) – R (c) – P (d) – R

PLM191

3. If the net force acting on a system is represented by $\vec{F}$ and its momentum is $\vec{P}$. Then match the entries of column I with the entries of Column II.

Column-I		Column-II	
(a)	If $\vec{F}$ is constant	(P)	$\vec{P}$ may change its direction
(b)	If $\vec{F}$ is changing in magnitude only	(Q)	$\vec{P}$ must change its magnitude
(c)	If $\vec{F}$ is changing in direction only	(R)	$\vec{P}$ may not change its direction
(d)	If $\vec{F}$ is zero	(S)	$\vec{P}$ may be constant in magnitude

(1) (a) – P,Q,R (b) – P,Q,R (c) – P,S (d) – S
(2) (a) – P,R (b) – P,Q (c) – P,R (d) – S
(3) (a) – P,Q (b) – Q,R (c) – P,R (d) –S
(4) (a) – P,R (b) – P,Q,R (c) – P (d) – S

PLM192

4. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : When a block is placed in a lift which is accelerating upwards then the body experiences the following three forces :
(a) Weight (mg)
(b) Normal reaction (N) and
(c) Pseudo force (ma) if 'a' is acceleration of lift

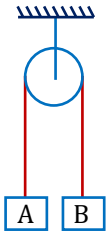
Reason (R) : Pseudo force is applied on a body only when body is seen from an accelerated observer.

In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PLM194

5. Equilibrium means zero acceleration while rest means zero velocity.
(1) True (2) False
(3) Partially true (4) None of the above
PLM195

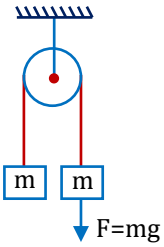
6. Two blocks of mass 2 kg and 3 kg are arranged as shown and released from rest. Assuming string and pulley to be ideal. Find the separation between them after 1 sec. [g =10 m/s²]



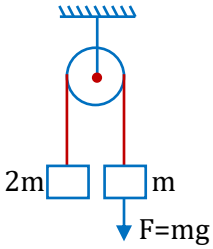
(1) 1 m (2) 2 m
(3) 4 m (4) None
PLM198

7. Match the situations in column I to the accelerations of blocks in the column II (acceleration due to gravity is g and F is an additional force applied to one of the blocks ?

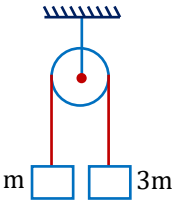
Column I **Column II**

(A)


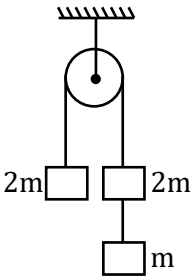
(P) $\frac{g}{5}$

(B)


(Q) $\frac{g}{3}$

(C)


(R) $\frac{g}{2}$

(D)


(S) 0

Options :-
(1) A → R, B → S, C → R, D → P
(2) A → R, B → S, C → Q, D → P
(3) A → Q, B → P, C → R, D → S
(4) A → P, B → Q, C → R, D → S
PLM199

8. If a body is placed on a rough inclined plane, the nature of forces acting on the body is(are)
(A) gravitational (B) electromagnetic
(C) nuclear (D) weak
(1) Only A (2) A and D
(3) A, B, C, D (4) A and B
PLM200

9. **Statement-I** : When a cloth with a lot of dirt in it, is jerked the dirt gets away from it.
Statement-II : When an object applies force on another object the other object also applies an equal force but opposite in direction.
(1) Statement–1 is True, Statement–2 is True ; Statement–2 is a correct explanation for Statement–1.
(2) Statement–1 is True, Statement–2 is True ; Statement–2 is not a correct explanation for Statement–1.
(3) Statement–1 is True, Statement–2 is False.
(4) Statement–1 is False, Statement–2 is True.
PLM201

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	1	2	1	4	1	2	1	4	2

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Exercise - I (Conceptual Questions) Build Up Your Understanding

DISTANCE & DISPLACEMENT

1. If displacement of a particle is zero, the distance covered :
(1) must be zero
(2) may or may not be zero
(3) cannot be zero
(4) depends upon the particle

PML001

2. If the distance covered is zero, the displacement :
(1) must be zero
(2) may or may not be zero
(3) cannot be zero
(4) depends upon the particle

PML002

3. The location of a particle is changed. What can we say about the displacement and distance covered by the particle :
(1) Both cannot be zero
(2) One of the two may be zero
(3) Both must be zero
(4) If one is positive, the other is negative and vice-versa

PML003

4. The numerical ratio of distance to the displacement covered is always :–
(1) less than one
(2) equal to one
(3) equal to or less than one
(4) equal to or greater than one

PML004

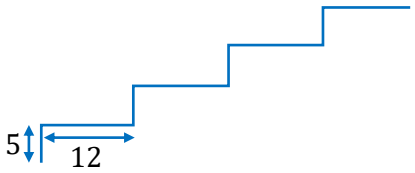
5. Milkha Singh can cover one round of a circular park in 40 second, After 1 minute and 40 second, he will cover a distance and displacement respectively
(R = Radius of circle):-
(1) Zero, Zero (2) $4\pi R$, R
(3) $5\pi R$, 2R (4) $6\pi R$, 2R

PML005

6. A hall has the dimensions 10 m × 10 m × 10 m. A fly starting at one corner ends up at a diagonally opposite corner. The magnitude of its displacement is nearly
(1) $5\sqrt{3}$ m (2) $10\sqrt{3}$ m
(3) $20\sqrt{3}$ m (4) $30\sqrt{3}$ m

PML006

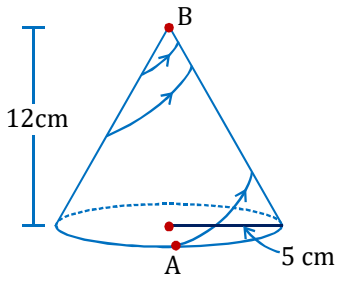
7. An ant is scramping on the stairs as shown in the figure. There are '4' stairs and each stair has width of 12 cm and height of 5 cm. The distance travelled by the ant to scramp the stairs is :-



(1) 52 cm (2) 68 cm
(3) 48 cm (4) 20 cm

PML007

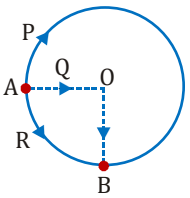
8. An insect starts climbing a conical birthday hat of radius 5 cm at base. It starts from point A and reaches point B, taking spiral path on the hat. Find out its displacement if height is 12 cm:-



(1) 12 cm (2) 8 cm
(3) 13 cm (4) 25 cm

PML008

9. Three particles P, Q and R are situated at point A on the circular path of radius 10 m. All three particles move along different paths and reach point B as shown in figure. Then the ratio of distance traversed by particles P and R is :



(1) $\frac{3}{4}$ (2) $\frac{3}{1}$ (3) $\frac{3\pi}{4}$ (4) $\frac{\pi}{3}$

PML009

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10. A body moves along the curved path of a semi circle. Calculate the ratio of distance to displacement:
(1) 11 : 7 (2) 7 : 11
(3) 11 : $\sqrt{2} \times 7$ (4) 7 : $11\sqrt{2}$
PML010

11. Distance travelled by the tip of minute hand of length 10 cm in 100 sec is
(1) $\frac{\pi}{180}$m (2) $\frac{\pi}{360}$m
(3) $\frac{\pi}{1200}$m (4) $\frac{3\pi}{2160}$m
PML011

12. If a particle moves from point P(2, 3, 5) to point Q(3, 4, 5). Its displacement vector be :-
(1) $\hat{i} + \hat{j} + 10\hat{k}$ (2) $\hat{i} + \hat{j} + 5\hat{k}$
(3) $\hat{i} + \hat{j}$ (4) $2\hat{i} + 4\hat{j} + 6\hat{k}$
PML012

13. A man walks 30 m towards north, then 20 m towards east and the last $30\sqrt{2}$ m towards south-east. The displacement from origin is :
(1) 10 m towards west
(2) 50 m towards east
(3) $60\sqrt{2}$ m towards north west
(4) $60\sqrt{2}$ m towards east north
PML013

14. A person walks 80 m east, then turns right through angle 143° walks further 50 m and stops. His position relative to the starting point is
(1) 50 m, 53° east of south
(2) 50 m, 53° south of east
(3) 30 m, 37° south of east
(4) 30 m, 53° south of east
PML014

15. A drunkard is walking along a straight road. He takes 5 steps forward and 3 steps backward, followed by 5 steps forward and 3 steps backward and so on. Each step is one meter long and takes one second. There is a pit on the road 13 meters away from the starting point. The drunkard will fall into the pit after :
(1) 29 s (2) 21 s (3) 37s (4) 31 s
PML015

SPEED & VELOCITY, AVERAGE SPEED & AVERAGE VELOCITY

16. A train covers the first half of the distance between two stations with a speed of 30 km/h and the other half with 70 km/h. Then its average speed is :
(1) 50 km/h (2) 48 km/h
(3) 42 km/h (4) 100 km/h
PML016

17. Select the correct statements from the following.
S1. Average velocity is path length divided by time interval.
S2. In general, average speed $\geq$ |average velocity|
S3. A particle moving in a given direction with a non-zero velocity can have zero speed.
S4. The magnitude of average velocity is the average speed.
(1) S_1 (2) S_2 (3) S_3 (4) S_4
PML017

18. The magnitude of average velocity is equal to the average speed when a particle moves :
(1) on a curved path
(2) in the same direction
(3) with constant acceleration
(4) with constant retardation
PML018

19. A man walks for some time 't' with velocity (v) due east. Then he walks for same time 't' with velocity (v) due north. The average speed of the man is :
(1) 2v (2) $\sqrt{2}$v (3) v (4) $\frac{v}{\sqrt{2}}$
PML019

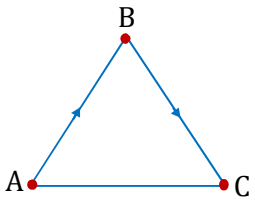
20. A car travels a distance d on a straight road in two hours and then returns to the starting point in next three hours. Its average speed and average velocity is :
(1) $\frac{d}{5}$, 0 (2) $\frac{2d}{5}$, 0
(3) $\frac{5d}{6}$, $\frac{d}{5}$ (4) none of these
PML020

21. A particle moves in the east direction with 15 m/sec for 2 sec then northwards with 5 m/s for 8 sec. Average velocity of the particle is :–

(1) 1 m/s (2) 5 m/s
(3) 7 m/s (4) 10 m/s

PML021

22. A man walks on an equilateral triangle along path ABC with constant speed then the ratio of average speed and magnitude of average velocity for A to C :-



(1) 1 (2) 2 (3) $\frac{1}{2}$ (4) None

PML022

23. A car runs at constant speed on a circular track of radius 10 m taking 6.28s on each lap (i.e. round). The average speed and average velocity for half lap is :

(1) Velocity $20/\pi$ m/s, speed 10 m/s
(2) Velocity zero, speed 10 m/s
(3) Velocity zero, speed zero
(4) Velocity 10 m/s, speed zero

PML023

24. A particle moves in straight line in same direction for 20 sec. with velocity 3 m/s and then moves with velocity 4 m/s for another 20 sec. and finally moves with velocity 5 m/s for next 20 sec. What is the average velocity of the particle?

(1) 3 m/s (2) 4 m/s
(3) 5 m/s (4) Zero

PML024

25. An object travels 10 km at a speed of 100 m/s and another 10 km at 50 m/s. The average speed over the whole distance is :-

(1) 75 m/s (2) 55 m/s
(3) 66.7 m/s (4) 33.3 m/s

PML025

26. A body has speed V, 2V and 3V in first 1/3 part of total travelled distance S, second 1/3 part of S and third 1/3 part of S respectively. Its average speed will be :-

(1) V (2) 2V (3) $\frac{18}{11}$V (4) $\frac{11}{18}$V

PML026

27. A body covers one-third of the time with a velocity v_1 the second one-third of the time with a velocity v_2, and the last one-third of the time with a velocity v_3. The average velocity is :

(1) $\frac{v_1+v_2+v_3}{3}$ (2) $\frac{3v_1v_2v_3}{v_1v_2+v_2v_3+v_3v_1}$

(3) $\frac{v_1v_2+v_2v_3+v_3v_1}{3}$ (4) $\frac{v_1v_2v_3}{3}$

PML027

28. If a car cover $2/5^{th}$ of total distance with v_1 speed and $3/5^{th}$ distance with v_2 speed then the average speed is :-

(1) $\frac{1}{2}\sqrt{v_1v_2}$ (2) $\frac{v_1+v_2}{2}$

(3) $\frac{2v_1+v_2}{v_1+v_2}$ (4) $\frac{5v_1v_2}{3v_1+2v_2}$

PML028

29. A particle moving in a straight line covers half the distance with speed of 10 m/s. The other half of the distance is covered in two equal time intervals with speed of 4.5 m/s and 7.5 m/s respectively. The average speed of the particle during this motion is :

(1) 8.0 m/s (2) 12.0 m/s
(3) 10.0 m/s (4) 7.5 m/s

PML029

30. A point object traverses half the distance with velocity v_0. The remaining part of the distance is covered with velocity v_1 for the half time and with velocity v_2 for the rest half. The average velocity of the object for the whole journey is

(1) $2v_1(v_0 + v_2) / (v_0 + 2v_1 + 2v_2)$
(2) $2v(v_0 + v_1) / (v_0 + v_1 + v_2)$
(3) $2v_0(v_1 + v_2) / (v_1 + v_2 + 2v_0)$
(4) $2v_2(v_0 + v_1) / (v_1 + 2v_2 + v_0)$

PML030

31. A scooter going due east at 10 ms^{-1} turns in right side through an angle of 90°. If the speed of the scooter remains unchanged in taking this turn, the change in the velocity of the scooter is :-

(1) 20.0 ms^{-1} in south-west direction

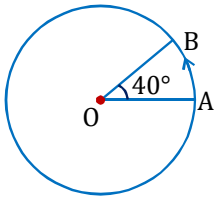
(2) Zero

(3) 10.0 ms^{-1} in south-east direction

(4) 14.14 ms^{-1} in south-west direction

PML031

32. A person is moving in a circle of radius r with constant speed V. The change in velocity in moving from A to B is



(1) 2V cos 40° (2) 2V sin 40°

(3) 2V cos 20° (4) 2V sin 20°

PML032

33. An insect crawls a distance of 4m along north in 10 s and then a distance of 3m along east in 5 s. The average velocity of the insect is :-

(1) $\frac{7}{15}$ m/s (2) $\frac{1}{5}$ m/s

(3) $\frac{1}{3}$ m/s (4) $\frac{4}{5}$ m/s

PML033

ACCELERATION, AVERAGE ACCELERATION & APPLICATION OF CALCULUS

34. The position x of a particle varies with time (t) as $x = at^2 - bt^3$. The velocity at time t of the particle will be equal to zero, where t is equal to :

(1) $\frac{2a}{3b}$ (2) $\frac{a}{b}$ (3) $\frac{a}{3b}$ (4) $\frac{a}{2b}$

PML034

35. If x denotes displacement in time t and x = a sint, then acceleration is :

(1) a cos t (2) – a cos t

(3) a sin t (4) –a sin t

PML035

36. The motion of a particle is described by the equation $x = a + bt^2$ where a = 15 cm and b = 3 cm/sec^2. Its acceleration at time 3 sec will be :-

(1) 36 cm/sec^2 (2) 18 cm/sec^2

(3) 6 cm/sec^2 (4) 32 cm/sec^2

PML036

37. Equation of displacement for a particle is $s = 3t^3 + 7t^2 + 14t + 8$ m. Its acceleration at time t = 2 sec is :-

(1) 10 m/s^2 (2) 16 m/s^2

(3) 25 m/s^2 (4) 50 m/s^2

PML037

38. A body is moving according to the equation $x = at^2 + bt - c$. Then its instantaneous speed is given by :–

(1) a + 2b + 3ct (2) $a + 2bt - 3ct^2$

(3) 2b – 6ct (4) 2at + b

PML038

39. The relation $t = \sqrt{x} + 3$ describes the position of a particle where x is in meters and t is in seconds. The acceleration of particle is :–

(1) 2 m/s^2 (2) 4 m/s^2

(3) 5 m/s^2 (4) zero

PML039

40. The displacement of a particle is given by $y = a + bt + ct^3$. The initial velocity and acceleration are respectively :

(1) b, 0 (2) –b, 2c

(3) b, 2c (4) 2c, – 4d

PML040

41. Equation of a particle moving along the x axis is $x = u(t - 2) + a(t - 2)^2$

(1) the initial velocity of the particle is u

(2) the acceleration of the particle is a

(3) the acceleration of the particle is 2a

(4) at t = 2 particle is not at origin

PML041

42. If for a particle position $x \propto t$ then :–

(1) velocity is constant

(2) acceleration is non zero

(3) acceleration is variable

(4) None of these

PML042

43. The velocity of a body depends on time according to the equation = 20 + 0.1t. The body has :
(1) uniform acceleration
(2) uniform retardation
(3) non-uniform acceleration
(4) zero acceleration

PML043

44. Which of the following relations representing velocity of a particle describes motion with constant acceleration ?
(1) $v = 6 - 7t$ (2) $v = 3t^2 + 5t^3 + 7$
(3) $v = 9t^2 + 8$ (4) $v = 4t^{-2} + 3t^{-1}$

PML044

45. Which of the following equations represents the motion of a body moving with constant finite velocity? in these equations, y denotes the displacement in time t and p, q and r are arbitary constants :
(1) $y = (p + qt)^2 (r + pt)$
(2) $y = p + tqr$
(3) $y = (p + t) (q + t) (r + 1)$
(4) $y = (p + qt)rt$

PML045

46. The displacement of a particle starting from rest (at t=0) is given by $s = 6t^2 - t^3$
The time when the particle will attain zero acceleration is :
(1) 2s (2) 8s (3) 12s (4) 16s

PML046

47. The displacement of a particle varies with time according to the relation $x = \frac{k}{b}[1 - e^{-bt}]$.
Then the velocity of the particle is :
(1) $k(e^{-bt})$ (2) $\frac{k}{b^2 e^{-bt}}$
(3) kbe^{-bt} (4) None of these

PML047

48. A particle moves along a straight line such that its displacement at any time t is given by $s = t^3 - 6t^2 + 3t + 4$ metres. The displacement when the acceleration is zero is :
(1) 3 m (2) –12 (3) 42 m (4) –6 m

PML048

49. Displacement x of a particle is related to time t as $x = at + bt^2 - ct^3$ where a, b and c are constants. The velocity of the particle when its acceleration is zero is given by :-
(1) $a + \frac{b^2}{c}$ (2) $a + \frac{b^2}{2c}$
(3) $a + \frac{b^2}{3c}$ (4) $a + \frac{b^2}{4c}$

PML049

50. A particle moves along the curve $y = \frac{x^2}{2}$. Here x varies with time as $x = \frac{t^2}{2}$. Where x and y are measured in metre and t in second. At t = 2s, the velocity of the particle (in ms^{-1}) is :
(1) $2\hat{i} - 4\hat{j}$ (2) $2\hat{i} + 4\hat{j}$
(3) $4\hat{i} + 2\hat{j}$ (4) $4\hat{i} - 2\hat{j}$

PML050

51. The velocity-time relation of an electron starting from rest is given by u = kt, where $k = 2\ m/s^2$. The distance traversed in 4 sec is :
(1) 9m (2) 16 m (3) 27 m (4) 36 m

PML051

52. The initial velocity of a particle is u (at t = 0) and the acceleration is given by $f = at^2$. Which of the following relations is valid?
(1) $v = u + at^2$ (2) $v = u + \frac{at^2}{2}$
(3) $v = u + \frac{at^3}{3}$ (4) $v = u + at$

PML052

53. A particle is moving with a velocity of 10m/s towards east. After 20 s its velocity changes to 10m/s towards north. Its average acceleration is:-
(1) zero
(2) $\sqrt{2}\ m/s^2$ towards N-W
(3) $\frac{1}{\sqrt{2}}\ m/s^2$ towards N-E
(4) $\frac{1}{\sqrt{2}}\ m/s^2$ towards N-W

PML053

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54. Starting from rest, the acceleration of a particle is $a = 2(t - 1) m/s^2$. The velocity of the particle at t = 5 s is :-

(1) 15 m/s (2) 25 m/s

(3) 5 m/s (4) None of these

PML054

55. If the velocity of a particle is $(10 + 2t^2)$ m/s, then the average acceleration of the particle between 2s and 5s is :-

(1) $2m/s^2$ (2) $4m/s^2$

(3) $12m/s^2$ (4) $14m/s^2$

PML055

56. If velocity of a particle is given by $v = (3t^2 + 2)m/s$, then average velocity in the interval $0 \le t \le 2$sec:-

(1) 6 m/s (2) 8 m/s

(3) 3 m/s (4) 4 m/s

PML056

57. A particle located at x = 0 at time t = 0, starts moving along the positive x–direction with a velocity 'v' which varies as $v = \alpha\sqrt{x}$, then velocity of particle varies with time as : (α is a constant)

(1) $v \propto t$ (2) $v \propto t^2$

(3) $v \propto \sqrt{t}$ (4) v = constant

PML057

58. If the velocity of a particle is given by $v = (180 - 16x)^{1/2}$ m/s, then its acceleration will be:-

(1) Zero (2) $8 m/s^2$

(3) $-8 m/s^2$ (4) $4 m/s^2$

PML058

59. The acceleration of a particle moving in a straight line varies with its displacement as a = 2S + 1. Velocity of the particle is zero at zero displacement. The relation between velocity and displacement is :-

(1) v = S(S + 1) (2) $v = \sqrt{2S(S+1)}$

(3) $v = \sqrt{S(S+1)}$ (4) None of these

PML059

60. The position vector of a particle is given as $\vec{r} = \left(t^2 - 4t + 6\right)\hat{i} + \left(t^2\right)\hat{j}$. The time after which the velocity vector and acceleration vector becomes perpendicular to each other is equal to:-

(1) 1 sec. (2) 2 sec.

(3) 1.5 sec. (4) 5 sec.

PML060

61. At any instant of time acceleration and velocity of a particle are given by $\vec{a} = \hat{i} + \hat{j}$ & $\vec{v} = 6\hat{i} + 8\hat{j}$, then rate of change of speed (component of acceleration along velocity) at the same instant will be:-

(1) $\frac{(3\hat{i} + 4\hat{j})}{25}$ (2) $\frac{6\hat{i} + 8\hat{j}}{25}$

(3) $\frac{7}{25}(3\hat{i} + 8\hat{j})$ (4) $\frac{7}{25}(3\hat{i} + 4\hat{j})$

PML061

CONSTANT ACCELERATION MOTION, FREE FALL

62. If a body starts from rest, the time in which it covers a particular displacement with uniform acceleration is :

(1) inversely proportional to the square root of the displacement

(2) inversely proportional to the displacement

(3) directly proportional to the displacement

(4) directly proportional to the square root of the displacement

PML062

63. A body at rest is imparted motion to move in a straight line. It is then obstructed by an opposite force, then:

(1) the body may necessarily change direction

(2) the body is sure to slow down

(3) the body will necessarily continue to move in the same direction at the same speed

(4) none of the above.

PML063

64. If a car at rest accelerates uniformly to a speed of 144 km/h in 40 seconds, it covers a distance of :

(1) 200 m (2) 800 m

(3) 1440 m (4) 2980 m

PML064

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65. If a car at rest accelerates uniformly and attains a speed of 54 km/h in 10s, then it covers a distance of

(1) 75 m (2) 100 m

(3) 200 m (4) 400 m

PML065

66. If a train travelling at 72 km/h is to be brought to rest in a distance of 100 m, then its retardation should be :

(1) 20 m/s^2 (2) 2 m/s^2

(3) 10 m/s^2 (4) 1 m/s^2

PML066

67. Initially a body is at rest. If its acceleration is $5ms^{-2}$ then the distance travelled in the 5th second is :–

(1) 86.6 m (2) 87.5 m

(3) 88 m (4) 22.5 m

PML067

68. A car starts from rest travelling with constant acceleration. If distance covered by it in 10th second of its journey is 19m, what will be the acceleration of car?

(1) 4 m/s^2 (2) 3 m/s^2

(3) 2 m/s^2 (4) 1 m/s^2

PML068

69. A car starts from rest and moves with constant acceleration. The ratio of the distance covered in the nth second to that covered in n seconds is :-

(1) $\frac{2}{n^2}-\frac{1}{n}$ (2) $\frac{2}{n^2}+\frac{1}{n}$

(3) $\frac{2}{n}-\frac{1}{n^2}$ (4) $\frac{2}{n}+\frac{1}{n^2}$

PML069

70. A body starts from rest. What is the ratio of the distance travelled by the body during the 4th and 5th second?

(1) $\frac{7}{5}$ (2) $\frac{5}{7}$

(3) $\frac{7}{9}$ (4) $\frac{3}{7}$

PML070

71. A car moving with a velocity of 10 m/s can be stopped by the application of a constant force F in a distance of 20m. If the velocity of the car is 40 m/s. It can be stopped by this force in :

(1) $\frac{20}{3}$ m (2) 320 m

(3) 60 m (4) 180 m

PML071

72. A car moving with a speed of 40 km/h can be stopped by applying brakes after at least 2m. If the same car is moving with a speed of 120 km/h., what is the minimum stopping distance?

(1) 2 m (2) 4 m (3) 6 m (4) 18 m

PML072

73. The velocity acquired by a body moving with uniform acceleration is 30 m/s in 2 seconds and 50 m/s in 4 seconds. The initial velocity is :

(1) zero (2) 2 m/s

(3) 4 m/s (4) 10 m/s

PML073

74. A body starts from rest and with a uniform acceleration of 5 ms^{-2} for 5 seconds. During the next 10 seconds it moves with uniform velocity. The total distance travelled by the body is :–

(1) 100 m (2) 312.5 m

(3) 500 m (4) 625 m

PML074

75. The velocity of a particle moving with constant acceleration at an instant t_0 is 10 m/s. After 5 seconds of that instant the velocity of the particle is 20m/s. The velocity at 2 second before t_0 is :

(1) 8 m/s (2) 4 m/s

(3) 6 m/s (4) 7 m/s

PML075

76. A bullet moving with a velocity of 200 cm/s penetrates a wooden block and comes to rest after traversing 4 cm inside it. What velocity is needed for travelling distance of 9 cm in same block :-

(1) 100 cm/s (2) 136.2 cm/s

(3) 300 cm/s (4) 250 cm/s

PML076

77. Which of the following four statements is true?
(1) A body can have zero velocity and still be accelerated
(2) A body can have a constant velocity and still have a varying speed
(3) A body cannot have a constant speed if it has a varying velocity
(4) The direction of the velocity of a body cannot change when its acceleration is constant.

PML077

78. If an iron ball and a wooden ball of same radii are released from a height h in vacuum then time taken by both of them to reach ground will be :
(1) unequal (2) exactly equal
(3) roughly equal (4) zero

PML078

79. Three different objects of masses m_1, m_2 and m_3 are allowed to fall from rest and from the same point 'O' along three different frictionless paths. The speeds of the three objects on reaching the ground, will be in the ratio of :–
(1) $m_1 : m_2 : m_3$ (2) $m_1 : 2m_2 : 3m_3$
(3) 1 : 1 : 1 (4) $\frac{1}{m_1} : \frac{1}{m_2} : \frac{1}{m_3}$

PML079

80. A body dropped from a tower reaches the ground in 5s. The height of the tower is about :
(1) 80 m (2) 125 m (3) 160 m (4) 40 m

PML080

81. A stone falls freely such that the distance covered by it in the last second of its motion is equal to the distance covered by it in the first 3 seconds. It remained in air for :–
(1) 2 s (2) 3 s (3) 5 s (4) 6 s

PML081

82. A body dropped from the top of a tower covers 5x distance in the last second of its fall. The time of fall is if x is the distance covered in first second of its fall-
(1) 2 sec (2) 4 sec.
(3) 3 sec. (4) $\left(\frac{50}{7}\right)$ sec

PML082

83. An object is dropped vertically down on earth. The change in its speed after falling through a distance 2d from its highest point is
(1) mgd (2) $\sqrt{2gd}$
(3) $2\sqrt{gd}$ (4) $2\sqrt{\frac{mg}{d}}$

PML083

84. A bullet enters in a thick wooden wall with speed u. If the bullet penetrates a distance 's' into the wood then retardation of the bullet is :
(1) $\frac{u^2}{s}$ (2) $\frac{u^2}{2s}$
(3) $\frac{u^2}{4s}$ (4) $\frac{2u^2}{s}$

PML084

85. A body dropped from the top of a tower covers a distance 9x in the last second of its journey, where x is the distance covered in first second. How much time does it take to reach the ground?
(1) 3s (2) 4s (3) 5s (4) 6s

PML085

86. A stone is dropped into a well in which the level of water is h below the top of the well. If v is velocity of sound, the time T after which the splash is heard is given by.
(1) $T = \frac{2h}{v}$ (2) $T = \sqrt{\frac{2h}{g}} + \frac{h}{v}$
(3) $T = \sqrt{\frac{2h}{v}} + \frac{h}{g}$ (4) $T = \sqrt{\frac{h}{2g}} + \frac{2h}{v}$

PML086

87. A body is released from the top of a tower of height H metres. It takes t time to reach the ground. Where is the body $\frac{t}{3}$ time after the release :
(1) At $\frac{H}{2}$ metres from ground
(2) At $\frac{H}{4}$ metres from ground
(3) At $\frac{8H}{9}$ metres from the ground
(4) At $\frac{H}{9}$ metres from the ground

PML087

88. A body falling from height 'h' takes t_1 time to reach the ground. The time taken to cover the first $\left(\frac{1}{4}\right)^{th}$ of the height is :

(1) $t_2 = \frac{t_1}{\sqrt{2}}$ (2) $t_1 = \frac{t_2}{\sqrt{2}}$

(3) $t_2 = \frac{t_1}{2}$ (4) None of these

PML088

89. A stone is dropped from a certain height which can reach the ground in 5 seconds. It is stopped after 3 seconds of its fall and is again released. The total time taken by the stone to reach the ground will be :

(1) 6 s (2) 6.5 s (3) 7 s (4) 7.5 s

PML089

90. A body released from a height falls freely towards earth. Another body is released from the same point exactly two second later. The separation between them two seconds after the release of the second body is :–

(1) 9.8 m (2) 49 m

(3) 58.8 m (4) 19.6 m

PML090

91. Two balls are dropped from different heights at different instants. Second ball is dropped 2 seconds after the first ball. If both balls reach the ground simultaneously after 5 seconds of dropping the first ball, then the difference between the initial heights of the two balls will be: ($g=9.8m/s^2$)

(1) 58.8 m (2) 78.4 m (3) 98.0 m (4) 117.6 m

PML091

92. A body is released from the top of a tower of height H m. After 2sec it is stopped and then instantaneously released. What will be its height from ground after next 2sec :-

(1) (H–5) m (2) (H–10) m

(3) (H–20)m (4) (H-40) m

PML092

93. The ratio of the distances traversed, in successive intervals of time by a body falling from rest, are

(1) 1 : 3 : 5 : 7 : 9 : .....

(2) 2 : 4 : 6 : 8 : 10 : .....

(3) 1 : 4 : 7 : 10 : 13 : .....

(4) None of these

PML093

94. A particle is dropped from a certain height. The time taken by it to fall through successive distances of 1 km each will be :

(1) all equal, being equal to $\sqrt{\frac{2}{g}}$ second.

(2) in the ratio of the square roots of the integers 1: $\sqrt{2}:\sqrt{3}$

(3) in the ratio of the difference in the square roots of the integers, i.e., $\sqrt{1},,(\sqrt{2}-\sqrt{1}),(\sqrt{3}-\sqrt{2}),(\sqrt{4}-\sqrt{3})$ .....

(4) in the ratio of the reciprocals of the square roots of the integers, ie., $\frac{1}{\sqrt{1}},\frac{1}{\sqrt{2}},\frac{1}{\sqrt{3}}$.....

PML094

95. Drops of water fall from the roof of a building 27m high at regular intervals of time. When the first drop reaches the ground, at the same instant fourth drop begins to fall. What are the distances of the second and third drops from the roof?

(1) 6 m and 12 m (2) 6 m and 3 m

(3) 12 m and 3 m (4) 8 m and 2 m

PML095

96. Water drops fall at regular intervals from a tap 6 m above the ground. The third drop is leaving the tap at the instant the first drop touches the ground. How far above the ground is the second drop at that instant?

(1) 1.25 m (2) 2.50 m

(3) 3.75 m (4) 4.5 m

PML096

97. Four marbles are dropped from the top of a tower one after the other with an interval of one second. The first one reaches the ground after 4 seconds. When the first one reaches the ground the distances between the first and second, the second and third and the third and forth will be respectively :-

(1) 35 m, 25 m and 15 m

(2) 30 m, 20 m and 10 m

(3) 20 m, 10 m and 5 m

(4) 40 m, 30 m and 20 m

PML097

AIM: FREE EDUCATION TO ALL APUL

98. A particle is thrown vertically upward. Its velocity at half of the maximum height is 10m/s. The maximum height attained by it is ($g=10\ ms^{-2}$) :–
(1) 8m (2) 20m (3) 10m (4) 16m
PML098

99. A ball is thrown upward with a velocity of 50 m/s. It will reach the ground after :–
(1) 10 s (2) 20 s (3) 5 s (4) 40 s
PML099

100. Two bodies are thrown vertically upwards with their initial speeds in the ratio 2 : 3. The ratio of the maximum heights reached by them and the ratio of time taken by them to return back to the ground respectively are :-
(1) 4 : 9 and 2 : 3 (2) 2 : 3 and $\sqrt{2}:\sqrt{3}$
(3) $\sqrt{2}:\sqrt{3}$ and 4 : 9 (4) $\sqrt{2}:\sqrt{3}$ and 2 : 3
PML100

101. A particle is thrown from the ground upward with velocity 40 m/s. Calculate maximum height:-
(1) 40 m (2) 80 m (3) 160 m (4) 8 m
PML101

102. If a ball is thrown vertically upwards with 40 m/s. its velocity after three seconds will be :
(1) 10 m/s (2) 20 m/s
(3) 30 m/s (4) 40 m/s
PML102

103. When a ball is thrown vertically up with velocity v_0, it reaches a maximum height 'h'. If one wishes to double the maximum height then the ball should be thrown with velocity –
(1) $\sqrt{2}\,v_0$ (2) $3v_0$
(3) $9v_0$ (4) $3/2v_0$
PML103

104. With what speed should a body be thrown upwards so that the distances traversed in 7th second and 8th second are equal?
(1) 58.4 m/s (2) 49 m/s
(3) $\sqrt{98}$ m/s (4) 68.6 m/s
PML104

105. If a ball is thrown vertically upwards with speed u, the distance covered during the last 't' seconds of its ascent is :–
(1) ut (2) $\frac{1}{2}gt^2$
(3) $ut - \frac{1}{2}gt^2$ (4) $(u + gt)t$
PML105

106. A body thrown vertically upwards with an initial velocity u reaches maximum height in 6 seconds. The ratio of the distance travelled by the body in the first second and the seventh second is :-
(1) 1 : 1 (2) 11 : 1 (3) 1 : 2 (4) 5 : 3
PML106

107. A ball is thrown vertically upwards with velocity 600 m/s. Calculate distance travelled in last 2 sec of its upward motion :-
(1) 20 m (2) 30 m (3) 10 m (4) 25 m
PML107

108. A body is projected vertically up at t = 0 with a velocity of 98 m/s. Another body is projected from the same point with same velocity after 4 seconds. Both bodies will meet at t =
(1) 6 s (2) 8 s (3) 10 s (4) 12 s
PML108

109. A ball is thrown vertically upwards. The ball was observed at a height h twice with a time interval Δt. The initial velocity of the ball is
(1) $\sqrt{8gh+(g\Delta t)^2}$ (2) $\sqrt{2gh+\left(\frac{g\Delta t}{2}\right)^2}$
(3) $\sqrt{8gh+(2g\Delta t)^2}$ (4) $\sqrt{2gh}$
PML109

110. A ball is thrown vertically upwards. Assuming the air resistance to be constant and considerable :–
(1) the time of ascent ≥ the time of descent
(2) the time of ascent < the time of descent
(3) the time of ascent > the time of descent
(4) the time of ascent = the time of descent
PML110

111. A particle is thrown up vertically with a speed 'v_1', in air. It takes time t_1 in upward journey and t_2 (> t_1) in the downward journey and returns to the starting point with a speed v_2.Then:
(1) $v_1 = v_2$
(2) $v_1 < v_2$
(3) $v_1 > v_2$
(4) Data is insufficient
PML111

112. A particle is thrown upwards from ground. It experiences a constant resistance force which produces a retardation of 2 m/s². The ratio of time of ascent to the time of descent is (g = 10 m/s²):-

(1) 1 (2) $\sqrt{\frac{2}{3}}$ (3) $\frac{2}{3}$ (4) $\sqrt{\frac{3}{2}}$

PML112

113. A stone is thrown straight upward with a speed of 20 m/sec from a tower 200 m high. The speed with which it strikes the ground is approximately (g = 9.8 m/s²)

(1) 60 m/sec (2) 65 m/sec
(3) 70 m/sec (4) 75 m/sec

PML113

114. A balloon is at a height of 81 m and is ascending upwards with a velocity of 12 m/s. A body of 2 kg weight is dropped from it. If g = 10 m/s², the body will reach the surface of the earth in:-

(1) 1.5 s (2) 4.025 s
(3) 5.4 s (4) 6.75 s

PML114

115. A ball is thrown vertically upwards from the top of a tower with velocity 4.9 ms⁻¹. It strikes the pond near the base of the tower after 3 seconds. The height of the tower is :-

(1) 73.5 m (2) 44.1 m
(3) 29.4 m (4) None of these

PML115

116. A stone is thrown upwards with a speed 'u' from the top of the tower reaches the ground with a velocity '3u'. The height of the tower is:-

(1) $\frac{3u^2}{g}$ (2) $\frac{4u^2}{g}$ (3) $\frac{6u^2}{g}$ (4) $\frac{9u^2}{g}$

PML116

117. A body dropped from a height h, strikes the ground with velocity 3 m/s. Another body of same mass is thrown downward from the height h with an initial velocity of 4 m/s. Find the final velocity with which it strikes the ground

(1) 3 m/s (2) 4 m/s
(3) 5 m/s (4) 12 m/s

PML117

118. From a building two balls A and B are thrown such that A is thrown upwards and B downwards (both vertically) with same speed. If V_A and V_B are their respective velocities on reaching the ground then :-

(1) $V_B > V_A$ (2) $V_A = V_B$
(3) $V_A > V_B$ (4) None of these

PML118

119. A stone falls from a balloon that is descending at a uniform rate of 10 ms⁻¹. The displacement of the stone from the point of release after 10 seconds is : (g = 10 m/s²)

(1) 490 m (2) 510 m
(3) 600 m (4) 725 m

PML119

120. A parachutist after bailing out falls 50 m without friction. When parachute opens, it decelerates at 2 m/s². He reaches the ground with a speed of 3 m/s. At what approximate height, did he bail out?

(1) 293 m (2) 111 m
(3) 91 m (4) 182 m

PML120

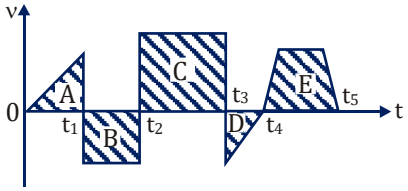
121. A rocket is fired vertically from the ground. It moves upwards with a constant acceleration 20 m/s² for 30 sec after which the fuel is finished. After what time from the instant of firing the rocket will attain the maximum height? (Take g = 10 m/s²)

(1) 90 s (2) 45 s (3) 60 s (4) 75 s

PML121

GRAPHICAL ANALYSIS

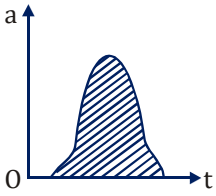
122. The velocity-time graph of an object is shown. The distance during the interval 0 to t_4 is :-



(1) Area A + Area B +Area C +Area D +Area E
(2) Area A + Area C – Area B – Area D
(3) Area A + Area B + Area C + Area D
(4) Area A + Area C + Area E –Area B +Area D

PML122

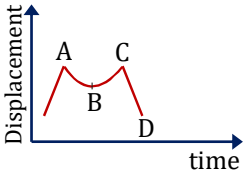
123. Figure below shows the acceleration-time graph of a one dimensional motion. Which of the following characteristics of the particle is represented by the shaded area?



(1) change in velocity
(2) change in position
(3) change in momentum
(4) velocity

PML123

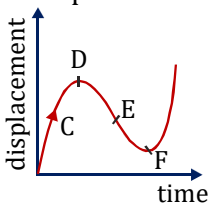
124. Fig. shows the displacement of a particle moving along x-axis as a function of time. The velocity of the particle is zero at :-



(1) A (2) B (3) C (4) D

PML124

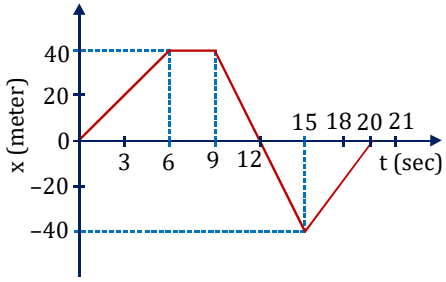
125. The displacement-time graph of a moving particle is shown. The instantaneous velocity of the particle is positive at the point :



(1) D (2) F (3) C (4) E

PML125

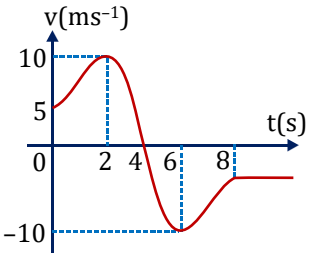
126. A person walks along an east-west street and a graph of his displacement from home is shown in figure. His average speed for the whole time interval is



(1) 0 (2) 23 m/s
(3) 8 m/s (4) None of these

PML126

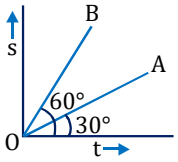
127. A particle is moving in a straight line y=3x. Its velocity time graph is shown in figure. Its speed is minimum at t =.....



(1) 2s (2) 4s (3) 6s (4) 8s

PML127

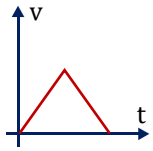
128. The displacement–time graph for two particles A and B are straight lines inclined at angles of 30° and 60° with the time axis. The ratio of velocity of particle A & B ($V_A : V_B$) is :-

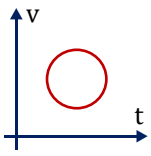


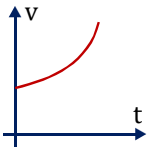
(1) 1 : 2 (2) 1 : $\sqrt{3}$ (3) $\sqrt{3}$: 1 (4) 1 : 3

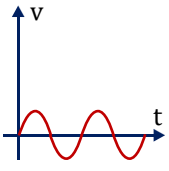
PML128

129. Which one of the following curves do not represent motion in one dimension :-

(1) 

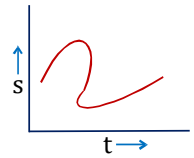
(2) 

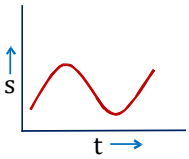
(3) 

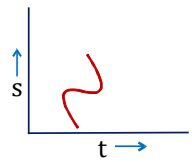
(4) 

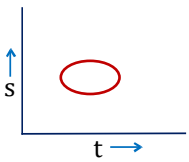
PML129

130. Which of the following displacement–time graphs shows a realistic situation for a body in motion ?

(1) 

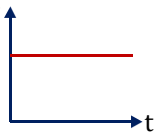
(2) 

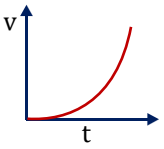
(3) 

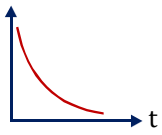
(4) 

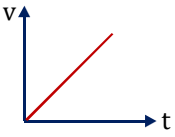
PML130

131. Which of the following velocity-time graphs represent uniformly accelerated motion ?

(1) 

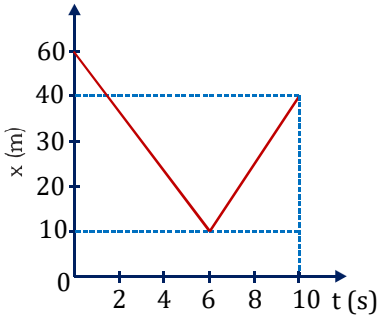
(2) 

(3) 

(4) 

PML131

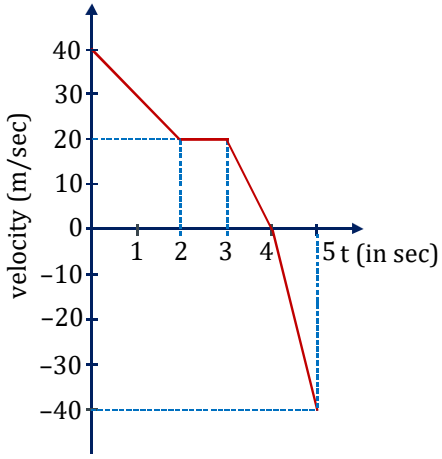
132. The fig. shows the position time graph of a particle moving on a straight line path. What is the magnitude of average speed of the particle over 10 seconds ?



(1) 2 m/s (2) 4 m/s
(3) 6 m/s (4) 8 m/s

PML132

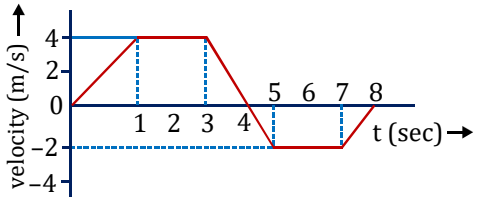
133. From the following velocity time graph of a body the distance travelled by the body and its displacement during 5 seconds in metres will be :



(1) 75, 75 (2) 110, 70
(3) 110, 110 (4) 110, 40

PML133

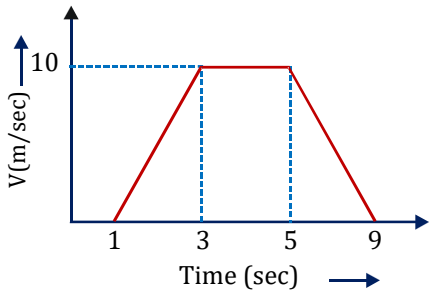
134. The v-t graph of linear motion of a particle starts its motion from origin is shown in adjoining figure. The distance of particle from origin after 8 sec. is :-



(1) 18 meters (2) 16 meters
(3) 8 meters (4) 6 meters

PML134

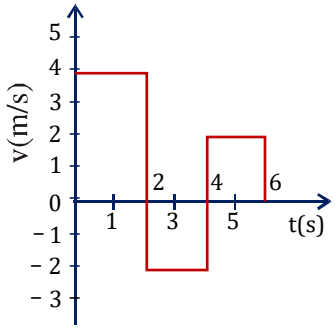
135. A particle moves according to given velocity–time graph. Then the ratio of distance travelled in last 4 seconds and 9 seconds is :-



(1) $\frac{1}{4}$ (2) $\frac{2}{5}$ (3) $\frac{1}{8}$ (4) $\frac{4}{11}$

PML135

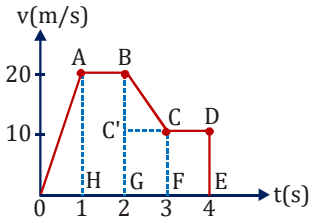
136. The velocity-time graph of a body moving in a straight line is shown in the figure. The displacement and distance travelled by the body in 6 s are, respectively :-



(1) 8 m, 16 m (2) 16 m, 8 m
(3) 16 m, 16 m (4) 8 m, 8m

PML136

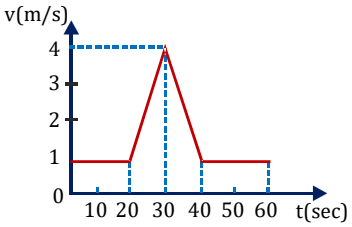
137. The variation of velocity of a particle moving along a straight line is illustrated in the figure. The distance transversed by the particle in 3 seconds is



(1) 60 m (2) 45 m (3) 55 m (4) 30 m

PML137

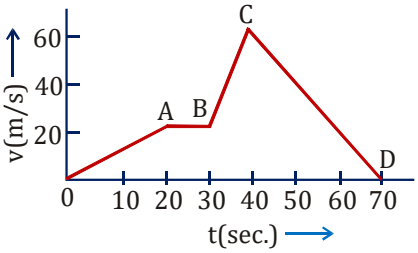
138. Velocity-time (v-t) graph for a moving object is shown in the figure. Total displacement of the object during the time interval when there is zero acceleration is:–



(1) 60 m (2) 50 m (3) 30 m (4) 40 m

PML138

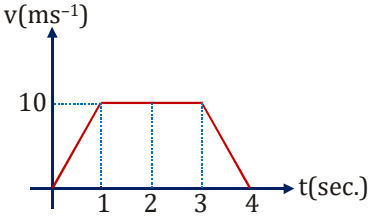
139. The velocity versus time curve of a moving particle is as shown in the following figure. The maximum acceleration is



(1) 1 m sec^{-2} (2) 2 m sec^{-2}
(3) 3 m sec^{-2} (4) 4 m sec^{-2}

PML139

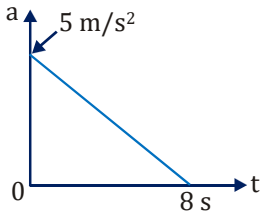
140. Find the average acceleration of the block from time t=2 sec to t=4 sec.



(1) 5 m/s^2 (2) 10 m/s^2
(3) –5 m/s^2 (4) –10 m/s^2

PML140

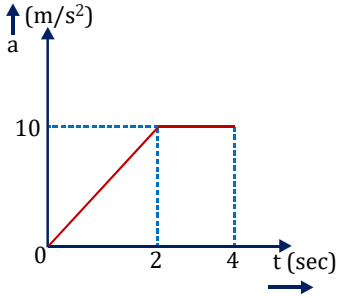
141. A particle starts from rest. Its acceleration at time t = 0 is 5 m/s^2 which varies with time as shown in the figure. The maximum speed of the particle will be :



(1) 7.5 m/s (2) 15 m/s
(3) 20 m/s (4) 37.5 m/s

PML141

142. A particle starts from rest, its acceleration-time graph is shown in figure.

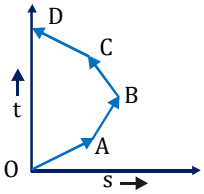


Find out velocity at t = 4 sec

(1) 20 m/s (2) 30 m/s
(3) 40 m/s (4) None of these

PML142

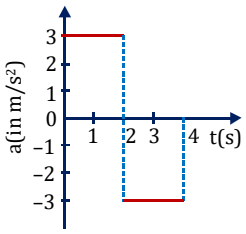
143. Which of the following options is correct for the object having a straight line motion represented by the following graph :-

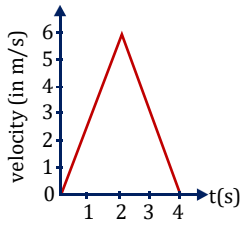


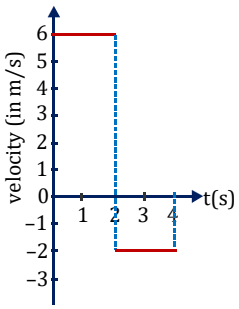
(1) The object moves with constantly increasing velocity from O to A and then it moves with constant velocity.
(2) Velocity of the object increases uniformly
(3) Average velocity is zero
(4) The graph shown is impossible

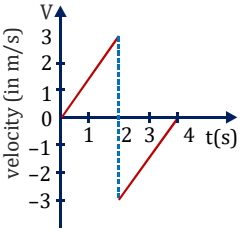
PML143

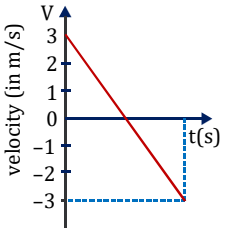
144. For the motion of a particle acceleration-time graph is shown in figure. The velocity time curve for the duration of 0 – 4 seconds is :



(1)


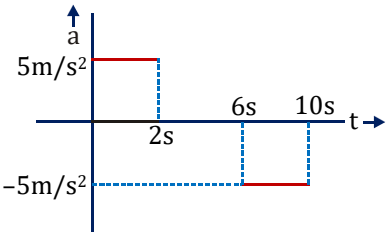
(2)


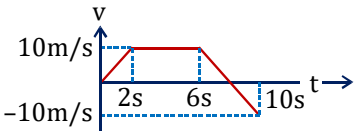
(3)


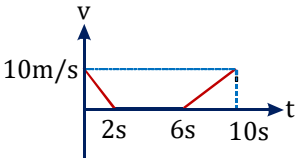
(4)


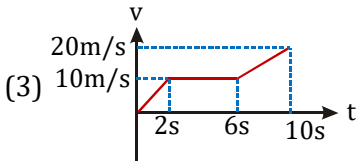
PML144

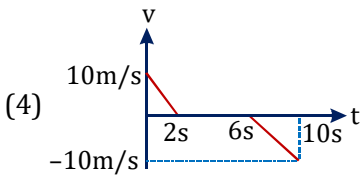
145. Acceleration-time graph of a body initially at rest is shown. The corresponding velocity-time graph of the same body is :-



(1)


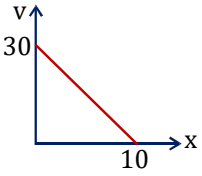
(2)


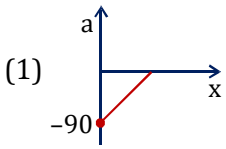
(3)


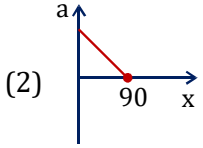
(4)


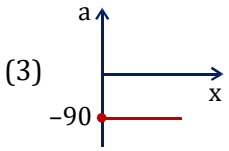
PML145

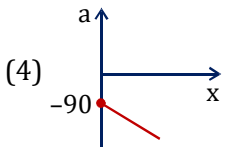
146. The given graph shows the variation of velocity with displacement. Which one of the graph given below correctly represents the variation of acceleration with displacement :-



(1)


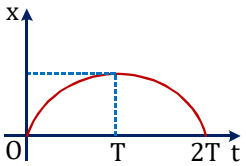
(2)


(3)


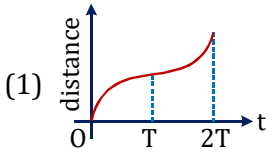
(4)


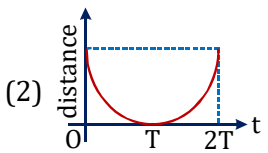
PML146

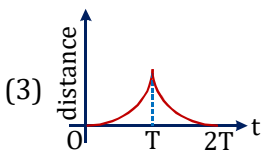
147. The x – t graph of a particle moving along a straight line is shown in figure.

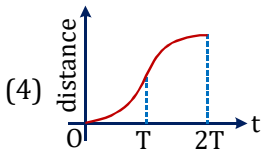


The distance-time graph of the particle is correctly shown by :

(1)


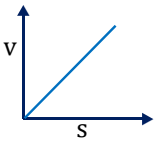
(2)


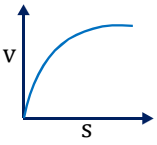
(3)


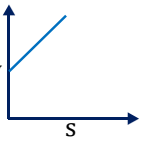
(4)


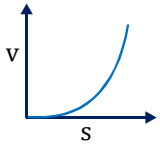
PML147

148. A particle starts from rest and move with constant acceleration. Its velocity-displacement curve is :

(1) 

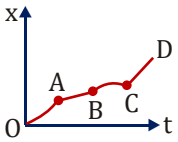
(2) 

(3) 

(4) 

PML148

149. The graph between the displacement x and time t for a particle moving in a straight line is shown in figure. During the interval OA, AB, BC and CD, the acceleration of the particle is :



	OA	AB	BC	CD
(1)	+	0	+	+
(2)	–	0	+	0
(3)	+	0	–	0
(4)	–	0	–	0

PML149

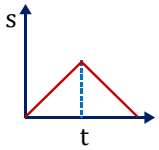
150. Acceleration-time curve for a body projected vertically upwards is a/an :–

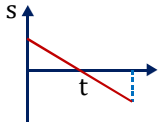
(1) Parabola (2) Ellipse

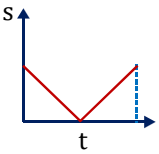
(3) Hyperbola (4) Straight line

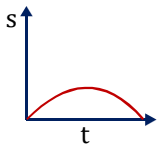
PML150

151. A body is projected vertically upward from the surface of the earth, its displacement-time graph is :

(1) 

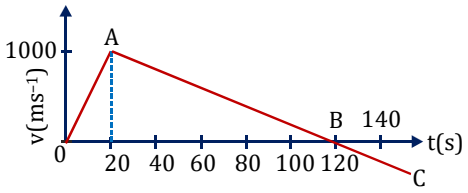
(2) 

(3) 

(4) 

PML151

152. A rocket is launched upward from the earth's surface whose velocity time graphs shown in figure. Then maximum height attained by the rocket is:



(1) 1 km (2) 10 km

(3) 100 km (4) 60 km

PML152

153. In above Q. no. 152, height covered by the rocket before retardation is :

(1) 1 km (2) 10 km

(3) 20 km (4) 60 km

PML153

154. In above Q. no. 152, mean velocity of rocket during the time it took to attain the maximum height :

(1) 100 m/s (2) 50 m/s

(3) 500 m/s (4) 25/3 m/s

PML154

155. In above Q. no. 152, the retardation of rocket is:

(1) 50 m/s^2 (2) 100 m/s^2

(3) 500 m/s^2 (4) 10 m/s^2

PML155

156. In above Q. no. 152, the acceleration of rocket is :

(1) 50 m/s^2 (2) 100 m/s^2

(3) 10 m/s^2 (4) 1000 m/s^2

PML156

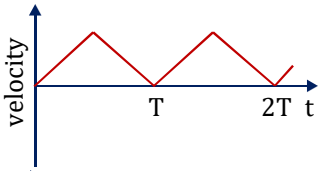
157. In above Q. no. 152, the rocket goes up and comes down on the following parts respectively :

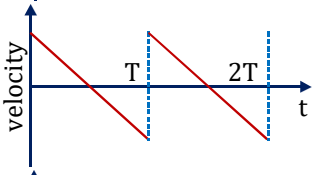
(1) OA and AB (2) AB and BC

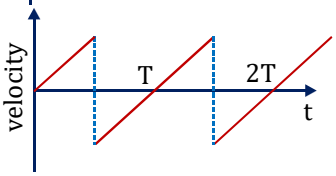
(3) OA and ABC (4) OAB and BC

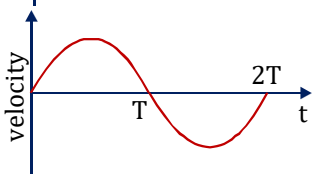
PML157

158. A ball is dropped from the certain height on the surface of glass. It collides elastically and comes back to its initial position. If this process it repeated then the velocity time graph is :
(Take downward direction as positive)

(1) 

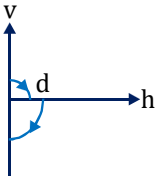
(2) 

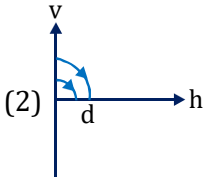
(3) 

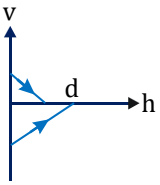
(4) 

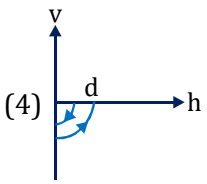
PML158

159. A ball is dropped vertically from a height d above the ground. It hits the ground and bounces up vertically to a height d/2. Neglecting subsequent motion and air resistance, The graph according to which its velocity V varies with the height h above the ground is :-

(1) 

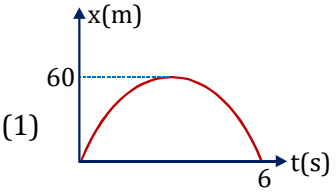
(2) 

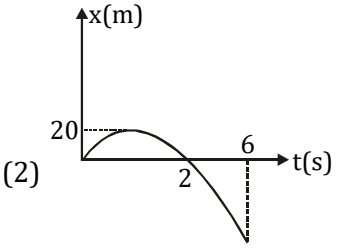
(3) 

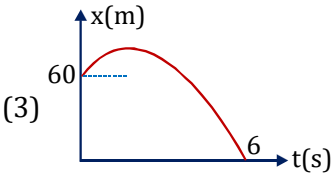
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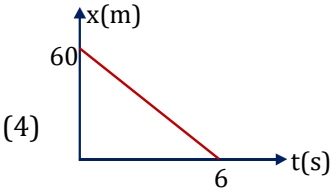
PML159

160. A stone is thrown upwards from top of a tower 60 m high at a speed of 20 m/s. The correct position-time graph for the time interval in which it reaches ground, is (g = 10 m/s² & take origin at the base of the tower) :-

(1) 

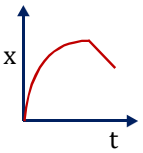
(2) 

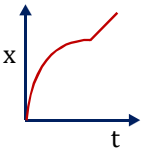
(3) 

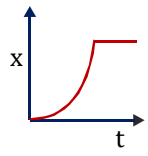
(4) 

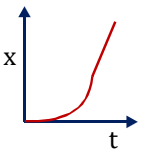
PML160

161. A car starts from rest and accelerates uniformly by for 4 seconds and then moves with uniform velocity which of the x-t graph represent the motion of the car ?

(1) 

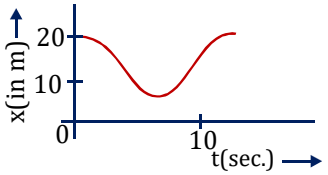
(2) 

(3) 

(4) 

PML161

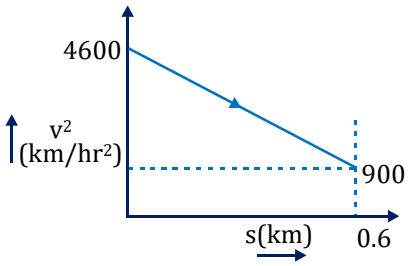
162. Figure shows x-t graph of a particle. Find the time t such that the average velocity of the particle during the period 0 to t is zero



(1) 6 sec. (2) 8 sec.
(3) 10 sec. (4) 12 sec.

PML162

163. Graph between the square of the velocity (v) of a particle and the distance (s) moved is shown in figure. The acceleration of the particle in kilometers per hour square is :



(1) 2250 (2) 3084
(3) – 2250 (4) – 3084

PML163

164. A train takes 4 min. to go between stations 2.25 km apart starting and finishing at rest. The acceleration is uniform for the first 40 sec and the deceleration is uniform for the last 20 sec. Assuming the velocity to be constant for the remaining time, then maximum speed of the train is :-

(1) 75 m/sec (2) 18.75 m/sec
(3) 37.5 m/sec (4) 10.7 m/sec

PML164

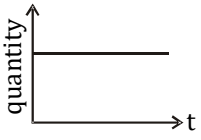
165. A car accelerates from rest at a constant rate of 2 m/s² for some time. Then, it retards at a constant rate of 4 m/s² and comes to rest. If it remains in motion for 3 seconds, then the maximum speed attained by the car is :

(1) 2 m/s (2) 3 m/s
(3) 4 m/s (4) 6 m/s

PML165

GROUND TO GROUND PROJECTION

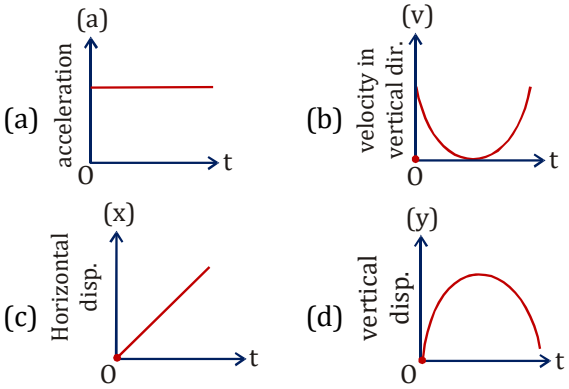
166. In the graph shown in fig. time is plotted along x-axis. Which quantity associated with a projectile motion is plotted along the y-axis ?



(1) kinetic energy (2) momentum
(3) horizontal velocity (4) none of the above

PML166

167. For ground to ground projection following curves are given :-



Find incorrect one.

(1) a (2) b (3) c (4) d

PML167

168. If air resistance is not considered in projectile motion, the horizontal motion takes place with

(1) constant velocity
(2) constant acceleration
(3) constant retardation
(4) variable velocity

PML168

169. In a projectile motion, the velocity :-

(1) is always perpendicular to the acceleration
(2) is never perpendicular to the acceleration
(3) is perpendicular to the acceleration for one instant only
(4) is perpendicular to the acceleration for two instants

PML169

170. In projectile motion, the modulus of rate of change of velocity–

(1) is constant
(2) first increases then decreases
(3) first decreases then increases
(4) None of the above

PML170

171. Which of the following quantities remains constant during projectile motion?

(A) Average velocity between two points.
(B) Average speed between two points
(C) $\frac{d\vec{v}}{dt}$
(D) $\frac{d^2}{dt^2}(\vec{v})$

(1) only A (2) only C
(3) A and C (4) C and D

PML171

172. A bomb is fired from a cannon with a velocity of 1000 m/s making an angle of 30° with the horizontal. What is the time taken by the bomb to reach at the highest point-

(1) 11 sec (2) 23 sec

(3) 38 sec (4) 50 sec

PML172

173. If time of flight of a projectile is 10 seconds. Range is 500 meters. The maximum height attained by it will be :-

(1) 125 m (2) 50 m

(3) 100 m (4) 150 m

PML173

174. A body is thrown with a velocity of 19.6 m/s making an angle of 30^0 with the horizontal. It will hit the ground after a time :–

(1) 3 s (2) 2 s (3) 1.5 s (4) 1 s

PML174

175. If a projectile is fired at an angle θ to the vertical with velocity u, then maximum height attained is given by :

(1) $\frac{u^2\cos\theta}{2g}$ (2) $\frac{u^2\sin^2\theta}{2g}$

(3) $\frac{u^2\sin^2\theta}{g}$ (4) $\frac{u^2\cos^2\theta}{2g}$

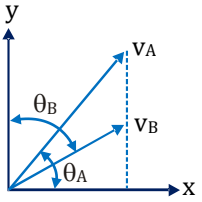
PML175

176. At the uppermost point of a projectile its velocity and acceleration are at an angle of :–

(1) 180° (2) 90° (3) 60° (4)45°

PML176

177. Two projectiles are projected with velocity v_A, v_B at angles θ_A (from horizontal) and θ_B (from vertical) as shown in the figure below, such that $v_A > v_B$ but having same horizontal component of velocity. Which of the following can not be correct ?



(1) $T_A > T_B$ (2) $H_A > H_B$

(3) $R_A > R_B$ (4) $R_B > R_A$

PML177

178. A player throws a ball that reaches to the another player in 4s. If the height of each player is 1.5m, the maximum height attained by the ball from the ground level is :

(1) 19.6 m (2) 21.1 m

(3) 23.6 m (4) 25.1 m

PML178

179. A body of mass m is thrown upwards at an angle θ with the horizontal with speed v. While rising up the magnitude of the velocity after t seconds will be:

(1) $\sqrt{(v\cos\theta)^2+(v\sin\theta)^2}$

(2) $\sqrt{(v\cos\theta-v\sin\theta)^2-gt}$

(3) $\sqrt{v^2+g^2t^2-(2v\sin\theta)gt}$

(4) $\sqrt{v^2+g^2t^2-(2v\cos\theta)gt}$

PML179

180. A particle is fired with velocity u making θ angle with the horizontal. What is the magnitude of change in velocity when it returns to the ground.

(1) u cos θ (2) u

(3) 2u sin θ (4) (u cosθ – u)

PML180

181. In the above, the change in speed is :

(1) u cos θ (2) 0

(3) u sin θ (4) (u cosθ – u)

PML181

182. The angle which the velocity vector of a projectile, will make with the vertical after time t of its being thrown with a velocity v at an angle θ to the horizontal, is :

(1) θ

(2) $\tan^{-1}\left(\frac{\theta}{t}\right)$

(3) $\tan^{-1}\left(\frac{v\cos\theta}{v\sin\theta-gt}\right)$

(4) $\tan^{-1}\left(\frac{v\sin\theta-gt}{v\cos\theta}\right)$

PML182

183. At a height 0.4 m from the ground, the velocity of a projectile in vector form is: $\vec{v}=(6\hat{i}+2\hat{j})\,m/s$. The angle of projection from horizontal is (g = 10 m/s^2) :-

(1) 45° (2) 60°

(3) 30° (4) $\tan^{-1}(3/4)$

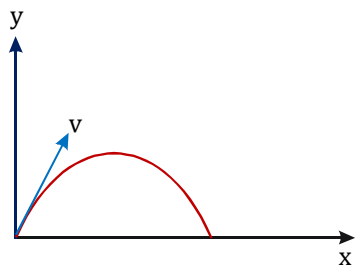
PML183

184. In case of a projectile fired at an angle 60° to the horizontal with velocity u, the horizontal range is:

(1) $\frac{u^2}{g}$ (2) $\frac{\sqrt{3}u^2}{2g}$ (3) $\frac{2u^2}{g}$ (4) $\frac{u^2}{g^2}$

PML184

185. If the instantaneous velocity of a particle projected as shown in figure is given by $\vec{v}=a\hat{i}+(b-ct)\hat{j}$ where a, b and c are positive constants, the range on the horizontal plane will be :



(1) 2ab/c (2) ab/c

(3) ac/b (4) a/2bc

PML185

186. A projectile is thrown from a point in a horizontal plane such that its horizontal and vertical velocity components are 9.8 m/s and 4.9 m/s respectively. Its horizontal range is :

(1) 4.9 m (2) 9.8 m

(3) 19.6 m (4) 39.2 m

PML186

187. A projectile is projected with initial velocity $(5\hat{i}+12\hat{j})$ m/s. If g = 10 ms^{-2}, then horizontal range is :

(1) 4.8 metre (2) 9.6 metre

(3) 19.2 metre (4) 12 metre

PML187

188. A shell is fired vertically upwards with a velocity v_1 from the deck of a ship moving with a speed v_2. A person on the shore observes the motion of the shell as a parabola. Its horizontal range is given by :

(1) $\frac{2v_1^2v_2}{g}$ (2) $\frac{2v_1v_2^2}{g}$

(3) $\frac{2v_1v_2}{g}$ (4) $\frac{2v_1^2v_2^2}{g}$

PML188

189. The range of a projectile when fired at 60° to the horizontal is 0.5 km. What will be its range when fired at 45° with the same speed ?

(1) 0.5 km (2) $\frac{1.0}{\sqrt{3}}$ km

(3) 1.5 km (4) 2.0 km

PML189

190. The speed of a projectile at its maximum height is $\frac{1}{2}$ times of its initial speed 'u' of projection. Its range on the horizontal plane is :

(1) $\frac{\sqrt{3}u^2}{2g}$ (2) $\frac{u^2}{2g}$ (3) $\frac{3u^2}{2g}$ (4) $\frac{3u^2}{g}$

PML190

191. A bullet is fired from a cannon with velocity 500 m/s. If the angle of projection is 15° and g = 10m/s^2. Then the range is :-

(1) 25×10^3 m (2) 12.5×10^3 m

(3) 50×10^2 m (4) 25×10^2 m

PML191

192. For a given angle of projection if the initial velocity is tripled the range of the projectile becomes :-

(1) 9 times (2) one-fourth

(3) two times (4) four times

PML192

193. A ball is thrown at an angle θ to the vertical and the range is maximum. The value of tanθ is :

(1) 1 (2) $\sqrt{3}$

(3) $\frac{1}{\sqrt{3}}$ (4) 2

PML193

194. The maximum horizontal range of a gun is 25 km. If g = 10 m/s², the muzzle velocity of the shell must be :–

(1) 1600 m/s (2) 500 m/s

(3) $200\sqrt{2}$ m/s (4) $160\sqrt{10}$ m/s

PML194

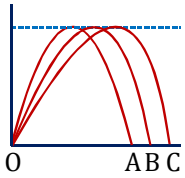
195. A grasshopper can jump maximum distance 1.6 m. It spends negligible time on the ground. How far can it go in 10 seconds :-

(1) $5\sqrt{2}$m (2) $10\sqrt{2}$m

(3) $20\sqrt{2}$m (4) $40\sqrt{2}$m

PML195

196. Three projectiles A, B and C are thrown from the same point in the same plane. Their trajectories are shown in the figure. Which of the following statement is incorrect?



(1) The time of flight is the same for all the three

(2) The launch speed is same for all the three

(3) The horizontal velocity component is largest for particle C

(4) The maximum height is same for all the three

PML196

197. A projectile is thrown with an initial velocity of $\vec{v}=a\hat{i}+b\hat{j}$. If range of the projectile is four times the maximum height attained by it then :

(1) a = 2 b (2) b = a

(3) b = 2a (4) b = 4a

PML197

198. Two projectiles are fired from the same point with the same speeds at angles of projection 60° and 30° respectively. Which one of the following is true ?

(1) Their horizontal ranges will be the same

(2) Their maximum heights will be the same

(3) Their landing velocities will be the same

(4) Their times of flight will be the same

PML198

199. Three particles A, B and C are projected from the same point with the same initial speeds making angles 30°, 45° and 60° respectively with the horizontal. Which of the following statements are correct ?

(1) A, B and C have unequal ranges

(2) Ranges of A and C are equal and less than that of B

(3) Ranges of A and C are equal and greater than that of B

(4) A, B and C have equal ranges

PML199

200. A number of bullets are fired in all possible directions with the same initial velocity u. The maximum area of ground covered by bullets is :–

(1) $\pi\left(\frac{2u^2}{g}\right)^2$ (2) $3\pi\left(\frac{u}{g}\right)^2$

(3) $5\pi\left(\frac{u}{2g}\right)^2$ (4) $\pi\left(\frac{u^2}{g}\right)^2$

PML200

201. Four bodies P, Q, R and S are projected with equal speed having angles of projection 15°, 30°, 45° and 60° with the horizontal respectively. The body having shortest range is :-

(1) P (2) Q (3) R (4) S

PMP001

202. A boy can throw a stone up to a maximum height of 10 m. The maximum horizontal distance up to which the boy can throw the same stone will be :

(1) $20\sqrt{2}$m (2) 10 m (3) $10\sqrt{2}$m (4) 20 m

PMP002

203. A cricketer can throw a ball to a maximum horizontal distance of 100m. How much high above the ground can the cricketer throw the same ball :-

(1) 200m (2) 400m

(3) 100m (4) 50m

PMP003

204. The maximum range of a projectile fired with some initial velocity is found to be 2000 m. The maximum height (H) reached by this projectile is:

(1) 250 metre (2) 500 metre

(3) 1000 metre (4) 2000 metre

PMP004

205. The cannon was set by mistake at 30° instead of 20° with the horizontal and was fired to hit the enemy target. The cannon-shell will now fall at :

(1) The enemy target

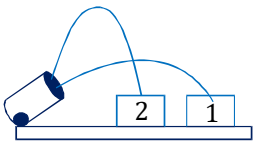
(2) Before the enemy target

(3) Beyond the enemy target

(4) May fall at any place

PMP005

206. Two similar cannon simultaneously fires two identical cannon balls at target 1 and 2 as shown in the figure. If the cannon balls have identical initial speeds, which of the following statements is true?



(1) Target 2 is hit before target 1

(2) Target 1 is hit before target 2

(3) Both are hit at the same time

(4) information is insufficient

PMP006

207. If R is the horizontal range of a particle projected at an angle 53° from horizontal, then the greatest height attained by it is :

(1) $\frac{R}{3}$ (2) 2R (3) $\frac{R}{2}$ (4) $\frac{R}{4}$

PMP007

208. A projectile can have the same range R for two angles of projection. If h_1 and h_2 be the maximum height in the two cases, then :–

(1) $h_1h_2 \propto R^2$ (2) $h_1h_2 \propto R$

(3) $h_1h_2 \propto \frac{1}{R}$ (4) $h_1h_2 \propto \frac{1}{R^2}$

PMP008

209. A body is thrown with some velocity from the ground. Maximum height attained when it is thrown at 37° to the horizontal is 90 m. What is the height attained when it is thrown at 53° to the horizontal?

(1) 90 m (2) 45 m

(3) 120 m (4) 160 m

PMP009

210. Two stones are projected with the same speed but making different angles with the horizontal. Their ranges are equal. If the angle of projection of one is $\frac{\pi}{6}$ and its maximum height is y_1, then the maximum height of the other will be :

(1) $3y_1$ (2) $2y_1$ (3) $\frac{y_1}{2}$ (4) $\frac{y_1}{3}$

PMP010

211. A ball is thrown at different angles with the same speed u and from the same point; it has the same range in both the cases. If y_1 and y_2 be the heights attained in the two cases, then $y_1 + y_2 = \ldots$:

(1) $\frac{u^2}{g}$ (2) $\frac{2u^2}{g}$ (3) $\frac{u^2}{2g}$ (4) $\frac{u^2}{4g}$

PMP011

212. A body is projected at such an angle that the horizontal range is four times the greatest height. The angle of projection is :–

(1) 25° (2) 33° (3) 45° (4) 53°

PMP012

213. The horizontal range of a projectile is $4\sqrt{3}$ times of its maximum height. Its angle of projection from horizontal will be:

(1) 45° (2) 60° (3) 90° (4) 30°

PMP013

214. A projectile have range double as compare to its maximum height attained. The angle of projection from horizontal is–

(1) $\tan^{-1}$ (2) (2) $\tan^{-1}$ (4)

(3) $\tan^{-1}$ (3) (4) $\tan^{-1}$ (5)

PMP014

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215. The speed of a projectile at its maximum height is $\frac{\sqrt{3}}{2}$ times its initial speed. If the range of the projectile is 'P' times the maximum height attained by it, then P equals to -

(1) $\frac{4}{3}$ (2) $2\sqrt{3}$

(3) $4\sqrt{3}$ (4) $\frac{3}{4}$

PMP015

216. A ball is projected to attain the maximum range. If the height attained is H, the range is

(1) H (2) 2H (3) 4H (4) H/2

PMP016

217. The equation of a projectile is $y = x - \frac{gx^2}{2}$. The angle of projection is :

(1) 30° (2) 60°

(3) 45° (4) none

PMP017

218. The equation of a projectile is $y = 4x - \frac{x^2}{4}$. The horizontal range is :

(1) 16 m (2) 8 m

(3) 64 m (4) 12.8 m

PMP018

219. A particle is projected at an angle of 45° from horizontal, 6m away from the foot of a wall, just touches the top of the wall and falls on the ground on the opposite side at a distance 3m from it. The height of wall is :

(1) 2m (2) $\frac{4}{3}$m (3) $\frac{8}{3}$m (4) $\frac{3}{4}$m

PMP019

220. A stone is projected from the ground with velocity 50 m/s at an angle of 30° from horizontal. It crosses a wall after 3 sec. How far beyond the wall the stone will strike the ground ($g = 10m/sec^2$)

(1) 90.2 m (2) 89.6 m

(3) 86.6 m (4) 70.2 m

PMP020

221. A shell fired from the ground is just able to cross the top of a wall 90m away and 45 m high moving horizontally. The direction of projection of the shell from horizontal will be :-

(1) 25° (2) 30° (3) 60° (4) 45°

PMP021

222. An arrow is shot into the air. Its range is 100 metres and its time of flight is 5 s. If the value of g is assumed to be 10 m/s^2, then the horizontal component of the velocity of arrow is:

(1) 40 m/s (2) 20 m/s

(3) 31.25 m/s (4) 12.5 m/s

PMP022

223. In the Q. 222, the maximum height attained by the arrow is :

(1) 25 m (2) 20 m/s

(3) 31.25 m (4) 12.5 m

PMP023

224. In the Q. 222, the vertical component of the projection velocity is :

(1) 25 m/s (2) 40 m/s

(3) 12.5 m/s (4) 31.25 m/s

PMP024

225. In the Q. 222, the angle of projection with the horizontal is :

(1) $\tan^{-1}\left(\frac{4}{5}\right)$ (2) $\tan^{-1}\left(\frac{5}{4}\right)$

(3) $\tan^{-1}\left(\frac{5}{8}\right)$ (4) $\tan^{-1}\left(\frac{8}{5}\right)$

PMP025

226. A projectile is thrown into space so as to have the maximum possible horizontal range equal to 200m. Taking the point of projection as the origin, the coordinates of the point where the velocity of the projectile is minimum are:

(1) (200, 100) m (2) (50, 200) m

(3) (100, 50) m (4) (50, 100) m

PMP026

227. If the range of a gun which fires a shell with muzzle speed v, is R, then the angle of elevation of the gun is :

(1) $\cos^{-1}\left(\frac{v^2}{Rg}\right)$ (2) $\cos^{-1}\left(\frac{Rg}{v^2}\right)$

(3) $\frac{1}{2}\sin^{-1}\left(\frac{v^2}{Rg}\right)$ (4) $\frac{1}{2}\sin^{-1}\left(\frac{Rg}{v^2}\right)$

PMP027

228. Two balls A and B are thrown with speeds u and u/2 respectively. Both the balls cover the same horizontal distance before returning to the plane of projection. If the angle of projection of ball B is 30° with the horizontal, then the angle of projection of A is :

(1) $\sin^{-1}\left(\frac{1}{8}\right)$ (2) $\frac{1}{2}\sin^{-1}\left(\frac{\sqrt{3}}{8}\right)$

(3) $\frac{1}{3}\sin^{-1}\left(\frac{1}{8}\right)$ (4) $\frac{1}{4}\sin^{-1}\left(\frac{1}{8}\right)$

PMP028

229. Two balls are projected from the same point on the ground simultaneously. First ball is projected vertically upwards and the second ball at an angle of projection 60° from the horizontal. Both the balls reach the ground simultaneously. The ratio of their velocities of projection are :

(1) 1 : 2 (2) 3 : 2 (3) $\sqrt{3}$: 2 (4) 2 : 3

PMP029

230. A body is projected at an angle θ with horizontal, another body is projected with the same speed at an angle θ with the vertical then the ratio of the maximum height is :-

(1) 1 : 1 (2) $\tan^2\theta$: 1

(3) 1 : cotθ (4) none of these

PMP030

231. A ball is thrown from a point on earth surface with a speed v at an angle of projection θ from horizontal. From the same point and at the same instant a person starts running with a constant speed $\frac{v}{2}$ to catch the ball. The angle of projection of the ball from horizontal is :-

(1) 60° (2) 30° (3) 90° (4) 45°

PMP031

232. For a projectile the ratio of maximum height reached to the square of flight time is (g=10 m/s²):-

(1) 5 : 4 (2) 5 : 2 (3) 5 : 1 (4) 10 : 1

PMP032

233. At what angle to the horizontal should a ball be thrown so that its range R is related to the time of flight T as $R = 5\sqrt{3}T^2$? Take g = 10 ms^{-2} :

(1) 30° (2) 45° (3) 60° (4) 90°

PMP033

234. A ball whose kinetic energy is E, is thrown at an angle of 60° to the horizontal. Its kinetic energy at the highest point of its flight will be :–

(1) $\frac{E}{4}$ (2) $\frac{E}{\sqrt{2}}$ (3) $\frac{E}{2}$ (4) zero

PMP034

235. A cricket ball is hit at an angle of 30° with the vertical with kinetic energy K. The kinetic energy of the ball at the highest point of the path is :-

(1) Zero (2) K/4 (3) K/2 (4) 3K/4

PMP035

236. A ball is projected vertically upwards with a certain speed. Another ball of the same mass is projected at an angle 30° to the vertical with the same initial speed. The ratio of their potential energies at highest points of their journey, will be

(1) 4 : 3 (2) 2 : 1 (3) 3 : 2 (4) 4 : 1

PMP036

237. A particle is projected with a velocity u making an angle θ with the horizontal. At any instant, its velocity v is at right angle to its initial velocity u; then v is:

(1) u cos θ (2) u tan θ

(3) u cot θ (4) u sec θ

PMP037

238. A force $\vec{F} = (6t^2\hat{i} + 4t\hat{j})$N acts on a particle of mass 1 kg. What will be velocity of the particle at t = 3 second if at t = 0, the particle was at rest :-

(1) $(54\hat{i} + 12\hat{j})$ m/s (2) $(54\hat{i} + 18\hat{j})$ m/s

(3) $(12\hat{i} + 6\hat{j})$ m/s (4) none

PMP038

PROJECTION FROM A HEIGHT

239. A stone is projected horizontally with a speed 10 m/s from a 80 m high building. The distance of the target on the ground from the foot of the building is :- (g = 10 m/s²)

(1) 80 m (2) 40 m (3) 20 m (4) 10 m

PMP039

240. An aeroplane moving horizontally with a speed of 90 km/h drops a food packet while flying at a height of 490 m. The horizontal range of the packet is :

(1) 180 m (2) 250 m

(3) 500 m (4) 670 m

PMP040

241. A stuntman plans to run along a roof top and then horizontally off it to land on the roof of next building. The roof of the next building is 19.6 metres below the first one and 6.2 metres away from it. What should be his minimum roof top speed in m/s, so that he can successfully make the jump ?

(1) 3.1 (2) 4.0 (3) 4.9 (4) 6.2

PMP041

242. Two stones are projected horizontally from the same height with speeds 100 m/s and 40 m/s. The ratio of their horizontal range is

(1) 1 : 1 (2) 5 : 2 (3) 2 : 5 (4) 3 : 4

PMP042

243. A body is thrown horizontally with a velocity 20 m/s from the top of a tower of height 20 m. It strikes the level ground through the foot of the tower at a distance x from the tower. The value of x is :-

(1) 20 m (2) 10 m (3) 40 m (4) 30 m

PMP043

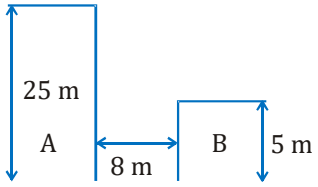
244. A bomber is flying horizontally with a constant speed of 150 m/s at a height of 19.6 m. The pilot has to drop a bomb at the enemy target. At what horizontal distance from the target should he release the bomb?

(1) 0 m (2) 300 m

(3) 600 m (4) 1000 m

PMP044

245. A boy wants to jump from building A to building B. Height of building A is 25 m and that of building B is 5m. Distance between buildings is 8m. Assume that the boy jumps horizontally, then calculate minimum velocity with which he has to jump to land safely on building B.

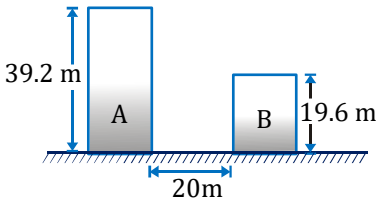


(1) 6 m/s (2) 8 m/s

(3) 4 m/s (4) 2 m/s

PMP045

246. A boy wants to jump from building A to building B. Height of building A is 39.2 m and building B is 19.6m. Distance between buildings is 20 m. Assuming boy jumps horizontally, then calculate minimum velocity with which boy has to jump to land safely on building B :-



(1) 10 m/s (2) 20 m/s

(3) 9.8 m/s (4) 19.6 m/s

PMP046

247. Two buildings are separated by 30 m. By what speed a ball is projected horizontally from a window at height 150 m in one building so as to enter in the window at height 27.5 m in the other building.

(1) 2 m/s (2) 6 m/s (3) 4 m/s (4) 8 m/s

PMP047

248. Two bodies of masses 100 kg and 50 kg are projected horizontally from same height with speeds 40 m/s and 20 m/s, simultaneously. The ratio of time taken by both the bodies to reach the ground is :-

(1) 1 : 1 (2) 1 : 2 (3) 2 : 1 (4) 1 : 4

PMP048

249. A ball is thrown horizontally from a height of 90 m with a speed of 4.5 m/s then find the time when it is at the height of 45 m from the ground:

(1) $3\sqrt{2}$ sec (2) 10 sec (3) 9 sec (4) 3 sec

PMP049

250. When a particle is thrown horizontally with speed u from top of tower, the displacement of the projectile at any time t is given by :

(1) $\sqrt{u^2t^2+g^2t^2}$ (2) $\sqrt{u^2t^2+\frac{1}{4}g^2t^4}$

(3) $\sqrt{u^2+g^2t^2}$ (4) $\sqrt{u^2-g^2t^2}$

PMP050

251. A plane is flying horizontally at $98\sqrt{3}$ m/s and releases an object which reaches the ground in 10 s. The angle made by it with horizontal while hitting the ground is :

(1) 55° (2) 45° (3) 60° (4) 30°

PMP051

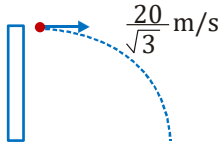
252. From the top of a tower 78.4 m high, a ball is thrown horizontally. If the line joining the point of projection to the point where it hits the ground makes an angle of 45° with the horizontal, then the initial velocity of the ball is :

(1) 9.8 m/s (2) 4.9 m/s

(3) 19.6 m/s (4) 2.8 m/s

PMP052

253. A particle is projected horizontally with a speed of $\frac{20}{\sqrt{3}}$ m/s, from some height at t = 0. At what time will its velocity make 30° angle with the initial velocity



(1) 1 sec (2) 2 sec

(3) 1.5 sec (4) 2/3 sec

PMP053

254. In the above question what will be the displacement of the particle in x-direction when its velocity makes 30° angle with the initial velocity

(1) $\frac{20}{\sqrt{3}}$ m (2) $\frac{40}{3\sqrt{3}}$ m

(3) $\frac{50}{\sqrt{3}}$ m (4) $\frac{10}{\sqrt{3}}$ m

PMP054

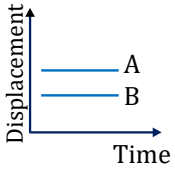
255. A ball is projected upwards from the top of a tower with a velocity of 50 m/s making an angle of 30^0 with the horizontal. The height of the tower is 70m. After how much time from the instant of throwing, will the ball reach the ground?

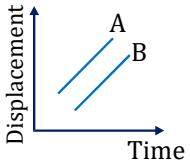
(1) 2 s (2) 5 s (3) 7 s (4) 9 s

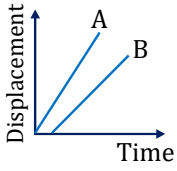
PMP055

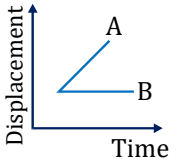
RELATIVE MOTION IN ONE DIMENSION

256. Which one of the following represents the displacement-time graph of two objects A and B moving with zero relative speed :-

(1) 

(2) 

(3) 

(4) 

PMP056

257. A monkey is climbing up a tree at a speed of 3 m/s. A dog runs towards the tree with a speed of 4 m/s. What is the relative speed of the dog as seen by the monkey ?

(1) > 7 m/s

(2) Between 5 m/s and 7 m/s

(3) 5 m/s

(4) < 5 m/s

PMP057

258. A train is moving towards east and a car is along north, both with same speed. The observed direction of car to the passenger in the train is:-

(1) East-north direction

(2) West-north direction

(3) South-east direction

(4) None of these

PMP058

259. Two car A & B start from rest (from the same point) in same direction with acceleration 8 m/s^2 & 4 m/s^2 respectively then acceleration of car B in frame of A(Take direction of motion of car is positive) :-

(1) 4 m/s^2 (2) –4 m/s^2

(3) 12 m/s^2 (4) None of these

PMP059

260. A motorcycle is moving with a velocity of 80 km/h and a car is moving with a velocity of 65 km/h in the opposite direction. What is the relative velocity of the motorcycle with respect to the car ?

(1) 15 km/h (2) 20 km/h

(3) 25 km/h (4) 145 km/h

PMP060

261. Two cars A and B start moving from the same point with same speed v = 5 km/minute. Car A moves towards south and car B is moving towards west. What is the relative velocity of B with respect to A ?
(1) $5\sqrt{2}$ km/min towards South-East
(2) $5\sqrt{2}$ km/min towards North-West
(3) $5\sqrt{2}$ km/min towards South-West
(4) $5\sqrt{2}$ km/min towards North-East

PMP061

262. A train is moving towards East with a speed 20 m/s. A person is running on the roof of the train with a speed 3 m/s in the direction of motion of train. Velocity of the person as seen by an observer on ground will be :
(1) 23 m/s towards East
(2) 17 m/s towards East
(3) 23 m/s towards West
(4) 17 m/s towards West

PMP062

263. A jet air plane travelling with a speed of 500 km/h ejects its products of combustion with a speed of 1600 km/h relative to the jet plane. The speed of the latter with respect to an observer on the ground is :–
(1) 1600 km/h (2) 2100 km/h
(3) 1100 km/h (4) 500 km/h

PMP063

264. A train moves in north direction with a speed of 54 km/h. A monkey is running on the roof of the train, against its motion with a velocity of 36 km/h. with respect to train. The velocity of monkey as observed by a man standing on the ground is:
(1) 5 m/s due north
(2) 25 m/s due south
(3) 10 m/s due south
(4) 10 m/s due north

PMP064

265. A lift is moving upwards with acceleration a. A man in the lift drops a ball within the lift. The acceleration of the ball as observed by the man in the lift and a man standing stationary on the ground are respectively :
(1) g, g (2) g – a, g – a
(3) g + a, g (4) a, g

PMP065

266. A stone is thrown upwards and it rises to a height of 200 m. The relative velocity of the stone with respect to the earth will be minimum at :–
(1) Height of 100 m (2) Height of 150 m
(3) Highest point (4) The ground

PMP066

267. A 100 m long train crosses a man travelling at 5 km/h, in opposite direction in 6 seconds, then the velocity of train is :–
(1) 40 km/h (2) 55 km/h
(3) 20 km/h (4) 45 km/h

PMP067

268. A 300 metres long train is moving due north with a speed of 20 m/s. A small bird is flying due south a little above the train with 5 m/s speed The time taken by the bird to cross the train is **:-**
(1) 6 s (2) 7 s (3) 9 s (4) 12 s

PMP068

269. Two trains each 100 m long, are travelling in opposite directions with respective velocities 35 m/s and 15 m/s. The time of crossing is :–
(1) 2 s (2) 4 s
(3) $2\sqrt{3}$ s (4) $4\sqrt{3}$ s

PMP069

270. A train 200 m long crosses a bridge 300 m long. It enters the bridge with a speed of 30 ms^{-1} and leaves it with a speed of 50 ms^{-1}. What is the time taken to cross the bridge ?
(1) 2.5s (2) 7.5s
(3) 12.5s (4) 15.0s

PMP070

271. Two cars get closer by 8 m in every second while travelling in the opposite directions. They get closer by 0.8 m in every second while travelling in the same direction. What are the speeds of the two cars?
(1) 4 ms^{-1} and 4.4 ms^{-1}
(2) 4.4 ms^{-1} and 3.6 ms^{-1}
(3) 4 ms^{-1} and 3.6 ms^{-1}
(4) 4 ms^{-1} and 3 ms^{-1}

PMP071

272. Velocity of a swimmer in still water is 5 m/s. If he takes 10 sec to swim upstream a distance of 30 m, then the speed of river is :-

(1) 3 m/s (2) 5 m/s

(3) 10 m/s (4) 2 m/s

PMP072

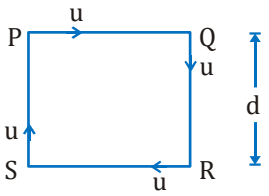
273. Two cars are moving in the same direction with the same speed of 60 km/h. They are separated by 10 km. What is the speed of a car moving in the opposite direction if it meets the two cars at an interval of 5 minute?

(1) 45 km/h (2) 60 km/h

(3) 105 km/h (4) None

PMP073

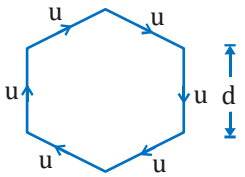
274. Four persons P, Q, R and S of same mass travel with same speed u along a square of side 'd' such that each one always faces the other. After what time will they meet each other?



(1) $\frac{d}{u}$ (2) $\frac{2d}{3u}$ (3) $\frac{2d}{u}$ (4) $d\sqrt{3}u$

PMP074

275. Six persons of same mass travel with same speed u along a regular hexagon of side 'd' such that each one always faces the other. After how much time will they meet each other ?



(1) $\frac{d}{u}$ (2) $\frac{2d}{3u}$ (3) $\frac{2d}{u}$ (4) $d\sqrt{3}u$

PMP075

276. A person walks up a stalled escalator in 45 sec. He is carried in 60s, when standing on the same escalator which is now moving. The time he would take to walk up the moving escalator will be :–

(1) 27 s (2) 72 s

(3) 18 s (4) 25.71 s

PMP076

277. A bus starts from rest moving with an acceleration of $2m/s^2$. A cyclist, 96 m behind the bus starts simultaneously towards the bus at 20 m/s. After what time will he be able to overtake the bus :–

(1) 8 s (2) 10 s (3) 12 s (4) 1 s

PMP077

278. A boat takes 2 hours to go 10 km and come back in still water lake. The time taken for going 10 km upstream and coming back with water velocity of 5 km/h is :

(1) 140 min (2) 150 min

(3) 160 min (4) 170 min

PMP078

279. A river 2 km wide flows at the rate of 2km/h. A boatman who can row a boat at a speed of 6 km/h in still water, goes a distance of 2 km upstream and then comes back. The time taken by him to complete his journey is

(1) 60 min (2) 45 min

(3) 80 min (4) 90 min

PMP079

280. Two bodies are held separated by 9.8 m vertically one above the other. They are released simultaneously to fall freely under gravity. After 2 s the relative distance between them is :-

(1) 4.9 m (2) 19.6 m

(3) 9.8 m (4) 39.2m

PMP080

281. Two balls are thrown simultaneously, (A) vertically upwards with a speed of 20 m/s from the ground and (B) vertically downwards from a height of 80 m with the same speed and along the same line of motion. At which point will the balls collide? (take g = $10 m/s^2$)

(1) 15 m above from the ground

(2) 15 m below from the top of the tower

(3) 20 m above from the ground

(4) 20 m below from the top of the tower

PMP081

282. While sitting on a tree branch 20m above the ground, you drop a walnut. When the walnut has fallen 5m, you throws a second walnut straight down. What initial speed must you give the second walnut if they are both to reach the ground at the same time?
($g=10ms^{-2}$)
(1) 5 ms^{-1} (2) 10 ms^{-1}
(3) 15 ms^{-1} (4) None of these

PMP082

283. A body A is thrown up vertically from the ground with velocity v_0 and another body B is simultaneously dropped from a height H. They meet at a height $\frac{H}{2}$ if v_0 is equal to

(1) $\sqrt{2gH}$ (2) $\sqrt{gH}$
(3) $\frac{1}{2}\sqrt{gH}$ (4) $\sqrt{\frac{2g}{H}}$

PMP083

284. An elevator is accelerating upward at a rate of 6 ft/sec^2 when a bolt from its ceiling falls to the floor of the lift (Distance = 19 feet). The time (in seconds) taken by the falling bolt to hit the floor is (take g = 32 ft/ sec^2)
(1) $\sqrt{2}$ (2) 1
(3) $2\sqrt{2}$ (4) $\frac{1}{2\sqrt{2}}$

PMP084

RELATIVE MOTION IN TWO DIMENSION

285. A boat is sent across a river with a velocity of 8km/hr. If the resultant velocity of boat is 10 km/hr, then velocity of the river is
(1) 10 km/hr (2) 8 km/hr
(3) 6 km/hr (4) 4 km/hr

PMP085

286. A bird is flying towards south with a velocity 40km/h and a train is moving with a velocity 40 km/h towards east. What is the velocity of the bird w.r.t. an observer in the train ?
(1) $40\sqrt{2}$ km/h. N-E (2) $40\sqrt{2}$ km/h. S-E
(3) $40\sqrt{2}$ km/h. S-W (4) $40\sqrt{2}$ km/h. N-W

PMP086

287. A bird is flying with a speed of 40 km/h in the north direction. A train is moving with a speed of 30 km/h in the west direction. A passenger sitting in the train will see the bird moving with velocity :
(1) 50 km/hr, 53° North of east
(2) 50 km/hr, 37° North of east
(3) 50 km/hr, 53° East of north
(4) 50 km/hr in North East direction

PMP087

288. A boat is sailing with a velocity $(3\hat{i}+4\hat{j})$ with respect to ground and water in river is flowing with a velocity $(-6\hat{i}-8\hat{j})$. Relative velocity of the boat with respect to water is :
(1) $8\hat{j}$ (2) $9\hat{i}+12\hat{j}$
(3) $6\hat{i}+8\hat{j}$ (4) $-6\hat{i}-8\hat{j}$

PMP088

289. Let $\vec{r}_1(t)=3t\hat{i}+4t^2\hat{j}$ and $\vec{r}_2(t)=4t^2\hat{i}+3t\hat{j}$ represent the positions of particles 1 and 2, respectively as functions of time t; $\vec{r}_1(t)$ and $\vec{r}_2(t)$ are in metres and t is in seconds. The relative speed of the two particles at the instant t = 2 sec, will be
(1) 1 m/s (2) $13\sqrt{2}$ m/s
(3) $5\sqrt{2}$ m/s (4) $7\sqrt{2}$ m/s

PMP089

290. Rain is falling vertically with a speed of 3 m/s. If a man is running with the same speed then the velocity of rain w.r.t. man is :-
(1) 3 m/s (2) 6 m/s
(3) 4.2 m/s (4) 0 m/s

PMP090

291. A man is walking on a road with a velocity of 5km/h. When suddenly it starts raining, velocity of rain is 10km/h in vertically downward direction, relative velocity of the rain with respect to man is :
(1) $\sqrt{13}$ km/hr (2) $\sqrt{7}$ km/hr
(3) $\sqrt{109}$ km/hr (4) $5\sqrt{5}$ km/hr

PMP091

292. If the rain is falling vertically downwards with velocity 20 m/s and if a bike is going with velocity 30 m/s. Calculate at what angle from the vertical a man on the bike must incline his umbrella so that he can save himself from rain :-

(1) $\tan^{-1}\left(\frac{2}{3}\right)$ (2) $\tan^{-1}\left(\frac{3}{2}\right)$

(3) $\tan^{-1}(1)$ (4) $\tan^{-1}\left(\frac{6}{5}\right)$

PMP092

293. If rain is falling at some angle from vertical and has horizontal velocity 2 m/s in east direction. With what velocity a man must move on the horizontal surface so that rain will appear vertical to him :-

(1) 4 m/s in east direction

(2) 2 m/s in east direction

(3) 2 m/s in west direction

(4) 2 m/s in a circular path

PMP093

294. A boy is running on a levelled road with velocity (u) with a long hollow tube in his hand. Water is falling vertically downwards with velocity (v). At what angle to the vertical, should he incline the tube so that the water drops enters without touching its side :

(1) $\tan^{-1}\left(\frac{v}{u}\right)$ (2) $\sin^{-1}\left(\frac{v}{u}\right)$

(3) $\tan^{-1}\left(\frac{u}{v}\right)$ (4) $\cos^{-1}\left(\frac{v}{u}\right)$

PMP094

295. A river is flowing at the rate of 8 km/h. A swimmer swims across the river with a velocity of 10 km/h w.r.t. water. The resultant velocity of the man will be in (km/h):-

(1) $\sqrt{117}$ (2) $\sqrt{340}$

(3) $\sqrt{164}$ (4) $3\sqrt{40}$

PMP095

296. A river is flowing from W to E with a speed of 5 m/min. A man can swim in still water with a velocity 10 m/min. In which direction should the man swim so as to take the shortest possible path to go to the north :-

(1) 30° with downstream

(2) 60° with downstream

(3) 120° with downstream

(4) South

PMP096

297. A river 4.0 miles wide is flowing at the rate of 2 miles/hr. The minimum time taken by a boat to cross the river with a speed v = 2 miles/hr (in still water) is approximately

(1) 1 hr and 0 minute

(2) 2 hr

(3) 1 hr and 12 minutes

(4) 2 hr and 25 minutes

PMP097

298. A river flows from east to west with a speed of 5m/min. A man on south bank of river, capable of swimming at the rate of 10 m/min in still water, wants to swim across the river in shortest time; he should swim :

(1) due north

(2) due north-east

(3) due north-east with double the speed of river

(4) none of the above

PMP098

299. A boat-man can row a boat to make it move with a speed of 10 km/h in still water. River flows steadily at the rate of 6 km/h. and the width of the river is 4 km. If the boat man cross the river along the minimum distance of approach then time elapsed in rowing the boat will be :

(1) $\frac{2\sqrt{3}}{5}$ h (2) $\frac{2}{5\sqrt{3}}$ h

(3) $\frac{3\sqrt{2}}{5}$ h (4) $\frac{1}{2}$ h

PMP099

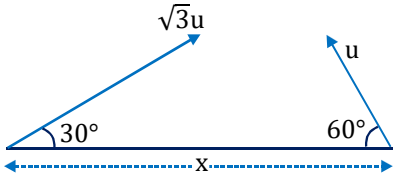
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300. A man wishes to swim across a river 0.5 km wide. If he can swim at the rate of $\sqrt{2}$ km/h in still water and the river flows at the rate of 1 km/h. The angle made by the direction (w.r.t. the flow of the river) along which he should swim so as to reach a point exactly opposite his starting point, should be :

(1) 60° (2) 120° (3) 135° (4) 90°

PMP100

301. Two particles are separated by a horizontal distance x as shown in figure. They are projected as shown in figure with different initial speeds. The time after which the horizontal distance between them becomes zero is :



(1) $\frac{x}{u}$

(2) $\frac{u}{2x}$

(3) $\frac{x}{2u}$

(4) none of these

PMP101

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EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	1	4	3	2	2	3	2	1	1	3	2	1	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	2	2	3	2	2	2	1	2	3	3	1	4	4	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	4	4	3	1	4	3	4	4	1	1	3	1	1	1	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	1	4	3	2	2	3	4	1	4	1	1	3	2	1
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	4	4	2	2	1	2	4	3	3	3	2	4	4	2	3
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	3	1	2	3	2	3	3	3	2	3	2	3	3	3	3
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	2	4	1	3	3	4	1	3	1	1	2	1	1	4	2
Question	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	2	1	4	2	2	3	2	2	3	3	2	3	2	3	1
Question	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Answer	1	3	1	2	3	3	2	4	2	2	4	4	2	4	2
Question	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Answer	1	2	4	4	3	3	2	3	1	1	1	1	2	3	4
Question	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
Answer	4	4	2	3	4	1	4	3	1	3	4	4	4	4	3
Question	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Answer	3	2	1	3	1	4	4	1	2	4	2	4	2	3	3
Question	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195
Answer	2	3	3	2	1	2	4	3	2	1	2	1	1	2	3
Question	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210
Answer	2	2	1	2	4	1	4	4	2	3	2	1	1	4	1
Question	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225
Answer	3	3	4	1	3	3	3	1	1	3	4	2	3	1	2
Question	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240
Answer	3	4	2	3	2	1	1	1	1	2	1	3	2	2	2
Question	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255
Answer	1	2	3	2	3	1	2	1	4	2	4	3	4	2	3
Question	256	257	258	259	260	261	262	263	264	265	266	267	268	269	270
Answer	2	3	2	2	4	2	1	3	1	3	3	2	4	2	3
Question	271	272	273	274	275	276	277	278	279	280	281	282	283	284	285
Answer	2	4	2	1	3	4	1	3	2	3	3	3	2	2	3
Question	286	287	288	289	290	291	292	293	294	295	296	297	298	299	300
Answer	3	1	2	2	3	4	2	2	3	3	3	2	1	4	3
Question	301														
Answer	3														

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. A projectile is fired from the surface of the earth with a velocity of 5 m/s and angle θ with the horizontal. Another projectile fired from another planet with a velocity of 3 m/s at the same angle follows a trajectory which is identical with the trajectory of the projectile fired from the earth. The value of the acceleration due to gravity on the planet is (in m/s^2) is: (given g = 9.8 m/s^2)

(1) 3.5 (2) 5.9

(3) 16.3 (4) 110.8

PMP130

2. A particle is moving such that its position coordinates (x, y) are

(2m, 3m) at time t = 0

(6m, 7m) at time t = 2 s and

(13m, 14m) at time t = 5s.

Average velocity vector ($\vec{V}_{av}$) from t = 0 to t = 5 s is

(1) $\frac{1}{5}\left(13\hat{i}+14\hat{j}\right)$ (2) $\frac{7}{3}\left(\hat{i}+\hat{j}\right)$

(3) $2\left(\hat{i}+\hat{j}\right)$ (4) $\frac{11}{5}\left(\hat{i}+\hat{j}\right)$

PMP131

AIPMT 2015

3. A particle of unit mass undergoes one-dimensional motion such that its velocity varies according to

$v(x) = \beta x^{-2n}$

where β and n are constants and x is the position of the particle. The acceleration of the particle as a function of x, is given by :

(1) $-2n\beta^2x^{-4n-1}$ (2) $-2\beta^2x^{-2n+1}$

(3) $-2n\beta^2e^{-4n+1}$ (4) $-2n\beta^2x^{-2n-1}$

PMP132

4. A ship A is moving Westwards with a speed of 10 km/h and a ship B 100 km South of A, is moving Northwards with a speed of 10 km/h. The time after which the distance between them becomes shortest, is :-

(1) 5 h (2) $5\sqrt{2}$ h

(3) $10\sqrt{2}$ h (4) 0 h

PMP133

Re-AIPMT 2015

5. Two particles A and B, move with constant velocities $\vec{v}_1$ and $\vec{v}_2$. At the initial moment their position vectors are $\vec{r}_1$ and $\vec{r}_2$ respectively. The condition for particle A and B for their collision is:-

(1) $\vec{r}_1-\vec{r}_2=\vec{v}_1-\vec{v}_2$ (2) $\frac{\vec{r}_1-\vec{r}_2}{|\vec{r}_1-\vec{r}_2|}=\frac{\vec{v}_2-\vec{v}_1}{|\vec{v}_2-\vec{v}_1|}$

(3) $\vec{r}_1\cdot\vec{v}_1=\vec{r}_2\cdot\vec{v}_2$ (4) $\vec{r}_1\times\vec{v}_1=\vec{r}_2\times\vec{v}_2$

PMP134

NEET-I 2016

6. If the velocity of a particle is $v = At + Bt^2$, where A and B are constants, then the distance travelled by it between 1s and 2s is :-

(1) $\frac{3}{2}A+4B$ (2) 3A+7B

(3) $\frac{3}{2}A+\frac{7}{3}B$ (4) $\frac{A}{2}+\frac{B}{3}$

PMP135

NEET-II 2016

7. Two cars P and Q start from a point at the same time in a straight line and their positions are represented by $x_p(t) = at + bt^2$ and $x_Q(t) = ft - t^2$. At what time do the cars have the same velocity?

(1) $\frac{a+f}{2(1+b)}$ (2) $\frac{f-a}{2(1+b)}$

(3) $\frac{a-f}{1+b}$ (4) $\frac{a+f}{2(b-1)}$

PMP136

NEET(UG) 2017

8. Preeti reached the metro station and found that the escalator was not working. She walked up the stationary escalator in time t_1. On other days, if she remains stationary on the moving escalator, then the escalator takes her up in time t_2. The time taken by her to walk up on the moving escalator will be

(1) $\frac{t_1t_2}{t_2-t_1}$ (2) $\frac{t_1t_2}{t_2+t_1}$

(3) $t_1 - t_2$ (4) $\frac{t_1+t_2}{2}$

PMP137

9. The x and y coordinates of the particle at any time are $x = 5t - 2t^2$ and $y = 10t$ respectively, where x and y are in meters and t in seconds. The acceleration of the particle at t = 2s is :-

(1) 5 m/s^2 (2) – 4 m/s^2

(3) – 8 m/s^2 (4) 0

PMP138

NEET(UG) 2019

10. The speed of a swimmer in still water is 20 m/s. The speed of river water is 10 m/s and is flowing due east. If he is standing on the south bank and wishes to cross the river along the shortest path, the angle at which he should make his strokes w.r.t. north is given by :

(1) 30° west (2) 0°

(3) 60° west (4) 45° west

PMP139

11. When an object is shot from the bottom of a long smooth inclined plane kept at an angle 60° with horizontal, it can travel a distance x_1 along the plane. But when the inclination is decreased to 30° and the same object the shot with the same velocity, it can travel x_2 distance. Then $x_1 : x_2$ will be

(1) $1:\sqrt{2}$ (2) $\sqrt{2}:1$

(3) $1:\sqrt{3}$ (4) $1:2\sqrt{3}$

PMP140

NEET(UG) 2019 (Odisha)

12. A person standing on the floor of an elevator drops a coin. The coin reaches the floor in time t_1 if the elevator is at rest and in time t_2 if the elevator is moving uniformly. Then :-

(1) $t_1 < t_2$ or $t_1 > t_2$ depending upon whether the lift is going up or down

(2) $t_1 < t_2$

(3) $t_1 > t_2$

(4) $t_1 = t_2$

PMP141

13. Two bullets are fired horizontally and simultaneously towards each other from roof tops of two buildings 100 m apart and of same height of 200m with the same velocity of 25 m/s. When and where will the two bullets collide. (g =10 m/s^2)

(1) after 2s at a height 180 m

(2) after 2s at a height of 20 m

(3) after 4s at a height of 120 m

(4) they will not collide

PMP142

14. A person travelling in a straight line moves with a constant velocity v_1 for certain distance 'x' and with a constant velocity v_2 for next equal distance. The average velocity v is given by the relation

(1) $\frac{1}{v}=\frac{1}{v_1}+\frac{1}{v_2}$ (2) $\frac{2}{v}=\frac{1}{v_1}+\frac{1}{v_2}$

(3) $\frac{v}{2}=\frac{v_1+v_2}{2}$ (4) $v=\sqrt{v_1v_2}$

PMP143

NEET(UG) 2020

15. A ball is thrown vertically downward with a velocity of 20 m/s from the top of a tower. It hits the ground after some time with a velocity of 80 m/s. The height of the tower is :

(g = 10 m/s^2)

(1) 300 m (2) 360 m

(3) 340 m (4) 320 m

PMP144

NEET(UG) 2020 (Covid-19)

16. A person sitting in the ground floor of a building notices through the window, of height 1.5 m, a ball dropped from the roof of the building crosses the window in 0.1 s. What is the velocity of the ball when it is at the topmost point of the window?

(g = 10 m/s^2)

(1) 15.5 m/s (2) 14.5 m/s

(3) 4.5 m/s (4) 20 m/s

PMP145

NEET(UG) 2021

17. A small block slides down on a smooth inclined plane, starting from rest at time t = 0. Let S_n be the distance travelled by the block in the interval t = n – 1 to t = n. Then, the ratio $\frac{S_n}{S_{n+1}}$ is :

(1) $\frac{2n-1}{2n}$ (2) $\frac{2n-1}{2n+1}$

(3) $\frac{2n+1}{2n-1}$ (4) $\frac{2n}{2n-1}$

PMP146

18. A car starts from rest and accelerates at 5 m/s². At t = 4 s, a ball is dropped out of a window by a person sitting in the car. What is the velocity and acceleration of the ball at t = 6 s ? (Take g = 10 m/s²)

(1) 20 m/s, 5 m/s² (2) 20 m/s, 0

(3) $20\sqrt{2}$ m/s, 0 (4) $20\sqrt{2}$ m/s, 10 m/s²

PMP147

19. A particle moving in a circle of radius R with a uniform speed takes a time T to complete one revolution.

If this particle were projected with the same speed at an angle 'θ' to the horizontal, the maximum height attained by it equals 4R. The angle of projection, θ, is then given by :

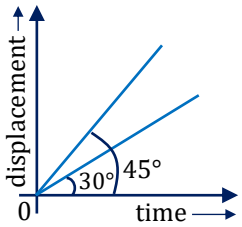
(1) $\theta = \cos^{-1}\left(\frac{gT^2}{\pi^2 R}\right)^{1/2}$ (2) $\theta = \cos^{-1}\left(\frac{\pi^2 R}{gT^2}\right)^{1/2}$

(3) $\theta = \sin^{-1}\left(\frac{\pi^2 R}{gT^2}\right)^{1/2}$ (4) $\theta = \sin^{-1}\left(\frac{2gT^2}{\pi^2 R}\right)^{1/2}$

PMP148

NEET(UG) 2022

20. The displacement-time graphs of two moving particles make angles of 30° and 45° with the x-axis as shown in the figure. The ratio of their respective velocity is :



(1) 1 : 1 (2) 1 : 2

(3) 1 : $\sqrt{3}$ (4) $\sqrt{3}$: 1

PMP149

21. The ratio of the distances travelled by a freely falling body in the 1st, 2nd, 3rd and 4th second :

(1) 1 : 4 : 9 : 16 (2) 1 : 3 : 5 : 7

(3) 1 : 1 : 1 : 1 (4) 1 : 2 : 3 : 4

PMP150

22. A ball is projected with a velocity, 10 ms⁻¹, at an angle of 60° with the vertical direction. Its speed at the highest point of its trajectory will be:

(1) $5\sqrt{3}$ ms⁻¹ (2) 5 ms⁻¹

(3) 10 ms⁻¹ (4) Zero

PMP151

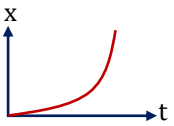
RE-NEET(UG) 2022

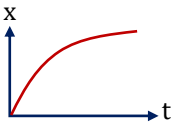
23. A cricket ball is thrown by a player at a speed of 20 m/s in a direction 30° above the horizontal. The maximum height attained by the ball during its motion is : (g = 10 m/s²)

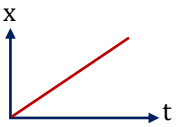
(1) 5 m (2) 10 m (3) 20 m (4) 25 m

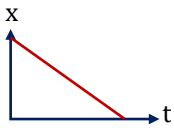
PMP152

24. The position-time (x – t) graph for positive acceleration is :

(1) 

(2) 

(3) 

(4) 

PMP153

NEET(UG) 2023

25. A bullet is fired from a gun at the speed of 280 ms⁻¹ in the direction 30° above the horizontal. The maximum height attained by the bullet is $\left(g = 9.8 ms^{-2}, \sin 30° = 0.5\right)$:-

(1) 2000 m (2) 1000 m

(3) 3000 m (4) 2800 m

PMP172

26. A vehicle travels half the distance with speed υ and the remaining distance with speed 2υ. Its average speed is :

(1) $\frac{2\upsilon}{3}$ (2) $\frac{4\upsilon}{3}$

(3) $\frac{3\upsilon}{4}$ (4) $\frac{\upsilon}{3}$

PMP173

27. A horizontal bridge is built across a river. A student standing on the bridge throws a small ball vertically upwards with a velocity 4 ms^{-1}. The ball strikes the water surface after 4s. The height of bridge above water surface is (Take g = 10 m s^{-2})

(1) 60 m (2) 64 m

(3) 68 m (4) 56 m

PMP174

28. A bullet from a gun is fired on a rectangular wooden block with velocity *u*. When bullet travels 24 cm through the block along its length horizontally, velocity of bullet becomes $\frac{u}{3}$. Then it further penetrates into the block in the same direction before coming to rest exactly at the other end of the block. The total length of the block is :

(1) 24 cm (2) 28 cm

(3) 30 cm (4) 27 cm

PMP175

RE-NEET(UG) 2023

29. The position of a particle is given by

$\vec{r}(t)=4t\,\hat{i}+2t^2\,\hat{j}+5\,\hat{k}$

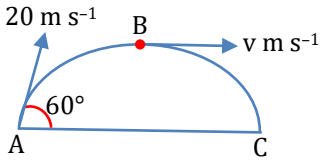
where t is in seconds and r in metre. Find the magnitude and direction of velocity v(t), at t = 1s, with respect to x-axis

(1) $4\sqrt{2}$ ms^{-1}, 45° (2) $4\sqrt{2}$ ms^{-1}, 60°

(3) $3\sqrt{2}$ ms^{-1}, 30° (4) $3\sqrt{2}$ ms^{-1}, 45°

PMP176

30. A ball is projected from point A with velocity 20 m s^{-1} at an angle 60° to the horizontal direction. At the highest point B of the path (as shown in figure), the velocity v m s^{-1} of the ball will be:

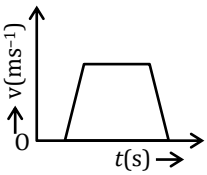


(1) 20 (2) $10\sqrt{3}$ (3) Zero (4) 10

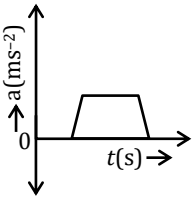
PMP177

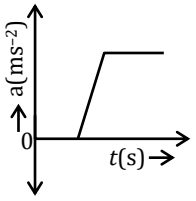
NEET(UG) 2024

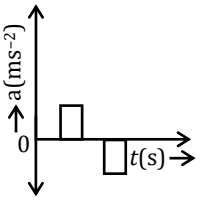
31. The velocity (v) – time (t) plot of the motion of a body is shown below :

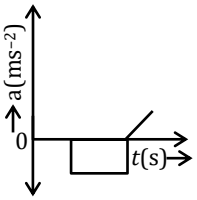


The acceleration (a) – time (t) graph that best suits this motion is :

(1) 

(2) 

(3) 

(4) 

PMP178

RE-NEET(UG) 2024

32. A particle is moving along x-axis with its position (x) varying with time (t) as $x = \alpha t^4 + \beta t^2 + \gamma t + \delta$. The ratio of its initial velocity to its initial acceleration, respectively is :

(1) $2\alpha : \delta$ (2) $\gamma : 2\delta$

(3) $4\alpha : \beta$ (4) $\gamma : 2\beta$

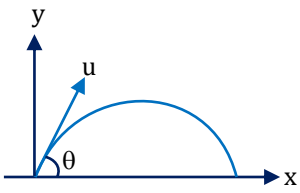
PMP179

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	1	1	2	3	2	2	2	1	3	4	1	2	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	2	4	4	3	2	1	1	1	2	2	2	4	1	4
Question	31	32													
Answer	3	4													

Exercise – III (Analytical Questions) Master Your Understanding

1. For ground to ground projectile motion with initial velocity u at angle θ from horizontal.



(A)	Change in speed from initial to final point	(i)	u–ucosθ
(B)	Magnitude of change in velocity from initial to final point	(ii)	usinθ
(C)	Change in speed from initial to top	(iii)	2usinθ
(D)	Magnitude of change in velocity from initial to top	(iv)	zero

Options :-

(1) (A)–(i), (B)–(ii), (C)–(iii), (D)–(iv)
(2) (A)–(iv), (B)–(ii), (C)–(i), (D)–(iii)
(3) (A)–(iv), (B)–(iii), (C)–(i), (D)–(ii)
(4) (A)–(iv), (B)–(iii), (C)–(ii), (D)–(i)

PMP154

2. For initial velocity $(\vec{v}_i) = 3\hat{i} - 4\hat{j}$ and final velocity $(\vec{v}_f) = 3\hat{i} + 4\hat{j}$:-

(A) $|\Delta\vec{v}|$ (i) $8\hat{j}$
(B) $\Delta|\vec{v}|$ (ii) $6\hat{i}$
(C) $\vec{v}_f - \vec{v}_i$ (iii) 0
(D) $\vec{v}_f + \vec{v}_i$ (iv) 8

(1) (A)–(iii), (B)–(iv), (C)–(ii), (D)–(i)
(2) (A)–(iii), (B)–(iv), (C)–(i), (D)–(ii)
(3) (A)–(iv), (B)–(iii), (C)–(ii), (D)–(i)
(4) (A)–(iv), (B)–(iii), (C)–(i), (D)–(ii)

PMP155

3. A particle is projected vertically upward from ground with initial velocity u such that it clears the top of a pole of height h after time t_1 in its path. It takes further time t_2 to reach the ground.

(A) u (i) $\frac{(t_1+t_2)^2}{8}g$
(B) h (ii) $(t_1 + t_2)$
(C) Time of flight (iii) $\frac{1}{2}g(t_1 + t_2)$
(D) Maximum height (iv) $\frac{1}{2}gt_1t_2$

(1) (A)–(ii), (B)–(i), (C)–(iii), (D)–(iv)
(2) (A)–(i), (B)–(ii), (C)–(iii), (D)–(iv)
(3) (A)–(iii), (B)–(iv), (C)–(ii), (D)–(i)
(4) (A)–(iv), (B)–(iii), (C)–(ii), (D)–(i)

PMP156

4. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Horizontal component of velocity is constant in projectile motion under gravity.

Reason (R) : Two projectiles having same horizontal range must have the same time of flight.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PMP158

5. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : A particle with constant acceleration always moves along a straight line.

Reason (R) : A particle with constant acceleration will not change direction of motion.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) Both **(A)** and **(R)** are false.

PMP159

6. An object may have-

(A) varying speed without having varying velocity

(B) varying velocity without having varying speed

(C) non-zero acceleration without having varying velocity

(D) non-zero acceleration without having varying speed

(1) Only B is correct

(2) Only D is correct

(3) Both B and D are correct

(4) All are correct

PMP160

7. The velocity of a particle is zero at t = 0, then -

(A) the acceleration at t = 0 must be zero

(B) the acceleration at t = 0 may be zero

(C) if the acceleration is zero from t = 0 to t = 10 s, speed is also zero in this interval

(D) if the speed is zero from t = 0 to t = 10 sec, the acceleration is also zero in the interval

(1) (A), (C) and (D) are correct

(2) (B), (C) and (D) are correct

(3) (A) and (D) are correct

(4) (B) and (C) are correct

PMP161

8. Instantaneous velocity of a particle -

(A) depends on instantaneous position

(B) depends on instantaneous speed

(C) independent of instantaneous position

(D) independent of instantaneous speed

(1) Both (A) & (B) are correct

(2) Both (C) & (D) are correct

(3) Both (A) & (D) are correct

(4) Both (B) & (C) are correct

PMP163

9. If the velocity of a body is constant -

(A) |Velocity| = speed

(B) |Average velocity| = speed

(C) Velocity = average velocity

(D) Speed = average speed

(1) Only (A) is correct

(2) (A) and (B) are correct

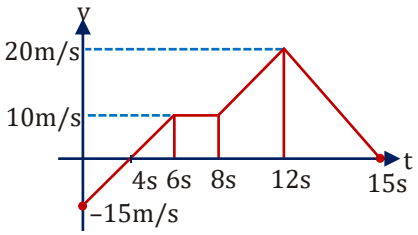
(3) (C) and (D) are correct

(4) All are correct

PMP165

Paragraph for Question Nos. 10 to 12

A moving particle is acted upon by three forces at different times to bring it to rest. Its velocity versus time graph is given below



10. The average speed for the first 6 s is

(1) zero (2) $\frac{5}{3}$ ms^{-1}

(3) $\frac{10}{3}$ ms^{-1} (4) $\frac{20}{3}$ ms^{-1}

PMP168

11. The average velocity for the first 12s is

(1) zero (2) 5 m/s

(3) $\frac{10}{3}$ m/s (4) 10 m/s

PMP169

12. The average acceleration from t = 5s to t = 15s is

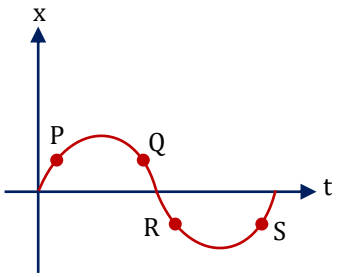
(1) zero (2) –0.5 m/s^2

(3) +0.5 m/s^2 (4) 1 m/s^2

PMP170

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13. A car is moving along a straight line. It's displacement (x) - time(t) graph is shown in column II. Match the entries in column I with points on graph.



Column I	Column II
(A) x → negative, v → positive, a → positive	(P) P
(B) x → positive, v → negative, a→ negative	(Q) Q
(C) x → negative, v → negative, a→ positive	(R) R
(D) x → positive, v → positive, a → negative	(S) S

Options :-
(1) A → S, B → Q, C → R, D → P
(2) A → P, B → R, C → Q, D → S
(3) A → S, B → R, C → Q, D → P
(4) A → S, B → Q, C → R, D → S

PMP171

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer	3	4	3	3	4	3	2	4	4	4	2	2	1

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Important Notes

Exercise - I (Conceptual Questions) Build Up Your Understanding

WORK

1. A force $\vec{F}=\left(4\hat{i}-4\hat{j}-2\hat{k}\right)N$ displaces a particle by $\vec{d}=\left(3\hat{i}+2\hat{j}-2\hat{k}\right)m$, calculate work done.

(1) Zero (2) 8 J (3) 4 J (4) 12 J

PWP001

2. A force $\vec{F}=\left(3\hat{i}+2\hat{j}\right)N$ acts on a body and produces a displacement $\vec{S}=\left(4\hat{i}-\hat{j}+3\hat{k}\right)m$. The work done will be

(1) 13 J (2) 30 J (3) 10 J (4) 20 J

PWP002

3. A particle is moved from a position $\vec{r}_1=(2\hat{i}+\hat{j}-2\hat{k})$ metre to a position $\vec{r}_2=(4\hat{i}+3\hat{j}-\hat{k})$ metre under the action of a force $\vec{F}=(3\hat{i}+2\hat{j}+2\hat{k})$ N.

The work done is equal to :-

(1) 32 J (2) 12 J (3) 24 J (4) 16 J

PWP003

4. A body of mass m is displaced from point A(3, 1, 3) m to point B(1, 2, 1) m under the effect of a force $\vec{F}=(5\hat{i}+2\hat{j}-4\hat{k})$, calculate work done by the force.

(1) 57 J (2) 11 J (3) 0 (4) 22 J

PWP004

5. A person of mass m is standing on one end of a plank of mass M and length L floating in water. The person moves from one end to another and stops. Work done by normal force is –

(1) MgL (2) mgL (3) $\frac{mMgL}{M+m}$ (4) 0

PWP005

6. The work done against gravity in taking 5 kg. mass to 2 m height in 1 s will be :-

(1) 49 J (2) 98 J

(3) 196 J (4) None of these

PWP006

7. A stone of mass m is tied to a string of length ℓ at one end and by holding second end it is whirled into a horizontal circle with constant speed v, then work done by the string on the stone will be :–

(1) 0 (2) $\left(\frac{mv^2}{\ell}\right)2\pi\ell$

(3) $(mg)\cdot 2\pi\ell$ (4) $\left(\frac{mv^2}{\ell}\right)\ell$

PWP007

8. A student of Allen is going from samarth to sabal building holding his bag in his hand. Direct distance between these buildings is 50m. Calculate work done by gravity if mass of bag is 3 kg.

(1) 750 J (2) 1500 J

(3) Zero (4) Data insufficient

PWP008

9. You lift a heavy book from the floor of the room and keep it in the book–shelf having a height 2 m. In this process you take 5 seconds. The work done by you will depend upon :-

(1) Mass of the book and time taken.

(2) Weight of the book and height of the book-shelf

(3) Height of the book-shelf and time taken.

(4) Mass of the book, height of the book-shelf and time taken.

PWP009

10. Work done by static friction is :

(1) Always zero

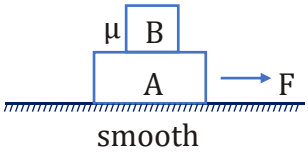
(2) Always non-zero

(3) May be zero or non-zero

(4) None of these

PWP010

11. For the system shown in figure both the block A and B are moving together. If coefficient of friction between both the blocks is μ, the work done by friction on block B is :-



(1) Zero (2) Negative

(3) Positive (4) None of these

PWP011

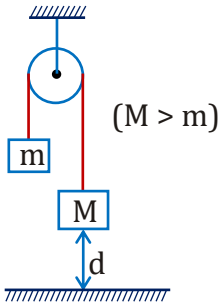
12. A body of mass M tied to a string is lowered at a constant acceleration of (3g/4) through a vertical distance h. The work done by the string will be.....

(1) $\frac{3}{4}$Mgh (2) $\frac{1}{4}$Mgh

(3) $\frac{-3}{4}$Mgh (4) $\frac{-1}{4}$Mgh

PWP012

13. The net work done by tension in the figure when the bigger block of mass M touches the ground is :-



(1) Mgd (2) –(M + m)gd

(3) –mgd (4) Zero

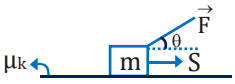
PWP013

14. The work done by the frictional force on a pencil in drawing a complete circle of radius r = 1/π metre on the surface by a pencil of negligible mass with a normal pressing force R = 5 N (μ = 0.5) is :

(1) + 4J (2) –3 J (3) – 2 J (4) – 5J

PWP014

15. Find work done by friction for displacement 'S' ?

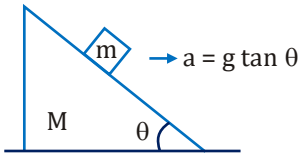


(1) $\mu_K(mg+F\sin\theta)S$ (2) $-\mu_K(mg+F\sin\theta)S$

(3) $\mu_K(mg-F\sin\theta)S$ (4) $-\mu_K(mg-F\sin\theta)S$

PWP015

16. The whole wedge and block system is moving as shown in the figure. Block of mass m is not slipping on the wedge. Then which of the following statement is correct (surface between M and m is rough) :-



(1) Work done by normal force on block is zero

(2) Work done by friction force on block is not zero

(3) Work done by normal force on block is not zero

(4) None of these

PWP016

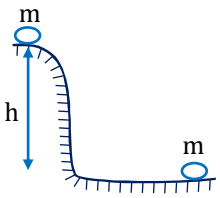
17. A particle is made to move from the origin in three spells of equal distances, first along the x-axis, second parallel to y-axis and third parallel to z-axis. Each time the force acting on it has constant magnitude of 50 N and always acts along the direction of motion. Work done by this force in the three spells of motion are equal and total work done in the three spells is 300 J. The final coordinates of the particle will be :-

(1) (2, 2, 2) m (2) (4, 4, 4) m

(3) (6, 6, 6) m (4) (10, 10, 10) m

PWP017

18. An object of mass m is sliding down a hill of arbitrary shape and after travelling a certain horizontal path stops because of friction. The work done that a force must perform to return the object to its initial position along the same path would be:-



(1) 0 (2) mgh

(3) 2mgh (4) None of these

PWP018

19. A force F = (x^2+3x–5) N acts on a 5 kg body along x-axis as a result of which the body gets displaced from x=0 to x=3 m. The work done by the force will be –

(1) 35 J (2) 15 J (3) 7.5 J (4) 70 J

PWP019

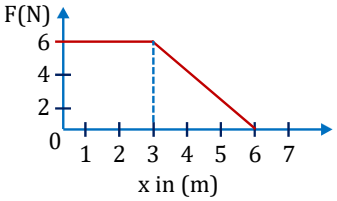
20. A force F = Kx^3 acts on a particle at an angle of 60° with the x–axis. The work done in displacing the particle from x_1 to x_2 will be –

(1) $\frac{kx^3}{3}$ (2) $\frac{k}{3}\left(x_2^2-x_1^3\right)$

(3) $\frac{k}{8}\left(x_2^4-x_1^4\right)$ (4) $\frac{k}{4}\left(x_2^4-x_1^4\right)$

PWP020

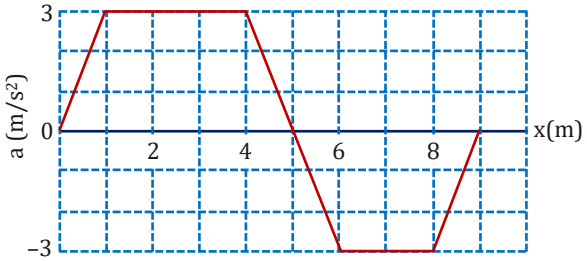
21. A force F acting on an object varies with distance x as shown here.
The work done by the force in moving the object from x = 0 to x = 6m is :-



(1) 18.0 J (2) 27 J
(3) 36 J (4) 9.0 J

PWP021

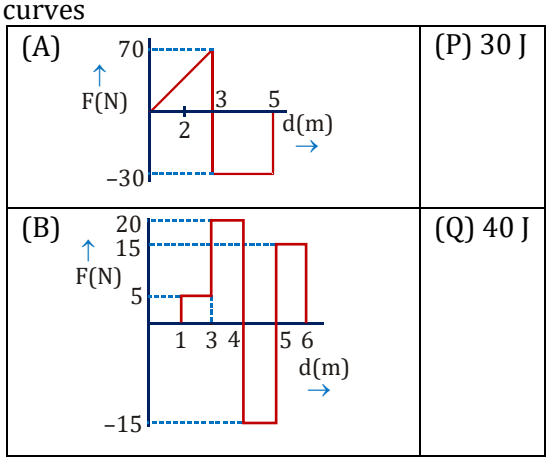
22. Figure gives the acceleration of a 4.0 kg body as it moves from rest along x axis while a variable force acts on it from x = 0m to x = 9m. The work done by the force on the body when it reaches
(i) x = 4m and
(ii) x = 7 m shall be as given below



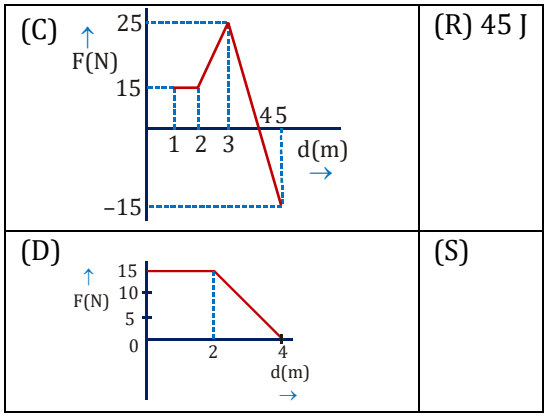
(1) 21 J and 33 J respectively
(2) 21 J and 15 J respectively
(3) 42 J and 60 J respectively
(4) 42 J and 30 J respectively

PWP022

23. Calculate the work done for following F-d curves



(A)	(P) 30 J
(B)	(Q) 40 J



(C)	(R) 45 J
(D)	(S)

(1) (A - R); (B - Q); (C - P); (D - R)
(2) (A - R); (B - P); (C - R); (D - Q)
(3) (A - R); (B - P); (C - Q); (D - R)
(4) (A - P); (B - Q); (C - R); (D - P)

PWP023

KINETIC ENERGY & WORK ENERGY THEOREM

24. The kinetic energy of a body becomes nine times of its initial value. The new linear momentum will be :-
(1) Four times the initial value
(2) Thrice the initial value
(3) Twice the initial value
(4) Same as the initial value

PWP024

25. If the momentum of a body is increased n times, its kinetic energy increases.
(1) n times (2) 2n times
(3) $\sqrt{n}$ times (4) n^2 times

PWP025

26. If the kinetic energy of a body becomes 4 times of its initial kinetic energy, then the momentum of the body will become :-
(1) $2\sqrt{2}$ times (2) $\sqrt{2}$ times
(3) 2 times (4) none of these

PWP026

27. Two bodies of mass 4kg and 9kg have equal K.E. then the ratio of their momentum is :-
(1) 3 : 2 (2) 2 : 3
(3) 4 : 9 (4) 16 : 8

PWP027

28. 2 particles of mass 3 Kg and 5 kg have same momentum, calculate ratio of their K.E.
(1) 5 : 3 (2) 3 : 5
(3) 25 : 9 (4) 9 : 25

PWP028

29. If K.E. of body is increased by 300%. Then % change in its 'P' will be

(1) 100% (2) 41.4%

(3) 17.3% (4) 200%

PWP029

30. If the kinetic energy of a body is increased by 800%, the momentum will increase by :-

(1) 100% (2) 200%

(3) 150% (4) 300%

PWP030

31. If K.E. of a body increases by 3%, then momentum will increase by :-

(1) 1.5% (2) 9%

(3) 3% (4) 2%

PWP031

32. If the momentum of a body increases by 2% then it's kinetic energy :-

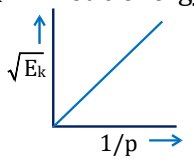
(1) increases by 2% (2) increases by 4%

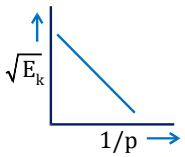
(3) increases by 8% (4) increases by 16%

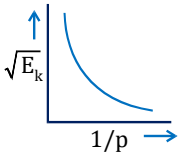
PWP032

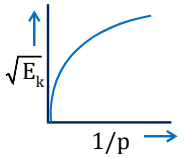
33. The graph between $\sqrt{E_k}$ and $\frac{1}{p}$ is

(E_K = kinetic energy and p = momentum) –

(1) 

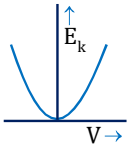
(2) 

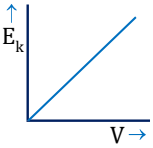
(3) 

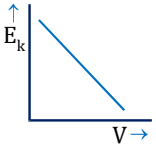
(4) 

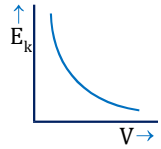
PWP033

34. The graph between kinetic energy E_K and velocity V is –

(1) 

(2) 

(3) 

(4) 

PWP034

35. A constant force F is applied to a body of mass m moving with initial velocity u. If after the body undergoes a displacement S, its velocity becomes v, then the total work done is :-

(1) $m[v^2 + u^2]$ (2) $\frac{m}{2}[u^2 + v^2]$

(3) $\frac{m}{2}[v^2 - u^2]$ (4) $m[v^2 - u^2]$

PWP035

36. A body of mass 6 kg is acted upon by a force which causes displacement in it given as $S = \frac{t^3}{4}$ metres where 't' is time. The work done by the force in 2 seconds is :-

(1) 18 J (2) 9 J (3) 27 J (4) 16 J

PWP036

37. A force acts on a 20 g particle in such a way that the position of the particle as a function of time is given by $x = 2t - 3t^2 + 2t^3$, where x is in metre and t is in second. The work done during the first 3 second is :-

(1) 5.28 J (2) 28.8 J (3) 14.4 J (4) 7.2 J

PWP037

38. A ball of mass 1 kg and another of mass 3 kg are dropped together from a 60 feet tall building. After a fall of 30 feet each towards earth, their respective kinetic energies will be in the ratio of :-

(1) 1 : 3 (2) 1 : 2 (3) $1 : \sqrt{3}$ (4) $\sqrt{3} : 1$

PWP038

39. A 2 kg ball is thrown up with an initial speed 16 m/s and reaches a maximum height of 10 m. How much energy is dissipated by air drag acting on the ball during the ascent ?

(1) 40 J (2) 56 J (3) 36 J (4) 72 J

PWP039

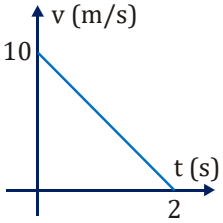
40. Three particles A, B and C are projected from the top of a tower with the same speed. A is thrown straight upwards, B straight down and C horizontally. They hit the ground with speeds V_A, V_B and V_C then:

(1) $V_A = V_B > V_C$ (2) $V_A = V_B = V_C$

(3) $V_A > V_B = V_C$ (4) $V_B > V_C > V_A$

PWP040

41. Velocity–time graph of a particle of mass 2 kg moving in a straight line is as shown in figure. Work done by all the forces acting on the particle is :



(1) 100 J (2) –100 J (3) –200 J (4) 200 J

PWP041

CONSERVATIVE FORCE & POTENTIAL ENERGY

42. Which of the following statements is not true for work done by conservative forces :–

(1) It does not depend on path
(2) It is equal to the difference of initial and final potential energy function
(3) It can be recovered completely
(4) None of the above

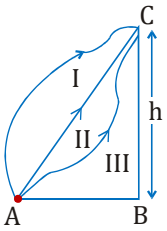
PWP042

43. Which of the following statement is incorrect for a conservative field ?

(1) Work done in going from initial to final position is equal to change in kinetic energy of the particle.
(2) Work done depends on path but not on initial and final positions.
(3) Work done does not depend on path but depends only on initial and final positions
(4) Work done on a particle in the field for a round trip is zero.

PWP043

44. As shown in the diagram a particle is to be carried from point A to C via paths (I), (II) and (III) in gravitational field, then which of the following statements is correct :–



(1) Work done is same for all the paths
(2) Work done is minimum for path (II)
(3) Work done is maximum for path (I)
(4) None of the above

PWP044

45. Which of the following is a non conservative force :–

(1) Electric force
(2) Gravitational force
(3) Spring force
(4) Viscous force

PWP045

46. The relation between conservative force and potential energy U is given by :–

(1) $F = \frac{dU}{dx}$ (2) $F = \int U dx$

(3) $F = -\frac{dU}{dx}$ (4) $F = \frac{dU}{dx}$

PWP046

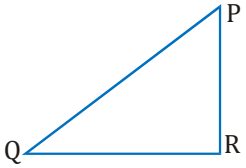
47. A 10 kg satellite completes one revolution around the earth at a height of 100 km in 108 minutes. The work done by the gravitational force of earth will be –

(1) 108 × 100 × 10 J (2) $\frac{108 \times 10}{100}$ J

(3) 0 J (4) $\frac{100 \times 10}{108}$ J

PWP047

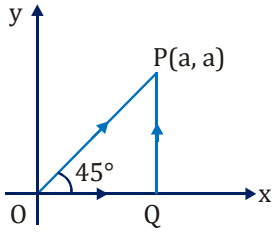
48. For the path PQR in a conservative force field the amount of work done in carrying a body from P to Q and from Q to R are 15 Joule and 5 Joule respectively. The work done in carrying the body from P to R will be –



(1) 20 J (2) 10 J (3) 5 J (4) Zero

PWP048

49. A particle is moved from (0, 0) to (a, a) under a force $\vec{F} = (3\hat{i} + 4\hat{j})$ from two paths. Path 1 is OP and path 2 is OQP. Let W_1 and W_2 be the work done by this force in these two paths. Then :



(1) $W_1 = W_2$ (2) $W_1 = 2W_2$
(3) $W_2 = 2W_1$ (4) $W_2 = 4W_1$

PWP049

50. A body of mass 3 kg falls from a height of 20 m. What is the loss in potential energy :-
(1) 400 J (2) 300 J
(3) 200 J (4) 600 J

PWP050

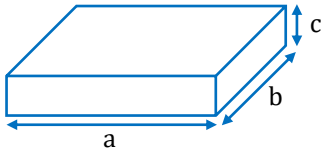
51. The mass of a bucket full of water is 13 kg. It is being pulled up from a 10 m deep well. Due to a hole in the bucket 4 kg water flows out of the bucket. The work done in drawing the bucket out of the well will be ($g = 10 m/s^2$)–
(1) 900 J (2) 1300 J
(3) 1100 J (4) 1700 J

PWP051

52. A uniform chain of length L and mass M is lying on a smooth table and $\frac{1}{3}$rd of its length is hanging down from the edge of the table. If g is the acceleration due to gravity, the work done to pull the hanging part on the table is :-
(1) MgL (2) $\frac{MgL}{18}$ (3) $\frac{MgL}{9}$ (4) $\frac{2MgL}{9}$

PWP052

53. A brick has 3 sides a, b, c {a > b > c}. The brick can be kept on ground on any face. The work required to be done to move slowly from minimum potential energy situation to maximum potential energy situation of brick is :



(1) $mg\left(\frac{a}{2}-\frac{b}{2}\right)$ (2) $mg\left(\frac{a}{2}-\frac{c}{2}\right)$
(3) $mg\left(\frac{b}{2}-\frac{c}{2}\right)$ (4) $mg\left(\frac{a+c}{2}-\frac{b}{2}\right)$

PWP053

54. If a spring extends by x on loading then energy stored in the spring is :- (T is tension in the spring, K = spring constant)
(1) $\frac{T^2}{2x}$ (2) $\frac{T^2}{2K}$ (3) $\frac{2K}{T^2}$ (4) $\frac{2T^2}{K}$

PWP054

55. A spring of force constant 1000 N/m has an extension of 15 cm. The work done in extending it from 15 cm to 25 cm is :-
(1) 16 J (2) 8 J (3) 32 J (4) 20 J

PWP055

56. In stretching a spring by 5 cm energy stored is given by U, then the energy stored if the spring is stretched by 10 cm will be :-
(1) U (2) 5U (3) 4U (4) 25U

PWP056

57. 5 J of work is required to stretch a spring through 5 cm beyond its unstretched length. The extra work required to stretch it through additional 5 cm shall be
(1) 5 J (2) 10 J (3) 15 J (4) 25 J

PWP057

CONSERVATION OF MECHANICAL ENERGY

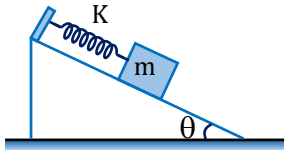
58. A body is dropped from a height h. When loss in its potential energy is U then its velocity is v. The mass of the body is –
(1) $\frac{U^2}{2v}$ (2) $\frac{2v}{U}$ (3) $\frac{2v}{U^2}$ (4) $\frac{2U}{v^2}$

PWP058

59. A block of mass 4 kg is moving on a frictionless horizontal surface with velocity 2m/s and comes to rest after pressing a spring. If the force constant of the spring is 100 N/m then the compression in the spring will be –
(1) 0·8 m (2) 1·6 m
(3) 0·4 m (4) 3·2 m

PWP059

60. A system of smooth inclined plane and block is shown in figure block is released when the spring in its natural length. Maximum elongation in the spring will be :-



(1) $\frac{2mg\sin\theta}{K}$ (2) $\frac{mg\sin\theta}{K}$
(3) $\frac{4mg\sin\theta}{K}$ (4) $\frac{mg\sin\theta}{2K}$

PWP060

61. A ball of mass 2 kg is released from the tower of Pisa. The kinetic energy generated in it after falling through 10m will be –
(1) 20 J (2) 196 J (3) 49 J (4) 98 J

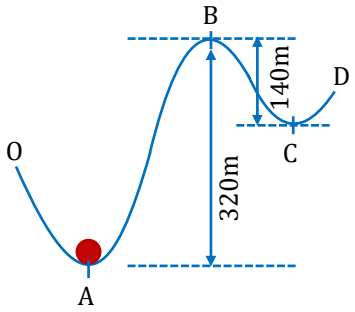
PWP061

62. A stone is projected vertically up to reach maximum height 'h'. The ratio of its kinetic energy to potential energy, at a height $\frac{h}{5}$ will be :-

(1) 5 : 4 (2) 4 : 5 (3) 1 : 4 (4) 4 : 1

PWP062

63. Track OABCD (as shown is figure) is smooth and fixed in vertical plane. What minimum speed has to be given to a particle lying at point A, so that it can reach point C ?



(1) 60 m/s (2) 100 m/s

(3) 70 m/s (4) 80 m/s

PWP063

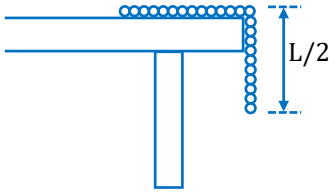
64. A projectile is fired at 30^0 with momentum p, neglecting friction, the change in kinetic energy, when it returns back to the ground, will be :-

(1) zero (2) 30 %

(3) 60 % (4) 100 %

PWP064

65. A uniform chain (mass M, length L) is released from rest from a smooth horizontal surface as shown in the figure. Velocity of the chain at the instant it completely comes out of the table will be.

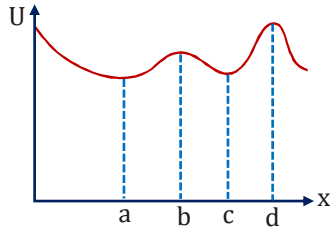


(1) $V = \sqrt{gL}$ (2) $V = \sqrt{\frac{3gL}{4}}$

(3) $V = \frac{1}{2}\sqrt{5gL}$ (4) $V = \frac{1}{2}\sqrt{gL}$

PWP065

66. Figure shows a plot of the potential energy as a function of x for a particle moving along the x-axis:-



Which of the following statement(s) is/are true?

(1) a, c, and d are points of equilibrium

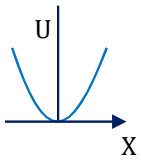
(2) a is a point of stable equilibrium

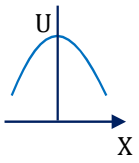
(3) b is a unstable equilibrium point

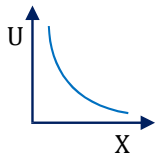
(4) All of the above

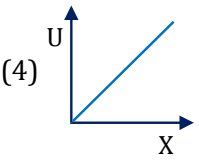
PWP066

67. The graph between potential energy U and displacement X in the state of stable equilibrium will be –

(1) 

(2) 

(3) 

(4) 

PWP067

68. A particle moves in a potential region given by $U = 2x^2 - 2x + 100$ J. Its state of equilibrium will be at :-

(1) x = 25 m (2) x = 0·25 m

(3) x = 0·05 m (4) x = 0·5 m

PWP068

69. If $F = 2x^2 - 3x - 2$, then choose correct option

(1) $x = -\frac{1}{2}$ is position of stable equilibrium

(2) x = 2 is position of stable equilibrium

(3) $x = -\frac{1}{2}$ is position of unstable equilibrium

(4) x = 2 is position of neutral equilibrium

PWP069

70. If the potential energy of system of two molecules is given by, $U = \frac{A}{r^{12}} - \frac{B}{r^6}$ then at equilibrium position, its potential energy is equal to :

(1) $\frac{A^2}{4B}$ (2) $-\frac{B^2}{4A}$ (3) $\frac{2B}{A}$ (4) 3A

PWP070

71. A particle in a certain conservative force field has a potential energy given by U = $\frac{2x^2y}{z^2}$. The force exerted on it is

(1) $\left(\frac{4xy}{z^2}\right)\hat{i} + \left(\frac{2x^2}{z^2}\right)\hat{j} + \left(\frac{4x^2y}{z^3}\right)\hat{k}$

(2) $-\left(\frac{4xy}{z^2}\right)\hat{i} - \left(\frac{2x^2}{z^2}\right)\hat{j} + \left(\frac{4x^2y}{z^3}\right)\hat{k}$

(3) $-\left(\frac{4xy}{z^2}\right)\hat{i} - \left(\frac{2x^2}{z^2}\right)\hat{j} - \left(\frac{4x^2y}{z^3}\right)\hat{k}$

(4) $\left(\frac{4xy}{z}\right)\hat{i} + \left(\frac{2x^2}{z}\right)\hat{j} - \left(\frac{4x^2y}{z^2}\right)\hat{k}$

PWP071

POWER

72. An electric motor produces a tension of 5000N in a load lifting cable and rolls it at the rate of 2 m/s. The power of the motor is –

(1) 10 kW (2) 15 kW
(3) 25 kW (4) 9×10^3 hp

PWP072

73. A motor of 50 hp is moving a car with a constant velocity of 36 km/hour. The forward force exerted by the engine on the car is – (1 hp = 746 W)

(1) $3{\cdot}73 \times 10^3$ N
(2) $3{\cdot}73 \times 10^2$ N
(3) $3{\cdot}73 \times 10^1$ N
(4) None of the above

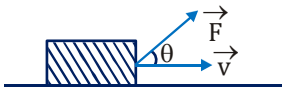
PWP073

74. A car is moving with a speed of 36 Km/hr. If the car engine generates 7 kilowatt power, then the resistance in the path of motion of the car will be –

(1) 300 newton (2) 700 newton
(3) Zero (4) 500 newton

PWP074

75. A constant force $\vec{F}$ is acting on a body of mass m moving with constant velocity $\vec{v}$ as shown in the figure. The power P exerted is



(1) Fv cosθ (2) $\frac{F\cos\theta}{mg}$
(3) $\frac{Fmg\cos\theta}{v}$ (4) $\frac{mg\sin\theta}{F}$

PWP075

76. A force $\vec{F} = \left(3\hat{i} + 4\hat{j}\right)$ N acts on a 2 kg movable object that moves from an initial position $\vec{d_i} = \left(-3\hat{i} - 2\hat{j}\right)$ m to final position $\vec{d_f} = \left(5\hat{i} + 4\hat{j}\right)$ in 6s. The average power delivered by the force during the interval is equal to :

(1) 8 watt (2) $\frac{50}{6}$ watt
(3) 15 watt (4) $\frac{50}{3}$ watt

PWP076

77. Two men with weights in the ratio 2 : 3 run up a staircase in times in the ratio 5 : 9. The ratio of power of first to that of second is –

(1) $\frac{6}{5}$ (2) $\frac{5}{6}$ (3) $\frac{10}{27}$ (4) $\frac{18}{15}$

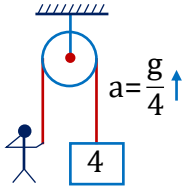
PWP077

78. A crane lifts 200 kg weight from earth's surface upto a height of 3m in 4 seconds. The average power generated by it will be – ($g = 9.8$ m/sec^2)

(1) 1470 W (2) 735 W
(3) 2940 W (4) 0 W

PWP078

79. Calculate power generated by tension in the string in first 2 seconds of motion :-



(1) 250 W (2) 125 W
(3) 500 W (4) 750 W

PWP079

80. What average horsepower is developed by a 50kg man while climbing in 10s flight of stairs that rises 10 m vertically ?

(1) 0.66 hp (2) 1.26 hp
(3) 1.8 hp (4) 2.1 hp

PWP080

81. A 2.0 hp motor pumps out water from a well of depth 8 m and fills a water tank of volume 1492 liters at a height of 16 m from the ground. The running time of the motor to fill the empty water tank is ($g = 10 m/s^2$) :-
(1) 50 minutes (2) 10 minutes
(3) 4 minutes (4) 20 minutes
PWP081

82. Water is falling on the blades of a turbine at a rate of 50 kg/s from a certain spring. If the height of the spring be 50 metres, the power transferred to the turbine will be :-
(1) 100 KW (2) 25 KW
(3) 5 KW (4) 250 KW
PWP082

83. If the force applied is F and the velocity gained is v, then the average power developed is :-
(1) $\frac{F}{v}$ (2) $\frac{v}{F}$ (3) $\frac{Fv}{2}$ (4) Fv^2
PWP083

84. A small body of mass 'm' is located in a horizontal plane at point O. The body accelerates uniformly and acquires horizontal velocity u then mean power developed by the friction force during the whole time of motion, if friction coefficient is k. :-
(1) $\frac{3mu}{4kg}$ (2) $\frac{5}{6}$mukg
(3) $\frac{1}{2}$mukg (4) $\frac{12}{7}$mukg
PWP084

85. A time varying power P = 2t is applied on a particle of mass 3kg. The change in kinetic energy of the particle at t = 4 sec is :-
(1) 4 J (2) 8 J (3) 16 J (4) 24 J
PWP085

86. Power supplied to a particle of mass 2 kg varies with time as $P = \frac{3t^2}{2}$ watt. Here t is in second. If velocity of particle at t = 0 is v = 0, the velocity of particle at time t = 2 s will be :-
(1) 1 m/s (2) 4 m/s
(3) 2 m/s (4) $2\sqrt{2}$ m/s
PWP086

87. A body of mass m starting from rest from origin moves along x-axis with constant power P. Calculate relation between velocity and distance :-
(1) $x \propto v^{1/2}$ (2) $x \propto v^2$
(3) $x \propto v$ (4) $x \propto v^3$
PWP087

88. A body of mass 2 kg is moving up an inclined plane rising 1 in 30 with velocity 60 m/sec. If efficiency is 40%, then calculate power required.
(1) 60 W (2) 40 W (3) 100 W (4) 20 W
PWP088

89. A pump is used to deliver water at a certain rate from a given pipe. To obtain n times water from the same pipe in the same time, by what factor, the force of the motor should be increased?
(1) n times (2) n^2 times
(3) n^3 times (4) $\frac{1}{n}$ times
PWP089

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	2	3	4	2	1	3	2	3	3	4	4	4	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	1	3	3	3	2	4	3	2	4	3	2	1	1	2
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	2	3	1	3	3	3	1	2	2	2	4	2	1	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	3	1	1	4	3	2	2	2	4	3	3	4	3	1
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	2	4	4	1	2	4	1	4	1	2	2	1	1	2	1
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	
Answer	1	1	1	2	1	3	2	3	3	3	3	4	3	2	

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2015

1. A block of mass 10 kg, moving in x direction with a constant speed of 10 m/s, is subjected to a retarding force F = 0.1x J/m during its travel from x = 20 m to 30 m. Its final KE will be :

(1) 450 J (2) 275 J (3) 250 J (4) 475 J

PWP102

2. A particle of mass m is driven by a machine that delivers a constant power k watts. If the particle starts from rest the force on the particle at time t is :-

(1) $\sqrt{mk}\, t^{-1/2}$ (2) $\sqrt{2mk}\, t^{-1/2}$

(3) $\frac{1}{2}\sqrt{mk}\, t^{-1/2}$ (4) $\sqrt{\frac{mk}{2}}\, t^{-1/2}$

PWP103

NEET-I 2016

3. A body of mass 1 kg begins to move under the action of a time dependent force $\vec{F}=(2t\hat{i}+3t^2\hat{j})N$, where $\hat{i}$ and $\hat{j}$ are unit vectors along x and y axis. What power will be developed by the force at the time t ?

(1) $(2t^2 + 3t^3)W$ (2) $(2t^2 + 4t^4)W$

(3) $(2t^3 + 3t^4)W$ (4) $(2t^3 + 3t^5)W$

PWP104

NEET-II 2016

4. A particle moves from a point $(-2\hat{i}+5\hat{j})$ to $(4\hat{j}+3\hat{k})$ when a force of $(4\hat{i}+3\hat{j})N$ is applied. How much work has been done by the force ?

(1) 5 J (2) 2 J (3) 8 J (4) 11 J

PWP105

NEET(UG) 2017

5. Consider a drop of rain water having mass 1 g falling from a height of 1 km. It hits the ground with a speed of 50 m/s. Take 'g' constant with a value 10 m/s^2. The work done by the (i) gravitational force and the (ii) resistive force of air is :-

(1) (i) 1.25 J (ii) – 8.25 J

(2) (i) 100 J (ii) 8.75 J

(3) (i) 10 J (ii) – 8.75 J

(4) (i) –10 J (ii) – 8.25 J

PWP106

NEET(UG) 2019

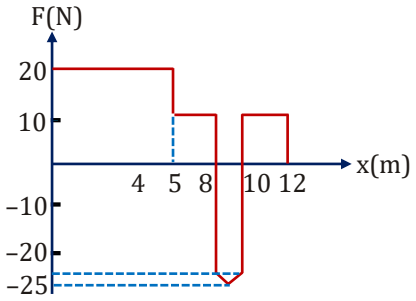
6. A force F = 20 + 10y acts on a particle in y-direction where F is in newton and y in meter. Work done by this force to move the particle from y = 0 to y = 1 m is :

(1) 30 J (2) 5 J (3) 25 J (4) 20 J

PWP107

NEET(UG) 2019 (Odisha)

7. An object of mass 500g, initially at rest is acted upon by a variable force, whose X component varies with x in the manner shown. The velocities of the object at point X = 8 m and X =12 m, would be the respective values of (nearly)



(1) 18 m/s and 24.4 m/s

(2) 23 m/s and 24.4 m/s

(3) 23 m/s and 20.6 m/s

(4) 18 m/s and 20.6 m/s

PWP108

NEET (UG) 2021

8. A particle is released from height S from the surface of the Earth. At a certain height its kinetic energy is three times its potential energy. The height from the surface of earth and the speed of the particle at that instant are respectively :

(1) $\frac{S}{4}, \frac{3gS}{2}$ (2) $\frac{S}{4}, \frac{\sqrt{3gS}}{2}$

(3) $\frac{S}{2}, \frac{\sqrt{3gS}}{2}$ (4) $\frac{S}{4}, \sqrt{\frac{3gS}{2}}$

PWP109

9. Water falls from a height of 60 m at the rate of 15 kg/s to operate a turbine. The losses due to frictional force are 10% of the input energy. How much power is generated by the turbine? (g = 10 m/s^2)

(1) 10.2 kW (2) 8.1 kW

(3) 12.3 kW (4) 7.0 kW

PWP110

NEET (UG) 2022

10. An electric lift with a maximum load of 2000 kg (lift + passengers) is moving up with a constant speed of 1.5 ms^{-1}. The frictional force opposing the motion is 3000 N. The minimum power delivered by the motor to the lift in watts is: ($g = 10\ ms^{-2}$)

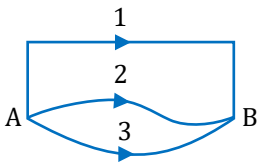
(1) 20000 (2) 34500

(3) 23500 (4) 23000

PWP111

RE-NEET (UG) 2022

11. A gravitational field is present in a region and a mass is shifted from A to B through different paths as shown. If W_1, W_2 and W_3 represent the work done by the gravitational force along the respective paths, then :



(1) $W_1 = W_2 = W_3$ (2) $W_1 > W_2 > W_3$

(3) $W_1 > W_3 > W_2$ (4) $W_1 < W_2 < W_3$

PWP112

NEET (UG) 2023

12. The potential energy of a long spring when stretched by 2 cm is U. If the spring is stretched by 8 cm, potential energy stored in it will be :

(1) 4U (2) 8U

(3) 16U (4) 2U

PWP125

RE-NEET (UG) 2023

13. A particle moves with a velocity $\left(5\hat{i}-3\hat{j}+6\hat{k}\right)$ m s^{-1} horizontally under the action of constant force $\left(10\hat{i}+10\hat{j}+20\hat{k}\right)$ N. The instantaneous power supplied to the particle is :

(1) 200 W (2) Zero

(3) 100 W (4) 140 W

PWP126

NEET (UG) 2024

14. At any instant of time t, the displacement of any particle is given by 2t –1 (SI unit) under the influence of force of 5N. The value of instantaneous power is (in SI unit) :

(1) 10 (2) 5

(3) 7 (4) 6

PWP127

RE-NEET (UG) 2024

15. An object moving along horizontal x-direction with kinetic energy 10 J is displaced through $x = \left(3\hat{i}\right)$ m by the force $\vec{F} = \left(-2\hat{i}+3\hat{j}\right)$N . The kinetic energy of the object at the end of the displacement x is :

(1) 10 J (2) 16 J (3) 4 J (4) 6 J

PWP128

16. An object falls from a height of 10 m above the ground. After striking the ground it loses 50% of its kinetic energy. The height upto which the object can rebounce from the ground is :

(1) 7.5 m (2) 10 m (3) 2.5 m (4) 5 m

PWP129

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	4	4	1	3	3	3	4	2	2	1	3	4	1	3
Question	16														
Answer	4														

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Exercise - III (Analytical Questions) Master Your Understanding

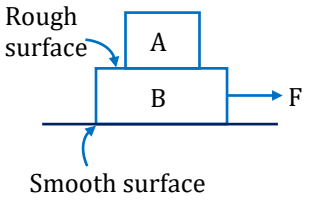
1. Select the correct statements –
(a) Work done by conservative force along closed path is zero.
(b) A body with negative energy can't have linear momentum
(c) Net work done by all the internal force of a system is independent of choice of reference frame
(d) Work done by a non conservative force is always negative
(1) a, b and c (2) a and c
(3) a, b and d (4) All

PWP115

2. Select the correct statements –
(a) Potential energy is defined only for gravitational force.
(b) Potential energy can be associated with frictional force.
(c) Work done by spring force is equal to change in potential energy of the spring.
(1) a and b
(2) a and c
(3) all
(4) none of these

PWP116

3.



If both blocks A and B are moving together, then select correct statements –
(a) Work done by the friction on block 'A' is zero
(b) Work done by the friction on block 'A' is positive
(c) Work done by the friction on block 'B' is negative
(d) Work done by friction forces on system (A) and (B) are zero
(1) a, c and d (2) b, c and d
(3) a and d (4) none of these

PWP117

4. For a system of particles –
(a) The work done by all external forces is equal to change in kinetic energy
(b) The work done by all forces is equal to change in mechanical energy
Select the correct statements :
(1) only a
(2) only b
(3) both a and b
(4) none of these

PWP118

5. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Kinetic energy of a system can be increased without applying external force on the system.
Reason (R) : If external forces are absent then work done by internal forces is equal to change in kinetic energy.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PWP119

6. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Work done is positive when force acts in the direction of displacement.
Reason (R) : Work done by frictional force cannot be positive.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PWP120

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7. Assume that in space potential energy U-depends on x-coordinate only by :

$$U = (x - 3)^2 (2 - x)$$

A mass m can be kept on x-axis. In each position indicated in column-I, comment on the situation from column-II.

Column I	**Column II**
(A) x = 7/3	(P) Stable equilibrium
(B) x = 2	(Q) Unstable equilibrium
(C) x = 3	(R) Neutral equilibrium
	(S) Not in equilibrium

Options :-

(1) A → P, B → S, C → R

(2) A → S, B → P, C → Q

(3) A → P, B → S, C → Q

(4) A → Q, B → S, C → P

PWP123

8. The potential energy (in joules) function of a particle in a region of space is given as :

$$U = (2x^2 + 3y^3 + 2z)$$

Here x, y and z are in metres. Find the magnitude of x component of force (in newton) acting on the particle at point P (1m, 2m, 3m).

(1) 4

(2) 2

(3) 6

(4) None

PWP124

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	2	4	2	4	1	3	3	1

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Exercise - I (Conceptual Questions) Build Up Your Understanding

KINEMATICS OF CIRCULAR MOTION

1. A particle of mass 'm' describes a circle of radius (r). The centripetal acceleration of the particle is $\frac{4}{r^2}$. The momentum of the particle :–

(1) $\frac{2m}{r}$ (2) $\frac{2m}{\sqrt{r}}$ (3) $\frac{4m}{r}$ (4) $\frac{4m}{\sqrt{r}}$

PCM001

2. A particle is moving around a circular path with uniform angular speed (ω). The radius of the circular path is (r). The acceleration of the particle is :–

(1) $\frac{\omega^2}{r}$ (2) $\frac{\omega}{r}$ (3) vω (4) vr

PCM002

3. A car moves on a circular road, describing equal angles about the centre in equal intervals of times. Which of the statements about the velocity of car is true :–

(1) velocity is constant
(2) magnitude of velocity is constant but the direction of velocity change
(3) both magnitude and direction of velocity change
(4) velocity is directed towards the centre of circle

PCM003

4. An insect trapped in a circular groove of radius 12 cm moves along the groove steadily and completes 7 revolutions in 100 s. What is the linear speed of the motion :–

(1) 2.3 cm/s (2) 5.3 cm/s
(3) 0.44 cm/s (4) none of these

PCM004

5. A particle moves in a circle of the radius 25 cm at two revolutions per second. The acceleration of the particle in m/sec^2 is :–

(1) π^2 (2) $8\pi^2$
(3) $4\pi^2$ (4) $2\pi^2$

PCM005

6. A particle moves in a circle describing equal angle in equal times, its velocity vector :–

(1) remains constant
(2) change in magnitude
(3) change in direction
(4) changes in magnitude and direction

PCM006

7. A mass of 2 kg is whirled in a horizontal circle by means of a string at an initial speed of 5 r.p.m. Keeping the radius constant the tension in the string is doubled. The new speed is nearly :–

(1) 7 r.p.m. (2) 14 r.p.m.
(3) 10 r.p.m. (4) 20 r.p.m.

PCM007

8. A particle moving along a circular path. The angular velocity, linear velocity, angular acceleration and centripetal acceleration of the particle at any instant respectively are $\vec{\omega}$, $\vec{v}$, $\vec{\alpha}$, $\vec{a}_c$. Which of the following relation is/are correct:–

(a) $\vec{\omega} \perp \vec{v}$ (b) $\vec{\omega} \perp \vec{\alpha}$
(c) $\vec{v} \perp \vec{a}_c$ (d) $\vec{\omega} \perp \vec{a}_c$

(1) a,b,d (2) b,c,d (3) a,b,c (4) a,c,d

PCM008

9. A particle is acted upon by a force of constant magnitude which is always perpendicular to the velocity of the particle. The motion of the particle takes place in a plane. It follows, that :–

(1) its velocity is constant
(2) its K.E. is constant
(3) its acceleration is constant
(4) it moves in a straight line

PCM009

10. If the equation for the displacement of a particle moving on a circular path is given by $(\theta) = 2t^3 + 0.5$, where θ is in radians and t in seconds, then the angular velocity of the particle after 2 s from its start is :–

(1) 8 rad/s (2) 12 rad/s
(3) 24 rad/s (4) 36 rad/s

PCM010

11. A sphere of mass m is tied to end of string of length ℓ and rotated through the other end along a horizontal circular path with speed v. The work done in full horizontal circle is :-

(1) 0 (2) $\left(\frac{mv^2}{\ell}\right).2\pi\ell$

(3) $mg.2\pi\ell$ (4) $\left(\frac{mv^2}{\ell}\right).\ell$

PCM011

12. A body moves with constant angular velocity on a circle. Magnitude of angular acceleration :-

(1) $r\omega^2$ (2) Constant

(3) Zero (4) None of the above

PCM012

13. A particle of mass (m) revolving in horizontal circle of radius (R) with uniform speed v. When particle goes from one end to other end of diameter, then:-

(1) K.E. changes by $\frac{1}{2}mv^2$

(2) K.E. change by mv^2

(3) no change in momentum

(4) change in momentum is 2 mv

PCM013

14. A stone is tied to one end of string 50 cm long and is whirled in a horizontal circle with constant speed. If the stone makes 10 revolutions in 20 s, then what is the magnitude of acceleration of the stone:-

(1) 493 cm/s^2 (2) 720 cm/s^2

(3) 860 cm/s^2 (4) 990 cm/s^2

PCM014

15. For a particle in a non-uniform accelerated circular motion :–

(1) velocity is radial and acceleration is tangential only

(2) velocity is tangential and acceleration is radial only

(3) velocity is radial and acceleration has both radial and tangential components

(4) velocity is tangential and acceleration has both radial and tangential components

PCM015

16. The angular velocity of a particle rotating in a circular orbit 100 times per minute is

(1) 1.66 rad / s (2) 10.47 rad / s

(3) 10.47 degree / s (4) 60 degree / s

PCM016

17. Two particles having mass 'M' and 'm' are moving in a circular path having radius R and r. If their time period are same then the ratio of angular velocity will be :–

(1) $\frac{r}{R}$ (2) $\frac{R}{r}$ (3) 1 (4) $\sqrt{\frac{R}{r}}$

PCM017

18. Angular velocity of minute hand of a clock is :–

(1) $\frac{\pi}{30}$ rad/s (2) 8π rad/s

(3) $\frac{2\pi}{1800}$ rad/s (4) $\frac{\pi}{1800}$ rad/s

PCM018

19. A car moving with speed 30 m/s on a circular path of radius 500 m. Its speed is increasing at the rate of $2m/s^2$. The acceleration of the car is :–

(1) 9.8 m/s^2 (2) 1.8 m/s^2

(3) 2 m/s^2 (4) 2.7 m/s^2

PCM019

20. If a particle is rotating uniformly in a horizontal circle, then –

(1) no force is acting on particle

(2) velocity of particle is constant

(3) particle has no acceleration

(4) no work is done

PCM020

21. A body of mass 1 kg tied to one end of string is revolved in a horizontal circle of radius 0.1 m with a speed of 3 revolution/sec, assuming the effect of gravity is negligible, then linear velocity, acceleration and tension in the string will be :–

(1) 1.88 m/s, 35.5 m/s^2, 35.5 N

(2) 2.88 m/s, 45.5 m/s^2, 45.5 N

(3) 3.88 m/s, 55.5 m/s^2, 55.5 N

(4) None of these

PCM021

22. A particle moves along a circle of radius $\left(\frac{20}{\pi}\right)$ m with constant tangential acceleration. If the velocity of the particle is 80 m/s at the end of the second revolution after motion has begun, the tangential acceleration is :–

(1) 40 m/s^2 (2) 640 π m/s^2

(3) 160 π m/s^2 (4) 40 π m/s^2

PCM022

23. The angular velocity of a wheel is 70 rad/s. If the radius of the wheel is 0.5 m, then linear velocity of the wheel is :–

(1) 70 m/s (2) 35 m/s

(3) 30 m/s (4) 20 m/s

PCM023

24. A stone tied to the end of a string of 1m long is whirled in a horizontal circle with a constant speed. If the stone makes 22 revolution in 44 seconds, what is the magnitude and direction of acceleration of the stone :–

(1) π^2 m/s^2 and direction along the tangent to the circle.

(2) π^2 m/s^2 and direction along the radius towards the centre.

(3) $\frac{\pi^2}{4}$ m/s^2 and direction along the radius towards the centre.

(4) π^2 m/s^2 and direction along the radius away from the centre.

PCM024

25. A fly wheel rotating at 600 rev/min is brought under uniform deceleration and stopped after 2 minutes, then what is angular deceleration in rad/sec^2 ?

(1) $\frac{\pi}{6}$ (2) 10 π (3) $\frac{1}{12}$ (4) 300

PCM025

26. The linear and angular acceleration of a particle are 10 m/s^2 and 5 rad/s^2 respectively. It will be at a distance from the axis of rotation.

(1) 50 m (2) 25m (3) 1 m (4) 2 m

PCM026

DYNAMICS OF HORIZONTAL CIRCULAR MOTION

27. The angular acceleration of particle moving along a circular path with uniform speed :–

(1) uniform but non zero

(2) zero

(3) variable

(4) as can not be predicted from given information

PCM027

28. If the speed and radius both are trippled for a body moving on a circular path, then the new centripetal force will be :–

(1) Doubled of previous value

(2) Equal to previous value

(3) Triple of previous value

(4) One third of previous value

PCM028

29. When a body moves with a constant speed along a circle :–

(1) no acceleration is present in the body

(2) no force acts on the body

(3) its velocity remains constant

(4) no work gets done on it

PCM029

30. A pendulum is suspended from the roof of a rail road car. When the car is moving on a circular track the pendulum inclines :

(1) Forward

(2) Backward

(3) Towards the centre of the path

(4) Away from the centre of the path

PCM030

31. A string of length 0.1 m cannot bear a tension more than 100N. It is tied to a body of mass 100g and rotated in a horizontal circle. The maximum angular velocity can be -

(1) 100 rad/s (2) 1000 rad/s

(3) 10000 rad/s (4) 0.1 rad/s

PCM031

32. The radius of the circular path of a particle is doubled but its frequency of rotation is kept constant. If the initial centripetal force be F, then the final value of centripetal force will be :–

(1) F (2) $\frac{F}{2}$ (3) 4F (4) 2F

PCM032

33. A 0.5 kg ball moves in a circle of radius 0.4 m at a speed of 4 m/s. The centripetal force on the ball is :–
(1) 10N (2) 20N (3) 40N (4) 80N
PCM033

34. A body is revolving with a constant speed along a circle. If its direction of motion is reversed but the speed remains the same then :–
(a) the centripetal force will not suffer any change in magnitude
(b) the centripetal force will have its direction reversed
(c) the centripetal force will not suffer any change in direction
(d) the centripetal force is doubled
(1) a,b (2) b,c (3) c,d (4) a, c
PCM034

35. a_r and a_t represent radial and tangential acceleration. The motion of a particle will be uniform circular motion, if :–
(1) $a_r = 0$ and $a_t = 0$ (2) $a_r = 0$ but $a_t \neq 0$
(3) $a_r \neq 0$ but $a_t = 0$ (4) $a_r \neq 0$ and $a_t \neq 0$
PCM035

36. In uniform circular motion, the velocity vector and acceleration vector are
(1) Perpendicular to each other
(2) Same direction
(3) Opposite direction
(4) Not related to each other
PCM036

37. A string of length 10 cm breaks if its tension exceeds 10 newton. A stone of mass 250 g tied to this string, is rotated in a horizontal circle. The maximum angular velocity of rotation can be :-
(1) 20 rad/s (2) 40 rad/s
(3) 100 rad/s (4) 200 rad/s
PCM037

38. The earth ($M_e = 6 \times 10^{24}$ kg) is revolving around the sun in an orbit of radius (1.5×10^8) km with angular velocity of (2×10^{-7}) rad/s. The force (in newton) exerted on the earth by the sun will be :-
(1) 36×10^{21} (2) 16×10^{24}
(3) 25×10^{16} (4) Zero
PCM038

39. A 500 kg car takes a round turn of radius 50 m with a velocity of 36 km/hr. The centripetal force is :-
(1) 250 N (2) 1000N
(3) 750N (4) 1200 N
PCM039

40. A motor cycle driver doubles its velocity when he is taking a turn. The force exerted towards the centre will become :-
(1) double (2) half
(3) 4 times (4) $\frac{1}{4}$ times
PCM040

41. The force required to keep a body in uniform circular motion is :-
(1) Centripetal force (2) Centrifugal force
(3) Resistance (4) None of the above
PCM041

BANKING OF TRACKS

42. A car moving on a horizontal road may be thrown out of the road in taking a turn :–
(1) by the gravitational force
(2) due to lack of proper centripetal force
(3) due to rolling friction between the tyres and the Road
(4) due to reaction of the road
PCM042

43. Radius of the curved road on national highway is R. Width of the road is b. The outer edge of the road is raised by h with respect to inner edge so that a car with velocity v can pass safely over it. The value of h is :–
(1) $\frac{v^2 b}{Rg}$ (2) $\frac{v}{Rgb}$ (3) $\frac{v^2 R}{bg}$ (4) $\frac{v^2 b}{R}$
PCM043

44. A boy holds a pendulum in his hand while standing at the edge of a circular platform of radius r rotating at an angular speed ω. The pendulum will hang at an angle θ with the vertical so that :–
(Neglect length of pendulum)
(1) $\tan\theta = 0$ (2) $\tan\theta = \frac{\omega^2 r^2}{g}$
(3) $\tan\theta = \frac{r\omega^2}{g}$ (4) $\tan\theta = \frac{g}{\omega^2 r}$
PCM044

VERTICAL CIRCULAR MOTION

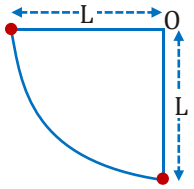
45. Let 'θ' denote the angular displacement of a simple pendulum oscillating in a vertical plane. If the mass of the bob is (m), then the tension in string is mg cosθ :–
(1) always
(2) never
(3) at the extreme positions
(4) at the mean position

PCM045

46. A pendulum bob has a speed 3 m/s while passing through its lowest position, length of the pendulum is 0.5 m then its speed when it makes an angle of 60° with the vertical is :–
($g = 10\ m/s^2$)
(1) 2 m/s (2) 1 m/s
(3) 4 m/s (4) 3 m/s

PCM046

47. The mass of the bob of a simple pendulum of length L is m. If the bob is left from its horizontal position then the speed of the bob and the tension in the thread in the lowest position of the bob will be respectively :–



(1) $\sqrt{2gL}$ and 3 mg (2) $\sqrt{3gL}$ and 2 mg
(3) $\sqrt{5gL}$ and 2 mg (4) $\sqrt{gL}$ and 3 mg

PCM047

48. A stone of mass 1 kg is tied to the end of a string of 1 m length. It is whirled in a vertical circle. If the velocity of the stone at the top be 4 m/s. What is the tension in the string (at that instant) ?
(1) 6 N (2) 16 N (3) 5 N (4) 10 N

PCM048

49. In a vertical circle of radius (r), at what point in its path a particle may have tension equal to zero :–
(1) highest point
(2) lowest point
(3) at any point
(4) at a point horizontal from the centre of radius

PCM049

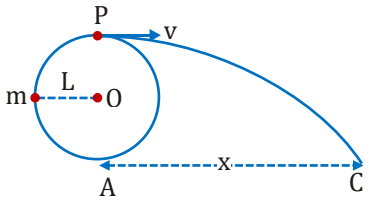
50. A stone attached to one end of a string is whirled in a vertical circle. The tension in the string is maximum when :–
(1) the string is horizontal
(2) the string is vertical with the stone at highest position
(3) the string is vertical with the stone at the lowest position
(4) the string makes an angle of 45° with the vertical

PCM050

51. A weightless thread can withstand tension upto 30 N. A stone of mass 0.5 kg is tied to it and is revolved in a circular path of radius 2m in a vertical plane. If $g = 10\ m/s^2$, then the maximum angular velocity of the stone can be :–
(1) 5 rad/s (2) $\sqrt{30}$ rad/s
(3) $\sqrt{60}$ rad/s (4) 10 rad/s

PCM051

52. A body tied to a string of length L is revolved in a vertical circle with minimum velocity, when the body reaches the upper most point the string breaks and the body moves under the influence of the gravitational field of earth along a parabolic path. The horizontal range AC of the body will be :–



(1) x = L (2) x = 2L
(3) $x = 2\sqrt{2}L$ (4) $x = \sqrt{2}L$

PCM052

53. A particle is moving in a vertical circle the tension in the string when passing through two position at angle 30° and 60° from vertical from lowest position are T_1 and T_2 respectively then :–
(1) $T_1 = T_2$ (2) $T_1 > T_2$
(3) $T_1 < T_2$ (4) $T_1 \geq T_2$

PCM053

54. A body crosses the topmost point of a vertical circle with critical speed. What will be its centripetal acceleration when the string is horizontal :–

(1) g (2) 2g

(3) 3g (4) 6g

PCM054

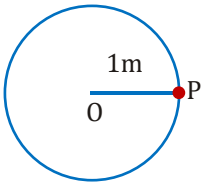
55. Stone tied at one end of light string is whirled round a vertical circle. If the difference between the maximum and minimum tension experienced by the string wire is 2 kg wt, then the mass of the stone must be :–

(1) 1 kg (2) 6 kg

(3) 1/3 kg (4) 2 kg

PCM055

56. A mass tied to a string moves in a vertical circle with a uniform speed of 5 m/s as shown. At the point P the string breaks. The mass will reach a height above P of nearly ($g = 10\ m/s^2$) :–



(1) 1 m (2) 0.5 m

(3) 1.75 m (4) 1.25 m

PCM056

57. If the overbridge is concave instead of being convex, then the reaction on the road at the lowest position will be :-

(1) $mg+\frac{mv^2}{r}$

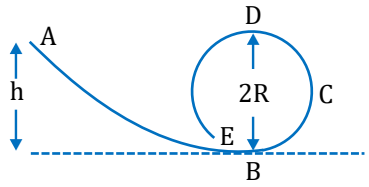
(2) $mg-\frac{mv^2}{r}$

(3) $\frac{m^2v^2g}{r}$

(4) $\frac{v^2g}{r}$

PCM057

58. A frictionless track ABCDE ends in a circular loop of radius R. A body slides down the track from point A which is at a height h = 5 cm. Maximum value of R for the body to successfully complete the loop is :–



(1) 5 cm.

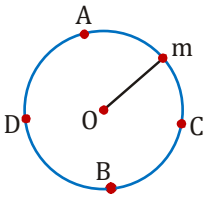
(2) 2 cm.

(3) $\frac{10}{3}$ cm.

(4) $\frac{15}{4}$ cm.

PCM058

59. A particle of mass m is performing vertical circular motion (see figure). If the average speed of the particle is increased, then at which point maximum breaking possibility of the string :–



(1) A (2) B

(3) C (4) D

PCM059

60. A stone tide with a string is moving in a vertical circle of radius 'r'. Its minimum velocity at the highest point of the circle will be :–

(1) $\sqrt{3gr}$

(2) $\sqrt{2gr}$

(3) $\sqrt{gr}$

(4) $\sqrt{\frac{gr}{2}}$

PCM060

61. A stone of mass 0.2 kg is tied to one end of a thread of length 0.1 m whirled in a vertical circle. When the stone is at the lowest point of circle, tension in thread is 52N, then velocity of the stone will be :–

(1) 4 m/s (2) 5 m/s

(3) 6 m/s (4) 7 m/s

PCM061

62. A suspended simple pendulum of length ℓ is making an angle θ with the vertical. On releasing, its velocity at lowest point will be :-

(1) $\sqrt{2g\ell(1+\cos\theta)}$ (2) $\sqrt{2g\ell\sin\theta}$

(3) $\sqrt{2g\ell(1-\cos\theta)}$ (4) $\sqrt{2g\ell}$

PCM062

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	2	2	3	3	1	4	2	3	1	3	4	1	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	3	4	4	4	1	1	2	2	1	4	2	3	4	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	4	2	4	3	1	1	1	2	3	1	2	1	3	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	1	1	1	3	1	2	2	3	3	4	1	2	2	3
Question	61	62													
Answer	2	3													

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Exercise - II (Previous Year Questions) AIPMT/NEET

Re-AIPMT 2015

1. Two stones of masses m and 2 m are whirled in horizontal circles, the heavier one in a radius $\frac{r}{2}$ and the lighter one in radius r. The tangential speed of lighter stone is n times that of the value of heavier stone when they experience same centripetal forces. The value of n is :

(1) 1 (2) 2 (3) 3 (4) 4

PCM070

2. The position vector of a particle $\vec{R}$ as a function of time is given by :-

$$\vec{R} = 4\sin(2\pi t)\hat{i} + 4\cos(2\pi t)\hat{j}$$

Where R is in meters, t is in seconds and $\hat{i}$ and $\hat{j}$ denote unit vectors along x and y-directions, respectively. Which one of the following statements is wrong for the motion of particle ?

(1) Path of the particle is a circle of radius 4 meter

(2) Acceleration vectors is along $-\vec{R}$

(3) Magnitude of acceleration vector is $\frac{v^2}{R}$ where v is the velocity of particle.

(4) Magnitude of the velocity of particle is 8 meter/second

PCM071

NEET-I 2016

3. A particle of mass 10 g moves along a circle of radius 6.4 cm with a constant tangential acceleration. What is the magnitude of this acceleration if the kinetic energy of the particle becomes equal to 8×10^{-4} J by the end of the second revolution after the beginning of the motion?

(1) 0.1 m/s^2 (2) 0.15 m/s^2

(3) 0.18 m/s^2 (4) 0.2 m/s^2

PCM072

4. What is the minimum velocity with which a body of mass m must enter a vertical loop of radius R so that it can complete the loop ?

(1) $\sqrt{gR}$ (2) $\sqrt{2gR}$

(3) $\sqrt{3gR}$ (4) $\sqrt{5gR}$

PCM073

5. A car is negotiating a curved road of radius R. The road is banked at an angle θ. The coefficient of friction between the tyres of the car and the road is μ_s. The maximum safe velocity on this road is:-

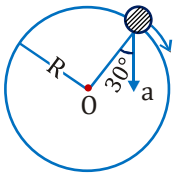
(1) $\sqrt{gR^2\frac{\mu_s+\tan\theta}{1-\mu_s\tan\theta}}$ (2) $\sqrt{gR\frac{\mu_s+\tan\theta}{1-\mu_s\tan\theta}}$

(3) $\sqrt{\frac{g}{R}\frac{\mu_s+\tan\theta}{1-\mu_s\tan\theta}}$ (4) $\sqrt{\frac{g}{R^2}\frac{\mu_s+\tan\theta}{1-\mu_s\tan\theta}}$

PCM074

NEET-II 2016

6. In the given figure, a = 15 m/s^2 represents the total acceleration of a particle moving in the clockwise direction in a circle of radius R = 2.5 m at a given instant of time. The speed of the particle is :-



(1) 5.7 m/s (2) 6.2 m/s

(3) 4.5 m/s (4) 5.0 m/s

PCM075

7. A particle moves so that its position vector is given by $\vec{r} = \cos\omega t\,\hat{x} + \sin\omega t\,\hat{y}$. Where ω is a constant.

Which of the following is true ?

(1) Velocity and acceleration both are perpendicular to $\vec{r}$.

(2) Velocity and acceleration both are parallel to $\vec{r}$

(3) Velocity is perpendicular to $\vec{r}$ and acceleration is directed towards the origin

(4) Velocity is perpendicular to $\vec{r}$ and acceleration is directed away from the origin

PCM076

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NEET(UG) 2017

8. One end of string of length l is connected to a particle of mass 'm' and the other end is connected to a small peg on a smooth horizontal table. If the particle moves in circle with speed 'v' the net force on the particle (directed towards centre) will be (T represents the tension in the string) :-

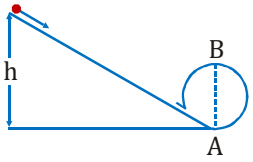
(1) $T+\frac{mv^2}{l}$ (2) $T-\frac{mv^2}{l}$

(3) Zero (4) T

PCM077

NEET(UG) 2018

9. A body initially at rest and sliding along a frictionless track from a height h (as shown in the figure) just completes a vertical circle of diameter AB = D. The height h is equal to :-



(1) $\frac{3}{2}D$ (2) D (3) $\frac{7}{5}D$ (4) $\frac{5}{4}D$

PCM078

NEET(UG) 2019

10. A block of mass 10 kg is in contact against the inner wall of a hollow cylindrical drum of radius 1m. The coefficient of friction between the block and the inner wall of the cylinder is 0.1. The minimum angular velocity needed for the cylinder to keep the block stationary when the cylinder is vertical and rotating about its axis, will be : ($g = 10\ m/s^2$)

(1) $\sqrt{10}$ rad/s (2) $\frac{10}{2\pi}$ rad/s

(3) 10 rad/s (4) 10π rad/s

PCM079

11. A mass m is attached to a thin wire and whirled in a vertical circle. The wire is most likely to break when :

(1) the mass is at the highest point

(2) the wire is horizontal

(3) the mass is at the lowest point

(4) inclined at an angle of 60° from vertical

PCM080

12. Two particles A and B are moving in uniform circular motion in concentric circles of radius r_A and r_B with speed υ_A and υ_B respectively. The time period of rotation is the same. The ratio of angular speed of A to that of B will be :

(1) $r_A : r_B$ (2) $\upsilon_A : \upsilon_B$

(3) $r_B : r_A$ (4) 1 : 1

PCM081

NEET(UG) 2019 (Odisha)

13. A particle starting from rest, moves in a circle of radius 'r'. It attains a velocity of V_0 m/s in the nth round. Its angular acceleration will be :-

(1) $\frac{V_0}{n}\text{rad}/s^2$ (2) $\frac{V_0^2}{2\pi nr^2}\text{rad}/s^2$

(3) $\frac{V_0^2}{4\pi nr^2}\text{rad}/s^2$ (4) $\frac{V_0^2}{4\pi nr}\text{rad}/s^2$

PCM082

NEET(UG) 2020 (COVID-19)

14. The angular speed of the wheel of a vehicle is increased from 360 rpm to 1200 rpm in 14 second. Its angular acceleration is

(1) 2π rad/s² (2) 28π rad/s²

(3) 120π rad/s² (4) 1 rad/s²

PCM083

15. A point mass 'm' is moved in a vertical circle of radius 'r' with the help of a string. The velocity of the mass is $\sqrt{7gr}$ at the lowest point. The tension in the string at the lowest point is :

(1) 6 mg (2) 7 mg

(3) 8 mg (4) 1 mg

PCM084

NEET(UG) 2022

16. The angular speed on a fly wheel moving with uniform angular acceleration changes from 1200 rpm to 3120 rpm in 16 seconds. The angular acceleration in rad/s² is :

(1) 4π (2) 12π

(3) 104π (4) 2π

PCM085

NEET(UG) 2023

17. The angular acceleration of a body, moving along the circumference of a circle, is :

(1) along the radius towards the centre

(2) along the tangent to its position

(3) along the axis of rotation

(4) along the radius, away from centre

PCM094

18. Calculate the maximum acceleration of a moving car so that a body lying on the floor of the car remains stationary. The coefficient of static friction between the body and the floor is 0.15
($g = 10\ m\ s^{-2}$).

(1) $150\ m\ s^{-2}$ (2) $1.5\ m\ s^{-2}$

(3) $50\ m\ s^{-2}$ (4) $1.2\ m\ s^{-2}$

PCM095

RE-NEET(UG) 2023

19. A particle is executing uniform circular motion with velocity $\vec{v}$ and acceleration $\vec{a}$. Which of the following is true ?

(1) $\vec{v}$ is a constant; $\vec{a}$ is not a constant

(2) $\vec{v}$ is not a constant; $\vec{a}$ is not a constant

(3) $\vec{v}$ is a constant; $\vec{a}$ is a constant

(4) $\vec{v}$ is not a constant; $\vec{a}$ is a constant

PCM096

NEET(UG) 2024

20. A bob is whirled in a horizontal plane by means of a string with an initial speed of ω rpm. The tension in the string is T. If speed becomes 2ω while keeping the same radius, the tension in the string becomes :

(1) T (2) 4T

(3) $\frac{T}{4}$ (4) $\sqrt{2}T$

PCM097

21. A particle moving with uniform speed in a circular path maintains :

(1) constant velocity

(2) constant acceleration.

(3) constant velocity but varying acceleration

(4) varying velocity and varying acceleration

PCM098

RE-NEET(UG) 2024

22. A bob is whirled in a horizontal circle by means of a string at an initial speed of 10 rpm. If the tension in the string is quadrupled while keeping the radius constant, the new speed is:

(1) 20 rpm (2) 40 rpm

(3) 5 rpm (4) 10 rpm

PCM099

23. Let ω_1, ω_2 and ω_3 be the angular speed of the second hand, minute hand and hour hand of a smoothly running analog clock, respectively. If x_1, x_2 and x_3 are their respective angular distances in 1 minute then the factor which remains constant (k) is:

(1) $\frac{\omega_1}{x_1} = \frac{\omega_2}{x_2} = \frac{\omega_3}{x_3} = k$

(2) $\omega_1 x_1 = \omega_2 x_2 = \omega_3 x_3 = k$

(3) $\omega_1 x_1^2 = \omega_2 x_2^2 = \omega_3 x_3^2 = k$

(4) $\omega_1^2 x_1 = \omega_2^2 x_2 = \omega_3^2 x_3 = k$

PCM100

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	4	1	4	2	1	3	4	4	3	3	4	3	1	3
Question	16	17	18	19	20	21	22	23							
Answer	1	3	2	2	2	4	1	1							

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Exercise - III (Analytical Questions) Master Your Understanding

1. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In circular motion, work done by the centripetal force is always zero.

Reason (R) : Centripetal force is always perpendicular to the instantaneous displacement.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PCM086

2. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In uniform circular motion speed of body is constant but its velocity is variable.

Reason (R) : In uniform circular motion tangential acceleration of body is zero, but body has normal acceleration

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PCM087

3. For a body moving in a circular path of radius R = 1m in anticlockwise direction with constant speed V.

	Column-I		Column-II
(A)	Angle between instantaneous velocity and instantaneous acceleration	(P)	Non-zero
(B)	Total acceleration of the body	(Q)	Zero
(C)	Tangential acceleration of the body	(R)	$\pi/2$
		(S)	$\pi/3$

(1) A $\rightarrow$ R, B $\rightarrow$ P, C $\rightarrow$ Q

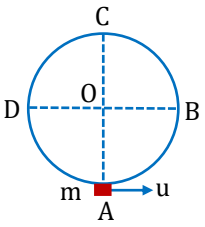
(2) A $\rightarrow$ P, B $\rightarrow$ Q, C $\rightarrow$ R

(3) A $\rightarrow$ Q, B $\rightarrow$ R, C $\rightarrow$ P

(4) A $\rightarrow$ R, B $\rightarrow$ Q, C $\rightarrow$ P

PCM088

4. A particle suspended from a string of length ℓ is given a horizontal speed $u = 3\sqrt{g\ell}$ at the bottom. Then for the particle match the following column –



	Column-I		Column-II
(A)	Speed at B	(P)	7 mg
(B)	Speed at C	(Q)	$\sqrt{5g\ell}$
(C)	Tension in string at B	(R)	$\sqrt{7g\ell}$
(D)	Tension in string at C	(S)	4 mg

(1) A $\rightarrow$ R, B $\rightarrow$ Q, C $\rightarrow$ P, D $\rightarrow$ S

(2) A $\rightarrow$ R, B $\rightarrow$ P, C $\rightarrow$ Q, D $\rightarrow$ S

(3) A $\rightarrow$ P, B $\rightarrow$ Q, C $\rightarrow$ S, D $\rightarrow$ R

(4) A $\rightarrow$ Q, B $\rightarrow$ P, C $\rightarrow$ S, D $\rightarrow$ R

PCM089

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5. A body is connected to a string of length r and revolved in vertical circle with one end of string as the centre of circle. Its velocity at bottom most point v_L is twice that of its value v_H at the top most point. Then
Choose correct statement -
(A) Velocity at highest point is $\sqrt{4gr/3}$
(B) $v_L^2 - v_H^2$ is equal to 4gr
(C) Tension in string at bottom most point is 6 mg
(D) Ratio of maximum tension to minimum tension in string is 10/9
(1) A & C (2) A & B
(3) B & D (4) B & C

PCM090

6. Speed of a body moving in a circular path changes with time as v = 2t, then –
(A) Magnitude of acceleration remains constant
(B) Magnitude of acceleration increases
(C) Angle between velocity and acceleration remains constant
(D) Angle between velocity and acceleration increases
(1) A & C
(2) A & D
(3) B & C
(4) B & D

PCM091

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6
Answer	1	1	1	1	2	4

Exercise - I (Conceptual Questions) Build Up Your Understanding

CALCULATION OF CENTRE OF MASS

1. Three identical spheres, each of mass 1 kg are placed touching each other with their centres on a straight line. Their centre are marked P, Q and R respectively. The distance of centre of mass of the system from P is :

(1) $\frac{PQ+PR+QR}{3}$ (2) $\frac{PQ+PR}{3}$

(3) $\frac{PQ+QR}{3}$ (4) $\frac{PR+QR}{2}$

PSC001

2. A uniform metal disc of radius R is taken and out of it a disc of diameter $\frac{R}{2}$ is cut off from the end. The centre of mass of the remaining part will be:

(1) $\frac{R}{10}$ from the centre

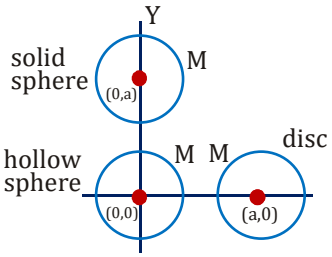
(2) $\frac{R}{15}$ from the centre

(3) $\frac{R}{5}$ from the centre

(4) $\frac{R}{20}$ from the centre

PSC002

3. The coordinate of the centre of mass of a system as shown in figure :–



(1) $\left(\frac{a}{3}, 0\right)$ (2) $\left(\frac{a}{2}, \frac{a}{2}\right)$

(3) $\left(\frac{a}{3}, \frac{a}{3}\right)$ (4) $\left(0, \frac{a}{3}\right)$

PSC003

4. The centre of mass of a system of particles does not depend on :

(1) masses of the particles

(2) Internal forces on the particles

(3) position of the particles

(4) relative distance between the particles

PSC004

5. The centre of mass of a system of two particles divides the distance between them

(1) In inverse ratio of square of masses of particles

(2) In direct ratio of square of masses of particles

(3) In inverse ratio of masses of particles

(4) In direct ratio of masses of particles

PSC005

6. The centre of mass of a body :-

(1) Lies always outside the body

(2) May lie within, outside of the surface of the body

(3) Lies always inside the body

(4) Lies always on the surface of the body

PSC006

7. Three identical metal balls, each of radius r, are placed touching each other on a horizontal surface such that an equilateral triangle is formed when the centres of the three balls are joined. The centre of mass of the system is located at :-

(1) horizontal surface

(2) centre of one of the balls

(3) line joining centres of any two balls

(4) point of intersection of their medians

PSC007

8. A system consists of mass M and m (<< M). The centre of mass of the system is :-

(1) at the middle

(2) nearer to M

(3) nearer to m

(4) at the position of larger mass.

PSC008

9. The centre of mass of a system of three particles of masses 1g, 2g and 3g is taken as the origin of a coordinate system. The position vector of a fourth particle of mass 4g such that the centre of mass of the four particle system lies at the point (1, 2, 3,) is $\alpha(\hat{i}+2\hat{j}+3\hat{k})$, where α is a constant. The value of α is :-

(1) $\frac{10}{3}$ (2) $\frac{5}{2}$ (3) $\frac{1}{2}$ (4) $\frac{2}{5}$

PSC009

MOTION OF CENTRE OF MASS

10. The law of conservation of momentum for a system is based on Newton's :–

(1) First law of motion
(2) Second law of motion
(3) Third law of motion
(4) Law of gravitation

PSC010

11. A person of mass m is standing on one end of a plank of mass M and length L and floating in water. The person moves from one end to another and stops. The displacement of the plank is –

(1) $\frac{Lm}{(m+M)}$ (2) $Lm(M+m)$

(3) $\frac{(M+m)}{Lm}$ (4) $\frac{LM}{(m+M)}$

PSC011

12. Bullets of mass 40 g each are fired from a machine gun with a velocity of 10^3 m/s. If the person firing the bullets experience an average force of 200N, then the number of bullets fired per minute will be–

(1) 300 (2) 600 (3) 150 (4) 75

PSC012

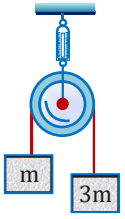
13. If the system is released, then the acceleration of the centre of mass of the system :-

(1) $\frac{g}{4}$

(2) $\frac{g}{2}$

(3) g

(4) 2g



PSC013

14. Initially two stable particles x and y start moving towards each other under mutual attraction. If at one time the velocities of x and y are V and 2V respectively, what will be the velocity of centre of mass of the system?

(1) V (2) Zero

(3) $\frac{V}{3}$ (4) $\frac{V}{5}$

PSC014

15. A 2 kg body and a 3 kg body are moving along the x-axis. At a particular instant the 2 kg body has a velocity of 3 m/s and the 3 kg body has the velocity of 2 m/s. The velocity of the centre of mass at that instant is :-

(1) 5 m/s (2) 1 m/s

(3) 0 (4) $\frac{12}{5}$ m/s

PSC015

16. Two objects of masses 200 gram and 500 gram possess velocities $10\hat{i}$ m/s and $3\hat{i}+5\hat{j}$ m/s respectively. The velocity of their centre of mass in m/s is :-

(1) $5\hat{i}-25\hat{j}$ (2) $\frac{5}{7}\hat{i}-25\hat{j}$

(3) $5\hat{i}+\frac{25}{7}\hat{j}$ (4) $25\hat{i}-\frac{5}{7}\hat{j}$

PSC016

MOMENTUM CONSERVATION

17. A bomb of mass 9 kg explodes into two pieces of 3kg and 6 kg. The velocity of 3 kg piece is 16 m/s. The kinetic energy of 6 kg piece is :–

(1) 768 J (2) 786 J
(3) 192 J (4) 687 J

PSC017

18. A bomb initially at rest explodes by it self into three equal mass fragments. The velocities of two fragments are $(3\hat{i}+2\hat{j})$ m/s and $(-\hat{i}-4\hat{j})$ m/s. The velocity of the third fragment is (in m/s) :-

(1) $2\hat{i}+2\hat{j}$ (2) $2\hat{i}-2\hat{j}$

(3) $-2\hat{i}+2\hat{j}$ (4) $-2\hat{i}-2\hat{j}$

PSC018

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19. A bomb of 50 Kg is fired from a cannon with a velocity 600 m/s. If the mass of the cannon is 10^3 kg, then its velocity will be –
(1) 30 m/s (2) –30 m/s
(3) 0·30 m/s (4) –0·30 m/s

PSC019

20. A metal ball does not rebound when struck on a wall, whereas a rubber ball of same mass when thrown with the same velocity on the wall rebounds. From this it is inferred that –
(1) Change in momentum is same in both
(2) Change in momentum in rubber ball is more
(3) Change in momentum in metal ball is more
(4) Initial momentum of metal ball is more than that of rubber ball

PSC020

21. A bomb of mass m = 1 kg thrown vertically upwards with a speed u = 100 m/s explodes into two parts after t = 5s. A fragment of mass m_1 = 400 g moves downwards with a speed v_1 = 25 m/s, then speed v_2 and direction of another mass m_2 will be :-
(1) 40 m/s downwards (2) 40 m/s upwards
(3) 60 m/s upwards (4) 100 m/s upwards

PSC021

22. A 1 kg stationary bomb is exploded in three parts having mass ratio 1 : 1 : 3. Parts having same mass move in perpendicular directions with velocity 30 m/s, then the velocity of bigger part will be :-
(1) $10\sqrt{2}$ m/s (2) $\frac{10}{\sqrt{2}}$ m/s
(3) $15\sqrt{2}$ m/s (4) $\frac{15}{\sqrt{2}}$ m/s

PSC022

23. A heavy nucleus at rest breaks into two fragments which fly off with velocities 8 : 1. The ratio of radii of the fragments is :-
(1) 1 : 2 (2) 1 : 4 (3) 4 : 1 (4) 2 : 1

PSC023

24. A stationary particle explodes into two particles of masses m_1 and m_2 which move in opposite directions with velocities v_1 and v_2. The ratio of their kinetic energies E_1/E_2 is :-
(1) m_2/m_1 (2) m_1/m_2
(3) 1 (4) m_1v_2/m_2v_1

PSC024

25. A body of mass 4 m at rest explodes into three pieces. Two of the pieces each of mass m move with a speed v each in mutually perpendicular directions. The total kinetic energy released is :-
(1) $\frac{1}{2}mv^2$ (2) mv^2
(3) $\frac{3}{2}mv^2$ (4) $\frac{5}{2}mv^2$

PSC025

26. A bomb of mass 3.0 kg explodes in air into two pieces of masses 2.0 kg and 1.0 kg. The smaller mass goes at a speed of 80 m/s. The total energy imparted to the two fragments is -
(1) 1.07 kJ (2) 2.14 kJ
(3) 2.4 kJ (4) 4.8 kJ

PSC026

27. A bomb of mass 30 kg at rest explodes into two pieces of masses 18 kg and 12 kg. The velocity of 18 kg mass is 6 m/s. The kinetic energy of the other mass is :-
(1) 524 J (2) 256 J (3) 486 J (4) 324 J

PSC027

28. A bullet of mass m is fired from a gun of mass M. The recoiling gun compresses a spring of force constant k by a distance d. Then the velocity of the bullet is :-
(1) $kd\sqrt{M/m}$ (2) $\frac{d}{M}\sqrt{km}$
(3) $\frac{d}{m}\sqrt{kM}$ (4) $\frac{kM}{m}\sqrt{d}$

PSC028

29. Identify the wrong statement.
(1) A body can have momentum without mechanical energy
(2) A body can have energy without momentum
(3) The momentum is conserved in an elastic collision only.
(4) Kinetic energy is not conserved in an inelastic collision

PSC029

COLLISION

30. A ball strikes the floor and after collision rebounds back. In this state –
(1) Momentum of the ball is conserved
(2) Mechanical energy of the ball is conserved
(3) Momentum of ball–earth system is conserved
(4) The kinetic energy of ball–earth system is conserved

PSC030

31. A bullet of mass P is fired with velocity Q in a large body of mass R. The final velocity of the system will be :–

(1) $\frac{R}{P+R}$ (2) $\frac{PQ}{P+R}$

(3) $\frac{(P+Q)}{R}$ (4) $\frac{(P+R)}{P}Q$

PSC031

32. A sphere of mass m moving with a constant velocity collides with another stationary sphere of same mass. The ratio of velocities of two spheres after collision will be, if the co-efficient of restitution is e-

(1) $\frac{1-e}{1+e}$ (2) $\frac{e-1}{e+1}$

(3) $\frac{1+e}{1-e}$ (4) $\frac{e+1}{e-1}$

PSC032

33. Two elastic bodies P and Q having equal masses are moving along the same line with velocities of 16 m/s and 10 m/s respectively. Their velocities after the elastic collision will be in m/s :-

(1) 0 and 25 (2) 5 and 20

(3) 10 and 16 (4) 20 and 5

PSC033

34. The unit of the coefficient of restitution is –

(1) m/s

(2) s/m

(3) m × s

(4) None of the above

PSC034

35. Two solid balls of rubber A and B whose masses are 200 gm and 400 gm respectively, are moving in mutually opposite directions. If the velocity of ball A is 0·3 m/s and both the balls come to rest after collision, then the velocity of ball B is –

(1) 0·15 m/s

(2) – 0·15 m/s

(3) 1·5 m/s

(4) None of the above

PSC035

36. A 1 Kg ball falls from a height of 25 cm and rebounds upto a height of 9 cm. The coefficient of restitution is –

(1) 0·6 (2) 0·32 (3) 0·40 (4) 0·56

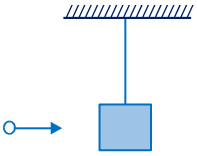
PSC036

37. A 50 gm bullet moving with a velocity of 10 m/s gets embedded into a 950 gm stationary body. The loss in kinetic energy of the system will be –

(1) 5% (2) 50% (3) 100% (4) 95%

PSC037

38. A bullet of mass m moving with a speed v strikes a wooden block of mass M and gets embedded into the block. The final speed is :-



(1) $\sqrt{\frac{M}{M+m}}v$ (2) $\sqrt{\frac{m}{M+m}}v$

(3) $\frac{m}{M+m}v$ (4) $\frac{v}{2}$

PSC038

39. A ball is dropped from height h on the ground level. If the coefficient of restitution is e then the height upto which the ball will go after n^{th} jump will be –

(1) $\frac{h}{e^{2n}}$ (2) $\frac{e^{2n}}{h}$

(3) he^{n} (4) he^{2n}

PSC039

40. Two bodies of same mass are moving with same speed V in mutually opposite directions. They collide and stick together. The resultant velocity of the system will be –

(1) Zero (2) $\frac{V}{2}$

(3) V (4) From Zero to ∞

PSC040

41. The bob (mass m) of a simple pendulum of length L is held horizontal and then released. It collides elastically with a block of equal mass lying on a frictionless table. The kinetic energy of the block will be :–

(1) Zero (2) mgL

(3) 2mgL (4) $\frac{mgL}{2}$

PSC041

42. Two particles each of mass m travelling with velocities u_1 and u_2 collide perfectly inelastically. The loss of kinetic energy will be –

(1) $\frac{1}{2}m(u_1-u_2)^2$ (2) $\frac{1}{4}m(u_1-u_2)^2$

(3) $m(u_1-u_2)^2$ (4) $2m(u_1-u_2)^2$

PSC042

43. A ball moving with velocity of 9m/s collides with another similar stationary ball. After the collision both the balls move in directions making an angle of 30° with the initial direction. After the collision their speed will be –

(1) 2·6 m/s (2) 5·2 m/s

(3) 0·52 m/s (4) 52 m/s

PSC043

44. A solid sphere is moving and it makes an elastic collision with another stationary sphere of half of its own radius. After collision it comes to rest. The ratio of the densities of materials of second sphere and first sphere is -

(1) 2 (2) 4 (3) 8 (4) 16

PSC044

45. A 5 kg body collides with another stationary body. After the collision, the bodies move in the same direction with one–third of the velocity of the first body. The mass of the second body will be –

(1) 5 kg (2) 10 kg

(3) 15 kg (4) 20 kg

PSC045

46. A 10 g bullet, moving with a velocity of 500 m/s. enters a stationary piece of ice of mass 10 kg and stops. If the piece of ice is lying on a frictionless plane, then its velocity will be

(1) 5 cm/s (2) 5 m/s

(3) 0·5 m/s (4) 0·5 cm/s

PSC046

47. A heavy body moving with a velocity 20 m/s and another small object at rest undergo an elastic collision. The latter will move with a velocity of :-

(1) 20 m/s. (2) 40 m/s.

(3) 60 m/s. (4) Zero

PSC047

48. A 5gm lump of clay, moving with a velocity of 10 cm/s towards east, collides head–on with another 2gm lump of clay moving with 15 cm/s towards west. After collision, the two lumps stick together. The velocity of the compound lump will be –

(1) 5 cm/s towards east

(2) 5 cm/s towards west

(3) 2·88 cm/s towards east

(4) 2·5 cm/s towards west

PSC048

49. In an inelastic collision between two bodies, the physical quantity that is conserved :-

(1) Kinetic energy

(2) Momentum

(3) Potential energy

(4) Kinetic energy and momentum

PSC049

50. A mass of 20 kg moving with a speed of 10 m/s collides with another stationary mass of 5 kg. As a result of the collision, the two masses stick together. The kinetic energy of the composite mass will be :-

(1) 600 J (2) 800 J

(3) 1000 J (4) 1200 J

PSC050

51. A body of mass m having an initial velocity v makes head on elastic collision with a stationary body of mass M. After the collision, the body of mass m comes to rest and only the body having mass M moves. This will happen only when :-

(1) $m >> M$ (2) $m << M$

(3) $m = M$ (4) $m = \frac{M}{2}$

PSC051

52. A body A experiences perfectly elastic collision with a stationary body B. If after collision the bodies fly apart in the opposite direction with equal speeds, the mass ratio of A and B is :-

(1) $\frac{1}{2}$ (2) $\frac{1}{3}$ (3) $\frac{1}{4}$ (4) $\frac{1}{5}$

PSC052

53. A collision is said to be perfectly inelastic when :-
(1) Coefficient of restitution = 0
(2) Coefficient of restitution = 1
(3) Coefficient of restitution = ∞
(4) Coefficient of restitution < 1

PSC053

54. A particle falls from a height 'h' upon a fixed horizontal plane and rebounds. If 'e' is the coefficient of restitution the total distance travelled before rebounding has stopped is :-
(1) $h\left(\frac{1+e^2}{1-e^2}\right)$ (2) $h\left(\frac{1-e^2}{1+e^2}\right)$
(3) $\frac{h}{2}\left(\frac{1-e^2}{1+e^2}\right)$ (4) $\frac{h}{2}\left(\frac{1+e^2}{1-e^2}\right)$

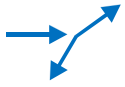
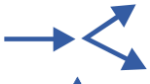
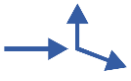
PSC054

55. If two masses m_1 and m_2 collide, the ratio of the changes in their respective velocities is proportional to :-
(1) $\frac{m_1}{m_2}$ (2) $\sqrt{\frac{m_1}{m_2}}$ (3) $\frac{m_2}{m_1}$ (4) $\sqrt{\frac{m_2}{m_1}}$

PSC055

56. Two particles of mass M_A and M_B and there velocities are V_A and V_B respectively collides. After collision they inter changes their velocities then ratio of $\frac{M_A}{M_B}$ is :-
(1) $\frac{V_A}{V_B}$ (2) $\frac{V_B}{V_A}$
(3) $\frac{(V_A+V_B)}{(V_B-V_A)}$ (4) 1

PSC056

57. In the diagrams given below the horizontal line represents the path of a ball coming from left and hitting another ball which is initially at rest. The other two lines represents the paths of the two balls after the collision. Which of the diagram shows a physically impossible situation ?
(1) 
(2) 
(3) 
(4) 

PSC057

58. Two identical balls, one moves with 12 m/s and second is at rest, collides elastically. After collision velocity of second and first ball will be :
(1) 6m/s, 6m/s (2) 12m/s, 12m/s
(3) 12m/s, 0m/s (4) 0m/s, 12m/s

PSC058

59. A sphere P of mass m and velocity $\vec{v}_i$ undergoes an oblique and perfectly elastic collision with an identical sphere Q initially at rest. The angle θ between the velocities of the spheres after the collision shall be :-
(1) 0 (2) 45° (3) 90° (4) 180°

PSC059

60. A ball is dropped from a height of 10 m. If 40% of its energy is lost on collision with the earth then after collision the ball will rebound to a height of–
(1) 10 m (2) 8 m (3) 4 m (4) 6 m

PSC060

61. A rubber ball is dropped from a height of 5m on a plane, where the acceleration due to gravity is not shown. On bouncing it rises to 1.8 m. The ball loses its velocity on bouncing by a factor of :-
(1) $\frac{16}{25}$ (2) $\frac{2}{5}$ (3) $\frac{3}{5}$ (4) $\frac{9}{25}$

PSC061

62. Which of the following is true :-
(1) Momentum is conserved in all collisions but kinetic energy is conserved only in inelastic collision
(2) Neither momentum nor kinetic energy is conserved in inelastic collisions.
(3) Momentum is conserved in all collisions but not kinetic energy
(4) Both momentum and kinetic energy are conserved in all collisions.

PSC062

63. A bullet of mass m is fired into a large block of wood of mass M with velocity v. The final velocity of the system is :-
(1) $\left(\frac{m}{M-m}\right)v$ (2) $\left(\frac{m+M}{M}\right)v$
(3) $\left(\frac{M-m}{M}\right)v$ (4) $\left(\frac{m}{m+M}\right)v$

PSC063

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64. A big ball of mass M, moving with velocity u strikes a small ball of mass m, which is at rest. Finally small ball attains velocity u and big ball v. What is the value of v :-

(1) $\frac{M-m}{M}u$ (2) $\frac{m}{M+m}u$

(3) $\frac{2m}{M+m}$ (4) $\frac{M}{M+m}v$

PSC064

65. A particle of mass m moving with speed v towards east strikes another particle of same mass moving with same speed v towards north. After striking, the two particles fuse together. With what speed this new particle of mass 2 m will move in north-east direction?

(1) v (2) $\frac{v}{2}$ (3) $\frac{v}{\sqrt{2}}$ (4) $v\sqrt{2}$

PSC065

66. Two ice skaters A and B approach each other at right angles. Skater A has a mass 30 kg and velocity 1 m/s and skater B has a mass 20 kg and velocity 2 m/s. They meet and cling together. Their final velocity of the couple is

(1) 2 m/s
(2) 1.5 m/s
(3) 1 m/s
(4) 2.5 m/s

PSC066

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	4	3	2	3	2	4	2	2	3	1	1	1	2	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	3	2	2	4	1	1	1	3	4	3	3	3	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	1	3	4	2	1	4	3	4	1	2	2	2	3	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	2	3	2	2	3	2	1	1	3	4	3	3	3	4
Question	61	62	63	64	65	66									
Answer	2	3	4	1	3	3									

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2015

1. Two particles of masses m_1, m_2 move with initial velocities u_1 and u_2. On collision, one of the particles get excited to higher level, after absorbing energy ε. If final velocities of particles be v_1 and v_2 then we must have :

(1) $\frac{1}{2}m_1u_1^2+\frac{1}{2}m_2u_2^2=\frac{1}{2}m_1v_1^2+\frac{1}{2}m_2v_2^2-\varepsilon$

(2) $\frac{1}{2}m_1u_1^2+\frac{1}{2}m_2u_2^2-\varepsilon=\frac{1}{2}m_1v_1^2+\frac{1}{2}m_2v_2^2$

(3) $\frac{1}{2}m_1^2u_1^2+\frac{1}{2}m_2^2u_2^2+\varepsilon=\frac{1}{2}m_1^2v_1^2+\frac{1}{2}m_2^2v_2^2$

(4) $m_1^2u_1 + m_2^2u_2 - \varepsilon = m_1^2v_1 + m_2^2v_2$

PSC074

2. Two spherical bodies of mass M and 5M and radii R and 2R are released in free space with initial separation between their centres equal to 12R. If they attract each other due to gravitational force only, then the distance covered by the smaller body before collision is :-

(1) 4.5R (2) 7.5R (3) 1.5R (4) 2.5R

PSC075

Re-AIPMT 2015

3. A ball is thrown vertically downwards from a height of 20 m with an initial velocity v_0. It collides with the ground, loses 50% of its energy in collision and rebounds to the same height. The initial velocity v_0 is :

(Take g = 10 m/s^2)

(1) 10 m/s (2) 14 m/s

(3) 20 m/s (4) 28 m/s

PSC076

4. On a frictionless surface, a block of mass. M moving at speed v collides elastically with another block of same mass M which is initially at rest. After collision the first block moves at an angle θ to its initial direction and has a speed $\frac{v}{3}$. The second block's speed after the collision is :-

(1) $\frac{\sqrt{3}}{2}v$ (2) $\frac{2\sqrt{2}}{3}v$ (3) $\frac{3}{4}v$ (4) $\frac{3}{\sqrt{2}}v$

PSC077

NEET-II 2016

5. A bullet of mass 10g moving horizontally with a velocity of 400 m/s strikes a wooden block of mass 2 kg which is suspended by a light inextensible string of length 5 m. As a result, the centre of gravity of the block is found to rise a vertical distance of 10 cm. The speed of the bullet after it emerges out horizontally from the block will be :-

(1) 120 m/s (2) 160 m/s

(3) 100 m/s (4) 80 m/s

PSC078

6. Two identical balls A and B having velocities of 0.5 m/s and –0.3 m/s respectively collide elastically in one dimension. The velocities of B and A after the collision respectively will be :-

(1) –0.3 m/s and 0.5 m/s

(2) 0.3 m/s and 0.5 m/s

(3) –0.5 m/s and 0.3 m/s

(4) 0.5 m/s and –0.3 m/s

PSC079

NEET(UG) 2018

7. A moving block having mass m, collides with another stationary block having mass 4m. The lighter block comes to rest after collision. When the initial velocity of the lighter block is v, then the value of coefficient of restitution (e) will be :-

(1) 0.5 (2) 0.25 (3) 0.8 (4) 0.4

PSC080

NEET(UG) 2019

8. Body A of mass 4m moving with speed *u* collides with another body B of mass 2m, at rest. The collision is head on and elastic in nature. After the collision the fraction of energy lost by the colliding body A is :

(1) $\frac{1}{9}$ (2) $\frac{8}{9}$

(3) $\frac{4}{9}$ (4) $\frac{5}{9}$

PSC081

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NEET(UG) 2019 (Odisha)

9. An object flying in air with velocity $\left(20\hat{i}+25\hat{j}-12\hat{k}\right)$ suddenly breaks in two pieces whose masses are in the ratio 1 : 5. The smaller mass flies off with a velocity $\left(100\hat{i}+35\hat{j}+8\hat{k}\right)$. The velocity of the larger piece will be :-

(1) $4\hat{i}+23\hat{j}-16\hat{k}$ (2) $-100\hat{i}-35\hat{j}-8\hat{k}$

(3) $20\hat{i}+15\hat{j}-80\hat{k}$ (4) $-20\hat{i}-15\hat{j}-80\hat{k}$

PSC082

10. A particle of mass 5 m at rest suddenly breaks on its own into three fragments. Two fragments of mass m each move along mutually perpendicular direction with speed v each. The energy released during the process is :

(1) $\frac{3}{5}mv^2$ (2) $\frac{5}{3}mv^2$

(3) $\frac{3}{2}mv^2$ (4) $\frac{4}{3}mv^2$

PSC083

NEET(UG) 2020

11. Two particles of mass 5 kg and 10 kg respectively are attached to the two ends of a rigid rod of length 1 m with negligible mass. The centre of mass of the system from the 5 kg particle is nearly at a distance of :

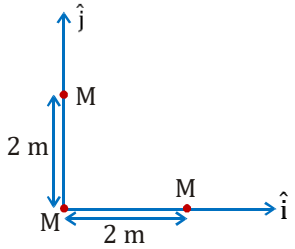
(1) 80 cm (2) 33 cm

(3) 50 cm (4) 67 cm

PSC084

NEET(UG) 2020 (COVID-19)

12. Three identical spheres, each of mass M, are placed at the corners of a right angle triangle with mutually perpendicular sides equal to 2 m (see figure). Taking the point of intersection of the two mutually perpendicular sides as the origin, find the position vector of centre of mass.



(1) $2(\hat{i}+\hat{j})$ (2) $(\hat{i}+\hat{j})$

(3) $\frac{2}{3}(\hat{i}+\hat{j})$ (4) $\frac{4}{3}(\hat{i}+\hat{j})$

PSC085

NEET(UG) 2022

13. Two objects of mass 10 kg and 20 kg respectively are connected to the two ends of a rigid rod of length 10 m with negligible mass. The distance of the center of mass of the system from the 10 kg mass is :

(1) $\frac{20}{3}$m (2) 10 m

(3) 5 m (4) $\frac{10}{3}$m

PSC086

14. A shell of mass m is at rest initially. It explodes into three fragments having mass in the ratio 2 : 2 : 1. If the fragments having equal mass fly off along mutually perpendicular directions with speed v, the speed of the third (lighter) fragment is

(1) $\sqrt{2}$ v (2) $2\sqrt{2}$ v

(3) $3\sqrt{2}$ v (4) v

PSC087

RE-NEET(UG) 2022

15. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R),

Assertion (A) :

When a fire cracker (rocket) explodes in mid air, its fragments fly in such a way that they continue moving in the same path, which the fire cracker would have followed, had it not exploded.

Reason (R) :

Explosion of cracker (rocket) occurs due to internal forces only and no external force acts for this explosion.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is not correct

(4) (A) is not correct but (R) is correct

PSC088

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RE-NEET(UG) 2023

16. A bullet of mass m hits a block of mass M elastically. The transfer of energy is the maximum, when :

(1) M = m (2) M = 2m

(3) M << m (4) M >> m

PSC097

17. Two particles A and B initially at rest, move towards each other under mutual force of attraction. At an instance when the speed of A is v and speed of B is 3v, the speed of centre of mass is :

(1) 2v (2) zero

(3) v (4) 4v

PSC098

NEET(UG) 2024

18. Two bodies A and B of same mass undergo completely inelastic one-dimensional collision. The body A moves with velocity v_1 while body B is at rest before collision. The velocity of the system after collision is v_2. The ratio $v_1 : v_2$ is :

(1) 1 : 2

(2) 2 : 1

(3) 4 : 1

(4) 1 : 4

PSC099

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	2	3	2	1	4	2	2	1	4	4	3	1	2	4
Question	16	17	18												
Answer	1	2	2												

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Exercise – III (Analytical Questions) Master Your Understanding

1. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : Two blocks of masses m_1 & m_2 are at rest. They are now moving towards each other under a mutual internal force. The velocity of centre of mass is zero.

Reason (R) : If no external force acts on the system, then velocity of centre of mass remains unchanged but can never be zero.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PSC089

2. Which of following statement is true.

(1) COM of a rigid body is always inside the body.

(2) Mass must be present physically at COM.

(3) COM of two particle system is closer to particle with greater mass.

(4) All above statements are true.

PSC091

3. Which of following statement is false.

(1) KE in an elastic collision between two bodies remains same before and after collision.

(2) There may or may not be physical contact between two bodies for collision to occur.

(3) KE of two object system remains conserved during collision in elastic collision.

(4) None of the above

PSC092

4. A projectile is projected at a speed u at an angle θ with horizontal. At the highest point projectile splits into two fragments of mass ratio 1 : 2. The smaller fragment coming to rest. Then consider following statements :-

(A) In a given time both parts will cover equal vertical displacement.

(B) COM of two fragments will follow the same path as the object would have followed if not exploded.

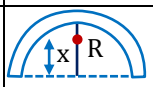
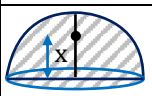
(C) Linear momentum is conserved in vertical direction only.

Which of the following is correct?

(1) A, B, C (2) A, B (3) B, C (4) A, C

PSC093

5. Match the following columns with respective positions of centre of mass -

Column – I		Column - II	
(A)	 uniform rod length = R	(P)	$x = \frac{3R}{8}$
(B)	 uniform thin semicircular ring of radius R	(Q)	$x = \frac{R}{2}$
(C)	 uniform semicircular disc of radius R	(R)	$x = \frac{4R}{3\pi}$
(D)	 uniform solid hemisphere of radius R	(S)	$x = \frac{2R}{\pi}$

(1) A : P, B : R, C : Q, D : S

(2) A : Q, B : P, C : R, D : S

(3) A : Q, B : S, C : R, D : P

(4) A : S, B : P, C : Q, D : R

PSC095

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5
Answer	3	3	3	2	3

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Important Notes

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Exercise - I (Conceptual Questions) Build Up Your Understanding

KINEMATICS OF ROTATIONAL MOTION

1. Which of the following pairs do not match :-
(1) rotational power–Joule/sec
(2) torque-Newton meter
(3) angular displacement-radian
(4) angular acceleration - radian/sec

PRM001

2. All the particles of a rigid rotating body move in a circular path when the axis of rotation:-
(1) passes through any point in the body
(2) is situated outside the body
(3) situated any where
(4) passes through the centre of mass

PRM002

3. On account of the earth rotating about its axis :-
(1) the linear velocity of objects at equator is greater than at other places.
(2) the angular velocity of objects at equator is more than that of objects at poles.
(3) the linear velocity of objects at all places at the earth is equal, but angular velocity is different.
(4) at all places the angular velocity and linear velocity are uniform.

PRM003

4. The quantity not involved directly in rotational motion of the body is
(1) moment of inertia (2) torque
(3) angular velocity (4) mass

PRM004

MOMENT OF INERTIA

5. The moment of inertia of a body about a given axis of rotation depends upon :-
(1) the distribution of mass
(2) distance of particle of body from the axis of rotation
(3) shape of the body
(4) all of the above

PRM005

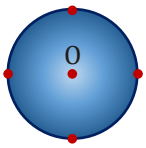
6. A fly wheel is so constructed that almost whole of its mass is concentrated at its rim, because:-
(1) it increases the moment of inertia of the fly-wheel
(2) it decreases the moment of inertia of the fly-wheel
(3) it increases the speed of the fly-wheel
(4) it increases the power of the fly-wheel

PRM006

7. The moment of inertia of a solid cylinder about its own axis is the same as its moment of inertia about an axis passing through its centre of gravity and perpendicular to its length. The relation between its length L and radius R is
(1) $L=\sqrt{2}R$ (2) $L=\sqrt{3}R$
(3) L = 3R (4) L = R

PRM007

8. Four similar point masses (each of mass m) are placed on the circumference of a disc of mass M and radius R. The M.I. of the system about the normal axis through the centre O will be:-



(1) $MR^2 + 4mR^2$ (2) $\frac{1}{2}MR^2 + 4mR^2$
(3) $MR^2 + \frac{8}{5}mR^2$ (4) none of these

PRM009

9. By the theorem of perpendicular axes, if a body be in X-Z-plane then :-
(1) $I_x - I_y = I_z$ (2) $I_x + I_z = I_y$
(3) $I_x + I_y = I_z$ (4) $I_y + I_z = I_x$

PRM010

10. The theorem of perpendicular axes is not applicable for determination of moment of inertia along the diameter, for which of the following body :-
(1) sphere (2) disc
(3) ring (4) blade

PRM011

11. The axis X and Z in the plane of a disc are mutually perpendicular and Y-axis is perpendicular to the plane of the disc. If the moment of inertia of the body about X and Y axes is respectively 30 kg m^2 and 40 kg m^2 then M.I. about Z-axis in kg m^2 will be :-
(1) 70 (2) 50 (3) 10 (4) Zero

PRM012

12. A solid sphere and a hollow sphere of the same mass have the same M.I. about their respective diameters. The ratio of their radii will be :-

(1) 1 : 2 (2) $\sqrt{3}:\sqrt{5}$ (3) $\sqrt{5}:\sqrt{3}$ (4) 5 : 4

PRM013

13. The moment of inertia of a square lamina about the perpendicular axis through its centre of mass is 20 kg-m^2. Then, its moment of inertia about an axis touching its side and in the plane of the lamina will be :-

(1) 10 kg-m^2 (2) 30 kg-m^2

(3) 40 kg-m^2 (4) 25 kg-m^2

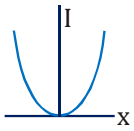
PRM014

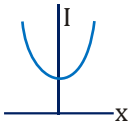
14. The M.I. of a thin rod of length ℓ about the perpendicular axis through its centre is I. The M.I. of the square structure made by four such rods about a perpendicular axis to the plane and through the centre will be :-

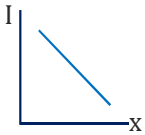
(1) 4I (2) 8I (3) 12I (4) 16I

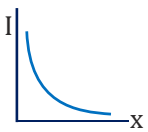
PRM015

15. The curve for the moment of inertia of a sphere of constant mass M versus distance of axis from its centre ?

(1) 

(2) 

(3) 

(4) 

PRM016

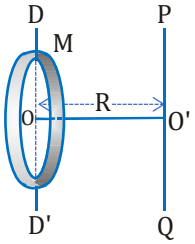
16. The moment of inertia of a ring of mass M and radius R about PQ axis will be :-

(1) MR^2

(2) $\frac{MR^2}{2}$

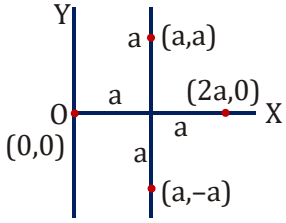
(3) $\frac{3}{2}$MR^2

(4) 2MR^2



PRM017

17. Four point masses (each of mass m) are arranged it the X-Y plane the moment of inertia of this array of masses about Y-axis is



(1) ma^2 (2) 2ma^2 (3) 4ma^2 (4) 6ma^2

PRM018

18. Which of the following has the highest moment of inertia when each of them has the same mass and the same radius ?

(1) a hollow sphere about one of its diameters

(2) a solid sphere about one of its diameters

(3) a disc about its central axis perpendicular to the plane of the disc

(4) a ring about its central axis perpendicular to the plane of the ring.

PRM019

19. If the radius of gyration of a solid disc of mass 10 kg about an axis is 0.40 m, then the moment of inertia of the disc about that axis is

(1) 1.6 kg m^2 (2) 3.2 kg m^2

(3) 6.4 kg m^2 (4) 9.5 kg m^2.

PRM020

20. If I_1, I_2 and I_3 are moments of inertia of solid sphere, hollow sphere and a ring of same mass and radius about geometrical axis, which of the following statement holds good ?

(1) $I_1 > I_2 > I_3$ (2) $I_3 > I_2 > I_1$

(3) $I_2 > I_1 > I_3$ (4) $I_2 > I_3 > I_1$

PRM021

21. Three thin uniform rods each of mass M and length L are placed along the three axis of a cartesian coordinate system with one end of each rod at the origin. The M. I. of the system about z-axis is

(1) $\frac{ML^2}{3}$ (2) $\frac{2ML^2}{3}$

(3) $\frac{ML^2}{6}$ (4) ML^2

PRM022

22. Four particles each of mass m are placed at the corners of a square of side length ℓ. The radius of gyration of the system about an axis perpendicular to the square and passing through centre is :–

(1) $\frac{\ell}{\sqrt{2}}$ (2) $\frac{\ell}{2}$

(3) ℓ (4) $\ell\sqrt{2}$

PRM023

23. The moment of inertia of a rod of mass M and length L about an axis passing through one edge and perpendicular to its length will be :–

(1) $\frac{ML^2}{12}$ (2) $\frac{ML^2}{6}$

(3) $\frac{ML^2}{3}$ (4) ML^2

PRM024

24. A circular disc is to be made by using iron and aluminium so that it acquired maximum moment of inertia about geometrical axis. It is possible with:

(1) aluminium at interior and iron surrounded to it.

(2) iron at interior and aluminium surrounded to it.

(3) using iron and aluminium layers in alternate order.

(4) sheet of iron is used at both external surface and aluminium sheet as internal layer.

PRM025

25. The moment of inertia in rotational motion is equivalent to :–

(1) angular velocity of linear motion

(2) mass of linear motion

(3) frequency of linear motion

(4) current

PRM026

26. Two rods each of mass m and length ℓ are joined at the centre to form a cross. The moment of inertia of this cross about an axis passing through the common centre of the rods and perpendicular to the plane formed by them, is :–

(1) $\frac{m\ell^2}{12}$ (2) $\frac{m\ell^2}{6}$ (3) $\frac{m\ell^2}{3}$ (4) $\frac{m\ell^2}{2}$

PRM027

27. The ratio of the radii of gyration of a circular disc about a tangential axis in the plane of the disc and of a circular ring of the same radius about a tangential axis in the plane of the ring is :–

(1) 2 : 1 (2) $\sqrt{5}:\sqrt{6}$

(3) 2 : 3 (4) $1:\sqrt{2}$

PRM028

28. Three particles, each of mass m are situated at the vertices of an equilateral triangle ABC of side ℓ (as shown in the figure).

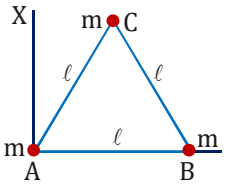
The moment of inertia of the system about a line AX perpendicular to AB and in the plane of ABC, will be :-

(1) $2\ m\ell^2$

(2) $\frac{5}{4}\ m\ell^2$

(3) $\frac{3}{2}\ m\ell^2$

(4) $\frac{3}{4}\ m\ell^2$



PRM029

29. A solid cylinder of mass 20kg has length 1 m and radius 0.2 m. Then its moment of inertia (in kg-m^2) about its geometrical axis is :–

(1) 0.8 kg-m^2 (2) 0.4 kg-m^2

(3) 0.2 kg-m^2 (4) 20.2 kg-m^2

PRM030

30. The moment of inertia of a circular ring (radius R, mass M) about an axis which passes through tangentially and perpendicular to its plane will be:–

(1) $\frac{MR^2}{2}$ (2) MR^2

(3) $\frac{3}{2}MR^2$ (4) $2MR^2$

PRM032

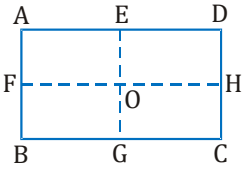
31. What is the moment of inertia of ring about its diameter ?

(1) MR^2 (2) $\frac{MR^2}{2}$

(3) $\frac{3}{4}MR^2$ (4) $\frac{5}{4}MR^2$

PRM033

32. In the rectangular lamina shown in the figure, AB = BC/2. The moment of inertia of the lamina is minimum along the axis passing through :-



(1) AB (2) BC (3) EG (4) FH

PRM034

33. Which of the following bodies of same mass and same radius has minimum moment of inertia ?

(1) Ring (2) Disc
(3) Hollow sphere (4) Solid sphere

PRM035

TORQUE, EQUILIBRIUM OF RIGID BODIES AND TOPPLING

34. A ring and a solid sphere of same mass and radius are rotating with the same angular velocity about their diametric axes then :-

(1) it is easier to stop the ring
(2) it is easier to stop the solid sphere
(3) it is equally difficult to stop both of them
(4) it is not possible to stop a rotating body

PRM036

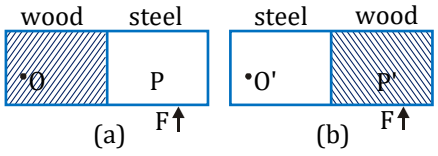
35. For rotational motion, the Newton's second law of motion is indicated by :-

(1) $\omega_2^2 = \omega_1^2 + 2\alpha\theta$ (2) $\vec{F} = \frac{d\vec{p}}{dt}$

(3) $\vec{\tau} = \frac{d\vec{L}}{dt}$ (4) $\vec{F}_{12} = \vec{F}_{21}$

PRM037

36. In the fig. (a) half of the meter scale is made of wood while the other half of steel. The wooden part is pivoted at O. A force F is applied at the end of steel part. In figure (b) the steel part is pivoted at O' and the same force is applied at the wooden end (In horizontal plane) :-



(1) more angular acceleration will be produced in (a)
(2) more angular acceleration will be produced in (b)
(3) same angular acceleration will be produced in both conditions
(4) information is incomplete

PRM038

37. The moment of inertia of a disc of radius 0.5 m about its geometric axis is 2kg-m^2. If a string is tied to its circumference and a force of 10 Newton is applied, the value of torque with respect to this axis will be :-

(1) 2.5 N-m (2) 5 N-m
(3) 10 N-m (4) 20 N-m

PRM039

38. In the above question, if the disc executes rotatory motion, its angular acceleration will be :-

(1) 2.5 rad/sec^2 (2) 5 rad/sec^2
(3) 10 rad/sec^2 (4) 20 rad/sec^2

PRM040

39. In the above question, the value of its angular velocity after 2 seconds will be :-

(1) 2.5 rad/sec (2) 5 rad/sec
(3) 10 rad/sec (4) 20 rad/sec

PRM041

40. In the above question, the change in angular momentum of disc in first 2 seconds (in Nm second) will be -

(1) 2.5 (2) 5
(3) 10 (4) 20

PRM042

41. In the above question, angular displacement of the disc, in first two second will be (in radian) :-

(1) 2.5 (2) 5 (3) 10 (4) 20

PRM043

42. A particle of mass m and radius of gyration K is rotating with an angular acceleration α. The torque acting on the particle is

(1) $\frac{1}{2}mK^2\alpha$ (2) $mK^2\alpha$

(3) mK^2/α (4) $\frac{1}{4}mK^2\alpha^2$

PRM044

43. The grinding stone of a flour mill is rotating at 600 rad/sec. For this, power of 1.2 k watt is used. the effective torque on stone in N–m will be :–
(1) 1 (2) 2 (3) 3 (4) 4
PRM045

44. A rigid body is rotating about an axis. To stop the rotation, we have to apply :-
(1) pressure (2) force
(3) momentum (4) torque
PRM046

45. A wheel has moment of inertia 5×10^{-3} kg m^2 and is making 20 rev/sec. The torque needed to stop it in 10 sec is ..... $\times 10^{-2}$ N-m :–
(1) 2π (2) 2.5π (3) 4π (4) 4.5π
PRM047

46. A wheel having moment of inertia 2 kg–m^2 about its vertical axis, rotates at the rate of 60 rpm about the axis. The torque which can stop the wheel's rotation in one minute would be :–
(1) $\frac{\pi}{12}$ N–m (2) $\frac{\pi}{15}$ N–m
(3) $\frac{\pi}{30}$ N–m (4) $\frac{2\pi}{15}$ N–m
PRM048

47. A constant torque acting on a uniform circular wheel changes its angular momentum from A_0 to 4 A_0 in 4 seconds. The magnitude of this torque is :–
(1) $\frac{3A_0}{4}$ (2) A_0 (3) $4A_0$ (4) 12 A_0
PRM049

48. When constant torque is acting on a body then :–
(1) body maintain its state or moves in straight line with same velocity
(2) acquire linear acceleration
(3) acquire angular acceleration
(4) rotates with a constant angular velocity
PRM051

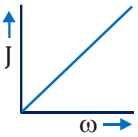
49. If torque on a body is zero, then which is conserved:
(1) force
(2) linear momentum
(3) angular momentum
(4) angular impulse
PRM052

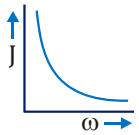
50. If I = 50 kg-m^2, then how much torque will be applied to stop it in 10 sec. Its initial angular speed is 20 rad/sec. :–
(1) 100 N-m (2) 150 N-m
(3) 200 N-m (4) 250 N-m
PRM053

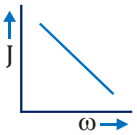
51. A force $\vec{F} = 2\hat{i} - 3\hat{k}$ acts on a particle at $\vec{r} = 0.5\hat{j} - 2\hat{k}$. The torque τ acting on the particle relative to a point with co-ordinates (2.0 m, 0, –3.0 m) is
(1) $(-3.0\hat{i} - 4.5\hat{j} - \hat{k})$N – m
(2) $(3\hat{i} + 6\hat{j} - \hat{k})$N – m
(3) $(-20\hat{i} + 4.0\hat{j} + \hat{k})$N – m
(4) $(-1.5\hat{i} - 4.0\hat{j} - \hat{k})$N – m
PRM054

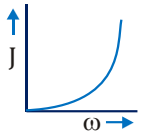
52. If a ladder is not in balance against a smooth vertical wall, then it can be made in balance by :-
(1) Decreasing the length of ladder
(2) Increasing the length of ladder
(3) Increasing the angle of inclination
(4) Decreasing the angle of inclination
PRM055

ANGULAR MOMENTUM

53. The graph between the angular momentum J and angular velocity ω for a body will be :-
(1) 

(2) 

(3) 

(4) 

PRM056

54. When a mass is rotating in a plane about a fixed point, its angular momentum is directed along
(1) the radius
(2) the tangent to orbit
(3) line at an angle of 45° to the plane of rotation
(4) the axis of rotation
PRM057

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55. A ring of mass 10 kg and diameter 0.4 meter is rotating about its geometrical axis at 1200 rotations per minute. Its moment of inertia and angular momentum will be respectively:-
(1) 0.4 kg-m^2 and 50.28 J-s
(2) 0.4 kg-m^2 and 0.4 J-s
(3) 50.28 kg-m^2 and 0.4 J-s
(4) 0.4 kg-m^2 and zero

PRM058

56. The rotational kinetic energy of two bodies of moments of inertia 9 kg-m^2 and 1kg-m^2 are same. The ratio of their angular momentum is :-
(1) 3 : 1 (2) 1 : 3 (3) 9 : 1 (4) 1 : 9

PRM059

57. A fan of moment of inertia 0.6 kgm^2 is to run upto a working speed of 0.5 revolution per second. Indicate the correct value of the angular momentum of the fan
(1) $0.6\pi \text{ kg} \times \frac{\text{metre}^2}{\text{sec}}$ (2) $6\pi \text{ kg} \times \frac{\text{metre}^2}{\text{sec}}$
(3) $3\pi \text{ kg} \times \frac{\text{metre}^2}{\text{sec}}$ (4) $\frac{\pi}{6}\text{kg} \times \frac{\text{metre}^2}{\text{sec}}$

PRM060

58. A body of mass m is moving with constant velocity parallel to x-axis. The angular momentum with respect to origin :–
(1) increases with time
(2) decreases with time
(3) does not change
(4) none of above

PRM061

59. In an orbital motion, the angular momentum vector is :–
(1) along the radius vector
(2) parallel to the linear momentum
(3) in the orbital plane
(4) perpendicular to the orbital plane

PRM062

60. Two bodies have their moments of inertia I and 2I respectively about their axis of rotation. If their kinetic energies of rotation are equal, their angular momentum will be in the ratio :–
(1) 1 : 2 (2) $\sqrt{2}:1$
(3) $1:\sqrt{2}$ (4) 2 : 1

PRM063

61. The rotational kinetic energy of a body is K_{rot} and its moment of inertia is I. The angular momentum of body is
(1) IK_{rot} (2) $2\sqrt{IK_{rot}}$
(3) $\sqrt{2IK_{rot}}$ (4) $2IK_{rot}$

PRM064

62. A body of mass 10 kg and radius of gyration 0.1 m is rotating about an axis. If angular speed is 10 rad/s, then angular momentum will be :-
(1) 1 kg m^2/s (2) 0.1 kg m^2/s
(3) 100 kg m^2/s (4) 10 kg m^2/s

PRM065

63. A particle of mass 1.0 kg is rotating on a circular path of diameter 2.0 m at the rate of 10 rotations in 31.4s. The angular momentum of the body,(in kgm^2/s) is :-
(1) 1.0 (2) 1.5 (3) 2.0 (4) 4.0

PRM066

64. A particle of mass m is rotating in a plane in a circular path of radius r. Its angular momentum is L. The centripetal force acting on the particle is
(1) $\frac{L^2}{mr}$ (2) $\frac{L^2m}{r}$ (3) $\frac{L^2}{mr^2}$ (4) $\frac{L^2}{mr^3}$

PRM067

CONSERVATION OF ANGULAR MOMENTUM

65. A stone attached to one end of string is revolved around a stick so that the string winds upon the stick and gets shortened. What is conserved ?
(1) angular momentum
(2) linear momentum
(3) kinetic energy
(4) none of the above

PRM068

66. A rotating table completes one rotation in 10 sec. and its moment of inertia is 100 kg-m^2. A person of 50 kg. mass stands at the centre of the rotating table. If the person moves 2m from the centre, the angular velocity of the rotating table (in rad/sec). will be:
(1) $\frac{2\pi}{30}$ (2) $\frac{20\pi}{30}$ (3) $\frac{2\pi}{3}$ (4) 2π

PRM069

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67. On melting of ice on the pole of the earth, its moment of inertia will :-
(1) increase
(2) decrease
(3) remain unchanged
(4) none of these

PRM070

68. If the earth loses its atmosphere suddenly, then the duration of day will :-
(1) increase
(2) decrease
(3) remain unchanged
(4) nothing can be definitely said

PRM071

69. A wheel is rotating about its axis at a constant angular velocity. If suddenly an object sticks to it on the rim, then its M.I. will :-
(1) increase
(2) decrease
(3) remain unchanged
(4) nothing can be said

PRM072

70. In the above question, the angular velocity will :–
(1) increase
(2) decrease
(3) not change
(4) nothing can be said

PRM073

71. An ant is sitting at the edge of a rotating disc. If the ant reaches the other end, after moving along the diameter, the angular velocity of the disc will:-
(1) remain constant
(2) first decreases and then increases
(3) first increases, then decrease
(4) Increase continuously

PRM074

72. The angular momentum of body remains conserved if :–
(1) applied force on body is zero.
(2) applied torque on body is zero.
(3) applied force on body is constant.
(4) applied torque on body is constant.

PRM075

73. A person is standing on the edge of a circular platform, which is moving with constant angular speed about an axis passing through its centre and perpendicular to the plane of platform. If person is moving along any radius towards axis of rotation then the angular velocity will :–
(1) decrease
(2) remain unchanged
(3) increase
(4) data is insufficient

PRM076

74. A thin circular ring of mass M and radius 'r' is rotating about its axis with a constant angular velocity ω. Four objects each of mass m, are kept gently to the opposite ends of two perpendicular diameters of the ring. The angular velocity of the ring will be:–
(1) $\frac{M\omega}{4m}$ (2) $\frac{M\omega}{M+4m}$
(3) $\frac{(M+4m)\omega}{M}$ (4) $\frac{(M+4m)\omega}{M+4m}$

PRM077

75. A round disc of moment of inertia I_2 about its axis perpendicular to its plane and passing through its centre is placed over another disc of moment of inertia I_1 rotating with an angular velocity ω about the same axis. The final angular velocity of the combination of discs is :–
(1) ω (2) $\frac{I_1\omega}{I_1+I_2}$
(3) $\frac{(I_1+I_2)\omega}{I_1}$ (4) $\frac{I_2\omega}{I_1+I_2}$

PRM078

76. Rate of change of angular momentum with respect to time is proportional to :–
(1) angular velocity
(2) angular acceleration
(3) moment of inertia
(4) torque

PRM079

77. A small steel sphere of mass m is tied to a string of length r and is whirled in a horizontal circle with a uniform angular velocity 2ω. The string is suddenly pulled, so that radius of the circle is halved. The new angular velocity will be
(1) 2ω (2) 4ω (3) 6ω (4) 8ω

PRM080

78. If the earth were to suddenly contract to half its present size, without any change in its mass, the duration of the new day will be
(1) 18 hours (2) 30 hours
(3) 6 hours (4) 12 hours
PRM081

79. A stone is attached to the end of a string and whirled in horizontal circle, then :–
(1) its linear and angular momentum are constant
(2) only linear momentum is constant
(3) its angular momentum is constant but linear momentum is variable
(4) both are variable
PRM082

PURE ROLLING AND KINETIC ENERGY

80. A thin rod of length L is suspended from one end and rotated with n rotations per second. The rotational kinetic energy of the rod will be :
(1) $2mL^2\pi^2n^2$ (2) $\frac{1}{2}mL^2\pi^2n^2$
(3) $\frac{2}{3}mL^2\pi^2n^2$ (4) $\frac{1}{6}mL^2\pi^2n^2$
PRM083

81. The rotational kinetic energy of a body is E. In the absence of external torque, if mass of the body is halved and radius of gyration doubled, then its rotational kinetic energy will be :-
(1) 0.5E (2) 0.25E (3) E (4) 2E
PRM084

82. A particle of mass m is describing a circular path of radius r with uniform speed. If L is the angular momentum of the particle (about the axis of the circle), then the kinetic energy of the particle is
(1) $\frac{L^2}{mr^2}$ (2) mr^2L (3) $\frac{L^2}{2mr^2}$ (4) $\frac{L^2r^2}{m}$
PRM085

83. A particle performs uniform circular motion with an angular momentum L. If the frequency of particle's motion is doubled and its kinetic energy halved, the angular momentum becomes
(1) 2L (2) 4L (3) $\frac{L}{2}$ (4) $\frac{L}{4}$
PRM086

84. A flywheel is making $\frac{3000}{\pi}$ revolutions per minute about its axis. If the moment of inertia of the flywheel about that axis is 400 kgm^2, its rotational kinetic energy is
(1) 2×10^6 J (2) 3×10^3 J
(3) $500\pi^2$ J (4) 12×10^3 J
PRM087

85. A ring is rolling without slipping. Its energy of translation is E. Its total kinetic energy will be :-
(1) E (2) 2E (3) 3E (4) 4E
PRM088

86. A thin hollow cylinder open at both ends slides without rotating and then rolls without slipping with the same speed. The ratio of the kinetic energies in the two cases is
(1) 1 : 1 (2) 1 : 2 (3) 2 : 1 (4) 1 : 4
PRM089

87. A solid sphere of mass M and radius R rolls on a horizontal surface without slipping. The ratio of rotational K.E. to total K.E. is :–
(1) $\frac{1}{2}$ (2) $\frac{3}{7}$ (3) $\frac{2}{7}$ (4) $\frac{2}{10}$
PRM090

88. A disc is rolling on an inclined plane without slipping then what fraction of its total energy will be in form of rotational kinetic energy :–
(1) 1 : 3 (2) 1 : 2 (3) 2 : 7 (4) 2 : 5
PRM091

89. A wheel is rolling along the ground with a speed of 2 m/s. The magnitude of the linear velocity of the points at the extremities of the horizontal diameter of the wheel is equal to
(1) $2\sqrt{10}$ m/s (2) $2\sqrt{3}$ m/s
(3) $2\sqrt{2}$ m/s (4) 2 m/s
PRM092

90. If rotational kinetic energy is 50% of total kinetic energy then the body will be :–
(1) ring (2) cylinder
(3) hollow sphere (4) solid sphere
PRM093

AIM: FREE EDUCATION TO ALL APUL

91. A ball rolls without slipping. The radius of gyration of the ball about an axis passing through its centre of mass is K. If radius of the ball be R, then the fraction of total energy associated with its rotational energy will be :–

(1) $\frac{K^2+R^2}{R^2}$ (2) $\frac{K^2}{R^2}$

(3) $\frac{K^2}{K^2+R^2}$ (4) $\frac{R^2}{K^2+R^2}$

PRM094

92. If the angular velocity of a body rotating about an axis is doubled and its moment of inertia halved, the rotational kinetic energy will changed by a factor of :-

(1) 4 (2) 2 (3) 1 (4) 1/2

PRM095

93. If a sphere is rolling, the ratio of its rotational energy to the total kinetic energy is given by

(1) 7 : 10 (2) 2 : 5 (3) 10 : 5 (4) 2 : 7

PRM096

94. A person, with outstretched arms, is spinning on a rotating stool. He suddenly brings his arms down to his sides. Which of the following is true about system kinetic energy K and angular momentum L ?

(1) Both K and L increase.

(2) Both K and L remain unchanged.

(3) K remains constant, L increases.

(4) K increases but L remains constant.

PRM097

95. A uniform thin ring of mass 0.4 kg rolls without slipping on a horizontal surface with a linear velocity of 10 cm/s. The kinetic energy of the ring is :-

(1) 4×10^{-3} J (2) 4×10^{-2} J

(3) 2×10^{-3} J (4) 2×10^{-2} J

PRM098

96. A body is rotating with angular momentum L. If I is its moment of inertia about the axis of rotation, its kinetic energy of rotation is :-

(1) $\frac{1}{2}IL^2$ (2) $\frac{1}{2}IL$

(3) $\frac{1}{2}(I^2/L)$ (4) $\frac{L^2}{2I}$

PRM099

ROLLING MOTION ON AN INCLINED PLANE

97. A disc rolls down a plane of length L and inclined at angle θ, without slipping. Its velocity on reaching the bottom will be :-

(1) $\sqrt{\frac{4gL\sin\theta}{3}}$ (2) $\sqrt{\frac{2gL\sin\theta}{3}}$

(3) $\sqrt{\frac{10gL\sin\theta}{7}}$ (4) $\sqrt{4gL\sin\theta}$

PRM100

98. A spherical shell and a solid cylinder of same radius rolls down an inclined plane. The ratio of their accelerations will be:-

(1) 15 : 14 (2) 9 : 10

(3) 2 : 3 (4) 3 : 5

PRM101

99. A ring takes time t_1 and t_2 for sliding down and rolling down an inclined plane of length L respectively for reaching the bottom. The ratio of t_1 and t_2 is :-

(1) $\sqrt{2}:1$ (2) $1:\sqrt{2}$

(3) 1 : 2 (4) 2 : 1

PRM102

100. A solid sphere is rolling down on inclined plane from rest and a rectangular block of same mass is also slipping down simultaneously from rest on a similar smooth inclined plane.

(1) both of them will reach the bottom simultaneously

(2) the sphere will reach the bottom first

(3) the rectangular block will reach the bottom first.

(4) depends on density of material

PRM103

101. Calculate the ratio of the times taken by a uniform solid sphere and a disc of the same mass and the same diameter to roll down through the same distance from rest on a inclined plane.

(1) 15 : 14 (2) $\sqrt{15}:\sqrt{14}$

(3) $15^2 : 14^2$ (4) $\sqrt{14}:\sqrt{15}$

PRM104

102. A body of mass m slides down an incline and reaches the bottom with a velocity v. If the same mass were in the form of a ring which rolls down this incline, the velocity of the ring at the bottom would have been

(1) v (2) $v\sqrt{2}$

(3) $\frac{v}{\sqrt{2}}$ (4) $\left(\sqrt{\frac{2}{5}}\right)v$

PRM105

103. When a sphere of moment of inertia I rolls down on an inclined plane the percentage of total energy which is rotational, is approximately

(1) 28 % (2) 72 %

(3) 100 % (4) none of these

PRM106

104. When a body starts to roll on an inclined plane, its potential energy is converted into

(1) translational kinetic energy only

(2) translational and rotational kinetic energy

(3) rotational energy only

(4) none

PRM107

105. A solid cylinder of mass M and radius R rolls without slipping down an inclined plane of length L and height h. What is the speed of its centre of mass when the cylinder reaches its bottom :–

(1) $\sqrt{2gh}$ (2) $\sqrt{\frac{3}{4}gh}$

(3) $\sqrt{\frac{4}{3}gh}$ (4) $\sqrt{4gh}$

PRM108

106. Which of the following is true about the angular momentum of a cylinder rolling down a slope without slipping ?

(1) Its magnitude changes but the direction remains same

(2) both magnitude and direction change

(3) only the direction change

(4) neither change

PRM109

107. A sphere and a disc of same radii and mass are rolling on an inclined plane without slipping. a_s & a_d are acceleration and g is acceleration due to gravity. Then which statement is correct ?

(1) $a_s > a_d > g$ (2) $g > a_s > a_d$

(3) $a_s > g > a_d$ (4) $a_d > a_s > g$

PRM110

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	2	1	4	4	1	2	2	2	1	3	3	3	4	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	4	4	1	2	2	1	3	1	2	2	2	2	2	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	4	4	2	3	2	2	1	2	3	2	2	2	4	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	1	3	3	1	4	3	1	4	1	1	1	3	4	3
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	3	1	3	4	1	1	1	2	1	2	3	2	3	2	2
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	4	4	3	3	3	1	3	4	1	2	2	3	1	3	1
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	3	2	4	4	1	4	1	2	2	3	4	3	1	2	3
Question	106	107													
Answer	1	2													

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. A solid cylinder of mass 50 kg and radius 0.5 m is free to rotate about the horizontal axis. A massless string is wound round the cylinder with one end attached to it and other hanging freely. Tension in the string required to produce an angular acceleration of 2 revolutions/s² is :-

(1) 25 N (2) 50 N
(3) 78.5 N (4) 157 N

PRM123

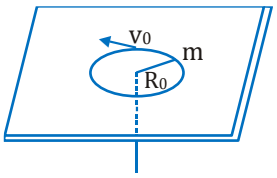
2. The ratio of the accelerations for a solid sphere (mass 'm' and radius 'R') rolling down an incline of angle 'θ' without slipping and slipping down the incline without rolling is :-

(1) 5 : 7 (2) 2 : 3 (3) 2 : 5 (4) 7 : 5

PRM124

AIPMT 2015

3. A mass m moves in a circle on a smooth horizontal plane with velocity v_0 at a radius R_0. The mass is attached to a string which passes through a smooth hole in the plane as shown.



The tension in the string is increased gradually and finally m moves in a circle of radius $\frac{R_0}{2}$. The final value of the kinetic energy is :-

(1) $\frac{1}{4}mv_0^2$ (2) $2mv_0^2$ (3) $\frac{1}{2}mv_0^2$ (4) mv_0^2

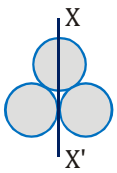
PRM125

4. A rod of weight W is supported by two parallel knife edges A and B and is in equilibrium in a horizontal position. The knives are at a distance d from each other. The centre of mass of the rod is at distance x from A. The normal reaction on A is :-

(1) $\frac{Wd}{x}$ (2) $\frac{W(d-x)}{x}$
(3) $\frac{W(d-x)}{d}$ (4) $\frac{Wx}{d}$

PRM126

5. Three identical spherical shells, each of mass m and radius r are placed as shown in figure. Consider an axis XX' which is touching to two shells and passing through diameter of third shell. Moment of inertia of the system consisting of these three spherical shells about XX' axis is :-



(1) $3\ mr^2$ (2) $\frac{16}{5}mr^2$
(3) $4\ mr^2$ (4) $\frac{11}{5}mr^2$

PRM127

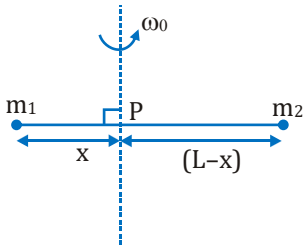
Re-AIPMT 2015

6. An automobile moves on a road with a speed of 54 km/h. The radius of its wheels is 0.45 m and the moment of inertia of the wheel about its axis of rotation is 3 kg-m². If the vehicle is brought to rest in 15s, the magnitude of average torque transmitted by its brakes to wheel is :-

(1) 2.86 kg-m²/s² (2) 6.66 kg-m²/s²
(3) 8.58 kg-m²/s² (4) 10.86 kg-m²/s²

PRM128

7. Point masses m_1 and m_2 are placed at the opposite ends of a rigid rod of length L, and negligible mass. The rod is to be set rotating about an axis perpendicular to it. The position of point P on this rod through which the axis should pass so that the work required to set the rod rotating with angular velocity ω_0 is minimum, is given by :-



(1) $x=\frac{m_2L}{m_1+m_2}$ (2) $x=\frac{m_1L}{m_1+m_2}$
(3) $x=\frac{m_1}{m_2}L$ (4) $x=\frac{m_2}{m_1}L$

PRM129

8. A force $\vec{F}=\alpha\hat{i}+3\hat{j}+6\hat{k}$ is acting at a point $\vec{r}=2\hat{i}-6\hat{j}-12\hat{k}$. The value of α for which angular momentum about origin is conserved is :-

(1) 1 (2) –1 (3) 2 (4) zero

PRM130

NEET-I 2016

9. From a disc of radius R and mass M, a circular hole of diameter R, whose rim passes through the centre is cut. What is the moment of inertia of the remaining part of the disc about a perpendicular axis, passing through the centre ?

(1) 15 MR²/32 (2) 13 MR²/32

(3) 11 MR²/32 (4) 9 MR²/32

PRM131

10. A uniform circular disc of radius 50 cm at rest is free to turn about an axis which is perpendicular to its plane and passes through its centre. It is subjected to a torque which produces a constant angular acceleration of 2.0 rad/s². Its net acceleration in m/s² at the end of 2.0 s is approximately :

(1) 8.0 (2) 7.0 (3) 6.0 (4) 3.0

PRM132

11. A disc and a sphere of same radius but different masses roll off on two inclined planes of the same altitude and length. Which one of the two objects gets to the bottom of the plane first ?

(1) Disk

(2) Sphere

(3) Both reach at the same time

(4) Depends on their masses

PRM133

NEET-II 2016

12. Two rotating bodies A and B of masses m and 2m with moments of inertia I_A and I_B ($I_B > I_A$) have equal kinetic energy of rotation. If L_A and L_B be their angular momenta respectively, then :-

(1) $L_B > L_A$ (2) $L_A > L_B$

(3) $L_A = \frac{L_B}{2}$ (4) $L_A = 2L_B$

PRM134

13. A solid sphere of mass m and radius R is rotating about its diameter. A solid cylinder of the same mass and same radius is also rotating about its geometrical axis with an angular speed twice that of the sphere. The ratio of their kinetic energies of rotation (E_{sphere} / $E_{cylinder}$) will be :-

(1) 1 : 4 (2) 3 : 1

(3) 2 : 3 (4) 1 : 5

PRM135

14. A light rod of length ℓ has two masses m_1 and m_2 attached to its two ends. The moment of inertia of the system about an axis perpendicular to the rod and passing through the centre of mass is :-

(1) $(m_1 + m_2)\ell^2$ (2) $\sqrt{m_1m_2}\ell^2$

(3) $\frac{m_1m_2}{m_1+m_2}\ell^2$ (4) $\frac{m_1+m_2}{m_1m_2}\ell^2$

PRM136

NEET(UG) 2017

15. A rope is wound around a hollow cylinder of mass 3 kg and radius 40 cm. What is the angular acceleration of the cylinder if the rope is pulled with a force of 30 N ?

(1) 0.25 rad/s² (2) 25 rad/s²

(3) 5 m/s² (4) 25 m/s²

PRM137

16. Two discs of same moment of inertia rotating about their regular axis passing through centre and perpendicular to the plane of disc with angular velocities ω_1 and ω_2. They are brought into contact face to face coinciding the axis of rotation. The expression for loss of energy during this process is:-

(1) $\frac{1}{4}I(\omega_1-\omega_2)^2$ (2) $I(\omega_1-\omega_2)^2$

(3) $\frac{I}{8}(\omega_1-\omega_2)^2$ (4) $\frac{1}{2}I(\omega_1+\omega_2)^2$

PRM138

17. Which of the following statements are **correct** ?
(a) Centre of mass of a body always coincides with the centre of gravity of the body
(b) Centre of mass of a body is the point at which the total gravitational torque on the body is zero
(c) A couple on a body produce both translational and rotation motion in a body
(d) Mechanical advantage greater than one means that small effort can be used to lift a large load
(1) (a) and (b) (2) (b) and (c)
(3) (c) and (d) (4) (b) and (d)

PRM139

NEET (UG) 2018

18. Three objects, A : (a solid sphere), B : (a thin circular disk) and C = (a circular ring), each have the same mass M and radius R. They all spin with the same angular speed ω about their own symmetry axes. The amounts of work (W) required to bring them to rest, would satisfy the relation :-
(1) $W_C > W_B > W_A$ (2) $W_A > W_B > W_C$
(3) $W_B > W_A > W_C$ (4) $W_A > W_C > W_B$

PRM140

19. The moment of the force, $\vec{F} = 4\hat{i} + 5\hat{j} - 6\hat{k}$ at (2, 0, –3), about the point (2, –2, –2), is given by:-
(1) $-8\hat{i} - 4\hat{j} - 7\hat{k}$ (2) $-4\hat{i} - \hat{j} - 8\hat{k}$
(3) $-7\hat{i} - 8\hat{j} - 4\hat{k}$ (4) $-7\hat{i} - 4\hat{j} - 8\hat{k}$

PRM141

20. A solid sphere is in rolling motion. In rolling motion a body possesses translational kinetic energy (K_t) as well as rotational kinetic energy (K_r) simultaneously. The ratio $K_t : (K_t + K_r)$ for the sphere is
(1) 7 : 10 (2) 5 : 7 (3) 10 : 7 (4) 2 : 5

PRM142

21. A solid sphere is rotating freely about its symmetry axis in free space. The radius of the sphere is increased keeping its mass same. Which of the following physical quantities would remain constant for the sphere ?
(1) Angular velocity
(2) Moment of inertia
(3) Rotational kinetic energy
(4) Angular momentum

PRM143

NEET(UG) 2019

22. A disc of radius 2m and mass 100 kg rolls on a horizontal floor. Its centre of mass has speed of 20 cm/s. How much work is needed to stop it ?
(1) 3J (2) 30 kJ (3) 2 J (4) 1 J

PRM144

23. A solid cylinder of mass 2 kg and radius 4 cm is rotating about its axis at the rate of 3 rpm. The torque required to stop after 2π revolutions is :
(1) 2×10^{-6} N m (2) 2×10^{-3} N m
(3) 12×10^{-4} N m (4) 2×10^{6} N m

PRM145

NEET(UG) 2019 (Odisha)

24. A solid cylinder of mass 2 kg and radius 50 cm rolls up an inclined plane of angle inclination 30°. The centre of mass of cylinder has speed of 4 m/s. The distance travelled by the cylinder on the incline surface will be : (Take g = 10 m/s^2)
(1) 2.2 m (2) 1.6 m
(3) 1.2 m (4) 2.4 m

PRM146

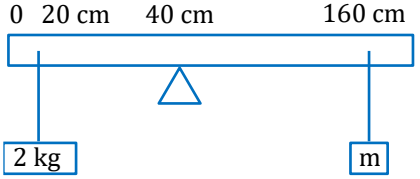
NEET(UG) 2020

25. Find the torque about the origin when a force of $3\hat{j}$ N acts on a particle whose position vector is $2\hat{k}$ m :
(1) $6\hat{k}$ Nm (2) $6\hat{i}$ Nm
(3) $6\hat{j}$ Nm (4) $-6\hat{i}$ Nm

PRM147

NEET(UG) 2021

26. A uniform rod of length 200 cm and mass 500 g is balanced on a wedge placed at 40 cm mark. A mass of 2 kg is suspended from the rod at 20 cm and another unknown mass 'm' is suspended from the rod at 160 cm mark as shown in the figure. Find the value of 'm' such that the rod is in equilibrium. (g= 10 m/s^2)



(1) $\frac{1}{2}$kg (2) $\frac{1}{3}$kg
(3) $\frac{1}{6}$kg (4) $\frac{1}{12}$kg

PRM148

27. From a circular ring of mass 'M' and radius 'R' an arc corresponding to a 90° sector is removed. The moment of inertia of the remaining part of the ring about an axis passing through the centre of the ring and perpendicular to the plane of the ring is 'K' times 'MR²'. Then the value of 'K' is :

(1) $\frac{3}{4}$ (2) $\frac{7}{8}$ (3) $\frac{1}{4}$ (4) $\frac{1}{8}$

PRM149

NEET(UG) 2022

28. The ratio of the radius of gyration of a thin uniform disc about an axis passing through its centre and normal to its plane to the radius of gyration of the dis about its diameter is :

(1) $\sqrt{2}:1$ (2) 4 : 1 (3) $1:\sqrt{2}$ (4) 2 : 1

PRM150

RE-NEET(UG) 2022

29. An energy of 484 J is spent in increasing the speed of a flywheel from 60 rpm to 360 rpm. The moment of inertia of the flywheel is :

(1) 0.7 kg-m² (2) 3.22 kg-m²
(3) 30.8 kg-m² (4) 0.07 kg-m²

PRM151

NEET(UG) 2023

30. The ratio of radius of gyration of a solid sphere of mass M and radius R about its own axis to the radius of gyration of the thin hollow sphere of same mass and radius about its axis is :-

(1) $\sqrt{5}:\sqrt{3}$ (2) $\sqrt{2}:\sqrt{5}$
(3) $\sqrt{5}:\sqrt{2}$ (4) $\sqrt{3}:\sqrt{5}$

PRM164

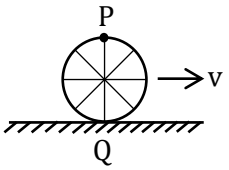
NEET(UG) 2024

31. The moment of inertia of a thin rod about an axis passing through its mid point and perpendicular to the rod is 2400 g cm². The length of the 400 g rod is nearly :

(1) 8.5 cm (2) 17.5 cm
(3) 20.7 cm (4) 72.0 cm

PRM165

32. A wheel of a bullock cart is rolling on a level road as shown in the figure below. If its linear speed is v in the direction shown, which one of the following options is correct (P and Q are any highest and lowest points on the wheel, respectively) ?

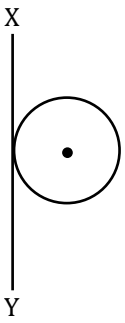


(1) Point P moves slower than point Q.
(2) Point P moves faster than point Q.
(3) Both the points P and Q move with equal speed.
(4) Point P has zero speed.

PRM166

RE-NEET(UG) 2024

33. The radius of gyration of a solid sphere of mass 5 kg about XY is 5m as shown in figure. The radius of the sphere is $\frac{5x}{\sqrt{7}}$ m, then the value of x is:



(1) 5 (2) $\sqrt{2}$ (3) $\sqrt{3}$ (4) $\sqrt{5}$

PRM167

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	1	2	3	3	2	1	2	2	1	2	1	4	3	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	4	1	4	2	4	1	1	4	4	4	1	1	1	4
Question	31	32	33												
Answer	1	2	4												

Exercise - III (Analytical Questions) Master Your Understanding

1. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : Moment of inertia of a particle is a scalar quantity.

Reason (R) : Moment of inertia of a particle about any axis is given by $m(\vec{r}\cdot\vec{r})$ where m is mass of particle and $\vec{r}$ is position of particle w.r.t. axis.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PRM152

2. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : Moment of inertia of a disc about an axis perpendicular to its plane passing through its centre is $\frac{mR^2}{2}$

Reason (R) : Moment of inertia of a disc about any axis passing through its plane remains equal.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

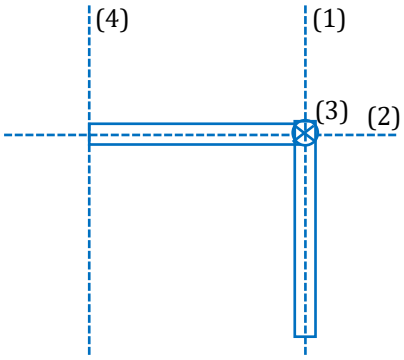
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PRM153

3. In diagram axis (3) is perpendicular to plane. Both rods are mass m and length 'ℓ' then moment of inertial of system



i	About axis (1)	P	$\frac{m\ell^2}{3}$
ii	About axis (2)	Q	$\frac{4m\ell^2}{3}$
iii	About axis (3)	R	$\frac{2m\ell^2}{3}$
iv	About axis (4)		

(1) i – P, ii – P, iii – R, iv – Q

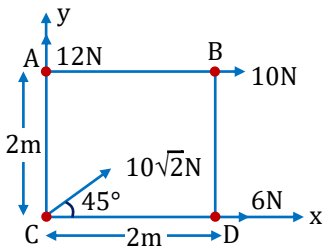
(2) i – Q, ii – Q, iii – P, iv – R

(3) i – R, ii – R, iii – P, iv – Q

(4) i – P, ii – R, iii – Q, iv – P

PRM155

4. Value of net torque about given points.



i	Torque about A	P	$-64\hat{k}$
ii	Torque about B	Q	$32\hat{k}$
iii	Torque about C	R	$-12\hat{k}$
iv	Torque about D	S	$-20\hat{k}$

(1) i – P, ii – Q, iii – R, iv – S

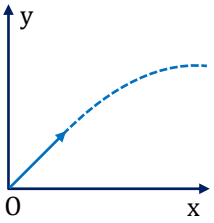
(2) i – Q, ii – P, iii – P, iv – S

(3) i – Q, ii – R, iii – S, iv – P

(4) i – R, ii – S, iii – P, iv – S

PRM156

5. A particle is projected from point 'O' having mass 'm' as shown in figure, moving on projectile path



i	Torque about 'O'	P	Zero
ii	Torque about x-axis	Q	Constant non zero
iii	Angular momentum about O	R	Increasing
iv	Angular momentum about x-axis	S	Decreasing

(1) i – P, ii – Q, iii – R, iv – S
(2) i – R, ii – P, iii – R, iv – P
(3) i – Q, ii – R, iii – R, iv – P
(4) i – P, ii – R, iii – R, iv – S

PRM157

6. Four identical rods each of mass m and length 'ℓ' are joined to form a rigid square, centre is at origin and sides are parallel to x and y axis then moment of inertia about

i	An axis parallel to z-axis and passing through a corner	P	$\frac{5}{3}m\ell^2$
ii	One side	Q	$\frac{2}{3}m\ell^2$
iii	The x-axis	R	$\frac{4}{3}m\ell^2$
iv	The z-axis passing through centre	S	$\frac{10}{3}m\ell^2$

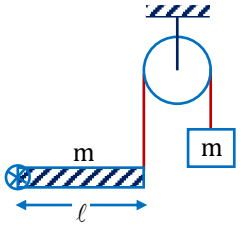
(1) i – S, ii – P, iii – Q, iv – R
(2) i – P, ii – Q, iii – R, iv – S
(3) i – Q, ii – R, iii – p, iv – S
(4) i – S, ii – P, iii – R, iv – Q

PRM158

7. If lamina is in x-y plane then from theorem of perpendicular axis
(A) $I_x - I_y = I_z$
(B) $I_x - I_z = I_y$
(C) $I_x + I_y = I_z$
(D) $I_y + I_z = I_x$
True statements is/are –
(1) B and C
(2) Only D
(3) Only C
(4) C and D

PRM160

8. Hinge is smooth and system is released from rest. Select correct statement –



(A) Acceleration of block is g/2 down ward
(B) Acceleration of block is 3g/8 down ward
(C) Tension in string is mg/2
(D) Tension in string is 5mg/8
True statements is/are
(1) A, C
(2) A, D
(3) B, C
(4) B, D

PRM161

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	1	3	1	3	2	1	3	4

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Important Notes

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Exercise - I (Conceptual Questions) Build Up Your Understanding

NEWTON'S LAW OF GRAVITATION & GRAVITATIONAL FIELD

1. Newton's law of gravitation :
(1) is not applicable outside the solar system
(2) is used to govern the motion of satellites only
(3) control the rotational motion of satellites and planets
(4) control the rotational motion of electrons in atoms

PGR001

2. Mass particles of 1 kg each are placed along x-axis at x = 1, 2, 4, 8, ...∞. Then gravitational force on a mass of 3kg placed at origin is (G = universal gravitational constant) :-

(1) 4G (2) $\frac{4G}{3}$ (3) 2G (4) ∞

PGR002

3. Gravitational force between two masses at a distance 'd' apart is 6N. If these masses are taken to moon and kept at same separation, then the force between them will become :

(1) 1 N (2) $\frac{1}{6}$N (3) 36 N (4) 6 N

PGR003

4. The value of universal gravitational constant G depends upon :
(1) Nature of material of two bodies
(2) Heat constant of two bodies
(3) Acceleration of two bodies
(4) None of these

PGR004

5. Three identical bodies (each mass M) are placed at vertices of an equilateral triangle of arm L, keeping the triangle as such by which angular speed the bodies should be rotated in their gravitational fields so that the triangle moves along circumference of circular orbit :

(1) $\sqrt{\frac{3GM}{L^3}}$ (2) $\sqrt{\frac{GM}{L^3}}$
(3) $\sqrt{\frac{GM}{3L^3}}$ (4) $3\sqrt{\frac{GM}{L^3}}$

PGR005

6. Four particles of masses m, 2m, 3m and 4m are kept in sequence at the corners of a square of side a. The magnitude of gravitational force acting on a particle of mass m placed at the centre of the square will be :

(1) $\frac{24m^2G}{a^2}$ (2) $\frac{6m^2G}{a^2}$
(3) $\frac{4\sqrt{2}Gm^2}{a^2}$ (4) Zero

PGR006

7. The tidal waves in the seas are primarily due to :
(1) The gravitational effect of the sun on the earth
(2) The gravitational effect of the moon on the earth
(3) The rotation of the earth
(4) The atmospheric effect of the earth itself

PGR007

8. During the journey of space ship from earth to moon and back, the maximum fuel is consumed :-
(1) Against the gravitation of earth in return journey
(2) Against the gravitation of earth in onward journey
(3) Against the gravitation of moon while reaching the moon
(4) None of the above

PGR008

9. If the distance between the centres of earth and moon is D and mass of earth is 81 times that of moon. At what distance from the centre of earth gravitational field will be zero :

(1) $\frac{D}{2}$ (2) $\frac{2D}{3}$
(3) $\frac{4D}{5}$ (4) $\frac{9D}{10}$

PGR009

10. An earth's satellite is moving in a circular orbit with a uniform speed v. If the gravitational force of the earth suddenly disappears, the satellite will:-

(1) vanish into outer space

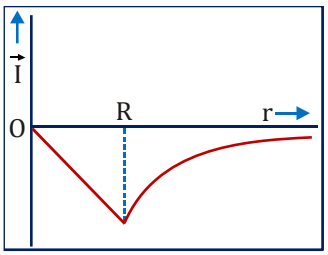
(2) continue to move with velocity v in original orbit

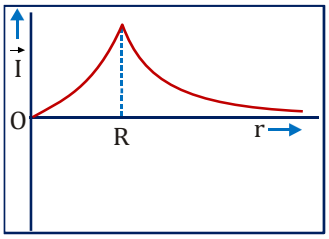
(3) fall down with increasing velocity

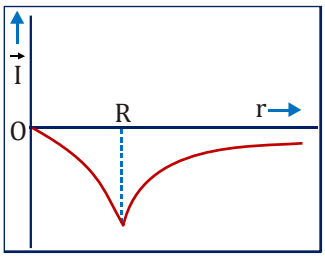
(4) fly off tangentially from the orbit with velocity v

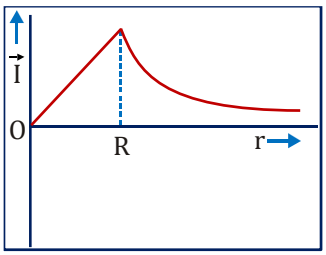
PGR010

11. Following curve shows the variation of intensity of gravitational field ($\vec{I}$) with distance from the centre of solid sphere(r) :

(1) 

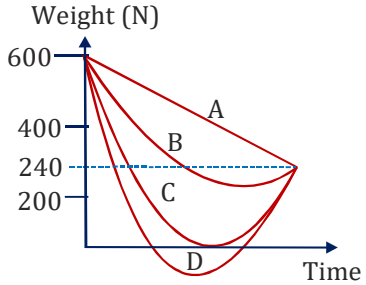
(2) 

(3) 

(4) 

PGR011

12. Suppose the acceleration due to gravity at the earth's surface is 10m/s² and at the surface of mars it is 4.0 m/s². A 60kg passenger goes from the earth to the mars in a spaceship moving with a constant velocity. Neglect all other objects in the sky. Which part of figure best represent the weight (Net gravitational force) of the passenger as a function of time :



(1) A (2) B (3) C (4) D

PGR012

13. Assume that a tunnel is dug through earth from North pole to south pole and that the earth is a non-rotating, uniform sphere of density ρ. The gravitational force on a particle of mass m dropped into the tunnel when it reaches a distance r from the centre of earth is

(1) $\left(\frac{3}{4\pi}mG\rho\right)r$ (2) $\left(\frac{4\pi}{3}mG\rho\right)r$

(3) $\left(\frac{4\pi}{3}mG\rho\right)r^2$ (4) $\left(\frac{4\pi}{3}m^2G\rho\right)r$

PGR013

14. Mars has a diameter of approximately 0.5 of that of earth, and mass of 0.1 of that of earth. The surface gravitational field strength on mars as compared to that on earth is a factor of –

(1) 0.1 (2) 0.2 (3) 2.0 (4) 0.4

PGR014

15. Three equal masses of 1 kg each are placed at the vertices of an equilateral triangle PQR and a mass of 2 kg is placed at the centroid O of the triangle which is at a distance of $\sqrt{2}$ m from each of the vertices of the triangle. The force, in newton, acting on the mass of 2 kg is :-

(1) 2 (2) $\sqrt{2}$ (3) 1 (4) zero

PGR015

16. One can easily "weigh the earth" by calculating the mass of earth using the formula (in usual notation)

(1) $\frac{G}{g}R_E^2$ (2) $\frac{g}{G}R_E^2$ (3) $\frac{g}{G}R_E$ (4) $\frac{G}{g}R_E^3$

PGR016

ACCELERATION DUE TO GRAVITY

17. Acceleration due to gravity at the centre of the earth is :-

(1) g (2) $\frac{g}{2}$

(3) zero (4) infinite

PGR017

18. The value of 'g' on earth surface depends :-

(1) only an earth's structure

(2) only an earth's rotational motion

(3) on above both

(4) on none these and is same

PGR018

19. The value of 'g' reduces to half of its value at surface of earth at a height 'h', then :-

(1) h = R (2) h = 2R

(3) $h=(\sqrt{2}+1)R$ (4) $h=(\sqrt{2}-1)R$

PGR019

20. At some planet 'g' is 1.96 m/sec^2. If it is safe to jump from a height of 2m on earth, then what should be corresponding safe height for jumping on that planet:-

(1) 5m (2) 2m (3) 10m (4) 20m

PGR020

21. If the earth stops rotating suddenly, the value of g at a place other than poles would :-

(1) Decrease

(2) Remain constant

(3) Increase

(4) Increase or decrease depending on the position of earth in the orbit round the sun

PGR021

22. Diameter and mass of a planet is double that earth. Then time period of a pendulum at surface of planet is how much times of time period at earth surface :-

(1) $\frac{1}{\sqrt{2}}$ times (2) $\sqrt{2}$ times

(3) Equal (4) None of these

PGR022

23. Gravitation on moon is 1/6th of that on earth. When a balloon filled with hydrogen is released on moon then, this :-

(1) Will rise with an acceleration less then $\left(\frac{g}{6}\right)$

(2) Will rise with acceleration $\left(\frac{g}{6}\right)$

(3) Will fall down with an acceleration less than $\left(\frac{5g}{6}\right)$

(4) Will fall down with acceleration $\left(\frac{g}{6}\right)$

PGR023

24. The acceleration due to gravity g and mean density of earth ρ are related by which of the following relations ? [G = gravitational constant and R = radius of earth] :

(1) $\rho=\frac{4\pi gR^2}{3G}$ (2) $\rho=\frac{4\pi gR^3}{3G}$

(3) $\rho=\frac{3g}{4\pi gR}$ (4) $\rho=\frac{3g}{4\pi gR^3}$

PGR024

25. More amount of sugar is obtained in 1kg weight:

(1) At North pole

(2) At equator

(3) Between pole and equator

(4) At South pole

PGR025

26. When you move from equator to pole, the value of acceleration due to gravity (g) :-

(1) increases

(2) decreases

(3) remains the same

(4) first increases then decreases

PGR026

27. When the radius of earth is reduced by 1% without changing the mass, then the acceleration due to gravity will

(1) increase by 2% (2) decrease by 1.5%

(3) increase by 1% (4) decrease by 1%

PGR027

28. Weight of a body of mass m decreases by 1% when it is raised to height h above the earth's surface. If the body is taken to a depth h in a mine, then in its weight will

(1) decrease by 0.5% (2) decrease by 2%

(3) increase by 0.5% (4) increase by 1%

PGR028

29. Acceleration due to gravity at earth's surface is 'g' m/s^2. Find the effective value of acceleration due to gravity at a height of 32 km from sea level : (R_e = 6400 Km)

(1) 0.5g m/s^2 (2) 0.99g m/s^2

(3) 1.01g m/s^2 (4) 0.90g m/s^2

PGR029

30. The mass of the moon is 1% of mass of the earth. The ratio of gravitational pull of earth on moon to that of moon on earth will be :

(1) 1 : 1 (2) 1 : 10 (3) 1 : 100 (4) 2 : 1

PGR030

31. Imagine a new planet having the same density as that of earth but its radius is 3 times bigger than the earth in size. If the acceleration due to gravity on the surface of earth is g and that on the surface of the new planet is g', then :

(1) g' = 3g (2) g' = g/9

(3) g' = 9g (4) g'=27 g

PGR031

32. The change in the value of 'g' at a height 'h' above the surface of the earth is same as at a depth 'd'. If 'd' and 'h' are much smaller than the radius of earth, then which one of the following is correct?

(1) d = h (2) d = 2h

(3) d = $\frac{3h}{2}$ (4) d = h/2

PGR032

33. If the rotational speed of earth is increased then weight of a body at the equator

(1) increases (2) decreases

(3) becomes double (4) does not changes

PGR033

34. A body weighs W newton at the surface of the earth. Its weight at a height equal to half the radius of the earth will be :

(1) $\frac{W}{2}$ (2) $\frac{2W}{3}$ (3) $\frac{4W}{9}$ (4) $\frac{W}{4}$

PGR034

35. The imaginary angular velocity of the earth for which the effective acceleration due to gravity at the equator shall be zero is equal to

(1) 1.25×10^{-3} rad/s (2) 2.50×10^{-3} rad/s

(3) 3.75×10^{-3} rad/s (4) 5.0×10^{-3} rad/s

[Take g = 10m/s^2 for the acceleration due to gravity if the earth were at rest and radius of earth equal to 6400 km.]

PGR035

GRAVITATIONAL POTENTIAL ENERGY & POTENTIAL

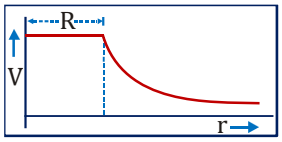
36. Two different masses are dropped from same heights. When these just strike the ground, the following is same :

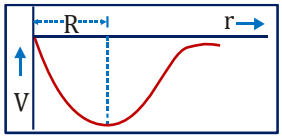
(1) kinetic energy (2) potential energy

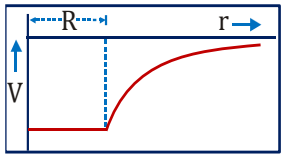
(3) linear momentum (4) acceleration

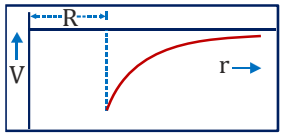
PGR036

37. Which of the following curve expresses the variation of gravitational potential with distance for a hollow sphere of radius R :

(1) 

(2) 

(3) 

(4) 

PGR037

38. Gravitational potential difference between surface of a planet and a point situated at a height of 20m above its surface is 2 J/kg. If gravitational field is uniform, then the work done in taking a 5kg body upto height 4 meter above surface will be :

(1) 2 J (2) 20 J (3) 40 J (4) 10 J

PGR038

39. If M_e is the mass of earth and M_m is the mass of moon ($M_e = 81\ M_m$). The potential energy of an object of mass m situated at a distance R from the centre of earth and r from the centre of moon, will be :-

(1) $-GmM_m\left(\frac{R}{81}+r\right)$ (2) $-GmM_e\left(\frac{81}{r}+\frac{1}{R}\right)$

(3) $-GmM_m\left(\frac{81}{R}+\frac{1}{r}\right)$ (4) $GmM_m\left(\frac{81}{r}-\frac{1}{R}\right)$

PGR039

40. The gravitational potential energy is maximum at:

(1) infinity
(2) the earth's surface
(3) The centre of the earth
(4) Twice the radius of the earth

PGR040

41. A missile is launched with a velocity less than the escape velocity. Sum of its kinetic energy and potential energy is :-

(1) Positive
(2) Negative
(3) May be negative or positive depending upon its initial velocity
(4) Zero

PGR041

42. A body attains a height equal to the radius of the earth when projected from earth' surface. The velocity of the body with which it was projected is :

(1) $\sqrt{\frac{GM}{R}}$ (2) $\sqrt{\frac{2GM}{R}}$

(3) $\sqrt{\frac{5}{4}\frac{GM}{R}}$ (4) $\sqrt{\frac{3GM}{R}}$

PGR042

43. The gravitational potential energy of a body at a distance r from the center of the earth is U. The force at that point is :

(1) $\frac{U}{r^2}$ (2) $\frac{U}{r}$ (3) Ur (4) Ur^2

PGR043

44. A particle falls from infinity to the earth. Its velocity on reaching the earth surface is :

(1) 2Rg (2) Rg
(3) $\sqrt{Rg}$ (4) $\sqrt{2Rg}$

PGR044

45. A projectile of mass m is thrown vertically up with an initial velocity v from the surface of earth (mass of earth = M). If it comes to rest at a height h, the change in its potential energy is

(1) GMmh/R(R + h) (2) $GMmh^2/R(R+h)^2$
(3) GMmhR/R(R + h) (4) GMm/hR(R+h)

PGR045

46. Two small and heavy spheres, each of mass M, are placed a distance r apart on a horizontal surface. The gravitational potential at the mid-point on the line joining the centre of the spheres is :-

(1) Zero (2) $-\frac{GM}{r}$

(3) $-\frac{2GM}{r}$ (4) $-\frac{4GM}{r}$

PGR046

47. An artificial satellite moving in a circular orbit around the earth has a total (kinetic + potential) energy E_0. Its potential energy is :-

(1) $-E_0$ (2) E_0
(3) $-2E_0$ (4) $2E_0$

PGR047

48. A particle of mass m is moving in a horizontal circle of radius R under a centripetal force equal to $-\frac{A}{r^2}$ (A = constant). The total energy of the particle is :-

(Potential energy at very large distance is zero)

(1) $\frac{A}{R}$ (2) $-\frac{A}{R}$ (3) $\frac{A}{2R}$ (4) $-\frac{A}{2R}$

PGR048

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ESCAPE VELOCITY

49. Potential energy of a 3kg body at the surface of a planet is – 54J, then escape velocity will be:

(1) 18 m/s (2) 162 m/s

(3) 36 m/s (4) 6 m/s

PGR049

50. Escape velocity of a 1kg body on a planet is 100 m/s. Potential energy of body at that planet is:

(1) –5000J (2) –1000J

(3) –2400J (4) –10000J

PGR050

51. The ratio of radii of two satellites is p and the ratio of their acceleration due to gravity is q. The ratio of their escape velocities will be :

(1) $\left(\frac{q}{p}\right)^{1/2}$ (2) $\left(\frac{p}{q}\right)^{1/2}$

(3) pq (4) $\sqrt{pq}$

PGR051

52. Escape velocity of a body from earth is 11.2 km/s. Escape velocity, when thrown at an angle of 45° from horizontal will be :-

(1) 11.2 km/s (2) 22.4 km/s

(3) $11.2/\sqrt{2}$ km/s (4) $11.2\sqrt{2}$ km/s

PGR052

53. The escape velocity from the earth is 11.2 km/s the mass of another planet is 100 times of mass of earth and its radius is 4 times the radius of earth. The escape velocity for the planet is :-

(1) 56.0 km/s (2) 280 km/s

(3) 112 km/s (4) 11.2 km/s

PGR053

54. Body is projected vertically upward from the surface of the earth with a velocity equal to half the escape velocity. If R is radius of the earth, the maximum height attained by the body is :-

(1) $\frac{R}{6}$ (2) $\frac{R}{3}$ (3) $\frac{2}{3}R$ (4) R

PGR054

PLANETARY MOTION & WEIGHTLESSNESS

55. Binding energy of moon and earth is :-

(1) $\frac{GM_eM_m}{r_{em}}$ (2) $\frac{GM_eM_m}{2r_{em}}$

(3) $-\frac{GM_eM_m}{r_{em}}$ (4) $-\frac{GM_eM_m}{2r_{em}}$

PGR055

56. Two artificial satellites A and B are at a distance r_A and r_B above the earth's surface. If the radius of earth is R, then the ratio of their speed will be :-

(1) $\left(\frac{r_B+R}{r_A+R}\right)^{1/2}$ (2) $\left(\frac{r_B+R}{r_A+R}\right)^{2}$

(3) $\left(\frac{r_B}{r_A}\right)^{2}$ (4) $\left(\frac{r_B}{r_A}\right)^{1/2}$

PGR056

57. The average radii of orbits of mercury and earth around the sun are 6×10^7 km and 1.5×10^8 km respectively. The ratio of their orbital speeds will be :-

(1) $\sqrt{5}:\sqrt{2}$ (2) $\sqrt{2}:\sqrt{5}$

(3) 2.5 : 1 (4) 1 : 25

PGR057

58. A body is dropped by a satellite in its geo-stationary orbit :

(1) it will burn on entering in to the atmosphere

(2) it will remain in the same place with respect to the earth

(3) it will reach the earth is 24 hours

(4) it will perform uncertain motion

PGR058

59. Two ordinary satellites are revolving round the earth in same elliptical orbit, then which of the following quantities is conserved :-

(1) Velocity

(2) Angular velocity

(3) Angular momentum

(4) None of above

PGR059

60. Kepler's second law is a consequence of :-

(1) conservation of kinetic energy

(2) conservation of linear momentum

(3) conservation of angular momentum

(4) conservation of speed

PGR060

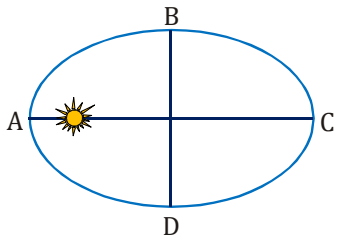
61. One projectile after deviating from its path starts moving round the earth in a circular path of radius equal to nine times the radius of earth R. Its time period will be :-

(1) $2\pi\sqrt{\frac{R}{g}}$ (2) $27\times2\pi\sqrt{\frac{R}{g}}$

(3) $\pi\sqrt{\frac{R}{g}}$ (4) $0.8\times3\pi\sqrt{\frac{R}{g}}$

PGR061

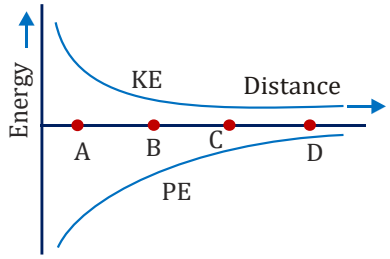
62. In adjoining figure earth goes around the sun in elliptical orbit on which point the orbital speed is maximum :



(1) On A (2) On B (3) On C (4) On D

PGR062

63. Potential energy and kinetic energy of a two particles system under imaginary force field are shown by curves KE and PE respectively in figure. This system is bound at :



(1) only point A
(2) only point D
(3) only point A, B, and C
(4) All points A, B, C and D

PGR063

64. A satellite of earth of mass 'm' is taken from orbital radius 2R to 3R, then minimum work done is :-

(1) $\frac{GMm}{6R}$ (2) $\frac{GMm}{12R}$

(3) $\frac{GMm}{24R}$ (4) $\frac{GMm}{3R}$

PGR064

65. If a graph is plotted between T^2 and r^3 for a planet then its slope will be :-

(1) $\frac{4\pi^2}{GM}$ (2) $\frac{GM}{4\pi^2}$ (3) 4π GM (4) Zero

PGR065

66. A planet is revolving round the sun. Its distance from the sun at Apogee is r_A and that at Perigee is r_p. The mass of planet and sun is m and M respectively, v_A and v_P is the velocity of planet at Apogee and Perigee respectively and T is the time period of revolution of planet round the sun.

(a) $T^2=\frac{\pi^2}{2Gm}(r_A+r_P)^2$

(b) $T^2=\frac{\pi^2}{2GM}(r_A+r_P)^3$

(c) $v_Ar_A = v_Pr_P$

(d) $v_A < v_P$, $r_A > r_P$

(1) a, b, c (2) a, b, d

(3) b, c, d (4) all

PGR066

67. A satellite launching station should be :

(1) near the equatorial region
(2) near the polar region
(3) on the polar axis
(4) all locations are equally good

PGR067

68. A space shuttle is launched in a circular orbit near the earth's surface. The additional velocity be given to the space - shuttle to get free from the influence of gravitational force, will be :

(1) 1.52 km/s (2) 2.75 km/s

(3) 3.28 km/s (4) 5.18 km/s

PGR068

69. A satellite is moving in a circular orbit around earth with a speed v. If its mass is m, then its total energy will be :

(1) $\frac{3}{4}mv^2$ (2) mv^2

(3) $\frac{1}{2}mv^2$ (4) $-\frac{1}{2}mv^2$

PGR069

AIM: FREE EDUCATION TO ALL APUL

70. If the length of the day is T, the height of that TV satellite above the earth's surface which always appears stationary from earth, will be :

(1) $h=\left[\frac{4\pi^2GM}{T^2}\right]^{1/3}$

(2) $h=\left[\frac{4\pi^2GM}{T^2}\right]^{1/2}-R$

(3) $h=\left[\frac{GMT^2}{4\pi^2}\right]^{1/3}-R$

(4) $h=\left[\frac{GMT^2}{4\pi^2}\right]^{1/3}+R$

PGR070

71. If two bodies of mass M and m are revolving around the centre of mass of the system in circular orbit of radii R and r respectively due to mutual interaction. Which of the following formula is applicable :-

(1) $\frac{GMm}{(R+r)^2}=m\omega^2r$ (2) $\frac{GMm}{R^2}=m\omega^2r$

(3) $\frac{GMm}{r^2}=m\omega^2r$ (4) $\frac{GMm}{R^2+r^2}=m\omega^2r$

PGR071

72. Two satellites of same mass m are revolving round of earth (mass M) in the same orbit of radius r. Rotational directions of the two are opposite therefore, they can collide. Total mechanical energy of the system (both satellites and earths) is (m << M) :-

(1) $-\frac{GMm}{r}$ (2) $-\frac{2GMm}{r}$

(3) $-\frac{GMm}{2r}$ (4) Zero

PGR072

73. A planet of mass m is moving in an elliptical orbit about the sun (mass of sun = M). The maximum and minimum distances of the planet from the sun are r_1 and r_2 respectively. The period of revolution of the planet will be proportional to :

(1) $r_1^{3/2}$ (2) $r_2^{3/2}$

(3) $(r_1-r_2)^{3/2}$ (4) $(r_1+r_2)^{3/2}$

PGR073

74. If the satellite is stopped suddenly in its orbit which is at a distance radius of earth from earth's surface and allowed to fall freely into the earth. The speed with which it hits the surface of earth will be :

(1) 7.919 m/s (2) 7.919 km/s

(3) 11.2 m/s (4) 11.2 km/s

PGR075

75. A planet is moving in an elliptical orbit. If T, U, E and L are its kinetic energy, potential energy, total energy and magnitude of angular momentum respectively, then which of the following statement is true :-

(1) T is conserved

(2) U is always positive

(3) E is always negative

(4) L is conserved but the direction of vector $\vec{L}$ will continuously change

PGR076

76. The gravitational force between two bodies is directly proportional to $\frac{1}{R}$ (not $\frac{1}{R^2}$), where 'R' is the distance between the bodies. Then the orbital speed for this force in circular orbit is proportional to :-

(1) $1/R^2$ (2) R° (3) R (4) 1/R

PGR077

77. What will be velocity of a satellite revolving around the earth at a height h above surface of earth if radius of earth is R :-

(1) $R^2\sqrt{\frac{g}{R+h}}$ (2) $R\frac{g}{(R+h)^2}$

(3) $R\sqrt{\frac{g}{R+h}}$ (4) $R\sqrt{\frac{R+h}{g}}$

PGR078

78. Two artificial satellites of masses m_1 and m_2 are moving with speeds v_1 and v_2 in orbits of radii r_1 and r_2 respectively. If $r_1 > r_2$ then which of the following statements in true :-

(1) $v_1 = v_2$ (2) $v_1 > v_2$

(3) $v_1 < v_2$ (4) $v_1/r_1 = v_2/r_2$

PGR079

79. Orbital radius of a satellite S of earth is four times that of a communication satellite C. Period of revolution of S is :-

(1) 4 days (2) 8 days

(3) 16 days (4) 32 days

PGR080

80. If a satellite is revolving very close to the surface of earth, then its orbital velocity does not depend upon:-

(1) Mass of satellite (2) Mass of earth

(3) Radius of earth (4) Orbital radius

PGR081

81. Two identical satellites are at the heights R and 7R from the earth's surface. Then which of the following statement is incorrect :–

(R = Radius of the earth)

(1) Ratio of total energy of both is 5

(2) Ratio of kinetic energy of both is 4

(3) Ratio of potential energy of both 4

(4) Ratio of total energy of both is 4

PGR082

82. The minimum projection velocity of a body from the earth's surface so that it becomes the satellite of the earth ($R_e = 6.4 \times 10^6$ m).

(1) 11×10^3 m/s (2) 8×10^3 m/s

(3) 6.4×10^3 m/s (4) 4×10^3 m/s

PGR083

83. The maximum and minimum distances of a comet from the sun are 8×10^{12} m and 1.6×10^{12} m respecting. If its velocity when it is nearest to the sun is 60 m/s then what will be its velocity in m/s when it is farthest?

(1) 12 (2) 60 (3) 112 (4) 6

PGR085

84. A satellite of mass m goes round the earth along a circular path of radius r. Let m_E be the mass of the earth and R_E its radius then the linear speed of the satellite depends on.

(1) m, m_E and r (2) m, R_E and r

(3) m_E only (4) m_E and r

PGR086

85. Near the earth's surface time period of a satellite is 1.4 hrs. Find its time period if it is at the distance '4R' from the centre of earth :-

(1) 32 hrs. (2) $\left(\frac{1}{8\sqrt{2}}\right)$ hrs.

(3) $8\sqrt{2}$ hrs. (4) 16 hrs.

PGR087

86. A communication satellite of earth which takes 24 hrs. to complete one circular orbit eventually has to be replaced by another satellite of double mass. If the new satellites also has an orbital time period of 24 hrs, then what is the ratio of the radius of the new orbit to the original orbit?

(1) 1 : 1 (2) 2 : 1 (3) $\sqrt{2}$:1 (4) 1 : 2

PGR088

87. Escape velocity for a projectile at earth's surface is V_e. A body is projected form earth's surface with velocity 2 V_e. The velocity of the body when it is at infinite distance from the centre of the earth is :-

(1) V_e (2) $2V_e$

(3) $\sqrt{2}\,V_e$ (4) $\sqrt{3}\,V_e$

PGR089

88. For a satellite moving in an orbit around the earth, the ratio of kinetic energy to potential energy is :-

(1) 2 (2) 1/2

(3) $\frac{1}{\sqrt{2}}$ (4) $\sqrt{2}$

PGR090

89. The orbital velocity of an artificial satellite in a circular orbit just above the earth's surface is v_0. The orbital velocity of satellite orbiting at an altitude of half of the radius is :-

(1) $\frac{3}{2}v_0$ (2) $\frac{2}{3}v_0$

(3) $\sqrt{\frac{2}{3}}v_0$ (4) $\sqrt{\frac{3}{2}}v_0$

PGR091

90. The earth revolves around the sun in one year. If distance between them becomes double, the new time period of revolution will be :-

(1) $4\sqrt{2}$ years (2) $2\sqrt{2}$ years

(3) 4 years (4) 8 years

PGR092

91. A satellite of mass m revolves in a circular orbit of radius R a round a planet of mass M. Its total energy E is :-

(1) $-\frac{GMm}{2R}$

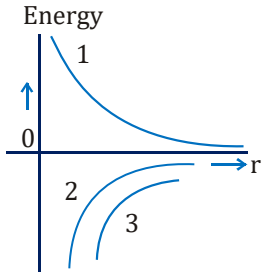
(2) $+\frac{GMm}{3R}$

(3) $-\frac{GMm}{R}$

(4) $+\frac{GMm}{R}$

PGR093

92. A satellite is orbiting earth at a distance r. Variations of its kinetic energy, potential energy and total energy, is shown in the figure. Of the three curves shown in figure, identify the type of mechanical energy they represent.



(1) 1 Potential, 2 Kinetic, 3 Total
(2) 1 Total, 2 Kinetic, 3 Potential
(3) 1 Kinetic, 2 Total, 3 Potential
(4) 1 Potential, 2 Total, 3 Kinetic

PGR094

93. The mean distance of mars from sun is 1.5 times that of earth from sun. What is approximately the number of years required by mars to make one revolution about sun ?

(1) 2.35 years (2) 1.85 years
(3) 3.65 years (4) 2.75 years

PGR095

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	1	4	4	1	3	2	2	4	4	1	3	2	4	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	3	3	4	3	3	2	4	3	2	1	1	1	2	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	2	2	3	1	4	3	1	3	1	2	1	2	4	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	4	4	4	4	1	4	1	1	2	2	1	1	2	3	3
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	2	1	3	2	1	3	1	3	4	3	1	1	4	2	3
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	2	3	3	2	1	1	2	1	4	3	1	4	2	3	2
Question	91	92	93												
Answer	1	3	2												

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

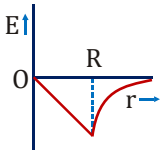
1. A black hole is an object whose gravitational field is so strong that even light cannot escape from it. To what approximate radius would earth (mass = 5.98×10^{24} kg) have to be compressed to be a black hole?

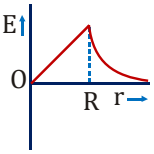
(1) 10^{-9} m (2) 10^{-6} m

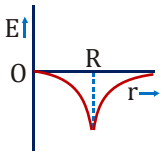
(3) 10^{-2} m (4) 100 m

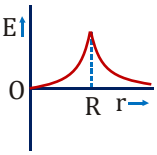
PGR108

2. Dependence of intensity of gravitational field (E) of earth with distance (r) from centre of earth is correctly represented by :-

(1) 

(2) 

(3) 

(4) 

PGR109

AIPMT 2015

3. Kepler's third law states that square of period of revolution (T) of a planet around the sun, is proportional to third power of average distance r between sun and planet i.e. $T^2 = Kr^3$ here K is constant.

If the masses of sun and planet are M and m respectively then as per Newton's law of gravitation force of attraction between them is $F = \frac{GMm}{r^2}$, here G is gravitational constant.

The relation between G and K is described as:

(1) $GMK = 4\pi^2$ (2) $K = G$

(3) $K = \frac{1}{G}$ (4) $GK = 4\pi^2$

PGR110

Re-AIPMT 2015

4. A satellite S is moving in an elliptical orbit around the earth. The mass of the satellite is very small compared to the mass of the earth. Then,

(1) the acceleration of S is always directed towards the centre of the earth.

(2) the angular momentum of S about the centre of the earth changes in direction, but its magnitude remains constant.

(3) the total mechanical energy of S varies periodically with time.

(4) the linear momentum of S remains constant in magnitude.

PGR111

5. A remote - sensing satellite of earth revolves in a circular orbit at a height of 0.25×10^6 m above the surface of earth. If earth's radius is 6.38×10^6 m and g = 9.8 m/s^2, then the orbital speed of the satellite is :

(1) 6.67 km/s (2) 7.76 km/s

(3) 8.56 km/s (4) 9.13 km/s

PGR112

NEET-I 2016

6. At what height from the surface of earth the gravitation potential and the value of g are -5.4×10^7J/kg and 6.0m/s^2 respectively?

Take the radius of earth as 6400 km :

(1) 2600 km (2) 1600 km

(3) 1400 km (4) 2000 km

PGR113

7. The ratio of escape velocity at earth (v_e) to the escape velocity at a planet (v_p) whose radius and mean density are twice as that of earth is :-

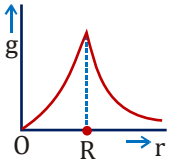
(1) 1 : 2 (2) $1 : 2\sqrt{2}$

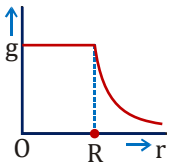
(3) 1 : 4 (4) $1 : \sqrt{2}$

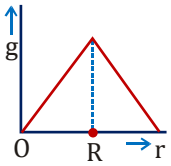
PGR114

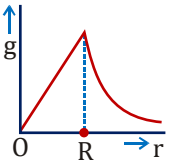
NEET-II 2016

8. Starting from the centre of the earth having radius R, the variation of g (acceleration due to gravity) is shown by :-

(1) 

(2) 

(3) 

(4) 

PGR115

9. A satellite of mass m is orbiting the earth (of radius R) at a height h from its surface. The total energy of the satellite in terms of g_0, the value of acceleration due to gravity at the earth's surface, is:-

(1) $\frac{2mg_0R^2}{R+h}$ (2) $-\frac{2mg_0R^2}{R+h}$

(3) $\frac{mg_0R^2}{2(R+h)}$ (4) $-\frac{mg_0R^2}{2(R+h)}$

PGR116

NEET(UG) 2017

10. The acceleration due to gravity at a height 1 km above the earth is the same as at a depth d below the surface of earth. Then :-

(1) d = 1 km (2) d = $\frac{3}{2}$ km

(3) d = 2 km (4) d = $\frac{1}{2}$ km

PGR117

11. Two astronauts are floating in gravitational free space after having lost contact with their spaceship. The two will :-

(1) Move towards each other.

(2) Move away from each other.

(3) Will become stationary

(4) Keep floating at the same distance between them.

PGR118

NEET(UG) 2018

12. If the mass of the Sun were ten times smaller and the universal gravitational constant were ten time larger in magnitude, which of the following is **not** correct?

(1) Raindrops will fall faster

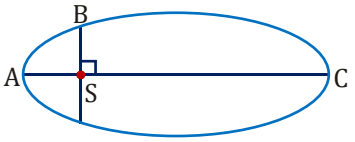
(2) Walking on the ground would become more difficult

(3) Time period of a simple pendulum on the Earth would decrease

(4) 'g' on the Earth will not change

PGR119

13. The kinetic energies of a planet in an elliptical orbit about the Sun, at positions A, B and C are K_A, K_B and K_C respectively. AC is the major axis and SB is perpendicular to AC at the position of the Sun S as shown in the figure. Then



(1) $K_A < K_B < K_C$ (2) $K_A > K_B > K_C$

(3) $K_B < K_A < K_C$ (4) $K_B > K_A > K_C$

PGR120

NEET(UG) 2019

14. A body weighs 200 N on the surface of the earth. How much will it weigh half way down to the centre of the earth ?

(1) 150 N (2) 200 N

(3) 250 N (4) 100 N

PGR121

15. The work done to raise a mass m from the surface of the earth to a height h, which is equal to the radius of the earth, is :

(1) mgR (2) 2 mgR

(3) $\frac{1}{2}$ mgR (4) $\frac{3}{2}$ mgR

PGR122

NEET(UG) 2019 (Odisha)

16. The time period of a geostationary satellite is 24 h, which is at a height $6R_E$ (R_E is radius of earth) from surface of earth. The time period of another satellite whose height is 2.5 R_E from surface will be,

(1) $6\sqrt{2}$ h (2) $12\sqrt{2}$ h

(3) $\frac{24}{2.5}$ h (4) $\frac{12}{2.5}$ h

PGR123

17. Assuming that the gravitational potential energy of an object at infinity is zero, the change in potential energy (final – initial) of an object of mass m, when taken to a height h from the surface of earth (of radius R), is given by,

(1) $-\frac{GMm}{R+h}$ (2) $\frac{GMmh}{R(R+h)}$

(3) mgh (4) $\frac{GMm}{R+h}$

PGR124

NEET (UG) 2020

18. A body weighs 72 N on the surface of the earth. What is the gravitational force on it, at a height equal to half the radius of the earth?

(1) 24 N (2) 48 N

(3) 32 N (4) 30 N

PGR125

NEET (UG) 2020 (Covid-19)

19. What is the depth at which the value of acceleration due to gravity becomes 1/n times the value that at the surface of earth? (radius of earth = R)

(1) R/n^2 (2) $R(n-1)/n$

(3) $Rn/(n-1)$ (4) R/n

PGR126

NEET (UG) 2021

20. The escape velocity from the Earth's surface is υ. The escape velocity from the surface of another planet having a radius, four times that of Earth and same mass density is:

(1) υ (2) 2υ (3) 3υ (4) 4υ

PGR127

21. A particle of mass 'm' is projected with a velocity $\upsilon = kV_e$ ($k < 1$) from the surface of the earth. (V_e = escape velocity) The maximum height above the surface reached by the particle is :

(1) $R\left(\frac{k}{1-k}\right)^2$ (2) $R\left(\frac{k}{1+k}\right)^2$

(3) $\frac{R^2k}{1+k}$ (4) $\frac{Rk^2}{1-k^2}$

PGR128

NEET (UG) 2022

22. A body of mass 60g experiences a gravitational force of 3.0 N, when placed at a particular point. The magnitude of the gravitational field intensity at that point is:

(1) 50 N/kg (2) 20 N/kg

(3) 180 N/kg (4) 0.05 N/kg

PGR129

23. Match List - I with List - II :

List - I		List – II	
(a)	Gravitational constant (G)	(i)	$[L^2T^{-2}]$
(b)	Gravitational potential energy	(ii)	$[M^{-1}L^3T^{-2}]$
(c)	Gravitational Potential	(iii)	$[LT^{-2}]$
(d)	Gravitational intensity	(iv)	$[ML^2T^{-2}]$

Choose the correct answer from the options given below :

(1) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(2) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

(3) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

(4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

PGR130

RE-NEET (UG) 2022

24. In a gravitational field, the gravitational potential is given by, $V = -\frac{K}{x}$ (J/kg). The gravitational field intensity at point (2, 0, 3) m is:

(1) $+\frac{K}{2}$ (2) $-\frac{K}{2}$ (3) $-\frac{K}{4}$ (4) $+\frac{K}{4}$

PGR131

NEET (UG) 2023

25. Two bodies of mass *m* and 9*m* are placed at a distance R. The gravitational potential on the line joining the bodies where the gravitational field equals zero, will be (*G* = gravitational constant) :

(1) $-\frac{12Gm}{R}$ (2) $-\frac{16Gm}{R}$

(3) $-\frac{20Gm}{R}$ (4) $-\frac{8Gm}{R}$

PGR140

26. A satellite is orbiting just above the surface of the earth with period T. If d is the density of the earth and G is the universal constant of gravitation, the quantity $\frac{3\pi}{Gd}$ represents :

(1) T^2 (2) T^3 (3) $\sqrt{T}$ (4) T

PGR141

RE-NEET (UG) 2023

27. The escape velocity of a body on the earth surface is 11.2 km/s. If the same body is projected upward with velocity 22.4 km/s, the velocity of this body at infinite distance from the centre of the earth will be:

(1) $11.2\sqrt{2}$ km/s (2) Zero

(3) 11.2 km/s (4) $11.2\sqrt{3}$ km/s

PGR142

28. If R is the radius of the earth and g is the acceleration due to gravity on the earth surface. Then the mean density of the earth will be :

(1) $\frac{\pi RG}{12g}$ (2) $\frac{3\pi R}{4gG}$ (3) $\frac{3g}{4\pi RG}$ (4) $\frac{4\pi G}{3gR}$

PGR143

NEET (UG) 2024

29. The mass of a planet is $\frac{1}{10}$th that of the earth and its diameter is half that of the earth. The acceleration due to gravity on that planet is :

(1) 19.6 m s^{-2} (2) 9.8 m s^{-2}

(3) 4.9 m s^{-2} (4) 3.92 m s^{-2}

PGR144

30. The minimum energy required to launch a satellite of mass m from the surface of earth of mass M and radius R in a circular orbit at an altitude of 2R from the surface of the earth is :

(1) $\frac{5GmM}{6R}$ (2) $\frac{2GmM}{3R}$

(3) $\frac{GmM}{2R}$ (4) $\frac{GmM}{3R}$

PGR145

RE-NEET (UG) 2024

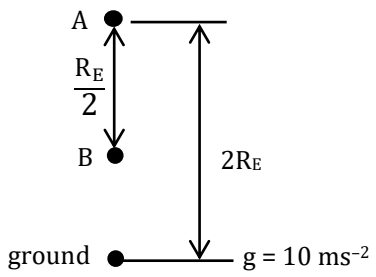
31. The escape velocity for earth is v. A planet having 9 times mass that of earth and radius, 16 times that of earth, has the escape velocity of :

(1) $\frac{v}{3}$ (2) $\frac{2v}{3}$

(3) $\frac{3v}{4}$ (4) $\frac{9v}{4}$

PGR146

32. An object of mass 100 kg falls from point A to B as shown in figure. The change in its weight, corrected to the nearest integer is (R_E is the radius of the earth) :



(1) 49 N (2) 89 N

(3) 5 N (4) 10 N

PGR147

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	1	1	1	2	1	2	4	4	3	4	4	2	4	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	2	3	2	4	4	1	1	3	2	1	4	3	4	1
Question	31	32													
Answer	3	1													

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Exercise - III (Analytical Questions) Master Your Understanding

1. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : An artificial satellite is moving in a circular orbit of the earth. If the gravitational pull suddenly disappears, then it moves with the same speed tangential to the original orbit.

Reason (R) : The orbital speed of a satellite decreases with the increase in radius of the orbit.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PGR132

2. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : Total energy is conserved in moving a satellite to higher orbit.

Reason (R) : For a satellite change in potential energy and kinetic energy is same in magnitude and opposite in nature.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) Both **(A)** and **(R)** are false.

PGR133

3. Choose the correct statement(s) from the following:-

(A) The magnitude of the gravitational force between two bodies of mass 1 kg each and separated by a distance of 1 m is 9.8 N.

(B) Higher the value of escape velocity for a planet, higher is the abundance of lighter gases in its atmosphere.

(C) Force of friction arises due to gravitational attraction.

(D) The gravitational force of attraction between two bodies of ordinary mass is not noticeable because the value of the gravitational constant is extremely small.

(1) A, B (2) B, C, D (3) D (4) B, D

PGR135

4. A satellite is moving round the earth in a circular orbit. The following statements are given :-

(i) It is moving with a constant velocity.

(ii) It has no acceleration.

(iii) It's angular momentum w.r.t. centre of the earth remains conserved.

(iv) Its distance from center must be equal to $\sqrt{2}$ times of earth's radius.

The correct option is :-

(1) i & ii are true

(2) i, iii & iv are true

(3) only iii is true

(4) i & iv are true

PGR136

5. Consider the following statement for a satellite S moving in an elliptical orbit around the earth (mass of satellite is negligibly small compared to earth)

I: The acc. of S is always towards earth.

II: The total mechanical energy of S varies periodically with time.

III: The linear momentum of 'S' remain constant in magnitude.

The correct sequence of true (T) and false (F) for the above statement is :-

(1) TTF (2) TFT (3) TFF (4) FFF

PGR137

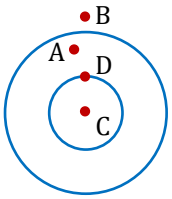
6. A satellite is in a circular equatorial orbit of radius 7000 km around the earth. If it is transferred to a circular orbit of double of radius :-

Column - I		Column - II	
(A)	Angular momentum	(P)	Increases
(B)	Area of earth covered by satellite signal	(Q)	Decreases
(C)	Potential energy	(R)	Becomes double
(D)	Kinetic energy	(S)	Becomes half

(1) (A) → Q; (B) → P; (C) → P, R; (D) → Q, S
(2) (A) → P; (B) → P; (C) → P; (D) → Q, S
(3) (A) → Q; (B) → Q; (C) → P, S; (D) → P, R
(4) (A) → P; (B) → Q; (C) → P; (D) → P, R

PGR138

7. Two concentric spherical shells of non-zero masses are as shown in the figure :-



Column - I		Column - II	
(A)	Potential at A	(P)	Greater than B
(B)	Gravitational field at A	(Q)	Less than B
(C)	As one moves from C to D	(R)	Potential remains constant
(D)	As one moves from D to A	(S)	Gravitational field decreases
		(T)	Data insufficient

(1) (A) → P; (B) → Q; (C) → R; (D) → S
(2) (A) → Q; (B) → T; (C) → R; (D) → S
(3) (A) → P; (B) → Q; (C) → S; (D) → R
(4) (A) → Q; (B) → P; (C) → S; (D) → R

PGR139

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	2	4	4	3	3	2	2

Important Notes

Exercise - I (Conceptual Questions) Build Up Your Understanding

ELASTICITY

1. The lower surface of a cube is fixed. On its upper surface, force is applied at an angle of 30° from its surface. The change will be in its
(1) shape
(2) size
(3) volume
(4) both shape and size

PPF001

2. One end of uniform wire of length L and of weight W is attached rigidly to a point in the roof and a weight W_1 is suspended from its lower end. If s is the area of cross-section of the wire, the stress in the wire at a height (L/4) from its lower end is

(1) $\frac{W_1}{s}$ (2) $\frac{\left[W_1+\frac{W}{4}\right]}{s}$

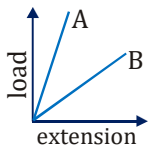
(3) $\frac{\left[W_1+\frac{3W}{4}\right]}{s}$ (4) $\frac{W_1+W}{4}$

PPF002

3. For steel, the breaking stress is 6×10^6 N/m^2 and the density is 8×10^3 kg/m^3. The maximum length of steel wire, which can be suspended without breaking under its own weight is [g = 10 m/s^2]
(1) 140 m (2) 120 m
(3) 75 m (4) 200 m

PPF003

4. The dimensions of two wires A and B are the same. But their materials are different. Their load-extension graphs are shown. If Y_A and Y_B are the values of Young's modulus of elasticity of A and B respectively then



(1) $Y_A > Y_B$
(2) $Y_A < Y_B$
(3) $Y_A = Y_B$
(4) $Y_B = 2Y_A$

PPF004

5. If the density of the material increase, the value of Young's modulus
(1) increases
(2) decreases
(3) first increases, then decreases
(4) first decreases, then increases

PPF005

6. A fixed volume of iron is drawn into a wire of length ℓ. The extension produced in this wire by a constant force F is proportional to -

(1) $\frac{1}{\ell^2}$ (2) $\frac{1}{\ell}$ (3) ℓ^2 (4) ℓ

PPF006

7. A wire elongates by ℓ mm when a load W is hanged from it. If the wire goes over a pulley and two weights W each are hung at the two ends, the elongation of the wire will be (in mm)-

(1) ℓ (2) 2ℓ (3) zero (4) $\ell/2$

PPF007

8. The Young's modulus of a rubber string 8 cm long and density 1.5 kg/m^3 is 5×10^8 N/m^2, is suspended on the ceiling in a room. The increase in length due to its own weight will be :
(1) 9.6×10^{-5} m (2) 9.6×10^{-11} m
(3) 9.6×10^{-3} m (4) 9.6 m

PPF008

9. A ball falling in a lake of depth 200 m shows 0.1% decrease in its volume at the bottom. What is the bulk modulus of the material of the ball :
(1) 19.6×10^8 N/m^2 (2) 19.6×10^{-10} N/m^2
(3) 19.6×10^{10} N/m^2 (4) 19.6×10^{-8} N/m^2

PPF009

10. The pressure of a medium is changed from 1.01×10^5 Pa to 1.165×10^5 Pa and change in volume is 10% keeping temperature constant. The bulk modulus of the medium is :–
(1) 204.8×10^5 Pa (2) 102.4×10^5 Pa
(3) 51.2×10^5 Pa (4) 1.55×10^5 Pa

PPF010

11. Two wires of the same material and length but diameters in the ratio 1 : 2 are stretched by the same force. The potential energy per unit volume for the two wires when stretched will be in the ratio.

(1) 16 : 1 (2) 4 : 1 (3) 2 : 1 (4) 1 : 1

PPF011

12. A weight is suspended from a long metal wire. If the wire suddenly breaks, its temperature

(1) rises

(2) falls

(3) remains unchanged

(4) attains a value 0 K

PPF012

13. A mass of 0.5 kg is suspended from wire, then length of wire increase by 3 mm then find out work done :

(1) 4.5×10^{-3} J (2) 7.3×10^{-3} J

(3) 9.3×10^{-2} J (4) 2.5×10^{-2} J

PPF013

14. If the strain in a wire is not more than 1/1000 and Y = 2×10^{11} N/m^2, Diameter of wire is 1mm. The maximum weight hung from the wire is:-

(1) 110 N (2) 125 N

(3) 157 N (4) 168 N

PPF014

15. When a tension F is applied in uniform wire of length ℓ and radius r, the elongation produced is e. When tension 2F is applied, the elongation produced in another uniform wire of length 2ℓ and radius 2r made of same material is :–

(1) 0.5 e (2) 1.0 e (3) 1.5 e (4) 2.0 e

PPF015

16. How much force is required to produce an increase of 0.2% in the length of a brass wire of diameter 0.6 mm ?

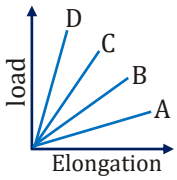
[Young's modulus for brass = 0.9×10^{11} N/m^2]

(1) Nearly 17 N (2) Nearly 34 N

(3) Nearly 51 N (4) Nearly 68 N

PPF016

17. The load versus elongation graph for four wires of the same material and same length is shown in the figure. The thinnest wire is represented by the line.



(1) OA

(2) OB

(3) OC

(4) OD

PPF017

18. Cross section area of a steel wire (Y = 2.0×10^{11} N/m^2) is 0.1 cm^2. The required force, to make its length double will be –

(1) 2×10^{12}N (2) 2×10^{11}N

(3) 2×10^{10}N (4) 2×10^{6}N

PPF018

19. The diameter of a brass rod is 4 mm and Young's modulus of brass is 9×10^{10} N/m^2. The force required to stretch by 0.1% of its length is :

(1) 360π N (2) 36 N

(3) $144\pi \times 10^3$ N (4) $36\pi \times 10^5$ N

PPF019

20. Poisson's ratio can not have the value :

(1) 0.1 (2) 0.7 (3) 0.2 (4) 0.5

PPF020

21. Two wires of the same length and material but different radii r_1 and r_2 are suspended from a rigid support both carry the same load at the lower end. The ratio of the stress developed in the second wire to that developed in the first wire is –

(1) $\frac{r_1}{r_2}$ (2) $\frac{r_1^2}{r_2^2}$

(3) $\left(\frac{r_1}{r_2}\right)^{5/2}$ (4) $\left(\frac{r_1}{r_2}\right)^{1/2}$

PPF021

22. An increases in pressure required to decreases the 200 litres volume of a liquid by 0.004% in container is :

(Bulk modulus of the liquid = 2100 MPa)

(1) 188 kPa (2) 8.4 kPa

(3) 18.8 kPa (4) 84 kPa

PPF022

23. If 'S' is stress and 'Y' is Young's modulus of material of a wire, the energy stored in the wire per unit volume is

(1) $\frac{S}{2Y}$ (2) $\frac{2Y}{S^2}$ (3) $\frac{S^2}{2Y}$ (4) $2S^2Y$

PPF023

24. For a constant hydraulic stress on an object, the fractional change in the object's volume $\left(\frac{\Delta V}{V}\right)$ and its bulk modulus (B) are related as :

(1) $\frac{\Delta V}{V} \propto B$ (2) $\frac{\Delta V}{V} \propto \frac{1}{B}$

(3) $\frac{\Delta V}{V} \propto B^2$ (4) $\frac{\Delta V}{V} \propto B^{-2}$

PPF024

25. A wire of length L and cross-sectional area A is made of a material of Young's modulus Y. The work done in stretching the wire by an amount x is given by :-

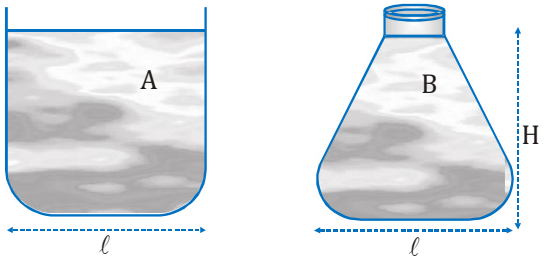
(1) $\frac{YAx^2}{L}$ (2) $\frac{YAx^2}{2L}$

(3) $\frac{YAL^2}{x}$ (4) $\frac{YAL^2}{2x}$

PPF025

FLUIDS STATICS

26. Two vessels A and B have the same base area and contain water to the same height, but the mass of water in A is four times that in B. The ratio of the liquid thrust at the base of A to that at the base of B is :-



(1) 4 : 1 (2) 2 : 1

(3) 1 : 1 (4) 16 : 1

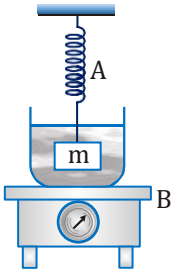
PPF026

27. A cylinder is filled with a liquid of density d upto a height h. If the beaker is at rest, then the mean pressure on the wall is :-

(1) Zero (2) hdg (3) $\frac{h}{2}$dg (4) 2 hdg

PPF027

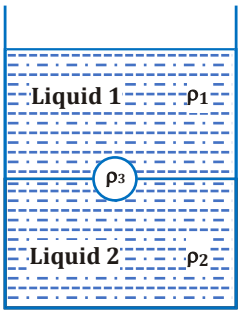
28. The spring balance A read 2 kg. with a block m suspended from it. A balance B reads 5 kg. when a beaker with liquid is put on the pan of the balance. The two balances are now so arranged that the hanging mass is inside the liquid in the beaker as shown in fig. In this situation :–



(1) The balance A will read more than 2 kg
(2) The balance B will read more than 5 kg
(3) The balance A will read less than 2 kg and B will read more than 5 kg
(4) The balance A and B will read 2 kg and 5 kg. respectively

PPF028

29. A jar is filled with two non-mixing liquids 1 and 2 having densities and, respectively.



A solid ball, made of a material of density ρ_3, is dropped in the jar. It comes to equilibrium in the position shown in the figure. Which of the following is true for ρ_1, ρ_2 & ρ_3

(1) $\rho_3 < \rho_1 < \rho_2$ (2) $\rho_1 > \rho_3 > \rho_2$

(3) $\rho_1 < \rho_2 < \rho_3$ (4) $\rho_1 < \rho_3 < \rho_2$

PPF029

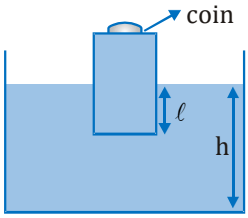
30. A boat having a length of 3 metre and breadth 2 metre is floating on a lake. The boat sinks by one cm when a man gets on it. The mass of the man is
(1) 60 kg (2) 62 kg
(3) 72 kg (4) 128 kg

PPF030

31. If the density of a block is 981 kg/m^3 then it shall
(1) Sink in water
(2) float with some part emmersed in water
(3) float completely immersed in water
(4) float completely out of water.

PPF031

32. A wooden block, with a coin placed on its top, floats in water as shown in figure. The distance ℓ and h are shown there. After sometime the coin falls into the water. Then :–



(1) ℓ decreases and h increases
(2) ℓ increases and h decreases
(3) both ℓ and h increase
(4) both ℓ and h decrease

PPF032

33. A piece of ice is floating in a jar containing water. When the ice melts, then the level of water :-
(1) Rises
(2) Falls
(3) Remains unchanged
(4) Changes erratically

PPF033

34. A sample of metal weights 210 gram in air, 180 gram in water and 120 gram in an unknown liquid. Then :-
(1) the density of metal is 3 g/cm^3
(2) the density of metal is 7 g/cm^3
(3) density of metal is 4 times the density of the unknown liquid
(4) the metal will float in water

PPF034

35. 'Torr' is the unit of :-
(1) Pressure (2) Density
(3) Volume (4) Flux

PPF035

36. A sphere is floating in water its 1/3rd part is outside the water and when sphere is floating in unknown liquid, its $\frac{3}{4}$th part is outside the liquid then density of liquid is
(1) 4/9 gm/c.c. (2) 9/4 gm/c.c.
(3) 8/3 gm/c.c. (4) 3/8 gm/c.c.

PPF036

37. Which of the following works on Pascal's law ?
(1) Sprayer (2) Venturimeter
(3) Hydraulic lift (4) Barometer

PPF037

38. An object of weight W and density ρ is submerged in a fluid of density ρ_1. Its appearent weight will be
(1) $W(\rho-\rho_1)$ (2) $\frac{(\rho-\rho_1)}{W}$
(3) $W\left(1-\frac{\rho_1}{\rho}\right)$ (4) $W(\rho_1-\rho)$

PPF038

39. Which law states that the magnitude of pressure within fluid is equal in all parts ?
(1) Pascal's law (2) Gay-Lusac's law
(3) Dalton's law (4) Boyle's law

PPF039

40. A body measures 5 N in air and 2 N when put in water. The buoyant force is
(1) 7 N (2) 9 N
(3) 3 N (4) None of these

PPF040

41. Hydraulic press is based upon
(1) Archimede's principle
(2) Bernoulli's theorem
(3) Pascal's law
(4) Zurin's law

PPF041

42. A wooden block is taken to the bottom of a lake of water and then released. it rise up with a
(1) Constant acceleration
(2) Decreasing acceleration
(3) Constant velocity
(4) Decreasing velocity

PPF042

FLUIDS DYNAMICS

43. Two water pipes P and Q having diameter 2×10^{-2} m and 4×10^{-2} m respectively are joined in series with the main supply line of water. The velocity of water flowing in pipe P is

(1) 4 times that of Q (2) 2 times that of Q

(3) $\frac{1}{2}$ times that of Q (4) $\frac{1}{4}$ times that of Q

PPF043

44. The cylindrical tube of a spray pump has a radius R, one end of which has n fine holes, each of radius r. If the speed of flow of the liquid in the tube is v, the speed of ejection of the liquid through the hole is :-

(1) $\frac{v}{n}\left(\frac{R}{r}\right)$ (2) $\frac{v}{n}\left(\frac{R}{r}\right)^{1/2}$

(3) $\frac{v}{n}\left(\frac{R}{r}\right)^{3/2}$ (4) $\frac{v}{n}\left(\frac{R}{r}\right)^{2}$

PPF044

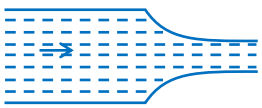
45. Water from a tap emerges vertically downwards with an initial speed of 1.0 m/s. The cross–sectional area of tap is 10^{-4} m^2. Assume that the pressure is constant throughout the stream of water and that the flow is steady, the cross–sectional area of stream 0.15 m below the tap is :–

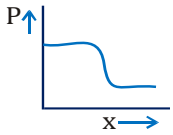
(1) 5.0×10^{-4} m^2 (2) 1.0×10^{-4} m^2

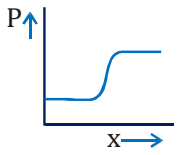
(3) 5.0×10^{-5} m^2 (4) 2.0×10^{-5} m^2

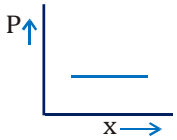
PPF045

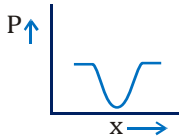
46. Water flows through a frictionless duct with a cross-section varying as shown in figure. Pressure P at points along the axis is represented by



(1) 

(2) 

(3) 

(4) 

PPF046

47. A tank of height 5 m is full of water. There is a hole of cross sectional area 1 cm^2 in its bottom. The initial volume of water that will come out from this hole per second is

(1) 10^{-3} m^3/s (2) 10^{-4} m^3/s

(3) 10 m^3/s (4) 10^{-2} m^3/s.

PPF047

48. The pressure of water in a water pipe when tap is opened and closed is respectively 3×10^5 N/m^2 and 3.5×10^5 N/m^2. With open tap, the velocity of water flowing is

(1) 10 m/s (2) 5 m/s

(3) 20 m/s (4) 15 m/s

PPF048

49. The flow speeds of air on the lower and upper surfaces of the wing of an aeroplane are v and $\sqrt{2}$ v respectively. The density of air is ρ and surface area of wing is A. The dynamic lift on the wing is :

(1) $\rho v^2 A$ (2) $\sqrt{2}\,\rho v^2 A$

(3) $(1/2)\,\rho v^2 A$ (4) $2\rho v^2 A$

PPF049

50. An incompressible fluid flows steadily through a cylindrical pipe which has radius 2 R at point A and radius R at point B farther along the flow direction. If the velocity at point A is v, its velocity at point B is:-

(1) 2v (2) v (3) $\frac{v}{2}$ (4) 4v

PPF050

51. Water is flowing through a non-uniform radius tube. If ratio of the radius of entry and exit end of the pipe is 3 : 2 then the ratio of velocities of entring and exit liquid is :-

(1) 4 : 9 (2) 9 : 4 (3) 8 : 27 (4) 1 : 1

PPF051

52. An aeroplane of mass 3×10^4 kg and total wing area of 120 m^2 is in a level flight at some height. The difference in pressure between the upper and lower surfaces of its wings in kilopascals is ($g = 10 m/s^2$)

(1) 2.5 (2) 5.0 (3) 10.0 (4) 12.5

PPF052

53. Scent sprayer is based on
(1) Charle's law
(2) Archimede's principle
(3) Boyle's law
(4) Bernoulli's theorem

PPF053

54. Bernoulli's equation for steady, non-viscous, in-compressible flow expresses the
(1) Conservation of angular momentum
(2) Conservation of density
(3) Conservation of momentum
(4) Conservation of mechanical energy.

PPF054

55. Application of Bernoulli's theorem can be seen in
(1) Dynamic lift to aeroplane
(2) Hydraulic press
(3) Speed Boat
(4) None of these

PPF055

56. The velocity of water flowing in a non-uniform tube is 20 cm/s at a point where the tube radius is 0.2 cm. The velocity at another point, where the radius is 0.1 cm is
(1) 80cm/s (2) 40cm/s
(3) 20cm/s (4) 5cm/s

PPF056

VISCOSITY

57. A small drop of water falls from rest through a large height h in air. The final velocity is
(1) almost independent of h
(2) proportional to $\sqrt{h}$
(3) proportional to h
(4) inversely proportional to h

PPF057

58. Two drops of equal radius are falling through air with a steady velocity of 5 cm/s. If the two drops coalesce, then its terminal velocity will be -
(1) $4^{1/3} \times 5 cm/s$ (2) $4^{1/3}$ cm/s
(3) $5^{1/3} \times 4 cm/s$ (4) $4^{2/3} \times 5 cm/s$

PPF058

59. If the terminal speed of a sphere of gold (density = 19.5 kg/m^3) is 0.2 m/s in a viscous liquid (density = 1.5 kg/m^3), find the terminal speed of a sphere of silver (density = 10.5 kg/m^3) of the same size in the same liquid.
(1) 0.4 m/s (2) 0.133 m/s
(3) 0.1 m/s (4) 0.2 m/s

PPF059

60. Speed of 2 cm radius ball in a viscous liquid is 20 cm/s. Then the speed of 1 cm radius ball in the same liquid is
(1) 5 cm/s (2) 10 cm/s
(3) 40 cm/s (4) 80 cm/s

PPF060

61. The velocity of falling rain drop attain limited value because of
(1) surface tension
(2) upthrust due to air
(3) viscous force exerted by air
(4) air current

PPF061

62. Poise is the unit of
(1) Pressure (2) Friction
(3) Surface tension (4) Viscosity

PPF062

63. Two rain drops falling through air have radii in the ratio 1:2. They will have terminal velocity in the ratio.
(1) 4 : 1 (2) 1 : 4 (3) 2 : 1 (4) 1 : 2

PPF063

64. A sphere of mass M and radius R is falling in a viscous fluid. The terminal velocity attained by the falling object will be proportional to
(1) MR^2 (2) M/R
(3) MR (4) M/R^2

PPF064

65. A drop of water of radius 0.0015 mm is falling in air. If the coefficient of viscosity of air is 2.0×10^{-5} kg / (m-s), the terminal velocity of the drop will be (The density of water = 1.0×10^3 kg/m^3 and g = 10 m/s^2)
(1) 1.0×10^{-4} m/s (2) 2.0×10^{-4} m/s
(3) 2.5×10^{-4} m/s (4) 5.0×10^{-4} m/s

PPF065

SURFACE TENSION

66. Spiders and insects move and run about on the surface of water without sinking because :
(1) Elastic membrane is formed on water due to propery of surface tension
(2) Spiders and insects are ligther
(3) Spiders and insects swim on water
(4) Spiders and insects experience up-thrust

PPF066

67. A thin liquid film formed between a U-shaped wire and a light slider supports a weight of 1.5×10^{-2} N (see figure). The length of the slider is 30 cm and its weight negligible. The surface tension of the liquid film is :-
(1) 0.025 N/m
(2) 0.0125 N/m
(3) 0.1 N/m
(4) 0.05 N/m



PPF067

68. A liquid drop of diameter D breaks into 27 tiny drops. The resultant change in energy is –
(1) $2\pi TD^2$ (2) $4\pi TD^2$
(3) πTD^2 (4) None of these

PPF068

69. The excess pressure inside an air bubble of radius r just below the surface of water is p_1. The excess pressure inside a drop of the same radius just outside the surface is p_2. If T is surface tension, then
(1) $p_1 = 2p_2$ (2) $p_1 = p_2$
(3) $p_2 = 2p_1$ (4) $p_2 = 0, p_1 \neq 0$

PPF069

70. A water drop is divided into 8 equal droplets. The pressure difference between inner and outer sides of the big drop
(1) will be the same as for smaller droplet
(2) will be half of that for smaller droplet
(3) will be one-forth of that for smaller droplet
(4) will be twice of that for smaller droplet.

PPF070

71. A false statement is :
(1) Angle of contact $\theta < 90°$, if cohesive force < adhesive force $\times \sqrt{2}$
(2) Angle of contact $\theta > 90°$, if cohesive force > adhesive force $\times \sqrt{2}$
(3) Angle of contact $\theta = 90°$, if cohesive force = adhesive force $\times \sqrt{2}$
(4) If the radius of capillary is reduced to half, the rise of liquid column becomes four times

PPF071

72. If a capillary of radius r is dipped in water, the height of water that rises in it is h and its mass is M. If the radius of the capillary is doubled the mass of water that rises in the capillary will be
(1) 4M (2) 2M (3) M (4) $\frac{M}{2}$

PPF072

73. On dipping a capillary of radius 'r' in water, water rises upto a height H and potential energy of water is u_1. If a capillary of radius 2r is dipped in water, then the potential energy is u_2. The ratio $\frac{u_1}{u_2}$ is
(1) 2 : 1 (2) 1 : 2 (3) 4 : 1 (4) 1 : 1

PPF073

74. A vessel, whose bottom has round holes with diameter of 0.1 mm, is filled with water. The maximum height to which the water can be filled without leakage is :–
(S.T. of water = 75 dyne/cm, g = 1000 cm/s^2)
(1) 100 cm (2) 75 cm
(3) 50 cm (4) 30 cm

PPF074

75. In a surface tension experiment with a capillary tube water rises up to 0.1 m. If the same experiment is repeated on an artificial satellite which is revolving round the earth, water will rise in the capillary tube up to a height of
(1) 0.1 m
(2) 0.98 m
(3) 9.8 m
(4) full length of capillary tube

PPF075

76. A soap bubble in vacuum has a radius of 3 cm and another soap bubble in vacuum has a radius of 4 cm. If the two bubbles coalesce under isothermal condition, then the radius of the new bubble is :
(1) 2.3 cm (2) 4.5 cm (3) 5 cm (4) 7 cm
PPF076

77. The spherical shape of rain-drop is due to
(1) Density of the liquid
(2) Surface tension
(3) Atmospheric pressure
(4) Gravity
PPF077

78. In a capillary tube, water rises by 1.2 mm. The height of water that will rise in another capillary tube having half the radius of the first, is :
(1) 1.2 mm (2) 2.4 mm
(3) 0.6 mm (4) 0.4 mm
PPF078

79. Water rises to a height h in a capillary at the surface of earth. On the surface of the moon the height of water column in the same capillary will be :
(1) 6h (2) 1/6 h
(3) h (4) Zero
PPF079

80. Shape of meniscus for a liquid of zero angle of contact is -
(1) plane (2) parabolic
(3) hemi-spherical (4) cylindrical
PPF080

81. Due to capillary action a liquid will rise in a tube if angle of contact is
(1) acute (2) obtuse
(3) 90° (4) 180°
PPF081

82. Two droplets merge with each other and form a large droplet. In this process :
(1) Energy is liberated
(2) Energy is absorbed
(3) Neither liberated nor absorbed
(4) Some mass is converted into energy
PPF082

83. Two capillary tubes of same diameter are put vertically one each in two liquids whose relative densities are 0.8 and 0.6 and surface tensions are 60 dyne/cm and 50 dyne/cm respectively. Ratio of heights of liquids in the two tubes h_1/h_2 is :
(1) $\frac{10}{9}$ (2) $\frac{3}{10}$ (3) $\frac{10}{3}$ (4) $\frac{9}{10}$
PPF083

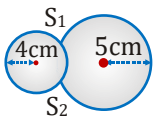
84. The property utilized in the manufacture of lead shots is :
(1) Specific weight of liquid lead
(2) Specific gravity of liquid lead
(3) Compressibility of liquid lead
(4) Surface tension of liquid lead
PPF084

85. Surface tension of a liquid is 5 N/m. If its thin film is made in a ring of area 0.02 m^2, then its surface energy will be –
(1) 5×10^{-2} Joule (2) 2.5×10^{-2} Joule
(3) 3×10^{-1} Joule (4) 2×10^{-1} Joule
PPF085

86. If one end of capillary tube is dipped into water then water rises up to 3cm. If the surface tension of water is 75×10^{-3} N/m then the diameter of capillary tube will be
(1) 0.1 mm (2) 0.5 mm
(3) 1 mm (4) 2 mm
PPF086

87. If the surface tension of a liquid is T and its surface area is increased by A, then the surface energy of that surface will be increased by -
(1) AT (2) A/T (3) A^2T (4) A^2T^2
PPF087

88. Two soap bubbles of radii r_1 and r_2 equal to 4cm and 5 cm are touching each other over a common surface S_1S_2 (shown in figure). Its radius will be :-
(1) 4 cm.
(2) 20 cm.
(3) 5 cm.
(4) 4.5 cm.



PPF088

89. The radius of a soap bubble is r. The surface tension of soap solution is T. Keeping temperature constant, the radius of the soap bubble is doubled, the energy necessary for this will be

(1) $24\pi r^2 T$ (2) $8\pi r^2 T$

(3) $12\pi r^2 T$ (4) $16\pi r^2 T$

PPF089

90. A liquid does not wet the sides of a solid, if the angle of contact is

(1) Zero

(2) Obtuse (more than 90°)

(3) Acute (less than 90°)

(4) 45°

PPF090

91. The excess of pressure inside a soap bubble than that of the outer pressure is :

(1) $\frac{2T}{r}$ (2) $\frac{4T}{r}$ (3) $\frac{T}{2r}$ (4) $\frac{T}{r}$

PPF091

92. In a capillary tube experiment, a vertical 30 cm long capillary tube is dipped in water. The water rises up to a height of 10 cm due to capillary action. If this experiment is conducted in a freely falling elevator, the length of the water column becomes :

(1) 10 cm (2) 20 cm (3) 30 cm (4) Zero

PPF092

93. Radius of a capillary is 2×10^{-3} m. A liquid of weight 6.2×10^{-4} N may remain in the capillary. Then the surface tension of liquid will be :

(1) 5×10^{-3} N/m (2) 5×10^{-2} N/m

(3) 5 N/m (4) 50 N/m

PPF093

94. A capillary tube of radius r can support a liquid of weight 6.28×10^{-4} N. If the surface tension of the liquid is 5×10^{-2} N/m. The radius of capillary must be :-

(1) 2×10^{-3} m (2) 2×10^{-4} m

(3) 1.5×10^{-3} m (4) 12.5×10^{-4} m

PPF094

95. Water rise in a capillary upto an extension height such that upward force of surface tension balances the force of 75×10^{-4} N. due to weight of water. If surface tension of water is 6×10^{-2} N/m. The internal circumference of the capillary must be :-

(1) 12.5×10^{-2} m (2) 6.5×10^{-2} m

(3) 0.5×10^{-2} m (4) 1.25×10^{-2} m

PPF095

96. Two small drops of mercury, each of radius R, coalesce to form a single large drop. The ratio of the total surface energies before and after the change is :–

(1) $1 : 2^{1/3}$ (2) $2^{1/3} : 1$

(3) 2 : 1 (4) 1 : 2

PPF096

97. Inside a drop excess pressure is maximum in :-

(1) 0.200 μm diameter

(2) 20.0 μm diameter

(3) 200 μm diameter

(4) 2.0 μm diameter

PPF097

98. The diameter of one drop of water is 0.2 cm. The work done in breaking one drop into 1000 equal droplets will be :-

(surface tension of water = 7×10^{-2} N/m)

(1) 7.9×10^{-6} J (2) 5.92×10^{-6} J

(3) 2.92×10^{-6} J (4) 1.92×10^{-6} J

PPF098

99. If two bubble of radii 0.03 cm and 0.04 cm come in contact with each other then the radius of curvature of the common surface 'r' is given by.

(1) 0.03 cm (2) 0.06 cm

(3) 0.12 cm (4) 0.24 cm

PPF099

100. Work done in forming a soap bubble of radius R is W. Then work done is forming a soap bubble of radius '2R' will be :

(1) 2W (2) 4W (3) W/2 (3) W/4

PPF100

101. At which angle liquid will not wet solid.

(1) Zero (2) acute

(3) 45° (4) obtuse

PPF101

102. Internal radius of a capillary tube is $\frac{1}{28}$ cm and surface tension of water 70 dyne/cm, if angle of contact is zero, then water will rise up in the tube up to height.

(1) 4 cm (2) 2 cm

(3) 14 cm (4) 18 cm

PPF102

103. Area of liquid film is 6 × 10 cm^2 and surface tension is T = 20 dyne/cm, what is the work done to change area up to 12 × 10 cm^2 :

(1) 120 joule (2) 120 erg

(3) 1200 joule (4) 2400 erg

PPF103

104. The work done in blowing a soap bubble of radius 0.2 m, given that the surface tension of soap solution is 60 × 10^{-3} N/m, is

(1) $24\pi \times 10^{-4}$ J

(2) $8\pi \times 10^{-4}$ J

(3) $96\pi \times 10^{-4}$ J

(4) $192\pi \times 10^{-4}$ J

PPF104

105. Adding detergents to water helps in removing dirty greasy stains. This is because

(a) It increases the oil-water surface tension

(b) It decreases the oil-water surface tension

(c) It increases the viscosity of the solution

(d) Dirt is held suspended surrounded by detergent molecules

(1) (b) and (d) (2) (a) only

(3) (c) and (d) (4) (d) only

PPF105

106. The excess pressure inside a soap bubble A is twice that in another soap bubble B. The ratio of volumes of A and B is

(1) 1 : 2 (2) 1 : 4 (3) 1 : 8 (4) 1 : 16

PPF106

107. Consider a soap film on a rectangular frame of wire of area 4× 4 cm^2. If the area of the soap film is increased to 4 × 5cm^2, the work done in the process will be (The surface tension of the soap film is 3 × 10^{-2} N/m)

(1) 12 × 10^{-6} J (2) 24 × 10^{-6} J

(3) 60 × 10^{-6} J (4) 96 × 10^{-6} J

PPF107

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	2	3	1	1	3	1	2	1	4	1	1	2	3	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	1	4	1	2	2	4	3	2	2	3	3	3	4	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	4	3	2	1	3	3	3	1	3	3	1	1	4	3
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	1	1	1	3	4	1	1	4	4	1	1	1	1	3	1
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	3	4	2	2	3	1	1	1	2	2	4	2	4	4	4
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	3	2	2	1	3	1	1	4	4	4	3	1	2	1	2
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	2	3	2	1	1	2	1	1	3	2	4	1	4	4	1
Question	106	107													
Answer	3	2													

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. Copper of fixed volume 'V; is drawn into wire of length 'ℓ'. When this wire is subjected to a constant force 'F', the extension produced in the wire is '$\Delta\ell$'. Which of the following graphs is a straight line ?

(1) $\Delta\ell$ versus $\frac{1}{\ell}$ (2) $\Delta\ell$ versus ℓ^2

(3) $\Delta\ell$ versus $\frac{1}{\ell^2}$ (4) $\Delta\ell$ versus ℓ

PPF108

2. A certain number of spherical drops of a liquid of radius 'r' coalesce to form a single drop of radius 'R' and volume 'V'. If 'T' is the surface tension of the liquid, then :

(1) energy = 4VT $\left(\frac{1}{r}-\frac{1}{R}\right)$ is released

(2) energy = 3VT $\left(\frac{1}{r}+\frac{1}{R}\right)$ is absorbed

(3) energy = 3VT $\left(\frac{1}{r}-\frac{1}{R}\right)$ is released

(4) Energy is neither released nor absorbed

PPF109

AIPMT 2015

3. The approximate depth of an ocean is 2700 m. The compressibility of water is 45.4×10^{-11} Pa^{-1} and density of water is 10^3 kg/m^3. What fractional compression of water will be obtained at the bottom of the ocean ?

(1) 1.0×10^{-2} (2) 1.2×10^{-2}

(3) 1.4×10^{-2} (4) 0.8×10^{-2}

PPF110

4. A wind with speed 40 m/s blows parallel to the roof of a house. The area of the roof is 250 m^2. Assuming that the pressure inside the house is atmospheric pressure, the force exerted by the wind on the roof and the direction of the force will be:

(ρ_{air} = 1.2 kg/m^3)

(1) 4.8×10^5 N, upwards

(2) 2.4×10^5 N, upwards

(3) 2.4×10^5 N, downwards

(4) 4.8×10^5 N, downwards

PPF111

Re-AIPMT 2015

5. The cylindrical tube of a spray pump has radius R, one end of which has n fine holes, each of radius r. If the speed of the liquid in the tube is V, the speed of the ejection of the liquid through the holes is :-

(1) $\frac{V^2R}{nr}$ (2) $\frac{VR^2}{n^2r^2}$ (3) $\frac{VR^2}{nr^2}$ (4) $\frac{VR^2}{n^3r^2}$

PPF112

6. The Young's modulus of steel is twice that of brass. Two wires of same length and of same area of cross section, one of steel and another of brass are suspended from the same roof. If we want the lower ends of the wires to be at the same level, then the weights added to the steel and brass wires must be in the ratio of :

(1) 1 : 1 (2) 1 : 2 (3) 2 : 1 (4) 4 : 1

PPF113

7. Water rises to height 'h' in capillary tube. If the length of capillary tube above the surface of water is made less than 'h', then -

(1) water does not rise at all.

(2) water rises upto the tip of capillary tube and then starts overflowing like a fountain.

(3) water rises upto the top of capillary tube and stays there without overflowing.

(4) water rises upto a point a little below the top and stays there.

PPF114

8. The heart of a man pumps 5 litres of blood through the arteries per minute at a pressure of 150 mm of mercury. If the density of mercury be 13.6×10^3 kg/m^3 and g = 10m/s^2 then the power of heart in watt is:

(1) 1.50 (2) 1.70 (3) 2.35 (4) 3.0

PPF115

NEET-I 2016

9. Two non-mixing liquids of densities ρ and $n\rho$ (n > 1) are put in a container. The height of each liquid is h. A solid cylinder of length L and density d is put in this container. The cylinder floats with its axis vertical and length pL(p < 1) in the denser liquid. The density d is equal to :-

(1) $\{1 + (n + 1)p\}\rho$ (2) $\{2 + (n + 1)p\}\rho$

(3) $\{2 + (n - 1)p\}\rho$ (4) $\{1 + (n - 1)p\}\rho$

PPF116

NEET-II 2016

10. A rectangular film of liquid is extended from (4 cm × 2 cm) to (5 cm × 4 cm). If the work done is 3×10^{-4} J, the value of the surface tension of the liquid is :-

(1) 0.2 Nm^{-1} (2) 8.0 Nm^{-1}

(3) 0.250 Nm^{-1} (4) 0.125 Nm^{-1}

PPF117

11. Three liquids of densities ρ_1, ρ_2 and ρ_3 (with $\rho_1 > \rho_2 > \rho_3$), having the same value of surface tension T, rise to the same height in three identical capillaries. The angles of contact θ_1, θ_2 and θ_3 obey:-

(1) $\frac{\pi}{2} < \theta_1 < \theta_2 < \theta_3 < \pi$

(2) $\pi > \theta_1 > \theta_2 > \theta_3 > \frac{\pi}{2}$

(3) $\frac{\pi}{2} > \theta_1 > \theta_2 > \theta_3 \geq 0$

(4) $0 \leq \theta_1 < \theta_2 < \theta_3 < \frac{\pi}{2}$

PPF118

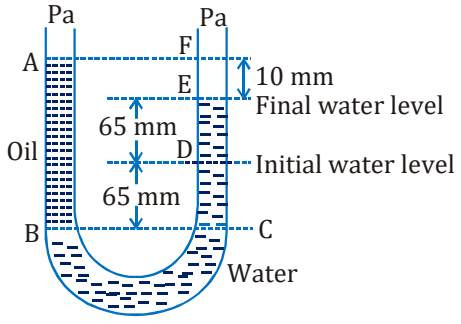
NEET(UG) 2017

12. The bulk modulus of a spherical object is 'B'. If it is subjected to uniform pressure 'p', the fractional decrease in radius is :-

(1) $\frac{B}{3p}$ (2) $\frac{3p}{B}$ (3) $\frac{p}{3B}$ (4) $\frac{p}{B}$

PPF119

13. A U tube with both ends open to the atmosphere, is partially filled with water. Oil, which is immiscible with water, is poured into one side until it stands at a distance of 10 mm above the water level on the other side. Meanwhile the water rises by 65 mm from its original level (see diagram). The density of the oil is :-



(1) 425 kg m^{-3} (2) 800 kg m^{-3}

(3) 928 kg m^{-3} (4) 650 kg m^{-3}

PPF120

NEET(UG) 2018

14. A small sphere of radius 'r' falls from rest in a viscous liquid. As a result, heat is produced due to viscous force. The rate of production of heat when the sphere attains its terminal velocity, is proportional to :-

(1) r^3 (2) r^2 (3) r^5 (4) r^4

PPF121

15. Two wires are made of the same material and have the same volume. The first wire has cross-sectional area A and the second wire has cross-sectional area 3A. If the length of the first wire is increased by Δl on applying a force F, how much force is needed to stretch the second wire by the same amount ?

(1) 9F (2) 6F (3) 4F (4) F

PPF122

NEET(UG) 2019

16. A soap bubble, having radius of 1 mm, is blown from a detergent solution having a surface tension of 2.5×10^{-2} N/m. The pressure inside the bubble equals at a point Z_0 below the free surface of water in a container. Taking g = 10 m/s^2 density of water = 10^3 kg/m^3, the value of Z_0 is :-

(1) 100 cm (2) 10 cm

(3) 1 cm (4) 0.5 cm

PPF123

17. When a block of mass M is suspended by a long wire of length L, the length of the wire become (L+*l*). The elastic potential energy stored in the extended wire is :-

(1) Mg*l* (2) MgL

(3) $\frac{1}{2}$ Mg*l* (4) $\frac{1}{2}$ MgL

PPF124

18. A small hole of area of cross-section 2 mm^2 is present near the bottom of a fully filled open tank of height 2 m. Taking g = 10 m/s^2, the rate of flow of water through the open hole would be nearly :

(1) 12.6×10^{-6} m^3/s (2) 8.9×10^{-6} m^3/s

(3) 2.23×10^{-6} m^3/s (4) 6.4×10^{-6} m^3/s

PPF125

NEET(UG) 2019 (Odisha)

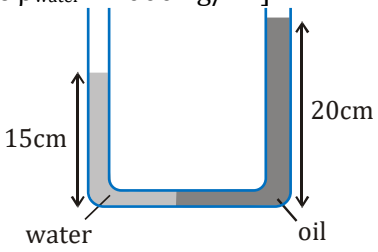
19. Two small spherical metal balls, having equal masses, are made from materials of densities ρ_1 and ρ_2($\rho_1 = 8\rho_2$) and have radii of 1mm and 2mm, respectively. They are made to fall vertically (from rest) in a viscous medium whose coefficient of viscosity equals η and whose density is 0.1 ρ_2. The ratio of their terminal velocities would be :-

(1) $\frac{79}{72}$ (2) $\frac{19}{36}$ (3) $\frac{39}{72}$ (4) $\frac{79}{36}$

PPF126

20. In a U-tube as shown in figure, water and oil are in the left side and right side of the tube respectively. The heights from the bottom for water and oil columns are 15 cm and 20 cm respectively. The density of the oil is :-

[take ρ_{water} = 1000 kg/m^3]



(1) 1200 kg/m^3 (2) 750 kg/m^3
(3) 1000 kg/m^3 (4) 1333 kg/m^3

PPF127

21. The stress-strain curves are drawn for two different materials X and Y. It is observed that the ultimate strength point and the fracture point are close to each other for material X but are far apart for material Y. We can say that materials X and Y are likely to be (respectively)

(1) ductile and brittle
(2) brittle and ductile
(3) brittle and plastic
(4) plastic and ductile

PPF128

NEET (UG) 2020

22. A capillary tube of radius r is immersed in water and water rises in it to a height h. The mass of the water in the capillary is 5g. Another capillary tube of radius 2r is immersed in water. The mass of water that will rise in this tube is :

(1) 20.0 g (2) 2.5 g
(3) 5.0 g (4) 10.0 g

PPF129

23. A wire of length L, area of cross section A is hanging from a fixed support. The length of the wire changes to L_1 when mass M is suspended from its free end. The expression for Young's modulus is :

(1) $\frac{MgL}{A(L_1 - L)}$ (2) $\frac{MgL_1}{AL}$
(3) $\frac{Mg(L_1 - L)}{AL}$ (4) $\frac{MgL}{AL_1}$

PPF130

NEET (UG) 2020 (Covid-19)

24. A liquid does not wet the solid surface if angle of contact is :

(1) equal to 45° (2) equal to 60°
(3) greater than 90° (4) zero

PPF131

25. A barometer is constructed using a liquid (density = 760 kg/m^3). What would be the height of the liquid column, when a mercury barometer reads 76 cm ?

(density of mercury = 13600 kg/m^3)

(1) 1.36 m (2) 13.6 m
(3) 136 m (4) 0.76 m

PPF132

NEET (UG) 2021

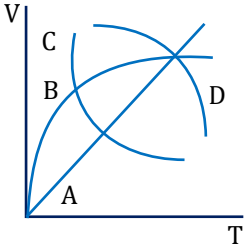
26. The velocity of a small ball of mass M and density d, when dropped in a container filled with glycerine becomes constant after some time. If the density of glycerine is $\frac{d}{2}$, then the viscous force acting on the ball will be :

(1) $\frac{Mg}{2}$ (2) Mg (3) $\frac{3}{2}$Mg (4) 2Mg

PPF133

NEET (UG) 2022

27. A spherical ball is dropped in a long column of a highly viscous liquid. The curve in the graph shown, which represents the speed of the ball (v) as a function of time (t) is :



(1) B (2) C (3) D (4) A

PPF134

28. If a soap bubble expands, the pressure inside the bubble :
(1) increases
(2) remains the same
(3) is equal to the atmospheric pressure
(4) decreases

PPF135

RE-NEET (UG) 2022

29. Two copper vessels A and B have the same base area but of different shapes. A lakes twice the volume of water as that B requires to fill upto a particular common height. Then the correct statement among the following is:
(1) Pressure on the base area of vessels A and B is same.
(2) Pressure on the base area of vessels A and B is not same.
(3) Both vessels A and B weigh the same.
(4) Vessel B weighs twice that of A.

PPF136

30. The terminal velocity of a copper ball of radius 5 mm falling through a tank of oil at room temperature is 10 cm s^{-1}. If the viscosity of oil at room temperature is 0.9 kg m^{-1} s^{-1}, the viscous drag force is :
(1) 8.48×10^{-3} N (2) 8.48×10^{-5} N
(3) 4.23×10^{-3} N (4) 4.23×10^{-6} N

PPF137

NEET (UG) 2023

31. The amount of energy required to form a soap bubble of radius 2 cm from a soap solution is nearly: (surface tension of soap solution = 0.03 N m^{-1})
(1) 5.06×10^{-4} J (2) 3.01×10^{-4} J
(3) 50.1×10^{-4} J (4) 30.16×10^{-4} J

PPF138

32. The venturi-meter works on :
(1) Bernoulli's principle
(2) The principle of parallel axes
(3) The principle of perpendicular axes
(4) Huygen's principle

PPF139

33. Let a wire be suspended from the ceiling (rigid support) and stretched by a weight W attached at its free end. The longitudinal stress at any point of cross-sectional area A of the wire is :
(1) W/A (2) W/2A
(3) Zero (4) 2W/A

PPF140

RE-NEET (UG) 2023

34. The viscous drag acting on a metal sphere of diameter 1 mm, falling through a fluid of viscosity 0.8 Pa s with a velocity of 2 m s^{-1} is equal to :
(1) 15×10^{-3} N (2) 30×10^{-3} N
(3) 1.5×10^{-3} N (4) 20×10^{-3} N

PPF141

35. Which of the following statement is not true ?
(1) Coefficient of viscosity is a scalar quantity
(2) Surface tension is a scalar quantity
(3) Pressure is a vector quantity
(4) Relative density is scalar quantity

PPF142

36. The amount of elastic potential energy per unit volume (in SI unit) of a steel wire of length 100 cm to stretch it by 1 mm is (if Young's modulus of the wire = 2.0×10^{11} Nm^{-2}):
(1) 10^{11} (2) 10^{17}
(3) 10^{7} (4) 10^{5}

PPF143

NEET (UG) 2024

37. The maximum elongation of a steel wire of 1m length if the elastic limit of steel and its Young's modulus, respectively, are 8×10^{8} N m^{-2} and 2×10^{11} N m^{-2} is :
(1) 4 mm (2) 0.4 mm
(3) 40 mm (4) 8 mm

PPF144

38. A thin flat circular disc of radius 4.5 cm is placed gently over the surface of water. If surface tension of water is 0.07 Nm^{-1}, then the excess force required to take it away from the surface is :
(1) 19.8 mN (2) 198 N
(3) 1.98 mN (4) 99 N

PPF145

39. A metallic bar of Young's modulus, 0.5×10^{11} N m^{-2} and coefficient of linear thermal expansion 10^{-5} °C^{-1}, length 1 m and area of cross-section 10^{-3} m^2 is heated from 0°C to 100°C without expansion or bending. The compressive force developed in it is:

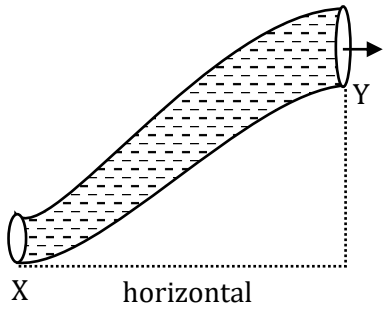
(1) 5×10^3 N (2) 50×10^3 N

(3) 100×10^3 N (4) 2×10^3 N

PPF146

RE-NEET (UG) 2024

40. An ideal fluid is flowing in a non-uniform cross sectional tube XY (as shown in figure) from end X to end Y. If K_1 and K_2 are the kinetic energy per unit volume of the fluid at X and Y respectively then the correct option is :



(1) $K_1 = K_2$

(2) $2K_1 = K_2$

(3) $K_1 > K_2$

(4) $K_1 < K_2$

PPF147

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	3	2	2	3	3	3	2	4	4	4	3	3	3	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	1	4	2	2	4	1	3	2	1	1	4	1	1
Question	31	32	33	34	35	36	37	38	39	40					
Answer	2	1	1	1	3	4	1	1	2	3					

Exercise - III (Analytical Questions) Master Your Understanding

1. Check the correctness of given statements :
Statement-I : Water wets glass capillary while mercury does not wet.
Statement-II : Excess pressure insides the soap bubble is inversely proportion to the radius of the bubble.
(1) Both statements are correct.
(2) Only statement-I is correct.
(3) Only statement-II is correct.
(4) Both statements are wrong.

PPF148

2. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
Assertion (A) : The shape of small liquid drop is spherical.
Reason (R) : Due to surface tension, liquid tends to occupy minimum surface area and for given volume a sphere has minimum surface area.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PPF149

3. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
Assertion (A) : When an ice cube, floating in a glass of water melts, the water level remains unchanged.
Reason (R) : The volume of ice on melting decreases.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PPF150

4. Match the column for followings

Column-A		Column-B	
i.	Bernoulli's theorem	A	$v=\sqrt{2gh}$
ii.	Law of continuity	B	$P+\frac{1}{2}\rho v^2+\rho gh=\text{constant}$
iii.	Torricelli's theorem	C	$Q=\frac{\pi Pr^4}{8\eta\ell}$
iv.	Poiseuilli's theorem	D	$Av=\text{constant}$

(1) i → B, ii → D, iii → A, iv → C
(2) i → B, ii → A, iii → D, iv → C
(3) i → C, ii → D, iii → A, iv → B
(4) i → A, ii → B, iii → C, iv → D

PPF151

5. There is a small hole near the bottom of an open tank filled with a liquid. The speed of the water ejected does not depend on (one or more options may be correct)
(1) Area of the hole
(2) Density of the liquid
(3) Height of the liquid from the hole
(4) Acceleration due to gravity

PPF152

6. A solid floats in a liquid in a partially dipped position. (Select incorrect option)
(1) The solid exerts a force, equal to its weight on the liquid.
(2) The liquid exerts a force of buoyancy on the solid which is equal to the weight of the solid.
(3) The weight of the displaced liquid equals the weight of the solid.
(4) The weight of the dipped part of the solid is equal to the weight of the displaced liquid.

PPF153

7. A copper wire ($Y = 10^{11}$ Nm^{-2}) of length 8m and steel wire ($Y = 2 \times 10^{11}$ Nm^{-2}) of length 4m and each of cross-sectional area 0.5 cm^2 are fastened end to end and stretched with a tension of 500 N. Match the given columns and select the correct option from the codes given below.

Column-I		Column-II	
(i)	Elongation in copper wire (in) mm)	(p)	0.2
(ii)	Elongation in steel wire (in mm)	(q)	1.0
(iii)	Total elongation (in mm)	(r)	0.8

Codes

(1) i – r, ii – p, iii – q

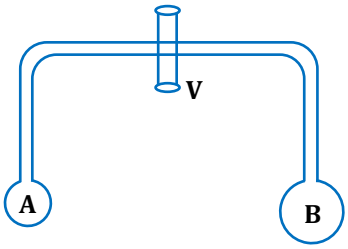
(2) i – p, ii – q, iii – r

(3) i – q, ii – r, iii – p

(4) i – p, ii – r, iii – p

PPF154

8. The valve V in the bent tubes initially kept closed. Two soap bubbles A (smaller) and B (larger) are formed at the two open ends of the tube. Valve V is now opened and air can flow freely between the bubbles.



(1) There will be no change in the size of the bubbles

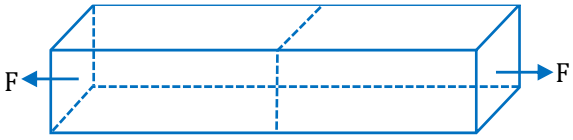
(2) The bubbles will becomes of equal size

(3) A will become smaller and B will become larger

(4) The sizes of A and B will be interchanged

PPF155

9. In the figure shown, forces of equal magnitude are applied to the two ends of a uniform rod. Consider A as the cross-sectional area of the rod. For this situation, mark out the correct statement.



(1) The rod is in compressive stress

(2) The numerical value of stress developed in the rod is equal to F/A

(3) The stress is defined as internal restoring force per unit length

(4) None of the above

PPF156

10. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : A needle placed carefully on the surface of water may float, whereas the ball of the same material will always sink.

Reason (R) : The buoyant force depends on the material and shape of body.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PPF157

11. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Spraying of water in form of small droplets causes cooling.

Reason (R) : For an isolated system, surface energy increase on the expense of internal energy.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PPF158

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12. Given below are two statements :
Statement I : Viscosity of gases is greater than that of liquids.
Statement II : Surface tension of a liquid decreases due to the presence of insoluble impurities.
In the light of the above statements, choose the most appropriate answer from the options given below :
(1) Both statement I and statement II are correct.
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statements I and statements II are incorrect.

PPF159

13. Given below are two statements :
Statement I : For a substance the product of bulk modulus of elasticity and compressibility is 1.
Statement II : The shear modulus for an ideal rigid body is infinite.
In the light of the above statements, choose the most appropriate answer from the options given below :
(1) Both statement I and statement II are correct.
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statements I and statements II are incorrect.

PPF160

14. Match the column:-

Column 1		Column 2	
(i)	For Concave meniscus	**(a)**	angle of contact > 90°
(ii)	For Convex meniscus	**(b)**	angle of contact = 90°
(iii)	For Plane meniscus	**(c)**	angle of contact < 90°

(1) (i) → (c); (ii) → (a); (iii) → (b)
(2) (i) → (a); (ii) → (b); (iii) → (c)
(3) (i) → (a); (ii) → (c); (iii) → (b)
(4) (i) → (b); (ii) → (a); (iii) → (c)

PPF161

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
Answer	1	1	2	1	2	4	1	3	2	3	1	3	1	1	

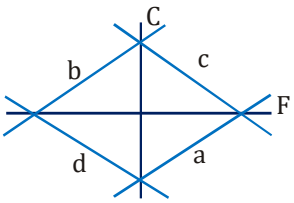
Exercise - I (Conceptual Questions) Build Up Your Understanding

TEMPERATURE & THERMAL EXPANSION

1. At what temperature does the temperature in Celsius and Fahrenheit equalise
(1) –40° (2) 40° (3) 36.6° (4) 38°
PTM001

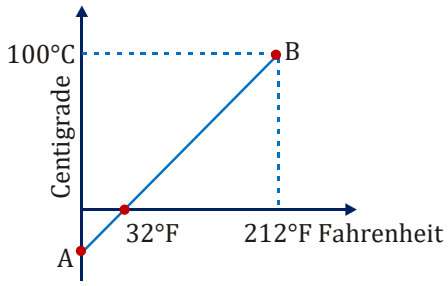
2. A difference of temperature of 25° C is equivalent to a difference of :
(1) 45° F (2) 72° F
(3) 32° F (4) 25° F
PTM002

3. Which of the curves in figure represents the relation between Celsius and Fahrenheit temperature?



(1) Curve a (2) Curve b
(3) Curve c (4) Curve d
PTM003

4. The graph AB shown in figure is a plot of temperature of a body in degree Celsius and degree Fahrenheit. Then



(1) slope of line AB is 9/5
(2) slope of line AB is 5/9
(3) slope of line AB is 1/9
(4) slope of line AB is 3/9
PTM004

5. Oxygen boils at –183°C. This temperature is approximately in Fahrenheit is :-
(1) –329°F (2) –261°F
(3) –215°F (4) –297°F
PTM005

6. Using which of the following instrument, the temperature of the sun can be determined?
(1) Platinum thermometer
(2) Gas thermometer
(3) Pyrometer
(4) Vapour pressure thermometer
PTM006

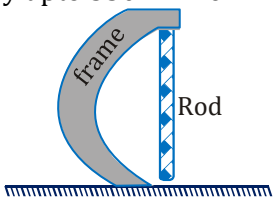
7. Two thermometers X and Y have ice points marked at 15° and 25° and steam points marked as 75° and 125° respectively. When thermometer X measures the temperature of a bath as 60° on it, what would thermometer Y read when it is used to measure the temperature of the same bath?
(1) 60° (2) 75° (3) 100° (4) 90°
PTM007

8. The figure below shows four isotropic solids having positive coefficient of thermal expansion. A student predicts that on heating the solid following things can happen. Mark true (T) or False (F) for comments made by the student.



(i) The angle α in figure (1) will not change.
(ii) The length of line in figure (2) will decrease.
(iii) The radius of inner hole will decrease.
(iv) The distance AB will increase.
(1) T F F T (2) F T T F (3) T T T T (4) F F T F
PTM008

9. At STP a rod is hung from a frame as shown in figure, leaving a small gap between the rod and floor. The frame and rod system is heated uniformly upto 350 K. Then



(1) The rod will never touch the floor in any case.
(2) If $\alpha_{rod} > \alpha_{frame}$, then rod may touch the floor.
(3) If $\alpha_{rod} < \alpha_{frame}$, then rod may touch the floor.
(4) None of the above
PTM009

10. The volume of a metal sphere increases by 0.15% when its temperature is raised by 24°C.The coefficient of linear expansion of metal is :
(1) 2.5×10^{-5}/°C (2) 2.0×10^{-5}/°C
(3) -1.5×10^{-5}/°C (4) 1.2×10^{-5}/°C
PTM010

11. Suppose there is a hole in a copper plate. On heating the plate, diameter of hole, would :
(1) always increase
(2) always decrease
(3) always remain the same
(4) none of these
PTM011

12. The table gives the initial length ℓ_0, change in temperature ΔT and change in length $\Delta\ell$ of four rods. Which rod has greatest coefficient of linear expansion

Rod	ℓ_0(m)	ΔT(°C)	$\Delta\ell$(m)
A_1	1	100	1
A_2	1	100	2
A_3	1.5	50	3
A_4	2.5	20	4

(1) A_1 (2) A_2 (3) A_3 (4) A_4
PTM012

13. An iron bar (Young's modulus = 10^{11} N/m^2, $\alpha = 10^{-6}$ /°C) 1 m long and 10^{-3} m^2 in area is heated from 0°C to 100°C without being allowed to bend or expand. Find the compressive force developed inside the bar.
(1) 10,000 N (2) 1000 N
(3) 5000 N (4) 10^5 N
PTM013

14. A rod of length 2m rests on smooth horizontal floor. If the rod is heated from 0°C to 20°C. Find the longitudinal strain developed?
($\alpha = 5 \times 10^{-5}$/°C)
(1) 10^{-3} (2) 2×10^{-3} (3) Zero (4) None
PTM014

CALORIMETRY

15. A body of mass 5 kg falls from a height of 30 metre. If its all mechanical energy is changed into heat, then heat produced will be:-
(1) 350 cal (2) 150 cal (3) 60 cal (4) 6 cal
PTM015

16. A bullet moving with velocity v collides against wall. consequently half of its kinetic energy is converted into heat. If the whole heat is acquired by the bullet, the rise in temperature will be:–
(1) $v^2/4S$ (2) $4v^2/2S$ (3) $v^2/2S$ (4) v^2/S
PTM016

17. The amount of heat required in converting 1 g ice at –10°C into steam at 100°C will be :–
(1) 3028 J (2) 6056 J (3) 721 J (4) 616 J
PTM017

18. 2 kg ice at – 20°C is mixed with 5 kg water at 20°C. Then final amount of water in the mixture would be;
Given specific heat of ice = 0.5 cal/g°C,
Specific heat of water = 1 cal/g°C,
Latent heat of fusion for ice = 80 cal/g.
(1) 6 kg (2) 5 kg (3) 4 kg (4) 2 kg
PTM018

19. Two identical masses of 5 kg each fall on a wheel from a height of 10m. The wheel disturbs a mass of 2kg water, the rise in temperature of water will be :
(1) 2.6° C (2) 1.2° C
(3) 0.32° C (4) 0.12° C
PTM019

20. A block of mass 2.5 kg is heated to temperature of 500°C and placed on a large ice block. What is the maximum amount of ice that can melt (approx.). Specific heat for the body = 0.1 cal/g°C.
(1) 1 kg (2) 1.5 kg (3) 2 kg (4) 2.5 kg
PTM020

21. 1 kg of ice at –10°C is mixed with 4.4 kg of water at 30°C. The final temperature of mixture is: (specific heat of ice = 2100 J/kg-k)
(1) 2.3°C (2) 4.4°C (3) 5.3°C (4) 8.7°C
PTM021

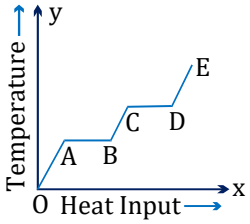
22. The amount of heat required to convert 1 g of ice at 0°C into steam at 100°C, is
(1) 716 cal. (2) 500 cal.
(3) 180 cal. (4) 100 cal.
PTM022

23. The latent heat for vapourisation for 1 g water is 536 cal. Its value in Joule/kg will be :-
(1) 2.25×10^6 (2) 2.25×10^3
(3) 2.25 (4) None of these
PTM023

24. If 10 g ice at 0°C is mixed with 10 g water at 20°C, the final temperature will be :-
(1) 50°C (2) 10°C (3) 0°C (4) 15°C
PTM024

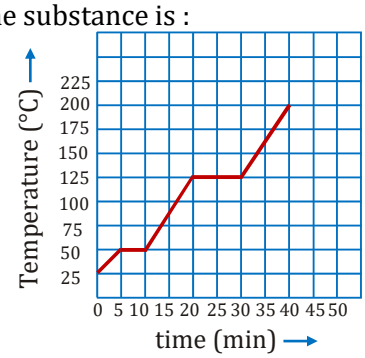
25. 420 joule of energy supplied to 10 g of water will raise its temperature by nearly :-
(1) 1°C (2) 4.2°C (3) 10°C (4) 42°C
PTM025

26. A solid material is supplied with heat at a constant rate. The temperature of material is changing with heat input as shown in the figure. What does slope DE represent.



(1) latent heat of liquid
(2) latent heat of vapour
(3) heat capacity of vapour
(4) inverse of heat capacity of vapour
PTM026

27. The graph shown in the figure represent change in the temperature of 5 kg of a substance as it absorbs heat at a constant rate of 42 kJ min^{-1}. The latent heat of vaporization of the substance is :

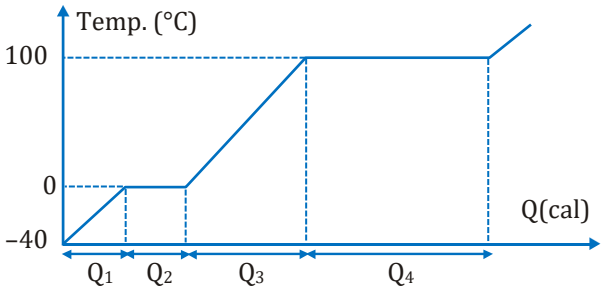


(1) 630 kJ kg^{-1} (2) 126 kJ kg^{-1}
(3) 84 kJ kg^{-1} (4) 12.6 kJ kg^{-1}
PTM027

28. A block of ice with mass m falls into a lake. After impact, a mass of ice m/5 melts. Both the block of ice and the lake have a temperature of 0°C. If L represents the heat of fusion, the minimum distance the ice fell before striking the surface is
(1) $\frac{L}{5g}$ (2) $\frac{5L}{g}$ (3) $\frac{gL}{5m}$ (4) $\frac{mL}{5g}$
PTM028

29. 10 g of ice at 0°C is kept in a calorimeter of water equivalent 10 g. How much heat should be supplied to the apparatus to evaporate the water thus formed? (Neglect loss of heat)
(1) 6200 cal (2) 7200 cal
(3) 13600 cal (4) 8200 cal
PTM029

30. Figure shows the temperature variation when heat is added continuously to a specimen of ice (10 g) at –40 °C at constant rate. (Specific heat of ice = 0.53 cal/g °C and L_{ice} = 80 cal/g, L_{water}= 540 cal/g)



Column–I	Column–II
(A) Value of Q_1 (in cal)	(P) 800
(B) Value of Q_2 (in cal)	(Q) 1000
(C) Value of Q_3 (in cal)	(R) 5400
(D) Value of Q_4 (in cal)	(S) 212
	(T) 900

(1) A→S; B→P; C→Q; D→T
(2) A→P; B→S; C→Q; D→R
(3) A→P; B→S; C→R; D→Q
(4) A→S; B→P; C→Q; D→R
PTM030

31. The thermal capacity of any body is
(1) a measure of its capacity to absorb heat
(2) a measure of its capacity to provide heat
(3) the quantity of heat required to raise its temperature by a unit degree
(4) the quantity of heat required to raise the temperature of a unit mass of the body by a unit degree
PTM031

32. 2 litre water at 27°C is heated by a 1 kW heater in an open container. On an average heat is lost to surroundings at the rate 160 J/s. The time required for the temperature to reach 77°C is
(1) 8 min 20 sec (2) 10 min
(3) 7 min (4) 14 min
PTM032

33. A 2100 W continuous flow geyser (instant geyser) has water inlet temperature = 10°C while the water flows out at the rate of 20 g/s. The outlet temperature of water must be about

(1) 20°C (2) 30°C (3) 35°C (4) 40°C

PTM033

34. The ratio of coefficient of thermal conductivity of two different materials is 5:3. If the thermal resistance of rods of same area of these material is same, then what is ratio of length of these rods-

(1) 3:5 (2) 5:3 (3) 25:9 (4) 9:25

PTM034

35. Rate of heat flow through a cylindrical rod is Q_1. Temperatures of ends of rod are T_1 and T_2. If all the linear dimensions of the rod become double and temperature difference remains same, it's rate of heat flow is Q_2, then :–

(1) $Q_1 = 2Q_2$ (2) $Q_2 = 2Q_1$

(3) $Q_2 = 4Q_1$ (4) $Q_1 = 4Q_2$

PTM035

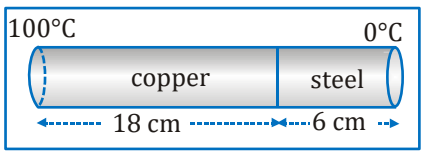
36. A heat flux of 4000 J/s is to be passed through a copper rod of length 10 cm and area of cross section 100 cm^2. The thermal conductivity of copper is 400 W/m°C. The two ends of this rod must be kept at a temperature difference of

(1) 1°C (2) 10°C

(3) 100°C (4) 1000°C

PTM036

37. The coefficient of thermal conductivity of copper is nine times that of steel. In the composite cylindrical bar shown in the figure what will be the temperature at the junction of copper and steel?

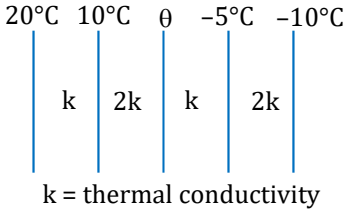


(1) 75°C (2) 67°C

(3) 33°C (4) 25°C

PTM037

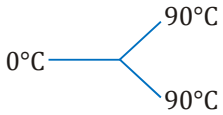
38. The figure shows the face and interface temperature of a composite slab containing of four layers of two materials having identical thickness. Under steady state condition, find the value of temperature θ.



(1) 5°C (2) 10°C (3) –15°C (4) 15°C

PTM038

39. Three rods made of the same material and having the same cross-section have been joined as shown in the figure. Each rod is of the same length. The left and right ends are kept at 0°C and 90°C respectively. The temperature of the junction of the three rods will be :



(1) 45°C (2) 60°C (3) 30°C (4) 20°C

PTM039

40. The coefficient of thermal conductivity depends upon-

(1) Temperature difference of two ends
(2) Area of the plate
(3) Thickness of the plate
(4) Material of the plate

PTM040

41. Which of the following cylindrical rods will conduct most heat, when their ends are maintained at the same steady temperature

(1) Length 1 m; radius 1 cm
(2) Length 2 m; radius 1 cm
(3) Length 2 m; radius 2 cm
(4) Length 1 m; radius 2 cm

PTM041

42. Gravitational force is required for –

(1) Stirring of liquid (2) Convection

(3) Conduction (4) Radiation

PTM042

43. The layers of atmosphere are heated through -

(1) Convection (2) Conduction

(3) Radiation (4) 2 and 3 both

PTM043

44. The lengths and radii of two rods made of same material are in the ratios 1 : 2 and 2 : 3 respectively. If the temperature difference between the ends for the two rods be the same then in the steady state. The amount of heat flowing per second through them will be in the ratio of
(1) 1 : 3 (2) 4 : 3 (3) 8 : 9 (4) 3 : 2
PTM044

45. Two metal rods, 1 & 2 of same length have same temperature difference between their ends, their thermal conductivities are K_1 & K_2 and cross sectional areas A_1 & A_2 respectively. What is required condition for same rate of heat conduction in them.
(1) $K_1 = K_2$ (2) $K_1 A_1 = K_2 A_2$
(3) $\frac{K_1}{A_1} = \frac{K_2}{A_2}$ (4) $\frac{K_1}{\ell_1^2} = \frac{K_2}{\ell_2^2}$
PTM045

46. The temperature of hot and cold end of a 20 cm long rod in thermal steady state are at 100°C and 20°C respectively. Temperature at the centre of the rod is
(1) 50°C (2) 60°C (3) 40°C (4) 30°C
PTM046

47. Consider a compound slab consisting of two different materials having equal thicknesses and thermal conductivities K and 2K, respectively. The equivalent thermal conductivity of the slab is
(1) 3K (2) $\frac{4}{3}$K (3) $\frac{2}{3}$K (4) $\sqrt{2}$K
PTM047

48. Under steady state, the temperature of a body
(1) Increases with time
(2) Decreases with time
(3) Does not change with time and is same at all the points of the body
(4) Does not change with time but is different at different points of the body
PTM048

49. The area of the glass of a window of a room is $10m^2$ and thickness 2 mm. The outer and inner temperature are 40°C and 20°C respectively. Thermal conductivity of glass in MKS system is 0.2 then heat flowing in the room per second will be -
(1) 3×10^4 joules (2) 2×10^4 joules
(3) 30 joules (4) 45 joules
PTM049

50. If the coefficient of conductivity of aluminium is 0.5cal/cm-sec-°C, then in order to conduct 10cal/sec-cm^2 in the steady state, the temperature gradient in aluminium must be
(1) 5°C/cm (2) 10°C/cm
(3) 20°C/cm (4) 10.5°C/cm
PTM050

51. The dimensional formula for thermal resistance is
(1) $M^{-1}L^{-2}T^{3}\theta$ (2) $M^{-1}L^{-2}T^{-3}\theta$
(3) $ML^2T^{-2}\theta$ (4) $ML^2T^2\theta^{-1}$
PTM051

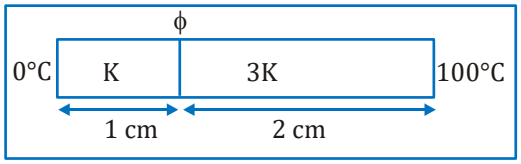
52. The material used in the manufacture of cooker must have (K-coefficient of thermal conductivity, S - specific heat of material used) :
(1) high K and low S (2) low K and low S
(3) high K and high S (4) low K and high S
PTM052

53. The cause of air currents from ocean to ground is example of
(1) The specific heat of water is more than that of sand
(2) Convection
(3) Radiation
(4) Diffraction
PTM053

54 On a cold morning, a person will feel metal surface colder to touch than a wooden surface because
(1) Metal has high specific heat
(2) Metal has high thermal conductivity
(3) Metal has low specific heat
(4) Metal has low thermal conductivity
PTM054

55. The ratio of the diameters of two metallic rods of the same material is 2 : 1 and their lengths are in the ratio 1 : 4. If the temperature difference between them are equal, the rate of flow of heat in them will be in the ratio of -
(1) 2 : 1 (2) 4 : 1 (3) 8 : 1 (4) 16 : 1
PTM055

56. Two bars of thermal conductivities K and 3K and lengths 1 cm and 2 cm respectively have equal cross-sectional area. they are joined length wise as shown in the figure. If the temperature at the ends of this composite bar is 0°C and 100°C respectively (see figure), then the temperature of the interface (ϕ) is



(1) 50°C (2) $\frac{100}{3}$ °C

(3) 60°C (4) $\frac{200}{3}$ °C

PTM056

57. Mud houses are cooler in summer and warmer in winter because
(1) Mud is super conductor of heat
(2) Mud is good conductor of heat
(3) Mud is bad conductor of heat
(4) None of these

PTM057

58. Two walls of thicknesses d_1 and d_2 and thermal conductivity K_1 and K_2 are in contact. In the steady state, if the temperatures at the outer surface are T_1 and T_2, the temperature at the common wall is-

(1) $\frac{K_1T_1d_2+K_2T_2d_1}{K_1d_2+K_2d_1}$ (2) $\frac{K_1T_1+K_2T_2}{d_1+d_2}$

(3) $\left(\frac{K_1d_1+K_2d_2}{T_1+T_2}\right)T_1T_2$ (4) $\frac{K_1d_1T_1+K_2d_2T_2}{K_1d_1+K_2d_2}$

PTM058

59. In which of the following phenomenon heat convection does not take place
(1) land and sea breeze
(2) boiling of water
(3) heating of glass surface due to filament of the bulb
(4) air around the furnace

PTM059

60. In natural convection, a heated portion of a liquid moves because :
(1) Its molecular motion becomes aligned
(2) Of molecular collisions within it
(3) Its density is less than that of the surrounding fluid
(4) Of currents of the surrounding fluid

PTM060

61. It is hotter at the same distance over the top of a fire than it is in the side of it, mainly because
(1) Air conducts heat upwards
(2) Heat is radiated upwards
(3) Convection takes more heat upwards
(4) Convection, conduction and radiation all contribute significantly transferring heat upward

PTM061

RADIATION

62. A spherical body of area A, and emissivity e = 0.6 is kept inside a black body. What is the rate at which energy is radiated per second at temperature T.

(1) $0.6\ \sigma AT^4$ (2) $0.4\ \sigma AT^4$

(3) $0.8\ \sigma AT^4$ (4) $1.0\ \sigma AT^4$

PTM062

63. Radius of two spheres of same material are 1 & 4 m respectively and their temperature are 4×10^3 and 2×10^3 K respectively. Then ratio of emitted energy of spheres per sec. will be -

(1) 1:2 (2) 2:1 (3) 1:1 (4) 4:1

PTM063

64. Cooling rate of a sphere of 600 K at external environment (200 K) is R. When the temperature of sphere is reduced to 400 K then cooling rate of the sphere becomes :

(1) $\frac{3}{16}$R (2) $\frac{16}{3}$R (3) $\frac{9}{27}$R (4) None

PTM064

65. If temperature of ideal black body increased by 10%, then percentage increase in quantity of radiation emitted from it's surface will be :-

(1) 10% (2) 40%

(3) 46% (4) 100%

PTM065

66. The rectangular surface of area 8cm × 4 cm of a black body at a temperature of 127°C emits energy at the rate of E. If the length and breadth of the surface are each reduced to half of the initial value and the temperature is raised to 327°C, the rate of emission of energy will become.

(1) $\frac{3}{8}$E (2) $\frac{81}{16}$E (3) $\frac{9}{16}$E (4) $\frac{81}{64}$E

PTM066

67. The rate of emission of radiation of a black body at 273°C is E, then the rate of emission of radiation of this body at 0°C will be

(1) $\frac{E}{16}$ (2) $\frac{E}{4}$ (3) $\frac{E}{8}$ (4) 0

PTM067

68. If a liquid takes 30 s in cooling from 95°C to 90°C and 70 s in cooling from 55°C to 50°C then temperature of room is -

(1) 16.5°C (2) 22.5°C
(3) 28.5°C (4) 32.5°C

PTM068

69. The thermal capacities of two bodies are in the ratio of 1:4. If the rate of loss of heat are equal for the two bodies under identical conditions of surroundings, then the ratio of rate of fall of temperature of the two bodies is -

(1) 1:4 (2) 4:1 (3) 1:8 (4) 8:1

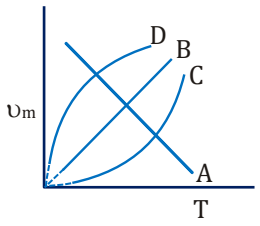
PTM069

70. Newton's law of cooling is used in laboratory for the determination of the

(1) Specific heat of the gases
(2) The latent heat of gases
(3) Specific heat of liquids
(4) Latent heat of liquids

PTM070

71. The υ_m – T curve for a perfect black body is – ($\upsilon_m \rightarrow$ frequency corresponding to maximum emission of radiation)



(1) A (2) B (3) C (4) D

PTM071

72. Two stars appear to be red and blue, what is true about them -

(1) The red star is nearer
(2) The blue star is nearer
(3) The temperature of red star is more
(4) The temperature of blue star is more

PTM072

73. The temperature of a furnace is 2324°C and the intensity is maximum in its radiation spectrum nearly at 12000A°. If the intensity in the spectrum of a star is maximum nearly at 4800A°, then the surface temperature of star is

(1) 8400°C (2) 7200°C
(3) 6219.5°C (4) 5900°C

PTM073

74. There is a black spot on a body. If the body is heated and carried in a dark room then it glows more. This can be explained on the basis of –

(1) Newton's law of cooling
(2) Wein's law
(3) Kirchhoff's law
(4) Stefan's law

PTM074

75. The colour of a star is an indication of its -

(1) Weight (2) Distance
(3) Temperature (4) Size

PTM075

76. If a carved black utensil is heated to high temperature and then brought in dark then :

(1) Both utensil and its carving will shine
(2) Only carving will shine
(3) Only utensil will shine
(4) None of the utensil and carving will shine

PTM076

77. According to Newton's law of cooling, the rate of cooling of a body is proportional to :–

(1) Temperature of the body
(2) Temperature of the surrounding
(3) Fourth power of the temperature of body
(4) Difference of the temperature of the body and the surrounding.

PTM077

78. The original temperature of a black body is 727°C. Calculate temperature at which total radiant energy from this black body becomes double : ($2^{1/4} = 1.19$)

(1) 971 K (2) 1190 K
(3) 2001 K (4) 1458 K

PTM078

79. Ratio of radius of curvature of cylindrical emitters of same material is 1:4 and their temperature are in ratio 2:1. Then ratio of amount of heat emitted by them is –
(For Cylinder length = radius)

(1) 2:1 (2) 1:1 (3) 4:1 (4) 1:4

PTM079

80. The ideal black body is :
(1) Hot coal at high temperature
(2) Surface of glass printed with coal tar
(3) Metal surface
(4) A hollow container painted with black colour

PTM080

81. The energy emitted per second by a black body at 27°C is 10 J. If temperature of the black body is increased to 327°C, the energy emitted per second will be :-
(1) 20 J (2) 40 J (3) 80 J (4) 160 J

PTM081

82. Energy is being emitted from the surface of black body at 127°C at the rate of 1.0×10^6 J/s m^2. The temperature of black body at which the rate of energy is 16.0×10^6 J/s m^2 will be :
(1) 754°C (2) 527°C
(3) 254°C (4) 508°C

PTM082

83. Solar constant for earth is 2 cal/min cm^2, if distance of mercury from sun is 0.4 times than distance of earth from sun then solar constant for mercury will be?
(1) 12.5 cal/min cm^2 (2) 25 cal/min cm^2
(3) 0.32 cal/min cm^2 (4) 2 cal/min cm^2

PTM083

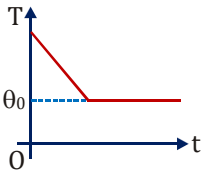
84. Two spherical bodies A (radius 6 cm) and B (radius 18 cm) are at temperature T_1 and T_2 respectively. The maximum intensity in the emission spectrum of A is at 500 nm and in that of B is at 1500 nm. Considering them to be black bodies, what will be the ratio of the rate of total energy radiated by A to that of B?
(1) 9 (2) 6 (3) 12 (4) 3

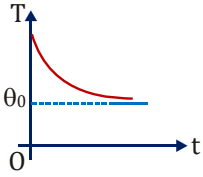
PTM084

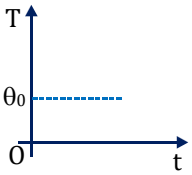
85. Star S_1 emits maximum radiation of wavelength 420 nm and the star S_2 emits maximum radiation of wavelength 560 nm, what is the ratio of the temperature of S_1 and S_2 :
(1) 4/3 (2) $(4/3)^{1/4}$
(3) 3/4 (4) $(3/4)^{1/2}$

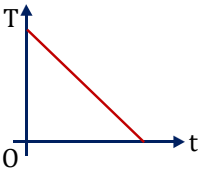
PTM085

86. If a piece of metal is heated to temperature θ and then allowed to cool in a room which is at temperature θ_0, the graph between the temperature T of the metal and time t will be closest to

(1) 

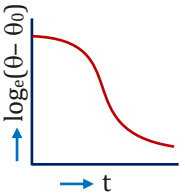
(2) 

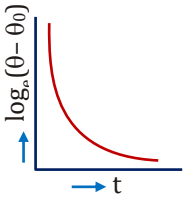
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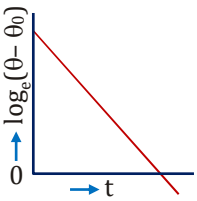
(4) 

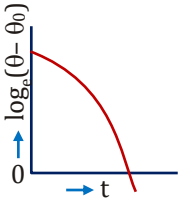
PTM086

87. A liquid in a beaker has temperature θ at time t and θ_0 is temperature of surroundings, then according to Newton's law of cooling, correct graph between $\log_e(\theta - \theta_0)$ and t is:

(1) 

(2) 

(3) 

(4) 

PTM087

88. A bucket full of hot water cools from 75°C to 70°C in time T_1, from 70°C to 65°C in time T_2 and from 65°C to 60°C in time T_3, then
(1) $T_1 = T_2 = T_3$ (2) $T_1 > T_2 > T_3$
(3) $T_1 < T_2 < T_3$ (4) $T_1 > T_2 < T_3$

PTM088

89. The Wein's displacement law express relation between :-
(1) Wavelength corresponding to maximum energy and temperature.
(2) Radiation energy and wavelength
(3) Temperature and wavelength
(4) Colour of light and temperature

PTM089

90. Four identical calorimeters painted in different colours, are heated to same temperature and then allowed to cool in vacuum. Which will cool fastest?
(1) One which is painted bright
(2) One which is painted thick white
(3) One which is painted thick black
(4) One which is painted bright white

PTM090

91. A body cools from 60°C to 50°C in 10 minutes. If the room temperature is 25°C and assuming Newton's cooling law holds good, the temperature of the body at the end of next 10 minutes is :
(1) 45°C (2) 42.85°C (3) 40°C (4) 38.5°C

PTM091

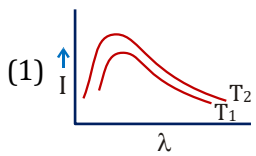
92. As compared to the person with white skin, the person with black skin will experience
(1) Less heat and more cold
(2) More heat and more cold
(3) More heat and less cold
(4) Less heat and less cold

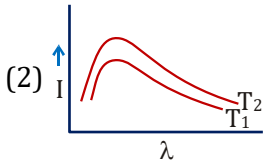
PTM092

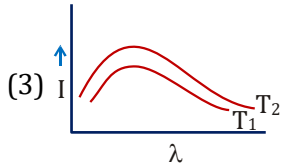
93. We consider the radiation emitted by the human body. Which of the following statements is true?
(1) The radiation is emitted during the summers and absorbed during the winters
(2) The radiation emitted lies in the ultraviolet region and hence is not visible
(3) The radiation emitted is in the infra-red region
(4) The radiation is emitted only during the day

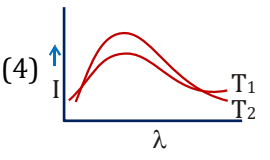
PTM093

94. Shown below are the black body radiation curves at temperatures T_1 and T_2 ($T_2 > T_1$). Which of the following plots is correct :–

(1) 

(2) 

(3) 

(4) 

PTM094

95. The radii of two spheres made of same metal are r and 2r. These are heated to the same temperature and placed in the same surrounding. The ratio of rates of decrease of their temperature will be
(1) 1 : 1 (2) 4 : 1 (3) 1 : 4 (4) 2 : 1

PTM095

96. If E is the total energy emitted by a body at a temperature T K and $E_{\lambda_{max}}$ is the maximum spectral energy emitted by it at the same temperature, then-
(1) $E \propto T^4; E_{\lambda_{max}} \propto T^5$ (2) $E \propto T^4; E_{\lambda_{max}} \propto T^{-5}$
(3) $E \propto T^{-4}; E_{\lambda_{max}} \propto T^4$ (4) $E \propto T^5; E_{\lambda_{max}} \propto T^4$

PTM096

97. If e_λ and a_λ be the emissive power and absorption power respectively of a body and E_λ be the emissive power of an ideal black body, then from Kirchhoff's laws
(1) $a_\lambda = E_\lambda / e_\lambda$ (2) $a_\lambda / e_\lambda = E_\lambda$
(3) $e_\lambda / a_\lambda = E_\lambda$ (4) $e_\lambda = E_\lambda / a_\lambda$

PTM097

98. A liquid takes 5 min. to cool from 80°C to 50°C. How much time it will take to cool from 60°C to 30°C. Temperature of surroundings is 20°C –
(1) 15 min. (2) 20 min.
(3) 100 min. (4) 9 min.

PTM098

99. A cup of tea cools from 80°C to 60°C in one minute. The ambient temperature is 30°C. In cooling from 60°C to 50°C, it will take :
(1) 50 sec. (2) 90 sec
(3) 60 sec. (4) 48 sec.

PTM099

100. If λ_m denotes the wavelength at which the radiation emission from a black body at a temperature T K is maximum then :
(1) $\lambda_m \alpha T$ (2) $\lambda_m \alpha T^2$
(3) $\lambda_m \alpha T^{-1}$ (4) $\lambda_m \alpha T^{-2}$

PTM100

101. A body is in thermal equilibrium with the surrounding:
(1) It will stop emitting heat radiation
(2) Amount of radiation emitted and absorbed by it will be equal
(3) It will emit heat radiation at faster rate
(4) It will emit heat radiation slowly

PTM101

102. Which of the following statement is correct for ideal black body :
(1) This absorbs visible radiation only.
(2) This absorbs infrared radiation only
(3) This absorbs half of radiation only and reflects the half
(4) This totally absorbs heat radiation of all the wavelengths

PTM102

103. Two spheres P and Q of same colour having radii 8 cm and 2 cm are maintained at temperatures 127°C and 527°C respectively. The ratio of energy radiated by P and Q is –
(1) 0.054 (2) 0.0034
(3) 1 (4) 2

PTM103

104. On increasing the temperature of a black body, wavelength for maximum emission.
(1) Shifts towards smaller wavelength
(2) Shifts towards greater wavelength
(3) Does not shift
(4) Depends on the shape of source.

PTM104

105. A solid cube and sphere are made of same substance and both have same surface area. If the temperature of both bodies 120° C then :
(1) Both will loss of Heat by same rate
(2) Rate of loss of Heat of cube will be more than that of the sphere
(3) Rate of loss of Heat of the sphere will be more than that of the cube
(4) Rate of loss of Heat will be more for that whose mass is more

PTM105

106. Two spheres of radii in the ratio 1 : 2 and densities in the ratio 2 : 1 and of same specific heat, are heated to same temperature and left in the same surrounding. Their rate of falling temperature will be in the ratio :
(1) 2 :1 (2) 1 : 1 (3) 1 : 2 (4) 1 : 4

PTM106

107. A black body at 200 K is found to emit maximum energy at a wavelength 14 μm. When its temperature is raised to 1000 K, then wavelength at which maximum energy emitted is:
(1) 14 μm (2) 15 μm
(3) 2.8 μm (4) 28 μm

PTM107

108. The spectrum from a black body radiation is a :
(1) line spectrum
(2) band spectrum
(3) continuous spectrum
(4) line & band both

PTM108

109. The temperature of a perfect black body is 727°C and its area is 0.1 m^2. If Stefan's constant is 5.67×10^{-8} watt/m^2–K^4, then heat radiated by it in 1 minute is :
(1) 8100 cal (2) 81000 cal
(3) 810 cal (4) 81 cal

PTM109

110. A black body radiates energy at the rate of E watt/m^2 at a high temperature T K. When the temperature is reduced to $\frac{T}{2}$ K, the radiant energy will be
(1) $\frac{E}{16}$ (2) $\frac{E}{4}$ (3) 4 E (4) 16 E

PTM110

111. In solar spectrum Fraunhofer's lines are presents because :
(1) Definite absorption takes place in photosphere of sun.
(2) Definite absorption takes place in chromosphere of sun.
(3) These wave lengths are not at all emitted by sun.
(4) Nuclear reactions take place in sun.

PTM111

112. On reducing temperature of surface to one third, amount of radiation becomes :-
(1) $\frac{1}{27}$ (2) $\frac{1}{81}$ (3) $\frac{1}{9}$ (4) $\frac{1}{3}$

PTM112

113. The absorptive power of a perfectly black body is
(1) zero (2) infinity (3) 1.5 (4) 1.0

PTM113

KTG & GAS LAWS AND IDEAL GAS EQUATION

114. Find the approximate number of molecules contained in a vessel of volume 7 litres at 0°C at 1.3×10^5 pascals

(1) 2.4×10^{23} (2) 3×10^{23}
(3) 6×10^{23} (4) 4.8×10^{23}

PTM114

115. A real gas behaves like an ideal gas if its

(1) pressure and temperature are both high
(2) pressure and temperature are both low
(3) pressure is high and temperature is low
(4) pressure is low and temperature is high

PTM115

116. Two gases of equal molar amount are in thermal equilibrium. If P_a, P_b and V_a, V_b are their respective pressures and volumes, then which relation is true:–

(1) $P_a \neq P_b, V_a = V_b$ (2) $V_a = V_b, V_a \neq V_b$
(3) $P_a/V_b = P_b/V_b$ (4) $P_aV_a = P_bV_b$

PTM116

117. Equal volume of H_2, O_2 and He gases are at same temperature and pressure. Which of these will have large number of molecules :-

(1) H_2
(2) O_2
(3) He
(4) All the gases will have same number of molecules

PTM117

118. Gases obey Vander - Waal's equation at :

(1) Only normal temperature and pressure
(2) Only high temperature and high pressure
(3) Only high temperature and low pressure
(4) All temperature and pressure

PTM118

119. A box contains N molecules of a gas. If the number of molecules is doubled, then the pressure will :-

(1) Decrease (2) Be same
(3) Be doubled (4) Get tripled

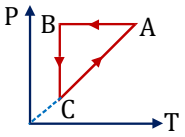
PTM119

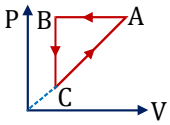
120. An ideal gas mixture filled inside a balloon expands according to the relation $PV^{2/3}$ = constant. The temperature inside the balloon is

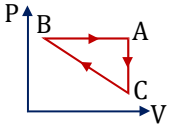
(1) increasing (2) decreasing
(3) constant (4) can't be said

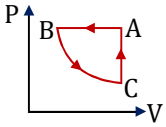
PTM120

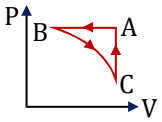
121. A cyclic process ABCA is shown in P–T diagram. When presented on P–V, it would



(1) 

(2) 

(3) 

(4) 

PTM121

122. During an experiment an ideal gas obeys an additional equation of state P^2V = constant. The initial temperature and volume of gas are T and V respectively. When it expands to volume 2V, then its temperature will be :

(1) T (2) $\sqrt{2}$ T (3) 2 T (4) $2\sqrt{2}$ T

PTM122

123. 250 litre of an ideal gas is heated at constant pressure from 27°C such that its volume becomes 500 liters. The final temperature is :

(1) 54°C (2) 300°C (3) 327°C (4) 600°C

PTM123

124. A balloon contains 500 m^3 of helium at 27°C and 1 atmosphere pressure. The volume of the helium at –3°C temperature and 0.5 atmosphere pressure will be-

(1) 500 m^3 (2) 700 m^3
(3) 900 m^3 (4) 1000 m^3

PTM124

125. A vessel has 6g of oxygen at pressure P and temperature 400 K. A small hole is made in it so that oxygen leaks out. How much oxygen leaks out if the final pressure is P/2 and temperature is 300 K?

(1) 3g (2) 2g (3) 4g (4) 5g

PTM125

126. Relation PV = RT is given for following condition for real gas -

(1) High temperature and high density
(2) Low temperature and low density
(3) High temperature and low density
(4) Low temperature and high density

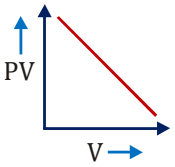
PTM126

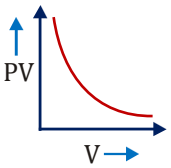
127. A container of 5 litre has a gas at pressure of 0.8 m column of Hg. This is joined to an evacuated container of 3 litre capacity. The resulting pressure will be:- (At constant temp.)
(1) 4/3 (2) 0.5m (3) 2.0m (4) 3/4m
PTM127

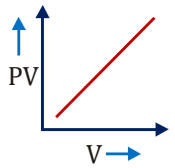
128. At a given temperature, the pressure of an ideal gas of density ρ is proportional to -
(1) $\frac{1}{\rho^2}$ (2) $\frac{1}{\rho}$ (3) ρ^2 (4) ρ
PTM128

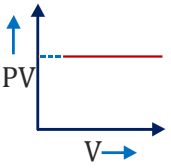
129. O_2 gas is filled in a cylinder. When pressure is increased 2 times, temperature becomes four times. then how much times their density will become :
(1) 2 (2) 4 (3) $\frac{1}{4}$ (4) $\frac{1}{2}$
PTM129

130. On increasing the temperature of a gas filled in a closed container by 1°C its pressure increases by 0.4%, then initial temperature of the gas is-
(1) 25°C (2) 250°C
(3) 250 K (4) 2500°C
PTM130

131. The variation of PV graph with V of a fixed mass of a ideal gas at constant temperature is graphically represented as shown in figure :
(1) 

(2) 

(3) 

(4) 

PTM131

132. The number of oxygen molecules in a cylinder of volume 1 m^3 at a temperature of 27°C and pressure 13.8 Pa is :
(Boltzmann's constant k = 1.38×10^{-23} JK^{-1})
(1) 6.23×10^{26} (2) 0.33×10^{28}
(3) 3.3×10^{21} (4) None of these
PTM132

133. A cylinder contains 10 kg of gas at pressure of 10^7 N/m^2. When final pressure is reduce to 2.5×10^6 N/m^2 then quantity of gas taken out of the cylinder will be : (temperature of gas is constant)
(1) 15.2 kg (2) 3.7 kg
(3) zero (4) 7.5 kg
PTM133

134. Hydrogen and helium gases of volume V at same temperature T and same pressure P are mixed to have same volume V. The resulting pressure of the mixture will be :
(1) P/2
(2) P
(3) 2P
(4) Depending on the relative mass of the gases
PTM134

135. The equation of state for 5g of oxygen at a pressure P and temperature T occupying a volume V, will be :– (where R is the gas constant)
(1) PV = 5 RT (2) PV = (5/2) RT
(3) PV = (5/16) RT (4) PV = (5/32) RT
PTM135

136. In kinetic theory of gases, it is assumed that molecules :-
(1) Have same mass but can have different volume
(2) Have same volume but masses can be different
(3) Have both mass and volume different
(4) Have same mass but negligible volume
PTM136

137. The volume of an ideal gas is V at pressure P and temperature T. The mass of each molecule of the gas in m. The density of gas will be : - (K is Boltzmann's constant)
(1) mKT (2) Pm / KT
(3) P / KTV (4) P / KT
PTM137

138. The thermodynamic variables of a jar filled with gas A are P, V and T and another jar B filled with another gas are 2P, V/4 and 2T, where the symbols have their usual meaning. The ratio of the number of molecules of jar A to those of jar B is :
(1) 4 : 1 (2) 2 : 1 (3) 1 : 2 (4) 1 : 1
PTM138

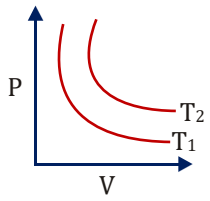
139. At N.T.P. volume of a gas is changed to one fourth volume, at constant temperature then the new pressure will be :

(1) 2 atm. (2) $2^{5/3}$ atm.

(3) 4 atm. (4) 1 atm.

PTM139

140. Find the correct relation in given P-V diagram:



(1) $T_1 = T_2$

(2) $T_1 > T_2$

(3) $T_1 < T_2$

(4) $T_1 \lesseqgtr T_2$

PTM140

141. Simple behaviour under all conditions of real gas is governed by the equation :-

(1) $PV = \mu RT$

(2) $\left(P+\frac{a}{V^2}\right)(V-b)=RT$

(3) PV = constant

(4) PV^{γ} = constant

PTM141

VARIOUS SPEEDS, DEGREE OF FREEDOM, SPECIFIC HEAT CAPACITIES OF GASES AND MEAN FREE PATH

142. The root mean square velocity of the molecules of an ideal gas is :-

(1) $\sqrt{RT/M_w}$ (2) $\sqrt{3RT/M_w}$

(3) $\sqrt{3RTM_w}$ (4) $\sqrt{RT/3M_w}$

PTM142

143. At constant pressure hydrogen is having temperature of 327° C. Till what temperature it is to be cooled so that the rms velocity of its molecules becomes half of the earlier value :-

(1) –123°C (2) 123°C (3) –100°C (4) 0°C

PTM143

144. The rms velocity of gas molecules of a given amount of a gas at 27°C and 1.0×10^5 N m^{-2} pressure is 200 m sec^{-1}. If temperature and pressure are respectively 127°C and 0.5×10^5 N m^{-2}, the rms velocity will be :-

(1) $400/\sqrt{3}$ ms^{-1} (2) $100\sqrt{2}$ ms^{-1}

(3) $100\sqrt{2}/3$ ms^{-1} (4) $50\sqrt{\frac{2}{3}}$ ms^{-1}

PTM144

145. Two containers of same volume are filled with atomic Hydrogen and Helium respectively at 1 and 2 atm pressure. If the temperature of both specimen are same then average speed $< C_H >$ for hydrogen atoms will be -

(1) $\langle C_H \rangle = \sqrt{2}\langle C_{He} \rangle$ (2) $\langle C_H \rangle = \langle C_{He} \rangle$

(3) $\langle C_H \rangle = 2\langle C_{He} \rangle$ (4) $\langle C_H \rangle = \frac{\langle C_{He} \rangle}{2}$

PTM145

146. The r.m.s. speed of a gas molecule is 300 m/s. Calculate the r.m.s. speed if the molecular weight is doubled while the temperature is halved-

(1) 300 m/s (2) 150 m/s

(3) 600 m/s (4) 75 m/s

PTM146

147. The root mean square velocity of hydrogen molecules at 300 K is 1930 m/s. Then the r.m.s. velocity of oxygen molecules at 1200 K will be :

(1) 482.5 m/s (2) 965.0 m/s

(3) 1930 m/s (4) 3860 m/s

PTM147

148. The rms velocity of H_2 is 2×10^3 m /s. What will be the rms velocity of O_2 molecules at the same temperature :-

(1) 10^3 m/s (2) 500 m/s

(3) 0.5×10^4 m/s (4) 3×10^3 m/s

PTM148

149. The temperature at which root mean square velocity of molecules of helium is equal to root mean square velocity of hydrogen at N.T.P is-

(1) 273°C (2) 273 K

(3) 546°C (4) 844 K

PTM149

150. If the pressure of a gas is doubled at constant temperature, then the mean square velocity will become :-

(1) No change (2) double

(3) Four times (4) None of the above

PTM150

151. The reason for the absence of atmosphere on moon is that the :

(1) Value of v_{rms} of the molecules of gas is more than the value of escape velocity

(2) Value of v_{rms} of gas is less than escape velocity

(3) Value of v_{rms} is negligible

(4) None of the above

PTM151

152. The speeds of 5 molecules of a gas (in arbitrary units) are as follows 2, 3, 4, 5, 6. The root mean square speed for these molecules is -
(1) 2.91 (2) 3.52 (3) 4.00 (4) 4.24
PTM152

153. The root mean square speed of the molecules of a gas is :
(1) Independent of its pressure but directly proportional to its Kelvin temperature
(2) Directly proportional to the square roots of both its pressure and its Kelvin temperature
(3) Independent of its pressure but directly proportional to the square root of its Kelvin temperature
(4) Directly proportional to both its pressure and its Kelvin temperature
PTM153

154. At 0°C temperature root mean square speed of which of the following gases will be maximum:-
(1) H_2 (2) N_2 (3) O_2 (4) SO_2
PTM154

155. The root mean square speed of hydrogen molecules of an ideal hydrogen gas kept in a gas chamber is 3180 m/s. The pressure of the hydrogen gas is :-
(Density of hydrogen gas = 8.99×10^{-2} Kg/m^3, 1 atmosphere = 1.01×10^5 N/m^2)
(1) 1.0 atmosphere (2) 1.5 atmosphere
(3) 2.0 atmosphere (4) 3.0 atmosphere
PTM155

156. If the rms velocity of molecules of a gas in a container is doubled then the pressure will:-
(1) Become four times (2) Also get doubled
(3) Be same (4) Become one half
PTM156

157. The root mean square velocity of a gas molecule of mass m at a given temperature is proportional to –
(1) m^0 (2) m
(3) $\sqrt{m}$ (4) $\frac{1}{\sqrt{m}}$
PTM157

158. v_{rms}, v_{av} and v_{mp} are root mean square, average and most probable speeds of molecules of a gas obeying Maxwell's velocity distribution. Which of the following statements is correct
(1) $v_{rms} < v_{av} < v_{mp}$ (2) $v_{rms} > v_{av} > v_{mp}$
(3) $v_{mp} < v_{rms} < v_{av}$ (4) $v_{mp} > v_{rms} > v_{av}$
PTM158

159. If the r.m.s. velocity of hydrogen becomes equal to the escape velocity from the earth surface, then the temperature of hydrogen gas would be-
(1) 1060 K (2) 5030 K
(3) 8270 K (4) 10^4 K
PTM159

160. The pressure exerted by a gas in P_0. If the mass of molecules becomes half and their velocities become double, then pressure will become
(1) $\frac{P_0}{2}$ (2) P_0 (3) $2P_0$ (4) $4P_0$
PTM160

161. The root mean square (rms) speed of oxygen molecules O_2 at a certain temperature T (absolute) is v. If the temperature is doubled and oxygen gas dissociates into atomic oxygen. The rms speed :
(1) becomes $v/\sqrt{2}$ (2) remains v
(3) becomes $\sqrt{2}v$ (4) becomes 2v
PTM161

162. If the root mean square speed of hydrogen molecules is equal to root mean square speed of oxygen molecules at 47°C, the temperature of hydrogen is -
(1) 20 K (2) 47 K (3) 50 K (4) 94 K
PTM162

163. The root mean square and most probable speed of the molecules in a gas are :
(1) same
(2) different
(3) cannot say
(4) depends on nature of the gas
PTM163

164. According to Maxwell's law of distribution of velocities of molecules, the most probable velocity is :-
(1) greater than the mean speed
(2) equal to the mean speed
(3) equal to the root mean square speed
(4) less than the root mean square speed
PTM164

165. The ratio of average translational kinetic energy to rotational kinetic energy of a diatomic molecule at temperature T is
(1) 3 (2) 7/5 (3) 5/3 (4) 3/2
PTM165

166. For hydrogen gas $c_P - c_V = a$ and for oxygen gas $c_P - c_V = b$ then the relation between a and b is (where c_P & c_V are gram specific heats)
(1) a = 16 b (2) b = 16 a
(3) a = b (4) None of these
PTM166

167. A gas mixture consists of 2 moles of oxygen and 4 moles of argon at temperature T. Neglecting all vibrational modes, the total internal energy of the system is
(1) 4 RT (2) 15 RT
(3) 9 RT (4) 11 RT
PTM167

168. The average kinetic energy of a gas molecule at 27°C is 6.21×10^{-21} J. Its average kinetic energy at 227°C will be
(1) 52.2×10^{-21} J (2) 5.22×10^{-21} J
(3) 10.35×10^{-21} J (4) 11.35×10^{-21} J
PTM168

169. Two containers A and B contain molecular gas at same temperature with masses of molecules are m_A and m_B, then relation of momentum P_A and P_B will be-
(1) $P_A = P_B$ (2) $P_A = \left(\frac{m_A}{m_B}\right)^{1/2} P_B$
(3) $P_A = \left(\frac{m_B}{m_A}\right)^{1/2} P_B$ (4) $P_A = \left(\frac{m_A}{m_B}\right) P_B$
PTM169

170. A cylinder of 200 litre capacity is containing H_2. The total translational kinetic energy of molecules is 1.52×10^5 J. The pressure of H_2 in the cylinder will be in N m^{-2} :-
(1) 2×10^5 (2) 3×10^5
(3) 4×10^5 (4) 5×10^5
PTM170

(171–172) Five moles of helium are mixed with two moles of hydrogen to form a mixture. Take molar mass of helium M_1 = 4g and that of hydrogen M_2 = 2g

171. The equivalent molar mass of the mixture is
(1) 6g (2) $\frac{13g}{7}$ (3) $\frac{18g}{7}$ (4) $\frac{24g}{7}$
PTM171

172. The equivalent value of γ in the above question is
(1) 1.59 (2) 1.53 (3) 1.56 (4) none
PTM172

173. Two monoatomic ideal gas at temperature T_1 and T_2 are mixed. There is no loss of energy. If the mass of molecules of the two gases are m_1 and m_2 and number of their molecules are n_1 and n_2 respectively, then temperature of the mixture will be :
(1) $\frac{T_1 + T_2}{n_1 + n_2}$ (2) $\frac{T_1}{n_1} + \frac{T_2}{n_2}$
(3) $\frac{n_2 T_1 + n_1 T_2}{n_1 + n_2}$ (4) $\frac{n_1 T_1 + n_2 T_2}{n_1 + n_2}$
PTM173

174. The total kinetic energy of 1 mole of N_2 at 27°C will be approximately :-
(1) 1500 J (2) 1500 calorie
(3) 1500 kilo calorie (4) 1500 erg.
PTM174

175. Mean kinetic energy (or average energy) per gm. molecule of a monoatomic gas is given by:
(1) 3RT/2 (2) kT/2
(3) RT/3 (4) 3kT/2
PTM175

176. Relation between the ratio of specific heats (γ) of gas and degree of freedom 'f' will be
(1) $\gamma = f + 2$ (2) $\frac{1}{\gamma} = \frac{1}{f} + \frac{1}{2}$
(3) $f = 2 / (\gamma - 1)$ (4) $f = 2(\gamma - 1)$
PTM176

177. Relation between pressure (P) and energy density (E) of an ideal gas is -
(1) $P = \frac{2E}{3}$ (2) $P = \frac{3E}{2}$
(3) $P = \frac{3E}{5}$ (4) $P = E$
PTM177

178. On mixing 1 g mole of a monoatomic with 1 g mole of a diatomic gas the specific heat of mixture at constant volume will be :–
(1) R (2) 3/2 R (3) 2R (4) 5/2R
PTM178

179. Absolute zero temperature is one at which-
(1) All liquids convert into solid
(2) All gases convert to solid
(3) All matter is in solid state
(4) The K.E. of molecules becomes zero
PTM179

180. For a gas $\frac{R}{C_v}$ = 0.67. This gas is made up of molecules which are :
(1) Diatomic
(2) Mixture of diatomic and polyatomic molecules
(3) Monoatomic
(4) Polyatomic
PTM180

181. If the total number of H_2 molecules is double that of the O_2 molecules then ratio of total kinetic energies of H_2 to that of O_2 at 300 K is:
(1) 1: 1 (2) 1 : 2 (3) 2 : 1 (4) 1 : 16
PTM181

182. At which of the following temperature any gas has average molecular kinetic energy double that of at 20°C
(1) 40°C (2) 80°C
(3) 313°C (4) 586°C
PTM182

183. When temperature is increased from 0°C to 273°C, what will be the ratio of final to initial the average kinetic energy of molecules?
(1) 1 (2) 3 (3) 4 (4) 2
PTM183

184. The kinetic energy associated with per degree of freedom of a molecule is -
(1) $\frac{1}{2}MC^2_{rms}$ (2) kT
(3) kT/2 (4) 3kT/2
PTM184

185. Which of the following statement is true according to kinetic theory of gases?
(1) The collision between two molecules is inelastic and the time between two collisions is less than the time taken during the collision.
(2) There is a force of attraction between the molecules
(3) All the molecules of a gas move with same velocity
(4) The average of the distances travelled between two successive collisions is mean free path.
PTM185

186. Gas exerts pressure on the walls of container because the molecules-
(1) Are loosing their Kinetic energy
(2) Are getting stuck to the walls
(3) Are transferring their momentum to walls
(4) Are accelerated towards walls.
PTM186

187. For a diatomic gas, change in internal energy for unit change in temperature at constant pressure and volume is U_1 and U_2 respectively then $U_1 : U_2$ is :
(1) 5 : 3 (2) 7 : 5 (3) 1 : 1 (4) 5 : 7
PTM187

188. The specific heat of an ideal gas depends on temperature is -
(1) $\frac{1}{T}$
(2) T
(3) $\sqrt{T}$
(4) Does not depends on temperature
PTM188

189. The specific heat of a gas :
(1) Has only two value Cp and Cv
(2) Has a unique value at a given temperature
(3) Can have any value between 0 and ∞
(4) Depends upon the mass of the gas
PTM189

190. 22 g of CO_2 at 27°C is mixed with 16 g of O_2 at 37°C. The temperature of the mixture is :–
(At room temperature, degrees of freedom of CO_2 = 7 and degrees of freedom of O_2 = 5)
(1) 31.16°C (2) 27°C
(3) 37°C (4) 30°C
PTM190

191. At 27°C temperature, the kinetic energy of an ideal gas is E_1. If the temperature is increased to 327°C, then kinetic energy would be

(1) $2\,E_1$ (2) $\frac{1}{2}E_1$

(3) $\sqrt{2}\,E_1$ (4) $\frac{1}{\sqrt{2}}E_1$

PTM191

192. Oxygen and hydrogen gases are at temperature T. Then average K.E of each molecule of oxygen gas is equal to how many times of average K.E. of each molecule of hydrogen gas :-

(1) 16 times (2) 8 times

(3) Equal (4) 1/16 times

PTM192

193. The average energy of the molecules of a monoatomic gas at temperature T is :-
(K = Boltzmann constant)

(1) $\frac{1}{2}kT$ (2) kT (3) $\frac{3}{2}kT$ (4) $\frac{5}{2}kT$

PTM193

194. A diatomic molecule has

(1) 1 degree of freedom

(2) 3 degree of freedom

(3) 5 degree of freedom

(4) 6 degree of freedom

PTM194

195. Two moles of monoatomic gas are mixed with 1 mole of a diatomic gas. Then γ for the mixture is:

(1) 1.4 (2) 1.55 (3) 1.62 (4) 1.67

PTM195

ZEROTH AND FIRST LAW OF THERMODYNAMICS, HEAT, WORK AND INTERNAL ENERGY

196. The first law of thermodynamics is based on:-

(1) Law of conservation of energy

(2) Law of conservation of mechanical energy

(3) Law of conservation of gravitational P.E.

(4) None of the above

PTM196

197. In a process, 500 calories of heat is given to a system and at the same time 100 joules of work is done on the system. The increase in the internal energy of the system is :-

(1) 40 calories (2) 1993 joules

(3) 2193 joules (4) 82 calories

PTM197

198. In a thermodynamic process pressure of a fixed mass of a gas is changed in such a manner that the gas releases 20 joules of heat and 8 joules of work was done on the gas. If the initial internal energy of the gas was 30 joules, then the final internal energy will be: -

(1) 2 J (2) 42 J (3) 18 J (4) 58 J

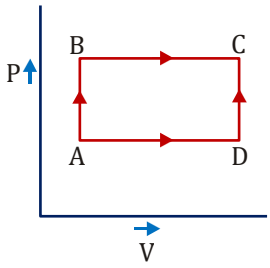
PTM198

199. 1 kg of a gas does 20 kJ of work and receives 16 kJ of heat when it is expanded between two states. A second kind of expansion can be found between the same initial and final state which requires a heat input of 9 kJ. The work done by the gas in the second expansion is :

(1) 32 kJ (2) 5 kJ (3) –4 kJ (4) 13 kJ

PTM199

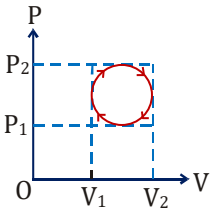
200. As shown in the figure the amount of heat absorbed along the path ABC is 90J and the amount of work done by the system is 30 J. If the amount of work done along the path ADC is 20 J then amount of heat absorbed will be :-



(1) 80 J (2) 90 J (3) 110 J (4) 120 J

PTM200

201. In a cyclic process shown on the P – V diagram, the magnitude of the work done is :

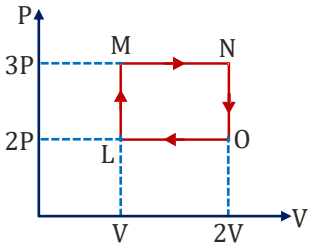


(1) $\pi\left(\frac{P_2-P_1}{2}\right)^2$ (2) $\pi\left(\frac{V_2-V_1}{2}\right)^2$

(3) $\frac{\pi}{4}(P_2-P_1)(V_2-V_1)$ (4) $\pi(P_2V_2-P_1V_1)$

PTM201

202. The work by an ideal monoatomic gas along the cyclic path LMNOL is



(1) PV (2) 2PV (3) 3 PV (4) 4 PV

PTM202

203. For a gas C_v = 4.96 cal/mole K, the increase in internal energy of 2 mole gas in heating from 340 K to 342 K will be :-

(1) 27.80 cal (2) 19.84 cal
(3) 13.90 cal (4) 9.92 cal

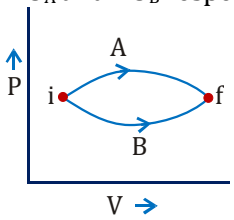
PTM203

204. When a system changes from one to another state the value of work done :-

(1) Depends on the force acting on the system
(2) Depends on the nature of material present in a system
(3) Does not depend on the path
(4) Depends on the path

PTM204

205. A system is taken along the paths A and B as shown. If the amounts of heat given in these processes are Q_A and Q_B and change in internal energy are ΔU_A and ΔU_B respectively then :-



(1) $Q_A = Q_B$; $\Delta U_A < \Delta U_B$ (2) $Q_A \geq Q_B$; $\Delta U_A = \Delta U_B$
(3) $Q_A < Q_B$; $\Delta U_A > \Delta U_B$ (4) $Q_A > Q_B$; $\Delta U_A = \Delta U_B$

PTM205

206. If the heat of 110 J is added to a gaseous system and change in internal energy is 40 J, then the amount of external work done is :

(1) 180 J (2) 70 J (3) 110 J (4) 30 J

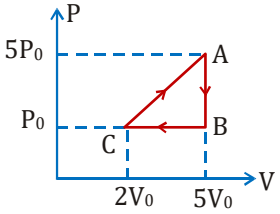
PTM206

207. If amount of heat supplied is Q, work done is W and change in internal energy is $mC_V\,dT$, then relation among them is. (C_v = gram specific heat)

(1) $mC_V\,dT = Q + W$ (2) $Q = W + mC_V\,dT$
(3) $Q + mC_V\,dT = W$ (4) None of these.

PTM207

208. The work done by a gas taken through the closed process ABCA is

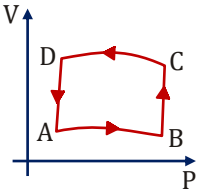


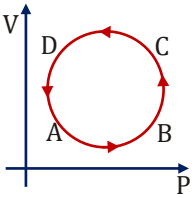
(1) $6P_0V_0$ (2) $4P_0V_0$
(3) P_0V_0 (4) zero

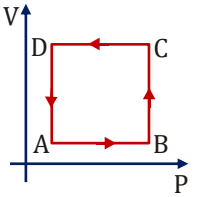
PTM208

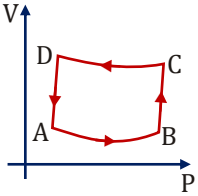
CYCLIC, ISOCHORIC, ISOBARIC, ISOTHERMAL, ADIABATIC AND POLYTROPIC PROCESS

209. In the diagrams (i) to (iv) of variation of volume with changing pressure is shown. A gas is taken along the path ABCDA. The change in internal energy of the gas will be: -

(i)


(ii)


(iii)


(iv)


(1) Positive in all cases (i) to (iv)
(2) Positive in cases (i), (ii) and (iii) but zero in case (iv)
(3) Negative in cases (i), (ii) and (iii) but zero in case (iv)
(4) Zero in all the four cases

PTM209

210. The temperature of 5 moles of a gas which was held at constant volume was changed from 100°C to 120°C. The change in internal energy was found to be 80 joules. The total heat capacity of the gas at constant volume will be equal to: -

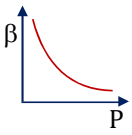
(1) 8 J/K (2) 0.8 J/K
(3) 4.0 J/K (4) 0.4 J/K

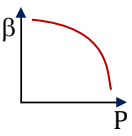
PTM210

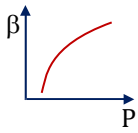
211. Monoatomic, diatomic and triatomic gases whose initial volume and pressure are same, are compressed till their volume become half the initial volume.
(1) If the compression is adiabatic then monoatomic gas will have maximum final pressure.
(2) If the compression is adiabatic then triatomic gas will have maximum final pressure.
(3) If the compression is adiabatic then their final pressure will be same.
(4) If the compression is isothermal then their final pressure will be different.

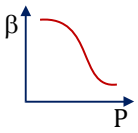
PTM211

212. Which of the following graphs correctly represents the variation of $\beta = -(dV/dP)/V$ with P for an ideal gas at constant temperature?

(1) 

(2) 

(3) 

(4) 

PTM212

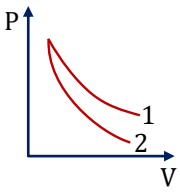
213. The adiabatic Bulk modulus of a diatomic gas at atmospheric pressure is
(1) 0 Nm^{-2} (2) 1 Nm^{-2}
(3) 1.4×10^4 Nm^{-2} (4) 1.4×10^5 Nm^{-2}

PTM213

214. A given quantity of an ideal gas is at pressure P and absolute temperature T. The isothermal bulk modulus of the gas is :
(1) 2P/3 (2) P (3) 3P/2 (4) 2P

PTM214

215. P-V plots for two gases during adiabatic processes are shown in the figure. Plots 1 and 2 should correspond respectively to



(1) He and O_2 (2) O_2 and He
(3) He and Ar (4) O_2 and N_2

PTM215

216. For an adiabatic expansion of a perfect gas, the value of $\Delta P/P$ is equal to: -
(1) $-\sqrt{\gamma}\,\Delta V/V$ (2) $-\Delta V/V$
(3) $-\gamma\,\Delta V/V$ (4) $-\gamma^2\,\Delta V/V$

PTM216

217. An ideal gas at 27°C is compressed adiabatically to 8/27 of its original volume. If $\gamma = 5/3$, then the rise in temperature is:-
(1) 450 K (2) 375 K
(3) 675 K (4) 405 K

PTM217

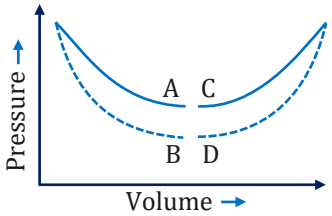
218. For monoatomic gas the relation between pressure of a gas and temperature T is given by $P \propto T^C$. Then value of C will be:
(For adiabatic process)
(1) $\frac{5}{3}$ (2) $\frac{2}{5}$ (3) $\frac{3}{5}$ (4) $\frac{5}{2}$

PTM218

219. A gas for which $\gamma = 5/3$ is heated at constant pressure. The percentage of total heat given that will be used for external work is :
(1) 40% (2) 30% (3) 60% (4) 20%

PTM219

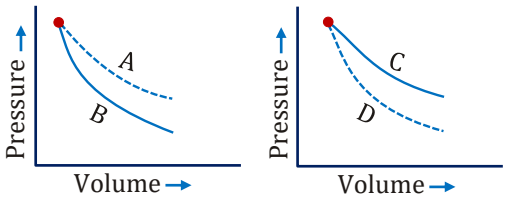
220. In which of the figure no heat exchange between the gas and the surroundings will take place, if the gas is taken along curve:
(curves are isothermal and adiabatic)



(1) A (2) B (3) C (4) D

PTM220

221. In the following figures, four curves A, B, C, D are shown the curves are :–



(1) Isothermal for A and B while adiabatic for C and D
(2) Isothermal for A and C while adiabatic for B and D
(3) Isothermal for A and D
(4) Adiabatic for A and C while isothermal for B and D

PTM221

222. Equal volumes of a perfect gas are compressed to half of their initial volumes. The first is brought about by isothermal process and the second by adiabatic process then :

(1) Both temperature and pressure will increase in the isothermal process.

(2) In the isothermal process, the temperature will decrease and pressure will increases

(3) Both temperature and pressure will increase in adiabatic process

(4) In the adiabatic process, the temperature will decrease and pressure will increase

PTM222

223. A vessel contains an ideal monoatomic gas which expands at constant pressure, when heat Q is given to it. Then the work done in expansion is :

(1) Q (2) $\frac{3}{5}Q$ (3) $\frac{2}{5}Q$ (4) $\frac{2}{3}Q$

PTM223

224. One mole of an ideal gas at temperature T_1 expends according to the law $\frac{P}{V^2} = a$ (constant). The work done by the gas till temperature of gas becomes T_2 is :

(1) $\frac{1}{2}R(T_2 - T_1)$ (2) $\frac{1}{3}R(T_2 - T_1)$

(3) $\frac{1}{4}R(T_2 - T_1)$ (4) $\frac{1}{5}R(T_2 - T_1)$

PTM224

225. When an ideal diatomic gas is heated at constant pressure, the fraction of the heat energy supplied which increases the internal energy of the gas is -

(1) 2/5 (2) 3/5 (3) 3/7 (4) 5/7

PTM225

226. Pressure-temperature relationship for an ideal gas undergoing adiabatic change is ($\gamma = C_p/C_v$)

(1) PT^{γ} = constant

(2) $PT^{-1+\gamma}$ = constant

(3) $PT^{\gamma-1}T^{\gamma}$ = constant

(4) $P^{1-\gamma}T^{\gamma}$ = constant

PTM226

227. The value of internal energy in an adiabatic process :-

(1) Remains unchanged

(2) Only increases

(3) Only diminishes

(4) May diminish and may also increase

PTM227

228. One mole of an ideal monoatomic gas is heated at a constant pressure of one atmosphere from 0°C to 100°C. Then the change in the internal energy is

(1) 20.80×10^2 J (2) 12.48×10^2 J

(3) 832×10^2 J (4) 6.25×10^2 J

PTM228

229. The specific heat of a gas at constant pressure is more than that of the same gas at constant volume because :-

(1) Work is done in the expansion of gas at constant pressure

(2) Work is done in the expansion of the gas at constant volume

(3) The molecular attraction increase under constant pressure

(4) The vibration of molecules increases under constant pressure

PTM229

230. When a gas is adiabatically compressed then it's temperature increase because :–

(1) Work done is zero

(2) Internal energy is increased

(3) Heat is supplied to it

(4) No change in pressure

PTM230

231. Air is filled in a tube of the wheel of a car at 27°C and 2 atm pressure if the tube is suddenly bursts, the final temperature of air will be :– ($\gamma = 1.5$, $2^{1/3} = 1.251$)

(1) –33° C (2) 0° C

(3) 21.6° C (4) 240° C

PTM231

232. Specific heat of a gas undergoing adiabatic change is :

(1) Zero (2) Infinite

(3) Positive (4) Negative

PTM232

233. A quantity of air (γ = 1.4) at 27°C is compressed suddenly, the temperature of the air system will :
(1) Fall
(2) Rise
(3) Remain unchanged
(4) First rise and then fall

PTM233

234. A gas specimen in one vessel is expended isothermally to double its volume and a similar specimen in the second vessel is expanded adiabatically the same extent, then:
(1) In the second vessel, both pressure and work done are more
(2) In the second vessel, pressure in more, but the work done is less.
(3) In the first vessel, both pressure & work done are more.
(4) In the first vessel, pressure is more, but work done is less

PTM234

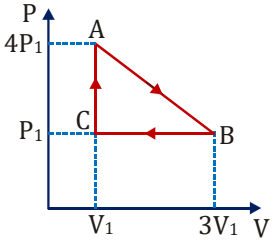
235. If an ideal gas is compressed during isothermal process then :-
(1) No work is done against gas
(2) heat is rejected by gas
(3) It's internal energy will increase
(4) Pressure does not change

PTM235

236. Graphs between P–V diagram for isothermal and adiabatic processes are drawn the relation between their slopes will be :–
(1) Slope of adiabatic curve = γ (slope of isothermal curve)
(2) Slope of isothermal curve = γ (slope of adiabatic curve)
(3) Slope of isothermal curve = slope of adiabatic curve
(4) Slope of adiabatic curve = γ^2 (slope of isothermal curve)

PTM236

237. An ideal gas is taken round the cycle ABCA. In the cycle the amount of work done involved is:-



(1) $12\ P_1V_1$ (2) $6\ P_1V_1$
(3) $3\ P_1V_1$ (4) P_1V_1

PTM237

238. One mole ideal gas is compressed adiabatically at 27°C. Its temperature becomes 102°C. The work done in this process will be :- (γ = 1.5)
(1) –625 J (2) 625 J
(3) 1245 J (4) –1245 J

PTM238

239. In an isometric change :
(1) $\delta Q = dU$ (2) $\delta W = dU$
(3) $\delta Q + \delta W = dU$ (4) None of these

PTM239

240. The volume of a gas expands by 0.25 m^3 at a constant pressure of 10^3 N/m^2. The work done is equal to
(1) 2.5 erg (2) 250 J
(3) 250 W (4) 250 N

PTM240

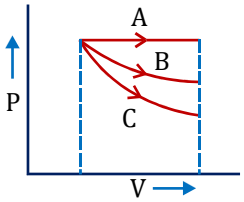
241. The process in which the heat given to a system is completely transformed into work is for ideal gas :–
(1) Isobaric process
(2) Isometric process
(3) Isothermal process
(4) Adiabatic process

PTM241

242. The volume of a poly-atomic gas $\left(\gamma = \frac{4}{3}\right)$ compressed adiabatically to $\frac{1}{8^{th}}$ of the original volume. If the original pressure of the gas is P_0 the new pressure will be: -
(1) $8\ P_0$ (2) $16\ P_0$
(3) $6\ P_0$ (4) $2\ P_0$

PTM242

243. During isothermal, isobaric and adiabatic processes, work done for same change in volume will be maximum for :-



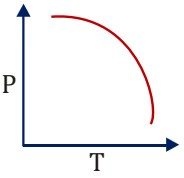
(1) Isothermal
(2) Isobaric
(3) Adiabatic
(4) None of the above

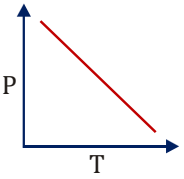
PTM243

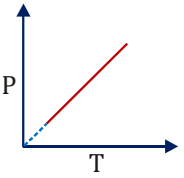
244. In an adiabatic process the quantity which remains constant is :-
(1) Temperature
(2) Pressure
(3) Total heat content of the system
(4) Volume

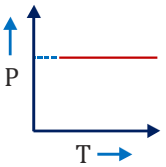
PTM244

245. Graph of isometric process is :-

(1) 

(2) 

(3) 

(4) 

PTM245

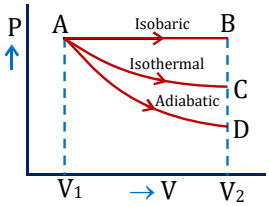
246. An ideal gas undergoes an adiabatic change in volume (V) with pressure (P). Then :-
(1) $P^{\gamma}V$ = constant (2) PV^{γ} = constant
(3) $(PV)^{\gamma}$ = constant (4) PV = constant

PTM246

247. 300 calories of heat is supplied to raise the temperature of 50 gm of air from 20°C to 30°C without any change in its volume. Change in internal energy per gram of air is
(1) zero (2) 0.6 calories
(3) 1.2 calories (4) 6.0 calories

PTM247

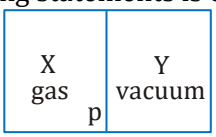
248. A gas is expanded from volume V_1 to volume V_2 in three processes, shown in the figure. If UA, UB, UC and UD represent the internal energies of the gas in state A,B,C and D respectively, the which of the following is not correct



(1) $U_B - U_A > 0$ (2) $U_C - U_A = 0$
(3) $U_D - U_A < 0$ (4) $U_B = U_C = U_D$

PTM248

249. A closed container is fully insulated from outside. One half of it is filled with an ideal gas X separated by a plate P from the other half Y which contains a vacuum as shown in figure. When P is removed, X moves into Y. Which of the following statements is correct?



(1) No work is done by X
(2) X decreases in temperature
(3) X increases in internal energy
(4) X doubles in pressure

PTM249

SECOND LAW OF THE THERMODYNMAMICS, HEAT ENGINES AND REFRIGERATORS

250. According to the second law of thermodynamics:
(1) heat energy cannot be completely converted to work
(2) work cannot be completely converted to heat energy
(3) for all cyclic processes we have $dQ/T < 0$
(4) the reason all heat engine efficiencies are less than 100% is friction, which is unavoidable

PTM250

251. "Heat cannot flow by itself from a body at lower temperature to a body at higher temperature" is a statement or consequence of :
(1) second law of thermodynamics
(2) conservation of momentum
(3) conservation of mass
(4) first law of thermodynamics

PTM251

252. A Carnot engine takes 3×10^6 cal of heat from reservoir at 627°C and gives it to a sink at 27°C. Then work done by the engine is
(1) 4.2×10^6 J (2) 8.4×10^6 J
(3) 16.8×10^6 J (4) zero

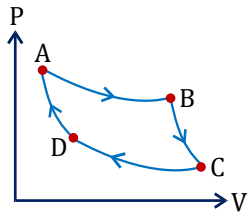
PTM252

253. A reversible refrigerator operates between a low temperature reservoir at T_C and a high temperature reservoir at T_H. Its coefficient of performance is given by :
(1) $(T_H - T_C)/T_C$ (2) $T_C/(T_H - T_C)$
(3) $(T_H - T_C)/T_H$ (4) $T_H/(T_H - T_C)$

PTM253

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254. In the given graph the isothermal curves are :-



(1) AB and CD (2) AB and AD
(3) AD and BC (4) BC and CD

PTM254

255. In the above question, the curve for which the heat is absorbed from the surroundings is :–

(1) AB (2) BC (3) CD (4) DA

PTM255

256. A Carnot engine shows efficiency of 40% on taking energy at 500 K. To increase the efficiency to 50%, at what temperature it should take energy?

(1) 400 K (2) 700 K
(3) 600 K (4) 800 K

PTM256

257. If the system takes 100 cal. heat, and releases 80 cal to sink, if source temperature is 127°C find the sink temperature :–

(1) 47° C (2) 127° C
(3) 67° C (4) 107° C

PTM257

258. A Carnot engine working between 300 K and 600 K has work output of 800 J per cycle. The amount of heat energy supplied to the engine from source per cycle will be :

(1) 800 J (2) 1600 J
(3) 1200 J (4) 900 J

PTM258

259. An ideal gas heat engine operates in Carnot cycle between 227°C and 127°C. It absorbs 6×10^4 cal of heat at higher temperature. Then amount of heat converted to work is :

(1) 2.4×10^4 cal (2) 6×10^4 cal
(3) 1.2×10^4 cal (4) 4.8×10^4 cal

PTM259

260. A refrigerator works between temperature –10°C & 27°C, the coefficient of performance is :

(1) 7.1 (2) 1
(3) 8.1 (4) 15.47

PTM260

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	1	2	4	3	3	1	2	2	1	4	1	3	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	1	1	1	4	2	4	1	1	3	3	4	3	1	4	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	1	3	2	2	3	1	1	2	4	4	2	1	3	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	2	4	2	3	1	1	2	2	4	3	3	1	3	3
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	3	1	3	1	3	4	1	2	2	3	2	4	3	3	3
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	3	4	2	2	4	4	2	1	1	1	2	3	3	1	3
Question	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105
Answer	2	2	3	1	4	1	3	4	4	3	2	4	3	1	1
Question	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	2	3	3	2	1	2	2	4	1	4	4	4	4	3	1
Question	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135
Answer	3	2	3	3	2	3	2	4	4	3	4	3	4	3	4
Question	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
Answer	4	2	1	3	3	2	2	1	1	3	2	2	2	1	1
Question	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165
Answer	1	4	3	1	4	1	4	2	4	3	4	1	2	4	4
Question	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
Answer	1	4	3	2	4	4	3	4	2	1	3	1	3	4	3
Question	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195
Answer	3	3	4	3	4	3	3	4	3	1	1	3	3	3	2
Question	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210
Answer	1	3	3	4	1	3	1	2	4	4	2	2	1	4	3
Question	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225
Answer	1	1	4	2	2	3	2	4	1	2	2	3	3	2	4
Question	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240
Answer	4	4	2	1	2	1	1	2	3	2	1	3	4	1	2
Question	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255
Answer	3	2	2	3	3	2	4	4	1	1	1	2	2	1	1
Question	256	257	258	259	260										
Answer	3	1	2	3	1										

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

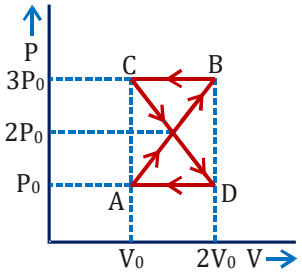
1. Steam at 100°C is passed into 20 g of water at 10°C. When water acquires a temperature of 80°C, the mass of water present will be :
[Take specific heat of water = 1 cal g^{-1} $°C^{-1}$ and latent heat of steam = 540 cal g^{-1}]
(1) 24 g (2) 31.5 g
(3) 42.5 g (4) 22.5 g
PTM293

2. Certain quantity of water cools from 70°C to 60°C in the first 5 minutes and to 54°C in the next 5 minutes. The temperature of the surroundings is:-
(1) 45°C (2) 20°C
(3) 42°C (4) 10°C
PTM294

3. The mean free path of molecules of a gas, (radius 'r') is inversely proportional to :-
(1) r^3 (2) r^2 (3) r (4) $\sqrt{r}$
PTM295

4. A monoatomic gas at a pressure P, having a volume V expands isothermally to volume 2V and then adiabatically to volume 16V. The final pressure of the gas is : (take $\gamma = \frac{5}{3}$)
(1) 64P (2) 32P (3) $\frac{P}{64}$ (4) 16P
PTM296

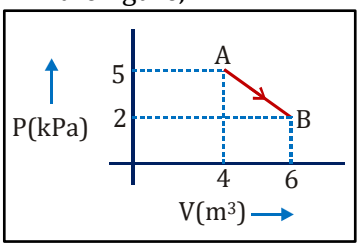
5. A thermodynamic system undergoes cyclic process ABCDA as shown in fig. The work done by the system in the cycle is :-



(1) P_0V_0 (2) $2P_0V_0$ (3) $\frac{P_0V_0}{2}$ (4) Zero
PTM297

AIPMT 2015

6. One mole of an ideal diatomic gas undergoes a transition from A to B along a path AB as shown in the figure,



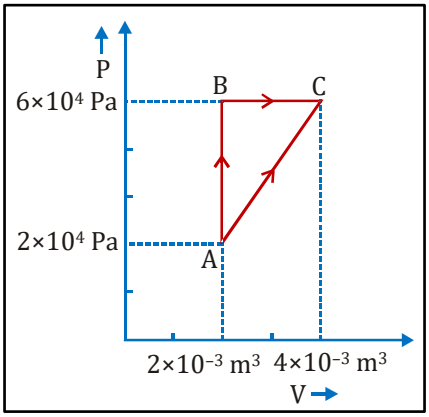
The change in internal energy of the gas during the transition is :
(1) –20 kJ (2) 20 J
(3) –12 kJ (4) 20 kJ
PTM298

7. On observing light from three different stars P, Q and R, it was found that intensity of violet color is maximum in the spectrum of P, the intensity of green colour is maximum in the spectrum of R and the intensity of red colour is maximum in the spectrum of Q. If T_P, T_Q and T_R are the respective absolute temperatures of P, Q and R, then it can be concluded from the above observation that :
(1) $T_P > T_R > T_Q$ (2) $T_P < T_R < T_Q$
(3) $T_P < T_Q < T_R$ (4) $T_P > T_Q > T_R$
PTM299

8. A Carnot engine, having an efficiency of $\eta = \frac{1}{10}$ as heat engine, is used as a refrigerator. If the work done on the system is 10 J, the amount of energy absorbed from the reservoir at lower temperature is :-
(1) 99 J (2) 90 J (3) 1 J (4) 100 J
PTM300

9. The ratio of the specific heats $\frac{C_P}{C_V} = \gamma$ in terms of degrees of freedom (n) is given by :
(1) $\left(1+\frac{n}{3}\right)$ (2) $\left(1+\frac{2}{n}\right)$
(3) $\left(1+\frac{n}{2}\right)$ (4) $\left(1+\frac{1}{n}\right)$
PTM301

10. Figure below shows two paths that may be taken by a gas to go from a state A to a state C.



In process AB, 400 J of heat is added to the system and in process BC, 100 J of heat is added to the system. The heat absorbed by the system in the process AC will be :

(1) 500 J (2) 460 J (3) 300 J (4) 380 J

PTM302

11. The two ends of a metal rod are maintained at temperatures 100°C and 110°C. The rate of heat flow in the rod is found to be 4.0 J/s. If the ends are maintained at temperatures 200°C and 210°C, the rate of heat flow will be :

(1) 16.8 J/s (2) 8.0 J/s
(3) 4.0 J/s (4) 44.0 J/s

PTM303

Re-AIPMT 2015

12. Two vessels separately contain two ideal gases A and B at the same temperature, the pressure of A being twice that of B. Under such conditions, the density of A is found to be 1.5 times the density of B. The ratio of molecular weight of A and B is:

(1) $\frac{1}{2}$ (2) $\frac{2}{3}$ (3) $\frac{3}{4}$ (4) 2

PTM304

13. 4.0 g of a gas occupies 22.4 litres at NTP. The specific heat capacity of the gas at constant volume is 5.0 $JK^{-1}\,mol^{-1}$. If the speed of sound in this gas at NTP is 952 ms^{-1}, then the heat capacity at constant pressure is

(Take gas constant R = 8.3 $JK^{-1}\,mol^{-1}$)

(1) 8.5 $JK^{-1}\,mol^{-1}$ (2) 8.0 $JK^{-1}\,mol^{-1}$
(3) 7.5 $JK^{-1}\,mol^{-1}$ (4) 7.0 $JK^{-1}\,mol^{-1}$

PTM305

14. The coefficient of performance of a refrigerator is 5. If the temperature inside freezer is –20°C, the temperature of the surroundings to which it rejects heat is :

(1) 21°C (2) 31°C (3) 41°C (4) 11°C

PTM306

15. An ideal gas is compressed to half its initial volume by means of several processes. Which of the process results in the maximum work done on the gas?

(1) Isothermal (2) Adiabatic
(3) Isobaric (4) Isochoric

PTM307

16. The value of coefficient of volume expansion of glycerine is 5 × 10^{-4} K^{-1}. The fractional change in the density of glycerine for a rise of 40°C in its temperature, is :-

(1) 0.010 (2) 0.015
(3) 0.020 (4) 0.025

PTM308

NEET-I 2016

17. A refrigerator works between 4°C and 30°C. It is required to remove 600 calories of heat every second in order to keep the temperature of the refrigerated space constant. The power required is:

(Take 1 cal = 4.2 Joules)

(1) 2.365 W (2) 23.65 W
(3) 236.5 W (4) 2365 W

PTM309

18. A black body is at a temperature of 5760 K. The energy of radiation emitted by the body at wavelength 250 nm is U_1, at wavelength 500 nm is U_2 and that at 1000 nm is U_3. Wien's constant, b = 2.88 × 10^6 nm K. Which of the following is correct?

(1) $U_1 = 0$ (2) $U_3 = 0$
(3) $U_1 > U_2$ (4) $U_2 > U_1$

PTM310

19. Coefficient of linear expansion of brass and steel rods are α_1 and α_2. Lengths of brass and steel rods are ℓ_1 and ℓ_2 respectively. If $(\ell_2 - \ell_1)$ is maintained same at all temperatures, which one of the following relations holds good?

(1) $\alpha_1\ell_2 = \alpha_2\ell_1$ (2) $\alpha_1\ell_2^2 = \alpha_2\ell_1^2$
(3) $\alpha_1^2\ell_2 = \alpha_2^2\ell_1$ (4) $\alpha_1\ell_1 = \alpha_2\ell_2$

PTM311

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20. The molecules of a given mass of a gas have r.m.s. velocity of 200 m/s at 27°C and 1.0×10^5 N/m^2 pressure. When the temperature and pressure of the gas are respectively, 127°C and 0.05×10^5 N/m^2, the r.m.s. velocity of its molecules in m/s is :

(1) $100\sqrt{2}$ (2) $\frac{400}{\sqrt{3}}$ (3) $\frac{100\sqrt{2}}{3}$ (4) $\frac{100}{3}$

PTM312

21. A gas is compressed isothermally to half its initial volume. The same gas is compressed separately through an adiabatic process until its volume is again reduced to half. Then: -

(1) Compressing the gas isothermally will require more work to be done.

(2) Compressing the gas through adiabatic process will require more work to be done.

(3) Compressing the gas isothermally or adiabatically will require the same amount of work.

(4) Which of the case (whether compression through isothermal or through adiabatic process) requires more work will depend upon the atomicity of the gas.

PTM313

22. A piece of ice falls from a height h so that it melts completely. Only one-quarter of the heat produced is absorbed by the ice and all energy of ice gets converted into heat during its fall. The value of h is :

[Latent heat of ice is 3.4×10^5 J/kg and g = 10 N/kg]

(1) 34 km (2) 544 km

(3) 136 km (4) 68 km

PTM314

NEET-II 2016

23. Two identical bodies are made of a material for which the heat capacity increases with temperature. One of these is at 100 °C, while the other one is at 0°C. If the two bodies are brought into contact, then, assuming no heat loss, the final common temperature is :-

(1) less than 50 °C but greater than 0 °C

(2) 0 °C

(3) 50 °C

(4) more than 50 °C

PTM315

24. A body cools from a temperature 3T to 2T in 10 minutes. The room temperature is T. Assume that Newton's law of cooling is applicable. The temperature of the body at the end of next 10 minutes will be: -

(1) $\frac{4}{3}$T (2) T (3) $\frac{7}{4}$T (4) $\frac{3}{2}$T

PTM316

25. One mole of an ideal monatomic gas undergoes a process described by the equation PV^3 = constant. The heat capacity of the gas during this process is

(1) 2 R (2) R (3) $\frac{3}{2}$R (4) $\frac{5}{2}$R

PTM317

26. The temperature inside a refrigerator is t_2°C and the room temperature is t_1°C. The amount of heat delivered to the room for each joule of electrical energy consumed ideally will be: -

(1) $\frac{t_2+273}{t_1-t_2}$ (2) $\frac{t_1+t_2}{t_1+273}$

(3) $\frac{t_1}{t_1-t_2}$ (4) $\frac{t_1+273}{t_1-t_2}$

PTM318

27. A given sample of an ideal gas occupies a volume V at a pressure P and absolute temperature T. The mass of each molecule of the gas is m. Which of the following gives the density of the gas?

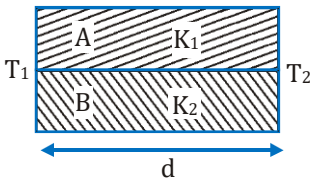
(1) P/(kTV) (2) mkT

(3) P/(kT) (4) Pm/(kT)

PTM319

NEET(UG)-2017

28. Two rods A and B of different materials are welded together as shown in figure. Their thermal conductivities are K_1 and K_2. The thermal conductivity of the composite rod will be: -

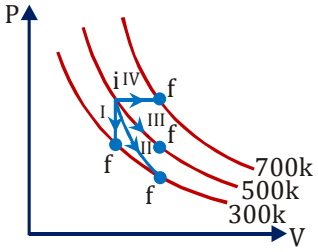


(1) $\frac{3(K_1+K_2)}{2}$ (2) $K_1 + K_2$

(3) $2(K_1 + K_2)$ (4) $\frac{K_1+K_2}{2}$

PTM320

29. Thermodynamic processes are indicated in the following diagram: -



Match the following

	Column-1		Column-2
P.	Process I	a.	Adiabatic
Q.	Process II	b.	Isobaric
R.	Process III	c.	Isochoric
S.	Process IV	d.	Isothermal

(1) P → c, Q → a, R → d, S → b
(2) P → c, Q → d, R → b, S → a
(3) P → d, Q → b, R → a, S → c
(4) P → a, Q → c, R → d, S → b

PTM321

30. A spherical black body with a radius of 12 cm radiates 450 watt power at 500 K. If the radius were halved and the temperature doubled, the power radiated in watt would be: -
(1) 450 (2) 1000 (3) 1800 (4) 225

PTM322

31. A carnot engine having an efficiency of $\frac{1}{10}$ as heat engine, is used as a refrigerator. If the work done on the system is 10 J, the amount of energy absorbed from the reservoir at lower temperature is :-
(1) 90 J (2) 99 J (3) 100 J (4) 1 J

PTM323

32. A gas mixture consists of 2 moles of O_2 and 4 moles of Ar at temperature T. Neglecting all vibrational modes, the total internal energy of the system is:-
(1) 15 RT (2) 9 RT (3) 11 RT (4) 4 RT

PTM324

NEET(UG)-2018

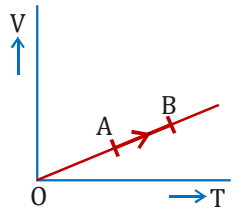
33. A sample of 0.1 g of water at 100°C and normal pressure (1.013×10^5 Nm^{-2}) requires 54 cal of heat energy to convert to steam at 100°C. If the volume of the steam produced is 167.1 cc, the change in internal energy of the sample, is :-
(1) 104.3 J (2) 208.7 J
(3) 42.2 J (4) 84.5 J

PTM325

34. The power radiated by a black body is P and it radiates maximum energy at wavelength λ_0. If the temperature of the black body is now changed so that it radiates maximum energy at wavelength $\frac{3}{4}\lambda_0$, the power radiated by it becomes nP. The value of n is: -
(1) $\frac{3}{4}$ (2) $\frac{4}{3}$ (3) $\frac{256}{81}$ (4) $\frac{81}{256}$

PTM326

35. The volume (V) of a monatomic gas varies with its temperature (T), as shown in the graph. The ratio of work done by the gas, to the heat absorbed by it, when it undergoes a change from state A to state B, is :-



(1) $\frac{2}{5}$ (2) $\frac{2}{3}$ (3) $\frac{1}{3}$ (4) $\frac{2}{7}$

PTM327

36. The efficiency of an ideal heat engine working between the freezing point and boiling point of water, is :-
(1) 26.8% (2) 20%
(3) 6.25% (4) 12.5%

PTM328

37. At what temperature will the rms speed of oxygen molecules become just sufficient for escaping from the Earth's atmosphere?
(Given: Mass of oxygen molecule (m) = 2.76×10^{-26} kg Boltzmann's constant $k_B = 1.38 \times 10^{-23}$ J K^{-1}) :-
(1) 2.508×10^4 K (2) 8.360×10^4 K
(3) 5.016×10^4 K (4) 1.254×10^4 K

PTM329

NEET(UG) 2019

38. In which of the following processes, heat is neither absorbed nor released by a system?
(1) isothermal (2) adiabatic
(3) isobaric (4) isochoric

PTM330

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39. Increase in temperature of a gas filled in a container would lead to :
(1) increase in its mass
(2) increase in its kinetic energy
(3) decrease in its pressure
(4) decrease in intermolecular distance

PTM331

40. A copper rod of 88 cm and an aluminum rod of unknown length have their increase in length independent of increase in temperature. The length of aluminum rod is :
(α_{Cu} = 1.7 × 10^{-5} K^{-1} and α_{Al} = 2.2 × 10^{-5} K^{-1})
(1) 6.8 cm (2) 113.9 cm
(3) 88 cm (4) 68 cm

PTM332

41. The unit of thermal conductivity is :
(1) J m K^{-1} (2) J m^{-1} K^{-1}
(3) W m K^{-1} (4) W m^{-1} K^{-1}

PTM333

NEET(UG) 2019 (Odisha)

42. An object kept in a large room having air temperature of 25°C takes 12 minutes to cool from 80°C to 70°C.
The time taken to cool for the same object from 70°C to 60°C would be nearly: -
(1) 10 min (2) 12 min
(3) 20 min (4) 15 min

PTM334

43. A deep rectangular pond of surface area A, containing water (density = ρ, specific heat capacity = s), is located in a region where the outside air temperature is at a steady value of –26°C. The thickness of the frozen ice layer in this pond, at a certain instant is x.
Taking the thermal conductivity of ice as K, and its specific latent heat of fusion as L, the rate of increase of the thickness of ice layer, at this instant would be given by :-
(1) 26K/ρr(L–4s) (2) 26K/ (ρx^2–L)
(3) 26K/(ρxL) (4) 26K/ρr(L+4s)

PTM335

44. The value of $\gamma\left(=\frac{C_p}{C_v}\right)$, for hydrogen, helium and another ideal diatomic gas X (whose molecules are not rigid but have an additional vibrational mode), are respectively equal to: -
(1) $\frac{7}{5},\frac{5}{3},\frac{9}{7}$ (2) $\frac{5}{3},\frac{7}{5},\frac{9}{7}$
(3) $\frac{5}{3},\frac{7}{5},\frac{7}{5}$ (4) $\frac{7}{5},\frac{5}{3},\frac{7}{5}$

PTM336

45. 1g of water, of volume 1 cm^3 at 100°C, is converted into steam at same temperature under normal atmospheric pressure (≃1×10^5Pa). The volume of steam formed equals 1671 cm^3. If the specific latent heat of vaporisation of water is 2256 J/g, then the change in internal energy is,
(1) 2423 J (2) 2089 J
(3) 167 J (4) 2256 J

PTM337

NEET(UG) 2020

46. The average thermal energy for a mono-atomic gas is : (k_B is Boltzmann constant and T, absolute temperature)
(1) $\frac{7}{2}k_BT$ (2) $\frac{1}{2}k_BT$
(3) $\frac{3}{2}k_BT$ (4) $\frac{5}{2}k_BT$

PTM338

47. The mean free path for a gas, with molecular diameter d and number density n can be expressed as :
(1) $\frac{1}{\sqrt{2}n^2\pi^2d^2}$ (2) $\frac{1}{\sqrt{2}n\pi d}$
(3) $\frac{1}{\sqrt{2}n\pi d^2}$ (4) $\frac{1}{\sqrt{2}n^2\pi d^2}$

PTM339

48. The quantities of heat required to raise the temperature of two solid copper spheres of radii r_1 and r_2 (r_1 = 1.5 r_2) through 1 K are in the ratio:
(1) $\frac{5}{3}$ (2) $\frac{27}{8}$ (3) $\frac{9}{4}$ (4) $\frac{3}{2}$

PTM340

49. A cylinder contains hydrogen gas at pressure of 249 kPa and temperature 27°C.
Its density is : (R = 8.3 J mol^{-1} K^{-1})
(1) 0.02 kg/m^3 (2) 0.5 kg/m^3
(3) 0.2 kg/m^3 (4) 0.1 kg/m^3

PTM341

50. Two cylinders A and B of equal capacity are connected to each other via a stop cock. A contains an ideal gas at standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. The stop cock is suddenly opened. The process is: -
(1) isobaric (2) isothermal
(3) adiabatic (4) isochoric

PTM342

NEET(UG) 2020 (COVID-19)

51. The efficiency of a Carnot engine depends upon

(1) the temperature of the sink only

(2) the temperatures of the source and sink

(3) the volume of the cylinder of the engine

(4) the temperature of the source only

PTM343

52. The mean free path ℓ for a gas molecule depends upon diameter, d of the molecule as :

(1) $\ell \propto \frac{1}{d^2}$ (2) $\ell \propto d$

(3) $\ell \propto d^2$ (4) $\ell \propto \frac{1}{d}$

PTM344

53. An ideal gas equation can be written as $P = \frac{\rho RT}{M_0}$ where ρ and M_0 are respectively,

(1) mass density, mass of the gas

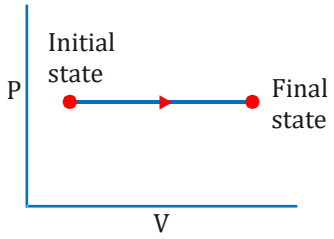
(2) number density, molar mass

(3) mass density, molar mass

(4) number density, mass of the gas

PTM345

54. The P-V diagram for an ideal gas in a piston cylinder assembly undergoing a thermodynamic process is shown in the figure. The process is



(1) adiabatic (2) isochoric

(3) isobaric (4) isothermal

PTM346

55. Three stars A, B, C have surface temperatures T_A, T_B, T_C respectively. Star A appears bluish, star B appears reddish and star C yellowish. Hence,

(1) $T_A > T_B > T_C$ (2) $T_B > T_C > T_A$

(3) $T_C > T_B > T_A$ (4) $T_A > T_C > T_B$

PTM347

NEET(UG) 2021

56. A cup of coffee cools from 90°C to 80°C in t minutes, when the room temperature is 20°C. The time taken by a similar cup of coffee to cool from 80°C to 60°C at a room temperature same at 20°C is: -

(1) $\frac{13}{10}t$ (2) $\frac{13}{5}t$ (3) $\frac{10}{13}t$ (4) $\frac{5}{13}t$

PTM348

57. Match **Column-I** and **Column-II** and choose the **correct** match from the given choices.

Column-I		Column-I	
(A)	Root mean square speed of gas molecules	(P)	$\frac{1}{3}nm\bar{v}^2$
(B)	Pressure exerted by ideal gas	(Q)	$\sqrt{\frac{3RT}{M}}$
(C)	Average kinetic energy of a molecule	(R)	$\frac{5}{2}RT$
(D)	Total internal energy of 1 mole of a diatomic gas	(S)	$\frac{3}{2}k_BT$

(1) (A) - (R), (B) - (P), (C) - (S), (D) - (Q)

(2) (A) - (Q), (B) - (R), (C) - (S), (D) - (P)

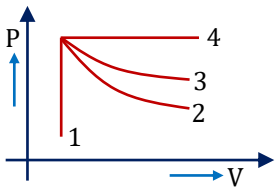
(3) (A) - (Q), (B) - (P), (C) - (S), (D) - (R)

(4) (A) - (R), (B) - (Q), (C) - (P), (D) - (S)

PTM349

NEET(UG) 2022

58. An ideal gas undergoes four different processes from the same initial state as shown in the figure below. Those processes are adiabatic, isothermal, isobaric and isochoric. The curve which represents the adiabatic process among 1,2,3 and 4 is: -



(1) 2 (2) 3 (3) 4 (4) 1

PTM350

59. The energy that will be ideally radiated by a 100 kW transmitter in 1 hour is: -

(1) 36×10^4 J (2) 36×10^5 J

(3) 1×10^5 J (4) 36×10^7 J

PTM351

60. The volume occupied by the molecules contained in 4.5 kg water at STP, if the intermolecular forces vanish away is: -

(1) $5.6 \times 10^3\ m^3$ (2) $5.6 \times 10^{-3}\ m^3$

(3) $5.6\ m^3$ (4) $5.6 \times 10^6\ m^3$

PTM352

RE-NEET(UG) 2022

61. An ideal gas follows a process described by the equation $PV^2 = C$ from the initial (P_1, V_1, T_1) to final (P_2, V_2, T_2) thermodynamics states, where C is a constant. Then: -

(1) If $P_1 > P_2$ then $T_1 < T_2$

(2) If $V_2 > V_1$ then $T_2 > T_1$

(3) If $V_2 > V_1$ then $T_2 < T_1$

(4) If $P_1 > P_2$ then $V_1 > V_2$

PTM353

62. Two rods one made of copper and other made of steel of the same length and same cross-sectional area are joined together. The thermal conductivity of copper and steel are $385 Js^{-1}K^{-1}\ m^{-1}$ and $50\ Js^{-1}K^{-1}m^{-1}$ respectively. The free ends of copper and steel are held at 100°C and 0°C respectively. The temperature at the junction is, nearly:

(1) 12°C (2) 50°C

(3) 73°C (4) 88.5°C

PTM354

NEET(UG) 2023

63. The temperature of a gas is –50°C. To what temperature the gas should be heated so that the rms speed is increased by 3 times?

(1) 3295°C (2) 3097 K

(3) 223 K (4) 669°C

PTM382

64. A Carnot engine has an efficiency of 50% when its source is at a temperature 327° C. The temperature of the sink is :-

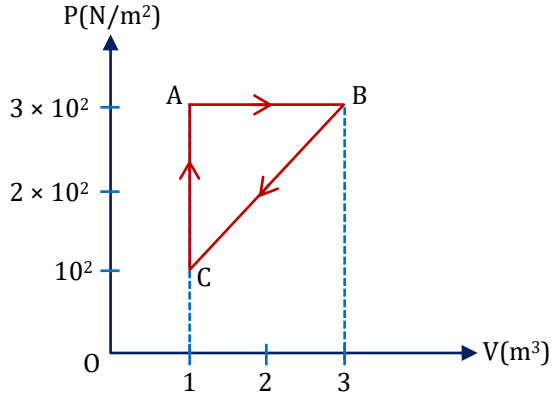
(1) 15°C (2) 100°C

(3) 200°C (4) 27°C

PTM383

RE-NEET(UG) 2023

65. For the given cycle, the work done during isobaric process is :



(1) 200 J (2) Zero

(3) 400 J (4) 600 J

PTM384

66. A container of volume 200 cm^3 contains 0.2 mole of hydrogen gas and 0.3 mole of argon gas. The pressure of the system at temperature 200 K ($R = 8.3\ JK^{-1}\ mol^{-1}$) will be:-

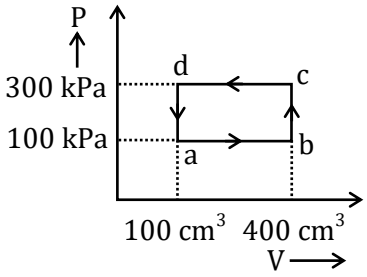
(1) 6.15×10^5 Pa (2) 6.15×10^4 Pa

(3) 4.15×10^5 Pa (4) 4.15×10^6 Pa

PTM385

NEET(UG) 2024

67. A thermodynamic system is taken through the cycle abcda. The work done by the gas along the path bc is :

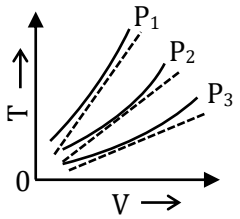


(1) zero (2) 30 J

(3) –90 J (4) –60 J

PTM386

68. The following graph represents the T-V curves of an ideal gas (where T is the temperature and V the volume) at three pressures P_1, P_2 and P_3 compared with those of Charles's law represented as dotted lines.



Then the correct relation is :

(1) $P_3 > P_2 > P_1$ (2) $P_1 > P_3 > P_2$

(3) $P_2 > P_1 > P_3$ (4) $P_1 > P_2 > P_3$

PTM387

RE-NEET(UG) 2024

69. Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R.**

Assertion A : Houses made of concrete roofs overlaid with foam keep the room hotter during summer.

Reason R : The layer of foam insulation prohibits heat transfer, as it contains air pockets.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) **A** is true but **R** is false.

(2) **A** is false but **R** is true.

(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.

(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

PTM388

70. Water is used as a coolant in a nuclear reactor because of its :

(1) high thermal expansion coefficient

(2) high specific heat capacity

(3) low density

(4) low boiling point

PTM389

71. The equilibrium state of a thermodynamic system is described by :

A. Pressure B. Total heat

C. Temperature D. Volume

E. Work done

Choose **the most appropriate** answer from the options given below :

(1) A, B and E only (2) B, C and D only

(3) A, B and C only (4) A, C and D only

PTM390

72. According to the law of equipartition of energy, the number of vibrational modes of a polyatomic gas of constant $\gamma = \frac{C_p}{C_v}$ is (where C_p and C_v are the specific heat capacities of the gas at constant pressure and constant volume, respectively) :

(1) $\frac{4+3\gamma}{\gamma-1}$ (2) $\frac{3+4\gamma}{\gamma-1}$

(3) $\frac{4-3\gamma}{\gamma-1}$ (4) $\frac{3-4\gamma}{\gamma-1}$

PTM391

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	1	2	3	4	1	1	2	2	2	3	3	2	2	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	3	4	4	2	2	3	4	4	2	4	4	4	1	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	1	3	2	3	1	1	2	2	2	4	4	4	3	1	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	3	2	3	3	2	1	3	3	4	2	3	1	4	3
Question	61	62	63	64	65	66	67	68	69	70	71	72			
Answer	3	4	1	4	4	4	1	4	2	2	4	3			

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Exercise - III (Analytical Questions) Master Your Understanding

THERMAL EXPANSION

1. Match the column with scale of temperature and its range -

	Column-1		Column-2
(A)	Celsius scale	(P)	32° to 212°
(B)	Fahrenheit scale	(Q)	x° to y°
(C)	Kelvin scale	(R)	0° to 100°
(D)	Unknown scale	(S)	273.15 to 373.15

(1) A → P, B → Q, C → R, D → S
(2) A → R, B → P, C → S, D → Q
(3) A → R, B → S, C → P, D → Q
(4) A → R, B → Q, C → S, D → P

PTM355

2. Choose the correct statement: -
(a) When a bimetallic strip is heated it bends in the form of an arc.
(b) It is used in thermostat
(c) In bimetallic strip the material having lower temperature expansion co-efficient always goes on outer side.
(d) A bimetallic strip on heating bends in the form of helix
(1) (a, b) (2) (b, c) (3) (a, c) (4) (b, d)

PTM356

3. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : A brass disc is just fitted in a hole in steel plate. The system must be cooled to loosen the disc from the hole.
Reason (R) : The coefficient of linear expansion for brass is greater than the coefficient of linear expansion for steel.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true and (R) is NOT the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

PTM357

CALORIMETRY

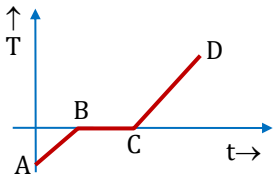
4. If heat is supplied to a solid, its temperature:-
(A) must increase
(B) may increase
(C) may remain constant
(D) may decrease
Choice the correct statement: -
(1) A (2) B,D (3) B,C,D (4) B,C

PTM358

5. A good cooking utensil should have: -
(A) low specific heat
(B) high specific heat
(C) low thermal capacity
(D) high coefficient of thermal conductivity
Which statement are correct: -
(1) A,C,D (2) B,C,D (3) A,B,D (4) B,C

PTM359

6. Heat is supplied to a ice at a constant rate temperature variation with time is as shown in figure, then –



(A) During AB volume of substance increases
(B) During BC volume of substance decreases
(C) Specific heat of substance in liquid phase is proportional to reciprocal of slope of portion AB of graph
(D) Latent heat of fusion of substance is independent of portion AB of graph.
Then correct option is -
(1) A,C (2) B,C (3) A,C,D (4) B,D

PTM360

7. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In change of state from solid to liquid the temperature decrease until the entire amount of the solid substance melts.

Reason (R) : The phenomenon of refreezing is called melting.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) Both **(A)** and **(R)** are false.

PTM361

8. Match the column :-

	Column-1		**Column-2**
(A)	Heat	(P)	Degree of hotness or coolness of a substance
(B)	Internal energy	(Q)	Amount of heat required to rise the temperature of unit mass of a substance by 1°C
(C)	Temperature	(R)	Energy in transit
(D)	Specific heat	(S)	The sum of energy due to attractive force between molecules and due to random motion

(1) A → R, B → S, C → P, D → Q

(2) A → R, B → P, C → S, D → Q

(3) A → R, B → Q, C → P, D → S

(4) A → R, B → S, C → Q, D → P

PTM362

HEAT TRANSFER

9. Assume transmittivity t = 0 for all the cases - Which of the following statement is correct -

(A) Bad absorber is bad emitter

(B) Bad absorber is good reflector

(C) Bad reflector is good emitter

(D) Bad emitter is good absorber

(1) A, B, C, D (2) A, C, D

(3) A, B, D (4) A, B, C

PTM363

10. Match the column in which column-1 gives the statement and column-2 its explanation by which law.

	Column-1		**Column-2**
(A)	When a person enters in an A.C. room then he feels cooler	(P)	Newtons law of cooling
(B)	Good absorber are good emitter of heat	(Q)	Wein's law
(C)	Colour of a star is an indication of temperature	(R)	Kirchhoff's law
(D)	A hot cup of tea when placed in a room is cooled in some time	(S)	Prevost theory of mutual heat exchange

(1) A → P, B → Q, C → R, D → S

(2) A → Q, B → S, C → P, D → R

(3) A → S, B → R, C → Q, D → P

(4) A → R, B → P, C → S, D → Q

PTM364

11. A spherical black body of radius r radiates power P and its rate of cooling is R then -

(a) $P \propto r$ (b) $P \propto r^2$

(c) $R \propto r^2$ (d) $R \propto \frac{1}{r}$

Choice the correct one :-

(1) a, b (2) a, c (3) b,d (4) a,b,d

PTM365

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12. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : For a given amount of an ideal gas, at constant temperature, the product of pressure and volume is constant.
Reason (R) : The root mean square velocity of the molecules is inversely proportional to molecular weight.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) Both **(A)** and **(R)** are false.

PTM366

13. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : For higher temperature, the peak emission wavelength of a black body shifts to lower wavelengths.
Reason (R) : Peak emission wavelength of a black body is proportional to the fourth power of temperature.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PTM367

KTG

14. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : The rms speed of helium gas is equal to rms speed of hydrogen gas when absolute temperature of helium is twice that of hydrogen gas.
Reason (R) : RMS speed of gas depends only on the temperature of gas.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PTM368

15. In a mixture of nitrogen and helium kept at room temperature, as compared to a helium molecule nitrogen molecule strikes into the wall –
(a) with greater average speed
(b) with smaller average speed
(c) with greater average kinetic energy
(d) with smaller average kinetic energy
Choose the correct statements :-
(1) a, c (2) b,d (3) a,d (4) b,c

PTM369

16. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : An ideal gas is enclosed within a container fitted with a piston. When volume of this enclosed gas is increased at constant temperature. The pressure exerted by gas on piston decreases.
Reason (R) : In the above situation, the rate of molecules per unit area striking on the piston decreases.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PTM370

17. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Evaporation has a cooling effect.

Reason (R) : At the surface of liquid, some of the faster upward moving molecules have enough kinetic energy to overcome the attraction from other molecules and escape from the liquid. With these faster molecules the average kinetic energy left behind is reduced.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PTM371

18. For a monoatomic gas at temperature T, match the following column –

	Column-1		**Column-2**
(A)	Speed of sound	(P)	$\sqrt{\frac{2RT}{M_w}}$
(B)	RMS speed of gas molecule	(Q)	$\sqrt{\frac{8}{\pi}\frac{RT}{M_w}}$
(C)	Average speed of gas molecule	(R)	$\sqrt{\frac{3RT}{M_w}}$
(D)	Most probable speed of gas molecule	(S)	$\sqrt{\frac{\gamma RT}{M_w}}$

(1) A → P, B → Q, C → R, D → S

(2) A → Q, B → S, C → P, D → R

(3) A → R, B → P, C → S, D → Q

(4) A → S, B → R, C → Q, D → P

PTM372

THERMODYNAMICS

19. Match the following column –

	Column-1		**Column-2**
(A)	Adiabatic bulk modulus	(P)	– P/V
(B)	Slope of PV curve in isothermal process	(Q)	$\frac{2}{\gamma-1}$
(C)	Degree of freedom	(R)	γP
(D)	Molar heat capacity at constant pressure divide by R	(S)	$\frac{\gamma}{\gamma-1}$

(1) A → R, B → P, C → Q, D → S

(2) A → P, B → R, C → S, D → Q

(3) A → Q, B → S, C → P, D → R

(4) A → S, B → Q, C → R, D → P

PTM373

20. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : In an adiabatic process, change in internal energy of a gas is equal to work done on the gas in the process.

Reason (R) : Temperature of gas remains constant in a adiabatic process.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

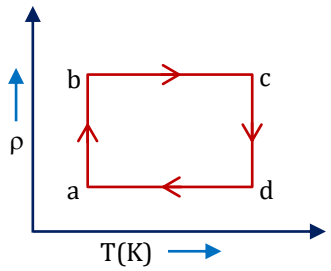
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

PTM374

21. In the ρ – T graph (for fixed mass of an ideal gas), shown in the figure match the following (ρ → density, T → temperature)



	Column-1		Column-2
(A)	Process ab	(p)	isochoric process
(B)	Process bc	(q)	$\Delta U = 0$
(C)	Process cd	(r)	P increasing
(D)	Process da	(s)	P decreasing

(1) A → q,s, B → p,q, C → p,s, D → r,q
(2) A → q,r, B → p, C → q,s, D → p
(3) A → p,s, B → q,r, C → r,s, D → s,r
(4) A → p,q, B → q,r, C → r,s, D → s,p

PTM375

22. Match the column –

	Column-1		Column-2
(A)	$dU = nC_V dT$ is valid for	(P)	Adiabatic process
(B)	Temperature of the system can change in	(Q)	Isothermal process
(C)	Q = dU + W is valid for	(R)	Polytopic process
(D)	The process in which heat exchange between the system and surrounding is zero.	(S)	Free expansion of ideal gas

(1) A→P,Q,R,S; B→P,R,S; C→P,Q,R,S; D→P,S
(2) A→P,Q,R,S; B→P,Q; C→P,Q,R,S; D→Q,R
(3) A→P,Q,R,S; B→P,S; C→P,Q,R,S; D→Q,R
(4) A→P,Q,R,S; B→Q,S; C→P,Q,R,S; D→Q,R

PTM376

23. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : For an ideal gas $C_P > C_V$.
Reason (R) : Work is done in expansion of the gas at constant pressure.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PTM377

24. The internal energy of a system remains constant when it undergoes –
(a) a cyclic process
(b) an isothermal process
(c) an adiabatic process
(d) any process in which heat released by the system is equal to the work done on the system.
Choose the correct statements :-
(1) a, b, c (2) b,c (3) a, b (4) a,b,d

PTM378

25. The molar heat capacity for an ideal gas -
(a) is zero for adiabatic process.
(b) is infinite for an isothermal process
(c) depends on the nature of gas for a process in which either volume or pressure is constant
(d) is equal to the product of the molecular weight and gram specific heat capacity for any process.
Choose the correct statements :-
(1) a, b (2) c, d
(3) a, b, c & d (4) b, c, d

PTM379

26. Match the following column –

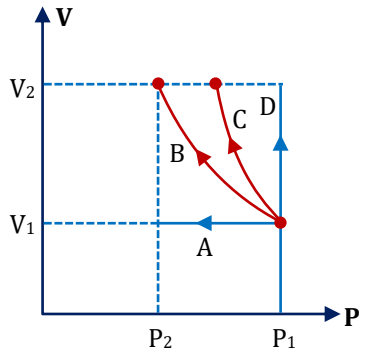
	Column-1		Column-2
(A)	Adiabatic expansion	(P)	No work done
(B)	isobaric expansion	(Q)	Constant internal energy
(C)	Isothermal expansion	(R)	Increase in internal energy
(D)	Isochoric process	(S)	Decrease in internal energy

(1) A → S, B → R, C → Q, D → P
(2) A → P, B → Q, C → R, D → S
(3) A → Q, B → P, C → S, D → R
(4) A → R, B → S, C → P, D → Q

PTM380

27. Volume versus pressure curves for one mole of an ideal gas are given for four processes as shown in figure.

(B → Adiabatic process, C → Isothermal process)



	Column-I		Column-II
(A)	For process A	(P)	Work done by the gas is positive.
(B)	For process B	(Q)	Temperature will increase.
(C)	For process C	(R)	Heat supplied is positive.
(D)	For process D	(S)	Change in internal energy is negative.

The correct option is :-

(1) (A) – S, (B) – PS, (C) – PQ, (D) – PQR

(2) (A) – PS, (B) – P, (C) – PR, (D) – PQQ

(3) (A) – PS, (B) – S, (C) – PR , (D) – PQ

(4) (A) – S, (B) – PS, (C) – PR, (D) – PQR

PTM381

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	2	1	1	4	1	4	4	1	4	3	3	2	3	3	4
Question	16	17	18	19	20	21	22	23	24	25	26	27			
Answer	1	1	4	1	3	2	1	1	4	3	1	4			

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Exercise - I (Conceptual Questions) Build Up Your Understanding

PERIODIC MOTION AND ITS CHARACTERISTICS

1. A particle of mass m is executing S.H.M. If amplitude is a and frequency n, the value of its force constant will be :
(1) mn^2 (2) $4mn^2a^2$
(3) ma^2 (4) $4\pi^2mn^2$
POS001

2. The equation of motion of a particle executing S.H.M. where letters have usual meaning is :
(1) $\frac{d^2x}{dt^2} = -\frac{k}{m}x$ (2) $\frac{d^2x}{dt^2} = +\omega^2x$
(3) $\frac{d^2x}{dt^2} = -\omega^2x^2$ (4) $\frac{d^2x}{dt^2} = -kmx$
POS002

3. The equation of motion of a particle executing SHM is $\left(\frac{d^2x}{dt^2}\right) + kx = 0$. The time period of the particle will be :
(1) $2\pi/\sqrt{k}$ (2) $2\pi/k$
(3) $2\pi k$ (4) $2\pi\sqrt{k}$
POS003

4. Which of the following equation does not represent a simple harmonic motion :
(1) $y = a\sin\omega t$ (2) $y = b\cos\omega t$
(3) $y = a\sin\omega t + b\cos\omega t$ (4) $y = a\tan\omega t$
POS004

SIMPLE HARMONIC MOTION (SHM) AND ITS EQUATION

5. The displacement of a particle in S.H.M. is indicated by equation $y = 10\sin(20t + \pi/3)$ where y is in metres. The value of time period of vibration will be (in seconds) :
(1) $10/\pi$ (2) $\pi/10$
(3) $2\pi/10$ (4) $10/2\pi$
POS005

6. The value of phase at maximum displacement from the mean position of a particle in S.H.M. is:
(1) $\pi/2$ (2) π (3) Zero (4) 2π
POS006

7. The equation of a simple harmonic motion is $x = 0.34\cos(3000t + 0.74)$. Where x and t are in mm and sec. respectively. The frequency of the motion is :
(1) 3000 (2) $3000/2\pi$
(3) $0.74/2\pi$ (4) $3000/\pi$
POS007

8. The acceleration of a particle executing S.H.M. is
(1) Always directed towards the equilibrium position
(2) Always towards the one end
(3) Continuously changing in direction
(4) Maximum at the mean position
POS008

9. The distance covered by a particle executing SHM, in one time period is equal to :
(1) Four times the amplitude
(2) Two times the amplitude
(3) One times the amplitude
(4) Eight times the amplitude
POS009

10. The phase of a particle in S.H.M. is $\pi/2$, then :
(1) Its velocity will be maximum.
(2) Its acceleration will be minimum.
(3) Restoring force on it will be minimum.
(4) Its displacement will be maximum.
POS010

11. The displacement of a particle in S.H.M. is indicated by equation $y = 10\sin(20t + \pi/3)$ where y is in metres. The value of maximum velocity of the particle will be :
(1) 100 m/sec. (2) 150 m/sec.
(3) 200 m/sec. (4) 400 m/sec.
POS011

12. In the above question, the value of phase constant will be :
(1) Zero (2) 45° (3) 60° (4) 30°
POS012

13. The phase of a particle in SHM at time t is $\pi/6$. The following inference is drawn from this:
(1) The particle is at $x = a/2$ and moving in + X-direction
(2) The particle is at $x = a/2$ and moving in – X-direction
(3) The particle is at $x = -a/2$ and moving in + X-direction
(4) The particle is at $x = -a/2$ and moving in – X-direction
POS013

14. Two particles execute S.H.M. along the same line at the same frequency. They move in opposite direction at the mean position. The phase difference will be :
(1) 2π (2) $2\pi/3$ (3) π (4) $\pi/2$
POS014

15. The displacement from mean position of a particle in SHM at 3 seconds is $\sqrt{3}/2$ of the amplitude. Its time period will be :
(1) 18 sec. (2) $6\sqrt{3}$ sec.
(3) 9 sec. (4) $3\sqrt{3}$ sec.
POS015

16. A particle executes SHM of type $x = a\sin\omega t$. It takes time t_1 from $x = 0$ to $x = \frac{a}{2}$ and t_2 from $x = \frac{a}{2}$ to $x = a$. The ratio of $t_1 : t_2$ will be :
(1) 1 : 1 (2) 1 : 2
(3) 1 : 3 (4) 2 : 1
POS016

17. The time taken by a particle in SHM for maximum displacement is :
(1) T/8 (2) T/6 (3) T/2 (4) T/4
POS017

18. A particle executes SHM with periodic time of 6 seconds. The time taken for traversing a distance of half the amplitude from mean position is :
(1) 3 sec. (2) 2 sec.
(3) 1 sec. (4) 1/2 sec.
POS018

19. The phase difference between the displacement and acceleration of particle executing S.H.M. in radian is :
(1) $\pi/4$ (2) $\pi/2$ (3) π (4) 2π
POS019

20. The phase difference in radians between displacement and velocity in S.H.M. is :
(1) $\pi/4$ (2) $\pi/2$ (3) π (4) 2π
POS020

21. If the maximum velocity of a particle in SHM is v_0. then its velocity at half the amplitude from position of rest will be :
(1) $v_0/2$ (2) v_0
(3) $v_0\sqrt{3/2}$ (4) $v_0\sqrt{3}/2$
POS021

22. At a particular position the velocity of a particle in SHM with amplitude a is $\sqrt{3}/2$ that at its mean position. In this position, its displacement is :
(1) a/2 (2) $\sqrt{3}$ a/2 (3) $a\sqrt{2}$ (4) $\sqrt{2a}$
POS022

23. The acceleration of a particle in SHM at 5 cm from its mean position is 20 cm/sec^2. The value of angular frequency in radians/sec will be :
(1) 2 (2) 4 (3) 10 (4) 14
POS023

24. The amplitude of a particle in SHM is 5 cm and its time period is π. At a displacement of 3 cm from its mean position the velocity in cm/sec will be :
(1) 8 (2) 12 (3) 2 (4) 16
POS024

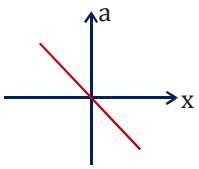
25. The maximum velocity and acceleration of a particle in S.H.M. are 100 cm/sec and 157 cm/sec^2 respectively. The time period in seconds will be :
(1) 4 (2) 1.57 (3) 0.25 (4) 1
POS025

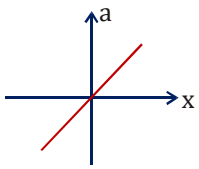
26. If the displacement, velocity and acceleration of a particle in SHM are 1 cm, 1cm/sec, 1cm/sec^2 respectively its time period will be (in seconds) :
(1) π (2) 0.5π (3) 2π (4) 1.5π
POS026

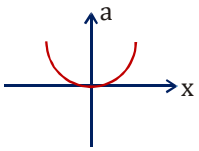
27. The particle is executing S.H.M. on a line 4 cm long. If its velocity at mean position is 12 cm/sec, its frequency in Hertz will be :
(1) $\frac{2\pi}{3}$ (2) $\frac{3}{2\pi}$ (3) $\frac{\pi}{3}$ (4) $\frac{3}{\pi}$
POS027

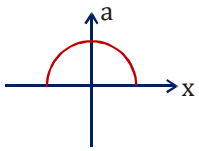
28. Which of the following statement is incorrect for an object executing S.H.M.:
(1) The value of acceleration is maximum at the extreme points
(2) The total work done for completing one oscillation is zero.
(3) The energy changes from one form to another
(4) The velocity at the mean position is zero
POS028

29. The variation of acceleration (a) and displacement (x) of the particle executing SHM is indicated by the following curve :

(1) 

(2) 

(3) 

(4) 

POS029

30. The time period of an oscillating body executing SHM is 0.05 sec and its amplitude is 40 cm. The maximum velocity of particle is :

(1) 16π ms^{-1} (2) 2π ms^{-1}

(3) 3.1 ms^{-1} (4) 4π m/s

POS030

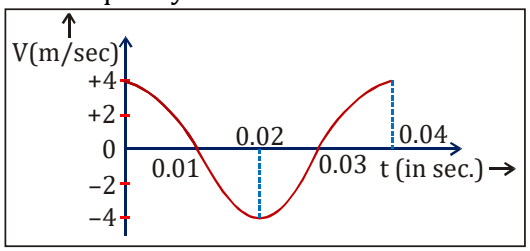
31. A body of mass 5 gm is executing S.H.M. about a point with amplitude 10 cm. Its maximum velocity is 100 cm/sec. Its velocity will be 50 cm/sec at a distance from mean position :

(1) 5 cm (2) $5\sqrt{2}$ cm

(3) $5\sqrt{3}$ cm (4) $10\sqrt{2}$ cm

POS031

32. The velocity-time diagram of a harmonic oscillator is shown in the adjoining figure. The frequency of oscillation is :



(1) 25 Hz (2) 50 Hz

(3) 12.25 Hz (4) 33.3 Hz

POS032

33. If amplitude of the particle which is executing S.H.M., is doubled, then which quantity will become double?

(1) Frequency

(2) Time period

(3) Energy

(4) Maximum velocity

POS033

34. Which one of the following statements is true for the speed 'v' and the acceleration 'a' of a particle executing simple harmonic motion

(1) Value of 'a' is zero, whatever may be the value of 'v'

(2) When 'v' is zero, 'a' is zero

(3) When 'v' is maximum, 'a' is zero

(4) When 'v' is maximum, 'a' is maximum

POS034

35. For a particle executing simple harmonic motion which of the following statement is not correct:

(1) The total energy of particle always remains the same

(2) The restoring force is always directed towards a fix point.

(3) The restoring force is maximum at the extreme positions.

(4) The acceleration of particle is maximum at the equilibrium positions.

POS035

36. In SHM velocity is maximum:

(1) At extreme position

(2) When displacement is half of amplitude

(3) At the central position

(4) When Displacement is $\frac{1}{\sqrt{2}}$ of amplitude

POS036

37. A body oscillates with SHM according to the equation $x = 5.0\cos(2\pi t + \pi)$. At time $t = 1.5$ s, its displacement, speed and acceleration respectively is :

(1) 0, -10π, $+20\pi^2$ (2) 5, 0, $-20\pi^2$

(3) 2.5, $+20\pi$, 0 (4) -5.0, $+5\pi$, $-10\pi^2$

POS037

38. The maximum velocity of simple harmonic motion represented by $y = 3\sin\left(100t + \frac{\pi}{6}\right)$ is given by

(1) 300 (2) $\frac{3\pi}{6}$ (3) 100 (4) $\frac{\pi}{6}$

POS038

39. The maximum velocity of a particle, executing simple harmonic motion with an amplitude 7 mm is 4.4 m/s. The period of oscillation is :
(1) 100 s (2) 0.01 s (3) 10 s (4) 0.1 s
POS039

40. Average velocity of a particle performing SHM in one time period is :-
(1) Zero (2) $\frac{A\omega}{2}$ (3) $\frac{A\omega}{2\pi}$ (4) $\frac{2A\omega}{\pi}$
POS040

41. A particle is executing S.H.M. with amplitude A and Time period T. Time taken by the particle to reach from extreme position to $\frac{A}{2}$
(1) $\frac{T}{6}$ (2) $\frac{T}{12}$ (3) $\frac{T}{3}$ (4) $\frac{T}{4}$
POS041

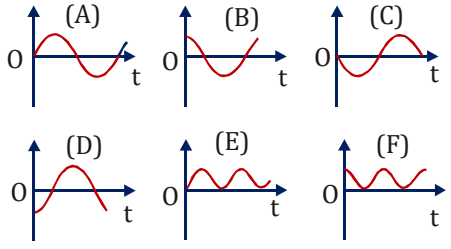
42. Total work done on a simple pendulum in one complete oscillation will be :-
(1) $\frac{1}{2}kx^2$ (2) $\frac{1}{2}kA^2$
(3) kA^2 (4) Zero
POS042

43. In S.H.M. Which one of the following quantities has constant ratio with acceleration :-
(1) Time (2) Displacement
(3) Velocity (4) Mass
POS043

44. The displacement y of a particle varies with time t, in seconds, as $y = 2\cos(\pi t + \pi/6)$. The time period of the oscillations is
(1) 2 s (3) 4 s (3) 1 s (4) 0.5 s
POS044

ENERGY IN SHM-KINETIC AND POTENTIAL ENERGIES

45. The displacement of a particle in S.H.M. is $x = a\sin\omega t$. Which of the following graph between displacement and time is correct :



(1) A (2) B (3) C (4) D
POS045

46. In previous question which of the graph between velocity and time is correct?
(1) A (2) B (3) C (4) D
POS046

47. In Question 45 which of the graph between kinetic energy and time is correct?
(1) A (2) B (3) E (4) F
POS047

48. In Question 45 which of the graph between potential energy and time is correct?
(1) A (2) B (3) E (4) F
POS048

49. In Question 45 which of the graph between acceleration and time is correct?
(1) A (2) B (3) C (4) D
POS049

50. In Question 45 if the displacement of a particle executing SHM is $x = a\cos\omega t$, which of the graph between displacement and time is correct?
(1) A (2) B (3) C (4) D
POS050

51. In question 50 which of the graph between velocity and time is correct?
(1) A (2) B (3) C (4) D
POS051

52. In question 50 which of the graph between acceleration and time is correct?
(1) A (2) B (3) C (4) D
POS052

53. In question 50 which of the graph between K.E. and time is correct?
(1) A (2) B (3) E (4) F
POS053

54. In question 50 which of the graph between P.E. and time is correct?
(1) A (2) B (3) E (4) F
POS054

55. The total energy of a particle executing SHM is directly proportional to the square of the following quantity :
(1) Acceleration (2) Amplitude
(3) Time period (4) Mass
POS055

56. The total energy of a vibrating particle in SHM is E. If its amplitude and time period are doubled, its total energy will be :
(1) 16E (2) 8E (3) 4E (4) E
POS056

57. The total vibrational energy of a particle in S.H.M. is E. Its kinetic energy at half the amplitude from mean position will be :
(1) E/2 (2) E/3 (3) E/4 (4) 3E/4
POS057

58. If total energy of a particle in SHM is E, then the potential energy of the particle at half the amplitude will be :
(1) E/2 (2) E/4 (3) 3E/4 (4) E/8
POS058

59. A particle executes SHM on a line 8 cm long. Its K.E. and P.E. will be equal when its distance from the mean position is :
(1) 4 cm (2) 2 cm
(3) $2\sqrt{2}$ cm (4) $\sqrt{2}$ cm
POS059

60. The average P.E. of a body executing S.H.M. is :
(1) $\frac{1}{2}ka^2$ (2) $\frac{1}{4}ka^2$ (3) ka^2 (4) Zero
POS060

61. The value of total mechanical energy of a particle in S.H.M. is :
(1) Always constant
(2) Depend on time
(3) $\frac{1}{2}kA^2\cos^2(\omega t+\phi)$
(4) $\frac{1}{2}mA^2\cos^2(\omega t+\phi)$
POS061

62. The maximum K.E. of a oscillating spring is 5 joules and its amplitude 10 cm. The force constant of the spring is :
(1) 100 Newton/m. (2) 1000 Newton–m
(3) 1000 Newton/m. (4) 1000 watts.
POS062

63. The force acting on a 4gm mass in the energy region $U = 8x^2$ at x = –2cm is :
(1) 8 dyne (2) 4 dyne
(3) 16 dyne (4) 32 dyne
POS063

64. Displacement between max. P.E. position and max. K.E. position for a particle executing simple harmonic motion is :
(1) $\pm\frac{a}{2}$ (2) + a (3) ± a (4) – 1
POS064

65. A particle is describing SHM with amplitude 'a'. When the potential energy of particle is one fourth of the maximum energy during oscillation, then its displacement from mean position will be:
(1) $\frac{a}{4}$ (2) $\frac{a}{3}$ (3) $\frac{a}{2}$ (4) $\frac{2a}{3}$
POS065

66. The ratio of K.E. of the particle at mean position to the point when distance is half of amplitude will be:
(1) $\frac{1}{3}$ (2) $\frac{2}{3}$ (3) $\frac{4}{3}$ (4) $\frac{3}{2}$
POS066

67. A particle is executing S.H.M., If its P.E. & K.E. is equal then the ratio of displacement & amplitude will be :
(1) $\frac{1}{\sqrt{2}}$ (2) $\sqrt{2}$ (3) $\frac{1}{2}$ (4) $\frac{3}{2}$
POS067

68. Which of the following is constant during SHM:
(1) Velocity (2) Acceleration
(3) Total energy (4) Phase
POS068

69. If <E> and <V> denotes the average kinetic and average potential energies respectively of mass describing a simple harmonic motion over one period then the correct relation is:
(1) <E> = <V> (2) <E> = 2<V>
(3) <E> = –2<V> (4) <E> = – <V>
POS069

70. The elongation of spring is 1 cm and its potential energy is U. If the spring is elongated by 3cm then potential energy will be :-
(1) 3U (2) $\frac{U}{3}$ (3) 9U (4) $\frac{U}{9}$
POS070

71. The potential energy of a spring when stretched by a distance x is E. The energy of the spring when stretched by x/2 is

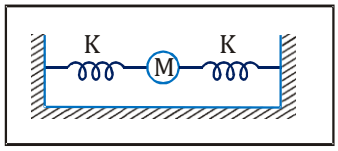
(1) E (2) E/2 (3) E/4 (4) E/6

POS071

OSCILLATIONS OF A SPRING

72. On suspending a mass M from a spring of force constant K, frequency of vibration f is obtained.

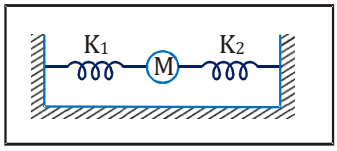
If a second spring as shown in the figure, is arranged then the frequency will be :



(1) $f\sqrt{2}$ (2) $f/\sqrt{2}$ (3) 2f (4) f

POS072

73. In the adjoining figure the frequency of oscillation for a mass M will be proportional to :



(1) K_1K_2 (2) $K_1 + K_2$

(3) $\sqrt{K_1+K_2}$ (4) $\sqrt{1/K_1+K_2}$

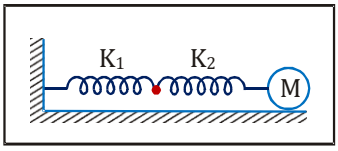
POS073

74. An object of mass m is suspended from a spring and it executes S.H.M. with frequency υ. If the mass is increased 4 times, the new frequency will be :

(1) 2υ (2) υ/2 (3) υ (4) υ/4

POS074

75. As shown in the figure, two light springs of force constant K_1 and K_2 oscillate a block of mass M. Its effective force constant will be :

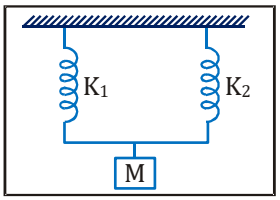


(1) K_1K_2 (2) $K_1 + K_2$

(3) $\frac{1}{K_1}+\frac{1}{K_2}$ (4) $\frac{K_1K_2}{K_1+K_2}$

POS075

76. The spring constants of two springs of same length are K_1 and K_2 as shown in figure. If an object of mass M is suspended and set vibration, the time period will be :



(1) $2\pi\sqrt{\frac{MK_1}{K_2}}$ (2) $2\pi\sqrt{\frac{M}{K_1K_2}}$

(3) $2\pi\sqrt{\frac{M}{K_1-K_2}}$ (4) $2\pi\sqrt{M/(K_1+K_2)}$

POS076

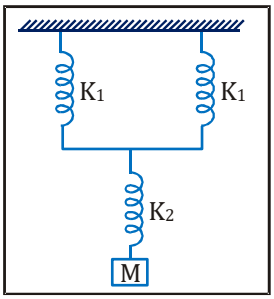
77. The total spring constant of the system as shown in the figure will be :

(1) $\frac{K_1}{2}+K_2$

(2) $\left[\frac{1}{2K_1}+\frac{1}{K_2}\right]^{-1}$

(3) $\frac{1}{2K_1}+\frac{1}{K_2}$

(4) $\left[\frac{2}{K_1}+\frac{1}{K_2}\right]^{-1}$



POS077

78. A spring is made to oscillate after suspending a mass m from one of its ends. The time period obtained is 2 seconds. On increasing the mass by 2 kg, the period of oscillation is increased by 1 second. The initial mass m will be :

(1) 2 kg (2) 1 kg

(3) 0.5 kg (4) 1.6 kg

POS078

79. The time period of a spring pendulum on earth is T. If it is taken on the moon, and made to oscillate, the period of vibration will be :

(1) Less than T (2) Equal to T

(3) More than T (4) None of these

POS079

80. On loading a spring with bob, its period of oscillation in a vertical plane is T. If this spring pendulum is tied with one end to the a friction less table and made to oscillate in a horizontal plane, its period of oscillation will be :
(1) T
(2) 2T
(3) T/2
(4) will not execute S.H.M.

POS080

81. In a winding (spring) watch, the energy is stored in the form of :
(1) Kinetic energy (2) Potential energy
(3) Electrical energy (4) None of these

POS081

82. In an artificial satellite, the object used is :
(1) Spring watch
(2) Pendulum watch
(3) Watches of both spring and pendulum
(4) None of these

POS082

83. Mass 'm' is suspended from a spring of force constant K. Spring is cut into two equal parts and same mass is suspended from it, then new frequency will be:

(1) 2ν (2) $\sqrt{2}\,\nu$ (3) ν (4) $\frac{\nu}{2}$

POS083

84. The spring constant of two springs are K_1 and K_2 respectively springs are stretch up to that limit when potential energy of both becomes equal. The ratio of applied force (F_1 and F_2) on them will be :
(1) $K_1 : K_2$ (2) $K_2 : K_1$
(3) $\sqrt{K_1} : \sqrt{K_2}$ (4) $\sqrt{K_2} : \sqrt{K_1}$

POS084

85. Force constant of a spring is K. If one fourth part is detach then force constant of remaining spring will be :

(1) $\frac{3}{4}K$ (2) $\frac{4}{3}K$ (3) K (4) 4K

POS085

86. The spring constant of a spring is K. When it is divided into n equal parts, then what is the spring constant of one part :
(1) nK (2) K/n
(3) $\frac{nK}{(n+1)}$ (4) $\frac{(n+1)K}{n}$

POS086

87. A mass of 10g is connected to a massless spring then time period of small oscillation is 10 second. If 10 g mass is replaced by 40 g mass in same spring, then its time period will be:-
(1) 5s (2) 10s (3) 20s (4) 40s

POS087

SIMPLE PENDULUM

88. The mass of a bob, suspended in a simple pendulum, is halved from the initial mass, its time period will :
(1) Be less (2) Be more
(3) Remain unchanged (4) None of these

POS088

89. If the amplitude of a simple pendulum is doubled, how many times will the value of its maximum velocity be that of the maximum velocity in initial case :

(1) $\frac{1}{2}$ (2) 2 (3) 4 (4) $\frac{1}{4}$

POS089

90. The length of a simple pendulum is $39.2/\pi^2$ m. If g = 9.8 m/sec^2, the value of time period is:
(1) 4 s (2) 8 s (3) 2 s (4) 3 s

POS090

91. The length of a simple pendulum is increased four times of its initial value, its time period with respect to its previous value will :
(1) Become twice (2) Not be different
(3) Be halved (4) Be $\sqrt{2}$ times

POS091

92. The time taken for a second pendulum from one extreme point to another is :
(1) 1 s (2) 2 s (3) 1/2 s (4) 4 s

POS092

93. The length of a seconds pendulum is (approximately) :
(1) 1 m (2) 1 cm (3) 2 m (4) 2 cm
POS093

94. The acceleration due to gravity at height R above the surface of the earth is g/4. The periodic time of a simple pendulum in an artificial satellite at this height will be :
(1) $T = 2\pi\sqrt{2l/g}$ (2) $T = 2\pi\sqrt{l/2g}$
(3) Zero (4) Infinity
POS094

95. An oscillating pendulum stops, because its energy
(1) Changes into kinetic energy
(2) Change into potential energy
(3) Change into heat energy
(4) Is destroyed
POS095

96. Simple pendulum of large length is made equal to the radius of the earth. Its period of oscillation will be :
(1) 84.6 min. (2) 59.8 min.
(3) 42.3 min. (4) 21.15 min.
POS096

97. The maximum time period of oscillation of a simple pendulum of large length is:
(1) Infinity (2) 24 hours
(3) 12 hours (4) 1½ hours
POS097

98. In a simple oscillating pendulum, the work done by the string in one oscillation will be:
(1) Equal to the total energy of the pendulum
(2) Equal to the K.E. of the pendulum
(3) Equal to the P.E. of the pendulum
(4) Zero
POS098

99. A lift is ascending with acceleration g/3. What will be the time period of a simple pendulum suspended from its ceiling if its time period in stationary lift is T?
(1) $\frac{T}{2}$ (2) $\frac{\sqrt{3}T}{2}$ (3) $\frac{\sqrt{3}T}{4}$ (4) $\frac{T}{4}$
POS099

100. A child swinging on a swing in sitting position, stands up, then the period of the swing will be:
(1) Increase
(2) Decrease
(3) Remain same
(4) Increase if child is long and decrease if child is short
POS100

101. A simple pendulum is suspended from the ceiling of a vehicle, its time period is T. Vehicle is moving with constant velocity, then time period of simple pendulum will be :
(1) Less than T (2) Equal to T
(3) More than T (4) Cannot predict
POS101

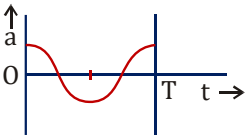
EXERCISE-I (Conceptual Questions) ANSWER KEY

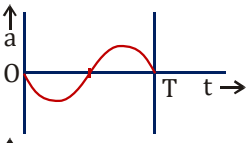
Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	1	1	4	2	1	2	1	1	4	3	3	1	3	1
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	2	4	4	3	2	4	1	1	1	1	3	4	4	1	1
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	1	4	3	4	3	2	1	2	1	1	4	2	1	1
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	4	3	3	2	3	4	3	4	2	4	4	2	3	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	1	3	4	3	3	3	1	3	1	3	3	1	3	2	4
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	4	2	4	2	1	2	1	2	3	2	1	3	3	2	1
Question	91	92	93	94	95	96	97	98	99	100	101				
Answer	1	1	1	4	3	2	4	4	2	2	2				

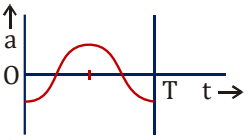
Exercise - II (Previous Year Questions) AIPMT/NEET

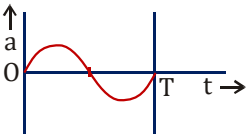
AIPMT 2014

1. The oscillation of a body on a smooth horizontal surface is represented by the equation,
$X = A\cos(\omega t)$
where X = displacement at time t
ω = angular frequency of oscillation
Which one of the following graphs shows correctly the variation acceleration 'a' with time 't'?

(1) 

(2) 

(3) 

(4) 

POS119

AIPMT 2015

2. When two displacements represented by $y_1 = a\sin(\omega t)$ and $y_2 = b\cos(\omega t)$ are superimposed the motion is :

(1) simple harmonic with amplitude $\frac{a}{b}$

(2) simple harmonic with amplitude $\sqrt{a^2+b^2}$

(3) simple harmonic with amplitude $\frac{(a+b)}{2}$

(4) not a simple harmonic

POS120

3. A particle is executing SHM along a straight line. Its velocities at distances x_1 and x_2 from the mean position are V_1 and V_2, respectively. Its time period is :-

(1) $2\pi\sqrt{\frac{x_2^2-x_1^2}{V_1^2-V_2^2}}$ (2) $2\pi\sqrt{\frac{V_1^2+V_2^2}{x_1^2+x_2^2}}$

(3) $2\pi\sqrt{\frac{V_1^2-V_2^2}{x_1^2-x_2^2}}$ (4) $2\pi\sqrt{\frac{x_1^2+x_2^2}{V_1^2+V_2^2}}$

POS121

4. Two similar springs P and Q have spring constants K_P and K_Q, such that $K_P > K_Q$. They are stretched, first by the same amount (case a) then by the same force (case b). The work done by the springs W_P and W_Q are related as, in case (a) and case (b), respectively :
(1) $W_P = W_Q$; $W_P = W_Q$
(2) $W_P > W_Q$; $W_Q > W_P$
(3) $W_P < W_Q$; $W_Q < W_P$
(4) $W_P = W_Q$; $W_P > W_Q$

POS122

RE-AIPMT 2015

5. A particle is executing a simple harmonic motion. Its maximum acceleration is α and maximum velocity is β. Then its time period of vibration will be :-

(1) $\frac{2\pi\beta}{\alpha}$ (2) $\frac{\beta^2}{\alpha^2}$ (3) $\frac{\alpha}{\beta}$ (4) $\frac{\beta^2}{\alpha}$

POS123

NEET-II 2016

6. A body of mass m is attached to the lower end of a spring whose upper end is fixed. The spring has negligible mass. When the mass m is slightly pulled down and released, it oscillates with a time period of 3s. When the mass m is increased by 1 kg, the time period of oscillations becomes 5 s. The value of m in kg is :-

(1) $\frac{16}{9}$ (2) $\frac{9}{16}$ (3) $\frac{3}{4}$ (4) $\frac{4}{3}$

POS124

NEET(UG) 2017

7. A spring of force constant k is cut into lengths of ratio 1 : 2 : 3. They are connected in series and the new force constant is k'. Then they are connected in parallel and force constant is k''. Then k' : k'' is:-

(1) 1 : 9 (2) 1 : 11 (3) 1 : 14 (4) 1 : 16

POS125

8. A particle executes linear simple harmonic motion with an amplitude of 3 cm. When the particle is at 2 cm from the mean position, the magnitude of its velocity is equal to that of its acceleration. Then its time period in seconds is :-

(1) $\frac{\sqrt{5}}{2\pi}$ (2) $\frac{4\pi}{\sqrt{5}}$ (3) $\frac{2\pi}{\sqrt{3}}$ (4) $\frac{\sqrt{5}}{\pi}$

POS126

AIM: FREE EDUCATION TO ALL APUL

NEET(UG) 2018

9. A pendulum is hung from the roof of a sufficiently high building and is moving freely to and fro like a simple harmonic oscillator. The acceleration of the bob of the pendulum is 20 m/s² at a distance of 5 m from the mean position. The time period of oscillation is :-

(1) 2π s (2) π s (3) 2 s (4) 1 s

POS127

NEET(UG) 2019

10. The displacement of a particle executing simple harmonic motion is given by

$y = A_0 + A\sin\omega t + B\cos\omega t$.

Then the amplitude of its oscillation is given by :

(1) $A_0 + \sqrt{A^2 + B^2}$ (2) $\sqrt{A^2 + B^2}$

(3) $\sqrt{A_0^2 + (A+B)^2}$ (4) A + B

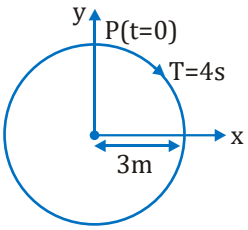
POS128

11. Average velocity of a particle executing SHM in one complete vibration is :

(1) $\frac{A\omega}{2}$ (2) $A\omega$ (3) $\frac{A\omega^2}{2}$ (4) Zero

POS129

12. The radius of circle, the period of revolution, initial position and sense of revolution are indicated in the fig.



y-projection of the radius vector of rotating particle P is :

(1) $y(t) = -3\cos 2\pi t$, where y in m

(2) $y(t) = 4\sin\left(\frac{\pi t}{2}\right)$, where y in m

(3) $y(t) = 3\cos\left(\frac{3\pi t}{2}\right)$, where y in m

(4) $y(t) = 3\cos\left(\frac{\pi t}{2}\right)$, where y in m

POS130

NEET(UG) 2019 (Odisha)

13. The distance covered by a particle undergoing SHM in one time period is (amplitude = A) :-

(1) zero (2) A (3) 2A (4) 4A

POS131

14. A mass falls from a height 'h' and its time of fall 't' is recorded in terms of time period T of a simple pendulum. On the surface of earth it is found that t = 2T. The entire set up is taken on the surface of another planet whose mass is half of that of earth and radius the same. Same experiment is repeated and corresponding times noted as t' and T', then

(1) $t' = \sqrt{2}T'$ (2) $t' > 2T'$

(3) $t' < 2T'$ (4) $t' = 2T'$

POS132

NEET(UG) 2020

15. The phase difference between displacement and acceleration of a particle in a simple harmonic motion is :

(1) Zero (2) π rad

(3) $\frac{3\pi}{2}$ rad (4) $\frac{\pi}{2}$ rad

POS133

NEET(UG) 2020 (Covid-19)

16. Identify the function which represents a periodic motion

(1) $e^{\omega t}$ (2) $\log_e(\omega t)$

(3) $\sin\omega t + \cos\omega t$ (4) $e^{-\omega t}$

POS134

NEET(UG) 2021

17. A body is executing simple harmonic motion with frequency 'n', the frequency of its potential energy is : -

(1) n (2) 2n

(3) 3n (4) 4n

POS135

18. A spring is stretched by 5 cm by a force 10 N. The time period of the oscillations when a mass of 2 kg is suspended by it is :

(1) 0.0628 s (2) 6.28 s

(3) 3.14 s (4) 0.628 s

POS136

NEET(UG) 2022

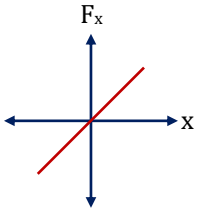
19. Two pendulums of length 121 cm and 100 cm start vibrating in phase. At some instant, the two are at their mean position in the same phase. The minimum number of vibrations of the shorter pendulum after which the two are again in phase at the mean position is :

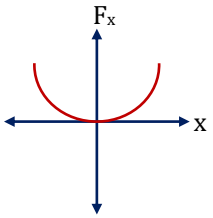
(1) 9 (2) 10 (3) 8 (4) 11

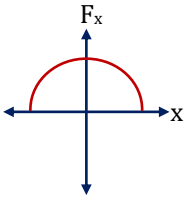
POS137

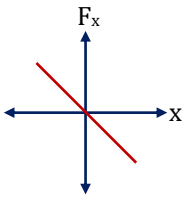
RE-NEET(UG) 2022

20. The restoring force of a spring with a block attached to the free end of the spring is represented by :

(1) 

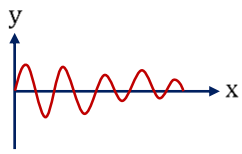
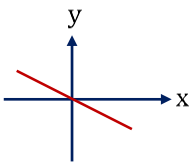
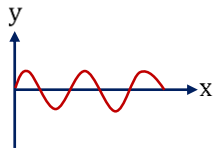
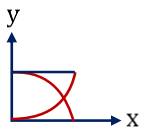
(2) 

(3) 

(4) 

POS138

21. Match List - I with List - II :

	List –I (x-y graphs)		List-II (Situations)
(a)		(i)	Total mechanical energy is conserved
(b)		(ii)	Bob of a pendulum is oscillating under negligible air friction
(c)		(iii)	Restoring force of a spring
(d)		(iv)	Bob of a pendulum is oscillating along with air friction

Choose the correct answer from the options given below :

(1) (a) - (iv), (b) - (ii), (c) - (iii), (d) - (i)

(2) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

(3) (a) - (i), (b) - (iv), (c) - (iii), (d) - (ii)

(4) (a) - (iii), (b) - (ii), (c) - (i), (d) - (iv)

POS139

22. Identify the function which represents a non-periodic motion.

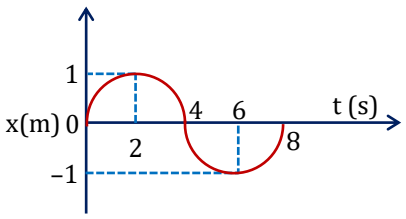
(1) $e^{-\omega t}$ (2) $\sin\omega t$

(3) $\sin\omega t + \cos\omega t$ (4) $\sin(\omega t + \pi/4)$

POS140

NEET(UG) 2023

23. The x - t graph of a particle performing simple harmonic motion is shown in the figure. The acceleration of the particle at t = 2 s is :



(1) $-\frac{\pi^2}{8} ms^{-2}$ (2) $\frac{\pi^2}{16} ms^{-2}$

(3) $-\frac{\pi^2}{16} ms^{-2}$ (4) $\frac{\pi^2}{8} ms^{-2}$

POS164

NEET(UG) 2024

24. If $x = 5\sin\left(\pi t + \frac{\pi}{3}\right)$ m represents the motion of a particle executing simple harmonic motion, the amplitude and time period of motion respectively, are :

(1) 5 cm, 2 s (2) 5 m, 2 s

(3) 5 cm, 1 s (4) 5 m, 1 s

POS165

AIM: FREE EDUCATION TO ALL APUL

25. If the mass of the bob in a simple pendulum is increased to thrice its original mass and its length is made half its original length, then the new time period of oscillation is $\frac{x}{2}$ times its original time period. Then the value of x is:

(1) $\sqrt{3}$ (2) $\sqrt{2}$

(3) $2\sqrt{3}$ (4) 4

POS166

RE-NEET(UG) 2024

26. A particle executing simple harmonic motion with amplitude A has the same potential and kinetic energies at the displacement :

(1) $2\sqrt{A}$ (2) $\frac{A}{2}$

(3) $\frac{A}{\sqrt{2}}$ (4) $A\sqrt{2}$

POS167

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	2	1	2	1	2	2	2	2	2	4	4	4	4	2
Question	16	17	18	19	20	21	22	23	24	25	26				
Answer	3	2	4	4	4	2	1	3	2	2	3				

Exercise - III (Analytical Questions) Master Your Understanding

1. The equation of motion for an oscillating particle is given by

$x = 3\sin 4\pi t + 4\cos 4\pi t$

where x is in cm and t is in sec. Choose the correct statements :

(a) the motion is simple harmonic

(b) amplitude of oscillationg is 5 cm

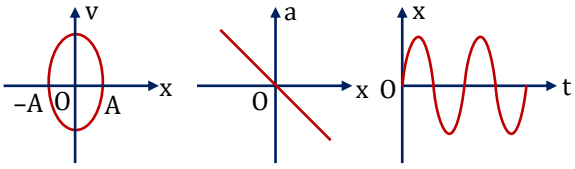
(c) motion is periodic

(1) a, b, c (2) Only a, b

(3) Only b, c (4) Only b

POS141

2. For a particle excuting S.H.M.



(a) (b) (c)

Select the correct graph.

(1) only a (2) only b

(3) both (a) & (b) (4) all (a), (b) & (c)

POS142

3. Match the column

Column-I		Column-II	
(i)	t=0 → M.P.	(a)	x = Acosωt
(ii)	t=0, M.P. +A	(b)	x = Asinωt
(iii)	t=0 ← M.P.	(c)	x = –Asinωt
(iv)	t=0, –A M.P.	(d)	x = –Acosωt

Choose the correct option :

(1) I → b, ii → d, iii → a, iv → c

(2) I → c, ii → d, iii → b, iv → a

(3) I → b, ii → a, iii → c, iv → d

(4) I → a, ii → c, iii → d, iv → b

POS143

4. A spring has natural length 40 cm and spring constant 500 N/m. A block of mass 1 kg is attached at one end of the spring and others end of the spring is attached to the wall in horizontal position. The block release from the position, where the spring has length 45 cm then choose the correct statements :

(A) The block will perform SHM of amplitude 5 cm

(B) The block will have maximum velocity $50\sqrt{5}$ cm/sec.

(C) The block will have maximum acceleration at released position

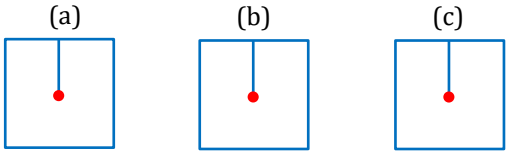
(D) The minimum P.E. of the spring will be zero

(1) only (A) (2) only (B)

(3) both (A) & (B) (4) (A), (B), (C), (D)

POS144

5. A simple pendulum is suspended from the celling of a lift. Three cases of lift are given



(a) Lift is at rest

(b) Lift is moving downward with constant speed

(c) Lift is moving downward with increasing speed

In which case time period of pendulum is $T = 2\pi\sqrt{\frac{\ell}{g}}$

(1) only in (a) (2) only in (b)

(3) only in (c) (4) in both (a) & (b)

POS145

6. A particle is executing linear SHM. For this match the following :-

i	If mass of the block is doubled (keeping other factor constant)	a	Energy of oscillation becomes 4 times
ii	If amplitude of oscillation is doubled (keeping other factor constant)	b	Max. speed of the particle becomes half times

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iii	If force constant is doubled (keeping other factor constant)	c	Time period become $\sqrt{2}$ times
iv	If angular freq. is halved (keeping mass constant)	d	Max. potential energy becomes 2 times

(1) i – c, ii – a, iii – d, iv – b
(2) i – c, ii – a, iii – b, iv – d
(3) i – a, ii – c, iii – b, iv – d
(4) i – c, ii – b, iii – d, iv – a

POS146

7. A particle performing SHM with time period T according to the equation $x = A\sin\omega t$ then match the following :-

i	time taken to displace from x = 0 to x = A/2	a	$\frac{T}{24}$
ii	time taken to displace from x = A/2 to x = A	b	$\frac{T}{8}$
iii	time taken to go form $x = A/\sqrt{2}$ to $x = \sqrt{3}A/2$	c	$\frac{T}{12}$
iv	time taken to displace from $x = A/\sqrt{2}$ to x = A	d	$\frac{T}{6}$

(1) i – c, ii – a, iii – d, iv – b
(2) i – c, ii – a, iii – b, iv – d
(3) i – c, ii – d, iii – a, iv – b
(4) i – c, ii – b, iii – d, iv – a

POS147

8. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : General vibration of a polyatomic molecule about its equilibrium position is periodic but not SHM.

Reason (R) : A periodic motion can always be expressed as a sum of infinite number of harmonic motion with appropriate amplitude.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

POS148

9. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : All simple harmonic motion are periodic.

Reason (R) : Periodic motion is the motion in which the particle repeats its motion in a fixed time internal.

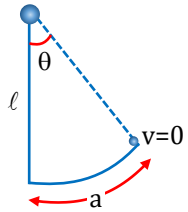
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

POS149

10. Given blow are two statements:

Statement I : The motion of a simple pendulum is simple harmonic only if $a << \ell$.



Statement II : The motion of a simple pendulum is simple harmonic for large angular displacement.

In light of above statement choose the most **appropriate** answer from the options given below:

(1) Both statement I and statement II are correct.
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statements I and statements II are incorrect.

POS150

11. Given blow are two statements:
Statement I : SHM is not an example of uniformly accelerated motion.
Statement II : SHM is the projection of uniform circular motion along one of the diameter of the circle.
In light of above statement choose the most **appropriate** answer from the options given below:
(1) Both statement I and statement II are correct.
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statements I and statements II are incorrect.

POS151

12. Given blow are two statements:
Statement I : In linear SHM net force always act in the direction of velocity.
Statement II : In linear SHM direction of velocity changes.
In light of above statement choose the most **appropriate** answer from the options given below:
(1) Both statement I and statement II are correct.
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statements I and statements II are incorrect.

POS152

13. Given blow are two statements:
Statement I : In damped oscillation amplitude of the particle decreases exponentially with time.
Statement II : In forced oscillation amplitude of a particle can maintained constant with time.
In light of above statement choose the most **appropriate** answer from the options given below:
(1) Both statement I and statement II are correct.
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statements I and statements II are incorrect.

POS153

14. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : A small body suspended by a light spring performing SHM. When the entire system is immersed in a non-viscous liquid period of oscillationg does not change.
Reason (R) : In simple harmonic motion the acceleration is always in a direction opposite to that of the displacement from mean position.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

POS154

15. Motion of a ball bearing inside a smooth fixed bowl, when released from a point slightly above the lower point is
(A) simple harmonic motion.
(B) non-periodic motion.
(C) periodic motion.
(D) non-oscillatory motion
(1) A, C (2) B, D (3) C, D (4) None

POS155

16. A particle moves such that its position from origin is given by $x = 2 + 3\sin^2\omega t$. Choose **CORRECT** statement(s) about its motion :-
(A) Particle performs S.H.M. of amplitude $\frac{3}{2}$m and frequency $\frac{\omega}{\pi}$
(B) Particle performs oscillations but no S.H.M.
(C) Particle performs S.H.M. about mean position at $x = \frac{7}{2}$m
(D) Particle performs oscillations of amplitude 3m
(1) B, D (2) A, C (3) A, C, D (4) None

POS156

17. **STATEMENT – I :** In SHM let x be the maximum speed, y the frequency of oscillation and z the maximum acceleration, then $\left(\frac{xy}{z}\right)$ is a constant quantity.

and

STATEMENT – II : This is because $\left(\frac{xy}{z}\right)$ becomes a dimensionless quantity

(1) Statement-I is true, statement-II is true and statement-II is correct explanation for statement-I.

(2) Statement-I is true, statement-II is true and statement-II is NOT the correct explanation for statement-I.

(3) Statement-I is true, statement-II is false.

(4) Statement-I is false, statement-II is true.

POS157

18. **Statement–I :** Total energy in SHM will be always more than or equal to potential energy.

and

Statement–II : Total energy in SHM will be always more than or equal to kinetic energy.

(1) Statement–I is True, Statement–II is True; Statement–II is a correct explanation for Statement–I.

(2) Statement–I is True, Statement–II is True; Statement–II is not a correct explanation for Statement–I.

(3) Statement–I is True, Statement–II is False.

(4) Statement–I is False, Statement–II is True.

POS158

19. A vertical spring block system is made to oscillate.

Statement I : Its time period on earth is more than that on the moon.

and

Statement II : Its extension on earth (in equilibrium) is more than that on the moon.

(1) Statement-I is true, statement-II is true and statement-II is correct explanation for statement-I.

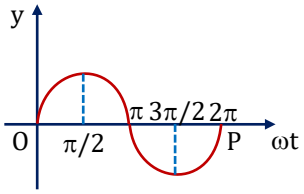
(2) Statement-I is true, statement-II is true and statement-II is NOT the correct explanation for statement-I.

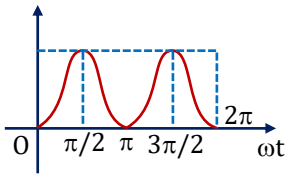
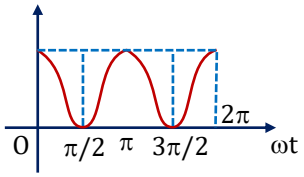
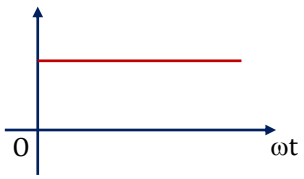
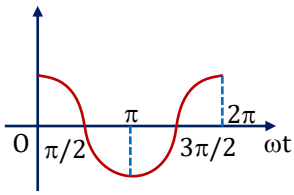
(3) Statement-I is true, statement-II is false.

(4) Statement-I is false, statement-II is true.

POS159

20. The graph plotted between phase angle (ϕ) and displacement of a particle from equilibrium position (y) is a sinusoidal curve as shown below. Then the best matching is



Column A		Column B	
(a)	K.E. versus phase angle curve	(i)	
(b)	P.E. versus phase angle curve	(ii)	
(c)	T.E. versus phase angle curve	(iii)	
(d)	Velocity versus phase angle curve	(iv)	

(1) (a)-(i), (b)-(ii), (c)-(iii) & (d)-(iv)

(2) (a)-(ii), (b)-(i), (c)-(iii) & (d)-(iv)

(3) (a)-(ii), (b)-(i), (c)-(iv) & (d)-(iii)

(4) (a)-(ii), (b)-(iii), (c)-(iv) & (d)-(i)

POS160

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21. **Statement I** : Displacement–time equation of a particle moving along x–axis is $x = 4 + 6\sin\omega t$. Under this situation, motion of particle is not simple harmonic.

and

Statement II : for the given equation $\frac{d^2x}{dt^2}$ is not proportional to –x.

(1) Statement–I is True, Statement–II is True ; Statement–II is a correct explanation for Statement–I.

(2) Statement–I is True, Statement–II is True ; Statement–II is not a correct explanation for Statement–I.

(3) Statement–I is True, Statement–II is False.

(4) Statement–I is False, Statement–II is True.

POS161

22. The angle made by the string of a simple pendulum with the vertical depends upon time as $\theta = \frac{\pi}{90}\sin \pi t$. Find the length of the pendulum if $g = \pi^2\ ms^{-1}$

(1) 1 m

(2) 1 cm

(3) 10 m

(4) 10 cm

POS162

23. **Statement–I :** Cavalry troops are asked not to march in cadence across a suspension bridge otherwise bridge may collapse.

Statement–II : If the soldiers march in cadence, in every step they step down at exactly the same instant producing a very large impulse on the bridge, which may cause it to collapse.

(1) Statement–I is true, statement–II is true; statement–II is a correct explanation for statement–I.

(2) Statement–I is true, statement–II is true; statement–II is not a correct explanation for statement–I.

(3) Statement–I is true, statement–II is false.

(4) Statement–I is false, statement–II is true.

POS163

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	3	4	4	1	3	2	1	2	1	3	1	2	1
Question	16	17	18	19	20	21	22	23							
Answer	2	2	3	4	2	4	1	3							

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Important Notes

Exercise - I (Conceptual Questions) Build Up Your Understanding

WAVE AND ITS CHARACTERISTICS

1. Water waves are of the nature :
(1) Transverse
(2) Longitudinal
(3) Sometimes longitudinal and sometimes transverse and longitudinal both
(4) Neither transverse nor longitudinal

PWA001

2. Sound wave are not polarized because :
(1) Their speed is less
(2) A medium is needed for their propagation
(3) These are longitudinal
(4) Their speed depends on temperature

PWA002

3. A thunder tap is heard 5.5 second after the lightening flash. The distance of the flash is (velocity of sound in air is 330 m/sec.): -
(1) 3560 m (2) 300 m
(3) 1780 m (4) 1815 m

PWA003

4. Transverse waves can propagate
(1) only in solids
(2) both in solids and gases
(3) neither in solids nor in gases
(4) only in gases

PWA004

5. Transverse elastic waves can be propagate in
(1) Both solid & gas
(2) In solid but not gas
(3) Neither solid nor gas
(4) None

PWA005

6. A wave of frequency 500 Hz travels between X and Y and travel a distance of 600 m in 2 sec. between X and Y. How many wavelength are there in distance XY :
(1) 1000 (2) 300 (3) 180 (4) 2000

PWA006

7. If at a place the speed of a sound wave of frequency 300 Hz is V, the speed of another wave of frequency 150 Hz at the same place will be:
(1) V (2) V/2 (3) 2V (4) 4V

PWA007

8. The equation of a progressive wave for a wire is: $Y = 4\sin\left[\frac{\pi}{2}\left(8t - \frac{x}{8}\right)\right]$. If x and y are measured in cm then velocity of wave is :
(1) 64 cm/s along – x direction
(2) 32 cm/s along – x direction
(3) 32 cm/s along + x direction
(4) 64 cm/s along + x direction

PWA008

9. The equation of progressive wave is $Y = 4\sin\left\{\pi\left(\frac{t}{5} - \frac{x}{9}\right) + \frac{\pi}{6}\right\}$ where x and y are in cm. Which of the following statement is true?
(1) λ = 18 cm
(2) amplitude = 0.04 cm
(3) velocity v = 50 cm/s
(4) frequency f = 20 Hz

PWA009

10. A plane progressive wave is represented by the equation y = 0.25 cos (2πt – 2πx).
The equation of a wave is with double the amplitude and half frequency but travelling in the opposite direction will be.
(1) y = 0.5 cos (πt – πx)
(2) y = 0.5 cos (2πt + 2πx)
(3) y = 0.25 cos (πt + 2πx)
(4) y = 0.5 cos (πt + πx)

PWA010

11. A plane wave is described by the equation $y = 3\cos\left(\frac{x}{4} - 10t - \frac{\pi}{2}\right)$. The maximum velocity of the particles of the medium due to this wave is
(1) 30 (2) $\frac{3\pi}{2}$ (3) 3/4 (4) 40

PWA011

12. The equation y = 4 + 2 sin (6t – 3x) represents a wave motion with
(1) amplitude 6 units
(2) amplitude 4 units
(3) wave speed 2 units
(4) wave speed 1/2 units

PWA012

13. Due to propagation of longitudinal wave in a medium, the following quantities also propagate in the same direction :
(1) Energy, Momentum and Mass
(2) Energy
(3) Energy and Mass
(4) Energy and Linear Momentum

PWA013

14. The waves in which the particles of the medium vibrate in a direction perpendicular to the direction of wave motion is known as :
(1) transverse waves
(2) propagated waves
(3) longitudinal waves
(4) stationary waves

PWA014

15. Two wave are represented by equation
$y_1 = a \sin \omega t \quad y_2 = a \cos \omega t$
the first wave –
(1) leads the second by π
(2) lags the second by π
(3) leads the second by $\frac{\pi}{2}$
(4) lags the second by $\frac{\pi}{2}$

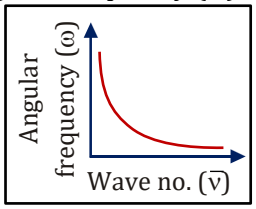
PWA015

16. The distance between two consecutive crests in a wave train produced in string is 5 m. If two complete waves pass through any point per second, the velocity of wave is
(1) 2.5 m/s (2) 5 m/s
(3) 10 m/s (4) 15 m/s

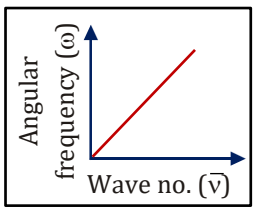
PWA016

17. The graph between wave number ($\bar{\nu}$) and angular frequency (ω) is :

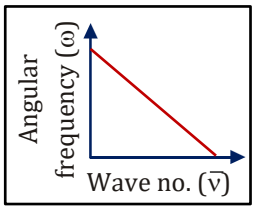
(1)



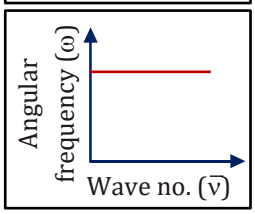
(2)



(3)



(4)



PWA017

18. The waves produced by a motorboat sailing on water are
(1) Transverse
(2) Longitudinal
(3) Longitudinal and Transverse
(4) Stationary

PWA018

PROGRESSIVE WAVE ON STRING

19. In a string the speed of wave is 10 m/s and its frequency is 100 Hz . The value of the phase difference at a distance 2.5 cm will be :
(1) $\pi/2$ (2) $\pi/8$ (3) $3\pi/2$ (4) 2π

PWA019

20. A uniform rope of mass 0.1 kg and length 2.5 m hangs from ceiling. The speed of transverse wave in the rope at upper end and at a point 0.5 m distance from lower end will be :
(1) 5 m/s, 2.24 m/s (2) 10 m/s, 3.23 m/s
(3) 7.5 m/s, 1.2 m/s (4) 2.25 m/s, 5 m/s

PWA020

21. The equation of a wave on a string of linear density 0.04 kg m^{-1} is given by

$$y = 0.02(m)\sin\left[2\pi\left(\frac{t}{0.04(s)} - \frac{x}{0.50(m)}\right)\right].$$

The tension in the string is :
(1) 6.25 N (2) 4.0 N (3) 12.5 N (4) 0.5 N

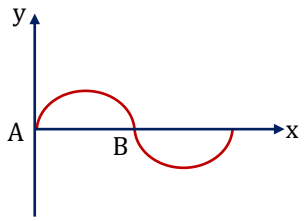
PWA021

22. The mathematical forms for three sinusoidal travelling waves are given by
Wave 1 : y(x,t) = (2cm) sin(3x–6t)
Wave 2 : y(x,t) = (3cm) sin(4x–12t)
Wave 3 : y(x,t) = (4cm) sin(5x–11t)
where x is in meters and t is in seconds. Of these waves :

(1) wave 1 has the greatest wave speed and the greatest maximum transverse string speed.
(2) wave 2 has the greatest wave speed and wave 1 has the greatest maximum transverse string speed.
(3) wave 3 has the greatest wave speed and the greatest maximum transverse string speed.
(4) wave 2 has the greatest wave speed and wave 3 has the greatest maximum transverse string speed.

PWA022

23. The figure shows an instantaneous profile of a rope carrying a progressive wave moving from left to right, then



(a) the phase at A is greater than the phase at B
(b) the phase at B is greater than the phase at A
(c) A is moving upwards
(d) B is moving upwards
(1) a & c (2) a & d (3) b & c (4) b & d

PWA023

24. Linear density of a string is 1.3×10^{-4} kg/m and wave equation is $y = 0.021\sin(x + 30t)$. Find the tension in the string where x in meter, t in sec.
(1) 1.17×10^{-2} N (2) 1.17×10^{-1} N
(3) 1.17×10^{-3} N (4) None

PWA024

SOUND WAVES AND ITS CHARACTERISTICS

25. The speed of sound in air at constant temperature
(1) is proportional to the atmospheric pressure.
(2) is proportional to the square of atmospheric pressure.
(3) is proportional to the square root of atmospheric pressure
(4) does not depend on atmospheric pressure.

PWA025

26. At the room temperature the velocity of sound in O_2 gas is V. Then in mixture of H_2 and O_2 gas the speed of sound at same temperature:
(1) will be less than V (2) will be more than V
(3) will be equal to V (4) nothing can be said

PWA026

27. The velocity of sound in a gas depends
(1) only on its wave length
(2) on the density and the elasticity of gas
(3) on intensity of the sound
(4) on the amplitude and the frequency.

PWA027

28. If at some point the amplitude of the sound becomes double and the frequency becomes one fourth then at that point the intensity of sound will be :-
(1) Become double (2) Be half
(3) Become one fourth (4) Remain unchanged

PWA028

29. A sound is produced in water and moves towards surface of water and some sound moves in air velocity of sound in water is 1450 m/s and that in air is 330 m/s. When sound moves from water to air then the effect on frequency f and wave length λ will be:
(1) f and λ will remain same
(2) f will remain same but λ will increase
(3) f will remain same but λ will decrease
(4) f will increase and λ will decrease

PWA029

30. When sound wave travels from air to water, which are of the following remain constant :
(1) wavelength (2) velocity
(3) frequency (4) intensity

PWA030

31. Newton's formula for the velocity of sound in gases is :
(1) $v = \sqrt{\frac{2p}{\rho}}$ (2) $v = \sqrt{\frac{p}{\rho}}$
(3) $v = \sqrt{\frac{\rho}{p}}$ (4) $v = \frac{3}{2}\sqrt{\frac{p}{\rho}}$

PWA031

32. Intensity level of a sound of intensity I is 30 dB. The ratio I/I_0 is (I_0 is the threshold of hearing)
(1) 1000 (2) 3000 (3) 300 (4) 30
PWA032

33. If υ_m is the velocity of sound in moist air and υ_d is the velocity of sound in dry air then :
(1) $\upsilon_m < \upsilon_d$ (2) $\upsilon_m > \upsilon_d$
(3) $\upsilon_d >> \upsilon_m$ (4) $\upsilon_m = \upsilon_d$
PWA033

34. A sine wave has an amplitude A and wavelength λ. Let V be wave velocity and v be the maximum velocity of a particle in medium then.
(1) V = v if $A = \frac{\lambda}{2\pi}$
(2) V can not be equal to v
(3) V = v if $\lambda = \frac{3A}{2\pi}$
(4) V= v if A = 2 $\pi\lambda$
PWA034

35. A sings with a frequency (n) and B sings with a frequency 1/8 that of A. If the energy remains the same and the amplitude of A is a, then amplitude of B will be :
(1) 2a (2) 8a (3) 4a (4) a
PWA035

36. The velocities of sound at the same pressure in two monoatomic gases of densities ρ_1 and ρ_2 are v_1 and v_2 respectively. If $\frac{\rho_1}{\rho_2} = 4$, then the value of $\frac{v_1}{v_2}$ is :
(1) $\frac{1}{4}$ (2) $\frac{1}{2}$ (3) 2 (4) 4
PWA036

37. The time period of SHM of a particle is 12 s. The phase difference between the position at t = 3s and t = 4s will be :
(1) $\pi/4$ (2) $3\pi/5$ (3) $\pi/6$ (4) $\pi/2$
PWA037

38. Velocity of sound in medium is V. If the density of the medium is doubled, what will be the new velocity of sound?
(1) $\sqrt{2}V$ (2) V (3) $V/\sqrt{2}$ (4) 2V
PWA038

REFLECTION OF WAVES AND ECHO

39. A man standing on a cliff claps his hand and hears its echo after one second. If the sound in reflected from another mountain then the distance between the man & reflection points is V_{sound} = 340 m/sec.
(1) 680 m (2) 340 m
(3) 170 m (4) 85 m
PWA039

PRINCIPLE OF SUPERPOSITION OF WAVES : INTERFERENCE, BEATS

40. At a particle two simple harmonic motion are acting along the same direction. These are $y_1 = a_1\sin\omega t$ and $y_2 = a_2\sin(\omega t+\phi)$. The resultant motion is also a simple harmonic motion whose amplitude will be:
(1) $a_1^2 + a_2^2 + 2a_1a_2\cos\phi$
(2) $\sqrt{a_1^2 + a_2^2 + 2a_1a_2\cos\phi}$
(3) $\sqrt{a_1^2 + a_2^2 - 2a_1a_2\cos\phi}$
(4) $a_1^2 + a_2^2 - 2a_1a_2\cos\phi$
PWA040

41. The energy in the superposition of waves: -
(1) Is lost
(2) Increase
(3) remain same, only redistribution occurs
(4) None of the above
PWA041

42. Waves from two sources superimpose on each other at a particular point. Amplitude and frequency of both the waves are equal. The ratio of intensities when both waves reach in the same phase and they reach with the phase difference of 90° will be
(1) 1:1 (2) $\sqrt{2}$:1 (3) 2:1 (4) 4:1
PWA042

43. Two waves whose intensity are same (I) move towards a point P in same phase, then the resultant intensity at point P will be:
(1) 4 I (2) 2 I (3) $\sqrt{2}$ I (4) None
PWA043

44. Ratio of amplitudes of two waves is 3:4. The ratio of maximum and minimum intensity obtained from them will be :
(1) 7:1 (2) 49:1 (3) 1:25 (4) 5:1
PWA044

45. Two coherent sources of intensities I_1 and I_2 produce an interference pattern, the maximum intensity in the interference pattern will be –

(1) $I_1 + I_2$ (2) $I_1^2 + I_2^2$

(3) $(I_1+I_2)^2$ (4) $\left(\sqrt{I_1}+\sqrt{I_2}\right)^2$

PWA045

46. Intensities ratio of two waves are 9:1 then the ratio of their maximum and minimum intensities will be:-

(1) 10 : 8 (2) 7 : 2 (3) 4 : 1 (4) 2 : 1

PWA046

47. When two tuning forks are sounded together x beats/sec are heard and frequency of A is n. Now when one prong of B is loaded with a little wax, the number of beats per second decreases. The frequency of fork B is :

(1) n + x (2) n – x

(3) $n - x^2$ (4) n – 2x

PWA047

48. A source x of unknown frequency produces 8 beats with a source of 250 Hz and 12 beats with a source of 270 Hz. The frequency of source x is:

(1) 258 Hz (2) 242 Hz

(3) 262 Hz (4) 282 Hz

PWA048

49. Two waves of wave length 2 m and 2.02 m respectively moving with the same velocity and superimpose to produce 2 beats per sec. The velocity of the waves is:

(1) 400.0 m/s (2) 402 m/s

(3) 404 m/s (4) 406 m/s

PWA049

50. A tuning fork produces 4 beats/sec. with another tuning fork B of frequency 288 Hz. If fork is loaded with little wax no. of beats per sec decreases. The frequency of the fork A, before loading is

(1) 290 Hz (2) 288 Hz

(3) 292 Hz (4) 284 Hz

PWA050

51.

Column-I		Column-II	
A	Longitudinal waves	P	Particles of the medium vibrate perpendicular to the wave propagation.
B	Transverse waves	Q	Two progressive waves of slightly different frequencies superimpose in the same direction
C	Beats	R	Two progressive waves of same frequency superimpose in the opposite directions
D	Stationary waves	S	Particles of the medium vibrate along the wave propagation.

(1) A-Q, B-R, C-Q, D-P (2) A-S, B-P, C-Q, D-R

(3) A-Q, B-S, C-P, D-R (4) A-P, B-Q, C-S, D-R

PWA051

52. What is the path difference for destructive interference?

(1) $n\lambda$ (2) $n(\lambda + 1)$

(3) $\frac{(n+1)\lambda}{2}$ (4) $\frac{(2n+1)\lambda}{2}$

PWA052

53. When beats are produced by two progressive waves of the same amplitude and of nearly the same frequency, the ratio of maximum intensity to the intensity of one of the waves will be n. Where n is

(1) 3 (2) 1 (3) 4 (4) 2

PWA053

54. What is the beat frequency produced when following two waves are sounded together?

$x_1 = 10 \sin (404\pi t - 5\pi x)$,

$x_2 = 10 \sin (400\pi t - 5\pi x)$.

(1) 4 (2) 1 (3) 3 (4) 2

PWA054

55. Two waves having equation

$x_1 = a \sin(\omega t + \phi_1)$ $x_2 = a \sin(\omega t + \phi_2)$

If in the resultant wave the frequency and amplitude remains equals to amplitude of superimposing waves. Then phase diff. between them –

(1) $\frac{\pi}{6}$ (2) $\frac{2\pi}{3}$ (3) $\frac{3\pi}{4}$ (4) $\frac{\pi}{4}$

PWA055

56. Two sources have frequency 256 Hz and 258 Hz, then time difference between two consecutive maxima is –
(1) 1 s (2) 0·5 s (3) 2 ms (4) None
PWA056

57. Two vibrating tuning forks produce progressive waves given by $Y_1 = 4 \sin 500\pi t$ and $Y_2 = 2 \sin 506\ \pi t$. Number of beats produced per minute is:
(1) 3 (2) 360 (3) 180 (4) 60
PWA057

58. Two plane progressive waves shows destructive interference at point P. Which of the following statement is true at point P :-
(1) Crest of one wave is superimposed on crest of another wave
(2) Trough of one wave is superimposed on crest of another wave
(3) Intensity of resultant wave is equal to the intensity difference of two waves
(4) Resultant amplitude is equal to the amplitude sum of two waves
PWA058

STATIONARY WAVES OR STANDING WAVES IN STRINGS

59. A uniform string of length L and mass M is fixed at both ends under tension T, then it can vibrate with frequency given by the formula.
(1) $f = \frac{1}{2}\sqrt{\frac{T}{ML}}$ (2) $f = \frac{1}{2L}\sqrt{\frac{T}{M}}$
(3) $f = \frac{1}{2}\sqrt{\frac{T}{M}}$ (4) $f = \frac{1}{2}\sqrt{\frac{M}{LT}}$
PWA059

60. The speed of transverse waves in a stretched string is 700 cm/s. If the string is 2 m long, the frequency with which it resonates in fundamental mode is:
(1) (7/2) Hz (2) (7/4) Hz
(3) (14) Hz (4) (2/7) Hz
PWA060

61. A wave represented by the equation y = acos(ωt–kx) is superposed by another wave to form a stationary wave such that the point x = 0 is a node. The equation for other wave is –
(1) y = a sin(ωt+kx) (2) y = – a cos(ωt–kx)
(3) y = – a cos(ωt+kx) (4) y = – a sin(ωt–kx)
PWA061

62. A stretched string is vibrating according to the equation $y = 5\sin\left(\frac{\pi x}{2}\right)\cos 4\pi t$, where y and a are in cm and t is in sec. The distance between two consecutive nodes on the strings is :-
(1) 2 cm (2) 4 cm
(3) 8 cm (4) 16 cm
PWA062

63. A wave of frequency 100 Hz travels along a string towards its fixed end. When this wave travels back, after reflection, a node is formed at a distance of 10 cm from the fixed end. The speed of the wave (incident and reflected) is :
(1) 5 m/s (2) 10 m/s
(3) 20 m/s (4) 40 m/s
PWA063

64. Stationary wave is represented by Y = A sin (100 t) cos (0.01 x) where y and A are in mm, t in sec and x in m. The velocity of the wave:
(1) 1 m/s (2) 10^2 m/s
(3) 10^4 m/s (4) zero
PWA064

65. If the tension in a sonometer wire is increased by a factor of four then fundamental frequency of vibration changes by a factor of :
(1) 4 (2) (1/4)
(3) 2 (4) (1/2)
PWA065

66. A sonometer wire, with a suspended mass of M = 1 kg., is in resonance with a given tuning fork. The apparatus is taken to moon where the acceleration due to gravity is 1/6 that of earth. To obtain resonance on the moon, the value of M should be
(1) 1 kg. (2) $\sqrt{6}$ kg
(3) 6 kg (4) 36 kg
PWA066

67. Stationary waves are produced in 10m long stretched string. If the string vibrates in 5 segments and wave velocity is 20m/sec, then the frequency is-
(1) 10 Hz (2) 5 Hz
(3) 4 Hz (4) 2Hz
PWA067

68. A standing wave having 3 nodes and 2 antinodes is formed between 1.21 Å distance then the wavelength is
(1) 1.21 Å (2) 2.42 Å
(3) 0.605 Å (4) 4.84 Å
PWA068

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69. A string under a tension of 129.6 N produces 10 beats/sec when it is vibrated along with a tuning fork. When the tension is the string is increased to 160 N it sounds in unison with same tuning fork. Calculate fundamental freq. of tuning fork.
(1) 100 Hz (2) 50 Hz
(3) 150 Hz (4) 200 Hz
PWA069

70. If vibrations of a string are to be increased to a factor of two, then tension in the string must be made :
(1) half (2) thrice
(3) four times (4) eight times
PWA070

71. Stationary waves are so called because in them –
(1) The particles of the medium are not disturbed at all
(2) The particles of the medium do not execute S.H. M.
(3) There occurs no flow of energy along the wave
(4) The interference effect can't be observed
PWA071

72. In a sonometer wire, the tension is maintained by suspending a mass M from free end of wire. The fundamental frequency of the wire is N Hz. If the suspended mass is completely immerged in water the fundamental frequency will
(1) increases (2) constant
(3) decrease (4) can't say
PWA072

73. A second harmonic has to generated in a string of length ℓ stretched between two rigid supports. The points where the string has to be plucked and touched are –
(1) Pluck at $\frac{\ell}{2}$ touch at $\frac{3\ell}{4}$
(2) Pluck at $\frac{\ell}{2}$ touch at $\frac{\ell}{4}$
(3) Pluck at $\frac{\ell}{4}$ touch at $\frac{3\ell}{4}$
(4) Pluck at $\frac{\ell}{4}$ touch at $\frac{\ell}{2}$
PWA073

74. If the tension and diameter of a sonometer wire of fundamental frequency (n) is doubled and density is halved then its fundamental frequency will become: -
(1) $\frac{n}{4}$ (2) $\sqrt{2}n$ (3) n (4) $\frac{n}{\sqrt{2}}$
PWA074

75. The tension in a piano wire is 10N. What should be the tension in the wire to produce a note of double the frequency?
(1) 10N (2) 20N (3) 40N (4) 80N
PWA075

76. Fundamental frequency of sonometer wire is n. If the length, tension and diameter of wire are tripled, the new fundamental frequency is: -
(1) $n/\sqrt{3}$ (2) $n/3$
(3) $n\sqrt{3}$ (4) $n/3\sqrt{3}$
PWA076

77. A string in a musical instrument is 50 cm long and its fundamental frequency is 800 Hz. If a frequency of 1000 Hz is to be produced, then required length of string is: -
(1) 62.5 cm (2) 50 cm
(3) 40 cm (4) 37.5
PWA077

78. Four wires of identical lengths, diameters and of the same material are stretched on a sonometer wire. The ratio of their tension is 1 : 4 : 9 : 16. The ratio of their fundamental frequencies is
(1) 1 : 2 : 3 : 4 (2) 16 : 9 : 4 : 1
(3) 1 : 4 : 9 : 16 (4) 4 : 3 : 2 : 1
PWA078

79. Given equation is related to
$$y = \cos\left(\frac{2\pi}{\lambda}x\right)\cos(2\pi vt)$$
(1) Transverse progressive
(2) Longitudinal progressive
(3) Longitudinal stationary wave
(4) Transverse stationary wave
PWA079

80. A stretched string is 1 m long. Its mass per unit length is 0.5 g/m. It is stretched with a force of 20 N. It plucked at a distance of 25 cm from one end. The frequency of note emitted by it will be:
(1) 400 Hz (2) 300 Hz
(3) 200 Hz (4) 100 Hz
PWA080

STATIONARY WAVE IN ORGAN PIPES

81. With the increase of temperature, the frequency of the organ pipe-
(1) increases (2) decreases
(3) remains unchanged (4) can not say
PWA081

82. An empty vessel is partially filled with water the frequency of vibration of air column in the vessel
(1) decreases
(2) increases
(3) depends on the purity of water
(4) remains the same
PWA082

83. The end correction of resonance tube is 1 cm. If lowest resonant length is 15 cm then next resonant length will be: -
(1) 36 cm (2) 45 cm
(3) 46 cm (4) 47 cm
PWA083

84. If the fundamental frequency for a COP is n, then the next three overtones will have ratio: -
(1) 2 : 3 : 4 (2) 3 : 4 : 5
(3) 3 : 7 : 11 (4) 3 : 5 : 7
PWA084

85. A tube closed at one end and containing air produces, when excited, the fundamental note of frequency 512 Hz. If the tube is open at both ends, the fundamental frequency that can be excited is (in Hz)
(1) 1024 (2) 512 (3) 256 (4) 128
PWA085

86. An air column in pipe, which is closed at one end will be in resonance with a vibrating tuning fork of frequency 264 Hz if the length of the column in cm is : [v = 330 m/s]
(1) 31.25 (2) 62.50 (3) 110 (4) 125
PWA086

87. Velocity of sound in air is 320 m/s. A pipe closed at one end has a length of 1 m neglecting end corrections, the air column in the pipe can resonant for sound of frequency.
(a) 80 Hz (b) 240 Hz
(c) 500 Hz (d) 400 Hz
(1) a (2) a,b (3) a,b,d (4) a,d
PWA087

88. The velocity of sound in air is 330 m/s. The fundamental frequency of an organ pipe open at both ends and of length 0.3 metre will be:
(1) 200 Hz (2) 550 Hz
(3) 300 Hz (4) 275 Hz
PWA088

89. An air column having one end closed contains minimum resonance length 50 cm. If it is vibrated by same tuning fork then its next resonance length will be –
(1) 250 cm (2) 200 cm
(3) 150 cm (4) 100 cm
PWA089

90. If the air column in a pipe which is closed at one end, is in resonance with a vibrating tuning fork of frequency 260 Hz, then the length of the air column is : (v_{sound} = 330 m/s)
(1) 35.7 cm (2) 31.7 cm
(3) 12.5 cm (4) 62.5 cm
PWA090

91. A sound wave of frequency 330Hz is incident normally at reflected wall then minimum distance from wall at which particle vibrate very much: - (V_{sound} = 330 m/s)
(1) 0.25 m (2) 0.125 m
(3) 1 m (4) 0.5 m
PWA091

92. An open organ pipe of length 33 cm, vibrates with frequency 1000 Hz. If velocity of sound is 330 m/s, then its frequency is: -
(1) Fundamental frequency
(2) First overtone of pipe
(3) Second overtone
(4) Fourth overtone
PWA092

93. If V is the speed of sound in air then the shortest length of the closed pipe which resonates to a frequency n :
(1) $\frac{V}{2n}$ (2) $\frac{V}{4n}$
(3) $\frac{4n}{V}$ (4) $\frac{2n}{V}$
PWA093

94. The lengths of two closed organ pipes are 0.750 m and 0.770 m. If they are sounded together, 3 beats per second are produced. The velocity of sound will be :-
(1) 350.5 m/sec (2) 335.5 m/sec
(3) 346.5 m/sec (4) None of these
PWA094

95. What is minimum length of a tube, open at both ends, that resonates with tuning fork of frequency 350 Hz? (velocity of sound in air = 350 m/s)
(1) 50 cm (2) 100 cm
(3) 75 cm (4) 25 cm
PWA095

96. An underwater sonar source operating at a frequency of 60 kHz directs its beam towards the surface. If velocity of sound in air is 330 m/s, wavelength and frequency of the waves in air are: -
(1) 5.5 mm, 60 kHz (2) 3.30 m, 60kHz
(3) 5.5 mm, 30 kHz (4) 5.5 mm, 80 kHz
PWA096

97. An organ pipe closed at one end has fundamental frequency of 1500 Hz. The maximum number of overtones generated by this pipe which a normal person can hear is
(1) 14 (2) 13 (3) 6 (4) 9
PWA097

98. Length of the close organ pipe is 1 m. At which frequency resonance will not occur (v = 320 m/sec.)
(1) 80 Hz (2) 240 Hz
(3) 300 Hz (4) 400 Hz
PWA098

99. An open resonating tube has fundamental frequency of n. When half of its length is dipped into water, then its fundamental frequency will be:
(1) n (2) n/2
(3) 2n (4) 3/2 n.
PWA099

100. A pipe is closed from one end and open from another end then which statement is true?
(1) Node is formed slightly above the open end.
(2) Node is formed slightly below the open end.
(3) Antinode is formed slightly above the open end.
(4) Antinode formed slightly below the open end.
PWA100

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	4	2	2	1	1	4	1	4	1	3	4	1	4
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	3	2	3	1	1	1	4	2	2	4	2	2	3	3	3
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	2	1	2	1	2	2	3	3	3	2	3	3	1	2	4
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	3	1	1	3	3	2	4	3	4	2	2	3	2	1	2
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	3	1	3	4	3	3	2	1	1	3	3	3	4	3	3
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	4	3	1	4	3	1	2	4	4	1	1	3	2	3	2
Question	91	92	93	94	95	96	97	98	99	100					
Answer	1	2	2	3	1	1	3	3	1	3					

Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. If n_1, n_2 and n_3 are the fundamental frequencies of three segments into which a string is divided, then the original fundamental frequency n of the string is given by :-

(1) $\frac{1}{n}=\frac{1}{n_1}+\frac{1}{n_2}+\frac{1}{n_3}$

(2) $\frac{1}{\sqrt{n}}=\frac{1}{\sqrt{n_1}}+\frac{1}{\sqrt{n_2}}+\frac{1}{\sqrt{n_3}}$

(3) $\sqrt{n}=\sqrt{n_1}+\sqrt{n_2}+\sqrt{n_3}$

(4) $n = n_1 + n_2 + n_3$

PWA137

2. The number of possible natural oscillations of air column in a pipe closed at one end of length 85 cm whose frequencies lie below 1250 Hz are: (velocity of sound = 340 ms^{-1})

(1) 4 (2) 5 (3) 7 (4) 6

PWA138

AIPMT 2015

3. The fundamental frequency of a closed organ pipe of length 20 cm is equal to the second overtone of an organ pipe open at both the ends. The length of organ pipe open at both the ends is :-

(1) 100 cm (2) 120 cm

(3) 140 cm (4) 80 cm

PWA139

RE-AIPMT-2015

4. A string is stretched between two fixed points separated by 75.0 cm. It is observed to have resonant frequencies of 420 Hz and 315 Hz. There are no other resonant frequencies between these two. The lowest resonant frequencies for this string is :-

(1) 105 Hz (2) 155 Hz

(3) 205 Hz (4) 10.5 Hz

PWA141

NEET-I 2016

5. A uniform rope of length L and mass m_1 hangs vertically from a rigid support. A block of mass m_2 is attached to the free end of the rope.

A transverse pulse of wavelength λ_1 is produced at the lower end of the rope. The wavelength of the pulse when it reaches the top of the rope is λ_2. The ratio λ_2/λ_1 is :

(1) $\sqrt{\frac{m_1}{m_2}}$ (2) $\sqrt{\frac{m_1+m_2}{m_2}}$

(3) $\sqrt{\frac{m_2}{m_1}}$ (4) $\sqrt{\frac{m_1+m_2}{m_1}}$

PWA143

6. An air column, closed at one end and open at the other, resonates with a tuning fork when the smallest length of the column is 50 cm. The next larger length of the column resonating with the same tuning fork is :

(1) 66.7 cm (2) 100 cm

(3) 150 cm (4) 200 cm

PWA144

NEET-II 2016

7. The second overtone of an open organ pipe has the same frequency as the first overtone of a closed pipe L metre long. The length of the open pipe will be

(1) $\frac{L}{2}$ (2) 4 L (3) L (4) 2 L

PWA145

8. Three sound waves of equal amplitudes have frequencies (n – 1), n, (n + 1). They superimpose to give beats. The number of beats produced per second will be :-

(1) 3 (2) 2 (3) 1 (4) 4

PWA146

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NEET(UG) 2017

9. The two nearest harmonics of a tube closed at one end and open at other end are 220 Hz and 260 Hz. What is the fundamental frequency of the system?

(1) 20 Hz (2) 30 Hz
(3) 40 Hz (4) 10 Hz

PWA147

NEET(UG) 2018

10. A tuning fork is used to produce resonance in a glass tube. The length of the air column in this tube can be adjusted by a variable piston. At room temperature of 27°C two successive resonances are produced at 20 cm and 73 cm column length. If the frequency of the tuning fork is 320 Hz, the velocity of sound in air at 27°C is:-

(1) 330 m/s (2) 339 m/s
(3) 350 m/s (4) 300 m/s

PWA149

11. The fundamental frequency in an open organ pipe is equal to the third harmonic of a closed organ pipe. If the length of the closed organ pipe is 20 cm, the length of the open organ pipe is :-

(1) 13.2 cm (2) 8 cm
(3) 12.5 cm (4) 16 cm

PWA150

NEET(UG) 2019 (Odisha)

12. A tuning fork with frequency 800 Hz produces resonance in a resonance column tube with upper end open and lower end closed by water surface. Successive resonance are observed at length 9.75 cm, 31.25 cm and 52.75 cm. The speed of sound in air is :-

(1) 500 m/s (2) 156 m/s
(3) 344 m/s (4) 172 m/s

PWA151

NEET(UG) 2020

13. In a guitar, two strings A and B made of same material are slightly out of tune and produce beats of frequency 6 Hz. When tension in B is slightly decreased, the beat frequency increases to 7 Hz. If the frequency of A is 530 Hz, the original frequency of B will be:

(1) 537 Hz (2) 523 Hz
(3) 524 Hz (4) 536 Hz

PWA152

NEET(UG) 2020 (Covid-19)

14. The length of the string of a musical instrument is 90 cm and has a fundamental frequency of 120 Hz. Where should it be pressed to produce fundamental frequency of 180 Hz?

(1) 75 cm (2) 60 cm
(3) 45 cm (4) 80 cm

PWA153

NEET(UG) 2022

15. If the initial tension on a stretched string is doubled, then the ratio of the initial and final speeds of a transverse wave along the string is:

(1) $\sqrt{2}:1$ (2) $1:\sqrt{2}$
(3) 1 : 2 (4) 1 : 1

PWA172

RE-NEET(UG) 2022

16. An organ pipe filled with a gas at 27°C resonates at 400 Hz in its fundamental mode. If it is filled with the same gas at 90°C, the resonance frequency at the same mode will be :-

(1) 420 Hz (2) 440 Hz
(3) 484 Hz (4) 512 Hz

PWA173

NEET(UG) 2023

17. The ratio of frequencies of fundamental harmonic produced by an open pipe to that of closed pipe having the same length is :

(1) 2 : 1 (2) 1 : 3
(3) 3 : 1 (4) 1 : 2

PWA174

RE-NEET(UG) 2024

18. Interference pattern can be observed due to superposition of the following waves :

A. $y = a \sin \omega t$ B. $y = a \sin 2\omega t$
C. $y = a \sin (\omega t - \phi)$ D. $y = a \sin 3\omega t$

Choose the **correct** answer from the options given below :

(1) B and C (2) B and D
(3) A and C (4) A and B

PWA175

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19. The displacement of a travelling wave is $y = C \sin \frac{2\pi}{\lambda}(at - x)$ where t is time, x is distance and λ is the wavelength, all in S.I. units. Then the frequency of the wave is :

(1) $\frac{2\pi\lambda}{a}$ (2) $\frac{2\pi a}{\lambda}$

(3) $\frac{\lambda}{a}$ (4) $\frac{a}{\lambda}$

PWA176

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	2	1	2	3	4	2	1	2	1	3	3	2	2
Question	16	17	18	19											
Answer	2	1	3	4											

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Exercise - III (Analytical Questions) Master Your Understanding

1. Given below are two statements:
Statement I : For same length but different gas COP fundamental frequency are different.
Statement II : For same length but different gas COP fundamental wavelength remains same.
In light of above statement choose the most **appropriate** answer from the options given below:
(1) Both statement I and statement II are correct.
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statements I and statements II are incorrect.

PWA154

2. Given below are two statements:
Statement I : Mechanical transverse waves cannot be generated in gaseous medium.
Statement II : Mechanical transverse waves can be produced only in such medium which have shearing property.
In light of above statement choose the most **appropriate** answer from the options given below:
(1) Both statement I and statement II are correct.
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statements I and statements II are incorrect.

PWA155

3. Consider the following statements :
Statement I : The velocity of sound in the air increases due to presence of moisture in it.
Statement II : The presence of moisture in air lowers the density of air.
In light of above statement choose the most **appropriate** answer from the options given below:
(1) Both statement I and statement II are correct.
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statements I and statements II are incorrect.

PWA156

4. The equation of travelling wave is given by (all quantities are in SI units)
$y = 0.02 \sin 2\pi (10t - 5x)$

	Column-I		Column-II
i	Speed of wave	a	20π
ii	Angular frequency of wave	b	0.4π
iii	Wavelength of wave	c	2
iv	Maximum particle speed	d	0.2

(1) i → c, ii → a, iii → d, iv → c
(2) i → a, ii → b, iii → c, iv → d
(3) i → c, ii → b, iii → a, iv → d
(4) i → c, ii → a, iii → d, iv → b

PWA157

5. In a resonance tube experiment, a close organ pipe of length 120 cm resonate when tuned with a tuning fork of frequency 340 Hz. If water is poured in the pipe then (given $v_{air} = 340$m/sec.):
(a) Minimum length of water column to have the resonance is 45 cm.
(b) The distance between two successive nodes is 50 cm.
(c) The maximum length of water column to create the resonance is 95 cm
(d) None of these
Choose the correct statements :
(1) a, b, c (2) b, c (3) a, c (4) None

PWA158

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6. Regarding speed of sound in gas match the following -

	Column-I		Column-II
i	Temperature of gas is made 4 times and pressure 2 times	a	Speed becomes $2\sqrt{2}$ times
ii	Only pressure is made 4 times without change in temperature	b	Speed becomes two times
iii	Only temperature changed to 4 times	c	Speed remain unchanged
iv	Molecular mass of the gas is made 4 times	d	Speed become half

(1) i → a, ii → c, iii → d, iv → b
(2) i → b, ii → c, iii → d, iv → a
(3) i → b, ii → c, iii → b, iv → d
(4) i → c, ii → b, iii → a, iv → c

PWA159

7. For a closed organ pipe match the following :

	Column-I		Column-II
i	Third overtone frequency is x times the fundamental frequency. Here, x is equal to	a	3
ii	No. of nodes in second overtone	b	4
iii	No. of antinode in third overtone	c	5
		d	None

(1) i → a, ii → d, iii → b
(2) i → d, ii → a, iii → b
(3) i → b, ii → a, iii → d
(4) i → c, ii → d, iii → b

PWA160

8. In a stationary wave pattern, all the particles -
(a) of the medium vibrate in the same phase
(b) in the region between two antinodes vibrate in the same phase
(c) in the region between two successive nodes vibrate in the same phase
(d) on either side of a node vibrate in opposite phase
Choose the correct statement :
(1) a, c, d (2) a, b, c, d
(3) c, d (4) a, b, c

PWA161

9. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
Assertion (A) : Sound travels faster on a rainy day than on a dry day.
Reason (R) : Moisture increases the pressure.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PWA162

10. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
Assertion (A) : For a closed pipe, the first resonance length is 60 cm. The second resonance position will be obtained at 120 cm.
Reason (R) : In a closed pipe $\ell_2 = 3\ell_1$.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PWA163

11. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
Assertion (A) : In stationary waves, energy is confined within the wave region.
Reason (R) : Every particle is stationary in a stationary wave.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PWA164

12. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**

Assertion (A) : When two vibrating tuning forks having frequencies 256 and 512 are held near each other, beats cannot be heard.

Reason (R) : The principle of superposition is valid only if frequencies of oscillators are nearly equal.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PWA165

13. The tension in a stretched string fixed at both ends in changed by 2%, the fundamental frequency is found to get changed by 15 Hz.

(a) Wavelength of the string of fundamental frequency does not change
(b) Velocity of propagation of wave changes by 2%
(c) Velocity of propagation of wave changes by 1%
(d) Original frequency is 1500 Hz

Choose the correct statement(s) -

(1) a, b, c, d (2) a, b, d
(3) a, c, d (4) a, b, c

PWA166

14. Velocity of sound in air is 320 m/s. A pipe closed at one end has a length of 1 m. Neglecting end corrections, the air column in the pipe can resonate for sound of frequency.

(a) 80 Hz (b) 240 Hz
(c) 320 Hz (d) 400 Hz

Choose the correct statements:

(1) a, b, c (2) a, b, d
(3) b, c, d (4) a, b, c, d

PWA167

15. The equation for the displacement of a stretched string is given by $y = 8 \sin^2\pi \left[\frac{t}{0.02} - \frac{x}{100}\right]$ where y and x are in cm and t in second

Column I		Column II	
(A)	Amplitude of wave in cm	(P)	50
(B)	Frequency of wave in Hz	(Q)	4
(C)	Velocity of wave in ms^{-1}	(R)	4π
(D)	Maximum particle velocity in cm/s	(S)	5000
		(T)	400π

Options :-

(1) (A) → (Q); (B) → (P); (C) → (P); (D) → (T)
(2) (A) → (R); (B) → (S); (C) → (P); (D) → (T)
(3) (A) → (Q); (B) → (P); (C) → (S); (D) → (T)
(4) (A) → (R); (B) → (S); (C) → (S); (D) → (T)

PWA168

16. Which of the following actions would make a pulse travel faster along a stretched string?

(A) Move your hand up and down more quickly as you generate the pulse
(B) Use a heavier string of the same length, under the same tension
(C) Use a lighter string of the same length, under the same tension
(D) Stretch the string tighter to increase the tension.

(1) A,B,D (2) B,D (3) C,D (4) A,C

PWA169

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17. Sinusoidal waves travel on five identical strings. Use the mathematical forms of the waves given below. In the expressions given below x is in m and y is in millimeters and t is in seconds.
(a) y = (2mm) sin(2x - 4t)
(b) y = (3mm) sin(4x - 10t)
(c) y = (2.5mm) sin(6x - 12t)
(d) y = (1mm) sin(8x - 16t)
(e) y = (4mm) sin(10x - 20t)
Four of the strings have the same tension, but any one has a different tension. Identify it.
(1) a (2) b (3) d (4) e

PWA170

18. A wave disturbance in a medium is described by $y(x, t) = 0.02 \cos(50\pi t + \pi/2) \cos(10\pi x)$, where x and y are in metres and t in seconds.
(A) A node occurs at x = 0.15 m
(B) An antinode occurs at x = 0.3 m
(C) The speed of the wave is 5.0 m/s
(D) The wavelength is 0.2 m
(1) A,C,D
(2) B,C,D
(3) A,B,C,D
(4) C,D

PWA171

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	1	1	4	1	3	2	3	3	4	3	3	3	2	1
Question	16	17	18												
Answer	3	2	3												

Exercise - I (Conceptual Questions) Build Up Your Understanding

UNIT

1. Which of the following system of units is not based on units of mass, length and time alone ?
(1) SI (2) MKS
(3) FPS (4) CGS
PPM001

2. Which of the following is not a fundamental physical quantity ?
(1) Mass (2) Length
(3) Temperature (4) Density
PPM002

3. Which of the following is not the name of a physical quantity :-
(1) kilogram (2) impulse
(3) energy (4) density
PPM003

4. Which of the following is not a unit of time ?
(1) microsecond (2) leap year
(3) lunar month (4) light year
PPM004

5. Unit of energy in SI system is :-
(1) erg (2) calorie
(3) joule (4) electron volt
PPM005

6. In the SI system, the unit of temperature is :-
(1) Degree centrigrade
(2) Degree Celsius
(3) Kelvin
(4) Degree Fahrenheit
PPM006

7. The SI unit of momentum is :-
(1) $\frac{kg}{m}$ (2) $\frac{kg.m}{sec}$
(3) $\frac{kg.m^2}{sec}$ (4) kg × newton
PPM007

8. SI unit of pressure is :-
(1) pascal (2) Dyne/cm^2
(3) cm of Hg (4) atmosphere
PPM008

9. kg m/s^2 stand for the unit of :-
(1) Energy (2) acceleration
(3) Force (4) Momentum
PPM009

10. N kg^{-1} is the unit of :-
(1) velocity (2) energy
(3) acceleration (4) force
PPM010

11. If the unit of length is micrometre and unit of time is microsecond, the unit of velocity will be :-
(1) 100 m/s (2) 10 m/s
(3) micrometre/s (4) m/s
PPM011

12. The unit of power is :-
(1) kilowatt (2) kilowatt-hour
(3) dyne (4) joule
PPM012

13. Kilowatt-hour is a unit of :-
(1) electrical charge (2) energy
(3) power (4) force
PPM013

14. One nanometer is equal to :-
(1) 10^9 mm (2) 10^{-6} cm
(3) 10^{-7} cm (4) 10^{-9} cm
PPM014

15. The units nanometre, fermi, angstrom and attometre, arranged in decreasing order will read as :-
(1) angstrom, nanometre, fermi, attometre
(2) fermi, attometre, angstrom, nanometre
(3) nanometre, angstrom, fermi, attometre
(4) attometre, angstrom, fermi, nanometer
PPM015

16. Which is the correct unit for measuring nuclear radii:-
(1) micron (2) millimetre
(3) agnstrom (4) fermi
PPM016

17. Which of the following is usually a derived quantity?
(1) Mass (2) Velocity
(3) Length (4) Time
PPM017

18. Temperature can be expressed as a derived quantity in terms of which of the following ?
(1) Length and mass
(2) Mass and time
(3) Length. mass and time
(4) None of these
PPM018

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19. Which of the following sets can not enter into the list of fundamental quantities in any system of units?
(1) length, mass, velocity
(2) length, time, velocity
(3) length, time, mass
(4) mass, time, velocity

PPM019

20. The density of a liquid is 1000 kg m^{-3}. Its value in CGS system :-
(1) 1 gcm^{-3} (2) 0.1 gcm^{-3}
(3) 10 gcm^{-3} (4) 100 gcm^{-3}

PPM020

21. The surface tension of a liquid is 70 dynes/cm. It may be expressed in M.K.S. system as :-
(1) 70 N/m (2) 7×10^{-2} N/m
(3) 7×10^{2} N/m (4) 7×10^{3} N/m

PPM021

22. Units of length, velocity and force are doubled. Which of the following is correct :-
(1) Units of time is doubled
(2) Unit of mass is doubled
(3) Unit of momentum is doubled
(4) Unit of energy is doubled

PPM022

23. If the units of length, velocity and force are half, then the units of Power will be :-
(1) doubled
(2) halved
(3) one fourth
(4) remains unaffected

PPM023

DIMENSIONS

24. A dimensionless quantity
(1) Never has a unit (2) Always has a unit
(3) May have a unit (4) Does not exist

PPM024

25. A unitless quantity
(1) Does not exist
(2) Always has a nonzero dimension
(3) Never has a nonzero dimension
(4) May have a nonzero dimension

PPM025

26. Which of the following physical quantities has neither dimensions nor unit ?
(1) angle
(2) Luminous intensity
(3) coefficient of friction
(4) Electric current

PPM026

27. Dimensions of relative density is
(1) kg m^{-3} (2) $[ML^{-3}]$
(3) dimensionless (4) $[M^2 L^{-6}]$

PPM027

28. Which of the following does not have dimensions of length ?
(1) fermi (2) micron
(3) angstrom (4) radian

PPM028

29. The quantity having dimension –2 in the time is :-
(1) force
(2) pressure
(3) gravitational constant
(4) all of these

PPM029

30. The dimensional formula $[ML^2T^{-2}]$ represents :-
(1) pressure
(2) linear momentum
(3) power
(4) energy

PPM030

31. Which pair has the same dimensions ?
(1) Work and power
(2) Density and relative density
(3) Momentum and impulse
(4) Latent heat and specific heat

PPM031

32. Dimensions of one pair is same. Identify the pair:-
(1) Torque and energy
(2) Angular momentum and work
(3) Energy and pressure
(4) Light year and solar day

PPM032

33. The method of dimensional analysis can be used to derive which of the following relations ?
(1) $N_0 e^{-\lambda t}$
(2) $A \sin(\omega t + kx)$
(3) $\frac{1}{2} mv^2 + \frac{1}{2} I\omega^2$
(4) None of the above

PPM033

34. If dimensions of A and B are different, then which of the following operation is valid ?
(1) $\frac{A}{B}$ (2) $e^{-A/B}$ (3) A – B (4) A + B

PPM034

35. Which of the following is incorrect statement
(1) A dimensionally correct equation may be correct
(2) A dimensionally correct equation may be incorrect
(3) A dimensionally incorrect equation may be correct
(4) A dimensionally incorrect equation is incorrect

PPM035

36. Which of the following is incorrect ?
(1) All derived quantities may be represented dimensionally in terms of the base quantities
(2) A base quantity cannot be represented dimensionally in terms of other base quantities
(3) The dimension of a derived quantity is never zero in any base quantity
(4) The dimension of a base quantity in other base quantities is always zero.

PPM036

37. If $x = at + bt^2$, where x is in metre and t is in hour(hr), then unit of b will be :-
(1) m^2/hr (2) m
(3) m/hr (4) m/hr^2

PPM037

38. A force is given by $F = at + bt^2$, where t is time. The dimensions of a and b are
(1) $[M\ L\ T^{-3}]$ and $[M\ L\ T^{-4}]$
(2) $[M\ L\ T^{-4}]$ and $[M\ L\ T^{-3}]$
(3) $[M\ L\ T^{-1}]$ and $[M\ L\ T^{-2}]$
(4) $[M\ L\ T^{-2}]$ and $[M\ L\ T^{0}]$

PPM038

39. If $S = \frac{1}{3}ft^3$, 'f' has the dimensions of - (S = distance, t = time)
(1) $[M^0L^{-1}T^3]$ (2) $[M^1L^1T^{-3}]$
(3) $[M^0L^1T^{-3}]$ (4) $[M^0L^{-1}T^{-3}]$

PPM039

40. The time dependence of a physical quantity P is given by $P = P_0 \exp(-\alpha t^2)$, where α is a constant and t is time. The constant α
(1) is dimensionless
(2) has dimensions $[T^{-2}]$
(3) has dimensions of P
(4) has dimensions $[T^2]$

PPM040

41. Given the force $F = Pt^{-1} + Qt$. Here t is time. The unit of P is same as that of :-
(1) displacement (2) velocity
(3) acceleration (4) momentum

PPM041

42. The force F, on a sphere of radius 'a' moving in a medium with velocity v is given by $F = 6\pi\eta av$. The dimensions of η are :-
(1) $[ML^{-1}T^{-1}]$ (2) $[MT^{-1}]$
(3) $[MLT^{-2}]$ (4) $[ML^{-3}]$

PPM042

43. The force is given in terms of time (t) and displacement (x) by the equation :

$$F = A\cos Bx + C\sin Dt$$

Then, the dimensions of $\frac{D}{B}$ are :-
(1) $M^oL^oT^o$ (2) $M^oL^oT^{-1}$
(3) $M^oL^{-1}T^o$ (4) M^oLT^{-1}

PPM043

44. For a body moving along x-axis, the distance travelled by body from a reference point is given as function of time t as $x = at^2 + b$, where a and b are constants, then the dimension of $\sqrt{ab}$ is same as :-
(1) speed (2) distance travelled
(3) acceleration (4) none of these

PPM044

45. In the relation $y = r\sin(\omega t - kx)$ the dimensions of $\frac{\omega}{k}$ are :-
(1) $[M^0L^1T^0]$ (2) $[M^0L^1T^{-1}]$
(3) $[M^0L^0T^3]$ (4) $[M^0L^1T^0]$

PPM045

46. A wave is represented by
$y = a\sin(At - Bx + C)$
where A, B, C are constants. The dimensions of A, B, C are
(1) T^{-1}, L, $M^0L^0T^0$ (2) T^{-1}, L^{-1}, $M^0L^0T^0$
(3) T, L, M (4) T^{-1}, L^{-1}, M^{-1}

PPM046

47. For $10^{(at+3)}$, the dimensions of a is :-
(1) $M^0L^0T^0$ (2) $M^0L^0T^1$
(3) $M^0L^0T^{-1}$ (4) None of these

PPM047

48. The equation $\frac{dv}{dt} = AT - Bv$ is describing the rate of change of velocity of a body falling from rest in a resisting medium. The dimensions of A and B are :-
(1) LT^{-3}, T (2) LT^{-3}, T^{-1}
(3) LT, T (4) LT, T^{-1}
PPM048

49. If y represents distance and x-represents time, dimensions of $\frac{d^2y}{dx^2}$ are :-
(1) LT^{-1} (2) L^2T^2 (3) L^2T^{-1} (4) LT^{-2}
PPM049

50. If energy (E), velocity (v) and force (F), be taken as fundamental quantities, then what are the dimensions of mass :-
(1) EV^2 (2) EV^{-2} (3) FV^{-1} (4) FV^{-2}
PPM050

51. A system has basic dimensions as density [D], velocity [V] and area [A]. The dimensional representation of force in this system is :-
(1) AV^2D (2) A^2VD (3) AVD^2 (4) A^0VD
PPM051

ERRORS

52. The significant digits in 200.40 are :-
(1) 4 (2) 5 (3) 2 (4) 3
PPM052

53. Number of significant digits in 0.01230 is :-
(1) 3 (2) 4 (3) 5 (4) 6
PPM053

54. Which of the following has the highest number of significant figures ?
(1) 0.007 m^2 (2) 2.64×10^{24} kg
(3) 0.0006032 m^2 (4) 6.3200 J
PPM054

55. Given P = 0.0030 m, Q = 2.40 m and R = 3000m, the number of significant figures in P, Q and R are respectively :-
(1) 1, 2, 1 (2) 2, 3, 4
(3) 4, 2, 1 (4) 4, 2, 4
PPM055

56. After rounding off the number 4621 to 2 significant digits the value becomes :-
(1) 4600 (2) 4500
(3) 4700 (4) 4720
PPM056

57. If L = 2.331 cm and B = 2.1 cm, then L + B = (According to significant figures)
(1) 4.431 cm (2) 4.43 cm
(3) 4.4 cm (4) 4 cm
PPM057

58. Subtract 0.2 J from 5.27 J and express the result with correct number of significant figures :-
(1) 5.1 J (2) 5.06 J (3) 5.0 J (4) 5 J
PPM058

59. In arithmetic 17.8 × 3.1143 = 55.4354. But as a result of experimental measurements the best way to express the product is :-
(1) 55.3454 (2) 55.4
(3) 55.44 (4) 55.435
PPM059

60. The order of magnitude of the number 1013 is:-
(1) 1 (2) 2 (3) 3 (4) 4
PPM060

61. The length of a rod is (11.05 ± 0.05) cm. What is the sum of lengths of two such rods :-
(1) (22.1 ± 0.05) cm (2) (22.10 ± 0.05) cm
(3) (22.1 ± 0.15) cm (4) (22.10 ± 0.10) cm
PPM061

62. The external and internal radius of a hollow cylinder are measured to be (5.23 ± 0.015) cm and (4.89 ± 0.005) cm. The thickness of the wall of the cylinder is :-
(1) (0.34 ± 0.02) cm (2) (0.17 ± 0.02) cm
(3) (0.17 ± 0.01) cm (4) (0.34 ± 0.01) cm
PPM062

63. The resistance is $R = \frac{V}{I}$ where V = (100 ± 5) volt and I = (10 ± 0.2) ampere. What is the total error in R ?
(1) 5 % (2) 7 %
(3) 5.2 % (4) $\left(\frac{5}{2}\right)$%
PPM063

64. When a copper sphere is heated, maximum percentage change will be observed in–
(1) radius (2) area
(3) volume (4) none of these
PPM064

65. If error in measuring diameter of a circle is 2%, the error in circumference of the circle would be :-
(1) 2% (2) 8% (3) 4% (4) 1%
PPM065

66. The error is the measurement of volume of sphere is 1.5%. The error in the measurement of its diameter is :-
(1) 1% (2) 2% (3) 0.5% (4) 1.5%
PPM066

67. There is an error of 2% in the measurement of side of a cube. The percentage error in the calculation of its volume will be :-
(1) 1% (2) 2% (3) 3 (4) 6%
PPM067

68. A thin copper wire of length ℓ metre increases in length by 2% when heated through 10°C. What is the percentage increase in area when a square copper sheet of length ℓ metre is heated through 10°C ?
(1) 4% (2) 8%
(3) 16% (4) None of these
PPM068

69. If $f = x^2$, then the relative error in f is :-
(1) $\frac{2\Delta x}{x}$ (2) $\frac{(\Delta x)^2}{x}$
(3) $\frac{\Delta x}{x}$ (4) $(\Delta x)^2$
PPM069

70. The percentage error in the measurement of mass and speed are 2% and 3% respectively. How much will be the maximum error in the estimate of kinetic energy obtained by measuring mass and speed :-
(1) 11% (2) 8% (3) 5% (4) 4%
PPM070

71. The heat generated in a circuit is dependent upon the resistance, current and time for which the current is flown. If the error in measuring the above are as 1%, 2% and 1% the maximum error in measuring heat will be
(1) 2% (2) 3%
(3) 6% (4) 1%
PPM071

72. Percentage error in measuring the radius (R) and mass (M) of a solid sphere are 2% & 1% respectively. Then error in measurement of moment of inertia about to its diameter $\left(\text{i.e. } I=\frac{2}{5}MR^2\right)$ is :-
(1) 3 % (2) 6 % (3) 5 % (4) 4 %
PPM072

73. The resistance R of a wire is given by the relation $R = \frac{\rho\ell}{\pi r^2}$. Percentage error in the measurement of ρ, ℓ and r is 1%, 3 % and 2% respectively. Then the percentage error in the measurement of R is
(1) 6% (2) 9% (3) 8% (4) 10%
PPM073

74. The pressure on a square plate is measured by measuring the force on the plate and the length of the sides of the plate. If the maximum error in the measurement of force and length are respectively 4% and 2%, the maximum error in the measurement of pressure is –
(1) 1% (2) 2% (3) 6% (4) 8%
PPM074

75. What is the fractional error in g calculated from $T=2\pi\sqrt{\ell/g}$? Given that fractional errors in T and ℓ are ±x and ±y respectively.
(1) x + y (2) x – y
(3) 2x + y (4) 2x – y
PPM075

76. While measuring acceleration due to gravity by a simple pendulum a student makes a positive error of 1% in the length of the pendulum and a negative error of 3% in the value of the time period. His percentage error in the measurement of the value of g will be (Use time period of a simple pendulum, $T=2\pi\sqrt{\frac{\ell}{g}}$) -
(1) 2 % (2) 4 %
(3) 7 % (4) 10 %
PPM076

77. An experiment measures quantities a, b and c, and X is calculated from $X = ab^2/c^3$. If the percentage error in a, b and c are ±1%, ±3% and ±2% respectively, the percentage error in X will be –
(1) ±13% (2) ±7%
(3) ±4% (4) ±1%

PPM077

78. A physical quantity X is given by $X = \frac{2k^3\ell^2}{m\sqrt{n}}$

The percentage error in the measurements of k,ℓ, m and n are 1%, 1%, 3% and 4% respectively. The value of X is uncertain by
(1) 8 % (2) 10 % (3) 12 % (4) None

PPM078

79. A quantity is represented by $X = M^a L^b T^c$. The percentage error in measurement of M, L and T are α%, β % and γ % respectively. The percentage error in X would be
(1) (α a + βb + γc) %
(2) (αa – βb + γc) %
(3) (α a – βb– γc) %
(4) None of these

PPM079

80. the dimensional formula of a physical quantity x is $[M^{-1}L^3T^{-2}]$. The error in measuring the quantities M, L, and T are 2%, 3% and 4%. The maximum percentage of error that occurs in measuring the quantity x is :-
(1) 9 (2) 10 (3) 14 (4) 19

PPM080

81. The pressure P and volume V of a gas are related as $PV^{3/2} = K$, where K is a constant. The percentage change in the pressure for a decrease of 0.5% in the volume is :-
(1) –0.75% (2) 0.75%
(3) 1.50% (4) –1.50%

PPM081

MEASUREMENT

82. Zero error of an instrument introduces
(1) Systematic errors (2) Random errors
(3) Both (4) None of these

PPM082

83. A pendulum clock, designed to give correct time in planes, gives incorrect time at hill station. Type of error in measurement of time can be :-
(1) Systematic error (2) Random error
(3) Gross error (4) None of these

PPM083

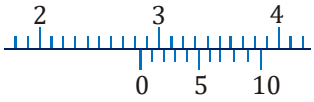
84. A vernier callipers has 40 divisions on the vernier scale which coincide with 38 divisions on the main scale. The least count of the instrument is 0.1 mm. The main scale divisions are of
(1) 0.5 mm (2) 1 mm
(3) 2 mm (4) 1/4 mm

PPM084

85. One centimetre on the main scale of vernier callipers is divided into ten equal parts. If 20 divisions of vernier scale coincide with 16 small divisions of the main scale, the least count of the callipers is
(1) 0.01 cm (2) 0.02 cm
(3) 0.05 cm (4) 0.005 cm

PPM085

86. In main scale of a vernier callipers 1 cm divided into 10 equal divisions. The vernier scale has 10 equal divisions that correspond to 9 main scale divisions. The measured value by this calliper is :-



(1) 2.82 cm (2) 2.83 cm
(3) 2.87 cm (4) 2.89 cm

PPM086

87. A student measured the thickness of a metal sheet using a screw gauge with least count 0.001 cm and listed the measurements. The correct measurement is –
(1) 4.2 cm (2) 4.22 cm
(3) 4.220 cm (4) 4.2200 cm

PPM087

88. One full rotation of the cap of a screw gauge is equivalent to 5 mm. The cap has 100 division. Find the least count.
(1) 0.5 m (2) 0.05 mm
(3) 0.005 m (4) 5×10^{-4} m

PPM088

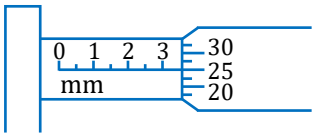
89. Distance moved by a screw in 5 rotation is 5 mm and total number of circular divisions is 100. The least count of the screw gauge is :-

(1) 1 mm (2) 0.001 cm

(3) 0.0001 cm (4) 0.001 mm

PPM089

90. The number of circular division on screwgauge is 100. It moves 1 mm on main scale for one complete revolution. Find reading of micrometer.

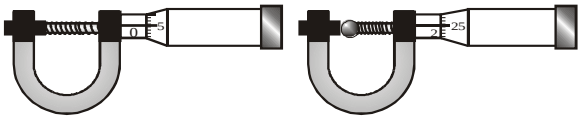


(1) 3.25 mm (2) 3.24 mm

(3) 3.30 mm (4) 3.41 mm

PPM090

91. The number of circular division on the shown screw gauge is 100. It moves 1 mm on main scale for one complete rotation and main scale has 1 mm marks. The diameter of the ball is :-



(1) 2.25 mm

(2) 2.20 mm

(3) 1.20 mm

(4) 1.25 mm

PPM091

EXERCISE-I (Conceptual Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	1	4	1	4	3	3	2	1	3	3	4	1	2	3	3
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	2	4	2	1	2	3	3	3	3	3	3	4	4	4
Question	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
Answer	3	1	4	1	3	3	4	1	3	2	4	1	4	1	2
Question	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Answer	2	3	2	4	2	1	2	2	4	2	1	3	1	2	3
Question	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
Answer	4	1	2	3	1	3	4	1	1	2	3	3	3	4	3
Question	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	3	1	2	1	4	2	1	1	3	2	3	3	2	2	1
Question	91														
Answer	2														

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Exercise - II (Previous Year Questions) AIPMT/NEET

AIPMT 2014

1. If force (F), velocity (V) and time (T) are taken as fundamental units, then the dimensions of mass are:

(1) $[F\ V\ T^{-1}]$ (2) $[F\ V\ T^{-2}]$

(3) $[F\ V^{-1}\ T^{-1}]$ (4) $[F\ V^{-1}\ T]$

PPM103

AIPMT 2015

2. If energy (E), velocity (V) and time (T) are chosen as the fundamental quantities, the dimensional formula of surface tension will be :

(1) $[EV^{-1}T^{-2}]$ (2) $[EV^{-2}T^{-2}]$

(3) $[E^{-2}V^{-1}T^{-3}]$ (4) $[EV^{-2}T^{-1}]$

PPM104

Re-AIPMT 2015

3. If dimension of critical velocity v_c, of liquid flowing through a tube is expressed as $(\eta^x \rho^y r^z)$, where η, ρ and r the coefficient of viscosity of liquid, density of liquid and radius of the tube respectively, then the values of x, y and z are given by :

(1) 1, 1, 1 (2) 1, –1, –1

(3) –1, –1, 1 (4) –1, –1, –1

PPM105

NEET-II 2016

4. Planck's constant (h), speed of light in vacuum (c) and Newton's gravitational constant (G) are three fundamental constants. Which of the following combinations of these has the dimension of length?

(1) $\sqrt{\frac{hc}{G}}$ (2) $\sqrt{\frac{Gc}{h^{3/2}}}$ (3) $\frac{\sqrt{hG}}{c^{3/2}}$ (4) $\frac{\sqrt{hG}}{c^{5/2}}$

PPM106

NEET(UG) 2017

5. A physical quantity of the dimensions of length that can be formed out of c, G and $\frac{e^2}{4\pi\varepsilon_0}$ is [c is velocity of light, G is universal constant of gravitation and e is charge] :-

(1) $c^2\left[G\frac{e^2}{4\pi\varepsilon_0}\right]^{1/2}$ (2) $\frac{1}{c^2}\left[\frac{e^2}{G\,4\pi\varepsilon_0}\right]^{1/2}$

(3) $\frac{1}{c}G\frac{e^2}{4\pi\varepsilon_0}$ (4) $\frac{1}{c^2}\left[G\frac{e^2}{4\pi\varepsilon_0}\right]^{1/2}$

PPM107

NEET(UG) 2018

6. A student measured the diameter of a small steel ball using a screw gauge of least count 0.001 cm. The main scale reading is 5 mm and zero of circular scale division coincides with 25 divisions above the reference level. If screw gauge has a zero error of – 0.004 cm, the correct diameter of the ball is :-

(1) 0.521 cm (2) 0.525 cm

(3) 0.053 cm (4) 0.529 cm

PPM108

NEET(UG) 2019

7. In an experiment, the percentage of error occurred in the measurement of physical quantities A, B, C and D are 1%, 2%, 3% and 4% respectively. Then the maximum percentage of error in the measurement X, where $X=\frac{A^2B^{1/2}}{C^{1/3}D^3}$, will be :

(1) $\left(\frac{3}{13}\right)\%$ (2) 16% (3) –10% (4) 10%

PPM109

8. The unit of thermal conductivity is :

(1) J m K^{-1} (2) J m^{-1} K^{-1}

(3) W m K^{-1} (4) W m^{-1} K^{-1}

PPM110

NEET(UG) 2019 (Odisha)

9. The main scale of a vernier calliper has n divisions/cm. n divisions of the vernier scale coincide with (n – 1) divisions of main scale. The least count of the vernier calliper is,

(1) $\frac{1}{(n+1)(n-1)}$ cm (2) $\frac{1}{n}$ cm

(3) $\frac{1}{n^2}$ cm (4) $\frac{1}{n(n+1)}$ cm

PPM111

NEET(UG) 2020

10. Taking into account of the significant figures, what is the value of 9.99 m – 0.0099 m ?

(1) 9.9 m (2) 9.9801 m

(3) 9.98 m (4) 9.980 m

PPM112

11. Dimensions of stress are :

(1) $[M\ L^{-1}\ T^{-2}]$ (2) $[M\ L\ T^{-2}]$

(3) $[M\ L^2\ T^{-2}]$ (4) $[M\ L^0\ T^{-2}]$

PPM113

12. A screw gauge has least count of 0.01 mm and there are 50 divisions in its circular scale. The pitch of the screw gauge is :
(1) 1.0 mm (2) 0.01 mm
(3) 0.25 mm (4) 0.5 mm

PPM114

NEET(UG) 2020 (COVID-19)

13. The angle of 1' (minute of arc) in radian is nearly equal to
(1) 2.91×10^{-4} rad (2) 4.85×10^{-4} rad
(3) 4.80×10^{-6} rad (4) 1.75×10^{-2} rad

PPM115

14. Time intervals measured by a clock give the following readings :
1.25 s, 1.24 s, 1.27 s, 1.21 s and 1.28 s.
What is the percentage relative error of the observations ?
(1) 2 % (2) 4 %
(3) 16 % (4) 1.6 %

PPM116

NEET(UG) 2021

15. If force [F], acceleration [A] and time [T] are chosen as the fundamental physical quantities. Find the dimensions of energy.
(1) [F] [A] [T] (2) [F] [A] [T^2]
(3) [F] [A] [T^{-1}] (4) [F] [A^{-1}] [T]

PPM117

16. A screw gauge gives the following readings when used to measure the diameter of a wire
Main scale reading : 0 mm
Circular scale reading : 52 divisions
Given that 1 mm on main scale corresponds to 100 divisions on the circular scale. The diameter of the wire from the above data is :
(1) 0.52 cm (2) 0.026 cm
(3) 0.26 cm (4) 0.052 cm

PPM118

17. If E and G respectively denote energy and gravitational constant, then $\frac{E}{G}$ has the dimensions of :
(1) [M^2] [L^{-1}] [T^0] (2) [M] [L^{-1}] [T^{-1}]
(3) [M] [L^0] [T^0] (4) [M^2] [L^{-2}] [T^{-1}]

PPM119

NEET(UG) 2022

18. Plane angle and solid angle have :
(1) Dimensions but no units
(2) No units and no dimensions
(3) Both units and dimensions
(4) Units but no dimensions

PPM120

19. The dimensions [MLT^{-2} A^{-2}] belong to the :
(1) self inductance
(2) magnetic permeability
(3) electric permittivity
(4) magnetic flux

PPM121

RE-NEET(UG) 2022

20. The physical quantity that has the same dimensional formula as pressure is :
(1) Force
(2) Momentum
(3) Young's modulus of elasticity
(4) Coefficient of viscosity

PPM122

21. The ratio of coulomb's electrostatic force to the gravitational force between an electron and a proton separated by some distance is 2.4×10^{39}. The ratio of the proportionality constant, $K = \frac{1}{4\pi\varepsilon_0}$ to the Gravitational constant G is nearly (Given that the charge of the proton and electron each $=1.6 \times 10^{-19}$ C, the mass of the electron = 9.11×10^{-31} kg, the mass of the proton = 1.67×10^{-27} kg):
(1) 10^{20} (2) 10^{30} (3) 10^{40} (4) 10

PPM123

NEET(UG) 2023

22. The errors in the measurement which arise due to unpredictable fluctuations in temperature and voltage supply are :
(1) Personal errors
(2) Least count errors
(3) Random errors
(4) Instrumental errors

PPM142

23. A metal wire has mass (0.4 ± 0.002) g, radius (0.3 ± 0.001) mm and length (5 ± 0.02) cm. The maximum possible percentage error in the measurement of density will nearly be :
(1) 1.3% (2) 1.6% (3) 1.4% (4) 1.2%

PPM143

RE-NEET(UG) 2023

24. The diameter of a spherical bob, when measured with vernier callipers yielded the following values : 3.33 cm, 3.32 cm, 3.34 cm, 3.33 cm and 3.32 cm. The mean diameter to appropriate significant figures is :
(1) 3.328 cm (2) 3.3 cm
(3) 3.33 cm (4) 3.32 cm

PPM144

25. The mechanical quantity, which has dimensions of reciprocal of mass (M^{-1}) is :
(1) angular momentum
(2) coefficient of thermal conductivity
(3) torque
(4) gravitational constant

PPM145

NEET(UG) 2024

26. In a vernier calipers, (N+1) divisions of vernier scale coincide with N divisions of main scale. If 1 MSD represents 0.1 mm, the vernier constant (in cm) is :
(1) $\frac{1}{10N}$ (2) $\frac{1}{100(N+1)}$
(3) 100N (4) 10(N+1)

PPM146

27. The quantities which have the same dimensions as those of solid angle are :
(1) strain and angle
(2) stress and angle
(3) strain and arc
(4) angular speed and stress

PPM147

28. A force defined by $F = \alpha t^2 + \beta t$ acts on a particle at a given time t. The factor which is dimensionless, if α and β are constants, is :
(1) $\frac{\beta t}{\alpha}$ (2) $\frac{\alpha t}{\beta}$
(3) $\alpha\beta t$ (4) $\frac{\alpha\beta}{t}$

PPM148

RE-NEET(UG) 2024

29. In an electrical circuit, the voltage is measured as V = (200±4) volt and the current is measured as I = (20±0.2)A. The value of the resistance is :
(1) (10±4.2)Ω (2) (10±0.3)Ω
(3) (10±0.1)Ω (4) (10±0.8)Ω

PPM149

30. The pitch of an error free screw gauge is 1 mm and there are 100 divisions on the circular scale. While measuring the diameter of a thick wire, the pitch scale reads 1 mm and 63rd division on the circular scale coincides with the reference line. The diameter of the wire is :
(1) 1.63 cm (2) 0.163 cm
(3) 0.163 m (4) 1.63 m

PPM150

31. The potential energy of a particle moving along x-direction varies as $V = \frac{Ax^2}{\sqrt{x}+B}$. The dimensions of $\frac{A^2}{B}$ are :
(1) $[M^{3/2} L^{1/2} T^{-3}]$
(2) $[M^{1/2} L T^{-3}]$
(3) $[M^2 L^{1/2} T^{-4}]$
(4) $[M L^2 T^{-4}]$

PPM151

EXERCISE-II (Previous Year Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	4	2	2	3	4	4	2	4	3	3	1	4	1	4	2
Question	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	4	1	4	2	3	1	3	2	3	4	2	1	2	2	2
Question	31														
Answer	3														

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Exercise - III (Analytical Questions) Master Your Understanding

1. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Absolute error in a physical quantity may be +ve, –ve or zero.
Reason (R) : Absolute error is the difference of measured value and true value of a physical quantity.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PPM124

2. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Maximum possible error in a reading is taken as least count of the measuring instrument.
Reason (R) : Error in a measurement can not be greater than least count of the measuring instrument.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PPM125

3. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : Higher is the accuracy of measurement, if instrument has smaller least count.
Reason (R) : Smaller the percentage error, higher is the accuracy of measurement.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.
(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.
(3) **(A)** is true but **(R)** is false.
(4) **(A)** is false but **(R)** is true.

PPM126

4. Given below are two statements :
Statement I : Absolute error is unitless and dimensionless.
Statement II : All types of errors are unitless and dimensionless.
In the light of the above statements, choose the most appropriate answer from the options given below :
(1) Both statement I and statement II are correct.
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statements I and statements II are incorrect.

PPM128

5. Given below are two statements :
Statement I : Least count of a screw gauge is directly proportional to the number of divisions on circular scale.
Statement II : A screw gauge having a smaller value of pitch has greater accuracy.
In the light of the above statements, choose the most appropriate answer from the options given below :
(1) Both statement I and statement II are correct.
(2) Statement I is correct and statement II is incorrect.
(3) Statement I is incorrect and statement II is correct.
(4) Both statements I and statements II are incorrect.

PPM131

6. Match the quantities having same dimensions in column-1 and column-2 :

	Column-1		Column-2
A	Stress	p	Pressure
B	Strain	q	Energy density
C	Modulus of elasticity	r	Angle
D	Torque	s	Energy

(1) A → r ; B → p ; C → s ; D → q
(2) A → q ; B → r ; C → p ; D → s
(3) A → p ; B → s ; C → r ; D → q
(4) A → q ; B → s ; C → r ; D → q

PPM132

7. Match the column :

	Column-1		Column-2
A	Electrical resistance	p	$M^{-1}L^{-2}T^{4}A^{2}$
B	Capacitance	q	$ML^{2}T^{-2}A^{-2}$
C	Magnetic field	r	$ML^{2}T^{-3}A^{-2}$
D	Inductance	s	$MT^{-2}A^{-1}$

(1) A → r ; B → p ; C → s ; D → q
(2) A → s ; B → p ; C → r ; D → q
(3) A → p ; B → s ; C → r ; D → q
(4) A → q ; B → s ; C → r ; D → q

PPM133

8. Match the column :

	Column-1		Column-2
A	Boltzmann constant	p	$ML^{2}T^{-1}$
B	Coefficient of viscosity	q	$ML^{-1}T^{-1}$
C	Planck constant	r	$MLT^{-3}K^{-1}$
D	Thermal conductivity	s	$ML^{2}T^{-2}K^{-1}$

(1) A → r ; B → s ; C → p ; D → q
(2) A → p ; B → s ; C → r ; D → q
(3) A → p ; B → r ; C → s ; D → q
(4) A → s ; B → q ; C → p ; D → r

PPM135

9. Which two of the following five physical quantities have the same dimensions?
(a) Energy density (b) Refractive index
(c) Dielectric constant (d) Young's modulus
(e) Magnetic field
(1) (a) and (e) (2) (b) and (d)
(3) (c) and (e) (4) (a) and (d)

PPM136

10. Photon is quantum of radiation with energy E = hν where ν is frequency and h is Planck's constant. The dimensions of h are the same as that of
(a) Linear impulse
(b) Angular impulse
(c) Linear momentum
(d) Angular momentum
(1) (a) and (c) (2) (b) and (d)
(3) (b) and (c) (4) (a) and (d)

PPM138

11. Which of the following ratios express pressure?
(a) Force / area (b) Energy / volume
(c) Energy / area (d) Force / volume
(1) (a) and (c) (2) (b) and (d)
(3) (b) and (c) (4) (a) and (b)

PPM139

12. **Statement–1 :** The displacement cannot be added with distance travelled.
Statement–2 : Addition of scalar and vector of same dimensions is not a meaningful algebraic operation.
(1) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1.
(2) Statement–1 is True, Statement–2 is True; Statement–2 is not a correct explanation for Statement–1.
(3) Statement–1 is True, Statement–2 is False.
(4) Statement–1 is False, Statement–2 is True.

PPM140

13. Match the column – I with column – II.

	Column-I (Number)		Column-II (Number of significant digits)
(A)	1001	(P)	5
(B)	0100.1	(Q)	4
(C)	100.100	(R)	6
(D)	0.001001	(S)	3

Options :-
(1) A → Q, B → Q, C → R, D → Q
(2) A → Q, B → P, C → R, D → R
(3) A → Q, B → S, C → S, D → Q
(4) A → Q, B → Q, C → P, D → Q

PPM141

EXERCISE-III (Analytical Questions) ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer	1	3	1	4	3	2	1	4	4	2	4	1	1

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Important Notes

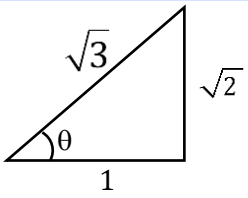
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Basic Mathematics Used in Physics and Vectors

SOLUTIONS

Exercise-I (Conceptual Questions)

2.



$\sin\theta = \frac{\sqrt{2}}{\sqrt{3}}$

$\tan\theta = \frac{P}{B} = \frac{\sqrt{2}}{1}$

3. If $\sin\theta = \cos\theta$

$\tan\theta = 1$

For $180° < \theta < 360° \Rightarrow \theta = 225°$

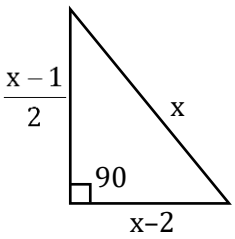
4. If $\sin\theta_1 + \sin\theta_2 + \sin\theta_3 = 3$

then $\theta_1 = \theta_2 = \theta_3 = 90°$

so $\cos\theta_1 + \cos\theta_2 + \cos\theta_3 = 0$

5. Max. value $= \sqrt{7^2 + (-24)^2} = \sqrt{49 + 576} = 25$

6.



Hypotenuse (H) = x

Base (B) = x – 2

Perpendicular (P) $= \frac{x-1}{2}$

$x^2 = (x-2)^2 + \left(\frac{x-1}{2}\right)^2$

$4x^2 = 4(x-2)^2 + (x-1)^2$

$x^2 - 18x + 17 = 0$

x = 17 cm and x = 1 cm (not considerable)

So, H = 17 cm

B = 17 – 2 = 15 cm

$P = \frac{17-1}{2} = 8$ cm

7. $x = at^4$; $y = bt^3$

$\frac{dx}{dt} = 4at^3$ & $\frac{dy}{dt} = 3bt^2$

So $\frac{dy}{dt} = \frac{3bt^2}{4at^3} = \frac{3b}{4at}$

8. $A = 5t^2 + 4t + 8$

$\frac{dA}{dt} = 10t + 4$

At t = 3 sec

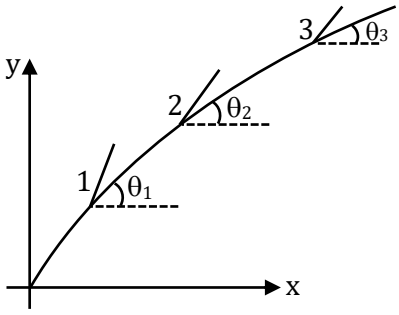
$\frac{dA}{dt} = 34\ m^2/s$

9. If side = a, then rate of increase of perimeter w.r.t. time $= 4\left(\frac{da}{dt}\right) = 4(0.1) = 0.4\,cm/s$

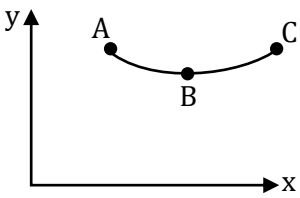
10. $3y = x + 5$

$\frac{3dy}{dt} = \frac{dx}{dt}$ (x changes at a faster rate)

11. $\theta_1 > \theta_2 > \theta_3 \Rightarrow m_1 > m_2 > m_3$



12.



From A to B : θ is obtuse and increases, so slope (tan θ) will decreases.

From B to C : θ is acute and increases

So, B to C magnitude will increases.

13. Area under the curve $\int_2^3 f(x)dx = \int_2^3 x^2dx$

$= \frac{1}{3}[x^3]_2^3 = \frac{19}{3}$

14. $\int_0^4 |1-x|dx = \int_0^1 (1-x)dx + \int_1^4 -(1-x)dx$

$= [x]_0^1 - \left[\frac{x^2}{2}\right]_0^1 - [x]_1^4 + \left[\frac{x^2}{2}\right]_1^4$

$= 1 - \frac{1}{2} - 3 + \frac{15}{2} = 5$

15. $y = x^2 + 2 - 3x$

on y-axis $x = 0$

$\Rightarrow \quad y = 2$

16. at t = 2 $\quad x_A = 6, \quad y_A = 2$

$x_B = 6, \quad y_B = 2 + 3\,(2)^2 = 14$

distance $= \sqrt{(6-6)^2 + (14-2)^2} = 12$

17. distance d $= \sqrt{\{a-b-a-b\}^2 + \{c-b-b-c\}^2}$

$d = \sqrt{4b^2 + 4b^2} = 2\sqrt{2}\,b$

18. Minimum distance

$= \sqrt{(5-0)^2 + (3-3)^2 + (-8-4)^2}$

$= \sqrt{5^2 + 12^2} = 13$

19. $\because y = x^2 + 2x - 3$

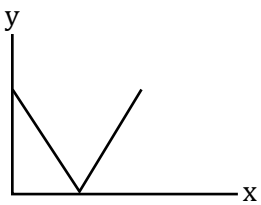
$\therefore y = (x + 3)(x - 1)$

For $y = 0 \Rightarrow x = -3$ and $x = 1$

For $x = 0 \Rightarrow y = -3$

20. For $0 \le x \le 1, \quad y = -(x - 1) = -x + 1$

For $x \ge 1, \quad y = x - 1$



22. $f = \frac{1}{2\pi}\sqrt{\frac{g}{\ell}}$

$f^2 = \frac{1}{4\pi^2}\frac{g}{\ell}$

$\Rightarrow f^2 \propto \frac{1}{\ell}$ graph between f & $\frac{1}{\ell}$ is parabolic

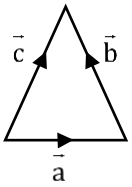
23. $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + ...\infty$

$S_\infty = \frac{a}{1-r} = \frac{1}{1-1/3} = \frac{3}{2}$

29. Angular velocity is an axial vector

30. $\hat{n} = \frac{\vec{A}}{|\vec{A}|}$

31.



Resultant of $\vec{a}$ & $\vec{b}$ is $\vec{c}$

$(\vec{a} + \vec{b} = \vec{c})$

so $\quad \vec{a} + \vec{b} + \vec{c} = \vec{c} + \vec{c} = 2\vec{c}$

32. $10 - 5 \le R \le 10 + 5$

$5N \le R \le 15N$

33. $10 - 8 \le R \le 10 + 8$

$2N \le R \le 18N$, So 20N cannot be the resultant of these forces.

34. $(A-B) \le R \le (A+B)$

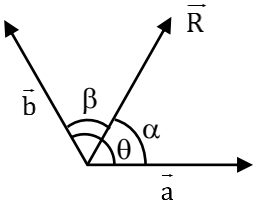
35. $R = 2a\cos(\theta/2) \Rightarrow a = 2a\cos(\theta/2)$

$\Rightarrow \cos(\theta/2) = 1/2 \Rightarrow \theta/2 = 60° \Rightarrow \theta = 120°$

36. If R = a = b

then angle between a & b will be 120° ($\theta = 120°$) & resultant will be bisector

$\alpha = \beta = \frac{\theta}{2} = \frac{\pi}{3}$



37. $\sqrt{a^2 + b^2 + 2ab\cos 60°} = \sqrt{7}\,b$

$\Rightarrow \quad a^2 + b^2 + ab = 7b^2$

$\Rightarrow \quad a^2 + ab - 6b^2 = 0$

$\Rightarrow \quad (a - 2b)(a + 3b) = 0$

$\Rightarrow \quad \boxed{\frac{b}{a} = \frac{1}{2}}$ or $\boxed{\frac{b}{a} = -\frac{1}{3}}$ (not considerable)

38. $\vec{A}+\vec{B}=\vec{C} \Rightarrow \sqrt{A^2+B^2+2AB\cos\theta}=C$

and $A + B = C \Rightarrow \sqrt{A^2+B^2+2AB\cos\theta} = A + B$

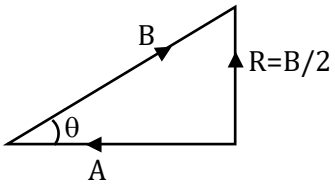
$\Rightarrow A^2 + B^2 + 2AB\cos\theta = A^2 + B^2 + 2AB$

$\Rightarrow \cos\theta = 1 \Rightarrow \theta = 0°$

39. $\vec{A}+\vec{B}=\vec{C}$ and $|\vec{C}|=A-B$

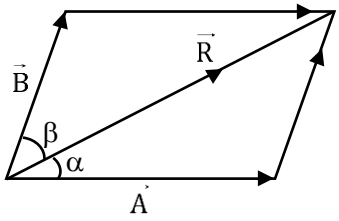
$\Rightarrow$ angle between $\vec{A}$ & $\vec{B}$ is 180°

40.



$\sin\theta = \frac{R}{B}=\frac{B/2}{B}=\frac{1}{2} \Rightarrow \theta = 30°$

41.



Resultant is inclined towards vector of large magnitude thus. If A > B then $\alpha < \beta$.

42. If two vectors have equal magnitude then resultant will be bi-sector of angle between them.

43. $\tan\beta=\frac{A\sin\theta}{B+A\cos\theta}=\tan 90°=\infty$

thus, $B + A\cos\theta = 0 \Rightarrow \cos\theta = -B/A$

44. $\tan 30°=\frac{(Q)\sin(90°)}{P+Q\cos(90°)}$ ($\because$ 30° + 60° = 90°)

$\frac{1}{\sqrt{3}}=\frac{Q}{P}\Rightarrow \boxed{P=\sqrt{3}(Q)}$

As $\sqrt{(\sqrt{3}Q)^2+Q^2+2(\sqrt{3}Q)(Q)\cos 90°}=40$

$\Rightarrow$ Q = 20N and $P=20\sqrt{3}N$

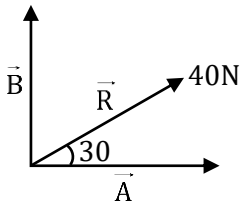
OR

$A = R\cos 30°$

$A=40\times\frac{\sqrt{3}}{2}$

$A=20\sqrt{3}N$

$B=R\cos 60°=40\times\frac{1}{2}$



B = 20N

46. $R_1^2 = A^2 + B^2 + 2AB\cos\theta$

$|\vec{R}_2|=|\vec{A}-\vec{B}|\Rightarrow R_2^2 = A^2 + B^2 - 2AB\cos\theta$

thus $R_1^2+R_2^2 = 2(A^2 + B^2)$

$\Rightarrow 2\left(R_1^2+R_2^2\right)=4(A^2+B^2)$

47. If $|\vec{A}+\vec{B}|=|\vec{A}|=|\vec{B}|$

then angle between them $\theta = 120°$

Let $|\vec{A}|=|\vec{B}|$ = a

So $|\vec{A}-\vec{B}|=\sqrt{A^2+B^2-2AB\cos\theta}$

$=\sqrt{a^2+a^2-2a^2\left(-\frac{1}{2}\right)}=\sqrt{3}|\vec{A}|$

48. $|\hat{n}_1-\hat{n}_2|=1$

$\sqrt{1^2+1^2-2\cos\theta}=1\Rightarrow 2-2\cos\theta=1$

$\Rightarrow\cos\theta=\frac{1}{2}\Rightarrow\theta=60°$

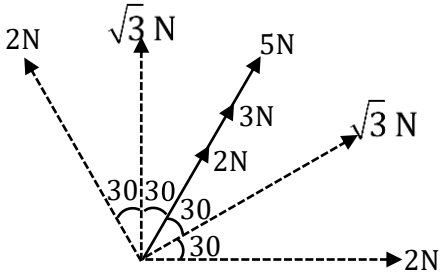
$|\hat{n}_1+\hat{n}_2|=\sqrt{1^2+1^2+2\times1\times1\cos 60°}=\sqrt{3}$

49. If vectors of equal magnitude are acting at equal angle of separation then resultant will be zero.

51. Sum of all small forces should be greater than largest force then resultant can be zero.

52. Resultant of 2N & 2N

$= (2)(2)\cos\left(\frac{120°}{2}\right) = 2N$

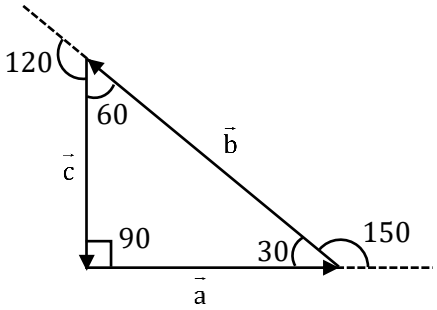


Resultant of $\sqrt{3}N$ & $\sqrt{3}N$

$= (2)(\sqrt{3})\cos\left(\frac{60°}{2}\right) = 3N$

Net resultant = 5 + 3 + 2 = 10 N

53.

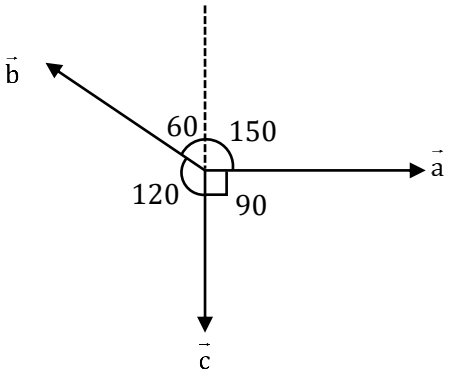


According to this

$\boxed{b^2 = a^2 + c^2}$

a : b : c = $\sqrt{3}$: 2 : 1

OR



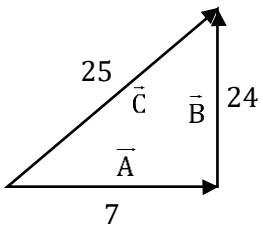
bcos60° = c

$\frac{b}{c} = \frac{2}{1}$

bsin60° = a

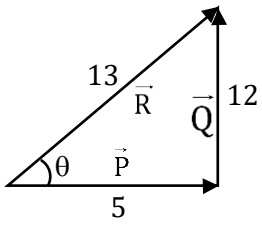
$\frac{b}{a} = \frac{2}{\sqrt{3}}$ so, ratio of a, b & c is $\sqrt{3}$: 2 : 1

54. $\vec{A} + \vec{B} = \vec{C}$ & $A^2 + B^2 = C^2$



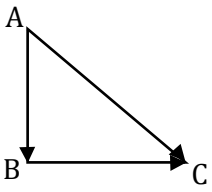
Angle between $\vec{A}$ & $\vec{B}$ is 90°

55. $\vec{P} + \vec{Q} = \vec{R}$ and $P^2 + Q^2 = R^2$



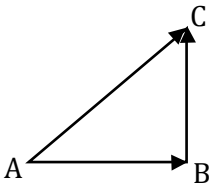
$\cos\theta = \frac{5}{13} \Rightarrow \theta = \cos^{-1}\left(\frac{5}{13}\right)$

56.



$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = (\hat{i} + \hat{j} + \hat{k}) + 3\hat{i}$

$= 4\hat{i} + \hat{j} + \hat{k}$



$= \sqrt{16+1+1} = \sqrt{18}$ unit

$\overrightarrow{BC} = \overrightarrow{AC} - \overrightarrow{AB} = 3\hat{i} - (\hat{i} + \hat{j} + \hat{k})$

$= 2\hat{i} + \hat{j} + \hat{k}$

$|\overrightarrow{BC}| = \sqrt{4+1+1} = \sqrt{6}$ unit

60. The direction of a vector remains unchanged on multiplying the vector by a positive scalar.

64. $\vec{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{\hat{i} - 2\hat{j}}{\sqrt{5}}$

65. $\vec{A} + \vec{B} = \hat{j}$

$\Rightarrow \vec{B} = \hat{j} - \vec{A} = \hat{j} - (\hat{i} - \hat{j} + \hat{k}) = -\hat{i} + 2\hat{j} - \hat{k}$

66. $\hat{n} = 0.3\hat{i} - 0.4\hat{j} + c\hat{k}$

$\sqrt{(0.3)^2 + (-0.4)^2 + c^2} = 1$

$\Rightarrow$ $c^2 = 1 - 0.09 - 0.16$

$\Rightarrow$ $c^2 = 0.75$

$\Rightarrow$ $c = \sqrt{0.75}$

67. $\hat{e}_r = |\hat{e}_r|\cos\theta\hat{i} + |\hat{e}_r|\sin\theta\hat{j}$

$\because |\hat{e}_r| = 1$

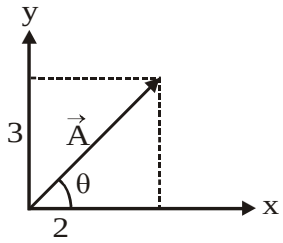
$\Rightarrow \hat{e}_r = (\cos\theta)\hat{i} + (\sin\theta)\hat{j}$

68. $\vec{R} = 7\hat{i} + 24\hat{j} + 25\hat{k}$

$|\vec{R}| = \sqrt{7^2 + 24^2 + 25^2} = \sqrt{49 + 576 + 625}$

$= 25\sqrt{2}\,N$

69.



$\vec{A} = 2\hat{i} + 3\hat{j}$

$\tan\theta = \frac{3}{2}$

$\theta = \tan^{-1}\left(\frac{3}{2}\right)$

70. Let $\vec{A}$ has to the added

$(\hat{i} - 2\hat{j} + 2\hat{k}) + (2\hat{i} + \hat{j} - \hat{k}) + (\vec{A}) = \hat{j}$

$\vec{A} = -3\hat{i} + 2\hat{j} - \hat{k}$

71. $\vec{R} = \vec{A} + \vec{B} = 4\hat{i} + 3\hat{j} + 6\hat{k} + 2\hat{i} - 3\hat{j} + 2\hat{k}$

$\vec{R} = 6\hat{i} + 8\hat{k}$

$\vec{R} = \frac{6\hat{i} + 8\hat{k}}{\sqrt{6^2 + 8^2}} = \frac{6\hat{i} + 8\hat{k}}{10}$

72. $\vec{c} = 3|\vec{b}|\hat{a}$

$|\vec{b}| = \sqrt{3}; \hat{a} = \frac{2\hat{i} + 2\hat{j} - \hat{k}}{3}$

so $\vec{c} = (3)(\sqrt{3})\left(\frac{2\hat{i} + 2\hat{j} - \hat{k}}{3}\right) = 2\sqrt{3}\hat{i} + 2\sqrt{3}\hat{j} - \sqrt{3}\hat{k}$

73. Unit vector $\vec{v} = \frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}$

Force in the direction of velocity

$\vec{F} = |\vec{F}|\hat{v}$

$\vec{F} = (10\sqrt{3})\left(\frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}\right) = (10\hat{i} + 10\hat{j} - 10\hat{k})N$

74. $\vec{A} = \sqrt{2}\hat{i} + \sqrt{2}\hat{j} + \hat{k}$

$\cos\alpha = \frac{A_x}{A},\ \cos\beta = \frac{A_y}{A},\ \cos\gamma = \frac{A_z}{A}$

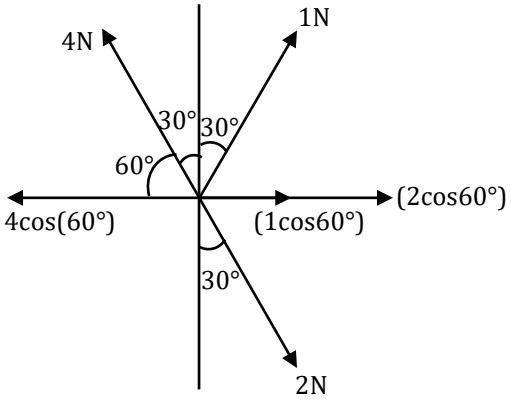
$\cos\alpha = \frac{\sqrt{2}}{\sqrt{5}},\ \cos\beta = \frac{\sqrt{2}}{\sqrt{5}},\ \cos\gamma = \frac{1}{\sqrt{5}}$

75. $\vec{a} - \vec{b} = \hat{j} - 2\hat{k}$

$|\vec{a} - \vec{b}| = \sqrt{1^2 + (-2)^2} = \sqrt{5}$

$\cos\alpha = 0\ ;\ \cos\beta = \frac{1}{\sqrt{5}}; \cos\gamma = -\frac{2}{\sqrt{5}}$

76. To have the resultant force along the y-direction the force along x-axis must be zero.



Let F is the require force then,

–4cos(60°) + (1)cos(60°) + 2cos(60°) + F = 0

F = 0.5N

77. If a vector rotates through any angle then its magnitude remain same.

so, $\sqrt{1^2 + b^2 + 3^2} = \sqrt{2^2 + (2b-1)^2 + 1^2}$

$\Rightarrow$ $10 + b^2 = 5 + 4b^2 + 1 - 4b$

$\Rightarrow$ $3b^2 - 4b - 4 = 0$

$\Rightarrow$ $(3b + 2)(b - 2) = 0$

$\Rightarrow$ $b = 2$ and $b = -\frac{2}{3}$

78. $\vec{P}\cdot\vec{Q} = -PQ$

$\vec{P}\vec{Q}\cos\theta = -PQ$

$\cos\theta = -1$

$\Rightarrow \theta = 180°$

79. $w=\vec{F}\cdot\vec{S}$

$w=(2\hat{i}+3\hat{j})\cdot(3\hat{i}+4\hat{j})$

w = 6 + 12 = 18J

80. $W=\vec{F}.\vec{S}=(2\hat{i}+2\hat{j}).(2\hat{i}-3\hat{j})$

W = 4 – 6 = –2J

81. $\vec{F}=14\left(\frac{3\hat{i}+2\hat{j}-6\hat{k}}{\sqrt{9+4+36}}\right)=6\hat{i}+4\hat{j}-12\hat{k}$

$\vec{S}=2\hat{i}+4\hat{j}-2\hat{k}$

$W=\vec{F}.\vec{S}=12+16+24=52J$

82. $\vec{A}=2\hat{i}+3\hat{j}+8\hat{k}$ & $\vec{B}=4\hat{i}-4\alpha\hat{j}+\alpha\hat{k}$

$\because \quad \vec{A}\perp\vec{B}\Rightarrow \vec{A}\cdot\vec{B}=0$

$\Rightarrow$ 8 – 12α + 8α = 0

$\Rightarrow$ 4α = 8

$\Rightarrow$ α = 2

83. $\vec{P}.\vec{Q}=0\Rightarrow(a\hat{i}+a\hat{j}+3k).(a\hat{i}-2\hat{j}-k)=0$

$a^2 - 2a - 3 = 0 \Rightarrow (a-3)(a+1)=0$

a = 3 and a = –1

84. $\therefore \vec{A}\perp\vec{B}$

$\vec{A}.\vec{B}=0$

85. $\because$ $\vec{A}$ is normal to $\vec{B}$ $\Rightarrow \vec{A}\cdot\vec{B}=0$

86. $\hat{n}=a\hat{i}+b\hat{j}$ is perpendicular to $\hat{i}-\hat{j}$

$\Rightarrow(a\hat{i}+b\hat{j}).(\hat{i}-\hat{j})=0$

$\Rightarrow$ a – b = 0

$\Rightarrow$ a = b

88. $(\vec{A}+\vec{B}).(\vec{A}-\vec{B})=0$

$\Rightarrow A^2-\vec{A}.\vec{B}+\vec{B}.\vec{A}-B^2=0$

$\Rightarrow$ A = B ($\because \vec{A}.\vec{B}=\vec{B}.\vec{A}$)

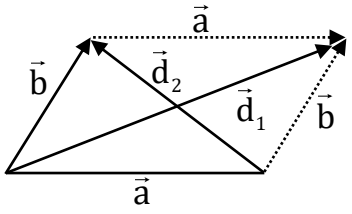
89. $(\vec{A}+\vec{B}).(\vec{A}-\vec{B})=A^2-\vec{A}.\vec{B}+\vec{B}.\vec{A}-B^2$

$= A^2 - B^2$

$= 0\ [\because |\vec{A}|=|\vec{B}|]$

Hence $\vec{A}+\vec{B}$ & $\vec{A}-\vec{B}$ are perpendicular to each other

90. According to triangle law



$\vec{a}+\vec{b}=\vec{d}_1$

$\Rightarrow \quad a^2+b^2+2ab\cos\theta=d_1^2$ ....(1)

& $\vec{a}-\vec{b}=-\vec{d}_2$

$\Rightarrow \quad a^2+b^2-2ab\cos\theta=d_2^2$ ....(2)

By (1) + (2)

$a^2+b^2=\frac{d_1^2+d_2^2}{2}$

91. $\vec{A}=3\hat{i}+4\hat{j}+5\hat{k}$, $\vec{B}=6\hat{i}+8\hat{j}+10\hat{k}$

$\vec{B}=2[3\hat{i}+4\hat{j}+5\hat{k}]$

$\vec{B}=2\vec{A}$

$\Rightarrow$ $\vec{B}$ is parallel to $\vec{A}$

92. $\vec{R}=\vec{A}+2\vec{B}+2\vec{A}-2\vec{B}$

$\vec{R}=3\vec{A}$ $\Rightarrow \vec{R}$ is parallel to $\vec{A}$

93. $\cos\theta=\frac{\vec{A}.\vec{B}}{AB}=\frac{(\hat{i}+\hat{j}).(\hat{i}+\hat{k})}{\sqrt{2}\sqrt{2}}=\frac{1}{2}\Rightarrow\theta=60°$

94. $\because \cos\theta=\frac{\vec{A}.\vec{B}}{|\vec{A}||\vec{B}|}$

$\Rightarrow \cos\theta=\frac{(2\hat{i}+\hat{j}+2\hat{k}).(\hat{i}-\hat{j}+\hat{k})}{\left(\sqrt{2^2+1^2+2^2}\right)\sqrt{1^2+(-1)^2+(1)^2}}$

$\Rightarrow \cos\theta=\frac{1}{\sqrt{3}}\Rightarrow \boxed{\theta=\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)}$

95. Projection $=\frac{(3\hat{i}+4\hat{k}).\hat{k}}{|\hat{k}|}=4$

96. Projection of $\vec{B}$ on $\vec{A}=B\cos\theta$

$=B\left[\frac{\vec{A}\cdot\vec{B}}{AB}\right]=\vec{B}\cdot\hat{A}$

100. $\vec{P}\times\vec{Q}=PQ\sin\theta\hat{n}\Rightarrow\hat{n}=\frac{\hat{P}\times\hat{Q}}{\sin\theta}=\frac{\vec{P}\times\vec{Q}}{|\vec{P}\times\vec{Q}|}$

102. $(\vec{A}+\vec{B})\times(\vec{A}-\vec{B})$

$=(\vec{A}\times\vec{A})-(\vec{A}\times\vec{B})+(\vec{B}\times\vec{A})-(\vec{B}\times\vec{B})$

$=2(\vec{B}\times\vec{A})=-2(\vec{A}\times\vec{B})$

103. $\vec{A}\times\vec{B}=-(\vec{B}\times\vec{A})$

$\therefore$ angle between these vectors is zero

104. Vertically upward = $\hat{k}$

South = $-\hat{j}$

$\vec{A}\times\vec{B}\Rightarrow\hat{k}\times(-\hat{j})=\hat{i}$ $\quad \hat{i}\Rightarrow$ east

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105. $\frac{3}{4}AB\sin\theta = AB\cos\theta$

$\frac{\sin\theta}{\cos\theta} = \frac{4}{3}$

$\frac{\sin\theta}{\cos\theta} = \frac{4}{3}$

$\theta = 53°$

106. $\vec{A}\times\vec{B} = (3\hat{i}+4\hat{j})\times(6\hat{i}+8\hat{j}) = \vec{O}$

107. $|\vec{A}+\vec{B}| = |\vec{A}-\vec{B}|$

$\sqrt{A^2+B^2+2A\cos\theta} = \sqrt{A^2+B^2-2AB\cos\theta}$

$\Rightarrow \quad 4AB\cos\theta = 0$

$\Rightarrow \quad \cos\theta = 0$

$\Rightarrow \quad \theta = 90°$

$\Rightarrow \quad \vec{A}.\vec{B} = 0$

108. Cross product of two || vector are zero.

$\therefore \vec{F}_1, \vec{F}_2$ are || vectors

109. Cross product of two parallel or anti parallel vectors are zero.

110. $\vec{A}\times\vec{B} = 0$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 2 \\ 6 & 2 & S \end{vmatrix} = 0$$

$\hat{i}(S-4)-\hat{j}(3S-12)+\hat{k}(6-6)$

$(S-4)\hat{i}-\hat{j}(3S-12) = 0\hat{i}+0\hat{j}$

$S-4=0$ or $3S-12=0$

$S=4$ or $S=4$

112. $\vec{v} = \vec{\omega}\times\vec{r}$

$$\vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

$= \hat{i}(-1-1)-\hat{j}(1-1)+\hat{k}(1+1)$

$= -2\hat{i}+2\hat{k}$

113. $\vec{a}\times\vec{b} = \begin{vmatrix} i & j & k \\ 2 & -2 & 1 \\ 1 & 2 & 2 \end{vmatrix}$

$= \hat{i}(-4-2)-\hat{j}(4-1)+\hat{k}(4+2)$

$\vec{a}\times\vec{b} = -6\hat{i}-3\hat{j}+6\hat{k}$

$\hat{n} = \pm\frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|} = \pm\left(\frac{-6\hat{i}-3\hat{j}+6\hat{k}}{3\times3}\right)$

$= \pm\left(-\frac{2}{3}\hat{i}-\frac{1}{3}\hat{j}+\frac{2}{3}\hat{k}\right)$

Exercise-II (Previous Year Questions)

1. $\vec{A}\cdot\vec{B} = 0$

$\cos\omega t\cos\frac{\omega t}{2}+\sin\omega t\sin\frac{\omega t}{2} = 0$

$\cos\left(\omega t-\frac{\omega t}{2}\right) = 0 \Rightarrow \cos\frac{\omega t}{2} = 0$

2. $|\vec{A}+\vec{B}| = |\vec{A}-\vec{B}|$

$A^2 + B^2 + 2AB\cos\theta = A^2 + B^2 - 2AB\cos\theta$

$\cos\theta = 0 \Rightarrow \theta = 90°.$

3. $\vec{F}_{net} = \vec{F}_1+\vec{F}_2+\vec{F}_3 = \vec{0} \Rightarrow \vec{a} = 0 \Rightarrow \vec{v} = \text{constant}$

4. $\vec{F} = 2\hat{i}+\hat{j}-\hat{k}$ $\qquad$ $\vec{r} = 3\hat{i}+2\hat{j}-2\hat{k}$

$\vec{F}.\vec{r} = 6+2+2 = 10$

$$\vec{F}\times\vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 3 & 2 & -2 \end{vmatrix} = \hat{i}(0)-\hat{j}(-1)+\hat{k}(1)$$

$= \hat{j}+\hat{k}$

$|\vec{F}\times\vec{r}| = \sqrt{2}$

Exercise-III (Analytical Questions)

2. $|\vec{A}| = |\vec{B}| = Z, \theta = 60°$

$|\vec{A}+\vec{B}| = 2a\cos\left(\frac{\theta}{2}\right) = \sqrt{3}Z$

$|\vec{A}-\vec{B}| = 2a\sin\left(\frac{\theta}{2}\right) = Z$

$\vec{A}\cdot\vec{B} = AB\cos\theta = \frac{1}{2}\cdot Z^2$

$|\vec{A}\times\vec{B}| = AB\sin\theta = \frac{\sqrt{3}}{2}\cdot Z^2$

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3. According to triangle law.

5. Direction of null vector is not defined and direction of $\vec{A}\times\vec{B}$ is given by right hand thumb rule which lies in $\perp$ plane of $\vec{A}, \vec{B}, (\vec{A}+\vec{B})$ and $(\vec{A}-\vec{B})$.

6. $\vec{A}\cdot\vec{B}$ in a scalar quantity that's why it is known as scalar product.

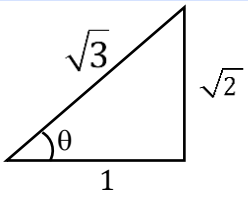
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Basic Mathematics Used in Physics and Vectors

SOLUTIONS

Exercise-I (Conceptual Questions)

2.



$\sin\theta = \frac{\sqrt{2}}{\sqrt{3}}$

$\tan\theta = \frac{P}{B} = \frac{\sqrt{2}}{1}$

3. If $\sin\theta = \cos\theta$

$\tan\theta = 1$

For $180° < \theta < 360° \Rightarrow \theta = 225°$

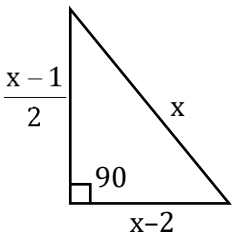
4. If $\sin\theta_1 + \sin\theta_2 + \sin\theta_3 = 3$

then $\theta_1 = \theta_2 = \theta_3 = 90°$

so $\cos\theta_1 + \cos\theta_2 + \cos\theta_3 = 0$

5. Max. value $= \sqrt{7^2 + (-24)^2} = \sqrt{49 + 576} = 25$

6.



Hypotenuse (H) = x

Base (B) = x – 2

Perpendicular (P) $= \frac{x-1}{2}$

$x^2 = (x-2)^2 + \left(\frac{x-1}{2}\right)^2$

$4x^2 = 4(x-2)^2 + (x-1)^2$

$x^2 - 18x + 17 = 0$

x = 17 cm and x = 1 cm (not considerable)

So, H = 17 cm

B = 17 – 2 = 15 cm

$P = \frac{17-1}{2} = 8$ cm

7. $x = at^4$; $y = bt^3$

$\frac{dx}{dt} = 4at^3$ & $\frac{dy}{dt} = 3bt^2$

So $\frac{dy}{dt} = \frac{3bt^2}{4at^3} = \frac{3b}{4at}$

8. $A = 5t^2 + 4t + 8$

$\frac{dA}{dt} = 10t + 4$

At t = 3 sec

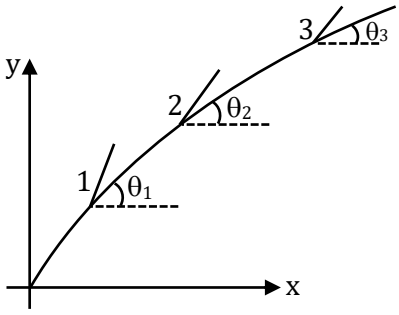
$\frac{dA}{dt} = 34\ m^2/s$

9. If side = a, then rate of increase of perimeter w.r.t. time $= 4\left(\frac{da}{dt}\right) = 4(0.1) = 0.4\,cm/s$

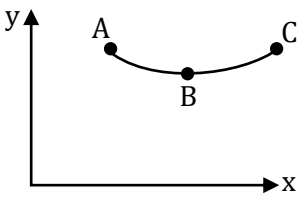
10. $3y = x + 5$

$\frac{3dy}{dt} = \frac{dx}{dt}$ (x changes at a faster rate)

11. $\theta_1 > \theta_2 > \theta_3 \Rightarrow m_1 > m_2 > m_3$



12.



From A to B : θ is obtuse and increases, so slope (tan θ) will decreases.

From B to C : θ is acute and increases

So, B to C magnitude will increases.

13. Area under the curve $\int_2^3 f(x)dx = \int_2^3 x^2dx$

$= \frac{1}{3}[x^3]_2^3 = \frac{19}{3}$

14. $\int_0^4 |1-x|dx = \int_0^1 (1-x)dx + \int_1^4 -(1-x)dx$

$= [x]_0^1 - \left[\frac{x^2}{2}\right]_0^1 - [x]_1^4 + \left[\frac{x^2}{2}\right]_1^4$

$= 1 - \frac{1}{2} - 3 + \frac{15}{2} = 5$

15. $y = x^2 + 2 - 3x$

on y-axis $x = 0$

$\Rightarrow \quad y = 2$

16. at t = 2 $\quad x_A = 6, \quad y_A = 2$

$x_B = 6, \quad y_B = 2 + 3\,(2)^2 = 14$

distance $= \sqrt{(6-6)^2 + (14-2)^2} = 12$

17. distance d $= \sqrt{\{a-b-a-b\}^2 + \{c-b-b-c\}^2}$

$d = \sqrt{4b^2 + 4b^2} = 2\sqrt{2}\,b$

18. Minimum distance

$= \sqrt{(5-0)^2 + (3-3)^2 + (-8-4)^2}$

$= \sqrt{5^2 + 12^2} = 13$

19. $\because y = x^2 + 2x - 3$

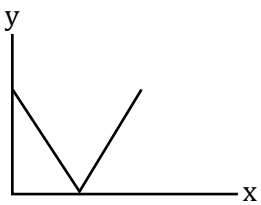
$\therefore y = (x + 3)(x - 1)$

For $y = 0 \Rightarrow x = -3$ and $x = 1$

For $x = 0 \Rightarrow y = -3$

20. For $0 \le x \le 1, \quad y = -(x - 1) = -x + 1$

For $x \ge 1, \quad y = x - 1$



22. $f = \frac{1}{2\pi}\sqrt{\frac{g}{\ell}}$

$f^2 = \frac{1}{4\pi^2}\frac{g}{\ell}$

$\Rightarrow f^2 \propto \frac{1}{\ell}$ graph between f & $\frac{1}{\ell}$ is parabolic

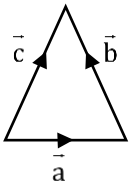
23. $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + ...\infty$

$S_\infty = \frac{a}{1-r} = \frac{1}{1-1/3} = \frac{3}{2}$

29. Angular velocity is an axial vector

30. $\hat{n} = \frac{\vec{A}}{|\vec{A}|}$

31.



Resultant of $\vec{a}$ & $\vec{b}$ is $\vec{c}$

$(\vec{a} + \vec{b} = \vec{c})$

so $\quad \vec{a} + \vec{b} + \vec{c} = \vec{c} + \vec{c} = 2\vec{c}$

32. $10 - 5 \le R \le 10 + 5$

$5N \le R \le 15N$

33. $10 - 8 \le R \le 10 + 8$

$2N \le R \le 18N$, So 20N cannot be the resultant of these forces.

34. $(A-B) \le R \le (A+B)$

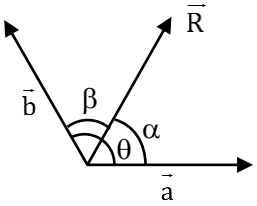
35. $R = 2a\cos(\theta/2) \Rightarrow a = 2a\cos(\theta/2)$

$\Rightarrow \cos(\theta/2) = 1/2 \Rightarrow \theta/2 = 60° \Rightarrow \theta = 120°$

36. If R = a = b

then angle between a & b will be 120° ($\theta = 120°$) & resultant will be bisector

$\alpha = \beta = \frac{\theta}{2} = \frac{\pi}{3}$



37. $\sqrt{a^2 + b^2 + 2ab\cos 60°} = \sqrt{7}\,b$

$\Rightarrow \quad a^2 + b^2 + ab = 7b^2$

$\Rightarrow \quad a^2 + ab - 6b^2 = 0$

$\Rightarrow \quad (a - 2b)(a + 3b) = 0$

$\Rightarrow \quad \boxed{\frac{b}{a} = \frac{1}{2}}$ or $\boxed{\frac{b}{a} = -\frac{1}{3}}$ (not considerable)

38. $\vec{A}+\vec{B}=\vec{C} \Rightarrow \sqrt{A^2+B^2+2AB\cos\theta}=C$

and $A + B = C \Rightarrow \sqrt{A^2+B^2+2AB\cos\theta} = A + B$

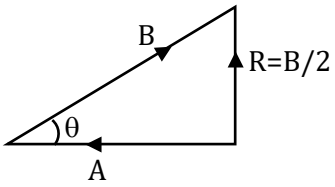
$\Rightarrow A^2 + B^2 + 2AB\cos\theta = A^2 + B^2 + 2AB$

$\Rightarrow \cos\theta = 1 \Rightarrow \theta = 0°$

39. $\vec{A}+\vec{B}=\vec{C}$ and $|\vec{C}|=A-B$

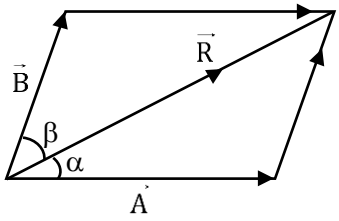
$\Rightarrow$ angle between $\vec{A}$ & $\vec{B}$ is 180°

40.



$\sin\theta = \frac{R}{B}=\frac{B/2}{B}=\frac{1}{2} \Rightarrow \theta = 30°$

41.



Resultant is inclined towards vector of large magnitude thus. If A > B then $\alpha < \beta$.

42. If two vectors have equal magnitude then resultant will be bi-sector of angle between them.

43. $\tan\beta=\frac{A\sin\theta}{B+A\cos\theta}=\tan 90°=\infty$

thus, $B + A\cos\theta = 0 \Rightarrow \cos\theta = -B/A$

44. $\tan 30°=\frac{(Q)\sin(90°)}{P+Q\cos(90°)}$ ($\because$ 30° + 60° = 90°)

$\frac{1}{\sqrt{3}}=\frac{Q}{P}\Rightarrow \boxed{P=\sqrt{3}(Q)}$

As $\sqrt{(\sqrt{3}Q)^2+Q^2+2(\sqrt{3}Q)(Q)\cos 90°}=40$

$\Rightarrow$ Q = 20N and $P=20\sqrt{3}N$

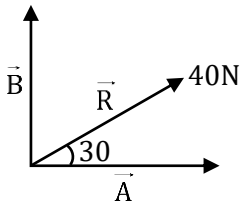
OR

$A = R\cos 30°$

$A=40\times\frac{\sqrt{3}}{2}$

$A=20\sqrt{3}N$

$B=R\cos 60°=40\times\frac{1}{2}$



B = 20N

46. $R_1^2 = A^2 + B^2 + 2AB\cos\theta$

$|\vec{R}_2|=|\vec{A}-\vec{B}|\Rightarrow R_2^2 = A^2 + B^2 - 2AB\cos\theta$

thus $R_1^2+R_2^2 = 2(A^2 + B^2)$

$\Rightarrow 2\left(R_1^2+R_2^2\right)=4(A^2+B^2)$

47. If $|\vec{A}+\vec{B}|=|\vec{A}|=|\vec{B}|$

then angle between them $\theta = 120°$

Let $|\vec{A}|=|\vec{B}|$ = a

So $|\vec{A}-\vec{B}|=\sqrt{A^2+B^2-2AB\cos\theta}$

$=\sqrt{a^2+a^2-2a^2\left(-\frac{1}{2}\right)}=\sqrt{3}|\vec{A}|$

48. $|\hat{n}_1-\hat{n}_2|=1$

$\sqrt{1^2+1^2-2\cos\theta}=1\Rightarrow 2-2\cos\theta=1$

$\Rightarrow\cos\theta=\frac{1}{2}\Rightarrow\theta=60°$

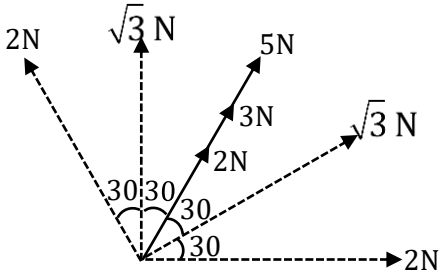
$|\hat{n}_1+\hat{n}_2|=\sqrt{1^2+1^2+2\times1\times1\cos 60°}=\sqrt{3}$

49. If vectors of equal magnitude are acting at equal angle of separation then resultant will be zero.

51. Sum of all small forces should be greater than largest force then resultant can be zero.

52. Resultant of 2N & 2N

$= (2)(2)\cos\left(\frac{120°}{2}\right) = 2N$

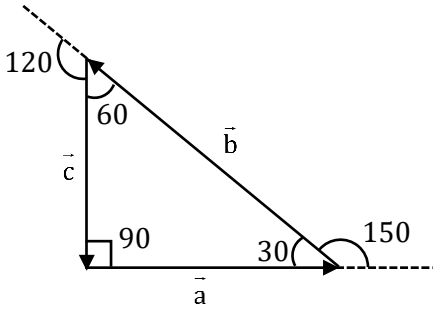


Resultant of $\sqrt{3}N$ & $\sqrt{3}N$

$= (2)(\sqrt{3})\cos\left(\frac{60°}{2}\right) = 3N$

Net resultant = 5 + 3 + 2 = 10 N

53.

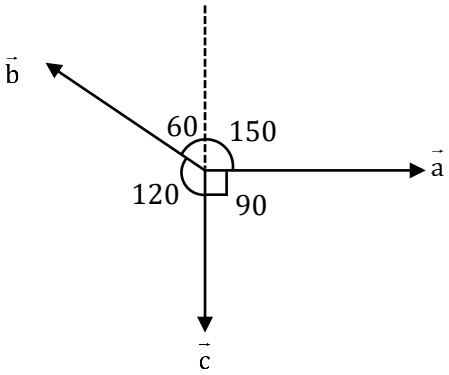


According to this

$\boxed{b^2 = a^2 + c^2}$

a : b : c = $\sqrt{3}$: 2 : 1

OR



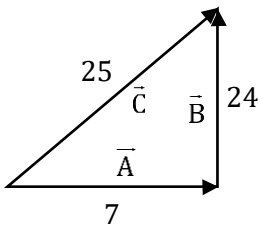
bcos60° = c

$\frac{b}{c} = \frac{2}{1}$

bsin60° = a

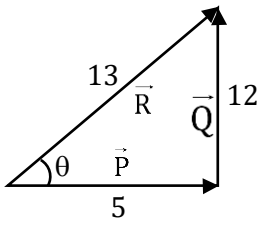
$\frac{b}{a} = \frac{2}{\sqrt{3}}$ so, ratio of a, b & c is $\sqrt{3}$: 2 : 1

54. $\vec{A} + \vec{B} = \vec{C}$ & $A^2 + B^2 = C^2$



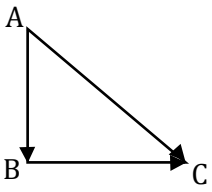
Angle between $\vec{A}$ & $\vec{B}$ is 90°

55. $\vec{P} + \vec{Q} = \vec{R}$ and $P^2 + Q^2 = R^2$



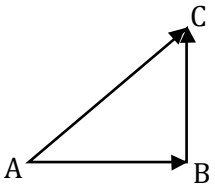
$\cos\theta = \frac{5}{13} \Rightarrow \theta = \cos^{-1}\left(\frac{5}{13}\right)$

56.



$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = (\hat{i} + \hat{j} + \hat{k}) + 3\hat{i}$

$= 4\hat{i} + \hat{j} + \hat{k}$



$= \sqrt{16+1+1} = \sqrt{18}$ unit

$\overrightarrow{BC} = \overrightarrow{AC} - \overrightarrow{AB} = 3\hat{i} - (\hat{i} + \hat{j} + \hat{k})$

$= 2\hat{i} + \hat{j} + \hat{k}$

$|\overrightarrow{BC}| = \sqrt{4+1+1} = \sqrt{6}$ unit

60. The direction of a vector remains unchanged on multiplying the vector by a positive scalar.

64. $\vec{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{\hat{i} - 2\hat{j}}{\sqrt{5}}$

65. $\vec{A} + \vec{B} = \hat{j}$

$\Rightarrow \vec{B} = \hat{j} - \vec{A} = \hat{j} - (\hat{i} - \hat{j} + \hat{k}) = -\hat{i} + 2\hat{j} - \hat{k}$

66. $\hat{n} = 0.3\hat{i} - 0.4\hat{j} + c\hat{k}$

$\sqrt{(0.3)^2 + (-0.4)^2 + c^2} = 1$

$\Rightarrow$ $c^2 = 1 - 0.09 - 0.16$

$\Rightarrow$ $c^2 = 0.75$

$\Rightarrow$ $c = \sqrt{0.75}$

67. $\hat{e}_r = |\hat{e}_r|\cos\theta\hat{i} + |\hat{e}_r|\sin\theta\hat{j}$

$\because |\hat{e}_r| = 1$

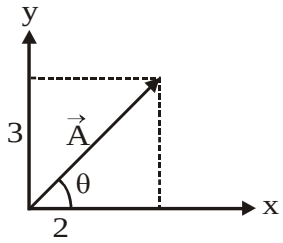
$\Rightarrow \hat{e}_r = (\cos\theta)\hat{i} + (\sin\theta)\hat{j}$

68. $\vec{R} = 7\hat{i} + 24\hat{j} + 25\hat{k}$

$|\vec{R}| = \sqrt{7^2 + 24^2 + 25^2} = \sqrt{49 + 576 + 625}$

$= 25\sqrt{2}\,N$

69.



$\vec{A} = 2\hat{i} + 3\hat{j}$

$\tan\theta = \frac{3}{2}$

$\theta = \tan^{-1}\left(\frac{3}{2}\right)$

70. Let $\vec{A}$ has to the added

$(\hat{i} - 2\hat{j} + 2\hat{k}) + (2\hat{i} + \hat{j} - \hat{k}) + (\vec{A}) = \hat{j}$

$\vec{A} = -3\hat{i} + 2\hat{j} - \hat{k}$

71. $\vec{R} = \vec{A} + \vec{B} = 4\hat{i} + 3\hat{j} + 6\hat{k} + 2\hat{i} - 3\hat{j} + 2\hat{k}$

$\vec{R} = 6\hat{i} + 8\hat{k}$

$\vec{R} = \frac{6\hat{i} + 8\hat{k}}{\sqrt{6^2 + 8^2}} = \frac{6\hat{i} + 8\hat{k}}{10}$

72. $\vec{c} = 3|\vec{b}|\hat{a}$

$|\vec{b}| = \sqrt{3}; \hat{a} = \frac{2\hat{i} + 2\hat{j} - \hat{k}}{3}$

so $\vec{c} = (3)(\sqrt{3})\left(\frac{2\hat{i} + 2\hat{j} - \hat{k}}{3}\right) = 2\sqrt{3}\hat{i} + 2\sqrt{3}\hat{j} - \sqrt{3}\hat{k}$

73. Unit vector $\vec{v} = \frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}$

Force in the direction of velocity

$\vec{F} = |\vec{F}|\hat{v}$

$\vec{F} = (10\sqrt{3})\left(\frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}\right) = (10\hat{i} + 10\hat{j} - 10\hat{k})N$

74. $\vec{A} = \sqrt{2}\hat{i} + \sqrt{2}\hat{j} + \hat{k}$

$\cos\alpha = \frac{A_x}{A},\ \cos\beta = \frac{A_y}{A},\ \cos\gamma = \frac{A_z}{A}$

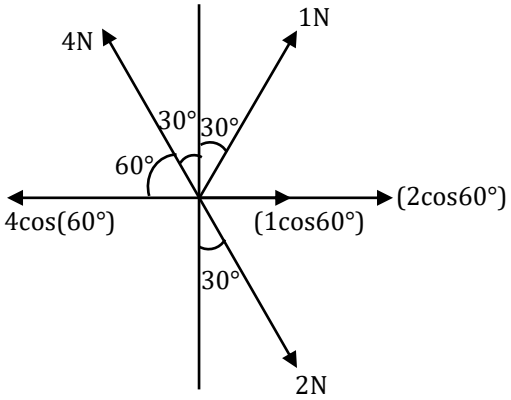
$\cos\alpha = \frac{\sqrt{2}}{\sqrt{5}},\ \cos\beta = \frac{\sqrt{2}}{\sqrt{5}},\ \cos\gamma = \frac{1}{\sqrt{5}}$

75. $\vec{a} - \vec{b} = \hat{j} - 2\hat{k}$

$|\vec{a} - \vec{b}| = \sqrt{1^2 + (-2)^2} = \sqrt{5}$

$\cos\alpha = 0\ ;\ \cos\beta = \frac{1}{\sqrt{5}}; \cos\gamma = -\frac{2}{\sqrt{5}}$

76. To have the resultant force along the y-direction the force along x-axis must be zero.



Let F is the require force then,

–4cos(60°) + (1)cos(60°) + 2cos(60°) + F = 0

F = 0.5N

77. If a vector rotates through any angle then its magnitude remain same.

so, $\sqrt{1^2 + b^2 + 3^2} = \sqrt{2^2 + (2b-1)^2 + 1^2}$

$\Rightarrow$ $10 + b^2 = 5 + 4b^2 + 1 - 4b$

$\Rightarrow$ $3b^2 - 4b - 4 = 0$

$\Rightarrow$ $(3b + 2)(b - 2) = 0$

$\Rightarrow$ $b = 2$ and $b = -\frac{2}{3}$

78. $\vec{P}\cdot\vec{Q} = -PQ$

$\vec{P}\vec{Q}\cos\theta = -PQ$

$\cos\theta = -1$

$\Rightarrow \theta = 180°$

79. $w=\vec{F}\cdot\vec{S}$

$w=(2\hat{i}+3\hat{j})\cdot(3\hat{i}+4\hat{j})$

w = 6 + 12 = 18J

80. $W=\vec{F}.\vec{S}=(2\hat{i}+2\hat{j}).(2\hat{i}-3\hat{j})$

W = 4 – 6 = –2J

81. $\vec{F}=14\left(\frac{3\hat{i}+2\hat{j}-6\hat{k}}{\sqrt{9+4+36}}\right)=6\hat{i}+4\hat{j}-12\hat{k}$

$\vec{S}=2\hat{i}+4\hat{j}-2\hat{k}$

$W=\vec{F}.\vec{S}=12+16+24=52J$

82. $\vec{A}=2\hat{i}+3\hat{j}+8\hat{k}$ & $\vec{B}=4\hat{i}-4\alpha\hat{j}+\alpha\hat{k}$

$\because \quad \vec{A}\perp\vec{B}\Rightarrow \vec{A}\cdot\vec{B}=0$

$\Rightarrow$ 8 – 12α + 8α = 0

$\Rightarrow$ 4α = 8

$\Rightarrow$ α = 2

83. $\vec{P}.\vec{Q}=0\Rightarrow(a\hat{i}+a\hat{j}+3k).(a\hat{i}-2\hat{j}-k)=0$

$a^2 - 2a - 3 = 0 \Rightarrow (a-3)(a+1)=0$

a = 3 and a = –1

84. $\therefore \vec{A}\perp\vec{B}$

$\vec{A}.\vec{B}=0$

85. $\because$ $\vec{A}$ is normal to $\vec{B}$ $\Rightarrow \vec{A}\cdot\vec{B}=0$

86. $\hat{n}=a\hat{i}+b\hat{j}$ is perpendicular to $\hat{i}-\hat{j}$

$\Rightarrow(a\hat{i}+b\hat{j}).(\hat{i}-\hat{j})=0$

$\Rightarrow$ a – b = 0

$\Rightarrow$ a = b

88. $(\vec{A}+\vec{B}).(\vec{A}-\vec{B})=0$

$\Rightarrow A^2-\vec{A}.\vec{B}+\vec{B}.\vec{A}-B^2=0$

$\Rightarrow$ A = B ($\because \vec{A}.\vec{B}=\vec{B}.\vec{A}$)

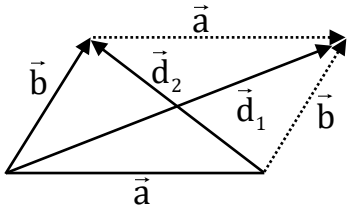
89. $(\vec{A}+\vec{B}).(\vec{A}-\vec{B})=A^2-\vec{A}.\vec{B}+\vec{B}.\vec{A}-B^2$

$= A^2 - B^2$

$= 0\ [\because |\vec{A}|=|\vec{B}|]$

Hence $\vec{A}+\vec{B}$ & $\vec{A}-\vec{B}$ are perpendicular to each other

90. According to triangle law



$\vec{a}+\vec{b}=\vec{d}_1$

$\Rightarrow \quad a^2+b^2+2ab\cos\theta=d_1^2$ ....(1)

& $\vec{a}-\vec{b}=-\vec{d}_2$

$\Rightarrow \quad a^2+b^2-2ab\cos\theta=d_2^2$ ....(2)

By (1) + (2)

$a^2+b^2=\frac{d_1^2+d_2^2}{2}$

91. $\vec{A}=3\hat{i}+4\hat{j}+5\hat{k}$, $\vec{B}=6\hat{i}+8\hat{j}+10\hat{k}$

$\vec{B}=2[3\hat{i}+4\hat{j}+5\hat{k}]$

$\vec{B}=2\vec{A}$

$\Rightarrow$ $\vec{B}$ is parallel to $\vec{A}$

92. $\vec{R}=\vec{A}+2\vec{B}+2\vec{A}-2\vec{B}$

$\vec{R}=3\vec{A}$ $\Rightarrow \vec{R}$ is parallel to $\vec{A}$

93. $\cos\theta=\frac{\vec{A}.\vec{B}}{AB}=\frac{(\hat{i}+\hat{j}).(\hat{i}+\hat{k})}{\sqrt{2}\sqrt{2}}=\frac{1}{2}\Rightarrow\theta=60°$

94. $\because \cos\theta=\frac{\vec{A}.\vec{B}}{|\vec{A}||\vec{B}|}$

$\Rightarrow \cos\theta=\frac{(2\hat{i}+\hat{j}+2\hat{k}).(\hat{i}-\hat{j}+\hat{k})}{\left(\sqrt{2^2+1^2+2^2}\right)\sqrt{1^2+(-1)^2+(1)^2}}$

$\Rightarrow \cos\theta=\frac{1}{\sqrt{3}}\Rightarrow \boxed{\theta=\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)}$

95. Projection $=\frac{(3\hat{i}+4\hat{k}).\hat{k}}{|\hat{k}|}=4$

96. Projection of $\vec{B}$ on $\vec{A}=B\cos\theta$

$=B\left[\frac{\vec{A}\cdot\vec{B}}{AB}\right]=\vec{B}\cdot\hat{A}$

100. $\vec{P}\times\vec{Q}=PQ\sin\theta\hat{n}\Rightarrow\hat{n}=\frac{\hat{P}\times\hat{Q}}{\sin\theta}=\frac{\vec{P}\times\vec{Q}}{|\vec{P}\times\vec{Q}|}$

102. $(\vec{A}+\vec{B})\times(\vec{A}-\vec{B})$

$=(\vec{A}\times\vec{A})-(\vec{A}\times\vec{B})+(\vec{B}\times\vec{A})-(\vec{B}\times\vec{B})$

$=2(\vec{B}\times\vec{A})=-2(\vec{A}\times\vec{B})$

103. $\vec{A}\times\vec{B}=-(\vec{B}\times\vec{A})$

$\therefore$ angle between these vectors is zero

104. Vertically upward = $\hat{k}$

South = $-\hat{j}$

$\vec{A}\times\vec{B}\Rightarrow\hat{k}\times(-\hat{j})=\hat{i}$ $\quad \hat{i}\Rightarrow$ east

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105. $\frac{3}{4}AB\sin\theta = AB\cos\theta$

$\frac{\sin\theta}{\cos\theta} = \frac{4}{3}$

$\frac{\sin\theta}{\cos\theta} = \frac{4}{3}$

$\theta = 53°$

106. $\vec{A}\times\vec{B} = (3\hat{i}+4\hat{j})\times(6\hat{i}+8\hat{j}) = \vec{O}$

107. $|\vec{A}+\vec{B}| = |\vec{A}-\vec{B}|$

$\sqrt{A^2+B^2+2A\cos\theta} = \sqrt{A^2+B^2-2AB\cos\theta}$

$\Rightarrow \quad 4AB\cos\theta = 0$

$\Rightarrow \quad \cos\theta = 0$

$\Rightarrow \quad \theta = 90°$

$\Rightarrow \quad \vec{A}.\vec{B} = 0$

108. Cross product of two || vector are zero.

$\therefore \vec{F}_1, \vec{F}_2$ are || vectors

109. Cross product of two parallel or anti parallel vectors are zero.

110. $\vec{A}\times\vec{B} = 0$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 2 \\ 6 & 2 & S \end{vmatrix} = 0$$

$\hat{i}(S-4)-\hat{j}(3S-12)+\hat{k}(6-6)$

$(S-4)\hat{i}-\hat{j}(3S-12) = 0\hat{i}+0\hat{j}$

$S-4=0$ or $3S-12=0$

$S=4$ or $S=4$

112. $\vec{v} = \vec{\omega}\times\vec{r}$

$$\vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

$= \hat{i}(-1-1)-\hat{j}(1-1)+\hat{k}(1+1)$

$= -2\hat{i}+2\hat{k}$

113. $\vec{a}\times\vec{b} = \begin{vmatrix} i & j & k \\ 2 & -2 & 1 \\ 1 & 2 & 2 \end{vmatrix}$

$= \hat{i}(-4-2)-\hat{j}(4-1)+\hat{k}(4+2)$

$\vec{a}\times\vec{b} = -6\hat{i}-3\hat{j}+6\hat{k}$

$\hat{n} = \pm\frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|} = \pm\left(\frac{-6\hat{i}-3\hat{j}+6\hat{k}}{3\times3}\right)$

$= \pm\left(-\frac{2}{3}\hat{i}-\frac{1}{3}\hat{j}+\frac{2}{3}\hat{k}\right)$

Exercise-II (Previous Year Questions)

1. $\vec{A}\cdot\vec{B} = 0$

$\cos\omega t\cos\frac{\omega t}{2}+\sin\omega t\sin\frac{\omega t}{2} = 0$

$\cos\left(\omega t-\frac{\omega t}{2}\right) = 0 \Rightarrow \cos\frac{\omega t}{2} = 0$

2. $|\vec{A}+\vec{B}| = |\vec{A}-\vec{B}|$

$A^2 + B^2 + 2AB\cos\theta = A^2 + B^2 - 2AB\cos\theta$

$\cos\theta = 0 \Rightarrow \theta = 90°.$

3. $\vec{F}_{net} = \vec{F}_1+\vec{F}_2+\vec{F}_3 = \vec{0} \Rightarrow \vec{a} = 0 \Rightarrow \vec{v} = \text{constant}$

4. $\vec{F} = 2\hat{i}+\hat{j}-\hat{k}$ $\qquad$ $\vec{r} = 3\hat{i}+2\hat{j}-2\hat{k}$

$\vec{F}.\vec{r} = 6+2+2 = 10$

$$\vec{F}\times\vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 3 & 2 & -2 \end{vmatrix} = \hat{i}(0)-\hat{j}(-1)+\hat{k}(1)$$

$= \hat{j}+\hat{k}$

$|\vec{F}\times\vec{r}| = \sqrt{2}$

Exercise-III (Analytical Questions)

2. $|\vec{A}| = |\vec{B}| = Z, \theta = 60°$

$|\vec{A}+\vec{B}| = 2a\cos\left(\frac{\theta}{2}\right) = \sqrt{3}Z$

$|\vec{A}-\vec{B}| = 2a\sin\left(\frac{\theta}{2}\right) = Z$

$\vec{A}\cdot\vec{B} = AB\cos\theta = \frac{1}{2}\cdot Z^2$

$|\vec{A}\times\vec{B}| = AB\sin\theta = \frac{\sqrt{3}}{2}\cdot Z^2$

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3. According to triangle law.

5. Direction of null vector is not defined and direction of $\vec{A}\times\vec{B}$ is given by right hand thumb rule which lies in $\perp$ plane of $\vec{A}, \vec{B}, (\vec{A}+\vec{B})$ and $(\vec{A}-\vec{B})$.

6. $\vec{A}\cdot\vec{B}$ in a scalar quantity that's why it is known as scalar product.

Kinematics

SOLUTIONS

Exercise-I (Conceptual Questions)

4. Displacement ≤ distance

5. Total time = 100 sec.

distance = $2\pi R + 2\pi R + \pi R = 5\pi R$

displacement = 2R

6. displacement (diagonal of cube)

$= \sqrt{(10)^2 + (10)^2 + (10)^2} = 10\sqrt{3}\,m$

7. distance = 5 × 4 + 12 × 4 = 68 cm

8. displacement

$S = \sqrt{(12)^2 + (5)^2}$

S = 13 cm

9. $\frac{s_P}{s_R} = \frac{\frac{3}{2}\pi R}{\frac{\pi R}{2}} = \frac{3}{1}$

10. Distance = πR

Displacement = 2R

$\frac{\text{Distance}}{\text{Displacement}} = \frac{11}{7}$

11. Distance = $\frac{2\pi R}{3600} \times 100$

$= \frac{2\pi \times 10}{3600 \times 100} \times 100 = \frac{\pi}{180}\,m$

12. $\vec{S} = (3-2)\hat{i} + (4-3)\hat{j} + (5-5)\hat{k}$

$\vec{S} = \hat{i} + \hat{j}$

13. $\vec{d} = \vec{d}_1 + \vec{d}_2 + \vec{d}_3$

$\vec{d} = 30\hat{j} + 20\hat{i} + 30\sqrt{2}\left(\frac{\hat{i} - \hat{j}}{\sqrt{2}}\right)$

$\vec{d} = 30\hat{j} + 20\hat{i} + 30\hat{i} - 30\hat{j}$

$\vec{d} = +50\hat{i}$

$\vec{d} = 50\,m$ towards east

14. $\vec{d}_1 = 80\hat{i}$

$\vec{d}_2 = -30\hat{j} - 40\hat{i}$

Position from starting point

$\vec{d} = \vec{d}_1 + \vec{d}_2$

$\vec{d} = 40\hat{i} - 30\hat{j}$

$|\vec{d}| = \sqrt{40^2 + 30^2} = 50\,m$

$\tan\theta = -\left(\frac{40}{30}\right) = \left(-\frac{4}{3}\right)$

θ = 53° east of south

15. 1 cycle = 5 step forward & 3 step backward

then net displacement in 1 cycle = (+5) + (−3)

= +2m

Net displacement in 4 cycle = 8 m

So, he will fall down in the pit after completing 4 cycles & 5 forward steps.

Total time = 4(5 + 3) + 5(1) = 37 sec.

16. $v_{avg} = \frac{2v_1v_2}{v_1 + v_1} = \frac{2 \times 30 \times 70}{30 + 70} = 42\,km/h$

17. Average velocity = $\frac{\text{Displacement}}{\text{Time interval}}$

A particle moving in a given direction with non-zero velocity cannot have zero speed. In general, average speed is not equal to magnitude of average velocity. However, it can be so if the motion is along a straight line without change in direction.

19. $v_{avg} = \frac{d_1 + d_2}{2t} = \frac{vt + vt}{2t}$

$v_{avg} = v$

20. Average speed = $\frac{\text{total distance}}{\text{total time}} = \frac{2d}{2+3} = \frac{2d}{5}$

21. Average velocity = $\frac{(15\hat{i} \times 2) + 5\hat{j} \times 8}{2+8} = 3\hat{i} + 4\hat{j}$

$|\vec{v}| = 5\,m/s$

22. Let the constant speed is V, then

average speed = V

average velocity = $\frac{a}{\frac{2a}{V}} = \frac{V}{2}$

$\frac{\text{average speed}}{\text{average velocity}} = \frac{V}{V/2} = \frac{2}{1}$

23. Average speed

$$= \frac{\text{Total distance}}{\text{Total time}} = \frac{\pi R}{3.14} = 10\,m/s$$

Average velocity

$$= \frac{\text{Total displacement}}{\text{Total time}} = \frac{2R}{3.14} = \frac{20}{\pi}\,m/s$$

24. Average velocity $= \frac{3\times20+4\times20+5\times20}{20+20+20}$

$$= \frac{240}{60} = 4\ m/s$$

25. $t_1 = \frac{10\times1000}{100} = 100$ sec.

$t_2 = \frac{10\times1000}{50} = 200$ sec.

Avg. speed $= \frac{\text{Total distance}}{\text{Total time}}$

$$= \frac{10000+10000}{300}$$

$= 66.66$

$\simeq 66.7$ m/s

26. Average speed

$$= \frac{3V_1V_2V_3}{V_1V_2+V_2V_3+V_1V_3}$$

$$= \frac{3\times V\times2V\times3V}{2V^2+6V^2+3V^2} = \frac{18}{11}V$$

27. $v_{avg} = \frac{v_1t+v_2t+v_3t}{3t}$

$v_{avg} = \frac{v_1+v_2+v_3}{3}$

28. Average speed $= \frac{S}{\frac{2S}{5}+\frac{3S}{5}}$

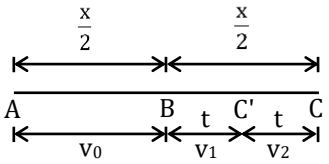
$$= \frac{5}{\frac{2}{V_1}+\frac{3}{V_2}} = \frac{5V_1V_2}{3V_1+2V_2}$$

29. Average speed in last half $= \frac{4.5+7.5}{2} = 6$ m/s

Average speed in total journey $= \frac{2\times10\times6}{10+6}$

$= 7.5$ m/s

30.



for second half $v_{avg} = \frac{v_1+v_2}{2}$

for total $v_{avg} = \frac{2(v_0)\left(\frac{v_1+v_2}{2}\right)}{v_0+\frac{v_1+v_2}{2}}$

$v_{avg} = \frac{2v_0(v_1+v_2)}{2v_0+v_1+v_2}$

31. $\Delta\vec{V} = \vec{V}_f - \vec{V}_i$

$\Delta\vec{V} = (-10\hat{j}-10\hat{i})\,m/s$

$\Delta\vec{V} = (-10\hat{i}-10\hat{j})\,m/s$

$|\Delta\vec{V}| = 10\sqrt{2}\,m/s$, in south-west direction

$|\Delta\vec{V}| = 14.14\,m/s$, in south-west direction

32. $\Delta V = 2V\sin\frac{\theta}{2}$

$\Delta V = 2V\sin\frac{40°}{2}$

$\Delta V = 2V\sin20°$

33. $\vec{V}_{avg} = \frac{4\hat{j}+3\hat{i}}{10+5}\,m/s$

$\vec{V}_{avg} = \left(\frac{4}{15}\hat{j}+\frac{3}{15}\hat{i}\right)m/s$

$V_{avg} = \frac{1}{3}\ m/s$

34. $x = at^2 - bt^3$

$v = \frac{dx}{dt} = 2at - 3bt^2 = 0 \Rightarrow t = \frac{2a}{3b}$

35. $x = a\sin t$

$v = \frac{dx}{dt} = a\cos t \Rightarrow a = \frac{dv}{dt} = -a\sin t$

36. $x = a + bt^2$, $v = \frac{dx}{dt} = 2bt$

$a = \frac{dv}{dt} = 2b = 2\times3\ cm/s^2 = 6\ cm/s^2$

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37. $s = 3t^3 + 7t^2 + 14t + 8$

$v = \frac{ds}{dt} = 9t^2 + 14t + 14$

$a = \frac{dv}{dt} = 18t + 14$

at $t = 2$ $\quad a = 50\ m/s^2$

38. $v = \frac{dx}{dt} = 2at + b$

39. $t = \sqrt{x} + 3 \Rightarrow \sqrt{x} = t - 3$

$x = (t-3)^2 = t^2 - 6t + 9$

$v = \frac{dx}{dt} = 2t - 6 \Rightarrow a = \frac{dv}{dt} = 2\ m/s^2$

40. $y = a + bt + ct^3$

$v = \frac{dy}{dt} = b + 3ct^2$

at $t = 0$ $v = b$

$a = \frac{dv}{dt} = 6ct$

at $t = 0$ $a = 0$

41. $x = u(t - 2) + a(t - 2)2$

$v = \frac{dx}{dt} = u + 2a(t - 2)$

acceleration $= \frac{dv}{dt} = 2a$

42. $x = kt$

$v = \frac{dx}{dt} = k$ (constant) $\Rightarrow a = \frac{dv}{dt} = 0$

43. $v = 20 + 0.1\,t$

$a = \frac{dv}{dt} = 0.1$

it is uniform acceleration.

44. Acceleration $= \frac{dv}{dt}$, It is the first derivative where t terms can only yield a constant value.

$v = 6 - 7t$

$a = \frac{dv}{dt} = -7$

45. Velocity $= \frac{dy}{dt}$

It is the first derivative where t terms can only yield a constant value.

46. $s = 6t^2 - t^3$

$v = 12t - 3t^2$

$a = 12 - 6t = 0 \Rightarrow t = 2$ sec.

47. $v = \frac{dx}{dt}$

$v = \frac{d}{dt}\left[\frac{k}{b} - \frac{ke^{-bt}}{b}\right] = \frac{d}{dt}\left(\frac{k}{b}\right) - \frac{d}{dt}\left[\frac{k}{b}e^{-bt}\right]$

$v = -\frac{d}{dt}\left[\frac{k}{b} \cdot e^{-bt}\right]$

$v = k(e^{-bt})$

48. $s = t^3 - 6t^2 + 3t + 4$

$v = \frac{ds}{dt} = 3t^2 - 12t + 3$

$a = \frac{dv}{dt} = 6t - 12$ If $a = 0$ Þ $t = 2$

$s = 2^3 - 6(2)^2 + 3(2) + 4 = 8 - 24 + 6 + 4 = -6m$

49. $x = at + bt^2 - ct^3$

$v = \frac{dx}{dt} = a + 2bt - 3ct^2$

acceleration $a' = 2b - 6ct$

if $a' = 0$ then $t = \frac{b}{3c}$

So at $t = \frac{b}{3c}$

$V = a + 2b \times \frac{b}{3c} - 3c \times \frac{b^2}{9c^2}$

$V = a + \frac{2b^2}{3c} - \frac{b^2}{3c}$

$V = a + \frac{b^2}{3c}$

50. $v_x = \frac{dx}{dt} = t$

At $t = 2$ sec. $\quad v_x = 2\ m/s$

$v_y = \frac{dy}{dt} = \frac{d}{dt}\left(\frac{t^4}{8}\right) = \frac{t^3}{2}$

At $t = 2$ sec. $\quad v_y = 4\ m/s$

$\boxed{\vec{v} = (2\hat{i} + 4\hat{j})\ m/s}$

51. $u = kt \Rightarrow \frac{ds}{dt} = kt \ (\because k = 2\ m/s^2)$

$\Rightarrow \int_0^s ds = 2\int_0^4 tdt \quad \Rightarrow s = 2\left|\frac{t^2}{2}\right|_0^4 \Rightarrow s = 16m$

52. $f = at^2 = \frac{dv}{dt}$

$\Rightarrow \int_u^v dv = \int_0^t at^2\ dt \Rightarrow v - u = \frac{at^3}{3} \Rightarrow v = u + \frac{at^3}{3}$

53. Acceleration $\vec{a} = \frac{\Delta\vec{v}}{\Delta t} = \frac{10\hat{j} - 10\hat{i}}{20}\ m/s^2$

$= \frac{-\hat{i} + \hat{j}}{2}$ (west-north)

$|\vec{a}| = \sqrt{\frac{1}{4} + \frac{1}{4}} = \frac{1}{\sqrt{2}}\ m/s^2$ (north-west)

54. $a = 2t - 2$

$\int_0^v dv = \int_0^5 (2t-2)dt$

$V = \left[t^2\right]_0^5 - 2\left[t\right]_0^5$

$V = 25 - 10 = 15\ m/s$

55. $V = (10 + 2t^2)\ m/s$

$a_{avg} = \frac{V_2 - V_1}{t_2 - t_1}$

at $t = 2$ sec, $v_1 = 18$ m/s

at $t = 5$ sec, $v_2 = 60$ m/s

$a_{avg} = \frac{60-18}{5-2} = 14\ m/s^2$

56. $V = (3t^2 + 2)\ m/s$

$S = (t^3 + 2t)\ m$

at $t = 0 \Rightarrow S_1 = 0$

at $t = 2 \Rightarrow S_2 = 12$ m

$v_{avg} = \frac{S_2 - S_1}{t_2 - t_1} = \frac{12-0}{2-0} = 6\ m/s$

57. $v = \alpha\sqrt{x} \Rightarrow \frac{dx}{dt} = \alpha\sqrt{x} \Rightarrow \frac{dx}{\sqrt{x}} = \alpha dt$

Integrating, $\int_{x=0}^{x} x^{-1/2}dx = \int_{t=0}^{t} \alpha\, dt$

$2\sqrt{x} = \alpha t \Rightarrow \sqrt{x} = \alpha t/2$

Put this value of $\sqrt{x}$ in the original given eqn.

$v = \alpha\sqrt{x} = \alpha(\alpha.t/2) = \alpha^2 t/2$

$\therefore v \propto t$

58. $v = (180 - 16x)^{1/2}$

$\because\ a = \frac{(v)dv}{dx}$

$a = (180-16x)^{1/2} \cdot \frac{d}{dx}(180-16x)^{1/2}$

$a = \frac{1}{2}\left[\frac{(180-16x)^{1/2}}{(180-16x)^{1/2}}\right][0-16]$

$= -8\ m/s^2$

59. $a = 2s + 1$

$\frac{(v)dv}{ds} = 2s + 1$

$v(dv) = (2s + 1)ds$

$\int_0^v vdv = \int_0^s (2s+1)ds$

$\Rightarrow \frac{v^2}{2} = \left(\frac{2s^2}{2}\right) + s$

$\Rightarrow v = \sqrt{2s(s+1)}$

60. $\vec{r} = (t^2 - 4t + 6)\hat{i} + (t^2)\hat{j}$

$\vec{v} = \frac{d\vec{r}}{dt} = (2t-4)\hat{i} + (2t)\hat{j}$

$\vec{a} = \frac{d\vec{v}}{dt} = 2\hat{i} + 2\hat{j}$

For $\vec{v} \perp \vec{a}$

$\vec{v} \cdot \vec{a} = 0$

$4t - 8 + 4t = 0$

$t = 1$ sec.

61. Rate of change of speed = component of acceleration along velocity

$= \left(\frac{\vec{a}\cdot\vec{v}}{|\vec{v}|}\right)(\hat{v})$

$= \left(\frac{6+8}{10}\right)\left(\frac{6\hat{i}+8\hat{j}}{10}\right)$

$= \frac{7}{25}(3\hat{i} + 4\hat{j})$

62. $s = ut + \frac{1}{2}at^2$

$u = 0$

$s = \frac{1}{2}at^2$

$t^2 = \frac{2s}{a}$

$t \propto \sqrt{s}$

63. As force is opposite to velocity so body will be retarded and slow down.

64. $v = 144 \times \frac{5}{18} = 40$ m/s

$a = \frac{40-0}{40} = 1 m/s^2$

$s = \frac{1}{2}at^2 = \frac{1}{2} \times 1 \times 40 \times 40 = 800$ m

65. $s = \left(\frac{u+v}{2}\right) \times t = \left(\frac{0+15}{2}\right) \times 10 = 75$ m

66. $u = 72$ km/h $= 20$ m/s;

$v = 0$; $S = 100$ m

$a = \frac{v^2 - u^2}{2s} = -2$ m/s^2

67. $S = 0 + \frac{5}{2}(2 \times 5 - 1) = \frac{45}{2} = 22.5$m

68. $S_{n^{th}} = U + \frac{a}{2}(2n-1)$

$19 = 0 + \frac{a}{2}(2 \times 10 - 1)$

$a = 2$ m/s^2

69. $\frac{S_{n^{th}}}{S_n} = \frac{0 + \frac{a}{2}(2n-1)}{0 + \frac{1}{2}an^2}$

$\frac{S_{n^{th}}}{S_n} = \frac{2}{n} - \frac{1}{n^2}$

70. $\frac{S_{4^{th}}}{S_{5^{rd}}} = \frac{2 \times 4 - 1}{2 \times 5 - 1} = \frac{7}{9}$

71. $S \propto u^2 \Rightarrow \frac{S_1}{S_2} = \frac{u_1^2}{u_2^2}$

$\Rightarrow \frac{20}{s_2} = \frac{10^2}{40^2} = s_2 = 320$m

72. $s = \frac{u^2}{2a}$ so $s \propto u^2$

So if speed is tripled, stopping distance will be nine times i.e. 9×2 m $= 18$ m

73. $u + 2a = 30$

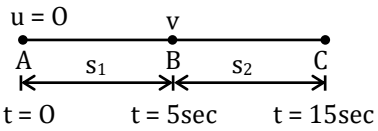
$u + 4a = 50$

$\Rightarrow a = 10$ m/s^2, $u = 10$ m/s

74. $s_1 = \frac{1}{2} \times 5 \times (5)^2 = \frac{125}{2}$m

$v = 0 + 5(5) = 25$ m/s

$s_2 = 25 \times 10 = 250$ m



Total distance travelled = $s_1 + s_2 = 312.5$ m

75. $a = \frac{20-10}{5} = 2 m/s^2$

for 2 seconds before t_0

$10 = v + 2(2)$

$v = 10 - (2)(2) = 6$ m/s

76. Retardation remains same in both situations

$\frac{0-(200)^2}{2 \times 4} = \frac{0-V^2}{2 \times 9}$

$V = 300$ cm/s

78. Time of free fall does not depend on mass.

79. $v^2 = 2as$

$v = \sqrt{2gh}$ [does not depend upon mass]

80. $h = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times 5^2 = 125$ m

81. Let total time of fall = n, $u = 0$

$S_{last} = S_3$

$\frac{g}{2}(2n-1) = \frac{1}{2}g(3)^2$

$2n - 1 = 9 \Rightarrow t = 5$ sec

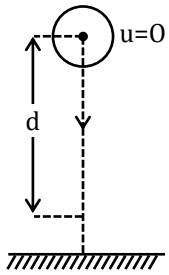
82. $x = \frac{1}{2}g(1)^2 = \frac{1}{2}g$

$5x = \frac{g}{2}(2n-1)$

$5 \times \frac{g}{2} = \frac{g}{2}(2n-1)$

$n = 3$ sec.

83.



$v^2 = u^2 + 2as$

$v^2 = 2g(2d)$

$v = 2\sqrt{gd}$

84. use $v^2 = u^2 - 2as$

$\Rightarrow 0 = u^2 - 2as$

$\Rightarrow a = \frac{u^2}{2s}$

85. Distance covered in 1st sec is $x = \frac{1}{2} g (1)^2 = \frac{g}{2}$

Let it falls for n sec then last sec is n^{th} sec.

$\therefore \quad 9x = 0 + \frac{g}{2}(2n - 1)$

$\Rightarrow \quad 9 \times \frac{g}{2} = \frac{g}{2}(2n - 1)$

$\Rightarrow \quad 2n = 10$

$\Rightarrow \quad n = 5$ sec

86. Time taken by stone to go down $t_1 = \sqrt{\frac{2h}{g}}$

Time taken by sound to come up $t_2 = \frac{h}{v}$

$\therefore \quad T = t_1 + t_2 = \sqrt{\frac{2h}{g}} + \frac{h}{v}$

87. $H = \frac{1}{2}gt^2$

$h = \frac{1}{2} g \left(\frac{t}{3}\right)^2 = \frac{H}{9}$ from releasing point

so distance from the ground

$= H - H/9 = 8H/9$

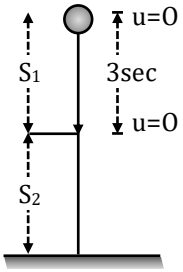
88. $h = 0 + \frac{1}{2}gt_1^2$

$\Rightarrow \quad t_1 = \sqrt{\frac{2h}{g}}$

$\frac{h}{4} = 0 + \frac{1}{2}gt_2^2 \Rightarrow t_2^2 = \frac{2}{g} \times \frac{h}{4}$

$\Rightarrow \quad t_2 = \sqrt{\frac{2}{g} \times \frac{h}{4}} = \frac{1}{2}\sqrt{\frac{2h}{g}} = \frac{t_1}{2}$

89.



Total height to fall

$h = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times (5)^2 = 125\,m$

In first 3 sec.

$s_1 = \frac{1}{2}gt_1^2 = \frac{1}{2} \times 10 \times (3)^2 = 45$ m

Now remaining height to fall

$\therefore \quad s_2 = h - s_1 = 125 - 45 = 80$ m

$t_2 = \sqrt{\frac{2s_2}{g}} = \sqrt{\frac{2 \times 80}{10}} = 4$ sec

$t = t_1 + t_2 = 3 + 4 = 7$ sec

90. Separation = $S_1 - S_2$

$= \frac{1}{2} g(4)^2 - \frac{1}{2} g(2)^2$

$= \frac{12}{2} g = 58.8$ m

91. $h_1 - h_2 = \frac{1}{2}g(t_1^2 - t_2^2)$

$= \frac{1}{2} \times 9.8 [(5)^2 - (3)^2]$

$= 8 \times 9.8 = 78.4$ m

92. In first 2 sec the distance travelled is

$h_1 = 0 + \frac{1}{2}(10)(2)^2 = 20$ m

After stopped & released the distance travelled in next 2 sec $h_2 = 0 + \frac{1}{2} 10 (2)^2 = 20$ m

So height from ground = (H – 40)m

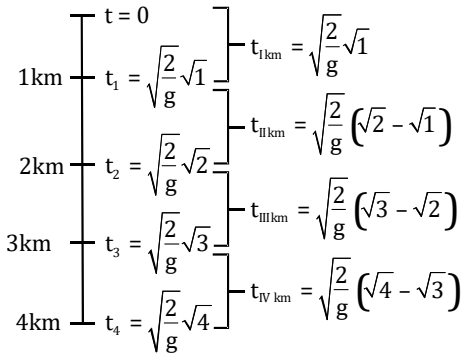
93. $S_{n^{th}} = u + \frac{a}{2}(2n-1)$

$u = 0$ and $a = g$ (const.)

$S_{n^{th}} \propto (2n - 1)$ (here n = 1,2,3.....)

$D_1 : D_2 : D_3 : ..... = 1 : 3 : 5 : .....$

94. $\because\ h = \frac{1}{2}gt^2$ $\quad\therefore\ t = \sqrt{\frac{2h}{g}}$



$\therefore\ t_{I km}\ ;\ t_{II km}\ ;\ t_{III km}\ ;....$

$= \sqrt{1} : (\sqrt{2}-\sqrt{1}) : (\sqrt{3}-\sqrt{2}) : ...$

95. By galileo's law of odd number

$x + 3x + 5x = 27m$

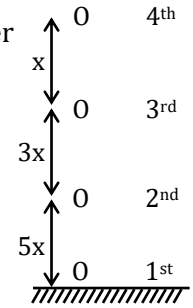
$x = 3$ m

Position of 2nd drop from top

$x + 3x = 4x = 12m$

Position of 3rd drop from top

$x = 3$ m



96. By galileo's law of odd number

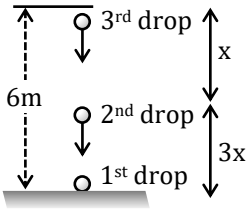
$x + 3x = 6m$

$4x = 6$

$x = \frac{6}{4} = 1.5m$

Height of 2nd drop from ground

$= 3x = 3 \times \frac{6}{4} = 4.5m$



97. $S_1 = \frac{1}{2}g(4)^2$

$S_1 = 80$ m

$S_2 = \frac{1}{2}g(3)^2 = 45$ m

$S_3 = \frac{1}{2}g(2)^2 = 20$ m

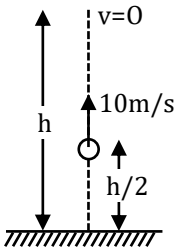
$S_4 = \frac{1}{2}g(1)^2 = 5$ m

$S_1 - S_2 = 80 - 45 = 35$ m

$S_2 - S_3 = 45 - 20 = 25$ m

$S_3 - S_4 = 20 - 5 = 15$ m

98.



$v^2 = u^2 - 2g\frac{h}{2}$

$0 = (10)^2 - 10h$

$h = 10$ m

99. $T = \frac{2u}{g} = \frac{2\times50}{10} = 10$ sec

100. $\because$ Height $H = \frac{u^2}{2g}$

$H \propto u^2$

$\frac{H_1}{H_2} = \frac{(2u)^2}{(3u)^2} = \frac{4}{9}$

$\because$ Time $T = \frac{2u}{g}$

$T \propto u$

$\frac{T_1}{T_2} = \left(\frac{2u}{3u}\right) = \frac{2}{3}$

101. $H_{max.} = \frac{u^2}{2g} = \frac{40\times40}{2\times10}$

$H_{max.} = 80$ m

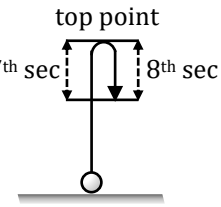
102. $v = u + at$,

$v = 40 - 10(3) = 10$ m/s.

103. $\frac{h_2}{h_1} = \frac{\frac{u_2^2}{2g}}{\frac{u_1^2}{2g}} \Rightarrow \frac{2h_1}{h_1} = \frac{u_2^2}{v_0^2}$

$u_2^2 = 2v_0^2 \Rightarrow u_2 = \sqrt{2}v_0$

104. It is only possible when 7th sec is the last second for its ascending motion and 8th sec is first second of its descending motion, which is clear from figure.



$\therefore \quad 0 = u - gt$

so $u = gt = 9.8 \times 7 = 68.6$ m/sec

105. Distance covered in last t second of ascent = distance covered in first t second of descent

$$= 0 + \frac{1}{2}gt^2 = \frac{1}{2}gt^2$$

106. Distance travelled in 1st second during upward motion = distance travelled in last second during downward motion.

$$S_{1^{st}} = \frac{g}{2}(2\times 6 - 1) = \frac{11g}{2}$$

$$S_{7^{th}} = \frac{1}{2}g(1)^2 = \frac{g}{2}$$

$$\frac{S_{1^{st}}}{S_{7^{th}}} = \frac{11}{1}$$

107. $S = \frac{1}{2}gt^2$

$$S = \frac{1}{2}g(2)^2 = 20\,m$$

108. $S_1 = S_2$

$$98(t) - \frac{1}{2}(9.8)t^2 = 98(t-4) - \frac{1}{2}(9.8)(t-4)^2$$

$$10t - \frac{t^2}{2} = 10(t-4) - \frac{1}{2}(t-4)^2$$

$$\Rightarrow \quad 4t - 48 = 0$$

$$\Rightarrow \quad t = \frac{48}{4} = 12s$$

109. For distance

'S' time should be $\Delta t/2$

$$S = \frac{1}{2}g\left(\frac{\Delta t}{2}\right)^2$$

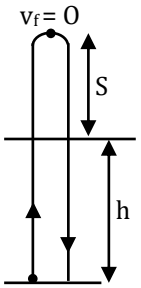
For total height

$$(h + S) = \frac{u^2}{2g}$$

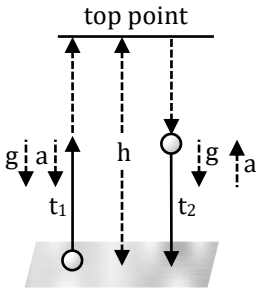
$$u = \sqrt{2g(h+S)}$$

$$= \sqrt{2g\left(h + \frac{1}{2}g\left(\frac{\Delta t}{2}\right)^2\right)}$$

$$= \sqrt{2gh + \left(\frac{g\Delta t}{2}\right)^2}$$



110.



Let retardation due to air resistance = a

then for upward journey

net retardation = (g + a)

$$\therefore \quad t_{asc} = \sqrt{\frac{2h}{g+a}}$$

and for downward journey

net acceleration = (g – a)

$$\therefore \quad t_{desc} = \sqrt{\frac{2h}{g-a}}$$

$\therefore \quad t_{asc} < t_{desc}$

111. There is loss of kinetic energy during the motion so it is clear that $v_1 > v_2$

112. $$\frac{t_{ascent}}{t_{descent}} = \sqrt{\frac{g-a}{g+a}} = \sqrt{\frac{10-2}{10+2}} = \sqrt{\frac{2}{3}}$$

113. $\because v^2 = u^2 + 2g(H)$

$v^2 = (20)^2 + 2(9.8)(200)$

$v = \sqrt{4320}$

$\boxed{v \simeq 65\ m/s}$

114. h = 81 m

$$h = -ut + \frac{1}{2}gt^2$$

$$81 = -12(t) + \frac{1}{2}(10)t^2$$

$5t^2 - 12t - 81 = 0$

$t = 5.4$ sec

115. $S = ut + \frac{1}{2}at^2$

$-h = 4.9 \times 3 - \frac{1}{2} \times 9.8 \times (3)^2$

$h = 29.4$ m

116. $V^2 = u^2 + 2as$

$(-3u)^2 = u^2 + 2(-g)(-h)$

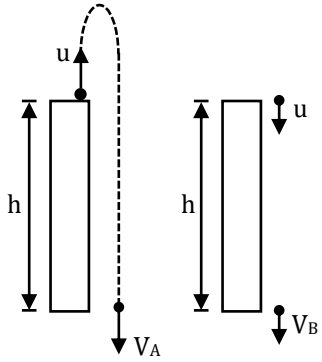
$h = \frac{4u^2}{g}$

117. Because acceleration remains same in both cases, then :

$\frac{(3)^2-(0)^2}{2h} = \frac{V^2-(-4)^2}{2h}$

$V = 5$ m/s

118.



Taking upward direction as negative and downward direction positive.

For A : $V_A^2 = (-u)^2 + 2gh$

$V_A = \sqrt{u^2+2gh}$

For B : $V_B^2 = u^2 + 2gh$

$V_B = \sqrt{u^2+2gh}$

119. Initial velocity of stone u = 10 m/sec downwards

$\therefore$ displacement $h = ut + \frac{1}{2}gt^2$

$= 10 \times 10 + \frac{1}{2} \times 10 \times (10)^2$

$= 100 + 500 = 600$ m

120. Speed after fall of 50 m

$V = \sqrt{2g(50)} = \sqrt{980}$

After parachute opens

$(3)^2 = \left(\sqrt{980}\right)^2 - 2(2)(S)$

$S = \frac{971}{4} = 242.75$ m

So total height

$242.75 + 50 \simeq 293$ m

121. Velocity at the time when fuel is finished :-

$V = U + at = 0 + 20 \times 30$

$V = 600$ m/s

Now, the time taken by the rocket to reach at max. height after this instant :-

$0 = 600 - 10 \times t$

$t = 60$ sec.

The total time taken by the rocket to reach at max. height = 30 + 60 = 90 sec.

124. Slope of s-t curve is equal to velocity.

When slope of s-t curve is zero, velocity must be zero. So it is at point B.

125. Slope of displacement-time graph gives instantaneous velocity.

126. Avg. speed $= \frac{\text{Total distance}}{\text{Total time}}$

$= \frac{40+40+40+40}{20} = 8$ m/s

127. Speed at t = 4 sec. is zero and it is the minimum value.

128. $\frac{V_A}{V_B} = \frac{\tan 30^\circ}{\tan 60^\circ} = \frac{\frac{1}{\sqrt{3}}}{\sqrt{3}} = \frac{1}{3}$

130. Particle cannot have different displacements at same time

131. Uniformly accelerated motion $\Rightarrow$ constant acceleration

132. Avg. speed $= \frac{\text{Total distance}}{\text{Total time}}$

$= \frac{(60-10)+(40-10)}{10} = 8$ m/s

133. distance

$= \frac{1}{2}(40 + 20) \times 2 + (20 \times 1) + \frac{1}{2}(20 \times 1) + \frac{1}{2}(40 \times 1)$

$= 110$ m

displacement

$= \frac{1}{2}(40 + 20) \times 2 + (20 \times 1) + \frac{1}{2}(20 \times 1)$

$- \frac{1}{2}(40 \times 1)$

= 70 m

134. Let particle is moving along x-axis then, distance travelled in first 4 sec. $= \frac{1}{2}(4 + 2) \times 4 = 12$ m

distance travelled in next 4 sec. (along –x axis)

$= \frac{1}{2}(4 + 2) \times 2 = 6$ m

distance of particle from origin = 12 – 6 = 6 m

135. $\frac{\text{distance in last 4 sec.}}{\text{distance in 9 sec.}} = \frac{\frac{1}{2}\times 4\times 10}{\frac{1}{2}(8+2)\times 10} = \frac{2}{5}$

136. displacement = 2 × 4 – 2 × 2 + 2 × 2 = 8 m

distance = 2 × 4 + 2 × 2 + 2 × 2 = 16 m

137. Distance travelled in 3 sec

= area under the curve

= area of ΔAOH + area of rectangle ABGH

+ area of ΔBCC' + area of rectangle C' CFG

$= \frac{20\times 1}{2} + 20\times 1 + \frac{10\times 1}{2} + 10\times 1 = 45\text{m}$

138. Distance = area under v-t curve

= 20 × 1 + (60 – 40) × 1 = 40 m

139. Acceleration will be maximum when slope will be maximum

$a_{max} = \frac{60-20}{40-30} = \frac{40}{10} = 4\ \text{m/s}^2$

140. $a_{avg} = \frac{V_2 - V_1}{t_2 - t_1} = \frac{0-10}{4-2}$

$a_{avg} = -5\ \text{m/s}^2$

141. Area under a-t graph gives the change in velocity during given time interval.

$\therefore \Delta V = \frac{1}{2} \times 5 \times 8 = 20$ m/s

Since initial velocity = 0

$\therefore$ Maximum speed of the particle = 20 m/s

142. Area under the acceleration-time graph = change in velocity

$\frac{1}{2}(4 + 2) \times 10 = v - 0$

v = 30 m/s

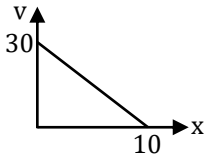
143. because the displacement is zero, hence the average velocity is also zero.

144. from 0 – 2s a = +ve $\Rightarrow$ slope of v-t curve =+ve

from 2 – 4s a = – ve $\Rightarrow$ Slope of v-t curve =–ve

Hence correct option (1)

146. v = –3x + 30



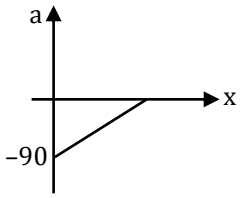
$\frac{dv}{dx} = -3$

$\because a = \frac{vdv}{dx} = (-3x+30)(-3)$

a = 9x – 90 {equation of straight line}

when x = 0 ; a = –90

& when a = 0 ; x = 10



148. $v^2 = 2as$

hence graph will be parabola

so correct option (2)

151. The slope of s-t curve gives velocity.

152. Area of (v–t) curve = displacement (height)

$= \frac{1}{2}\times 120\times 1000 = 60{,}000\text{m} = 60\text{km}$

153. Height before retardation

$= \frac{1}{2}\times(20)\times 1000 = 10{,}000 = 10$ km

154. Mean velocity = $\frac{\text{Displacement}}{\text{Total time}} = \frac{\text{Total height}}{\text{Total time}}$

$= \frac{60{,}000\text{m}}{120\text{sec}} = 500\ \text{m/s}$

155. Retardation = $\frac{1000}{(120-20)} = \frac{1000}{100} = 10\ \text{m/s}^2$

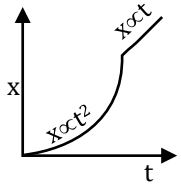
156. Acceleration = slope of OA line

$\tan\theta = \frac{1000}{20} = 50\ m/s^2$

158. Acceleration is always downward i.e. positive so slope of v – t curve will always be positive
Hence correct option (3)

159. Direction of motion is changed so velocity will be first positive & then negative and $v^2 \propto h$

161. a = constant, $\Rightarrow x \propto t^2$



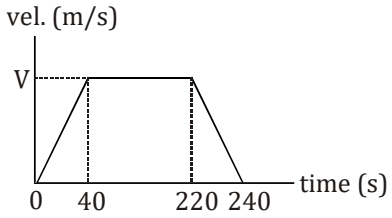
v = constant, $\Rightarrow x \propto t$

163. $v^2 = u^2 + 2as$

$a = \frac{v^2-u^2}{2s} = \frac{(900)-(4600)}{2\times0.6} = -\frac{3700}{1.2}$

$= -3084\ km/hr^2$

164.



$2250 = \frac{1}{2}(240+180)\times V$

V = 10.71 m/s

165. $v_{max} = \frac{abt}{a+b} = \frac{2\times4\times3}{6} = 4\ m/s$

166. Horizontal component of velocity remain same.

168. If air resistance is zero then for horizontal motion
$F_x = 0$; $a_x = 0$
So u_x = constant

169. At highest point velocity is in horizontal direction only & acceleration (g) is vertically downwards.

170. If air resistance is not considered but acceleration 'g' is considered, which is constant, $|\vec{a}| = \left|\frac{d\vec{v}}{dt}\right| = g$

171. For projectile motion acceleration is constant

$\frac{d\vec{v}}{dt}$ = constant & $\frac{d^2\vec{v}}{dt^2} = 0$

172. $t = \frac{U\sin\theta}{g} = \frac{1000\sin30^\circ}{10}$

t = 50 sec.

173. $\because \frac{2U\sin\theta}{g} = 10$

$\therefore U\sin\theta = 50$

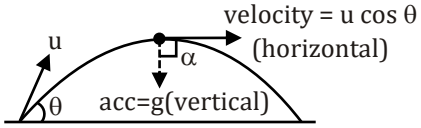
$\because H = \frac{U^2\sin^2\theta}{2g} = \frac{(50)^2}{2\times10}$

$\therefore$ H = 125 m

174. $T = \frac{2u\sin\theta}{g} = \frac{2\times19.6\sin30^\circ}{9.8} = 2\ sec$

175. $H = \frac{u_y^2}{2g} = \frac{u^2\cos^2\theta}{2g}$ (θ from vertical)

176. $\therefore \quad \angle\alpha = 90^\circ$



177. $\because$ Range R = (u_x) (T)
Here $(v_A)_x = (v_B)_x$
but $(T)_A > (T_B)$
So $R_A > R_B$ is true

178. $\frac{2U\sin\theta}{g} = 4$

$U\sin\theta = 2g$

$H = \frac{U^2\sin^2\theta}{2g} = 2g = 19.6\ m$

Height of the ball from ground = 19.6 + 1.5
= 21.1 m

179. After time 't'

velocity $\vec{v} = (v_x)\hat{i} + (v_y)\hat{j}$

$\vec{v} = (v\cos\theta)\hat{i} + (v\sin\theta - gt)\hat{j}$

$|\vec{v}| = \sqrt{(v\cos\theta)^2 + (v\sin\theta - gt)^2}$

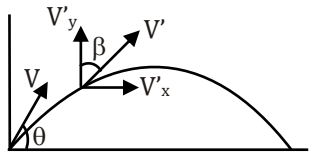
$= \sqrt{v^2 + g^2t^2 - (2v\sin\theta)gt}$

180. $\Delta v_x = 0$ (as $\vec{v}_x$ = const)

$\Delta\vec{v}_y$ = – u sin θ – (u sin θ) = –2 u sin θ

magnitude of change in velocity $\Delta v_y = 2u\sin\theta$

182. $\tan\beta = \frac{v'_x}{v'_y} = \frac{v\cos\theta}{v\sin\theta - gt}$



183. For vertical motion

$v_y^2 = u_y^2 - 2g(h)$

$(2)^2 = u_y^2 - 2(10)(0.4)$

$u_y = 2\sqrt{3}$

$\because \vec{u} = (u_x)\hat{i} + (u_y)\hat{j}$

$\therefore \vec{u} = 6\hat{i} + 2\sqrt{3}\hat{j}$

$\tan\theta = \frac{2\sqrt{3}}{6} = \frac{1}{\sqrt{3}} \Rightarrow \boxed{\theta = 30^\circ}$

184. $\theta = 60^\circ$; $R = \frac{u^2\sin(2\times 60^\circ)}{g} = \frac{\sqrt{3}u^2}{2g}$

185. $\vec{v} = a\hat{i} + (b - ct)\hat{j}$

$v_x = a$; $v_y = (b - ct)$

$\because$ At maximum height

$v_y = 0$

$b - ct = 0$

$t = \frac{b}{c}$

So time of Flight $T = \frac{2b}{c}$

Range $R = (v_x)(T)$

$\boxed{R = \frac{2ab}{c}}$

186. $R = \frac{2u_x u_y}{g} = \frac{2\times 9.8\times 4.9}{9.8} = 9.8\text{m}$

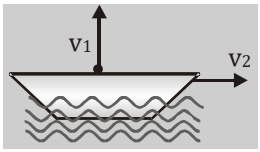
187. $\vec{v} = 5\hat{i} + 12\hat{j}$

$\Rightarrow$ $u\cos\theta = 5$ and $u\sin\theta = 12$

$R = \frac{2(u\sin\theta)(u\cos\theta)}{g} = \frac{2\times 5\times 12}{10} = 12\text{m}$

188. For vertical motion, time of flight $t = \frac{2v_1}{g}$

for horizontal motion $R = v_2 \times t = \frac{2v_1v_2}{g}$



189. $0.5 = \frac{u^2\sin(2\times 60^\circ)}{g} \Rightarrow 0.5 = \frac{u^2}{g}\times\frac{\sqrt{3}}{2}$

$\frac{u^2}{g} = \frac{1}{\sqrt{3}} \Rightarrow R' = \frac{u^2\sin(2\times 45^\circ)}{g} = \frac{u^2}{g} = \frac{1}{\sqrt{3}}$ km

190. $u\cos\theta = \frac{1}{2}u \quad \Rightarrow \quad \theta = 60^\circ$

$\Rightarrow R = \frac{u^2\sin(2\times 60^\circ)}{g} = \frac{u^2\sin 120^\circ}{g} = \frac{\sqrt{3}u^2}{2g}$

191. $R = \frac{u^2\sin(2\theta)}{g}$

$R = \frac{500\times 500\times\sin(30^\circ)}{10}$

$R = 12500$ m

192. Range of projectile $= \frac{u^2\sin 2\theta}{g}$

$\because$ Range $\propto u^2$

$\therefore$ When u is tripled, range becomes 9 times.

193. For maximum range $\theta = 45^\circ$

So $\tan\theta = 1$

194. $R_{max} = \frac{u^2}{g} \Rightarrow u^2 = Rg = 25000\times 10$

$\therefore$ $u = 500$ m/s

195. For maximum range

$R_{max} = \frac{u^2}{g} = 1.6$

$u = 4$ m/s

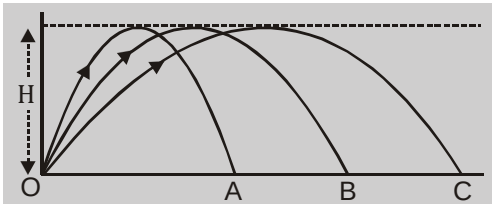
So horizontal distance in 10 sec. will be

$d = (u_x)(t)$

$d = (4)\cos(45^\circ)(10)$

$= 20\sqrt{2}$ m

196.



$H = \frac{u^2 \sin^2\theta}{2g}$ which is same for all

$\therefore$ u sinθ is same for all

$T = \frac{2u\sin\theta}{g}$ {same for all}

for large θ value of sinθ is large and hence u will be less

$\therefore \quad u_A < u_B < u_C$

i.e. launch speed is maximum for particle C

horizontal range R = $u_x \times T$

T is same for all and R_C is maximum for C

$\therefore$ u_x is maximum for C

197. $\vec{v} = a\hat{i} + b\hat{j} \Rightarrow$ u cos θ = a and u sin θ = b

$R = 4H_{max}$

$\tan\theta = \frac{b}{a}$ also $\frac{u^2\sin 2\theta}{g} = \frac{4u^2\sin^2\theta}{2g}$

$\Rightarrow$ tanθ = 1

$\therefore \quad \frac{b}{a} = 1 \Rightarrow b = a$

198. $\because \quad \theta_1 + \theta_2 = 90°$

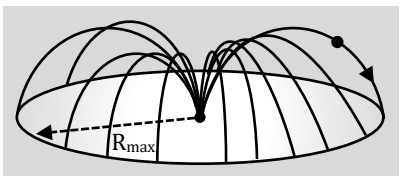
$\therefore \quad R_1 = R_2$

199. $\because \quad \theta_A + \theta_C = 90°$

$\therefore \quad R_A = R_C = \frac{u^2\sin 2\times 30°}{g} = \frac{u^2\sqrt{3}}{2g}$

$R_B = \frac{u^2\sin 2\times 45°}{g} = \frac{u^2}{g} \Rightarrow R_B > R_A = R_C$

200. $R_{max} = \frac{u^2}{g}$



$\text{Area} = \pi R_{max}^2 = \pi\left[\frac{u^2}{g}\right]^2$

201. $R = \frac{U^2\sin 2\theta}{g}$

For 'Q' and 'S' range will be same (complementary angles)

For 'R' it will be maximum

For 'P' it will be minimum

202. For maximum height

$\because \quad H_{max} = \frac{u^2}{2g}$

$(10) = \frac{u^2}{2g}$

$U = 10\sqrt{2}$ m/s

For maximum range

$R_{max} = \frac{u^2}{g} = \frac{(10\sqrt{2})^2}{10} = 20$ m

203. For maximum range

$R_{max} = \frac{u^2}{g}$

$100 = \frac{u^2}{g}$

For maximum height

$H_{max} = \frac{u^2}{2g} = \frac{100}{2}$

$\boxed{H_{max} = 50 \text{ m}}$

204. $R = \frac{u^2\sin 2\theta}{g}$ At θ = 45°

$R_{max} = \frac{u^2}{g} = 2000$

$\Rightarrow \quad h = \frac{u^2\sin^2 45°}{2g} = \frac{2000}{2\times 2} = 500\,\text{m}$

205. For θ < 45°, with increase in θ, Range also increases.

206. $\because$ Time of flight

$T = \frac{2(u)\sin\theta}{g}$

$\boxed{T \propto \sin\theta}$

For projectile '1' angle of projection is small so time of flight will be less.

207. $R=\frac{2u^2\sin\theta\cos\theta}{g}=\frac{2u^2\sin53°\cos53°}{g}$

$R=\frac{24u^2}{25g}$

$H=\frac{u^2\sin^2\theta}{2g}=\frac{u^2(\sin53°)^2}{2g}=\frac{16u^2}{2\times25g}$

$H=\frac{8u^2}{25g}=\frac{R}{3}$

208. R is same for two angles θ & 90 – θ

$h_1=\frac{u^2\sin^2\theta}{2g}, h_2=\frac{u^2\sin^2(90-\theta)}{2g}$

$h_1h_2=\frac{u^4\sin^2\theta\cos^2\theta}{4g^2}$

$R^2=\frac{4u^4\sin^2\theta\cos^2\theta}{g^2}\Rightarrow h_1h_2\propto R^2$

209. $h=\frac{u^2\sin^2\theta}{2g}\Rightarrow h_1=90=\frac{(u\sin37°)^2}{2g}$

$90=\frac{u^2}{2g}\times\left[\frac{3}{5}\right]^2\Rightarrow\frac{u^2}{2g}=250$

$h_2=\frac{(u\sin53°)^2}{2g}=\frac{u^2}{2g}\times\left[\frac{4}{5}\right]^2=250\times\frac{16}{25}=160m$

210. For same range $\theta_1+\theta_2=90°$

$\theta_1=\frac{\pi}{6}rad=30°$

$\therefore\quad \theta_2=90°-\theta_1=60°$

$y_1=\frac{u^2\sin^230°}{2g}=\frac{u^2\times1}{2g\times4}=\frac{u^2}{8g}$

$y_2=\frac{u^2\sin^260°}{2g}=\frac{u^2\times3}{2g\times4}=\frac{3u^2}{8g}=3y_1$

211. Range(R) = same

then angles of projection are θ & (90 – θ)

$y_1=\frac{u^2\sin^2\theta}{2g}$ and

$y_2=\frac{u^2\sin^2(90°-\theta)}{2g}=\frac{u^2\cos^2\theta}{2g}$

$y_1+y_2=\frac{u^2}{2g}(\sin^2\theta+\cos^2\theta)=\frac{u^2}{2g}$

212. R = 4H

$\tan\theta=\frac{4H}{R}=\frac{4}{4}=1\Rightarrow\theta=45°$

213. $R=4\sqrt{3}H$

$\tan\theta=\frac{4}{4\sqrt{3}}=\frac{1}{\sqrt{3}}$

θ = 30°

214. R = 2H

$\tan\theta=\frac{4}{2}=2$

$\theta=\tan^{-1}(2)$

215. Given

$u_x=\left(\frac{\sqrt{3}}{2}\right)(u)$

$u\cos\theta=\frac{\sqrt{3}}{2}(u)\Rightarrow\theta=30°$ & R = (P) H

$\frac{u^2 2\sin(\theta)(\cos\theta)}{g}=(P)\frac{u^2\sin^2\theta}{2g}$

4 = (P) tan(θ)

$P=\frac{4}{\tan\theta}=\frac{4}{\tan(30°)}$

$P=\frac{4}{\frac{1}{\sqrt{3}}}=4\sqrt{3}$

216. Maximum range for a given velocity is attained at θ = 45°

$R=\frac{u^2\sin2\theta}{g}\Rightarrow R_{max}=\frac{u^2\sin90°}{g}=\frac{u^2}{g}$

$H=\frac{u^2\sin^2\theta}{2g}=\frac{u^2.\sin^245°}{2g}=\frac{u^2}{4g}$

$\therefore R_{max}=4H$

217. $y=x\tan\theta-\frac{1}{2}\frac{gx^2}{u^2\cos^2\theta}$ (standard equation)

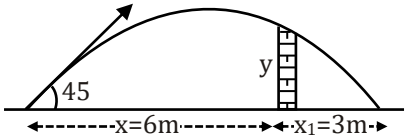
$y=x-\frac{1}{2}gx^2\Rightarrow\tan\theta=1\Rightarrow\theta=45°$

218. $y = 4x - \frac{x^2}{4}; y = 4x\left(1 - \frac{x}{16}\right)$

Comparing with $y = x\tan\theta\left(1 - \frac{x}{R}\right)$

R = 16m

219.



R = 6 + 3 = 9 m

$y = (x\tan\theta)\left[1 - \frac{x}{R}\right] = 6 \times 1 \times \left[1 - \frac{6}{9}\right] = 2\,m$

220. Time of flight $T = \frac{2u\sin\theta}{g}$

$T = \frac{(2)(50)\sin(30°)}{10} = 5$ sec

Horizontal range $R = \frac{u^2\sin 2\theta}{g}$

$R = \frac{(50)^2\sin(60°)}{10} = 216.5$ m

After 3 sec. the distance travelled is $d = U_x \times 3$

= 50 cos (30°) (3) = 129.9m

So, remaining horizontal distance after wall is

R – d = 216.5 – 129.9

$\simeq$ 86.6 m

221. Range 'R' = 2(90) = 180 m

So $\frac{R}{H} = \frac{180}{45} = 4$

$\therefore$ R = 4H, $\tan\theta = \frac{4H}{R} = \frac{4}{4} = 1$

$\boxed{\theta = 45°}$

222. $R = u_xT \Rightarrow u_x = \frac{R}{T} = \frac{100}{5} = 20$

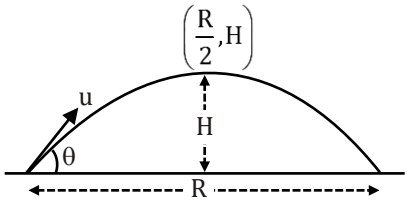
223. $H = \frac{1}{2}gt^2 = \frac{1}{2} \times 10 \times \left[\frac{5}{2}\right]^2 = 31.25\,m$

224. $u_y = gt_{up} = 10 \times 2.5 = 25$ m/s

225. $\because$ $u_x = 20$ and $u_y = 25$

$\therefore$ $\tan\theta = \frac{u_y}{u_x} = \frac{25}{20} = \frac{5}{4}$

226.



for maximum range θ = 45°

$\because$ $\tan\theta = \frac{4H}{R}$

$H = \frac{R}{4}\tan 45° = \frac{200}{4} = 50$

$\therefore$ required coordinates are $\left(\frac{R}{2}, H\right) = (100, 50)$

227. $R = \frac{v^2\sin 2\theta}{g} \Rightarrow \sin 2\theta = \frac{gR}{v^2}$

$\Rightarrow$ $2\theta = \sin^{-1}\frac{gR}{v^2}$ $\Rightarrow$ $\theta = \frac{1}{2}\sin^{-1}\frac{gR}{v^2}$

228. R = const. $\Rightarrow u^2 \sin 2q$ = const.

$\Rightarrow u_1^2\sin 2\theta_1 = u_2^2\sin 2\theta_2$

$\Rightarrow \sin 2\theta_1 = \frac{u_2^2\sin 2\theta_2}{u_1^2} = \frac{u^2}{4}\frac{\sin 60°}{u^2} = \frac{\sqrt{3}}{8}$

$2\theta_1 = \sin^{-1}\frac{\sqrt{3}}{8}$ $\Rightarrow$ $\theta_1 = \frac{1}{2}\sin^{-1}\frac{\sqrt{3}}{8}$

229. $T_1 = T_2$

$\frac{2U_1}{g} = \frac{2U_2\sin 60°}{g}$

$\frac{U_1}{U_2} = \frac{\sqrt{3}}{2}$

230. $H_1 = \frac{U^2\sin^2\theta}{2g}$

$H_2 = \frac{U^2[\sin(90-\theta)]^2}{2g}$

$H_2 = \frac{U^2\cos^2\theta}{2g}$

$\frac{H_1}{H_2} = \frac{\tan^2\theta}{1}$

231. Horizontal component of velocity of projection of projectile should be equal to speed of person

$v_x = v\cos\theta = \frac{v}{2}$

$\cos\theta = \frac{1}{2}$ $\boxed{\theta = 60°}$

232. $\frac{H}{T^2}=\frac{\frac{U^2\sin^2\theta}{2g}}{\left(\frac{2U\sin\theta}{g}\right)^2}$

$=\frac{g}{8}=\frac{10}{8}=\frac{5}{4}$

233. $R=5\sqrt{3}T^2$

$\Rightarrow \frac{2u^2\sin\theta\cos\theta}{g}=5\left[\frac{2u\sin\theta}{g}\right]^2\sqrt{3}$

$\Rightarrow \frac{2u^2\sin\theta\cos\theta}{g}=5\times\frac{4u^2\sin^2\theta}{g^2}\sqrt{3}$

$\Rightarrow \tan\theta=\frac{1}{\sqrt{3}}$

$\Rightarrow \theta = 30°$

234. $E=\frac{1}{2}mu^2$

and $E'=\frac{1}{2}mu^2\cos^2 60°=E\times\frac{1}{4}=\frac{E}{4}$

235. Kinetic energy $K=\frac{1}{2}mu^2$

At highest point :

$v = u_x = u\cos60° = \frac{u}{2}$

K.E. $= \frac{1}{2}m\left(\frac{u}{2}\right)^2$

$=\frac{1}{4}\left(\frac{1}{2}mu^2\right)=\frac{K}{4}$

236. In vertical upwards. PE = mgh

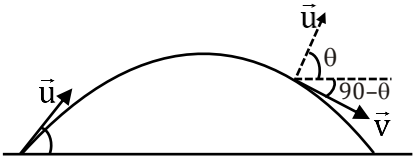
$=mg\left(\frac{v^2}{2g}\right)=\frac{mv^2}{2}$ ... (i)

at an angle of 30° with vertical PE = mgh

$=mg\left(\frac{v^2\sin^2 60°}{2g}\right)=3\frac{mv^2}{8}$ ... (ii)

$\frac{(PE)_{up}}{(PE)_{\text{at }\theta=30°\text{ from vt.}}}=\frac{\frac{mv^2}{2}}{3\frac{mv^2}{8}}=\frac{4}{3}$

237.



∵ horizontal component of velocity remains same

∴ u cosθ = v cos (90 – θ)

u cosθ = v sinθ $\Rightarrow$ v = u cot θ

238. $\vec{a}=\frac{\vec{F}}{m}=\frac{(6t^2\hat{i}+4t\hat{j})}{1}=6t^2\hat{i}+4t\hat{j}$

$\frac{d\vec{v}}{dt}=6t^2\hat{i}+4t\hat{j}$

$\Rightarrow\int_0^{\vec{v}}d\vec{v}=\int_0^3(6t^2\hat{i}+4t\hat{j})dt$

$\vec{v}=\left[\frac{6t^3}{3}\hat{i}+\frac{4t^2}{2}\hat{j}\right]_0^3=54\hat{i}+18\hat{j}$

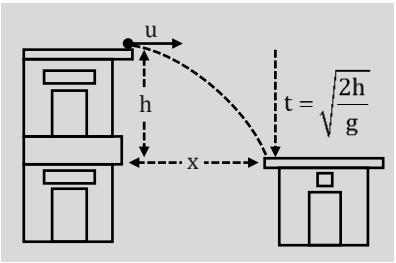
239. Time taken

$T=\sqrt{\frac{2h}{g}}=\sqrt{\frac{2\times80}{10}}=4\text{ sec.}$

distance 'd' = (u_x) (T) = (10) (4) = 40 m

240. $x=t\times u=\sqrt{\frac{2h}{g}}\times u=\sqrt{\frac{2\times490}{9.8}}\times25=250\,m$

241.



$x=\sqrt{\frac{2h}{g}}\times u$

$\Rightarrow u=x\sqrt{\frac{g}{2h}}=6.2\times\sqrt{\frac{9.8}{2\times19.6}}=3.1\text{ m/s}$

242. Horizontal range

R = (u_x) (T)

If projected horizontal from same height then time of flight will be same

$R\propto u_x$

$\frac{R_1}{R_2}=\frac{100}{40}=\frac{5}{2}$

243. Time of flight

$T = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2\times20}{10}} = 2$ sec.

Range 'R' = (u_x) (T)

= (20) (2) = 40 m

244. $t = \sqrt{\frac{2h}{g}}$; $x = v_x.t = 150\times\sqrt{\frac{2\times19.6}{9.8}} = 300$m

245. $t = \sqrt{\frac{2\times20}{g}} = 2$ sec

$x = v_x t \Rightarrow v_x = \frac{8}{2} = 4$ m/s

246. Time taken

$T = \sqrt{\frac{2(h)}{g}}$

h = 39.2 – 19.6 = 19.6 m

$T = \sqrt{\frac{2(19.6)}{9.8}} = 2$sec.

$\because$ Horizontal range

R = (u_x) (T)

20 = (u_x) (2)

u_x = 10 m/s

247. Difference in height

h = 150 – 27.5 = 122.5 m

time taken $T = \sqrt{\frac{2h}{g}}$

$T = \sqrt{\frac{2(122.5)}{9.8}} = 5$ sec.

Range R = (u_x) (T)

$u_x = \frac{30}{5} = 6$ m/s

248. For vertical motion

H = $(u_yT) + \frac{1}{2}g(T)^2$

Here 'H' & 'u_y' are same for both bodies so time will be same.

$\boxed{\frac{T_1}{T_2} = \frac{1}{1}}$

249. $\because$ $h = u_y(t) + \frac{1}{2}gt^2$

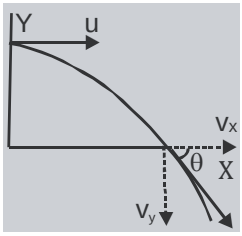
$45 = 0 + \frac{1}{2}(g)\,t^2$

t = 3 sec.

250. x = ut, $y = \frac{1}{2}gt^2$

Displacement $= \sqrt{x^2+y^2} = \sqrt{u^2t^2 + \frac{g^2t^4}{4}}$

251.



$v_x = 98\sqrt{3}$ m/s

$v_y = 98$ m/s

$v_y = u_y + a_yt$

$v_y = 0 + 9.8 \times 10 = 98$ m/s

$\tan\theta = \frac{v_y}{v_x} = \frac{98}{98\sqrt{3}} = \frac{1}{\sqrt{3}} \Rightarrow \theta = 30°$

252. $\tan 45° = \frac{H}{x} = 1$

x = H = 78.4m

time of flight $t = \sqrt{\frac{2H}{g}} = \sqrt{\frac{2\times78.4}{9.8}} = 4$ sec

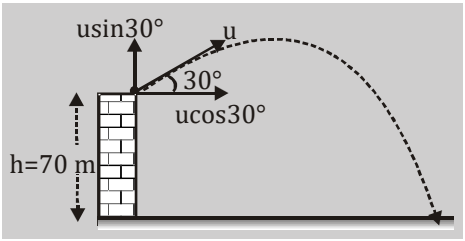
$u = \frac{x}{t} = \frac{78.4}{4} = 19.6$ m/s

253. $\tan 30° = \frac{v_y}{v_x} \Rightarrow v_y = \frac{v_x}{\sqrt{3}} = \frac{20}{3}$ m/s

$v_y = 0 + gt$ $t = \frac{20/3}{g} = \frac{2}{3}$ sec.

254. $x = \frac{20}{\sqrt{3}}\times\frac{2}{3} = \frac{40}{3\sqrt{3}}$ m

255.

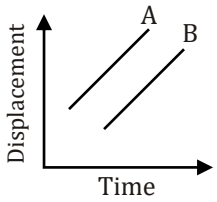


by $S_y = u_yt + \frac{1}{2}a_yt^2$

$\Rightarrow$ $h = -u\sin30°t + \frac{1}{2}gt^2$

$\Rightarrow$ $70 = -50\times\frac{1}{2}t + \frac{1}{2}\times10t^2 \Rightarrow t = 7$ sec

256. Slope of displacement-time graph gives velocity



Slope of A & B is same

257. ∵ Both velocities are perpendicular

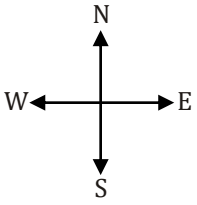
$\vec{v}_{rel} = 4\hat{i} - 3\hat{j}$

$|\vec{v}_{rel}| = \sqrt{4^2 + 3^2} = 5\ m/s$

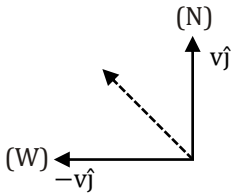
258. $\vec{v}_{Train} = v\hat{i}$

$\vec{v}_{Car} = v\hat{j}$

Velocity of car relative to train



$\vec{v}_{rel} = v\hat{j} - v\hat{i}$



West-north direction

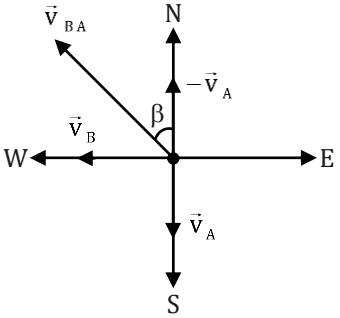
259. $a_A = 8\ m/s^2$ & $a_B = 4\ m/s^2$

$a_{BA} = a_B - a_A$

$= 4 - 8 = -4 m/s^2$

260. $v_{rel} = 80 + 65 = 145$ km/hr

261.



The velocity of car A and car B are represented by the vectors in the direction as given in the question as shown in the figure.

Given : $v_A = v_B = v = 5$ km/min

The relative velocity of car B w.r. t to car A is given by : $\vec{v}_{BA} = \vec{v}_B - \vec{v}_A$

$\vec{v}_{BA} = 5(-\hat{i}) - (-5\hat{j}) = -5\hat{i} + 5\hat{j}$

$|\vec{v}_{BA}| = \sqrt{5^2 + 5^2} = 5\sqrt{2}\ km/min$

$\tan\beta = \frac{5}{5} = 1$

β = 45° in north-west.

262. $\vec{v}_{MT} = \vec{v}_M - \vec{v}_T$

$\vec{v}_M = \vec{v}_{MT} + \vec{v}_T$

$\vec{v}_M = +3\hat{i} + 20\hat{i} = 23\hat{i}$

23 m/s towards east

263. $v_{gas,\ plane} = v_{gas} + v_{plane}$

$v_{gas} = v_{gas,\ plane} - v_{plane}$

$= 1600 - 500 = 1100$ km/hr

264. $\vec{v}_{TG} = 54 \times \frac{5}{18}\hat{j} = 15\hat{j}\ m/s$

$\vec{v}_{MT} = -36 \times \frac{5}{18}\hat{j} = -10\hat{j}\ m/s$

$\vec{v}_{MT} = \vec{v}_{MG} - \vec{v}_{TG}$

$\vec{v}_{MG} = \vec{v}_{MT} + \vec{v}_{TG} = -10\hat{j} + 15\hat{j} = 5\hat{j}\ m/s$

$= 5$ m/s along north

267. Length of train L = 100 m

Let speed of train = v_t

and speed of man $v_m = 5$ km/hr

relative speed $v_{rel} = v_t + v_m$

$v_{rel} = \frac{Distance}{Time} = \frac{100}{6}\ m/s$

$= \frac{100}{6} \times \frac{18}{5} = 60$ km/hr

$\therefore \quad v_t + 5 = 60 \quad \Rightarrow \quad v_t = 55$ km/hr

268. Relative velocity of the bird (v_{rel}) = 20 – (–5)

= 25 m/s

Distance = 300 m

$\therefore$ Time = $\frac{Distance}{v_{rel}} = \frac{300}{25} = 12$ sec.

269. $t = \frac{100+100}{35+15} = 4s$

270. ∵ $v^2 = u^2 + 2as$

$(50)^2 = (30)^2 + 2(a)(500)$

$a = 1.6\ m/s^2$

So, time $t = \frac{v-u}{a}$

$t = \frac{50-30}{1.6} = 12.5\ sec$

271. For opposite direction

$v_1 + v_2 = 8$ ...(1)

For same direction

$v_1 - v_2 = 0.8$ ...(2)

From (1) & (2)

$v_1 = 4.4\ m/s$ & $v_2 = 3.6\ m/s$

272. $v_M - v_R = \frac{30}{10}$

$5 - v_R = 3$

$\boxed{v_R = 2\ m/s}$

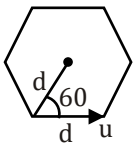
273. $\frac{5}{60} = \frac{10}{v+60} \Rightarrow v + 60 = 120$

$v = 60\ km/hr$

274. $t = \frac{d}{u + u\cos 90} = \frac{d}{u}$

275. $t = \frac{d}{u - u\cos 60°}$

$= \frac{d}{u - \frac{u}{2}} = \frac{d}{\frac{u}{2}} = \frac{2d}{u}$



Alternative method: -

$t = \frac{d}{u\cos 60°}$

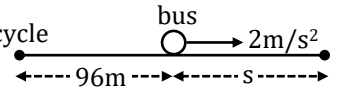
$t = \frac{2d}{u}$

276. speed of man $(v_m) = \frac{d}{45}$

speed of escalator $(v_s) = \frac{d}{60}$

$t = \frac{d}{v_m + v_s} = \frac{d}{\frac{d}{45} + \frac{d}{60}} = \frac{60 \times 45}{105} = 25.71\ sec$

277.



For cycle $96 + s = 20 \times t$

for bus $s = \frac{1}{2}at^2 = \frac{1}{2} \times 2 \times t^2 = t^2$

$\therefore 96 + t^2 = 20\ t \Rightarrow t^2 - 20t + 96 = 0$

$(t - 8)(t - 12) = 0 \therefore t = 8\ s$ or $12\ s$

cyclist over takes bus in 8 s

Note : At t = 12s, bus overtakes cyclist.

278. $\frac{10}{v_{mr}} + \frac{10}{v_{mr}} = 2 \Rightarrow v_{mr} = 10\ km/hr.$

$\Rightarrow t = \frac{10}{v_{mr} + v_r} + \frac{10}{v_{mr} - v_r}$

$= \frac{10}{10+5} + \frac{10}{10-5} = \frac{10}{15} + \frac{10}{5} = \frac{8}{3} hr$

$\Rightarrow t = \frac{8}{3} \times 60 = 160\ min.$

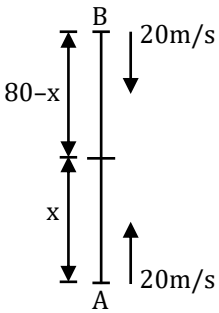
279. The velocity upstream is (6 – 2) km/hr and downstream is (6 + 2) km/hr.

$\therefore$ Total time taken $= \frac{2}{4} + \frac{2}{8} = \frac{6}{8} = \frac{3}{4} hr$

= 45 minutes

280. Distance will remain the same as before because acceleration of both the bodies is same and initial velocity is zero.

281.



Let us assume they are meeting at x distance from ground

$80 - x = 20\ t + \frac{1}{2}gt^2$ ... (i)

$x = 20\ t - \frac{1}{2}gt^2$ ... (ii)

From eq. (i) and (ii)

$80 = 40\ t$

$t = 2\ s$

$x = 20\ m$ from ground.

282. Total time taken by the first walnut to reach at ground :-

$h = \frac{1}{2} gt^2$

$20 = \frac{1}{2} \times 10 \times t^2$

$t = 2$ sec.

time taken by it to cover a distance of 5 m :

$5 = \frac{1}{2} gt^2 \Rightarrow t = 1$ sec.

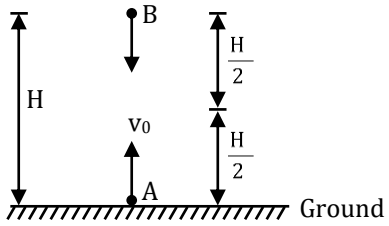
So, For 2nd walnut :

$S = Ut + \frac{1}{2} at^2$

$20 = U \times 1 + \frac{1}{2} \times 10 \times (1)^2$

$U = 15$ m/s

283. Let the two bodies A and B respectively meet at a time t, at a height $\frac{H}{2}$ from the ground.



Using $S = ut + \frac{1}{2} at^2$

For a body A, $u = v_0$, $a = -g$, $S = \frac{H}{2}$

$\therefore \frac{H}{2} = v_0 t - \frac{1}{2} gt^2$ ... (i)

For a body B, $u = 0$, $a = +g$, $S = \frac{H}{2}$

$\therefore \frac{H}{2} = \frac{1}{2} gt^2$

Equating equations (i) and (ii), we get

$v_0 t - \frac{1}{2} gt^2 = \frac{1}{2} gt^2$

$v_0 t = gt^2$ or $t = \frac{v_0}{g}$

Substituting the value of t is equation (i), we get

$$\frac{H}{2} = v_0 \times \left(\frac{v_0}{g}\right) - \frac{1}{2} g \left(\frac{v_0}{g}\right)^2 = \frac{v_0^2}{g} - \frac{1}{2}\frac{v_0^2}{g}$$

$$\frac{H}{2} = \frac{1}{2}\frac{v_0^2}{g} \text{ or } v_0^2 = gH$$

$v_0 = \sqrt{gH}$

Alternate solution: -

$U_{AB} = U_A - U_B = v_0 - 0 = v_0$

$a_{AB} = a_A - a_B = -g - (-g) = 0$, $S_{AB} = H$

$\therefore t = \frac{S_{AB}}{U_{AB}} = \frac{H}{v_0}$, for body B :- $S = 1/2\ gt^2$

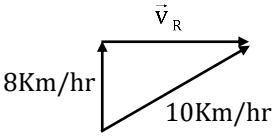
$\Rightarrow \frac{H}{2} = \frac{1}{2} g \times \frac{H^2}{v_0^2} \Rightarrow v_0 = \sqrt{gH}$

284. Effective acceleration in ascending lift = (g + a)

$t = \sqrt{\frac{2h}{g+a}} = \sqrt{\frac{2\times 19}{32+6}}$

$t = \sqrt{\frac{2\times 19}{38}} = 1$ sec

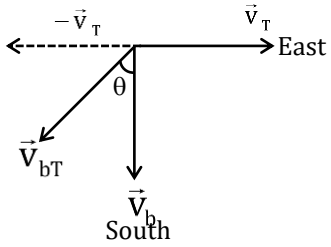
285.



$v_R^2 + 8^2 = 10^2$

$v_R = 6$ Km/hr

286.

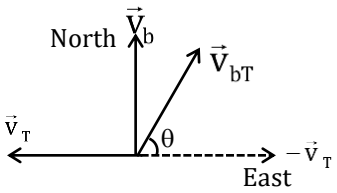


$\vec{v}_{bT} = -40\hat{j} - 40\hat{i}$

$|\vec{v}_{bT}| = \sqrt{40^2 + 40^2} = 40\sqrt{2}$ (south-west)

$\tan\theta = 1$; $\theta = 45°$

287.



$\vec{v}_{bT} = \vec{v}_b - \vec{v}_T$

$\vec{v}_{bT} = 40\hat{j} - (-30\hat{i})$

$\vec{v}_{bT} = (30\hat{i} + 40\hat{j})$ km/h

$|\vec{v}_{bT}| = \sqrt{(30)^2 + (40)^2} = 50$ km/h

$\tan\theta = \frac{40}{30} = \frac{4}{3}$

$\theta = 53°$

53° North of east

288. $\vec{v}_{BW} = \vec{v}_{BG} - \vec{v}_{WG} = 9\hat{i} + 12\hat{j}$

289. Here,

$\vec{r}_1(t) = 3t\,\hat{i} + 4t^2\,\hat{j}$

$\vec{r}_2(t) = 4t^2\,\hat{i} + 3t\,\hat{j}$

Velocity,

$\vec{v}_1(t) = \frac{d\vec{r}_1}{dt} = \frac{d}{dt}\left(3t\,\hat{i} + 4t^2\,\hat{j}\right) = 3\hat{i} + 8t\,\hat{j}$

$\vec{v}_2(t) = \frac{d\vec{r}_2}{dt} = \frac{d}{dt}\left(4t^2\,\hat{i} + 3t\,\hat{j}\right) = 8t\hat{i} + 3\hat{j}$

The relative speed of particle 1 with respect to particle 2 is : $\vec{v}_{12} = \vec{v}_1 - \vec{v}_2$

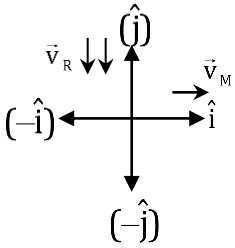
$= \left(3\hat{i} + 8t\,\hat{j}\right) - \left(8t\,\hat{i} + 3\hat{j}\right) = (3 - 8t)\hat{i} + (8t - 3)\hat{j}$

At t = 2 s,

$\vec{v}_{12} = (3-16)\hat{i} + (16-3)\hat{j} = -13\hat{i} + 13\hat{j}$

$|\vec{v}_{12}| = \sqrt{(13)^2 + (13)^2} = 13\sqrt{2}$ m/s

290.



$\vec{v}_{RM} = \vec{v}_R - \vec{v}_M = -3\hat{j} - 3\hat{i}$

$\vec{v}_{RM} = \sqrt{3^2 + 3^2} = 4.2$ m/s

291. $\vec{V}_M = 5\hat{i}$

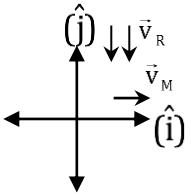
$\vec{V}_R = -10\hat{j}$

$\vec{V}_{RM} = \vec{V}_R - \vec{V}_M = -10\hat{j} - 5\hat{i}$

$\left|\vec{V}_{RM}\right| = \sqrt{100 + 25} = 5\sqrt{5}$ km/hr

292. $\vec{v}_{RM} = \vec{v}_R - \vec{v}_M$

$= -20\hat{j} - 30\hat{i}$



Angle with vertical

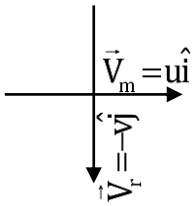
$\tan\theta = \left(\frac{30}{20}\right)$

$\boxed{\theta = \tan^{-1}\left(\frac{3}{2}\right)}$

293. If man has horizontal velocity equal to horizontal component of rain velocity, then relative velocity in horizontal direction becomes zero, so rain will appear vertically.

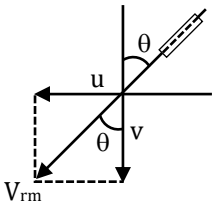
294. $\vec{v}_{rm} = \vec{v}_r - \vec{v}_m$

$\vec{v}_{rm} = -v\hat{j} - u\hat{i}$



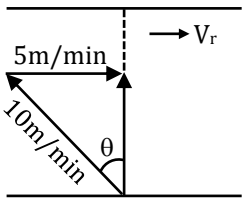
$\tan\theta = \frac{u}{v}$

$\theta = \tan^{-1}\left(\frac{u}{v}\right)$



295. $v_{res} = \sqrt{v_P^2 + v_r^2} = \sqrt{164}$ km/hr

296.



$\sin\theta = \frac{5}{10} = \frac{1}{2}$

$\theta = 30°$

Angle with downstream = 90° + 30° = 120°

297. $t_{min} = \frac{D}{v} = \frac{4}{2} = 2$hr

298. For shortest time he has to cross the river with swimming perpendicular to flow i.e. due north.

299. $t=\frac{d}{\sqrt{v^2-v_R^2}}=\frac{4}{\sqrt{10^2-6^2}}=\frac{4}{8}=\frac{1}{2}\text{ hr}$

300. $\sin\theta=\frac{v_R}{v}=\frac{1}{\sqrt{2}}\Rightarrow\theta=45°$

Angle with river flow = 90° + 45° = 135°

301. $u_{x_1}t+u_{x_2}t=x$

$\left[\sqrt{3}u\cos30°+u\cos60°\right]t=x$

$\left[\sqrt{3}u\times\frac{\sqrt{3}}{2}+\frac{u}{2}\right]t=x\Rightarrow t=\frac{x}{2u}$

Exercise-II (Previous Year Questions)

1. As Range = $\frac{u^2\sin2\theta}{g}$ so $g\propto u^2$

Therefore $g_{planet}=\left(\frac{3}{5}\right)^2(9.8\text{ m/s}^2)=3.5\text{ m/s}^2$

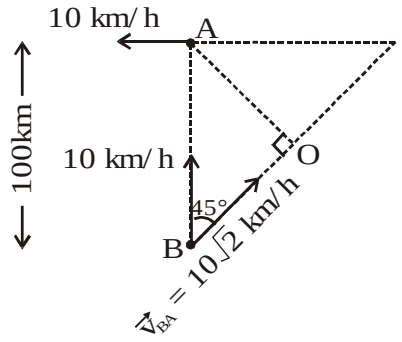
2. $\vec{v}_{avg}=\frac{\Delta\vec{r}}{\Delta t}=\frac{(13-2)\hat{i}+(14-3)\hat{j}}{5-0}=\frac{11}{5}(\hat{i}+\hat{j})$

3. $v=\beta x^{-2n}$ so $\frac{dv}{dx}=-2n\beta x^{-2n-1}$

Now $a=v\frac{dv}{dx}=(\beta x^{-2n})(-2n\beta x^{-2n-1})$

$\Rightarrow\quad a=-2n\beta^2x^{-4n-1}$

4.



$|\vec{v}_{BA}|=\sqrt{10^2+10^2}=10\sqrt{2}\text{ kmph}$

distance OB = 100 cos45° = $50\sqrt{2}$ km

Time taken to reach the shortest distance between A & B = $\frac{50\sqrt{2}}{|\vec{v}_{BA}|}=\frac{50\sqrt{2}}{10\sqrt{2}}$

t = 5 hrs.

5. For two particles to collide, the direction of the relative velocity of one with respect to other should be directed towards the relative position of the other particle

i.e. $\frac{\vec{r}_1-\vec{r}_2}{|\vec{r}_1-\vec{r}_2|}$ → direction of relative position of 1 w.r.t. 2.

& $\frac{\vec{v}_2-\vec{v}_1}{|\vec{v}_2-\vec{v}_1|}$ → direction of velocity of 2 w.r.t. 1

so for collision of A & B $\frac{\vec{r}_1-\vec{r}_2}{|\vec{r}_1-\vec{r}_2|}=\frac{\vec{v}_2-\vec{v}_1}{|\vec{v}_2-\vec{v}_1|}$

6. $v=At+Bt^2\Rightarrow\frac{ds}{dt}=At+Bt^2$

$\int_0^s ds=\int_1^2(At+Bt^2)dt$

$s=\frac{A}{2}(2^2-1^2)+\frac{B}{3}(2^3-1^3)=\frac{3A}{2}+\frac{7B}{3}$

7. $x_P(t)=at+bt^2$ $\quad x_Q(t)=ft-t^2$

$v_P=a+2bt$ $\quad v_Q=f-2t$

as $v_P=v_Q$

$a+2bt=f-2t\Rightarrow t=\frac{f-a}{2(1+b)}$

8. $V_1\rightarrow$ velocity of Preeti

$V_2\rightarrow$ velocity of escalator

$\ell\rightarrow$ distance

$t=\frac{\ell}{V_1+V_2}=\frac{\ell}{\frac{\ell}{t_1}+\frac{\ell}{t_2}}=\frac{t_1t_2}{t_1+t_2}$

9. $v_x=5-4t$, $v_y=10$

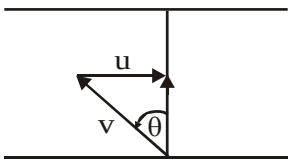
$a_x=-4$, $a_y=0$

$\vec{a}=a_x\hat{i}+a_y\hat{j}$

$\vec{a}=-4\hat{i}\text{ m/s}^2$

10. v = 20 m/s

u = 10 m/s



$\sin\theta=\frac{u}{v}=\frac{10}{20}=\frac{1}{2}$

$\Rightarrow\theta=30°$ west

11. $v^2 = u^2 - 2as$

$$\Rightarrow s = \frac{u^2}{2a} = \frac{u^2}{2g\sin\theta}$$

$$\frac{x_1}{x_2} = \frac{\sin\theta_2}{\sin\theta_1} = \frac{\sin 30^\circ}{\sin 60^\circ} = \frac{1/2}{\sqrt{3}/2} \Rightarrow \frac{x_1}{x_2} = \frac{1}{\sqrt{3}}$$

12. $t = \sqrt{\frac{2h}{a_{rel}}}$

In both cases $a_{rel} = g - 0 = g$.

Hence $t_1 = t_2$

13. $t = \frac{d}{v_{rel}} = \frac{100}{50} = 2s$

distance travelled in downward direction

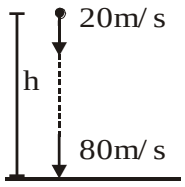
$$s_y = \frac{1}{2}gt^2 = \frac{1}{2}\times 10\times 4 = 20$$

$\therefore$ Height = 200 – 20 = 180 m

14. Average velocity $= \frac{\text{Total distance}}{\text{Total time}} = \frac{x+x}{\frac{x}{v_1}+\frac{x}{v_2}}$

$$\Rightarrow v = \frac{2}{\frac{1}{v_1}+\frac{1}{v_2}} \Rightarrow \frac{2}{v} = \frac{1}{v_1} + \frac{1}{v_2}$$

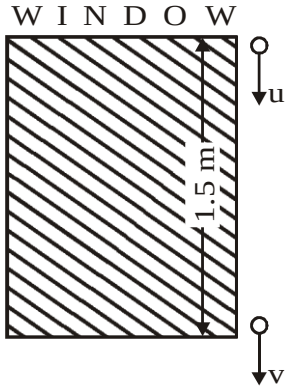
15.



$\because v^2 = u^2 + 2gh$

$\therefore 80^2 = 20^2 + 2 \times 10h \Rightarrow h = 300m$

16.



By using $s = ut + \frac{1}{2}at^2$

$1.5 = u\,(0.1) + \frac{1}{2}(10)\,(0.1)^2$

$\Rightarrow 15 = u + 0.5$

$\Rightarrow u = 14.5\ ms^{-1}$

17. S_n = Distance in n^{th} sec. i.e. t = n – 1 to t = n

S_{n+1} = Distance in $(n+1)^{th}$ sec.

i.e. t = n to t = n + 1

So as we know

$S_n = \frac{a}{2}(2n-1)$ a = acceleration

$$\frac{S_n}{S_{n+1}} = \frac{\frac{a}{2}(2n-1)}{\frac{a}{2}(2(n+1)-1)} = \frac{2n-1}{2n+1}$$

$$\frac{S_n}{S_{n+1}} = \frac{2n-1}{2n+1}$$

18. velocity of car at t = 4 sec is

$v = u + at$

$v = 0 + 5\,(4) = 20$ m/s

At t = 6 sec

Acceleration due to gravity $\therefore a = g = 10\ m/s^2$

$v_x = 20$ m/s (due to car)

$v_y = u + a\,(t-4)$

$= 0 + g\,(2)$ (downward)

$= 20$ m/s (downward)

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{20^2 + 20^2} = 20\sqrt{2}\ m/s$$

19. $T = \frac{2\pi R}{v} \Rightarrow v = \frac{2\pi R}{T}$ ... (1)

$$H_{max} = \frac{v^2\sin^2\theta}{2g} = \frac{2\pi^2R^2\sin^2\theta}{gT^2} = 4R$$

$$\sin\theta = \left(\frac{2gT^2}{\pi^2R}\right)^{1/2}$$

$$\theta = \sin^{-1}\left[\frac{2gT^2}{\pi^2R}\right]^{1/2}$$

20. Velocity is slope of x–t graph

$$V = \frac{dx}{dt} = \tan\theta$$

$$\frac{V_1}{V_2} = \frac{\tan\theta_1}{\tan\theta_2} = \frac{\tan 30^\circ}{\tan 45^\circ} = \frac{1}{\sqrt{3}}$$

21. $S_{nth} = u + \frac{a}{2}(2n-1)$

$= 0 + \frac{a}{2}(2n-1)$

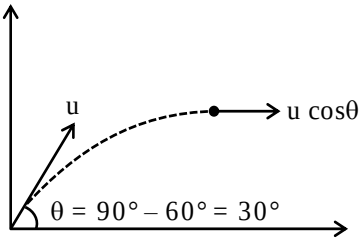
$S_{nth} \propto (2n-1)$

$\Rightarrow S_{1st}, S_{2nd}, S_{3rd}, S_{4th}$

$= [2(1)-1] : [2(2)-1] : [2(3)-1] : [2(4)-1]$

$= 1 : 3 : 5 : 7$

22. At highest point only horizontal component of velocity remains $\Rightarrow u_x = u\cos\theta$



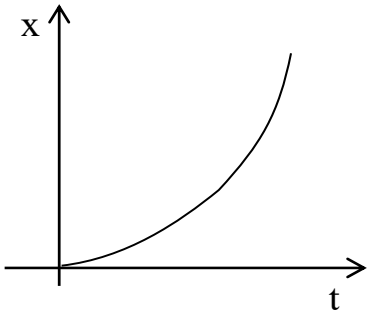
$u_x = u\cos\theta = 10\cos 30°$

$= 5\sqrt{3}\,ms^{-1}$

23. $H = \frac{u^2\sin^2\theta}{2g} = \frac{(20)^2\sin^2 30°}{2(10)}$

$= 5m$

24.



$x = \frac{1}{2}at^2$ (if coefficients of t^2 is positive ($a > 0$) upward opening parabola)

25. $H_{max} = \frac{u^2\sin^2\theta}{2g}$

$= \frac{(280)^2(\sin 30°)^2}{2(9.8)}$

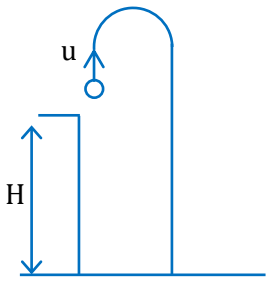
$= 1000$ m

26.



$V_{avg} = \frac{2\upsilon_1\upsilon_2}{\upsilon_1+\upsilon_2} = \frac{2(\upsilon)(2\upsilon)}{\upsilon+2\upsilon} = \frac{4\upsilon^2}{3\upsilon} = \frac{4\upsilon}{3}$

27. $S = ut + \frac{1}{2}at^2$



$-H = 4\times 4 - \frac{1}{2}\times 10\times 4^2$

$-H = 16 - 80$

$-H = -64$

$H = 64$ m

28. By $v^2 = u^2 + 2as$

$\left(\frac{u}{3}\right)^2 = u^2 - 2ax$

$2ax = u^2 - \frac{u^2}{9}$

$\boxed{2ax = \frac{8u^2}{9}}$ ... (1)

Similarly from starting

$v^2 = u^2 + 2ax$

$0 = u^2 - 2ax_2$

$\boxed{2ax_2 = u^2}$ ... (2)

By (1) / (2)

$\frac{x}{x_2} = \frac{8}{9}$

$\frac{24}{x_2} = \frac{8}{9}$

$x_2 = 27$ cm

29. $\vec{V} = \frac{d\vec{r}}{dt} = 4\hat{i} + 4t\hat{j} + 0\hat{k}$

at $t = 1$sec

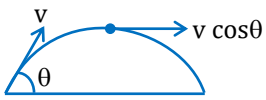
$\vec{V} = 4\hat{i} + 4(1)\hat{j}$

$|\vec{V}| = \sqrt{4^2+4^2} = 4\sqrt{2}$

$\tan\alpha = \frac{4}{4} = 1$

$\alpha = 45°$

30.



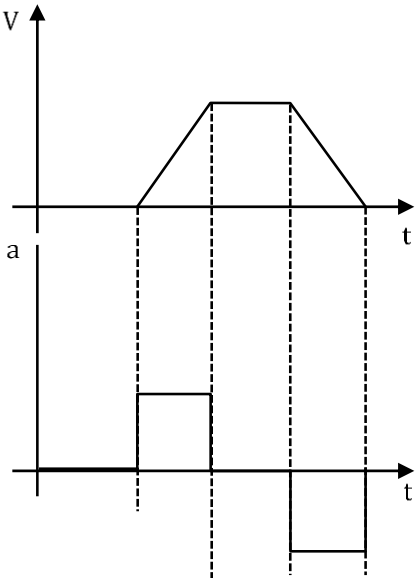
At the top most point of its trajectory particle will have only horizontal component of velocity

V at top = v cos θ

$= 20 \times \frac{1}{2}$

= 10 m/s

31.



32. $x = \alpha t^4 + \beta t^2 + \gamma t + \delta$

$V = \frac{dx}{dt} = 4\alpha t^3 + 2\beta t + \gamma$

$\Rightarrow$ at t = 0, v = γ

$a = \frac{dv}{dt} = 12\alpha t^2 + 2\beta$

$\Rightarrow$ at t = 0, a = 2β

$\therefore \frac{\text{Initial velocity}}{\text{Initial acceleration}} = \frac{\gamma}{2\beta}$

Exercise-III (Analytical Questions)

1. $(\vec{V}_i) = (u\cos\theta)\hat{i} + (u\sin\theta)\hat{j}$

$(\vec{V}_f) = (u\cos\theta)\hat{i} - (u\sin\theta)\hat{j}$

$(\vec{V}_{top}) = (u\cos\theta)\hat{i}$

$|\vec{V}_i|$ = Initial speed = u

$|\vec{V}_f|$ = final speed = u

$|\vec{V}_{top}|$ = speed at top = u cos θ (A)

(B) $\overrightarrow{\Delta V} = \vec{V}_f - \vec{V}_i$ & $\Delta|\vec{V}| = |\vec{V}_f| - |\vec{V}_i| = 0$

$|\overrightarrow{\Delta V}| = (2u\sin\theta)\hat{j}$

$|\overrightarrow{\Delta V}| = 2u\sin\theta$

(C) $|\vec{V}_i| - |\vec{V}_{top}|$ = u – u cos θ

(D) $|\vec{V}_i - \vec{V}_{top}| = u\sin\theta$

2. $\vec{v}_i = 3\hat{i} + 4\hat{j} \quad \Rightarrow \quad |\vec{v}_i| = 5$

$\vec{v}_f = 3\hat{i} + 4\hat{j} \quad \Rightarrow \quad |\vec{v}_f| = 5$

(A) $|\Delta\vec{v}| = |\vec{v}_f - \vec{v}_i| = |3\hat{i} + 4\hat{j} - 3\hat{i} + 4\hat{j}|$

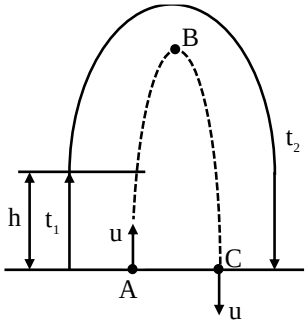
$= |8\hat{j}| = 8$

(B) $\Delta|\vec{v}| = |\vec{v}_f| - |\vec{v}_i| = 5 - 5 = 0$

(C) $\vec{v}_f - \vec{v}_i = 8\hat{j}$

(D) $\vec{v}_f + \vec{v}_i = 6\hat{i}$

3.



from the figure, it is clear that time of flight is $(t_1 + t_2)$

for maximum Ht :-

time taken from B to C is $\left(\frac{t_1 + t_2}{2}\right)$

Applying 2nd eqn of motion from B to C

$H_{max} = 0 + \frac{1}{2}g\left(\frac{t_1 + t_2}{2}\right)^2$

for (u) $\Rightarrow$ Speed at A = Speed at C

Apply 1st eqn of motion between B & C

$u = 0 + g\left(\frac{t_1 + t_2}{2}\right)$

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for (h) $\Rightarrow h = ut_1 - \frac{1}{2}gt_1^2$

$$h = g\left(\frac{t_1 + t_2}{2}\right)t_1 - \frac{1}{2}gt_1^2$$

$$\boxed{h = \frac{1}{2}gt_1t_2}$$

4. Two projectiles having same range may have different time of flight.

5. Projectile motion is an example of motion under constant acceleration.

9. When body moves with constant velocity then it moves with constant speed in same direction,

So |Inst. velocity| = |Avg. vel.|

= Inst. speed = Avg speed

10. Average speed for first 6 s

$$= \frac{\text{distance travelled in 6 s}}{6\text{ s}}$$

$$= \frac{\frac{1}{2}\times 15\times 4 + \frac{1}{2}\times 2\times 10}{6} = \frac{20}{3}\ \text{ms}^{-1}$$

11. Average velocity for first 12 s

$$= \frac{\text{displacement in first 12 s}}{12\text{ s}}$$

$$= \frac{-\frac{1}{2}\times 15\times 4 + \frac{1}{2}\times 2\times 10 + 2\times 10 + \left(\frac{20+10}{2}\right)(12-8)}{12}$$

$$= \frac{60}{12} = 5\ \text{ms}^{-1}$$

12. Average acceleration from t = 5 s to t = 15 s

$$= \frac{v(t=15) - v(t=5)}{15-5} = \frac{0-5}{10} = -\frac{1}{2}\text{ms}^{-2}$$

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Laws of Motion and Friction

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Newton's I law.

2. $F_{net} = 0$, $a = 0$

3. $F_{net} = ma$, F_{net} = const, a = const.

4. $V = u + at$

$30 = a(2)$

$a = 15\ m/s^2$

$\therefore \quad F = ma = 10\ (15)$

$F = 150\ N$

5. F = 0 (as acceleration = 0)

6. $v = u + at$

$0 = 8 + a \times 3$

$a = -\frac{8}{3} m/s^2$

$F = 6 \times \frac{8}{3} = 16N$

7. $a = \frac{-4-8}{6} = -2\ m/s^2$

8. $a = \frac{F}{m}$

$v^2 - 0 = 2\frac{F}{m} \times d \quad \therefore \quad v \propto \frac{1}{\sqrt{m}}$

9. $a = \frac{F}{m}$

$v^2 - 0 = 2\frac{F}{m} \times d$

$v^2 = 2\frac{F}{m} \times d$

$v^2 = 2\left(\frac{25 \times 10^4}{5 \times 10^7}\right) \times 100$

$v^2 = 1$

$v = 1\ m/s$

10. $F = mg = 40\ (980)$

$F = 39200$ dyne

11. $\vec{F}_{net} = 0 \Rightarrow \vec{a} = 0$

$\therefore \quad \vec{v}$ remains unchanged

12. $\vec{a} = \frac{\vec{F}}{m} = -\frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}$

$\vec{v} = \vec{u} + \vec{a}t$

$\vec{v} = 6\hat{i} - 12\hat{j} + \left(-\frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}\right)t$

$\vec{v} = \left(6 - \frac{3t}{5}\right)\hat{i} + \left(\frac{4t}{5} - 12\right)\hat{j}$

To have a velocity along y-axis,
its x-component must be zero.

$6 - \frac{3t}{5} = 0$

$t = 10$ sec

13. Since, $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = \vec{0}$

$\vec{F}_2 + \vec{F}_3 = -\vec{F}_1$

$\vec{a} = \frac{\vec{F}_2 + \vec{F}_3}{m} = -\frac{\vec{F}_1}{m}$

$a = \frac{F_1}{m}$

14. $F_{net} = ma$

$F_{net} = 1\ N$

15. Force from road causes acceleration of car.

16. $F = \frac{dp}{dt}$

17. Inertia of stone causes reaction to get hurt.

18. $a = \frac{F}{m} = 2\ m/s^2$

$v = u + at$

$v - u = 2(20)$

$v - u = 40$

$\Delta P = m(v-u) = 10(40)$

$\Delta P = 400$ kg-m/s

or

$\because \quad F = \frac{\Delta P}{\Delta t} \quad \therefore \quad \Delta P = F\Delta t = 20 \times 20$

$\therefore \quad \Delta P = 400$ kg-m/s

19. $F_{avg} = \frac{\Delta p}{\Delta t} = \frac{100 \times 10^{-3} \times 5}{0.25}$

$F_{avg} = 2N$

20. $F_{avg} = \frac{\Delta p}{\Delta t} = \frac{0.5(15-(-10))}{0.02}$

$F_{avg} = 625N$

21. Force F = Rate of change in the momentum

$= \frac{m[v_2 - v_1]}{t}$

$= \frac{(0.1)[40-(-20)]}{(5\times10^{-3})} = 1200N$

22. $F_{avg} = \frac{2\times\frac{50}{1000}\times25}{0.05} = 50N$

23. $a = \frac{v-u}{t}$

$a = \frac{\sqrt{2gh_2} - \left(-\sqrt{2gh_1}\right)}{t}$

$a = \frac{\sqrt{2\times10\times5} + \sqrt{2\times10\times10}}{0.01}$

$a = \frac{10+14}{0.01} = 2400\ m/s^2$

24. $F_{avg} = \frac{2mv}{t}$

$\frac{2\times150\times10^{-3}\times40}{1.20\times10^{-3}} = 10000N$

25. $F = \frac{\Delta p}{\Delta t} = 2mnv$

26. Mg = n × 2mv

2 × 9.8 = 5 × 2 × 0.1 × v

$v = \frac{9.8\times2}{1} = 19.6\ m/s$

27. $F = m \times a = 25 \times \frac{dv}{dt} = 25 \times 2 = 50\ N$

28. Δp = Area under F-t graph

$m(v-u) = \frac{1}{2} \times 8 \times 4 - \frac{1}{2} \times 1 \times 2 - 1 \times 2$

2(v–0) = 16 – 1 – 2

$v = \frac{13}{2} = 6.5\ m/s$

29. $\Delta p = \int F dt = 0$ (area of F-t graph is zero)

30. Δp = Area under F-t graph

m(v–u) = 4 + 8 + 1.625 + 5

2(v–5) = 18.625

v – 5 = 9.31

v = 14.3 m/s

32. $I = \int_0^t F dt = 50t - \frac{20t^2}{2} = 50t - 10t^2$

33. $\vec{I} = \Delta\vec{p} = \vec{p}_f - \vec{p}_i$

$= 0 - 0.2 = -0.2\ kg\ ms^{-1}$

34. $F = \frac{dm}{dt} \times v = 1.5 \times 3 = 4.5N$

35. $F = \frac{dm}{dt} \times V = 3 \times 10 = 30$

F = 30 N

$\therefore\ 1.5 \times a = 30$

$a = \frac{30}{1.5} = 20\ m/s^2$

36. $F = \frac{vdm}{dt} = 5 \times 1 = 5N$

$a = \frac{5}{2} = 2.5\ m/s^2$

37. $\frac{dm}{dt} = \frac{300}{400} = 0.75\ kg/s$

38. $F = -\frac{vdm}{dt} = -\alpha v^3 \ \therefore \quad a = \frac{-\alpha v^3}{M}$

39. F = 2 × 300 = 600

$\therefore \quad a = \frac{600}{60} = 10\ m/s^2$

40. $800\times9.8 + 800\times4.2 = \frac{dm}{dt} \times 1600$

$\frac{dm}{dt} = \frac{14}{2} = 7\ kg/s$

41. $\frac{dm}{dt} = \frac{210}{300} = 0.7\ kg/s$

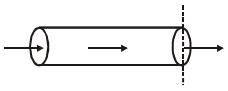
42. Minimum rate of burning fuel is $\frac{dm}{dt}$

$\therefore\ mg = u\frac{dm}{dt}$

or, $\frac{dm}{dt} = \frac{mg}{u} = \frac{800\times10}{40} = 200\ kg/s.$

44. $\Delta p = 2\ mv\sin\theta$

$F = \frac{dp}{dt} = (2v\sin\theta)\frac{dm}{dt}$



The mass crossing an area a in time t would be

m = v × t × a × d ...(1)

For small element

$\therefore \quad \frac{dm}{dt} = vad \quad \therefore \quad F = 2av^2d\sin\theta$

46. Action $\rightleftharpoons$ Reaction

47. Forces from ground causes horse motion.

48. Action $\rightleftharpoons$ Reaction

49. $m_Aa_A = m_Ba_B$

$ma_A = 2m \times a$

$a_A = 2a$

51. $m_1a_1 = m_2a_2$

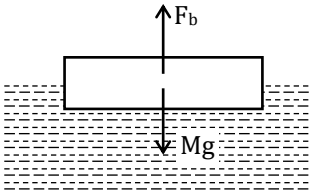
$$a_2 = \frac{m_1a_1}{m_2} = \frac{5\times2}{8}$$

$a_2 = 1.25\ m/s^2$

52. None of the above

53. Weight is gravitational force of earth on the body.

56. Free body diagram



Net force = $Mg - F_b = 0$

(F_b = Buoyant force)

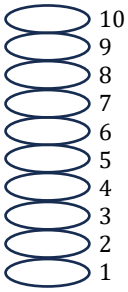
57. (1) is wrong 3rd coin has seven coins on its top which exert a force 7 mg on it.

(2) is correct. 6th coin has four coins, placed over it. Thus 7th coin exerts a force 4mg on 6th coin (downwards)

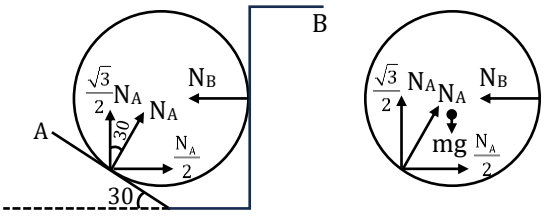
(3) is wrong,

As what is explained in (2), the reaction of 6th coin on the 7th coin is 4 mg (upwards)

(4) is wrong. 10th coin, which is the topmost coin, experiences a reaction force of mg (upwards) from all the coins below it.



59.



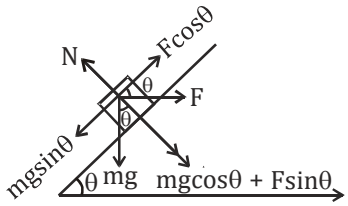
$$\because \quad \frac{\sqrt{3}}{2}N_A = mg$$

$$\therefore \quad N_A = \frac{2}{\sqrt{3}}\times50\times10$$

$$\therefore \quad N_A = \frac{1000}{\sqrt{3}}N$$

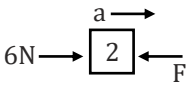
$$\therefore \quad N_B = \frac{N_A}{2} = \frac{500}{\sqrt{3}}N$$

60.



From the figure, the normal reaction on the block is $N = mg\cos\theta + F\sin\theta$.

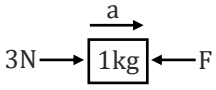
61. $a = \frac{6-3}{2+1} = 1\ m/s^2$



$6 - F = 2 \times 1$

$F = 4\ N$

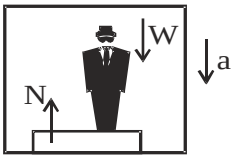
62. $a = \frac{3}{2+1} = 1\ m/s^2$



$3 - F = 1 \times 1$

$F = 2N$

64. $W - N = ma$



$\therefore \quad N = W - ma$

$\therefore \quad$ Reading $= 80 \times 10 - 80 \times 5 = 400N$

65. $\frac{W_{app_1}}{W_{app_2}} = \frac{m(g+a)}{m(g-a)} = \frac{12}{8} = \frac{3}{2}$

66. Horizontal force can not balance vertical forces.

67. Arms should be vertical for least tension.

68. $T_3 = 4g$



$T_2 = T_3 + 3g$

$T_2 = 7g$



$T_1 = T_2 + 2g$

$T_1 = 9g$

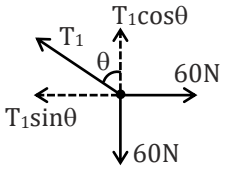


$\therefore \quad \frac{T_1}{T_3} = \frac{9g}{4g} = \frac{9}{4}$

69. $T_1\cos\theta = 60$

$T_1\sin\theta = 60$

$\therefore \tan\theta = \frac{6}{6} = 1$

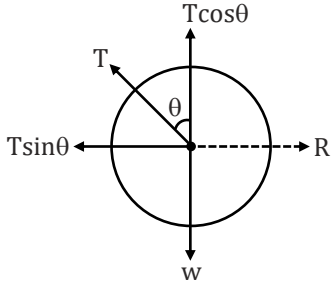


$\therefore \theta = 45°$

70. $T = mg, \quad 2T\cos\theta = mg$

$\therefore \cos\theta = \frac{1}{2}; \theta = 60°$

71.



$T\cos\theta = w$

$T = \frac{w}{\cos\theta} = \frac{w}{\frac{4}{5}}$

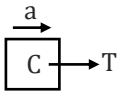
$T = \frac{5w}{4}$

72. $T\cos\theta = W \; ; \; T\sin\theta = R \; ; \; \tan\theta = \frac{R}{W}$

$T^2 = R^2 + W^2$

also as for equilibrium $\vec{R}+\vec{T}+\vec{W}=0$

73. $a = \frac{10.2}{2+2+2} = 1.7 \text{ m/s}^2$



$T = 2a = 2 \times 1.7$

$T = 3.4 \text{ N}$

74. $a = \frac{F}{(M_1+M_2+M_3)}$ ...(iv)

$T = (M_1 + M_2)a$

$= (M_1 + M_2)\, F/M_1 + M_2 + M_3 = \frac{3MF}{6M}$

$T = \frac{F}{2}$

75. $a = \frac{36}{36} = 1 \text{ m/s}^2$

$\therefore T' = (M_2 + M_3)a$

$\therefore T' = 35 \times 1 = 35 \text{ N}$

76. The common acceleration of the blocks is

$a = \frac{40-10}{10+6+4} = \frac{30}{20} = 1.5 \text{ m/s}^2$

$\therefore 40 - T_2 = m_1 a$

$\therefore T_2 = 40 - 10 \times 1.5$

$T_2 = 25\text{N}$

77. For figure 1 :-

$a = \frac{500}{25} = 20 \text{ m/s}^2$

$T = 15 \times 20 = 300 \text{ N}$

For figure 2 :-

$a = \frac{750}{25} = 30 \text{ m/s}^2$

$T' = 10 \times 30 = 300 \text{ N}$

Hence $T = T^1 = 300 \text{ N}$

78. $T_1 = (12 + 3)a$

$30 = 15a$ ...(i)

$a = 2 \text{ m/s}^2$

$F\cos 60° = 30 + (15 \times 2)$

$F = 120 \text{ N}$

79. $a = \frac{25}{10} = 2.5\ m/s^2$

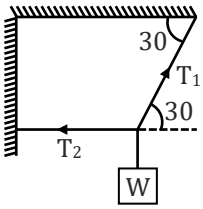
$T_A = 5 \times 2.5 = 12.5N$

$T_B = 4 \times 2.5 = 10N$

$T_C = 3 \times 2.5 = 7.5N$

80. $a = \frac{210-150}{15} = 4\ m/s^2$

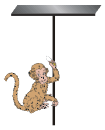
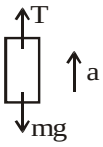
81.



$T_1 \sin 30° = W$

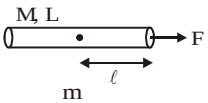
$T_1 \times \frac{1}{2} = 40$

$T_1 = 80N$

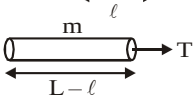
82.  FBD of monkey 

$\therefore\ T - mg = ma \qquad \because\ T_{max} = 300$

$\therefore\ 300 - 150 = ma \therefore\ a = \frac{150}{15} = 10\ m/s^2$

83. 

$a = \frac{F}{M}$



$\because\ T = m \times a$

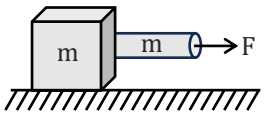
$\therefore\ T = \frac{M}{L}(L-\ell) \times \frac{F}{M}$

$\therefore\ T = F\left(1 - \frac{\ell}{L}\right)$

84. $T = \frac{M}{L} \times x \times \frac{F}{M}$

$T = \frac{Fx}{L}$

85. $a = \frac{F}{m+m} = \frac{F}{2m}$



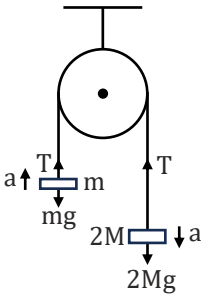
Tension at mid point $= \left(m + \frac{m}{2}\right) \times \frac{F}{2m} = \frac{3F}{4}$

86. For the particle of mass m,

$T - mg = ma$ ...(i)

For the particle of mass 2M,

$2Mg - T = 2Ma$ ...(ii)



Add (i) and (ii) to eliminate T.

$2Mg - mg = 2Ma + ma$

$g(2M - m) = a(2M + m)$

or $a = \left(\frac{2M-m}{2M+m}\right)g$ ...(iii)

Now $T - mg = m \times \left(\frac{2M-m}{2M+m}\right)g$

$T = mg + mg\left(\frac{2M-m}{2M+m}\right)$

$T = \frac{2mgM + m^2g + 2mgM - m^2g}{(2M+m)}$

$T = \frac{4mM}{(2M+m)}g$ ...(iv)

87. $a = \frac{(10+5)g - 5g}{20}$

$a = \frac{10g}{20} = \frac{g}{2}$

88. $a = \frac{10g-5g}{15} = \frac{g}{3}$

For block A :-

$T - 5g = \frac{5g}{3}$

$T = \frac{20g}{3} = 66.67\ N$

89. $T = \frac{2m_1m_2}{m_1+m_2}g$

$\therefore\ T = \frac{2 \times m \times 2m}{m+2m}g$

$\therefore\ T = \frac{4mg}{3}$

$\therefore\ F = 2T + 3mg$

$\therefore\ F = \frac{17}{3}mg = 5.67\ mg$

90. In case 'a' :- $a_a = \frac{2mg - mg}{3m} = \frac{g}{3}$

In case 'b' :- $a_b = \frac{2mg - mg}{2m} = \frac{g}{2}$

hence, $a_a < a_b$

91. $mg - T = ma$ ...(i)

$T = m'a$ ...(ii)

$\therefore \quad a = \frac{mg}{(m + m')}$

92. For case A

$15g - T = 15a$ ...(i)

$T = 5a$ ...(ii)

$\therefore \quad a = \frac{3g}{4}$

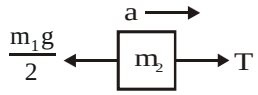
For case B

$15g - T = 15a$ ...(iii)

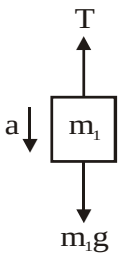
$T - 5g = 5a$ ...(iv)

$a = \frac{g}{2}$

93.



$T - \frac{m_1 g}{2} = m_2 a$ ...(1)

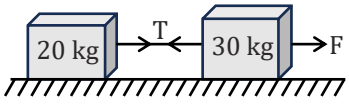


$m_1 g - T = m_1 a$ ...(2)

equation (1) + equation (2)

$a = \frac{m_1 g}{2(m_1 + m_2)}$

94.



$T = 20 \times 3 = 60N$

95. $a = \frac{1g}{1+1} = 5\ m/s^2$

$T = 1a = 1 \times 5$

$T = 5N$

96. $3g - T_2 = 3a$ ...(i)

$T_2 - T_1 = 6a$ ...(ii)

$T_1 - g = a$ ...(iii)

$\therefore$ By solving eq (i), (ii) and (iii)

$2g = 10a$

$a = 2\ m/s^2$

So, $T_1 = 10 + 2$

$T_1 = 12\ N$

and $T_2 = 3 \times 10 - 3 \times 2$

$T_2 = 24\ N$

have $\frac{T_1}{T_2} = \frac{12}{24} = \frac{1}{2}$

97. Force on block A in downward direction :-

$F_A = 3\ Mg \sin 53° = \frac{12Mg}{5}$

Force on block B in downward direction :-

$F_B = 4\ Mg \sin 37° = \frac{12Mg}{5}$

$\because \quad F_A = F_B$

hence, both the blocks remains in rest.

98. $2Mg \sin\theta - T = 2Ma$

$T = Ma$

$\therefore \quad a = \frac{2g \sin\theta}{3}$

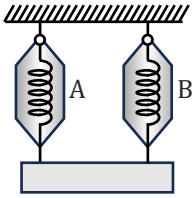
99. $Mg \times \frac{1}{2} = mg$

$\frac{M}{2} = m$

$M = 2m$

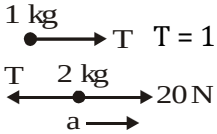
$M = 2 \times 10 = 20\ kg$

100. Both A and B reads 2 kg



101. The system, as a whole, will fall towards ground under gravity. The spring will neither be compressed nor stretched regardless of the values of m_1 and m_2.

102.

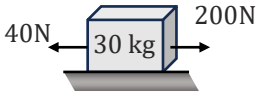


$T = 1 \times 10 = 10\ N$

$20 - T = 2 \times a$

$a = \frac{20 - 10}{2} = 5\ m/s^2$

103. $F = 20 \times 2 = 40N$



$a = \frac{160}{30} = 5.3\ m/s^2$

104. $T = 5g\sin30°$

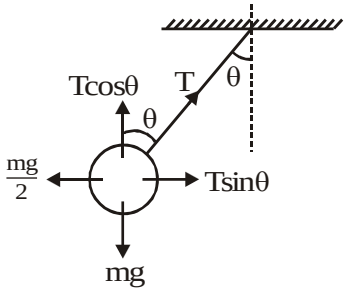
$T = 25\ N$

108. $T\cos\theta = mg$ ...(1)

$T\sin\theta = \frac{mg}{2}$ ...(2)

eqn. $(1)^2 + (2)^2$

$T^2 = (mg)^2 + \frac{(mg)^2}{4}$



$\therefore \quad T = \frac{\sqrt{5}}{2}mg$

109. Because of train's acceleration, pseudo forces acts in opposite to the direction of motion of train.

110. Force exerted by a person on the floor is less than his weight when lift has an downward acceleration and upward retardation ($g_{eff.} = g - a$).

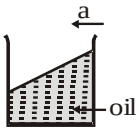
112. $\frac{m(g+a)}{m(g-a)} = \frac{3}{1}$

$(g + a) = 3(g - a) \quad \Rightarrow a = \frac{g}{2}$

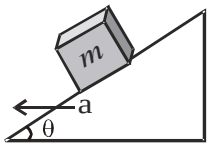
113. Zero (weightlessness)

114. $N = m(g + a) = 0.5(10 + 2) = 6N$

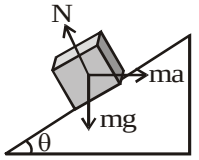
115.

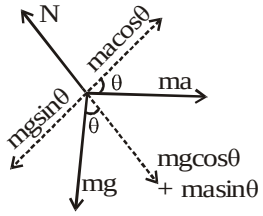


116.



FBD of m w.r.t. incline plane.





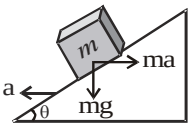
$\therefore$ $mg\sin\theta = ma\cos\theta$ ...(i)

$N = mg\cos\theta + ma\sin\theta$ ...(ii)

$\therefore N = mg\cos\theta + m\sin\theta \times g \times \frac{\sin\theta}{\cos\theta}$

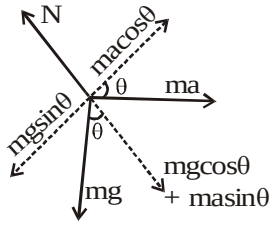
$N = \frac{mg}{\cos\theta}$

117.



According to question $\sin\theta = 1/x$ (1 in x)

So $\tan\theta = \frac{1}{\sqrt{x^2 - 1}}$



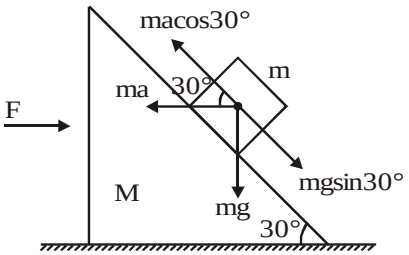
To keep the block stationary relative to the inclined plane

$mg\sin\theta = ma\cos\theta$

$a = g\tan\theta$

$\Rightarrow a = \frac{g}{\sqrt{x^2 - 1}}$

119.



$\because$ $ma\cos30° = mg\sin30°$

$\therefore$ $a = g\tan30°$

$\because$ $F = (M + m)a$

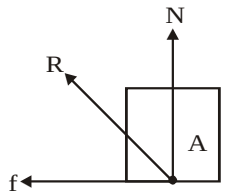
$\therefore$ $F = (M + m)g\tan30°$

120. $M.A. = \frac{Load}{Effort} = \frac{W}{F} = 1$

121. $M.A. = \frac{Load}{Effort}$

$M.A. = \frac{400}{80} = 5$

124.



125. Static friction is self adjusting force
so, $f = F_{ext} = F$

126. $f_{max} = 40 \times 0.4 = 16N$
$\therefore$ f = 8N (Applied force)

127. $f_L = \mu_s mg$
$f_L = 0.4 \times 3 \times 10 = 12$ N
F = 8.7 N
$F < f_L$
f = F = 8.7 N

128. mg – f = ma
f = m(g–a)
= 0.5 × (10–9)
= 0.5N

129. $v^2 = u^2 + 2as$
$0^2 = 100^2 - 2 \times 0.5 \times 10 \times S$ ($\because$ a = μg)
S = 1000 m

130. Pseudo force = 10 × 0.5 = 5 N
$f_{max} = 10 \times 10 \times 0.2 = 20N$
$\therefore$ f = 5 N

131. $F = f_L$
$75 = \mu_s \times 20 \times 10$
$\mu_s = 0.375$

132. $f_L = \mu_s$ mg
$f_L = 0.6 \times 1 \times 10 = 6$ N
$F = 1 \times 5 = 5N$
$\because$ $f_L > F$
hence f = F = 5N

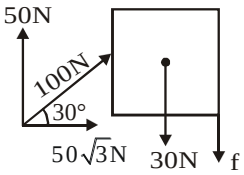
133. F – f = 10 × 5 ...(1)
2F – f = 10 × 18 ...(2)
from equation (1) & equation (2)
f = 80 N

$$\mu = \frac{f}{mg} = \frac{80}{10\times10} = 0.8$$

134. $f_L = 0.15 \times 20 = 3N$
for block to be stationary
f_L = weight of block
$\therefore$ weight of block = 3N

135. $f_{max} = 0.5 \times 25 = 12.5$ N
F = mg = 1 × 9.8 = 9.8 N
$F < f_{max}$
So, f = F = 9.8 N

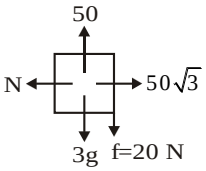
137.



$$f_L = \mu N = \frac{1}{\sqrt{3}} \times 50\sqrt{3}$$

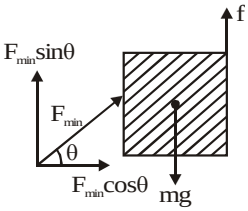
$f_L = 50$ N
$F_{ext} = 20$ N
$F_{ext} < f_L$
hence $f = F_{ext} = 20$ N

138.



$F_{net} = 0$
So, a = 0

139.



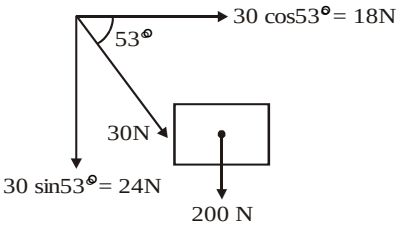
$F_{min}\sin\theta + f = mg$
$F_{min}\sin\theta + \mu F_{min}\cos\theta = mg$
$F_{min}(\sin\theta + \mu\cos\theta) = mg$

$$F_{min} = \frac{mg}{\sin\theta + \mu\cos\theta}$$

140. For, f = 0
$F\sin\theta = mg$

$$F = \frac{mg}{\sin\theta}$$

141.



$f_L = \mu_s N = 0.2 \times 224$

$f_L = 44.8$ N

$\because$ $f_L > F$

hence $f = F = 18$ N

142. $N = mg - F\sin45°$

$= 50 - 28.2 \times \frac{1}{\sqrt{2}}$

$N = 50 - 20 = 30$N

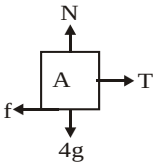
$f_s = \mu N$

$f_s = 0.8 \times 30 = 24$ N

$f_s > F_{ext}$ $\{\because F_{ext} = 28.2\cos 45° = 20N\}$

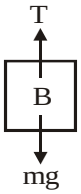
So, f = 20N

143.



$N = 40$

$f = 0.3 \times 40 = 12$ N $= T$



$mg = T$

$m = \frac{12}{10} = 1.2$ kg

144. $\mu \times \frac{4m}{5}g = \frac{m}{5}g \Rightarrow \mu = \frac{1}{4}$

145. $f_L = \mu_s mg$

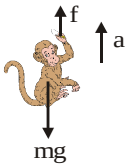
$f_L = 0.2 \times 2 \times 10 = 4$N

$F = 2 \times 1 = 2$N

$\because$ $F < f_L$

hence $f = F = 2$N

146.



$f - mg = ma$

$f = m(g + a)$

147. For constant speed, acceleration is zero

$mg\sin\theta = \mu mg\cos\theta$

$\mu = \tan\theta$

148. $0^2 - (2v_0)^2 = 2(-\mu g) \times s \Rightarrow s = \frac{2v_0^2}{\mu g}$

149. $f_{max} = \mu mg\cos\theta = 0.8 \times 2 \times 10 \times = 13.6$N

$F = mg\sin\theta = 2 \times 10 \times \frac{\sqrt{3}}{2} = 10$ N

$\because$ $F < f_{max}$, So $f = F = 10$ N

150. $F_1 = mg\sin\theta + \mu mg\cos\theta$

$F_2 = mg\sin\theta - \mu mg\cos\theta$

$F_1 = 3F_2$

$\therefore$ $\tan\theta + \mu = 3(\tan\theta - \mu)$

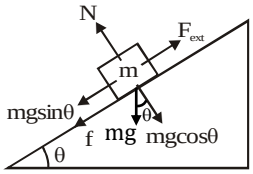
$2\mu = \tan\theta \Rightarrow \theta = \tan^{-1}(2\mu)$

151. $a = g\sin\theta - \mu g\cos\theta$

$g\left(\frac{1}{\sqrt{2}} - 0.25 \times \frac{1}{\sqrt{2}}\right) = \frac{10 \times 0.75}{\sqrt{2}} = \frac{7.5}{\sqrt{2}}$

$= 5.35$ m/s^2

152.



$f = \mu N$

$f = \mu mg\cos\theta$

$F_{ext} = mg\sin\theta + f$

$F_{ext} = mg\sin\theta + \mu mg\cos\theta$

153. $mg\sin\theta = \mu mg\cos\theta$

$\sin\theta = \mu\cos\theta$ ...(i)

when pushed up acceleration

$a = -(g\sin\theta + \mu g\cos\theta)$

$= -(g\sin\theta + g\sin\theta) = -2g\sin\theta$

$\therefore$ $v^2 - v_0^2 = 2(a) \times s$

Final velocity $v = 0$

$-v_0^2 = 2as \Rightarrow -v_0^2 = 2(-2g\sin\theta)s$

$\therefore$ $s = \frac{v_0^2}{2(2g\sin\theta)} = \frac{v_0^2}{4g\sin\theta}$

154. Let the length of incline is d

Case I : for rough incline plane

$a_r = g\sin45° - \mu g\cos45° = \frac{g - \mu g}{\sqrt{2}} = \left(\frac{1-\mu}{\sqrt{2}}\right)g$

time taken to slide down $(t_r) = \sqrt{\frac{2d}{a_r}}$

Case II : For smooth incline plane

$a_s = g\sin45° = \frac{g}{\sqrt{2}}$

$\Rightarrow \quad t_s = \sqrt{\frac{2d}{a_s}}$

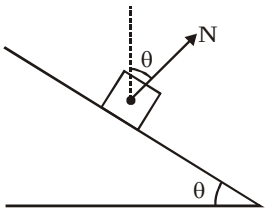
According to question

$nt_s = t_r \quad \Rightarrow \quad n^2t_s^2 = t_r^2$

$$n^2\left(\frac{2d}{g/\sqrt{2}}\right) = \frac{2d}{\left(\frac{1-\mu}{\sqrt{2}}\right)g}$$

$n^2 = \frac{1}{1-\mu} \quad \Rightarrow \quad \mu = 1 - \frac{1}{n^2}$

155.



as angle made by normal reaction with vertical = angle of inclined plane = θ
and block has to be pushed down,
So, θ < angle of repose.

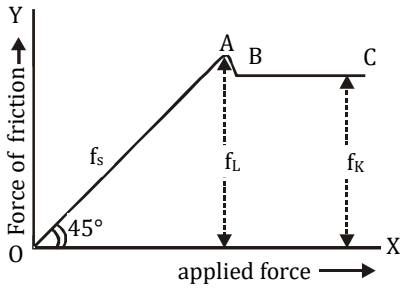
156. Angle of repose is independent of mass.

157. $\tan\theta = \mu_s$

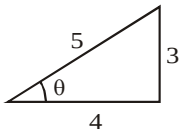
$\tan\theta = \sqrt{3}$

$\theta = 60°$

158.



159.



$\therefore \quad \tan\theta = \mu_s$

$3/4 = \mu_s$

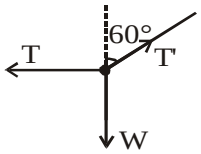
$\mu_s = 0.75$

160. $mg - Q\cos\theta = N$

$P + Q\sin\theta = \mu N$

$\therefore \quad \mu = \frac{P + Q\sin\theta}{mg - Q\cos\theta}$

161.



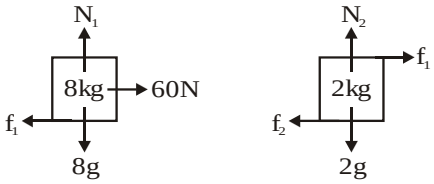
$T'\cos60° = W$

$T'\sin60° = T$

$\therefore \ \tan60° = \frac{T}{W} \Rightarrow W = \frac{T}{\sqrt{3}}$

$W = \frac{346}{1.7} = 200N$

162.



$f_{1max} = 0.5 \times 80 = 40$ N

$f_{2max} = 0.2 \times 100 = 20$ N

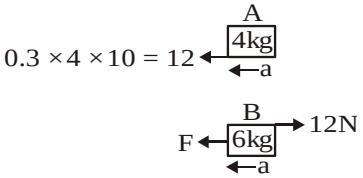
$\therefore$ combined acceleration (a) = $\frac{60-20}{10} = 4$ m/s²

from FBD of 8 kg :-

Friction between A & B

$f_1 = 60 - 8 \times 4 = 28$ N

163.



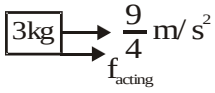
$a = \frac{12}{4} = 3$ m/s²

$F - 12 = 6 \times a$

$F = 12 + 6 \times 3 = 30$ N

164. Acceleration of system = $\frac{27}{9+3} = \frac{9}{4}$ m/s²

For upper block w.r.t lower block



$f_{max} = \frac{1}{3} \times 30 = 10N$

$f_{acting} = 3 \times \frac{9}{4} = 6.75N$

$f_{acting} < f_{max}$

So, $f = f_{acting} = 6.75$ N

Exercise-II (Previous Year Questions)

1. Acceleration

$$= \frac{\text{Net force in the direction of motion}}{\text{Total mass of system}}$$

$$= \frac{m_1 g - \mu(m_2 + m_3)g}{m_1 + m_2 + m_3}$$

$$= \frac{g}{3}(1 - 2\mu)$$

2. Change in momentum,

$\Delta p = \int F dt$

= Area of F-t graph

$= \left(\frac{1}{2} \times 2 \times 6\right) - (3 \times 2) + (4 \times 3)$

= 12 N-s

3. Let upthrust of air be F_a then

for downward motion

$mg - F_a = ma$ ...(1)

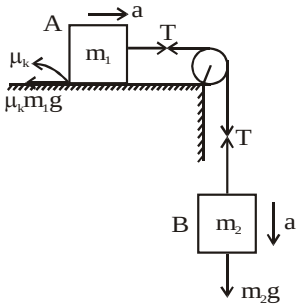
for upward motion (when Δm is removed)

$F_a - (m - \Delta m)g = (m - \Delta m)a$ ...(2)

After solving equation (1) & (2)

Therefore $\Delta m = \frac{2ma}{g + a}$

4.



For the motion of both blocks

$m_2 g - T = m_2 a$

$T - \mu_k m_1 g = m_1 a$

$$a = \frac{(m_2 - \mu_k m_1)g}{m_1 + m_2}$$

For the block of mass 'm_2'

$$m_2 g - T = m_2\left[\frac{m_2 - \mu_k m_1}{m_1 + m_2}\right]g$$

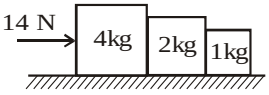
$$T = m_2 g - \left[\frac{m_2 - \mu_k m_1}{m_1 + m_2}\right]m_2 g$$

$$T = m_2 g\left[\frac{m_1 + \mu_k m_1}{m_1 + m_2}\right]$$

$$T = \frac{m_1 m_2(1 + \mu_k)g}{m_1 + m_2}$$

5. Acceleration of system $= \frac{F_{net}}{M_{total}}$

$$= \frac{14}{4 + 2 + 1} = 2 \text{ m/s}^2$$



The contact force between 4 kg & 2 kg block will move 2 kg & 1 kg block with the same acceleration

so $F_{contact} = (2 + 1)a = 3(2) = 6N$.

6. Coefficient of static friction,

$$\mu_s = \tan 30° = \frac{1}{\sqrt{3}} = 0.6$$

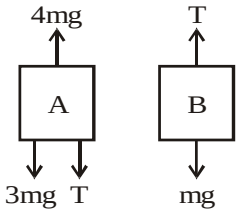
$a = g\sin 30° - \mu_K g\cos 30°$

$S = ut + \frac{1}{2}at^2$

$$4 = \frac{1}{2}\left[\frac{g}{2} - \frac{\mu_k g\sqrt{3}}{2}\right] \times 16 \Rightarrow \mu_k = 0.5$$

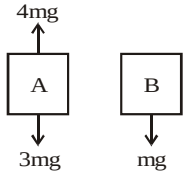
7. $\text{Impulse} = \left|\overrightarrow{\Delta p}\right| = m\left|\overrightarrow{\Delta V}\right| = m(2V\cos 60°) = mV$

8. Before cutting the string :-



$\therefore$ $T = mg$

After cutting the string :-

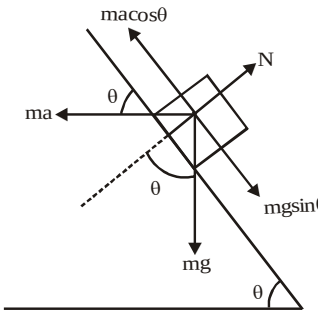


$$a_A = \frac{4mg - 3mg}{3m} = \frac{g}{3}$$

$$a_B = \frac{mg}{m} = g$$

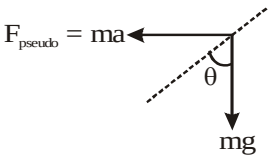
9. Coefficient of sliding friction has no dimension

10. FBD w.r.t. wedge



$ma\cos\theta = mg\sin\theta \Rightarrow a = g\tan\theta$

11.



$\tan\theta = \frac{F_{pseudo}}{mg} = \frac{a}{g}$ towards left

12. N = mg, F = f

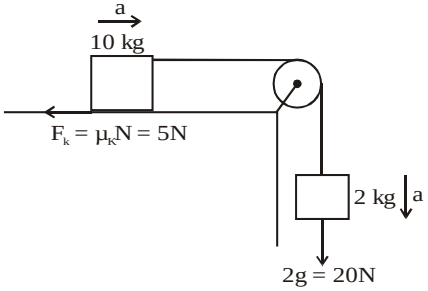
$\text{Resultant} = \sqrt{N^2 + f^2} = \sqrt{(mg)^2 + f^2}$

$\le mg\sqrt{1+\mu^2}$

13. Acceleration of system

$a = \frac{(m_2 - m_1)g}{m_1 + m_2} = \frac{(6-4)g}{6+4} \frac{2g}{10} = \frac{g}{5}$

14.

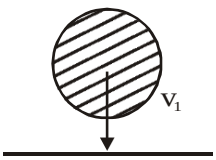


$a = \frac{\text{net force}}{\text{total mass}} = \frac{20-5}{12} = 1.25 m/s^2$

15. Velocity just before striking the ground

$v_1 = \sqrt{2gh}$

$v_1 = \sqrt{2(10)(10)} = 10\sqrt{2}$ m/s



$\vec{v}_1 = -10\sqrt{2}\,\hat{j}$

If it reaches the same height, speed remains same after collision only the direction changes.

$v_2 = 10\sqrt{2}$ m/s

$\vec{v_2} = 10\sqrt{2}\,\hat{j}$

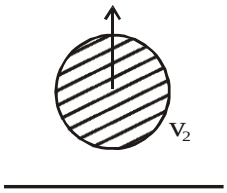
$|\text{Impulse}| = m|\Delta\vec{v}|$

$= m|10\sqrt{2}\,j - (-10\sqrt{2}\,j)|$

$= 0.15\left[2(10\sqrt{2})\right]$

$= 3\sqrt{2}$ kg m/s

$= 4.2$ kg m/s

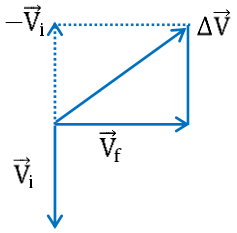


19. $\vec{V}_i = (V)$ southward

$\vec{V}_F = (V)$ Eastward

$\overrightarrow{\Delta V} = \vec{V}_F - \vec{V}_i$

= Along North – East



20. $F_S = ma$

$f_L = ma_{max}$

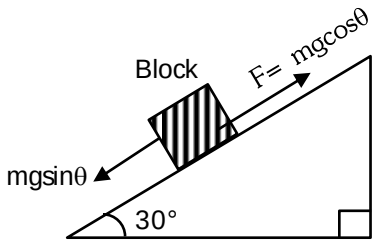
$\mu mg = ma_{max}$

$a_{max} = \mu g$

$= 0.15(10)$

$= 1.5$ m/s²

21.



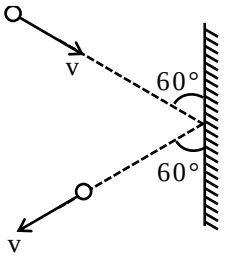
$\mu = \frac{1}{\sqrt{3}}$

$\tan 30^\circ = \frac{1}{\sqrt{3}}$

as $\mu = \tan\theta$

the block is at rest and net force on it must be zero

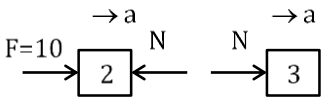
22.



$$F=\left|\frac{\Delta\vec{p}}{\Delta t}\right|=\frac{2mv\sin\theta}{t}=\frac{2(1)(1)\sin 60^\circ}{0.1}=10\sqrt{3}\,N$$

23. From Newton's II[nd] law.

$F_{net} = ma$



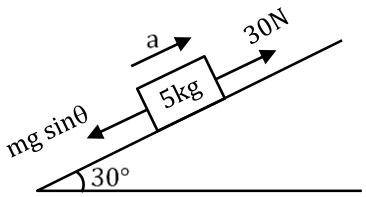
Block A : F – N = 2a or 10 – N = 2a .....(i)

Block B : N = 3a .....(ii)

On solving (i) & (ii)

$a = 2\ m/s^2$ and $N = 6\ N$

24.



$30 - Mg\sin\theta = ma$

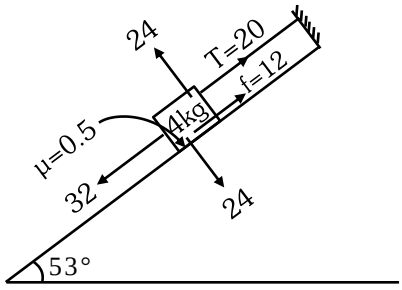
$30 - 5(10)\sin 30^\circ = 5a$

$5 = 5a$

$a = 1\ ms^{-2}$

Exercise-III (Analytical Questions)

1.



$f = f_{max} = \mu N = 0.5 \times 24$

$\Rightarrow \boxed{f = 12N}$

& $T + f = 32$

$\Rightarrow \boxed{T = 20N}$

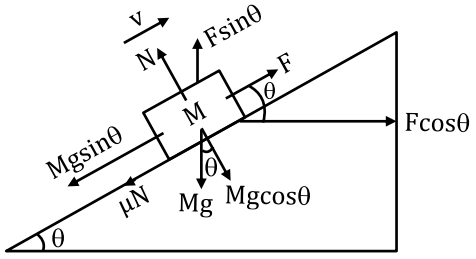
Contact force $= \sqrt{f^2+N^2} = \sqrt{12^2+24^2}$

$= 12\sqrt{5}$

$\simeq 26.83\ N$

$\simeq 27\ N$

2. $F = Mg\sin\theta + \mu N = Mg\sin\theta + \mu Mg\cos\theta$



Net force in horizontal direction = $F\cos\theta$

$= (Mg\sin\theta + \mu Mg\cos\theta)\cos\theta$

$= Mg\sin\theta\cos\theta + \mu Mg\cos^2\theta$

Net force in vertical direction = $F\sin\theta$

$= (Mg\sin\theta + \mu Mg\cos\theta)\sin\theta$

$= Mg\sin^2\theta + \mu Mg\cos\theta\sin\theta$

Net force along the plane

$= F = Mg\sin\theta + \mu Mg\cos\theta$

Net force perpendicular to plane = 0

3. $\vec{F}_{net\ ext} = \frac{d\vec{P}}{dt}$ (Newton's second law)

4. Pseudo force does not actually act on a body. It is only drawn in an F.B.D. when the frame of reference is non-inertial.

5. By definition of 'equilibrium' & 'rest'

6. Acceleration of blocks is $2\ m/s^2$.

Displacement $= \frac{1}{2} \times 2 \times 1^2 = 1$ m.

One moves 1 m upwards and the other 1 m downwards. Separation = 2 m

8. When a body is placed on a rough inclined plane, it is acted upon by a reactional force due to plane [electromagnetic in nature], a frictional force due to roughness of plane [electromagnetic in nature] and a gravitational force (due to its weight).

9. Statement (I) follows Newton's first law, i.e. law of inertia.

Statement (II) follows Newton's third law, i.e. action-reaction pair.

AIM: FREE EDUCATION TO ALL APUL

Work, Energy and Power

SOLUTIONS

Exercise-I (Conceptual Questions)

1. $W = \vec{F}.\vec{d}$

$= (4\hat{i} - 4\hat{j} - 2\hat{k}) \cdot (3\hat{i} + 2\hat{j} - 2\hat{k})$

= 12 – 8 + 4 = 8 J

2. $\vec{F} = (3\hat{i} + 2\hat{j}) \cdot (4\hat{i} - \hat{j} + 3\hat{k})$

= 12 – 2 = 10 J

3. $W = \vec{F} \cdot \vec{S}$

$\vec{S} = \vec{r}_2 - \vec{r}_1 = (4\hat{i} + 3\hat{j} - \hat{k}) - (2\hat{i} + \hat{j} - 2\hat{k})$

$\vec{S} = 2\hat{i} + 2\hat{j} + \hat{k}$

$W = (3\hat{i} + 2\hat{j} + 2\hat{k}) \cdot (2\hat{i} + 2\hat{j} + \hat{k})$

= 6 + 4 + 2 = 12 J

4. $\vec{d} = \vec{r}_2 - \vec{r}_1$

$= (\hat{i} + 2\hat{j} + \hat{k}) - (3\hat{i} + \hat{j} + 3\hat{k})$

$= (-2\hat{i} + \hat{j} - 2\hat{k})$

$W = \vec{F}.\vec{d}$

$= (5\hat{i} + 2\hat{j} - 4\hat{k}) \cdot (-2\hat{i} + \hat{j} - 2\hat{k})$

= –10 + 2 + 8 = 0

5. $W_N = Fd\cos 90°$

= 0

6. $W_F = FS\cos\theta$

$= (5g) \times 2 \times \cos 0°$

$= 5 \times 9.8 \times 2 \times 1$

= 98 J

7. Work will be zero as angle between force & displacement is 90°

8. Angle between Force and displacement is 90°

$\therefore$ W = 0

9. W = mgH

independent of time

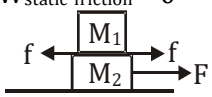
10.



(Body at rest)

F = f

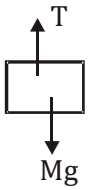
$W_{\text{static friction}} = 0$



If blocks move together then work done by static friction between blocks on upper block = +ve & lower block = –ve.

11. Block B will accelerate due to friction and gain speed therefore work by friction on block B will be positive.

12.



$Mg - T = M \times \frac{3g}{4}$

$Mg - \frac{3Mg}{4} = T$

$\frac{Mg}{4} = T$

$W = T \times h\cos 180° = -\frac{Mgh}{4}$

13. Work done by T on M = – Work done by T on m

$\therefore$ net work done by T = 0

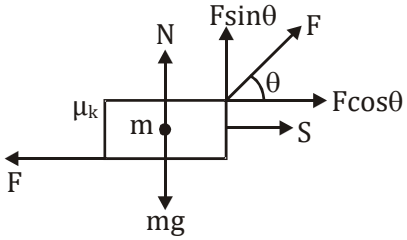
14. $W = F\Delta S\cos\theta$

$= \mu R \times 2\pi r \times \cos 180$

$= -0.5 \times 5 \times 2\pi \times \frac{1}{\pi}$

= – 5 J

15.



N = mg – Fsinθ

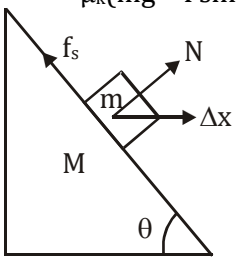
$f = \mu_k N$

$f = \mu_k(mg - F\sin\theta)$

$W_f = f \times S\cos 180°$

$= -\mu_k(mg - F\sin\theta)S$

16.



angle between N & Δx is not 90°

$\therefore W_N \neq 0$

17. Work in each direction = 100 J

$F\Delta S = 100$

$\Delta S = 2m$

displacement in each direction = x = y = z = 2m

Final coordinates = (2, 2, 2)m

18. Work done by external agent = $-(W_{friction} + W_{mg})$

$= -(-mgh - mgh)$

$= 2mgh$

19. $\int_0^3 (x^2+3x-5)dx = \left[\frac{x^3}{3}+\frac{3x^2}{2}-5x\right]_0^3$

$= \frac{27}{3}+\frac{3\times 9}{2}-5\times 3$

$= 9+\frac{27}{2}-15$

$= \frac{18+27-30}{2} = 7.5\ J$

20. $W = \int_{x_1}^{x_2} F dx \cos 60°$

$= \int_{x_1}^{x_2} kx^3 \cos 60° dx$

$= \left[\frac{kx^4}{4}\right]_{x_1}^{x_2} \times \frac{1}{2}$

$= \frac{k}{8}(x_2^4 - x_1^4)$

21. Work done = Area under F-x curve

$W = \frac{1}{2}(6+3)\times 6$

$W = 27\ J$

22. Work done is the area under F-x curve

$W = m$[area under a-x curve]

(i) $W = 4\left[\frac{1}{2}(4+3)\times 3\right]$

$W = 42\ J$

(ii) $W = 4\left[\frac{1}{2}(5+3)\times 3\right] - 4\left[\frac{1}{2}(2+1)\times 3\right]$

$W = 48 - 18 = 30\ J$

23. Area under the graph

24. $KE \propto P^2$

$\frac{P_1}{P_2} = \sqrt{\frac{K_1}{K_2}}$

$\therefore \quad P_2 = 3P_1$

25. $K \propto P^2$

So if P becomes 'n' times then K becomes n^2 times.

26. $KE \propto P^2$

$\frac{P_1}{P_2} = \sqrt{\frac{K_1}{K_2}}$

$P_2 = 2P_1$

27. $\frac{P_1^2}{2m_1} = \frac{P_2^2}{2m_2}$

$\frac{P_1^2}{P_2^2} = \frac{m_1}{m_2} = \frac{4}{9} \Rightarrow \frac{P_1}{P_2} = \frac{2}{3}$

28. P = constant

So $K \propto \frac{1}{m}$

$K_1 : K_2 \Rightarrow m_2 : m_1$

$= 5 : 3$

29. $P \propto \sqrt{K}$

$\frac{P_1}{P_2} = \sqrt{\frac{K_1}{K_2}}$

$P_2 = 2P_1$

$\therefore$ momentum is increased by 100%

30. $\frac{P_1}{P_2} = \sqrt{\frac{K_1}{K_2}}$

$P_2 = 3P_1$

$\therefore$ Momentum increased by 200%

31. $P \propto K^{1/2}$

% change in P = $\frac{1}{2}$ {% change in K}

% change in P = $\left(\frac{1}{2}\times 3\right)\% = 1.5\%$

32. $\because K \propto P^2$

% change in K = 2 (% change in P)

% change in K = (2 × 2) % = 4%

33. $E_k = \frac{p^2}{2m}$

$\sqrt{E_K} \times \frac{1}{p}$ = constant

$\therefore$ graph is rectangular hyperbola

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34. $E_K = \frac{1}{2}mv^2$ $\therefore$ Graph is parabola

35. W = ΔKE

$= \frac{1}{2}m[v^2 - u^2]$

36. $s = \frac{t^3}{4}$

$v = \frac{3t^2}{4}$

at t = 0, v = 0

at t = 2sec, $v = \frac{3(2)^2}{4} = 3$ m/s

By W.E.T.

W = ΔKE

$W = \frac{1}{2} \times 6 \times 3^2 - \frac{1}{2} \times 6 \times 0^2$

W = 27 J

37. $x = 2t - 3t^2 + 2t^3$

$v = 2 - 6t + 6t^2$

W = ΔKE

$= \frac{1}{2}m(v_2^2 - v_1^2)$

t = 0 sec $\Rightarrow$ $v_1 = 2$

t = 3 sec $\Rightarrow$ $v_2 = 38$

$W = \frac{1}{2} \times \frac{20}{1000}[1444 - 4]$

$= \frac{1440}{100} = 14.40$ J

38. $\frac{KE_1}{KE_2} = \frac{\frac{1}{2}m_1v^2}{\frac{1}{2}m_2v^2} = \frac{m_1}{m_2} = \frac{1}{3}$

39. $E = \frac{1}{2}mv^2 - mgh$

$= \frac{1}{2} \times 2 \times 256 - 2 \times 10 \times 10$

= 256 – 200 = 56 J

40. $W_{all} = \Delta KE$

$mgH = \frac{1}{2}m(v^2 - u^2)$

$2gH = v^2 - u^2$

$u \rightarrow$ same and $H \rightarrow$ same $\Rightarrow$ $v \rightarrow$ same

41. W = ΔKE

$= \frac{1}{2}m[0^2 - 10^2]$

$= \frac{1}{2} \times 2[-100] = -100$ J

42. Work done by a conservative force is independent from the path

$W_C = -\Delta U = U_i - U_f$

43. Work done is independent from the path for conservative forces.

44. Gravitation is conservative force, work done is path independent

45. Viscous force (fluid friction) $\Rightarrow$ non conservative force

47. Here gravitation force of earth is centripetal force θ = 90°, Work done = 0

48. $W_{PQ} + W_{QR} + W_{RP} = 0$

$15 + 5 + W_{RP} = 0$

$W_{RP} = -20$ J

$W_{PR} = 20$ J

49. Since force is constant so work done is path independent. Hence $W_1 = W_2$

50. W = –ΔU

W = mgh

= 3 × 10 × 20 = 600 J

51. $W = m_{avg}gh$

$= \left(\frac{13+9}{2}\right) \times 10 \times 10$

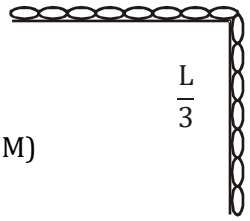
= 22 × 50 = 1100 J

52. W = mgh

$m = \frac{M}{L} \times \left(\frac{L}{3}\right) = \frac{M}{3}$

$h = \left(\frac{L}{3}\right) \times \frac{1}{2} = \frac{L}{6}$ (By COM)

$WD = \frac{M}{3} \times \frac{L}{6} \times g = \frac{MgL}{18}$



53. $W = \Delta U = U_{max} - U_{min} = mg\left(\frac{a}{2} - \frac{c}{2}\right)$

54. T = Kx for spring

$Energy = \frac{1}{2}Kx^2 = \frac{1}{2}K\frac{T^2}{K^2} = \frac{T^2}{2K}$

55. $W=\frac{1}{2}k\left(x_2^2-x_1^2\right)=\frac{1}{2}\times\frac{1000}{10^4}\left(25^2-15^2\right)$

$=\frac{1}{2}\times\frac{1}{10}[625-225]=\frac{1}{2}\times\frac{1}{10}\times400=20\text{ J}$

56. $\frac{1}{2}K(5)^2=U$

$\frac{1}{2}K(10)^2=U'$

$\frac{U}{U'}=\frac{5^2}{10^2}=\frac{25}{100}=\frac{1}{4}\Rightarrow U'=4U$

57. $W_1=\frac{1}{2}Kx_1^2$ ...(1)

$W_2=\frac{1}{2}Kx_2^2$ ...(2)

$\frac{W_2}{W_1}=\left(\frac{x_2}{x_1}\right)^2=\left(\frac{10}{5}\right)^2$

$W_2=4W_1$

Extra work done = $W_2 - W_1$

$= 4W_1 - W_1 = 3W_1$

$= 3 \times 5 = 15$ J

58. $U=\frac{1}{2}mv^2$

$\therefore\ m=\frac{2U}{v^2}$

59. By COME

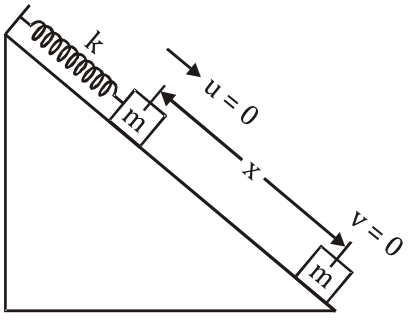
$\frac{1}{2}mv^2=\frac{1}{2}kx^2$

$4 \times (2)^2 = 100 \times x^2$

$\frac{4\times4}{100}=x^2\Rightarrow\frac{16}{100}=x^2$

$x=0.4\text{ m}$

60.



$W_{all} = \Delta KE$

$W_{mg} + W_{sp} = 0$

$mgx\sin\theta - \frac{1}{2}Kx^2 = 0$

$x=\frac{2mg\sin\theta}{K}$

61. $W = mgh = 2 \times 9.8 \times 10 = 196$ J

62. At $H=\frac{h}{5}$

$PE=\frac{mgh}{5}$

$\therefore\quad KE=\frac{4mgh}{5}$

KE : PE = 4 : 1

63. To reach point C it have to reach at point B

By COME

$\frac{1}{2}mv^2+0=mgH+0$

$v=\sqrt{2gH}$

$=\sqrt{2\times10\times320}=80\text{ m/s}$

64. In projectile motion, there is no change in kinetic energy of projectile while landing to the ground and projected from ground. Because speed is same at the time of projection and striking.

65. By COME

$-\frac{Mg}{2}\frac{L}{4}+0=-\frac{MgL}{2}+\frac{1}{2}Mv^2$

$\frac{1}{2}Mv^2=\frac{MgL}{2}-\frac{MgL}{8}$

$\frac{v^2}{2}=\frac{gL}{2}\left(\frac{3}{4}\right)$

$v=\sqrt{\frac{3gL}{4}}$

66. For equilibrium $\frac{dU}{dx}=0$, point a, b, c, d

For stable equilibrium $\frac{d^2U}{dx^2}>0$, point a, c

For unstable equilibrium $\frac{d^2U}{dx^2}<0$, point b, d

67. For stable equilibrium

$F = 0, \frac{dU}{dx}=0$

$\frac{d^2U}{dx^2}>0$

68. $F=-\frac{dU}{dx}$

For equilibrium $\frac{dU}{dx}=0$

$4x - 2 = 0$

$x=\frac{2}{4}=0.5m$

69. $\because F = 2x^2 - 3x - 2$

For equilibrium

$F = 0$

$2x^2 - 3x - 2 = 0$

$2x^2 - 4x + x - 2 = 0$

$2x(x - 2) + 1(x - 2) = 0$

$x = 2 \qquad x=-\frac{1}{2}$

$\frac{dF}{dx}=4x-3$

at $x = 2$ $\qquad$ at $x=-\frac{1}{2}$

$\frac{dF}{dx}=5$ $\qquad$ $\frac{dF}{dx}=-5$

$\frac{dF}{dx}>0$ unstable equil. $\qquad$ $\frac{dF}{dx}<0$ stable equil.

70. $U=\frac{A}{r^{12}}-\frac{B}{r^6}$

$\frac{dU}{dr}=0$ at equilibrium

$\therefore \quad \frac{-12A}{r^{13}}-\frac{(-6)B}{r^7}=0 \; ; \; \frac{6}{r^7}\left[\frac{-2A}{r^6}+B\right]=0$

$r=\left(\frac{2A}{B}\right)^{1/6}$

$\therefore$ At equilibrium U is given by

$U=\frac{A}{\left(\frac{2A}{B}\right)^2}-\frac{B}{\frac{2A}{B}}=\frac{B^2}{4A}-\frac{B^2}{2A}=-\frac{B^2}{4A}$

71. $U=\frac{2x^2y}{z^2}$

$\vec{F}=-\left[\frac{\partial U}{\partial x}\hat{i}+\frac{\partial U}{\partial y}\hat{j}+\frac{\partial U}{\partial z}\hat{k}\right]$

$\vec{F}=-\left[\frac{2y}{z^2}(2x)\hat{i}+\frac{2x^2}{z^2}\hat{j}+2x^2y\left(\frac{-2}{z^3}\right)\hat{k}\right]$

$\vec{F}=-\frac{4xy}{z^2}\hat{i}-\frac{2x^2}{z^2}\hat{j}+\frac{4x^2y}{z^3}\hat{k}$

72. $P=\vec{F}\cdot\vec{v}$

$= 5000 \times 2 = 10$ KW

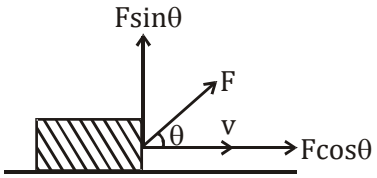
73. $F=\frac{P}{v}=\frac{50\times746}{36\times\frac{5}{18}}=3.73\times10^3$ N

74. $P = Fv$

$7 \times 10^3 = F \times 10$

$F = 700$ Newton

75.



Power $=\vec{F}.\vec{v} = Fv\cos\theta$

76. $\overrightarrow{\Delta x}=\vec{d}_f-\vec{d}_i$

$=8\hat{i}+6\hat{j}$

$P=\frac{W}{t}=\frac{\vec{F}\cdot\overrightarrow{\Delta x}}{t}=\frac{(3\hat{i}+4\hat{j})\cdot(8\hat{i}+6\hat{j})}{6}$

$=\frac{24+24}{6}=8$ Watt

77. $P_1=\frac{WD_1}{t_1}, \quad P_2=\frac{WD_2}{t_2}$

$\frac{P_1}{P_2}=\frac{m_1}{m_2}\times\frac{t_2}{t_1}=\frac{2}{3}\times\frac{9}{5}=\frac{6}{5}$

78. $P_{avg}=\frac{W}{t}=\frac{mgh}{t}=\frac{200\times9.8\times3}{4}=1470$

79. $T - 4g = 4 \times g/4$

$T = 4g + g = 5g = 50$ N

$S=\frac{1}{2}\times\left(\frac{10}{4}\right)\times(2)^2$

$S = 5m$

$P=\frac{TS}{t}=\frac{50\times5}{2}=125$ Watt

80. $P=\frac{mgh}{t}$

$=\frac{50\times9.8\times10}{10}=490$ watt

$P=\frac{490}{746}=0.66$ hp

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81. Mass of water
$= 1492 \times 10^{-3} \times 10^{3} = 1492$ kg
Energy = mgh
$= 1492 \times 10\ (16 + 8)$
$P = \frac{\text{Energy}}{\text{Time}}$
$2 \times 746 = \frac{1492 \times 10 \times 24}{t}$
t = 240 sec = 4 minutes

82. $P = \frac{mgh}{t}$
$= 50 \times 10 \times 50$
$= 25$ kW

83. $a = \frac{F}{m}$; $s = \frac{1}{2}at^2$ & $v = at$
Average power (P)
$= \frac{W}{t} = \frac{Fs}{t} = \frac{F\left(\frac{1}{2}at^2\right)}{t} = \frac{1}{2}Fv$

84. $P_{avg} = \frac{F\Delta x}{t} = FV_{avg} = kmg \times \left(\frac{u+0}{2}\right)$
$P_{avg} = \frac{1}{2}mukg$

85. $P = \frac{dK}{dt} = 2t$
$\int_{K_i}^{K_f} dK = \int_{0}^{4} 2t\ dt$
$\Delta K = \left[t^2\right]_0^4 = 16$ J

86. $FV = \frac{3t^2}{2}$
$\int_0^v mvdv = \int_0^2 \frac{3t^2}{2}dt$
$\frac{mv^2}{2} = \left[\frac{t^3}{2}\right]_0^2$
$2 \times v^2 = 8$
$v^2 = 4$
$v = 2$ m/s

87. $P = mav \Rightarrow P = m\left(v\frac{dv}{dx}\right).v$
$\Rightarrow mv^2dv = Pdx$
$\Rightarrow \frac{mv^3}{3} = Px \Rightarrow x \propto v^3$

88. $F = mg\sin\theta$
$F = 2 \times 10 \times \frac{1}{30} = \frac{2}{3}$N
$\text{Efficiency} = \frac{P_{out}}{P_{in}}$
$\frac{40}{100} = \frac{\frac{2}{3} \times 60}{P_{in}}$
$P_{in} = 100$ Watt

89. Amount of water flowing per unit time $\frac{dm}{dt} = Av\rho$
v = velocity of flow, A = is area of cross–section,
ρ = density of water
To get n times water in the same time,
$\left(\frac{dm}{dt}\right)' = n\ \frac{dm}{dt} \Rightarrow A\ v'\rho = nAv\rho \Rightarrow v' = nv$
$F = \frac{vdm}{dt}$ and $F' = v'\frac{dm'}{dt} = n^2\ v\frac{dm}{dt} = n^2F$
To get n times water, force must be increased n^2 times.

Exercise-II (Previous Year Questions)

1. $W = -\int Fdx$
$W = -\int_{20}^{30} 0.1x\,dx$
$W = -0.1\left[\frac{x^2}{2}\right]_{20}^{30}$
$W = -0.1\left[\frac{900-400}{2}\right] = -25$J
From work energy theorem $W = K_f - K_i$
$-25 = K_f - \frac{1}{2} \times 10(10)^2$
$K_f = 475$ J

2. $P = Fv = mav$
$k = mv\frac{dv}{dt}$
By integrating the equation
$\int v\,dv = \int \frac{k}{m}dt$
$\frac{v^2}{2} = \frac{k}{m}t \Rightarrow v = \sqrt{\frac{2k}{m}t}$

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$$a = \frac{dv}{dt} = \sqrt{\frac{2k}{m}}\left(\frac{1}{2}t^{-1/2}\right) = \sqrt{\frac{k}{2m}}\,t^{-1/2}$$

$$\therefore F = ma = m\sqrt{\frac{k}{2m}}\,t^{-1/2}$$

$$\therefore F = \sqrt{\frac{mk}{2}}\,t^{-1/2}$$

3. $\vec{F} = 2t\hat{i} + 3t^2\hat{j} \Rightarrow m\frac{d\vec{v}}{dt} = 2t\hat{i} + 3t^2\hat{j}$ {m = 1 kg}

$$\Rightarrow \int_0^{\vec{v}} d\vec{v} = \int_0^t (2t\hat{i} + 3t^2\hat{j})dt \Rightarrow \vec{v} = t^2\hat{i} + t^3\hat{j}$$

Power = $\vec{F}.\vec{v} = (2t^3 + 3t^5)W$

4. $\vec{s} = \vec{r}_f - \vec{r}_i = 2\hat{i} - \hat{j} + 3\hat{k}$

$$W = \vec{F}.\vec{s} = \left(4\hat{i} + 3\hat{j}\right).\left[2\hat{i} - \hat{j} + 3\hat{k}\right] = 8 - 3 = 5J$$

5. Work done by the gravity (W_g) = mgh
$= 10^{-3} \times 10 \times 10^3 = 10$ J
By work–energy theorem :- $W_g + W_{res} = \Delta KE$

$$10 + W_{res} = \frac{1}{2} \times 10^{-3} \times (50)^2$$

$W_{res} = -8.75$ J

6. $W = \int_{y_1}^{y_2} F\,dy$

$$W = \int_0^1 (20 + 10y)dy$$

$$W = 20\left[y\right]_0^1 + 10\left[\frac{y^2}{2}\right]_0^1$$

W = 25 J

7. W = ΔKE

for x = 8, $130 = \frac{1}{2}\left(\frac{1}{2}\right)v^2$

$v = 2\sqrt{130} = 22.8\ ms^{-1}$

For x=12 :-

$$130 - 40 - 5 + 20 = \frac{1}{2}\left(\frac{1}{2}\right)v^2$$

v = 20.6 m/s

8. U + KE = E and K.E = 3U
4U = E = mgS
4mgh = mgS

$$h = \frac{S}{4}$$

$$V = \sqrt{2g\left(\frac{3S}{4}\right)} = \sqrt{\frac{3gS}{2}}$$

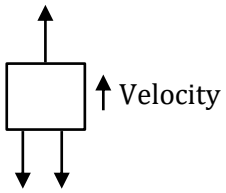
9. $P_{in} = \frac{mgh}{t} = \frac{15 \times 10 \times 60}{1} = 9000$ W

P_{out} = 90% of P_{in}

$\Rightarrow$ 8.1 kW

10. Constant velocity $\Rightarrow$ a = 0

$\Rightarrow$ T = W + f
= 20000 + 3000
= 23000 N



$\Rightarrow$ Power = Tv
= 23000 × 1.5
= 34500 watts

11. Gravitational force is a conservative force so W.D. is independent of path taken between two points.

12. $U = \frac{1}{2}kx^2$

for x = 2

$U = \frac{1}{2}k(2)^2$ ...(1)

$U' = \frac{1}{2}k(8)^2$ ...(2)

Eq. (2)/eq. (1)

$$\Rightarrow \frac{U'}{U} = \left(\frac{8}{2}\right)^2$$

$$\Rightarrow \boxed{U' = 16U}$$

13. $P = \vec{F}.\vec{V}$

$$P = (10\hat{i} + 10\hat{j} + 20\hat{k}).(5\hat{i} - 3\hat{j} + 6\hat{k})$$

P = 50 – 30 + 120

14. x = 2t –1

$$\frac{dx}{dt} = v = 2\ m/s$$

$$P = \vec{F}.\vec{v} = 5(2) = 10\ watt$$

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15. $W = K_f - K_i$

$\vec{F}.\vec{S} = K_f - K_i$

$\left(-2\hat{i}+3\hat{j}\right).3\hat{i} = K_f - 10$

$-6 = K_f - 10$

$\Rightarrow K_f = 4\ J$

16. Let final t_{height} be h_f

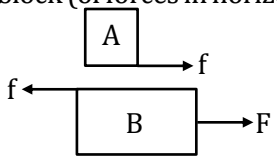
$E_f = \frac{1}{2}E_i$

$mg\,h_f = \frac{1}{2}\,mgH$

$\Rightarrow h_f = \frac{10}{2} = 5m$

Exercise-III (Analytical Questions)

3. F.B.D. of block (of forces in horizontal direction)



It can be seen that W.D. by friction (f) on block A is positive and on block B is negative.

4. Statements (a) & (b) are incorrect because according to work energy theorem, work done by all forces (Internal, External, Conservative, Non conservative) is equal to change in kinetic energy.

5. Assertion is correct because according to work energy theorem work done by all forces (Internal, External, Conservative, Non conservative) is equal to change in kinetic energy.

Therefore, kinetic energy of system may increase due to internal force only.

Reason is correct.

6. Assertion is correct, because work done is positive if angle between $\vec{F}$ & $\vec{S}$ is acute.

Reason is false:- Work done by friction may be positive, negative or zero.

8. $F = -\frac{\partial U}{\partial x}\hat{i} - \frac{\partial U}{\partial y}\hat{j} - \frac{\partial U}{\partial z}\hat{k} \Rightarrow -\left(4x\hat{i} + 9y^2\hat{j} + 2\hat{k}\right)$

Circular Motion

SOLUTIONS

Exercise-I (Conceptual Questions)

1. $a_C = \frac{v^2}{r} = \frac{4}{r^2} \Rightarrow v^2 = \frac{4}{r}$

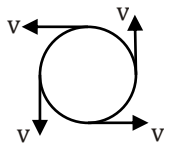
$\Rightarrow \quad v = \frac{2}{\sqrt{r}}$

Momentum $p = mv = \frac{2m}{\sqrt{r}}$

2. Uniform motion $a_t = 0$ and $a_C = \omega^2 r = v\omega$

So acceleration $a_C = v\omega$

3. $\omega = \frac{\theta}{t} = \text{const.}$



– Uniform circular motion

– Magnitude of velocity is constant but the direction of velocity change

4. $r = 12$ cm ; $n = \frac{7}{100} = 0.07$ rev./sec

$\because \quad \omega = 2\pi n \quad \therefore \quad v = \omega r = 2\pi nr$

$v = 2 \times \frac{22}{7} \times \frac{7}{100} \times 12 = 5.28$ cm/sec.

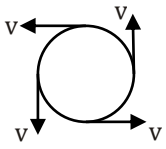
$\simeq 5.3$ cm/sec.

5. $\omega = 2\pi\, n = 2\pi \times 2 = 4\pi$ rad/sec

$\because \quad a_t = 0$ (because ω = const.)

$\therefore \quad a_{net} = a_C = \omega^2 r = 16\pi^2 \times 0.25 = 4\pi^2$

6.



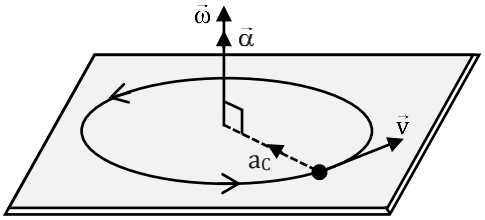
7. $T = m\omega^2 r$,

for constant r $= \frac{T_2}{T_1} = \frac{\omega_2^2}{\omega_1^2} = \frac{n_2^2}{n_1^2}$

$\frac{2T}{T} = \frac{n_2^2}{(5)^2} \quad \Rightarrow \quad n_2^2 = 25 \times 2$

$\Rightarrow \quad n_2 = 7$ r.p.m.

8.



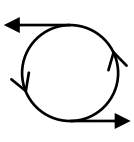
10. $\omega = \frac{d\theta}{dt} = \frac{d(2t^3 + 0.5)}{dt} \Rightarrow 6t^2 + 0$

At t = 2 $\quad \omega = 24$ rad/s.

11. WD = F.d cos 90° = 0

12. At constant angular velocity, angular acceleration i.e., $\alpha = 0$

13. $-mv\,\hat{i}$



$mv\,\hat{i}$

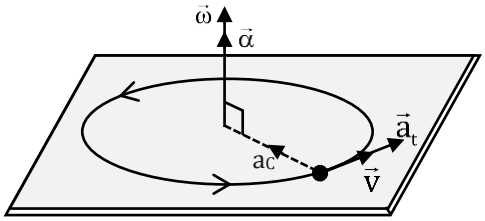
$|\Delta p| = \left|\left(\vec{p_f} - \vec{p_i}\right)\right| = 2mv$

14. Acceleration $= \frac{v^2}{r} = \frac{r^2\omega^2}{r} = r\omega^2$

r = 50 cm $\quad \omega = 2\pi n = 2\pi \times \frac{1}{2}$

acceleration $= \frac{50 \times 4\pi^2}{4} = 493$ cm/s²

15.



16. Angular velocity $(\omega) = 2\pi n = 2\pi \times \frac{100}{60}$

$= 10.47$ rad/sec

17. $T = \frac{2\pi}{\omega}$, For particle of mass M, $T_1 = \frac{2\pi}{\omega_1}$

For particle of mass m, $T_2 = \frac{2\pi}{\omega_2}$

since $T_1 = T_2$

$\frac{2\pi}{\omega_1} = \frac{2\pi}{\omega_2} \quad \text{i.e} \frac{\omega_1}{\omega_2} = 1$

18. $\omega = \frac{2\pi}{60\times60} = \frac{\pi}{1800}$ rad/sec.

19. $a = \sqrt{a_T^2 + a_c^2} = \sqrt{(2)^2 + \frac{(30)^2}{500}} = 2.7\ m/s^2$

20. $\theta = 90°$, WD = 0

21. $v = \omega r = 2\pi \times 3 \times 0.1 = 1.88$ m/s

$a_c = \frac{v^2}{r} = \frac{(1.88)^2}{0.1} = 35.5\ m/s^2$

$T = \frac{mv^2}{r} = 1\times35.5 = 35.5N$

22. Distance $s = 2(2\pi r) = 80m$

by $v^2 = u^2 + 2a_Ts$

$a_T = \frac{v^2 - u^2}{2s} = \frac{(80)^2 - 0}{2(80)}$

$\boxed{a_T = 40m/s^2}$

23. $v = r\omega \Rightarrow v = 0.5 \times 70 = 35$ m/s

24. $\omega = \frac{2\pi}{T} = 2\times\pi\times\frac{22}{44} = \pi$

centripetal acc. $a_C = \omega^2 r = \pi^2 \times 1 = \pi^2\ m/s^2$ (towards the center)

25. $\omega_0 = 2\pi n = 2\pi \times 10 = 20\pi$ rad/sec

$\omega = 0$

$t = 120$ sec

$\omega = \omega_0 + \alpha t$

$0 = 20\pi + \alpha \times 120$

$\alpha = -\frac{\pi}{6}\ rad/s^2$

26. $a = \alpha R$

27. In uniform speed angular acceleration (α) = 0

28. $F_1 = \frac{mv^2}{r}$, $F_2 = \frac{m(3v)^2}{(3r)} = \frac{9mv^2}{3r} = 3F_1$

31. $T = m\omega^2 r$

$\omega_{max}^2 = \frac{T_{max}}{mr} = \frac{100}{100\times10^{-3}\times0.1}$

$\Rightarrow$ $\omega_{max} = 100$ rad/sec

32. $F_C = m\omega^2 r = m(2\pi n)^2 r$

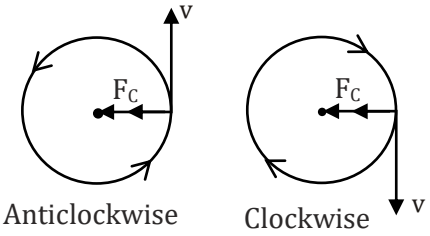
$\because$ Angular frequency n = constant

$\therefore$ $F_C \propto r$ so $\frac{F_{C_2}}{F_{C_1}} = \frac{2r}{r}$

$\Rightarrow$ $F_{C_2} = 2F_{C_1} = 2F$

33. $F_C = \frac{mv^2}{r} = \frac{0.5\times4\times4}{0.4} = 20N$

34.



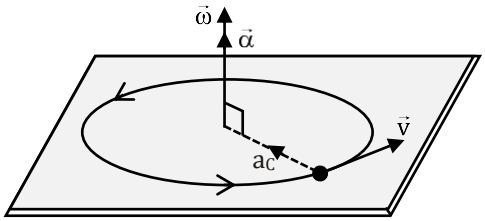
Centripetal force always toward centre.

35. In uniform circular motion.

magnitude of v = constant

so $a_t = 0$; $a_r \neq 0$

36.



37. $F_C = \frac{mv^2}{r} = \frac{mr^2\omega^2}{r} = mr\omega^2$ $T_{max} = 10$ N

At Balancing point $T_{max} = F_{cp} \Rightarrow 10 = mr\omega^2$

$\omega^2 = 400$

$\Rightarrow$ $\omega = 20$ rad/sec.

38. $F_{gravitation} = F_{centripetal}$

$F_{cen} = mr\omega^2$

$= 6 \times 10^{24} \times 1.5 \times 10^8 \times 10^3 \times 4 \times 10^{-14}$

$= 36 \times 10^{21}$ N

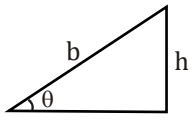
39. $F = \frac{mv^2}{r} = \frac{500\times(10)^2}{50} = 1000$ N

40. $F_C = \frac{mv^2}{r}$

$F_C' = \frac{mv'^2}{r'} = \frac{m(2v)^2}{r} = 4F_C$ $(r' = r)$

42. Lack of centripetal forces causes overturning.

43.

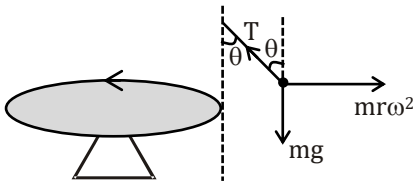


$\tan\theta \simeq \sin\theta = \frac{h}{b}$

$\therefore$ $\frac{h}{b} = \frac{v^2}{Rg}$

$\therefore$ $h = \frac{v^2 b}{Rg}$

44.



$T\cos\theta = mg$

$T\sin\theta = mr\omega^2$

$\therefore \quad \tan\theta = \frac{r\omega^2}{g}$

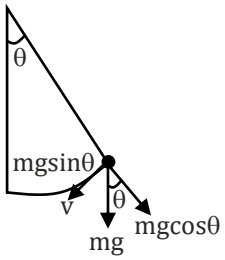
45. Tension at θ displacement

$T = mg\cos\theta + \frac{mv^2}{\ell}$

at extreme position v = 0

$\therefore$ T = mg cosθ

(at extreme position)



46. $h = \ell - \ell\cos 60° = \ell - \ell \times \frac{1}{2}$

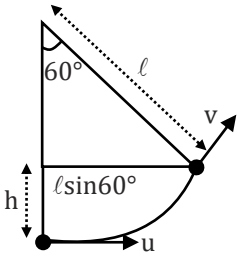
$= \ell/2$

by law of conservation of mechanical energy

$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 + mgh$

$\therefore \quad v^2 = u^2 - 2g\frac{\ell}{2}$

$v = \sqrt{(3)^2 - 10\times\frac{1}{2}} = 2\ \text{m/s}$



47. $0 + mgh = \frac{1}{2}mv^2 + 0$

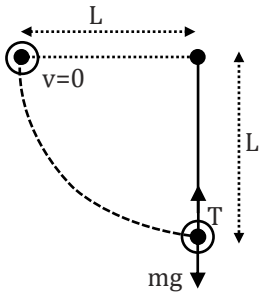
$v^2 = 2gL$

$\Rightarrow \quad v = \sqrt{2gL}$

$T = mg + \frac{mv^2}{L}$

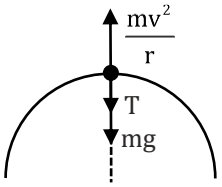
$T = mg + \frac{m}{L}\times 2gL$

$T = 3mg$



48. $T = \frac{mv^2}{\ell} - mg$

$= \frac{1\times(4)^2}{1} - 1\times 10 = 6\,\text{N}$



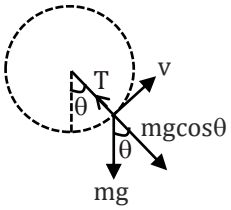
49. At highest point minimum possible value of tension is zero.

50. $T = \frac{mv^2}{r} + mg\cos\theta$

as θ increases cosθ and v both decreases

∵ v maximum at lowest point

∴ T maximum at lowest point



51. $\omega = \sqrt{\frac{T-mg}{mr}} = \sqrt{\frac{30-0.5\times 10}{0.5\times 2}} = 5\ \text{rad/sec}$

52. For looping the loop minimum velocity at top point $v = \sqrt{gL}$

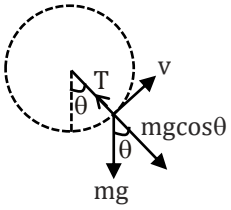
time taken by particle

$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2\times 2L}{g}} = 2\sqrt{\frac{L}{g}}$

$\therefore$ horizontal range $x = vt = \sqrt{gL}\times 2\sqrt{\frac{L}{g}} = 2L$

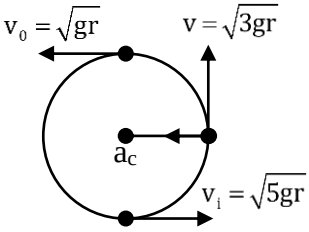
53. $T = \frac{mv^2}{r} + mg\cos\theta$

as θ increases cosθ and v both decreases hence for θ = 60°, T will be less i.e., $T_1 > T_2$



54. $a_C = \frac{v^2}{r} = \frac{3gr}{r}$

$a_C = 3g$



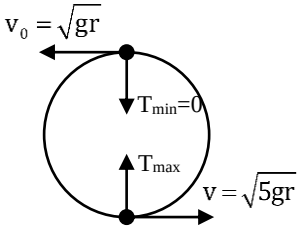
55. $T_{max} = mg + \frac{mv^2}{r} = mg + \frac{m}{r}\times 5gr = 6mg$

$T_{min} = 0$

$T_{max} - T_{min} = 6mg$

$\therefore \quad 6mg = 2g$

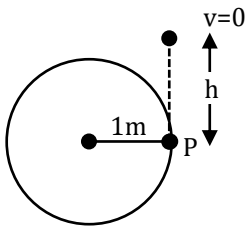
$m = \frac{1}{3}\,\text{kg}$



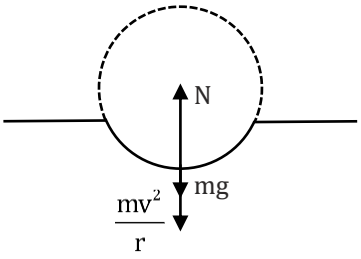
56. Acc to COME at point P
KE of stone = PE gained by stone.

$\frac{1}{2}m(5)^2 = m \times 10 \times h$

$h = \frac{25}{20} = 1.25m$



57. $N = mg + \frac{mv^2}{r}$



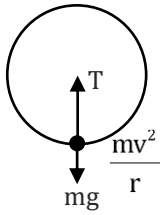
58. $\frac{1}{2}m(5Rg) = mgh$

$h = \frac{5R}{2}$

$R = \frac{2}{5} \times h = 2$ cm

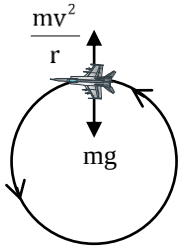
59. At position B tension is maximum ;

$T = mg + \frac{mv^2}{r}$



60. $\frac{mv^2}{r} = mg$

$\rightarrow v = \sqrt{gr}$

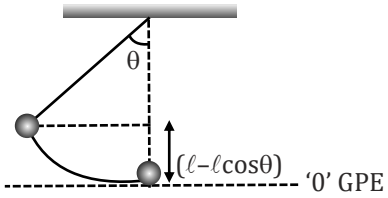


61. $T - mg = \frac{mv^2}{r}$

$\rightarrow \quad 52 - 2 = \frac{0.2v^2}{0.1}$

$\rightarrow \quad v = 5$ m/s

62.



$\therefore \quad mg\ell(1-\cos\theta) = \frac{1}{2}mv^2$

$v = \sqrt{2g\ell(1-\cos\theta)}$

Exercise-II (Previous Years Questions)

1. $(F_C)_{heavier} = (F_C)_{lighter}$

$\Rightarrow \quad \frac{2mv^2}{(r/2)} = \frac{m(nv)^2}{r}$

$\Rightarrow \quad n^2 = 4 \Rightarrow n = 2$

3. By using work-energy theorem, W=ΔKE

$\Rightarrow (ma_t)(4\pi R) = \frac{1}{2}mv^2 \Rightarrow a_t = \frac{\left(\frac{1}{2}mv^2\right)}{4\pi mR}$

$\Rightarrow a_t = \frac{8\times10^{-4}}{4\times3.14\times10\times10^{-3}\times6.4\times10^{-2}} = 0.1\ m/s^2$

OR

$\frac{1}{2}mv^2 = KE \Rightarrow \frac{1}{2}\left(\frac{10}{1000}\right)v^2 = 8 \times 10^{-4}$

$\Rightarrow v^2 = 16 \times 10^{-2} \Rightarrow v = 4 \times 10^{-1} = 0.4$ m/s

Now,

$v^2 = u^2 + 2a_t s \qquad (s = 4\pi R)$

$\Rightarrow \frac{16}{100} = 0^2 + 2a_t\left(4\times\frac{22}{7}\times\frac{6.4}{100}\right)$

$\Rightarrow a_t = \frac{16}{100}\times\frac{7\times100}{8\times22\times6.4} = 0.1\ m/s^2$

4. When minimum speed of body is $\sqrt{5gR}$, then no matter from where it enters the loop, it will complete full vertical loop.

5. $\frac{v^2}{Rg} = \tan(\phi+\theta) = \frac{\tan\phi+\tan\theta}{1-\tan\phi\tan\theta} = \left(\frac{\mu_s+\tan\theta}{1-\mu_s\tan\theta}\right)$

$\Rightarrow \quad v = \sqrt{Rg\left[\frac{\mu_s+\tan\theta}{1-\mu_s\tan\theta}\right]} \quad (\mu_s = \tan\phi)$

OR

Check by dimensions.

6. Centripetal acceleration $= \frac{v^2}{R} = a\cos30°$

$\Rightarrow v = \sqrt{aR\cos30°} = \sqrt{15\times2.5\times\frac{\sqrt{3}}{2}} = 5.7 m/s$

7. $\vec{r} = \cos\omega t\,\hat{x} + \sin\omega t\,\hat{y}$

$\vec{v} = -\omega\sin\omega t\,\hat{x} + \omega\cos\omega t\,\hat{y}$

$\vec{a} = -\omega^2\cos\omega t\,\hat{x} - \omega^2\sin\omega t\,\hat{y}$

$\Rightarrow \quad \vec{v}.\vec{r} = 0$

velocity is perpendicular to $\vec{r}$.

$\Rightarrow \quad \vec{a} = -\omega^2(\vec{r})$

acceleration is directed towards the origin.

8. Net force on the particle in uniform circular motion is centripetal force, which is provided by the tension in string.

9. To complete a vertical circle, speed at A should be $v_A = \sqrt{5gR} = \sqrt{\frac{5gD}{2}}$

using energy conservation $mgh = \frac{1}{2}mv_A^2$

$$h = \frac{1}{2}\frac{v_A^2}{g} = \frac{1}{2}\frac{5g}{g}\frac{D}{2} = \frac{5D}{4}$$

$\left(R = \frac{D}{2}\right)$

10. $f_L = \mu N = \mu mr\omega^2$

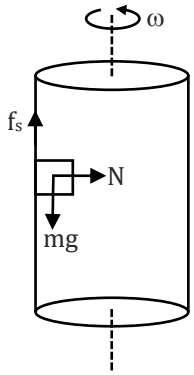
$f_s = mg$

As $f_s \le f_L$

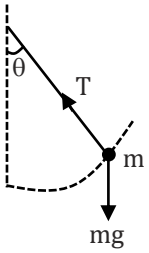
$\Rightarrow mg \le \mu mr\omega^2$

$\Rightarrow \quad \omega \ge \sqrt{\frac{g}{\mu r}}$

$\Rightarrow \omega_{min} = 10$ rad/s



11. $T - mg\cos\theta = \frac{mv^2}{R}$



T will be maximum when $\theta = 0°$,

When mass is at lowest point.

12. $T_A = T_B$

$\Rightarrow \quad \frac{2\pi}{\omega_A} = \frac{2\pi}{\omega_B}$

$\Rightarrow \quad \frac{\omega_A}{\omega_B} = 1:1$

13. $\theta = (2\pi n)$, $\omega_0 = 0$, $\omega = V_0/r$

$$\alpha = \frac{\omega^2 - \omega_0^2}{2\theta} == \frac{(V_0/r)^2 - 0}{2(2\pi n)} = \frac{V_0^2}{4\pi nr^2}$$

14. $\alpha = \frac{\Delta\omega}{\Delta t} = \frac{(1200-360)(2\pi)}{60\times 14}$

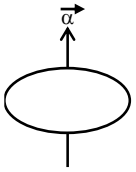
$\alpha = 2\pi$ rad/s²

15. $T - mg = \frac{mv^2}{r}$

$\Rightarrow \quad T - mg = \frac{m(7gr)}{r}$

$\Rightarrow \quad T = 8\,mg$

17.



Along the axis of rotation

18. $F_S = ma$

$f_L = ma_{max}$

$\mu\,mg = ma_{max}$

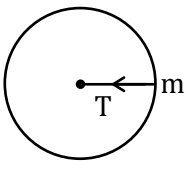
$a_{max} = \mu g$

$= 0.15(10)$

$= 1.5$ m/s²

19. Direction of velocity and centripetal acceleration changes continuously so $\vec{v}$ is not constant and $\vec{a}$ is not a constant.

20. $F_{cp} = ma_{cp}$



$F_{cp} = m\omega^2 r$

$T = m\omega^2 r$

Now speed becomes '2ω'

$T' = m(2\omega)^2 r$

$T' = 4\,m\omega^2 r$

$T' = 4T$

21. In uniform circular motion direction of velocity and acceleration keeps on changing

22. $T = m\omega^2 r$

$T \propto \omega^2$

$$\frac{T_2}{T_1} = \left(\frac{\omega_2}{\omega_1}\right)^2$$

$$\Rightarrow \frac{4T}{T} = \left(\frac{\omega_2}{10}\right)^2$$

$\Rightarrow \omega_2 = 20$ rpm

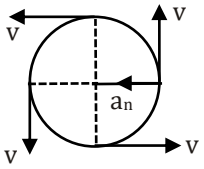
23. $\omega = \frac{2\pi}{T} \quad x = \frac{2\pi}{T} t$

$$\frac{\omega}{x} = \frac{1}{t} = \text{const.}$$

$\therefore$ 1 minute

Exercise-III (Analytical Questions)

4.



a_t : tangential acceleration

a_n : Normal acceleration

$a_t = 0$

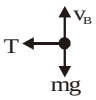
$$a_n \neq 0; \; a_n = \frac{v^2}{r}$$

For (A) $v_B^2 = v_A^2 - 2g\ell = 9g\ell - 2g\ell = 7g\ell$

$\Rightarrow v_B = \sqrt{7g\ell}$

For (B) $v_C^2 = v_A^2 - 2g(2\ell) = 5g\ell$

$\Rightarrow v_C = \sqrt{5g\ell}$

For (C) 

$T_B = \frac{mv_B^2}{\ell} = 7mg$

For (D) 

$T_C + mg = \frac{mv_C^2}{\ell}$

$\Rightarrow T_C = 4mg$

5. As $V_H^2 = V_L^2 - 2gh$ where $h = 2r$

$\therefore V_H^2 = V_L^2 - 4gr$

Now, $V_L = 2V_H$ (given)

$$\therefore V_H = \sqrt{\frac{4gr}{3}} \; \& \; V_L = 2V_H = 2\sqrt{\frac{4gr}{3}}$$

$$T_H = T_{min} = \frac{mv_H^2}{r} - mg = \frac{mg}{3}$$

$$T_L = T_{max} = \frac{mv_L^2}{r} + mg = \frac{19}{3}mg$$

6. As $v = 2t$ and let radius of circular path is r then

$$a_T = \frac{dv}{dt} = 2$$

$$a_r = \frac{v^2}{r} = \frac{4t^2}{r}$$

Therefore,

$$a = \sqrt{a_T^2 + a_r^2}$$

$$a = \sqrt{4 + \frac{16t^4}{r^2}}$$

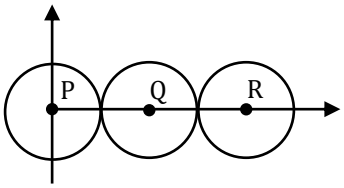
$\therefore$ (B) and (D) are correct.

Collisions and Centre of Mass

SOLUTIONS

Exercise-I (Conceptual Questions)

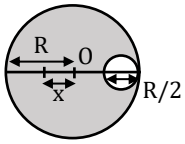
1.



Distance of C.O.M. of system from P is

$$=\frac{1\times PQ+1\times PR}{3}=\frac{PQ+PR}{3}$$

2. Mass of disc = M



Mass per unit area $=\frac{M}{\pi R^2}$

Mass of area of cutting part

$$=\frac{M}{\pi R^2}\times\pi\left(\frac{R}{4}\right)^2=\frac{M}{16}$$

Remaining mass $=\frac{15M}{16}$

$$\frac{15M}{16}\times x=\frac{M}{16}\times\frac{3R}{4}$$

$\Rightarrow x=\frac{R}{20}$ from centre O

3. $m_1 = M$ (Solid sphere)

$r_{x_1}=0$ $\quad r_{y_1}=a$

Hollow sphere $m_2 = M$

$r_{x_2}=0$ $\quad r_{y_2}=0$

Disc. $m_3 = M$

$r_{x_3}=a$ $\quad r_{y_3}=0$

$$r_{x_{cm}}=\frac{m_1r_{x_1}+m_2r_{x_2}+m_3r_{x_3}}{m_1+m_2+m_3}=\frac{0+0+aM}{3M}=\frac{a}{3}$$

$$r_{y_{cm}}=\frac{m_1r_{y_1}+m_2r_{y_2}+m_3r_{y_3}}{m_1+m_2+m_3}=\frac{Ma+0+0}{3M}=\frac{a}{3}$$

So co-ordinate of centre of mass of system will be

$$\left(\frac{a}{3},\frac{a}{3}\right)$$

4. The centre of mass of the system of particles does not depend on internal forces on the particles.

If $\vec{F}_{ext.}=\vec{0}$ then $\vec{v}_{CM}$ = constant

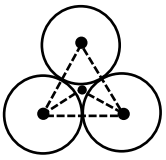
If no external force acts on a system the velocity of its centre of mass remains constant, i.e., velocity of centre of mass is unaffected by internal forces.

5. $m_1r_1 = m_2r_2$

$$\frac{r_1}{r_2}=\frac{m_2}{m_1}\quad\Rightarrow\quad r\propto\frac{1}{m}$$

6. COM may lie within (i.e. in solid sphere), outside of the surface of the body (L shaped Lamina)

7.



The COM of the system is located at the point of intersection of their median.

8. Centre of mass is towards heavier mass and $m << M$.

So centre of mass of the system in nearer to M.

9. $$\vec{r}_{cm}=\frac{m_1\vec{r}_1+m_2\vec{r}_2+m_3\vec{r}_3}{m_1+m_2+m_3}$$

$\vec{r}_{cm}=0$

$\Rightarrow\ m_1\vec{r}_1+m_2\vec{r}_2+m_3\vec{r}_3=0$

$$\vec{r}_{cm}=\frac{m_1\vec{r}_1+m_2\vec{r}_2+m_3\vec{r}_3+m_4\vec{r}_4}{m_1+m_2+m_3+m_4}$$

$$(\hat{i}+2\hat{j}+3\hat{k})=\frac{0+4\alpha(\hat{i}+2\hat{j}+3\hat{k})}{1+2+3+4}$$

$\Rightarrow\quad \alpha=\frac{5}{2}$

10. By third law for a system

$\vec{F}_{12} = -\vec{F}_{21}$ or $d\vec{P}_1 = -d\vec{P}_2$

11. Let plank shifted by x then

$\Delta x_{CM} = 0$, as there is no external force on the system.

$m(L - x_{plank}) - Mx_{plank} = 0$

$x_{plank} = \dfrac{mL}{M+m}$

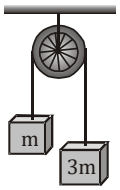
12. $200 = n \times \dfrac{40}{1000} \times 10^3$

n = 5 bullets/sec

$\therefore$ n = 300 bullets/min

13. $a = \dfrac{(3m-m)}{3m+m} g = \dfrac{g}{2}$

$\vec{a}_{cm} = \dfrac{3m\vec{a}_1 + m\vec{a}_2}{3m+m}$



Both mass have same magnitude of accleration but in opposite direction $\vec{a}_1 = -\vec{a}_2 = a$ Let)

$a_{cm} = \left(\dfrac{3m-m}{4m}\right) \times \dfrac{g}{2} = \dfrac{g}{4}$

14. Since in this system their is absence of external force which can move the centre of mass.

Here particles are moving due to mutual interaction between them that's why velocity of COM of the system will be zero.

15. $\vec{v}_{cm} = \dfrac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1+m_2}$

$m_1 = 2kg$ $m_2 = 3kg$

$|\vec{v}_1| = 3m/s$ $|\vec{v}_2| = 2m/s$

$\vec{v}_{cm} = \dfrac{2\times3+3\times2}{2+3} = \dfrac{12}{5} m/s$

16. $m_1 = 200$ gm $m_2 = 500$ gm

$\vec{v}_1 = 10\hat{i}$ $\vec{v}_2 = 3\hat{i}+5\hat{j}$

$\vec{v}_{cm} = \dfrac{m_1\vec{v}_1 + m_2\vec{v}_2}{m_1+m_2} = \dfrac{200\times10\hat{i} + 500(3\hat{i}+5\hat{j})}{200+500}$

$\vec{v}_{cm} = \dfrac{20\hat{i}+15\hat{i}+25\hat{j}}{7} = 5\hat{i} + \dfrac{25\hat{j}}{7}$

17. $3 \times 16 = 6 \times v$

$v = 8$ m/s $\therefore$ $KE = \dfrac{1}{2}\times6\times8\times8 = 192J$

18. $m\left(3\hat{i}+2j\right) + m\left(-\hat{i}-4\hat{j}\right) + m\vec{v} = 0$

$\therefore$ $\vec{v} = -2\hat{i}+2\hat{j}$

19. By COLM

$50 \times 600 = 10^3 \times v$

$v = -30$ m/s

negative sign show's opposite direction

20. For rubber ball $\Delta\vec{p}$ is more

as v_f will be in opp. direction

$\Delta\vec{p} = m\left(\vec{v}_f - \vec{v}_i\right)$

21. By $\vec{v} = \vec{u} + \vec{a}t$

$\vec{v} = 100\hat{j} - 10\hat{j}\times5 = 50\hat{j}\, m/s$

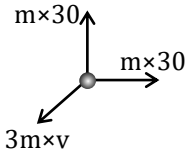
using COLM $\vec{p}_f = \vec{p}_i$

$0.6\vec{v}_2 - 0.4\times25\hat{j} = 1\times50\hat{j}$

$0.6\vec{v}_2 = 60\hat{j}$

$\vec{v}_2 = 100\hat{j}$ m/s

22. Total momentum = 0



$\therefore$ $m\times30\times\sqrt{2} = 3m\times v$

$v = 10\sqrt{2}\, m/s$

23. $M_1 \times 8v = M_2 v$

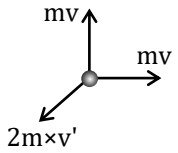
$M_2 = 8M_1$ $\left(\because M = \dfrac{4}{3}\pi r^3\rho, M \propto r^3\right)$

$\therefore$ $r_2 = 2r_1$ $\Rightarrow$ $\dfrac{r_1}{r_2} = \dfrac{1}{2}$

24. $KE = \dfrac{p^2}{2m}$

$\therefore$ $\dfrac{KE_1}{KE_2} = \dfrac{m_2}{m_1}$ $(\because p = \text{same})$

25.



$\therefore \quad 2mv' = mv\sqrt{2} \Rightarrow v' = \frac{v}{\sqrt{2}}$

$\therefore$ KE released

$$= \frac{1}{2}mv^2 \times 2 + \frac{1}{2}2m \times \frac{v^2}{2} = \frac{3}{2}mv^2$$

26. Total energy = $\frac{P^2}{2m_1} + \frac{P^2}{2m_2}$

$$= \frac{P^2}{2}\left[\frac{1}{m_1} + \frac{1}{m_2}\right]$$

$$= \frac{(1\times 80)^2}{2} \times \left[\frac{1}{1} + \frac{1}{2}\right] = 4.8 \text{ kJ}$$

27. $18 \times 6 = 12 \times v$

$v = 9$ m/s

$\therefore \quad KE = \frac{1}{2} \times 12 \times 9 \times 9 = 486 J$

28. $mv = MV$; $MV^2 = kd^2$

$\therefore V = \sqrt{\frac{kd^2}{M}} \quad \therefore v = \frac{M}{m}\sqrt{\frac{kd^2}{M}} = \frac{d}{m}\sqrt{kM}$

29. Momentum is conserved in all collision.

30. Consider ball and earth as a system.

31. By Conservation of linear momentum

$p_i = p_f$

$PQ + R(0) = (P + R)V$

$V = PQ/(P + R)$

32. By COLM

$mu + 0 = mv_1 + mv_2$ ...(1)

$e = \frac{v_2 - v_1}{u - 0}$ ...(2)

$\Rightarrow v_2 - v_1 = eu$ ...(3)

from (1)

$u = v_1 + v_2$ ...(4)

putting (4) in (1)

$v_2 - v_1 = e(v_1 + v_2)$

$\Rightarrow v_2 - ev_2 = v_1 + ev_1$

$$\frac{v_1}{v_2} = \frac{1-e}{1+e}$$

33. For same mass and e = 1 velocities exchange

34. 'e' is unit less

35. $m_1\vec{u}_1 + m_2\vec{u}_2 = 0 + 0$

$200 \times 0.3 + 400 \times \vec{u}_2 = 0$

$\vec{u}_2 = -0.15$ m/s

36. $\sqrt{2gh_f} = e\sqrt{2gh_i}$

$$\Rightarrow e = \sqrt{\frac{h_f}{h_i}} = \sqrt{\frac{9}{25}} = \frac{3}{5} = 0.6$$

37. $50 \times 10 = 1000 \times v \quad \therefore \quad v = \frac{1}{2}$ m/s

$$E_i = \frac{1}{2} \times \frac{50}{1000} \times 10 \times 10 = 2.5 J$$

$$E_f = \frac{1}{2} \times \frac{1000}{1000} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8} J$$

$$\% \text{ loss} = \frac{2.5 - 1/8}{2.5} \times 100 = 95\%$$

38. $mv = (M + m)v_f \quad \therefore \quad v_f = \frac{m}{M+m}v$

39. Let a ball fall from a height (h) and let it touch the ground with a velocity v taking time (t) to reach the ground.

Let v_1, v_2, v_3..... be the velocities immediately after first, second, third.....collisions with the ground.

Height attained by the Ball After the 'n'th Rebound

$v_1 = ev \Rightarrow \sqrt{2gh_1} = e\sqrt{2gh} \Rightarrow h_1 = e^2h$,

$v_2 = e^2v \Rightarrow \sqrt{2gh_2} = e^2\sqrt{2gh} \Rightarrow h_2 = e^4h$.

Similarly $\boxed{h_n = e^{2n}h}$

40. By COLM

$mv - mv = 0$

41. Velocity interchange when mass are equal

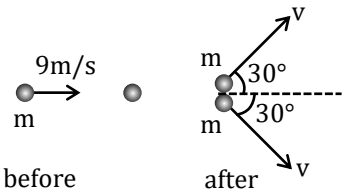
So KE = mgL

42. $mu_1 + mu_2 = 2mv$

$$KE_{loss} = \frac{1}{2}(2m)v^2 - \frac{1}{2}mu_1^2 - \frac{1}{2}mu_2^2$$

$$\frac{m}{2}\left[2\left(\frac{u_1+u_2}{2}\right)^2 - u_1^2 - u_2^2\right] = \frac{m}{4}(u_1 - u_2)^2$$

43.



$2mv\times\frac{\sqrt{3}}{2}=m\times 9$

$v=\frac{9}{\sqrt{3}}$ m/s = $3\sqrt{3}$ = 5.2 m/s

44. Mass must be same

$\therefore \quad \rho_1\times\frac{4}{3}\pi r^3=\rho_2\times\frac{4}{3}\pi\left(\frac{r}{2}\right)^3$

$\frac{\rho_2}{\rho_1}=8$

45. $5\times v=(5+m)\frac{v}{3}$

$\Rightarrow \quad m = 10$ kg

46. $\frac{10}{1000}\times 500=(10+0.01)\times v$

$\Rightarrow$ v = 0.5 m/s

47. $v_2 = -u_2 + 2u_1$ (using COLM & e = 1)

$v_2 = 0 + 2(20) = 40$ m/s

48. $5 \times 10 - 2 \times 15 = 7 \times v$

$v=\frac{20}{7}$ = 2.8 cm/s in east

49. Momentum is conserved.

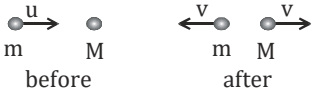
50. $20 \times 10 = 25 \times v$

$v=\frac{200}{25}$ = 8 m/s

$\therefore \quad KE=\frac{1}{2}\times 25\times 8\times 8=800J$

51. Velocity interchange when m = M

52.



$mu = (M - m)v$ ...(1)

$e=\frac{2v}{u}=1$

$\therefore \quad v=\frac{u}{2}$ ...(2)

$\therefore \quad m=(M-m)\times\frac{1}{2}$

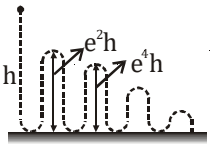
$2m = M - m$

$3m = M$

$\frac{m}{M}=\frac{1}{3}$

53. e = 0, for perfectly inelastic.

54.



Total distance = $h + 2e^2h + 2e^4h + ...$

$= h + 2e^2h(1 + e^2 + e^4+...)$

$= h+2e^2h\times\frac{1}{1-e^2}$

$= h\left[1+\frac{2e^2}{1-e^2}\right]=\left(\frac{1+e^2}{1-e^2}\right)h$

55. According to the law of conservation of momentum

$m_1\vec{u}_1+m_2\vec{u}_2=m_1\vec{v}_1+m_2\vec{v}_2$ .....(i)

Here $\vec{u}_1,\vec{u}_2$ = Initial velocities of m_1 and m_2

$\vec{v}_1,\vec{v}_2$ = Final velocities of m_1 and m_2

Change in velocity of m_1 = $\vec{v}_1-\vec{u}_1$

Change in velocity of m_2 = $\vec{v}_2-\vec{u}_2$

$m_1\vec{v}_1-m_1\vec{u}_1=-m_2\vec{v}_2+m_2\vec{u}_2$

or $m_1\left(\vec{v}_1-\vec{u}_1\right)=m_2\left(-\vec{v}_2+\vec{u}_2\right)$

or $\frac{\vec{v}_1-\vec{u}_1}{-\vec{v}_2+\vec{u}_2}=\frac{m_2}{m_1}$

$\therefore \quad \frac{\text{changes in velocity of } m_1}{\text{changes in velocity of } m_2}=\frac{m_2}{m_1}$

56. $\frac{M_A}{M_B}=1$

By COLM $M_AV_A = M_BV_B = \frac{v_A}{v_B}=\frac{M_B}{M_A}=1$

57. In option (3)

It is not possible that after collision one ball moves along the original line of motion while the other ball moves along some angle (α) with original line.The momuntum perpendicular to original line of motion cannot be conserved in this situation.

Initial momentum along perpendicular direction=zero

Final momentum along perpendicular direction = $m_2v_2 \sin\alpha$. Hence momentum is not conserved. Hence the situation is physically impossible. Rest option may be physically possible.

58. Velocity would interchange

59. For elastic oblique collision having object of same mass. After collision they move perpendicular to each other.

60. Initial energy = mgh = mg(10)

if 40% is lost

Remaining energy = 60% of mg(10)

$\Rightarrow \frac{60}{100} \times mg(10) = mg(h)$

h = 6m

61. $e = \sqrt{\frac{h_f}{h_i}} = \sqrt{\frac{1.8}{5}} = 0.6 = \frac{3}{5}$

loss in velocity = $1 - \frac{3}{5} = \frac{2}{5}$

62. $\vec{p}$ conserved in all collision

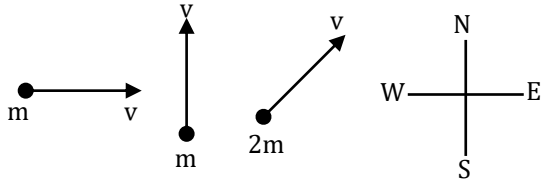
KE conserved only in elastic collision.

63. $mv = (M + m)v_f$

$v_f = \left(\frac{m}{m+M}\right)v$

64. $Mu = mu + Mv \Rightarrow v = \frac{u(M-m)}{M}$

65.



According to the law of conservation of linear momentum along horizontal direction,

$mv + 0 = 2mv'_x$

$v'_x = \frac{v}{2}$

According to the law of conservation of linear momentum along vertical direction,

$0 + mv = 2mv'_y$

$v'_y = \frac{v}{2}$

$\therefore$ Speed of the new mass 2m is

$v' = \sqrt{v'^2_x + v'^2_y} = \sqrt{\left(\frac{v}{2}\right)^2 + \left(\frac{v}{2}\right)^2} = \frac{v}{\sqrt{2}}$

66. By Conservation of linear momentum

$\vec{p}_i = \vec{p}_f$

in x direction

$m_A v_A = (m_A + m_B)v_x$

in y direction

$m_B v_B = (m_A + m_B)v_y$

$v_x = \frac{30}{50} m/s$

$v_y = \frac{40}{50} m/s$

$v_{net} = \sqrt{(v_x)^2 + (v_y)^2} = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2}$

= 1 m/s

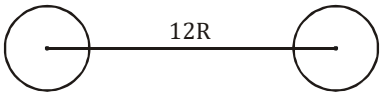
Exercise-II (Previous Year Questions)

1. Energy will always be conserved so

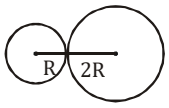
$K.E_{initial} = K.E_{final}$ + Excitation energy

$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 + \varepsilon$

2.



Initial distance between their centers = 12 R



At time of collision the distance between their centers = 3R

So total distance travelled by both

=12R–3R = 9R

Since the bodies move under mutual forces, center of mass will remain stationary so

$m_1x_1 = m_2x_2$

$mx = 5m(9R - x)$

$x = 45R - 5x$

$6x = 45R$

$x = \frac{45}{6}R$ $\qquad$ $\boxed{x = 7.5R}$

3. Let ball rebounds with speed v so

$$v = \sqrt{2gh} = \sqrt{2\times10\times20} = 20 \text{ m/s}$$

Energy just after rebound

$$E = \frac{1}{2}\times m\times v^2 = 200\ m$$

50% energy lost in collision means just before collision energy is 400 m

By using energy conservation

$$\frac{1}{2}mv_0^2 + mgh = 400m$$

$$\Rightarrow \frac{1}{2}mv_0^2 + m\times10\times20 = 400m$$

$$\Rightarrow v_0 = 20 \text{ m/s}$$

4. In elastic collision energy of system remains same so.

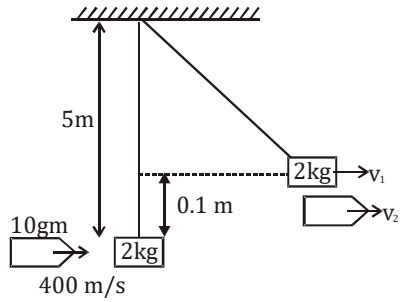
$(K.E)_{\text{before collision}} = (K.E)_{\text{After collision}}$

Let speed of second body after collision is v'

$$\frac{1}{2}mv^2 + 0 = \frac{1}{2}m\left(\frac{v}{3}\right)^2 + \frac{1}{2}m(v')^2$$

$$\Rightarrow v' = \frac{2\sqrt{2}}{3}v$$

5.



Applying momentum conservation

$$\frac{10}{1000}\times400 + 0 = 2\times v_1 + \frac{10}{1000}\times v_2$$

$\Rightarrow 4 = 2v_1 + 0.01v_2$ .....(1)

Applying work energy theorem for block

$W = \Delta KE$

$$\Rightarrow 2\times10\times0.1 = \frac{1}{2}\times2\times v_1^2$$

$$\Rightarrow v_1 = \sqrt{2} = 1.4 \text{ m/s}$$

Putting the value of v_1 in equation (1)

$4 = 2\times1.4 + 0.01\ v_2 \Rightarrow v_2 = 120$ m/s

6. Since both bodies are identical and collision is elastic.

Therefore velocities will be interchanged after collision.

$v_A = -0.3$ m/s and $v_B = 0.5$ m/s

7. By conservation of linear momentum

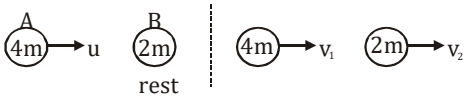
$$mv = 4mv' \Rightarrow v' = \frac{v}{4}$$

coefficient of restitution

$$(e) = \frac{\text{Velocity of separation}}{\text{Velocity of approach}}$$

$$= \frac{\frac{v}{4}-0}{v-0} = \frac{1}{4} = 0.25$$

8.



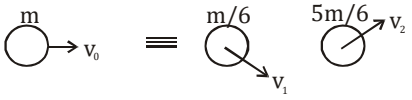
$$v_1 = \frac{4m-2m}{4m+2m}u = \frac{2mu}{6m} = \frac{u}{3}$$

Fraction of energy lost

$$= \frac{\frac{1}{2}(4m)u^2 - \frac{1}{2}(4m)\left(\frac{u}{3}\right)^2}{\frac{1}{2}(4m)u^2}$$

$$= 1 - \frac{1}{9} = \frac{8}{9}$$

9.



Conservation of linear momentum,

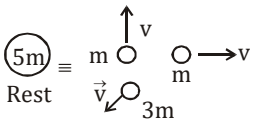
$$m\vec{v}_0 = \frac{m}{6}\vec{v}_1 + \frac{5m}{6}\vec{v}_2$$

$$\Rightarrow m(20\hat{i}+25\hat{j}-12\hat{k})$$

$$= \frac{m}{6}(100\hat{i}+35\hat{j}+8\hat{k}) + \frac{5m}{6}\vec{v}_2$$

$$\Rightarrow \vec{v}_2 = 4\hat{i}+23\hat{j}-16\hat{k}$$

10.



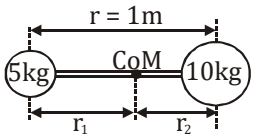
$$3m\vec{v} + mv\hat{i} + mv\hat{j} = 0$$

$$\Rightarrow \vec{v} = -\frac{v}{3}\hat{i} - -\frac{v}{3}\hat{j} \quad |\vec{v}| = \sqrt{2}\frac{v}{3}$$

Energy released

$$= \frac{1}{2}mv^2 + \frac{1}{2}mv^2 + \frac{1}{2}(3m)\left(\frac{2v^2}{9}\right) = \frac{4}{3}mv^2$$

11.

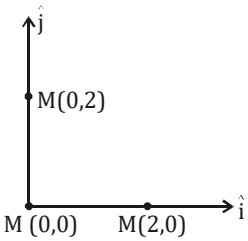


About CoM, mr = constant $\Rightarrow r \propto \frac{1}{m}$

$\Rightarrow \frac{r_1}{r_2} = \frac{m_2}{m_1} = \frac{10}{5} = \frac{2}{1}$

$\Rightarrow r_1 = \frac{2}{3} r = \frac{2}{3} \times 1\ m = 67\ cm$

12.



$X_{CM} = \frac{M\times0 + M\times2 + M\times0}{3M} = \frac{2}{3}$

$Y_{CM} = \frac{M\times0 + M\times2 + M\times0}{3M} = \frac{2}{3}$

Position vector = $\vec{r} = X_{CM}\hat{i} + Y_{CM}\hat{j} = \frac{2}{3}\hat{i} + \frac{2}{3}\hat{j}$

16. In elastic collision maximum energy is transfer when M = m

17. Final velocity of centre of mass = Initial velocity of centre of mass = 0 because net external force on system is zero.

18. By Conservation of linear momentum :-

$mv_1 = (m + m)\ v_2$

$\Rightarrow mv_1 = 2mv_2$

$\Rightarrow \frac{v_1}{v_2} = 2:1$

Exercise-III (Analytical Questions)

1. Acceleration of centre of mass $\vec{a}_{cm} = \frac{\vec{F}_{ext.}}{M}$

If no external force acts on the system

$\vec{a}_{cm} = 0$ (Assertion is correct)

Reason is false because; $\vec{a}_{cm} = 0$ implies velocity will not change, if C.O.M is initially at rest, then it will remain at rest in absence of external force.

3. (A) Both particles will touch the ground together.

(B) C.O.M will follow same path.

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Rotation Motion

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Physical quantity and its units.

2. Particles present on the axis of rotation are not performing circular motion.

3. During rotation angular velocity is constant linear velocity will be more if distance from the axis of rotation of earth will be more.

4. For rotational motion we use moment of inertia.

5. Moment of inertia depends on :

(a) mass of the body

(b) mass distribution of the body

(c) position of axis of rotation

6. M.I is proportional mass distribution of the body

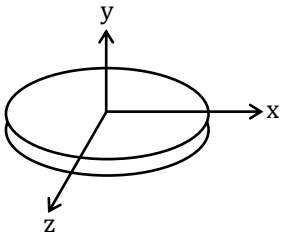
So, to increase M.I of a fly wheel its mass is concentrated at its rim.

7. $\frac{ML^2}{12}+\frac{MR^2}{4}=\frac{MR^2}{2}\Rightarrow\frac{L^2}{12}=\frac{R^2}{4}\Rightarrow L=\sqrt{3}\ R$

8. $I = I_{disc} + I_{Masses} = \frac{MR^2}{2} + 4mR^2$

9. For perpendicular axes

$I_y = I_x + I_z$



10. The theorem of perpendicular axis is applicable only for 2-D objects.

11. As disc is lying in the x-z plane, so applying perpendicular axis theorem :-

$I_x + I_z = I_y$

$30 + I_z = 40$

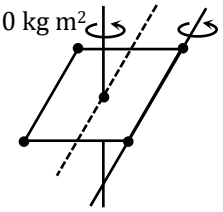
$\Rightarrow \quad I_z = 40 - 30 = 10\ kg\ m^2$

12. given $I_{solid\ sphere} = I_{hollow\ sphere}$

$\Rightarrow \quad \frac{2}{5}Mr_1^2=\frac{2}{3}Mr_2^2 \Rightarrow \frac{r_1^2}{r_2^2}=\frac{5}{3}$

$\Rightarrow \quad \frac{r_1}{r_2}=\sqrt{5}:\sqrt{3}$

13. $I = 20\ kg\ m^2$ $I' = ?$

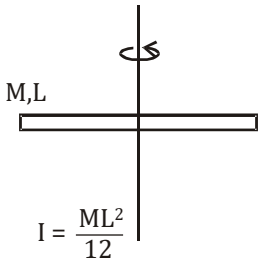


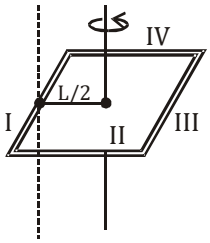
as, $I=\frac{ML^2}{6}=20 \quad \Rightarrow \quad ML^2 = 120$

$I'=I_c+M\left(\frac{L}{2}\right)^2=\frac{ML^2}{12}+\frac{ML^2}{4}$

$\Rightarrow \quad I'=\frac{ML^2}{3}=\frac{120}{3}=40\ kg\ m^2$

14.





for one rad, $I_1=\frac{ML^2}{12}+M\left(\frac{L}{2}\right)^2=\frac{4ML^2}{12}$

$I_{total} = 4I_1 = 4\times\frac{4ML^2}{12}=\frac{16ML^2}{12}$

so, $I_{total} = 16I$

15. Using parallel axis theorem

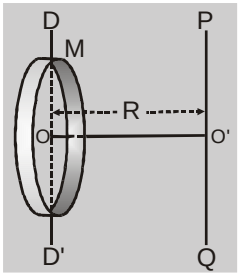
$I = \frac{2}{5}MR^2+MX^2$

Graph is parabola

For X = 0, $I = \frac{2}{5}MR^2$

$\therefore$ Correct option (2)

16.



Applying parallel axis theorem

$I_{PQ} = I_C + Md^2$

$\Rightarrow \quad I_{PQ} = \frac{MR^2}{2} + M(R)^2 = \frac{3}{2}MR^2$

17. $I = m(a)^2 + m(a)^2 + m(0)^2 + m(2a)^2$

$\Rightarrow \quad I = 6ma^2$

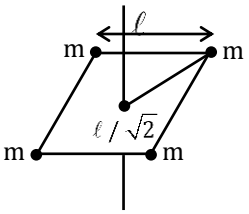
18. $I_{(1)} = \frac{2}{3}MR^2$, $I_{(2)} = \frac{2}{5}MR^2$, $I_{(3)} = \frac{MR^2}{2}$, $I_{(4)} = MR^2$

19. $I = MK^2 \Rightarrow I = 10(0.4)^2 = 1.6$ kg-m^2

20. $I_1 = \frac{2}{5}MR^2$; $I_2 = \frac{2}{3}MR^2$; $I_3 = MR^2$

21. $I_z = \frac{ML^2}{3} + \frac{ML^2}{3} + 0 \Rightarrow I_z = \frac{2}{3}ML^2$

22. so, $I = m \times \left(\frac{\ell}{\sqrt{2}}\right)^2 \times 4$



also, $I = MK^2$

{Where M = 4m}

so, $m \times \frac{\ell^2}{2} \times 4 = 4m \times K^2$

$\therefore \quad K = \frac{\ell}{\sqrt{2}}$

23. Using parallel axes theorem

$I = I_{CM} + Md^2$

$I = \frac{ML^2}{12} + M\left(\frac{L}{2}\right)^2$

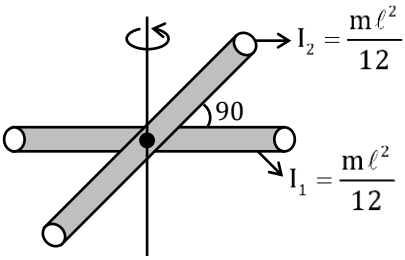
$I = \frac{ML^2}{3}$

24. M.I is proportional mass distribution of the body

So, to increase M.I of the disc its heavier mass is concentrated far from the axis of rotation.

25. The moment of inertia in rotational motion is equivalent to mass of linear motion.

26.

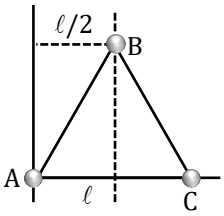


So $I = I_1 + I_2 = \frac{m\ell^2}{12} + \frac{m\ell^2}{12} = \frac{m\ell^2}{6}$

27. $K = \sqrt{\frac{I}{M}}$

$\frac{K_1}{K_2} = \sqrt{\frac{I_1}{I_2}} = \sqrt{\frac{\frac{5}{4}MR^2}{\frac{3}{2}MR^2}} = \sqrt{\frac{5}{6}}$

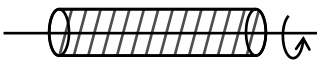
28.



$I = I_A + I_B + I_C$

$\Rightarrow \quad I = 0 + m\left(\frac{\ell}{2}\right)^2 + m\ell^2 = \frac{5}{4}m\ell^2$

29. $I = \frac{MR^2}{2}$

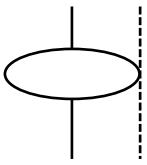


$= \frac{20 \times (0.2)^2}{2} = 0.4$ kg-m^2

30. Using parallel axis theorem

$I = MR^2 + MR^2$

$I = 2MR^2$



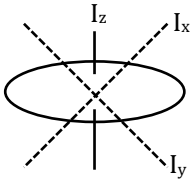
31. Along diameter in the plane.

Using Axis theorem

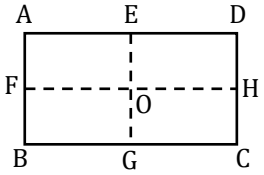
$I_x + I_y = I_z$

$2I_x = I_z$

$I_x = \frac{MR^2}{2}$



32. The moment of inertia is minimum about FH because mass distribution is at minimum distance from FH.



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33. $I_R = MR^2$, $I_D = MR^2/2$

$I_{HS} = \frac{2}{3}MR^2$ $I_{SS} = \frac{2}{5}MR^2$

34. $\tau = I\alpha$

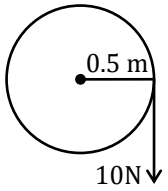
$\tau \propto I$ i.e, moment of inertia

35. $\vec{\tau} = I\vec{\alpha}$ like $\vec{F} = m\vec{a}$

$\vec{\tau} = \frac{d\vec{L}}{dt}$ like $\vec{F} = \frac{d\vec{p}}{dt}$

36. Angular acceleration will be more if M.I will be less. ($\vec{\tau} = I\vec{\alpha}$)

37. So, torque = 10 × 0.5 = 5 N-m



38. $\tau = I\alpha \Rightarrow \quad \alpha = \frac{\tau}{I}$

so, $\alpha = \frac{5}{2} = 2.5$ rad/sec²

39. $\omega = \omega_0 + \alpha t$

$\Rightarrow$ $\omega = 0 + 2.5 \times 2 = 5$ rad/sec

40. as, $\tau = \frac{\Delta L}{\Delta t}$

so, $\Delta L = \tau \times \Delta t$

$\Rightarrow$ ΔL = change in angular momentum = 5 × 2 = 10 N-m sec

41. $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$

$\Rightarrow \quad \theta = 0 \times 2 + \frac{1}{2} \times 2.5 \times (2)^2 = 5$ radian

42. as, torque = $I\alpha$

$\Rightarrow$ $\tau = mK^2\alpha$ {where K = radius of gyration}

43. Power, $P = \tau \times \omega$

so, $\tau = \frac{P}{\omega} = \frac{1200}{600} = 2$ N-m

44. $\vec{\tau} = \frac{d\vec{L}}{dt}$

45. Frequency, n = 20 Hz

$\omega_i = 2\pi n = 40\pi$ rad/sec

$\omega_f = 0$, t = 10 sec

so, $\omega_f = \omega_i + \alpha t$

$0 = 40\pi + \alpha \times 10$

so, $\alpha = -4\pi$ rad/sec²

Now, $\tau = I\alpha = 5 \times 10^{-3} \times 4\pi = 2\pi \times 10^{-2}$ N-m

46. $f = 60$ rpm $= \frac{60}{60}$ rps $= 1$ rps

$I = 2$ kg-m²

$\omega = \omega_0 + \alpha t$

$0 = 2\pi + \alpha(60)$

$\alpha = -\frac{2\pi}{60} = -\frac{\pi}{30}$ $\quad \therefore$ (Retardation)

$\tau = I\alpha$

$\tau = 2 \times \frac{\pi}{30} = \frac{\pi}{15}$

47. Torque = $\frac{\text{Change in angular momentum}}{\text{time}}$

$\tau = \frac{L_2 - L_1}{t} = \frac{4A_0 - A_0}{4}$

$\Rightarrow \quad \tau = \frac{3A_0}{4}$

48. $\vec{\tau} = I\vec{\alpha}$

$\vec{\tau}$ = constant

$\vec{\alpha}$ = constant, if I = constant

49. $\vec{\tau} = I\vec{\alpha} = \frac{d\vec{L}}{dt}$

$\vec{\tau} = 0 \Rightarrow \vec{L}$ = constant angular momentum

50. given, $\omega_0 = 20$ rad/sec

$\omega = 0$

$I = 50$ kg-m²

t = 10 sec

$\alpha = \frac{\omega - \omega_0}{t} = \frac{0 - 20}{10} = -2$ rad/sec²

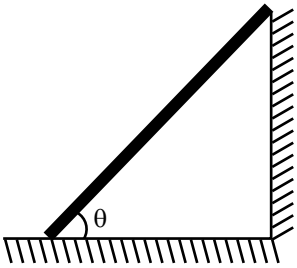
and $\tau = I\alpha = 50 \times 2 = 100$ kg-m²/s² = 100 N-m

51. Torque $\vec{\tau} = \vec{r} \times \vec{F}$

$\therefore \quad \vec{\tau} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 0.5 & 1 \\ 2 & 0 & -3 \end{vmatrix}$

$\vec{\tau} = \left(-1.5\hat{i} - 4\hat{j} - \hat{k}\right)$N − m

52.



Increasing angle causes equilibrium.

53. J = Iω where 'I' is constant. So graph will be straight line.

54. Angular momentum is an axial vector.

55. $I = mR^2 = 10(0.2)^2 = 0.4$ kg-m^2

$\omega = \frac{1200\times 2\pi}{60}$ rad/sec.

$\omega = 40\pi$ rad/s

Angular Momentum $L = I\omega = 16\pi$ J-s

≈ 50.25 J – s

56. given, $K_1 = K_2$

$\Rightarrow \quad \frac{L_1^2}{2I_1} = \frac{L_2^2}{2I_2} \quad \Rightarrow \quad \frac{L_1^2}{2\times 9} = \frac{L_2^2}{2\times 1}$

$\Rightarrow \quad \frac{L_1^2}{L_2^2} = \frac{9}{1} \quad$ or $\quad \frac{L_1}{L_2} = \frac{3}{1}$

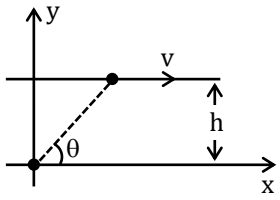
57. $f = 0.5$

$\omega = 2\pi f = \pi$

$L = I\omega$

$= 0.6\pi$ kg $\times \frac{m^2}{S}$

58. Constant velocity



Angular momentum = $m\vec{v}_1\vec{r}$

$= mvh =$ constant

59. Angular momentum is an axial vector.

60. Relation between, kinetic energy (K) and angular momentum (L) is :-

$L = \sqrt{2IK}$ {where, I = moment of inertia}

so, $\frac{L_1}{L_2} = \sqrt{\frac{I_1}{I_2}} = \sqrt{\frac{I}{2I}} = 1:\sqrt{2}$

61. Rotational kinetic energy of a body,

$K_{rot} = \frac{1}{2}I\omega^2$

or, $I\omega^2 = 2K_{rot}$ or $\omega = \sqrt{\frac{2K_{rot}}{I}}$ ....(i)

Angular momentum of a body $L = I\omega$

or, $L = I\sqrt{\frac{2K_{rot}}{I}} = \sqrt{2IK_{rot}}$ [From (i)]

62. $M = 10$ kg

$K = 0.1$ m

$\omega = 10$ rad/sec

angular momentum (L) = $I\omega$

$= MK^2\omega = 10 \times (0.1)^2 \times 10$

$\Rightarrow L = 1$ kg m^2/s

63. Here, Mass, M = 1.0 kg

Diameter, D = 2.0 m

$\therefore$ Radius, $R = \frac{D}{2} = 1.0$ m

The moment of inertia of the body,

$I = MR^2 = (1.0 \text{ kg})\ (1.0\text{m})^2 = 1.0$ kg m^2

The angular velocity of the body,

$\omega = 2\pi\upsilon = 2 \times 3.14 \times \frac{10}{31.4}$ rad / s

$= 2$ rad/s

The angular momentum of the body,

$L = I\omega = (1.0$ kg m$^2)\ (2$rad/s)

$= 2$ kg m^2/s

64. Centripetal force, $F_C = \frac{mv^2}{r}$ ...(i)

and angular momentum, $L = mvr$

so, $v = \frac{L}{mr}$

putting value of v in (i),

$F_C = \frac{m}{r}\times\left(\frac{L}{mr}\right)^2$

$\Rightarrow \quad F_C = \frac{m}{r}\times\frac{L^2}{m^2r^2} = \frac{L^2}{mr^3}$

65. Since torque of string on stone is zero, the angular momentum will be conserved

66. $\omega_1 = \frac{2\pi}{10} = \frac{\pi}{5}$ rad/s.

By conservation of Angular momentum

$I_1\omega_1 = I_2\omega_2$

$\Rightarrow \quad 100\left(\frac{\pi}{5}\right) = [100 + 50(2)^2]\omega$

$\Rightarrow \quad \omega = \frac{\pi}{15}$ or $\frac{2\pi}{30}$

67. On melting water will be more at equator which increase "r", $\therefore$ moment of inertia increases.

68. On loosing atmosphere, moment of inertia of earth decreases so, ω increases and time period T decreases.

$\left\{\because T = \frac{2\pi}{\omega}\right\}$

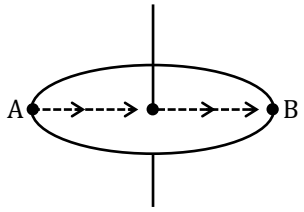
69. m ↑ I ↑

70. $I\omega$ = constant

I ↑ ω ↓

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71.



Initially as ant move towards axis I decreases so ω increases.

After crossing axis I increases and ω decreases.

so, first ω increases then ω decreases.

72. $\vec{\tau}=\frac{d\vec{L}}{dt}$

$\vec{\tau}=0$

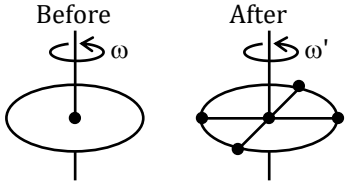
$\vec{L}$ = constant

73. As person is coming towards axis, it's distance from axis decreases. So M.I also decreases.

From conservation of angular momentum :-

as I decreases so, ω increases

74.



Applying conservation of angular momentum

$I_1\omega_1 = I_2\omega'$

$Mr^2 \times \omega = (Mr^2 + 4mr^2)\omega'$

so, $\omega'=\frac{Mr^2\omega}{r^2(M+4m)}=\frac{M\omega}{M+4m}$

75. Applying conservation of angular momentum

$I_1\omega = (I_1 + I_2)\omega'$

so, $\omega'=\frac{I_1\omega}{(I_1+I_2)}$

76. $\vec{\tau}=\frac{d\vec{L}}{dt}$

$\vec{L}$ = Angular momentum

77. As the angular momentum is conserved

$\therefore I_1\omega_1 = I_2\omega_2$

or, $mr^2 2\omega = m(r/2)^2\omega_2$ ($\omega_1 = 2\omega$)

$\omega_2 = 8\omega$.

78. Angular momentum is conserved,

i.e. $L = I\omega$ = constant

$\therefore I_1\omega_1 = I_2\omega_2$

$$\frac{2}{5}MR^2.\frac{2\pi}{T_1}=\frac{2}{5}M\left(\frac{R}{2}\right)^2.\frac{2\pi}{T_2}$$

$$\Rightarrow \frac{R^2}{T}=\frac{R^2}{4}\frac{1}{T_2}\Rightarrow T_2=\frac{T_1}{4}=\frac{24hr.}{4}$$

T_2 = 6hr.

79. Direction of linear momentum is variable.

80. Frequency of rotation = n Hz

so, $\omega = 2\pi n$

and kinetic energy, $K=\frac{1}{2}I\omega^2$

so, $K=\frac{1}{2}\times\frac{mL^2}{3}\times(4\pi^2\times n^2)$

$\Rightarrow K=\frac{2}{3}mL^2\pi^2n^2$

81. If torque is absent, Angular momentum will be conserved.

$E = \frac{L^2}{2I}\Rightarrow E\propto\frac{1}{I}$

$I \propto m$ & $I \propto K^2$

So $\frac{E_1}{E_2}=\frac{m_2}{m_1}\times\left(\frac{K_2}{K_1}\right)^2=2\Rightarrow E_2\ \frac{1}{2}$ = E_1 = 0.5 E

82. Kinetic energy of particle

$K=\frac{1}{2}mv^2$ ...(i)

and angular momentum, L = mvr

so, $v=\frac{L}{mr}$

putting v in (i), $K=\frac{1}{2}m\left(\frac{L}{mr}\right)^2=\frac{1}{2}m\times\frac{L^2}{m^2r^2}$

so, $K=\frac{L^2}{2mr^2}$

83. as, $K=\frac{1}{2}I\omega^2=\frac{1}{2}(I\times\omega)\times\omega$

$\Rightarrow K=\frac{1}{2}L\omega$ or $L=\frac{2K}{\omega}$

so, $\frac{L_2}{L_1}=\frac{K_2}{K_1}\times\frac{\omega_1}{\omega_2}=\frac{1}{2}\times\frac{1}{2}$

$\Rightarrow \frac{L_2}{L_1}=\frac{1}{4}$ or $L_2=\frac{L}{4}$

84. Revolutions per minute (N) $=\frac{3000}{\pi}$ rpm

so, frequency, $n=\frac{N}{60}=\frac{3000}{60\times\pi}=\frac{50}{\pi}$ Hz

and $\omega=2\pi n=2\pi\times\frac{50}{\pi}=100$ rad/sec

$I = 400\ kg\ m^2$

so rotational kinetic energy (K) is

$K=\frac{1}{2}I\omega^2=\frac{1}{2}\times400\times(100)^2=2\times10^6\ J$

85. Translational KE = $\frac{1}{2}mv^2 = E$

Total KE = $\frac{1}{2}mv^2\left(1+\frac{K^2}{R^2}\right)$

$=\frac{1}{2}mv^2\ (1+1)=2E$

86. $\frac{K_{Slide}}{K_{Roll}} = \frac{\frac{1}{2}mv^2}{\frac{1}{2}mv^2\left(1+\frac{K^2}{R^2}\right)}$

$=\frac{1}{1+1}=1:2$

87. For pure rolling

$K_{Rotation}=\frac{1}{2}mv^2\times\frac{K^2}{R^2}$

and $K_{Total}=\frac{1}{2}mv^2\left(1+\frac{K^2}{R^2}\right)$

so $\frac{K_{Rotation}}{K_{Total}}=\frac{\frac{K^2}{R^2}}{\left(1+\frac{K^2}{R^2}\right)}$

for solid sphere, $\frac{K^2}{R^2}=\frac{2}{5}$

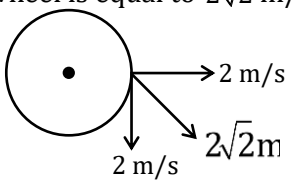
so, $\frac{K_{Rotation}}{K_{Total}}=\frac{\frac{2}{5}}{1+\frac{2}{5}}=\frac{2}{7}$

88. $\frac{K_{Rotation}}{K_{Total}}=\frac{\frac{K^2}{R^2}}{1+\frac{K^2}{R^2}}$

For disc, $\frac{K^2}{R^2}=\frac{1}{2}$,

so, fraction $=\frac{\frac{1}{2}}{1+\frac{1}{2}}=1:3$

89. Different parts of the rolloing wheel move with the same linear and angular speed. The magnitude of the linear velocity of the points at the extermities of the horizontal diameter of the wheel is equal to $2\sqrt{2}$ m/s



90. Given, $\frac{K_{Rotation}}{K_{Total}}\times100=50\%$

$\Rightarrow \frac{\frac{K^2}{R^2}}{1+\frac{K^2}{R^2}}\times100=50 \Rightarrow \frac{\frac{K^2}{R^2}}{1+\frac{K^2}{R^2}}=\frac{1}{2}$

so, $\frac{K^2}{R^2}=1$, so it is a ring.

91. Total energy = $\frac{1}{2}mv^2\left[1+\frac{K^2}{R^2}\right]$

Rotational energy $\frac{1}{2}I\omega^2$ = or $\frac{1}{2}mv^2\left[\frac{K^2}{R^2}\right]$

92. Rotational kinetic energy = $\frac{1}{2}I\omega^2$

$K_1=\frac{1}{2}I_1\omega_1^2; K_2=\frac{1}{2}I_2\omega_2^2$

$\therefore \frac{K_2}{K_1}=\left(\frac{I_2}{I_1}\right)\left(\frac{\omega_2}{\omega_1}\right)^2=\left(\frac{I_1}{2I_1}\right)\left(\frac{2\omega_1}{\omega_1}\right)^2=\frac{2}{1}$

or $K_2 = 2K_1$

Rotational KE will be doubled.

93. $\frac{K_{Rot.}}{K_{Total}}=\frac{\frac{K^2}{R^2}}{1+\frac{K^2}{R^2}}=\frac{\frac{2}{5}}{1+\frac{2}{5}}$

$= 2 : 7$

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94. as, $KE = \frac{L^2}{2I}$

if L = constant,

then $KE \propto \frac{1}{I}$

as $I_A > I_B$

so $(KE)_A < (KE)_B$

95. Kinetic energy of ring,

$$KE = \frac{1}{2}mv^2\left(1+\frac{K^2}{R^2}\right)$$

$$= \frac{1}{2}0.4\times\left(\frac{10}{100}\right)^2\times(1+1)\left[\because \frac{K^2}{R^2}=1\right]$$

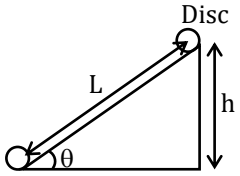
$\Rightarrow$ $KE = 4 \times 10^{-3}$ J

96. L = Angular momentum

$\Rightarrow L = I\omega$

Rot. KE = $\frac{1}{2}I\omega^2 = \frac{I^2\omega^2}{2I} = \frac{L^2}{2I}$

97.



Applying conservation of energy,

$$mgh = \frac{1}{2}mv^2\left(1+\frac{K^2}{R^2}\right)$$

also, $h = L\sin\theta$ and $\frac{K^2}{R^2}=\frac{1}{2}$

so, $mgL\sin\theta = \frac{1}{2}mv^2\left(1+\frac{1}{2}\right)$

on solving, $v = \sqrt{\frac{4gL\sin\theta}{3}}$

98. Acceleration of a purely rolling object on an inclined plane is :-

$$a = \frac{g\sin\theta}{\left(1+\frac{K^2}{R^2}\right)}$$

for spherical shell, $\frac{K^2}{R^2}=\frac{2}{3}$

for solid cylinder, $\frac{K^2}{R^2}=\frac{1}{2}$

so, $\frac{a_{shell}}{a_{cylinder}} = \frac{\frac{g\sin\theta}{\left(1+\frac{2}{3}\right)}}{\frac{g\sin\theta}{\left(1+\frac{1}{2}\right)}} = \frac{9}{10}$

99. t_1 = time for sliding = $\sqrt{\frac{2L}{a_s}}$

t_2 = time for rolling = $\sqrt{\frac{2L}{a_R}}$

$\Rightarrow t_1 : t_2 = \sqrt{a_R} : \sqrt{a_s}$

$a_R = \frac{mg\sin\theta}{m+\frac{I}{R^2}} = \frac{g\sin\theta}{2} = \frac{1}{2}a_s$

$\Rightarrow$ $t_1 : t_2 = 1 : \sqrt{2}$

100. $t_{rolling} > t_{sliding}$

101. As we know, $t = \frac{1}{\sin\theta}\sqrt{\frac{2h}{g}\left(1+\frac{K^2}{R^2}\right)}$

For sphere, $\frac{K^2}{R^2}=\frac{2}{5}$ & For disc, $\frac{K^2}{R^2}=\frac{1}{2}$

so, $\frac{t_{sphere}}{t_{disc}} = \frac{\frac{1}{\sin\theta}\sqrt{\frac{2h}{g}\left(1+\frac{2}{5}\right)}}{\frac{1}{\sin\theta}\sqrt{\frac{2h}{g}\left(1+\frac{1}{2}\right)}}$

102. $v \propto \sqrt{a}$

$\frac{v}{v_R} = \sqrt{\frac{a}{a_R}}$ & $a_R = \frac{g\sin\theta}{1+\frac{I}{mR^2}} = \frac{g\sin\theta}{2}$

$a = g\sin\theta$ $\Rightarrow v_R = \frac{v}{\sqrt{2}}$

103. Percentage of total energy which is rotational is

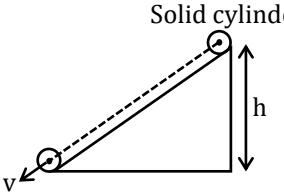
$$= \frac{\frac{K^2}{R^2}}{\left(1+\frac{K^2}{R^2}\right)}\times 100$$

$$= \frac{\frac{2}{5}}{1+\frac{2}{5}}\times 100 \left\{\because \text{for sphere}\ \frac{K^2}{R^2}=\frac{2}{5}\right\}$$

$$= \frac{2}{7}\times 100 \approx 28\%$$

104. For rolling we have both translational and rotational motion together.

105. Solid cylinder



Applying conservation of energy :-

$mgh = K_{Total}$

so $mgh = \frac{1}{2}mv^2\left(1+\frac{K^2}{R^2}\right)$

$mgh = \frac{1}{2}mv^2\left(1+\frac{1}{2}\right)$

on solving, $v = \sqrt{\frac{4}{3}gh}$

106. The magnitude changes and the direction of angular momentum does not change when the cylinder rolls down a slope.

107. For rolling motion on an inclined plain acceleration is given by $a_R = \frac{g\sin\theta}{\left(1+\frac{K^2}{R^2}\right)}$

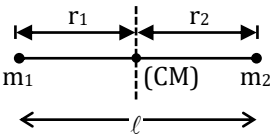
Exercise-II (Previous Year Questions)

13. $E_{sphere} = \frac{1}{2}I_s\omega^2 = \frac{1}{2}\times\frac{2}{5}MR^2\times\omega^2$

$E_{cylinder} = \frac{1}{2}I_c(2\omega)^2 = \frac{1}{2}\times\frac{MR^2}{2}\times 4\omega^2$

$\frac{E_{sphere}}{E_{cylinder}} = \frac{1}{5}$

14.



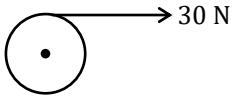
$r_1 = \frac{m_2\ell}{m_1+m_2}$, $r_2 = \frac{m_1\ell}{m_1+m_2}$

$I_{cm} = m_1r_1^2 + m_2r_2^2 = \frac{m_1m_2}{m_1+m_2}\ell^2$

OR

$I_{cm} = \mu\ell^2 = \frac{m_1m_2}{m_1+m_2}\ell^2$

15. $\tau = I\alpha$



$RF = mR^2\alpha$

$\alpha = \frac{F}{mR} = \frac{30}{3\times\frac{40}{100}} = 25\ rad/s^2$

16. COAM : $I\omega_1 + I\omega_2 = 2I\omega \Rightarrow \omega = \frac{\omega_1+\omega_2}{2}$

$(K.E.)_i = \frac{1}{2}I\omega_1^2 + \frac{1}{2}I\omega_2^2$

$(K.E.)_f = \frac{1}{2}\times 2I\omega^2 = I\left(\frac{\omega_1+\omega_2}{2}\right)^2$

Loss in K.E. = $(K.E.)_i - (K.E)_f = \frac{I}{4}(\omega_1-\omega_2)^2$

17. Centre of mass may lie on centre of gravity net torque of gravitational pull is zero about centre of mass.

Mechanical advantage = $\frac{Load}{Effort} > 1$

$\Rightarrow$ Load > Effort

18. W = loss in KE = $\frac{1}{2}I\omega^2 \propto I$

$I_A = \frac{2}{5}MR^2 = 0.4MR^2$

$I_B = \frac{1}{2}MR^2 = 0.5MR^2$

$I_C = MR^2$

$\therefore\ W_C > W_B > W_A$

19. Here $\vec{F} = 4\hat{i}+5\hat{j}-6\hat{k}$ and

$\vec{r} = (2-2)\hat{i}+(0-(-2))\hat{j}+(-3-(-2))\hat{k} = 2\hat{j}-1\hat{k}$

$\vec{r}\times\vec{F} = \begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 0 & 2 & -1\\ 4 & 5 & -6\end{vmatrix} = -7\hat{i}-4\hat{j}-8\hat{k}$

20. $K_t = \frac{1}{2}mv^2$

$K_t + K_r = \frac{1}{2}mv^2\left(1+\frac{K^2}{R^2}\right) = \frac{1}{2}mv^2\left(1+\frac{2}{5}\right)$

$\therefore\ \frac{K_t}{K_t+K_r} = \frac{1/2\ mv^2}{1/2\ mv^2\left(1+\frac{2}{5}\right)} = \frac{5}{7}$

21. $\tau_{ext} = 0 \Rightarrow \frac{d\vec{L}}{dt} = 0 \Rightarrow \vec{L}$ = constant

22. $W_{all} = \Delta KE$

$\Rightarrow \quad W = 0 - \frac{1}{2}mv_{cm}^2\left(1+\frac{K^2}{R^2}\right)$

$\Rightarrow$ W = –3J

23. $\theta = 2\pi \times 2\pi$ radian

$\omega_0 = 3$ rpm $\Rightarrow \frac{2\pi}{60}(3)\frac{rad}{sec}$

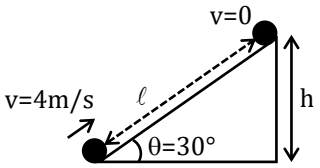
$\omega^2 = \omega_0^2 - 2\alpha\theta$

$0 = \left(\frac{3\times 2\pi}{60}\right)^2 - 2\alpha(4\pi^2)$

$\therefore \quad \alpha = \frac{1}{800}$ rad/s²

$\tau = \frac{mR^2}{2}\alpha = \frac{2}{2}\times\left(\frac{4}{100}\right)^2\times\frac{1}{800} = 2\times10^{-6}$ Nm

24.



$\frac{1}{2}mv^2\left(1+\frac{K^2}{R^2}\right) = mgh$

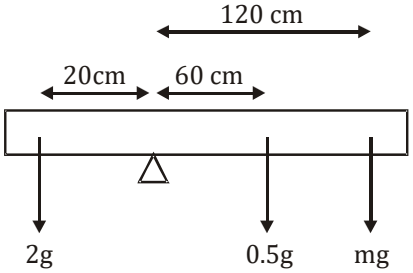
$\Rightarrow 8\left(1+\frac{1}{2}\right) = 10h \Rightarrow$ h = 1.2 m

$\frac{h}{\ell} = \sin 30° \Rightarrow \ell$ = 2.4 m

25. $\vec{F} = 3\hat{j}$N, $\vec{r} = 2\hat{k}$

$\vec{\tau} = \vec{r}\times\vec{F} = 2\hat{k}\times3\hat{j} = 6(\hat{k}\times\hat{j}) = 6(-\hat{i}) = -6\hat{i}$ Nm

26.



By balancing torque

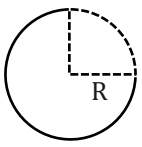
2g × 20 = 0.5 g × 60 + mg × 120

$m = \frac{0.5}{6}kg = \frac{1}{12}kg$

27. $M_{remain} = \frac{3}{4}M$

$I = M_{remain}R^2$

$= \frac{3}{4}MR^2$

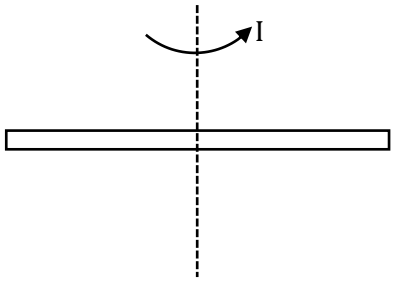


30. Radius of gyration : $K = \sqrt{\frac{I}{m}}$

$\frac{k_{solid\ sphere}}{k_{hollow\ sphere}} = \sqrt{\frac{2mR^2/5m}{2mR^2/3m}} = \sqrt{3}:\sqrt{5}$

31. I = 2400 gcm²

m = 400 g



$I = \frac{ML^2}{12}$

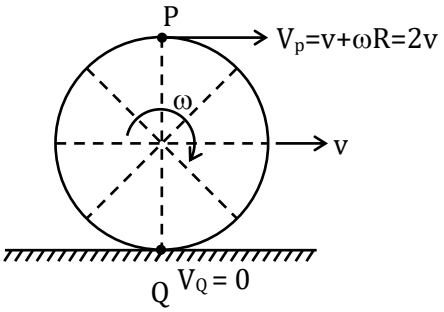
$2400 = \frac{400\times L^2}{12}$

$\Rightarrow L^2 = 72$

$\Rightarrow L = \sqrt{72} \approx 8.5$ cm

32. In case of Pure Rolling :-

$v = \omega R$



$\therefore V_p = 2V$

$V_Q = 0$

$\therefore$ Point 'P' moves faster than point Q.

33. $I = MK^2$

$\frac{7}{5}MR^2 = MK^2$

$\Rightarrow \frac{7}{5}\frac{25X^2}{7} = (5)^2$

$x = \sqrt{5}$

Exercise-III (Analytical Questions)

3. **Axis-1:**

$(M.I.)_{Rod\ Hz.} = \frac{m\ell^2}{3}$; $(M.I.)_{Rod\ Vert.} = 0$

Axis-2:

$(M.I.)_{Rod\ Hz.} = 0$; $(M.I.)_{Rod\ Vert.} = \frac{m\ell^2}{3}$

Axis-3:

$(M.I.)_{Rod\ Hz.} = \frac{m\ell^2}{3}$; $(M.I.)_{Rod\ Vert.} = \frac{m\ell^2}{3}$

Axis-4:

$(M.I.)_{Rod\ Hz.} = \frac{m\ell^2}{3}$; $(M.I.)_{Rod\ Vert.} = mL^2$

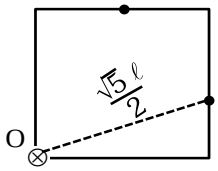
4. about (A) $10(2)\hat{k}+6(2)\hat{k}=32\hat{k}$

about (B) $10\times 2\hat{k}-10\times 2\hat{k}+6\times 2\hat{k}-12\times 2\hat{k}=-12\hat{k}$

about (C) $-(10)\ (2)\ \hat{k} = -20\hat{k}$

about (D) $-12(2)\hat{k}-10(2)\hat{k}-10(2)\hat{k}=-64\hat{k}$

6. (P)



M.O.I. about z-axis passing through 'O'

$= 2\left[\frac{m\ell^2}{3}\right]+2\left[\frac{m\ell^2}{12}+\frac{5}{4}m\ell^2\right] = \frac{10}{2}m\ell^2$

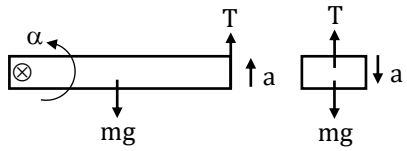
(Q) about are side $I=2\left(\frac{m\ell^2}{3}\right) + m\ell^2 = \frac{5m\ell^2}{3}$

(R) about x-axis $I=2\left(\frac{m\ell^2}{3}\right)+2(m)\left(\frac{\ell}{2}\right)^2 =\frac{2}{3}m\ell^2$

(S) about z-axis $I=4\left[\frac{m\ell^2}{12}+m\left(\frac{\ell}{2}\right)^2\right]$

$=\frac{m\ell^2}{3}+m\ell^2=\frac{4m\ell^2}{3}$

8.



$a = \ell\alpha$ ... (1)

$mg - T = ma$ ... (2)

$T\ell - mg\frac{\ell}{2} = I\alpha$ ... (3)

on solving (1), (2), (3)

$a = 3g/8$

$a = \frac{3g}{3\ell}$

$T = \frac{5mg}{8}$

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Gravitation

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Control the rotational motion of satellites and planets. Explained in Kepler's law.

2. $F_g = GMm\left(\frac{1}{r_1^2}+\frac{1}{r_2^2}+...\infty\right)$

$F_g = G(1)(3)\left(\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{4^2}+...\infty\right)$

$F_g = 3G\left(1+\frac{1}{4}+\frac{1}{16}+...\infty\right)$

$F_g = 3G\left(\frac{1}{1-\frac{1}{4}}\right) = 4G$

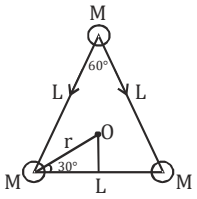
3. Gravitational force is medium independent

4. It is a universal constant.

5. Net force on any particle :

$= \sqrt{F^2+F^2+2F^2\cos 60°}$

$= \sqrt{3F^2} = F\sqrt{3}$



$\left(F = \frac{GM^2}{L^2}\right.$ is force of any one particle on another$\left.\right)$

$F\sqrt{3} = \frac{Mv^2}{r}$ (by centripetal force)

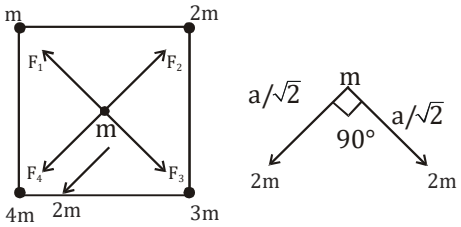
Where $\frac{L}{2r} = \cos 30° \Rightarrow r = \frac{L}{\sqrt{3}}$

$\sqrt{3}\frac{GM^2}{L^2} = \frac{Mv^2}{r} \Rightarrow \sqrt{3}\frac{GM}{L^2} = \sqrt{3}\frac{v^2}{L}$

$\Rightarrow v = \sqrt{\frac{GM}{L}}$ $\quad\because v = \omega r$

$\therefore \quad \omega = \frac{v}{r} = \sqrt{\frac{3GM}{L^3}}$

6.



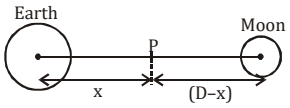
$F_{net} = \sqrt{2}\,F$

$= \sqrt{2}\frac{G\times 2m\times m}{(a/\sqrt{2})^2} = \frac{4\sqrt{2}Gm^2}{a^2}$

7. The tidal waves in seas are primarily due to gravitational force of the moon.

8. More work is done against the gravitational pull of the earth.

9.



At point 'p' for gravitational field to be zero field due to earth = field due to moon

$\Rightarrow \frac{GM_e}{x^2} = \frac{GM_m}{(D-x)^2} \Rightarrow \frac{81M_m}{x^2} = \frac{M_m}{(D-x)^2}$

$\Rightarrow \frac{x}{D-x} = 9 \Rightarrow 9(D-x) = x \Rightarrow x = \frac{9D}{10}$

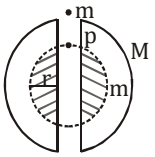
10. By first law of motion i.e. law of inertia, satellite will fly off tangentially from the orbit with speed V.

11. $I_g = \frac{-GMr}{R^3}$; $r < R$

$I_g = \frac{-GM}{r^2}$; $r \geq R$

12. Weight is due to gravitational pull of earth which varies with distance as $\frac{1}{r^2}$. Weight becomes zero when gravitational pull of both planets become equal & opposite. After that its weight again increase to 240 N due to mars.

13. The gravitational field at P is due to the mass enclosed. (m')



the force acting = $\frac{Gm'm}{r^2}$,

m = mass of the particle released.

$F = G\rho\frac{4}{3}\pi r^3 \cdot \frac{m}{r^2}$ $\quad \left\{m' = \frac{4}{3}\pi\rho r^3\right\}$

$F = \left[\frac{4}{3}\pi G\rho m\right] r$

14. $g = \frac{GM_e}{R_e^2}$

$g_{mass} = \frac{G(0.1)M_e}{(0.5)^2 R_e^2} = \frac{0.4GM_e}{R_e^2} = 0.4\,g$

15. By symmetrically all forces will cancel each other and resultant will be zero.

16. $g = \frac{GM_E}{R_E^2}$

where M_E and R_E is the mass and radius of the earth respectively.

$M_E = \frac{g}{G}R_E^2$

17. $g_d = g_s\left(1 - \frac{d}{Re}\right)$ $\quad$ d = R_e; g_d = 0

18. The value of 'g' on earth surface depends on both structure and rotational motion.

$g = \frac{GM_e}{R_e^2}$

$g' = g - R_e\omega^2\cos^2\lambda$

19. As $g' = \frac{g}{\left(1+\frac{h}{R}\right)^2}$

$\frac{g}{2} = \frac{g}{\left(1+\frac{h}{R}\right)^2} \Rightarrow \left(1+\frac{h}{R}\right)^2 = 2$

$\Rightarrow \frac{h}{R} = \sqrt{2}-1 \Rightarrow h = R\left(\sqrt{2}-1\right)$

20. Use $mg_1h_1 = mg_2h_2$

$\Rightarrow$ $(1.96)(h_1) = (9.8)(2)$

h_1 = 10 m

21. $g' = g - R_e\omega^2\cos^2\lambda \Rightarrow g' < g$

except for $\lambda = 90°$ i.e. for poles.

22. $g \propto \frac{M}{R^2}$

$\frac{g'}{g} = \left(\frac{M'}{M}\right)\left(\frac{R}{R'}\right)^2 = (2)\left(\frac{1}{2}\right)^2 = \frac{1}{2}$

As $T \propto \frac{1}{\sqrt{g}}$

Hence $\frac{T'}{T} = \sqrt{\frac{g}{g'}} = \sqrt{2}$

then $T' = \sqrt{2}T$

23. There is no atmosphere on the moon, means no buoyant force in upward direction so it will fall down with acceleration g/6.

24. $g = \frac{GM}{R^2}$

$\therefore$ $M = \rho \times \frac{4}{3}\pi R^3 \Rightarrow g = \frac{G\times 4\pi R^3 \times \rho}{3\times R^2}$

$\Rightarrow \rho = \frac{3g}{4\pi GR}$

25. $g' = g - \omega^2 R_e \cos^2\lambda$

at poles $\lambda = 90°$, $\cos 90° = 0$

at equator $\lambda = 0°$, $\cos 0° = 1$

$\therefore$ $g_{poles} > g_{equator}$

Given $m_pg_p = m_{eq}\,g_{eq}$

26. $g_{equator} = g - \omega^2 R_e$

$g_{pole} > g_{equator}$

27. As $g = \frac{GM}{R^2}$

$\frac{\Delta g}{g}\% = \frac{-2\Delta R}{R}\%$

$\frac{\Delta g}{g}\% = -2(-1)\% = +2\%$

28. g decreased by 1% for h height

$$g_h = g\left[1+\frac{h}{R}\right]^{-2}$$

$$\frac{\Delta g_h}{g} = \frac{-2h}{R}$$

for depth.

$$g_d = g\left[1-\frac{d}{R}\right]$$

$$\frac{\Delta g_d}{g} = \frac{-d}{R} \text{ (here d = h)}$$

$$\frac{\Delta g_h}{g} = \frac{2\Delta g_d}{g},\ \frac{\Delta g_d}{g}\times 100 = \frac{1}{2}\left[\frac{(\Delta g)_h}{g}\times 100\right]$$

$$\Delta g_d = \frac{1}{2}\Delta g_h$$

29. $g_{eff} = g\left(1-\frac{2h}{R}\right) = g\left(1-\frac{32\times 2}{6400}\right) = g(1-0.01)$

$g_{eff} = 0.99g\ m/s^2$

30. Action and reaction force are equal in magnitude.

31. As $g = \frac{4}{3}\pi\rho RG$

$g \propto R$ (As ρ = constant)

$$\frac{g'}{g} = \left(\frac{R'}{R}\right)$$

$$\frac{g'}{g} = 3 \quad \Rightarrow \quad g' = 3g$$

32. $$g_h = g\left[1-\frac{2h}{R_e}\right]$$

$$g_d = g\left[1-\frac{d}{R_e}\right]$$

put $g_h = g_d$

$$\frac{2h}{R_e} = \frac{d}{R_e} \Rightarrow d = 2h$$

33. The acceleration due to gravity at equator is given by

$g' = g - R\omega^2$

When rotational speed of earth i.e. ω increased, the acceleration due to gravity at equator (g') will decrease, it means weight of body will decrease.

34. At surface of earth,

$$g = \frac{GM}{R^2} \quad \text{....(i)}$$

At a height, h = R/2, the value of acceleration due to gravity is

$$g' = \frac{GM}{(h+R)^2} = \frac{GM}{\left(\frac{R}{2}+R\right)^2} = \frac{GM}{\left(\frac{3R}{2}\right)^2} = \frac{GM}{\left(\frac{3R}{2}\right)^2},$$

$$g' = \frac{4}{9}\frac{GM}{R^2} = \frac{4}{9}g$$

$\therefore$ weight at a height h = R/2 is

$$W' = mg' = \frac{4}{9}mg = \frac{4}{9}W.$$

35. At equator $g' = g - \omega^2 R_e$

For the case of weightlessness g' = 0

$\omega^2 R_e = g$

$$\omega = \sqrt{\frac{g}{R_e}} = \sqrt{\frac{10}{6.4\times 10^6}} = \frac{1}{8}\times 10^{-2}$$

$\omega = 0.125 \times 10^{-2}$ rad/sec

$\omega = 1.25 \times 10^{-3}$ rad/sec

36. Acceleration due to gravity is independent of the mass of object $a = \frac{GM_e}{r^2}$ [M_e = mass of earth]

37. Gravitational potential

$$V = \frac{-GM}{r} \text{ for } r > R$$

$$V = \frac{-GM}{R} \text{ for } r \le R$$

Potential remains constant inside the hollow sphere.

38. Gravitational field intensity

$$E_g = -\frac{dV}{dr} = \frac{2}{20} = 0.1$$

Hence gravitational force = mE_g

$= 5(0.1) = 0.5N$

Work = Force × displacement = (0.5)(4) = 2J

39. $$= -\frac{GM_e m}{R} - \frac{GM_m m}{r} = -\frac{G(81M_m)m}{R} - \frac{GM_m m}{r}$$

$$= -GmM_m\left(\frac{81}{R}+\frac{1}{r}\right)$$

40. $E_P = -\frac{GMm}{r}$, when $r \to \infty$, $E_p = 0$ (max.)

41. To escape from earth total energy would be zero. If body is unable to escape then total energy

(PE + KE) is negative, i.e. less than zero.

42. Apply energy conservation

Energy at surface = Energy at height

$$-\frac{GMm}{R}+\frac{1}{2}mv^2=\frac{-GMm}{2R}$$

$$\Rightarrow \quad \frac{1}{2}v^2=\frac{GM}{2R} \quad \Rightarrow \quad v=\sqrt{\frac{GM}{R}}$$

43. GPE of body is $|U|=\frac{GMm}{r}$

Force at that point is $F=\frac{GMm}{r^2}$

$$\frac{U}{F}=r \quad \Rightarrow \quad F=\frac{U}{r}$$

44. Apply energy conservation

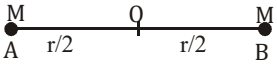
$$\frac{-GMm}{R}+\frac{1}{2}mv^2=0 \quad \Rightarrow \quad v^2=\frac{2GM}{R}$$

$$v=\sqrt{\frac{2GM}{R}} \quad \Rightarrow \quad v=\sqrt{2gR}$$

45. $\Delta U = U_f - U_i = -\frac{GMm}{R+h}-\left(-\frac{GMm}{R}\right)$

$$=GmM\left[\frac{1}{R}-\frac{1}{R+h}\right]=\frac{GmMh}{R(R+h)}$$

46.



Gravitational potential of A at

$$O=-\frac{GM}{r/2}=-\frac{2GM}{r}$$

due to B, potential at $O=-\frac{GM}{r/2}=-\frac{2GM}{r}$

$\therefore$ Total potential = $-\frac{4GM}{r}$

47. The potential energy of a satellite is equal to twice its total energy.

$\therefore$ Potential energy = $2E_0$

48. $\frac{mv^2}{R}=\frac{A}{R^2}$

$KE=\frac{1}{2}mv^2$ $\qquad$ $\boxed{KE=\frac{1}{2}\frac{A}{R}}$

$F=-\frac{dU}{dr}$

$-\frac{A}{r^2}=-\frac{dU}{dr}$

$\int_0^R \frac{A}{r^2}dr=\int dU=\frac{-A}{R}$

TE = PE + KE = $-\frac{A}{R}+\frac{A}{2R}=-\frac{A}{2R}$

49. $PE=\frac{-GMm}{R}$

$-54=\frac{-GM\times 3}{R}$

Hence $\frac{GM}{R}=18$

but $v_{esc}=\sqrt{\frac{2GM}{R}}=\sqrt{36}$ = 6 m/sec

50. $v_{esc}=\sqrt{\frac{2GM}{R}}$

$100=\sqrt{\frac{2GM}{R}} \Rightarrow \frac{GM}{R}=5000$

Gravitational Potential energy = $\frac{-GMm}{R}$

= –5000 × 1 = –5000J

51. $V_{escape} = \sqrt{2gR}$

$\frac{V_1}{V_2}=\sqrt{\frac{g_1R_1}{g_2R_2}}=\sqrt{pq}$

52. Escape velocity $v=\sqrt{\frac{2GM_e}{R}}$ is independent of angle at projection

53. As $v_{esc} \propto \sqrt{\frac{M}{R}}$

Hence $\frac{v_2}{v_1}=\sqrt{\frac{M_2}{M_1}}\sqrt{\frac{R_1}{R_2}}$

$$\frac{v_2}{v_1}=\sqrt{100}\sqrt{\frac{1}{4}}=10\times\frac{1}{2}=5$$

v_2 = 11.2 × 5 = 56 km/sec

54. Let a body is projected from the surface of the earth with the velocity V and reaches maximum height h.

According to law of conservation of mechanical energy, we get

$$\frac{1}{2}mV^2-\frac{GMm}{R}=0-\frac{GMm}{(R+h)}$$

or $\frac{1}{2}V^2-\frac{GM}{R}=-\frac{GM}{(R+h)}$

As per question,

$$V=\frac{V_e}{2}=\frac{1}{2}\sqrt{\frac{2GM}{R}} \qquad \left[\because \quad V_e=\sqrt{\frac{2GM}{R}}\right]$$

$$\frac{1}{2}\left(\frac{1}{2}\frac{GM}{R}\right)-\frac{GM}{R}=-\frac{GM}{(R+h)}$$

$$\frac{GM}{4R}-\frac{GM}{R}=-\frac{GM}{R+h}$$

$$-\frac{3}{4R}=-\frac{1}{(R+h)}$$

3 (R + h) = 4R or 3R + 3h = 4R

$3h = R$ or $h=\frac{R}{3}$

55. Binding energy of moon and earth is $\frac{GM_eM_m}{2r_{em}}$

56. $v\propto\frac{1}{\sqrt{R+h}}$

Hence $\frac{v_A}{v_B}=\sqrt{\frac{r_B+R}{r_A+R}}$

57. $\frac{v_1}{v_2}=\sqrt{\frac{R_2}{R_1}}=\sqrt{\frac{1.5\times10^8}{6\times10^7}}=\frac{\sqrt{5}}{\sqrt{2}}$

58. For a body in geo-stationary orbit gravitational force is equal to the centripetal force so it will remain stationary with respect to the earth like GSS.

59. According to kepler law angular momentum is conserved for planetary motion or satellite motion.

60. According to kepler law angular momentum is conserved for planetary motion or satellite motion.

61. As $T^2\propto r^3$

$$\frac{T_2}{T_1}=\left(\frac{r_2}{r_1}\right)^{3/2}=\left(\frac{9R}{R}\right)^{3/2}=27$$

$$T_2=27\times2\pi\sqrt{\frac{R}{g}}$$

62. Using conservation of angular momentum

$\Rightarrow mv_{max}\, r_{min} = m\, v_{min}\, r_{max}$

$\Rightarrow v_{max}\, r_{min} = v_{min}\, r_{max}$

63. For system to be bounded

Net energy i.e. (KE + PE) must be negative

TE = KE + PE = -ve at points A,B & C.

64. $W = E_2 - E_1 = \frac{GMm}{2}\left(\frac{1}{2R}-\frac{1}{3R}\right)=\frac{GMm}{12R}$

65. $T=\frac{2\pi r}{v}=\frac{2\pi r}{\sqrt{\frac{GM}{r}}}=\frac{2\pi r^{3/2}}{\sqrt{GM}}$

Hence $T^2=\frac{4\pi^2r^3}{GM}$

$$T^2=\left(\frac{4\pi^2}{GM}\right)r^3$$

Hence slope of T^2 Vs r^3 curve is = $\frac{4\pi^2}{GM}$

66. $T^2=\frac{4\pi^2}{GM}a^3$

$$T^2=\frac{4\pi^2}{GM}\left(\frac{r_p+r_A}{2}\right)^3=\frac{\pi^2}{2GM}\left(r_p+r_A\right)^3$$

mvr = constant $\Rightarrow$ vr = constant

$\therefore \quad v_Ar_A = v_pr_p$

$\therefore \quad r_A > r_p$

$\therefore \quad v_A < v_p$

67. Near equator 'g' is minimum due to rotation and shape of earth which help to minimize fuel consumption.

68. For escape from earths gravitational field T.E. must be zero.

KE + PE = 0

$$v_0=\sqrt{\frac{GM}{R}} \quad , \qquad v_e=\sqrt{\frac{2GM}{R}}$$

additional velocity = $v_e - v_0$

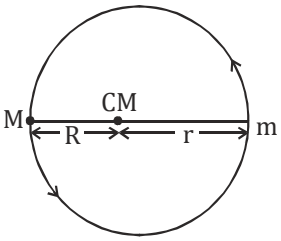
$= 11.2 - 8 \simeq 3.2$ km/s

69. $TE = -KE = -\frac{1}{2}mv^2$

70. $T = 2\pi\sqrt{\frac{r^3}{GM}} \Rightarrow T = 2\pi\sqrt{\frac{(R+h)^3}{GM}}$

$\Rightarrow h = \left[\frac{GMT^2}{4\pi^2}\right]^{1/3} - R$

71.



$\frac{GMm}{(R+r)^2} = m\omega^2 r$

72. $\frac{GMm}{r^2} = \frac{mv^2}{r} \Rightarrow v^2 = \frac{GM}{r}$

Now T.E. $= \frac{1}{2}mv^2 - \frac{GMm}{r}$

$= \frac{1}{2}\frac{mGM}{r} - \frac{GMm}{r} \Rightarrow -\frac{GMm}{2r}$

So total energy of system

$= -\frac{GMm}{2r} - \frac{GMm}{2r} = -\frac{GMm}{r}$

73. $T^2 \propto a^3$

$a = \frac{r_{max} + r_{min}}{2} = \frac{r_1 + r_2}{2}$

$T = \left(\frac{r_1 + r_2}{2}\right)^{3/2}$

$T \propto (r_1 + r_2)^{3/2}$

74. By conservation of mechanical energy

$-\frac{GMm}{2R} + 0 = \frac{1}{2}mv^2 - \frac{GMm}{R}$

$v = \sqrt{\frac{GM}{R}} = 7.919$ km/s

75. COME in conservative field is valid & total energy of bounded system is negative.

76. $F_{cp} = F_g$

$\frac{mv^2}{R} = \frac{K}{R} \Rightarrow v^2 = \frac{K/m}{R^0}$ So proportional to R^0

77. $v = \sqrt{\frac{GM}{R+h}} = \sqrt{\frac{gR^2}{R+h}}$

$v = R\sqrt{\frac{g}{R+h}}$

78. By $V = \sqrt{\frac{GM}{r}}$

$v \propto \frac{1}{\sqrt{r}}$

$\Rightarrow V_1 r_1 = V_2 r_2$

so, if $r_1 > r_2$, then $v_1 < v_2$

79. $T^2 \propto R^3$

$\left(\frac{T_1}{T_2}\right)^2 = \left(\frac{R_1}{R_2}\right)^3 = \left(\frac{R}{4R}\right)^3$

$\frac{T_1}{T_2} = \sqrt{\frac{1}{64}}$

$T_2 = 8T_1$

$T_1 = 1$ day for communication satellite

$T_2 = 8$ days

80. $v_0 = \sqrt{\frac{GM}{R}}$ where M = mass of earth

81. $TE_1 = -\frac{GMm}{2(R+R)} = -\frac{GMm}{4R}$

$TE_2 = -\frac{GMm}{2(R+7R)} = -\frac{GMm}{16R}$

$\frac{TE_1}{TE_2} = \frac{4}{1}$

$\frac{PE_1}{PE_2} = \frac{\frac{-GMm}{2R}}{-\frac{GMm}{8R}} = \frac{4}{1}$

$\frac{KE_1}{KE_2} = \frac{+\frac{GMm}{4R}}{+\frac{GMm}{16R}} = \frac{4}{1}$

82. For satellite net pulling force of gravity must be equal to centripetal force

$$\frac{GMm}{r^2}=\frac{mv^2}{r}$$

$$v=\sqrt{\frac{GM}{r}}$$

if $r \to R$

$$v=\sqrt{\frac{GM}{R}} \approx 8 \text{ km/s}$$

$= 8 \times 10^3$ m/s

83. Apply angular momentum conservation law

$$mv_1r_1 = mv_2r_2$$

$$\Rightarrow \quad 60 \times 1.6 \times 10^{12} = v_2 \times 8 \times 10^{12}$$

$$v_2 = 12 \text{ m/sec}$$

84. For satellite,

Centripetal force = gravitational force

$$\frac{mv^2}{r}=\frac{Gmm_E}{r^2}$$

$$\text{or } v^2=\frac{Gm_E}{r}$$

Hence v depends m_E and r.

85. As $T^2 \propto r^3$

$$\left(\frac{T_2}{T_1}\right)^2=\left(\frac{r_2}{r_1}\right)^3$$

$$\frac{T_2}{1.4}=\left(\frac{4R}{R}\right)^{3/2}=8$$

$T_2 = 1.4 \times 8 = 8\sqrt{2}$ hr.

86. $$T^2=\frac{4\pi^2}{GM}r^3$$

$$\left(\frac{T_1}{T_2}\right)^{2/3}=\frac{r_1}{r_2} \qquad [\because\ T_1 = T_2]$$

$$\Rightarrow \frac{r_1}{r_2}=1$$

87. Apply energy conservation

$$\frac{-GM_em}{R_e}+\frac{1}{2}m(2v_e)^2=\frac{1}{2}mv'^2=0$$

$$\Rightarrow \quad \frac{-GM_em}{R_e}+\frac{1}{2}m\left(2\sqrt{\frac{2GM_e}{R_e}}\right)^2=\frac{1}{2}mv'^2$$

$$\Rightarrow \quad \frac{3GM_em}{R_e}=\frac{1}{2}mv'^2$$

$$v'^2=\frac{6GM_e}{R_e}=3\times\frac{2GM_e}{R_e}$$

$$v'=\sqrt{3}\sqrt{\frac{2GM_e}{R_e}}=\sqrt{3}v_e$$

88. for a satellite

$$\text{K.E.} = \frac{GMm}{2r}$$

$$\text{PE} = \frac{-GMm}{r}$$

89. Orbital velocity of satellite, $v=\sqrt{\frac{GM}{R}}=v_0$

$$\therefore\ v'=\sqrt{\frac{GM}{R+\frac{R}{2}}}=\sqrt{\frac{2GM}{3R}}=\sqrt{\frac{2}{3}}v_0.$$

90. According to Kepler's law, $T^2 \propto r^3$

$$\text{or, } \frac{T_1}{T_2}=\left(\frac{r_1}{r_2}\right)^{3/2}$$

$$\text{or, } T_2=T_1\left(\frac{r_2}{r_1}\right)^{3/2}=T_1\left(\frac{2r_1}{r_1}\right)^{3/2}=T_1 2\sqrt{2}$$

$$T_2=2\sqrt{2}\ (1)$$

$$=2\sqrt{2} \text{ year}$$

91. The satellite revolving around a planet has two types of energies.

(i) Gravitational potential energy U due to position of the satellite is given by

$$U=-\frac{GMm}{R}.$$

(ii) Kinetic energy K due to orbital velocity of satellite is given by

$$K=\frac{1}{2}mv_0^2=\frac{1}{2}m\left(\frac{GM}{R}\right)=\frac{1}{2}\frac{GMm}{R}$$

$\therefore$ Total energy = U + K

$$=-\frac{GMm}{R}+\frac{GMm}{2R}=-\frac{GMm}{2R}$$

92. Kinetic energy of a satellite $K = \frac{GMm}{2r}$

where G = Universal gravitational constant

M = Mass of a earth

m = Mass of a satellite

Potential energy of a satellite $U = -\frac{GMm}{r}$

Total energy of a satellite $E = -\frac{GMm}{2r}$

In a graph 1 represents kinetic energy, 2 total energy and 3 represent potential energy.

93. According to Kepler's third law

$T^2 \propto r^3$ or $T \propto r^{3/2}$

$$\frac{T_E}{T_M} = \left(\frac{r_E}{r_M}\right)^{3/2}$$

where T_E = time of revolution of the earth around the sun = 1 year

r_E = mean distance of earth from the sun

T_M = time of revolution of the mars around the sun

r_M = mean distance of mars from the sun.

$$T_M = T_E\left(\frac{r_M}{r_E}\right)^{3/2} = (1)\left[\frac{1.5r_E}{r_E}\right]^{3/2} = 1.84 \text{ years}$$

Exercise-II (Previous Year Questions)

1. Escape velocity = $\sqrt{\frac{2GM}{R}}$ = c = speed of light

$$R = \frac{2GM}{c^2} = \frac{2\times6.6\times10^{-11}\times5.98\times10^{24}}{(3\times10^8)^2} = 10^{-2}\text{ m}$$

2. Gravitational field or intensity of gravitational field. w.r.t. distance

$I = \frac{-GM}{r^2}$ for $r > R$

$I = \frac{-GMr}{R^3}$; $r \le R$

3. $T = \frac{2\pi r}{v} = \frac{2\pi r}{\sqrt{GM}}\sqrt{r}$ $\left(\text{as } v = \sqrt{\frac{GM}{r}}\right)$

$T = \frac{2\pi}{\sqrt{GM}} r^{3/2} \Rightarrow$ $T^2 = \frac{4\pi^2}{GM} r^3$

Comparing $K = \frac{4\pi^2}{GM}$

4. For satellite S moving elliptical orbit around the earth net force will be towards centre of the earth.

(like centripetal force in circular motion)

5. For the satellite revolving around earth

$$v_0 = \sqrt{\frac{GM_e}{(R_e+h)}} = \sqrt{\frac{GM_e}{R_e\left(1+\frac{h}{R_e}\right)}} = \sqrt{\frac{gR_e}{1+\frac{h}{R_e}}}$$

substituting the values

$v_0 = \sqrt{60\times10^6}$ m/s

$v_0 = 7.76 \times 10^3$ m/s = 7.76 km/s

6. $V = \frac{-GM}{R+h} = -5.4 \times 10^7$ ...(1)

and $g = \frac{GM}{(R+h)^2} = 6$ ...(2)

dividing (1) and (2)

$\Rightarrow \frac{5.4\times10^7}{(R+h)} = 6$

$\Rightarrow$ R + h = 9000 km so h = 2600 km

7. Escape Velocity

$$= \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2G}{R}.\left(\frac{4}{3}\pi R^3\rho\right)} \propto R\sqrt{\rho}$$

$\therefore$ Ratio, $\frac{v_e}{v_p} = 1 : 2\sqrt{2}$

8. $g = \left(\frac{GM_e}{R_e^3}\right)r$ for $0 < r \le R_e \Rightarrow \mathbf{g \propto r}$

$g = \frac{GM_e}{r^2}$ for $r \ge R_e \Rightarrow \mathbf{g \propto \frac{1}{r^2}}$

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9. Total energy = $-\frac{GM_em}{2(R+h)}$

$\because \quad g_0 = \frac{GM_e}{R^2} \quad \Rightarrow M_e = \frac{g_0R^2}{G}$

$\therefore$ Energy = $-\frac{mg_0R^2}{2(R+h)}$

10. $g_h = g_d$

$g\left(1-\frac{2h}{R}\right) = g\left(1-\frac{d}{R}\right)$

$\frac{2h}{R} = \frac{d}{R}$

So $\boxed{d = 2 \text{ km}}$

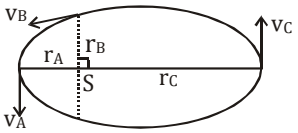
$2 \times 1 = d$

12. $m_s' = \frac{m_s}{10}$ & G' = 10 G

$g_E = \frac{G'M_E}{R^2} = \frac{10GM_E}{R^2} = 10g$

$\therefore$ Raindrops will fall faster, Walking on the ground would become more difficult, Time period of a simple pendulum on the earth would decrease.

13.



As L = mvr = contant

and $r_C > r_B > r_A$

so $v_A > v_B > v_C \Rightarrow K_A > K_B > K_C$

14. $g' = g\left(1-\frac{d}{R}\right)$

$g' = g\left(1-\frac{R/2}{R}\right)$

$mg' = mg\left(\frac{1}{2}\right)$

$W' = 200\left(\frac{1}{2}\right) = 100 \text{ N}$

15. $W = \frac{mgh}{1+h/R}$

at h = R, $W = \frac{mgR}{2}$

16. Kepler's Third Law :-

$T \propto r^{3/2}$

$\frac{T_2}{T_1} = \left(\frac{r_2}{r_1}\right)^{3/2} = \left(\frac{R+2.5R}{R+6R}\right)^{3/2} = \frac{1}{2\sqrt{2}}$

$\Rightarrow T_2 = \frac{24}{2\sqrt{2}} = 6\sqrt{2} \text{ hours}$

17. $\Delta U = -GMm\left[\frac{1}{r_f}-\frac{1}{r_i}\right]$

$= -GMm\left[\frac{1}{R+h}-\frac{1}{R}\right]$

$= \frac{GMmh}{R(R+h)}$

18. $W_s = mg_s = 72 \text{ N}$

$W_h = mg_h = \frac{mg_s}{\left(1+\frac{h}{R}\right)^2} = \frac{72N}{\left(1+\frac{R/2}{R}\right)^2} = \frac{72}{9/4}$

$W_h = 32 \text{ N}$

19. At depth : $g_{eff} = g\left(1-\frac{d}{R}\right)$

$\Rightarrow \frac{g}{n} = g\left(1-\frac{d}{R}\right)$

$\Rightarrow d = (n-1)\,R/n$

20. $v_e = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2G}{R}\times\frac{4}{3}\pi R^3\rho}$

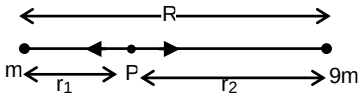
or $V_e = \sqrt{\frac{8\pi G\rho}{3}R^2}$

$\Rightarrow v_e \propto R$

$\Rightarrow \frac{v_e}{\upsilon} = \frac{4R}{R} \Rightarrow v_e = 4\upsilon$

21. $h = \frac{R}{\frac{2gR}{V^2}-1} = \frac{R}{\frac{V_e^2}{k^2V_e^2}-1} = \frac{Rk^2}{1-k^2}$

25.



Position of Neutral point (Zero Gravitational Field)

$$r_1 = \frac{\sqrt{m_1}\,R}{\sqrt{m_1}+\sqrt{m_2}} = \frac{\sqrt{m}\,R}{\sqrt{m}+\sqrt{9m}} = \frac{R}{4}$$

$r_2 = R - R/4 = 3R/4$

Now Gravitational potential at point P

$$V_p = -\frac{GM}{R/4} - \frac{9(GM)}{3R/4}$$

$$= \frac{-16GM}{R}$$

26. $$T = \frac{2\pi}{\sqrt{GM}} r^{3/2} \Rightarrow T^2 = \frac{4\pi^2 R^3}{G\left(\frac{4}{3}\pi R^3 d\right)} \quad (r = R)$$

$$T^2 = \frac{3\pi}{Gd}$$

27. $$\boxed{V_\infty = \sqrt{V^2 - V_e^2}}$$

Given than

$V = 2V_e$

So,

$$V_\infty = \sqrt{(2V_e)^2 - V_e^2}$$

$$V_\infty = \sqrt{3}V_e = 11.2\sqrt{3}\ \text{km/s}$$

28. $$g = \frac{4}{3}\pi G R \rho$$

$$\rho = \frac{3g}{4\pi G R}$$

29. At Earth surface

$$g = \frac{GM}{R^2} = 9.8\ \text{m/s}^2$$

At given planet

$$m' = \frac{m}{10}$$

$$R' = \frac{R}{2}$$

$$g' = \frac{G\left(\frac{m}{10}\right)}{\left(\frac{R}{2}\right)^2} = 0.4g$$

$$\boxed{g' = 3.92\,\text{m/s}^2}$$

30. Final Energy of satellite

$$TE_f = -\frac{GMm}{2(R+h)} = -\frac{GMm}{2(3R)} = -\frac{GMm}{6R}$$

Initial energy

$$PE_i = -\frac{GMm}{R}$$

Now by COME

$KE_i + PE_i = (KE_f + PE_f)$

$$KE_i - \frac{GMm}{R} = -\frac{GMm}{6R}$$

$$KE_i = -\frac{GMm}{6R} + \frac{GMm}{R}$$

$$\boxed{KE_i = \frac{5}{6}\frac{GMm}{R}}$$

31. escape velocity

$$v_e = \sqrt{\frac{2GM}{R}}$$

$$\frac{v_e}{v_p} = \sqrt{\frac{M_e}{M_p}\cdot\frac{R_p}{R_e}}$$

$$\frac{v}{v_p} = \sqrt{\frac{M_e}{9M_e}\cdot\frac{16Re}{R_e}}$$

$$\frac{v}{v_p} = \frac{4}{3}$$

$$v_p = \frac{3v}{4}$$

32. $\Delta W = m(g_B - g_A)$

$$= 100\left(\frac{g}{\left(1+\frac{h_B}{R}\right)^2} - \frac{g}{\left(1+\frac{h_A}{R}\right)^2}\right)$$

$$= 1000\left(\frac{1}{\left(1+\frac{3}{2}\right)^2} - \frac{1}{(1+2)^2}\right)$$

$$= 1000\left(\frac{4}{25} - \frac{1}{9}\right)$$

$$= 1000\frac{11}{(225)}$$

$$= 40\frac{(11)}{9} = \frac{440}{9} = 48.88$$

$$\simeq 49\,\text{N}$$

Exercise-III (Analytical Questions)

1. After the force disappears, it continue to move in a straight line due to inertia of direction and orbital speed $V_0 = \sqrt{\frac{GM}{r}}$.

So, $V_0 \propto \frac{1}{\sqrt{r}} \Rightarrow r \uparrow \Rightarrow V_0 \downarrow$

2. Work has to be done on a satellite by external agency to shift it in higher orbit. So total energy will not remain conserved.

3. (A) The magnitude of force between 1 Kg bodies is 6.67×10^{-11} N

(B) is correct, if escape velocity for a planet is high, the thermal speeds of lighter gases are less than the escape velocity.

$\therefore$ Lighter gases are not able to escape.

(C) is wrong; friction force is electromagnetic in nature.

(D) is correct.

4. Satellite in circular orbit has constant speed, but not constant velocity and it will have centripetal acceleration.

5. **I.** Gravitational force on satellite due to earth acts always towards the centre of earth.

II. Total mechanical energy remains conserve.

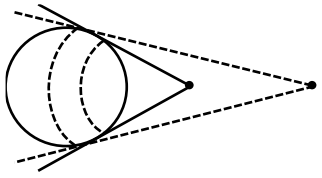
III. Since v changes in magnitude & direction.

$\therefore$ Linear momentum is not conserved.

6. $r \rightarrow 2r$

(A) Angular Momentum : $L = mvr = m\sqrt{\frac{GM}{r}}.r$

$L \propto \sqrt{r} \quad r \uparrow \Rightarrow L \uparrow$

(B) 

Area covered will $\uparrow$

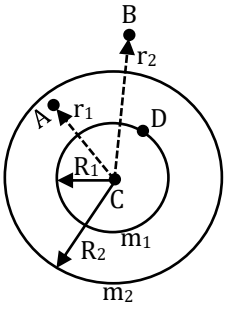
(C) $U = -\frac{GMm}{r} = -U_0$

$U' = -\frac{GMm}{2r} = -\left(\frac{U_0}{2}\right)$

P.E. will $\uparrow$

(D) $KE = \frac{GMm}{2r} \Rightarrow$ KE will $\downarrow$

7.



(A) $U_A = -\frac{Gm_1}{r_1} - \frac{Gm_2}{R_2}$; $U_B = -\frac{Gm_1}{r_2} - \frac{Gm_2}{r_2}$

$U_A < U_B$

(B) $I_A = \frac{Gm_1}{r_1^2}$; $I_B = \frac{G(m_1+m_2)}{r_2^2}$

Can't compare as data is insufficient.

(C) $U_C = -\frac{Gm_1}{R_1} - \frac{Gm_2}{R_2}$; $U_D = -\frac{Gm_1}{R_1} - \frac{Gm_2}{R_2}$

remains same

(D) $I_A = \frac{Gm_1}{r_1^2}$; $I_D = \frac{Gm_1}{R_1^2}$

$I_D > I_A$

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Properties of Matter and Fluid Mechanics

SOLUTIONS

Exercise-I (Conceptual Questions)

2. Total weight at height $\frac{L}{4}$ from its lower end = W_T

W_T = weight suspended + weight of $\frac{L}{4}$ of wire

$W_T = W_1 + \frac{W}{4}$

$\text{Stress} = \frac{F}{\text{Area}} = \frac{W_T}{S}$

$\text{Stress} = \frac{\left(W_1 + \frac{W}{4}\right)}{S}$

3. $\text{Stress} = \frac{F}{\text{Area}} = \frac{Mg}{A} \Rightarrow \frac{\rho LAg}{A} = \rho Lg$

$L_{max} = \frac{(\text{Stress})_{max}}{\rho g}$; $L_{max} = 75$ m

4. From graph

$\text{Slope} = \frac{F}{\Delta\ell}$

also $Y = \frac{FL}{A\Delta\ell}$

$\text{Slope} = \frac{F}{\Delta\ell} = Y\frac{A}{L}$

$\Rightarrow$ more slope more Y

So $Y_A > Y_B$

5. When density increase, mass of material increase, for same strain more stress required. So Young's modulus increases.

6. Volume = constant

$A \times \ell = \text{constant} \Rightarrow A \propto \frac{1}{\ell}$

$\Delta\ell = \frac{F\ell}{AY} \Rightarrow \Delta\ell \propto \frac{\ell}{A} \Rightarrow \Delta\ell \propto \ell^2 \quad \left(\text{as } A \propto \frac{1}{\ell}\right)$

7. In both case tension in wire are same and $\Delta\ell \propto T$

8. $\Delta L = \frac{\rho g L^2}{2Y} = \frac{1.5\times10\times64\times10^{-4}}{2\times5\times10^8}$

$= 9.6 \times 10^{-11}$ m.

9. Bulk modulus = $\frac{\Delta P}{\Delta V/V} = \frac{h\rho g}{0.1/100}$

$= \frac{200\times10^3\times9.8\times100}{0.1} = 19.6 \times 10^8$ N/m^2

10. (Bulk Modulus) $K = \frac{\Delta P}{\Delta V / V}$

$= \frac{0.155\times10^5}{10/100} = 1.55 \times 10^5$ Pa

11. Potential energy per unit volume is

$= \frac{1}{2}\times\frac{(\text{Stress})^2}{Y} \Rightarrow \frac{U_1}{U_2} = \frac{F_1^2}{A_1^2}\times\frac{A_2^2}{F_2^2} = \frac{r_2^4}{r_1^4} = 16 : 1$

12. If wire suddenly breaks then energy released on bond breaking appears in the form of heat.

13. $W_{ext} = \vec{F}.\vec{d}$ using COM

$= \frac{F\Delta\ell}{2}$

$= \frac{mg\Delta\ell}{2} = \frac{0.5\times9.8\times3\times10^{-3}}{2}$

$= 7.3\times10^{-3}$ Joule

14. $F = \frac{YA}{L}\Delta\ell = \pi\times\left(\frac{1}{2}\times10^{-3}\right)^2\times2\times10^{11}\times\frac{1}{1000}$

$= 157$ N

15. $\frac{F}{A} = \frac{Y\Delta L}{L}$

$\Delta L = \frac{FL}{YA}$ (as given $\Delta L = e$)

$e = \frac{FL}{YA} = \frac{F\ell}{Y\pi r^2}$

For second wire

$e' = \frac{(2F)(2\ell)}{Y\pi(2r)^2} = \frac{F\ell}{Y\pi r^2}$

$e' = e$

16. $F = \frac{YA}{L}\Delta\ell$

$= 0.9 \times 10^{11} \times \pi \times (0.3 \times 10^{-3})^2 \times \frac{0.2}{100} = 51$N

17. $\frac{F}{A}=\frac{Y\Delta L}{L}$ (Load = F)

$F=\frac{YA}{L}\cdot\Delta L$

$\boxed{\frac{F}{\Delta L}\propto A}$

more the slope, more will be area.

18. Here $\Delta\ell = \ell_0$

ℓ_0 = initial length

ℓ_f = final length = $2\ell_0$

$\frac{F}{A}=\frac{Y\Delta\ell}{\ell_0}$

$F=\frac{YA(\ell_0)}{\ell_0}$

$\boxed{F=YA}$

$= 2.0 \times 10^{11} \times 0.1 \times 10^{-4} = 2 \times 10^6$ N

19. $\frac{\Delta\ell}{L}=\frac{0.1}{100}$; Area = $\pi r^2 = \pi \times (2 \times 10^{-3})^2$

$F = YA.\ \frac{\Delta\ell}{L}$

$=9 \times 10^{10} \times (\pi \times 4 \times 10^{-6}) \times \frac{0.1}{100} = 360\ \pi$ N

21. Stress = $\frac{F}{A}$

$\Rightarrow \text{Stress}\propto\frac{1}{\text{Area}}$

$\frac{\text{Stress}_2}{\text{Stress}_1}=\frac{r_1^2}{r_2^2}$

22. $\left(-\frac{\Delta V}{V}\right)=\frac{0.004}{100}$

$\Delta P = \left(-\frac{\Delta V}{V}\right) K \times = 2100 \times 10^6 \times \frac{0.004}{100}$

$= 84$ kPa

23. Energy density = $\frac{1}{2}\times\frac{(\text{Stress})^2}{Y}=\frac{S^2}{2Y}$

24. For volume

Stress = ΔP

Strain = $-\frac{\Delta V}{V}$ ('–'represent decrease in volume)

$\frac{\text{Stress}}{\text{Strain}}=\frac{-\Delta P}{\frac{\Delta V}{V}}=B$

$\Rightarrow \quad \frac{\Delta V}{V}\propto\frac{1}{B}$

25. For a wire of length L stretched by a length x, the restoring elastic force is :

$F = \left[\frac{x}{L}\right] YA$

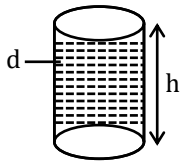
The work required to be done against the elastic restoring forces to elongate it further by a length dx is,

$dW = F.dx = \frac{YA}{L} x\ .\ dx$

The total work done in stretching the wire from x = 0 to x is,

$W=\int_0^x \frac{YA}{L}x.dx=\frac{YA}{L}\left[\frac{x^2}{2}\right]$

26. $\frac{F_A}{F_B}=\frac{P_A A_A}{P_B A_B}=\frac{\rho_A g h_A A_A}{\rho_B g h_B A_B}=1$



27. d = Density

h = Height of water

A = Base cross sectional area

P_{Bottom} (pressure due to liquid) = hdg

P_{Top}(pressure due to liquid) = dg(0) = 0

av. pressure = $\frac{P_{Bottom}+P_{Top}}{2}=\frac{hdg}{2}$

Alternate : Use concept of COM for a uniform solid cylinder.

28. Due to upthrust on the block, its apparent wt. will reduce & hence the reading of spring balance A will be less then 2 kg. But since the liquid in the beaker exerts buoyant force on the block, the block will also exert reaction force on the liquid in the downward direction and the reading of balance B will increase due to this additional force.

29. Liquid with low density float over high density.

30. Weight of man = upthrust $\Rightarrow$ $Mg = V_{in}\rho_w g$

$M = (3 \times 2 \times 1 \times 10^{-2}) \times 10^3 = 60$ kg

31. Water density = 1000 kg/m^3

block density = 981 kg/m^3

so it will float with some part immersed in water.

32. When coin is removed the downward force an block decrease, which will reduce the volume submerged also the removing volume will be filled by water so h $\downarrow$.

33. Same density

34. RD of metal = $\frac{W_A}{W_A - W_W} = \frac{210}{210-180} = 7$

$\therefore$ density of metal = $7 \times \rho_w = 7$ g/cm^3

35. Torr is the unit of pressure.

36. In water for floating

$W = Th$

$V\rho_B g = V_{in}\rho_w g$

$V\rho_B g = \frac{2}{3} V \times 1 \times g \Rightarrow \rho_B = \frac{2}{3}$ g/cm^3

in liquid

$W = Th$ (Buoyant force)

$V\rho_B g = \frac{V}{4} \times \rho_L g$

$\rho_L = 4\rho_B = \frac{8}{3}$ g/cm^3

37. Hydraulic lift, hydraulic jack, hydraulic piston works on Pascal's law.

38. $W_{APP} = W - F_B$ (F_B = force of buoyant)

$W_{APP} = V\rho g - V\rho_1 g$

$= V\rho g\left[1 - \frac{\rho_1}{\rho}\right]$

$W_{APP} = W\left[1 - \frac{\rho_1}{\rho}\right]$

39. Pascal's law states that the magnitude of pressure within fluid is equal in all parts.

40. $Th = W_A - W_W = 5 - 2 = 3$N

41. Hydraulic lift, hydraulic jack, hydraulic press works on Pascal's law.

42. For wooden block $\rho_{Block} < \rho_{Water}$

So $F_B > Mg$, hence it will rise up with an constant acceleration.

43. $A_P v_P = A_Q v_Q$

or $\frac{v_P}{v_Q} = \frac{A_Q}{A_P} = \frac{\pi \times (4\times10^{-2})^2/4}{\pi \times (2\times10^{-2})^2/4} = 4$

or $v_P = 4\, v_Q$

44. Writing equation of continuity for the tube & the holes

$\Rightarrow$ $A_{tube}\, v_{tube} = nA_{hole}\, v_{hole}$

$\pi R^2 \times v = n\pi r^2 v' \quad \Rightarrow \quad v' = \frac{vR^2}{nr^2}$

45. From Bernoulli equation

$\frac{1}{2}\rho v^2 = \rho gh \quad \Rightarrow v = \sqrt{2gh} = 2$ m/s

from equation of continuity $A_1 v_1 = A_2 v_2$

$A_2 = \left(\frac{v_1}{v_2}\right) \times A_1 = \left(\frac{1}{2}\right) \times 10^{-4} = 5 \times 10^{-5}$ m^2

46. According to Bernoulli's theorem.

$\frac{1}{2}\rho v^2 + \rho gh + P = \text{const.}$

So if A$\downarrow$ v$\uparrow$ and P$\downarrow$

and if A$\uparrow$ v$\downarrow$ and P$\uparrow$

47. Volume flowing per second = Av

$= A\sqrt{2gh} = 10^{-4}\sqrt{2\times10\times5} = 10^{-3}$ m^3/s

48. $P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$

$(P_1 - P_2) = \frac{1}{2}\rho v_2^2$

since tap is closed, $v_1 = 0$

$(3.5 \times 10^5 - 3 \times 10^5) = \frac{1}{2} \times 10^3 \times v^2$

$v = 10$ m/s

49. Dynamic lift = Pressure diff. $\times$ Area

$= (P_{lower} - P_{upper})\, A$

(According to Bernoullis Theorem)

$= \left(\frac{1}{2}\rho v_{upper}^2 - \frac{1}{2}\rho v_{lower}^2\right) A$

$= \frac{1}{2}\rho A\left[(\sqrt{2v})^2 - v^2\right] = \frac{1}{2}\rho A v^2$

50. $A_Av_A = A_Bv_B$

$$v_B = \frac{A_A}{A_B} \times v_A = \frac{\pi(2R)^2}{\pi R^2} \times v = 4v$$

51. Equation of continuity $A_1v_1 = A_2v_2$

Av = constant

$\pi r^2 v$ = constant

$$\boxed{v \propto \frac{1}{r^2}}$$

52. Level flight means net vertical force is zero

$\therefore \quad mg = \Delta P \times A$

$$\Delta P = \frac{mg}{A} = \frac{3\times10^4\times10}{120} = 2.5 \times 10^3 \text{ Pa} = 2.5 \text{ kPa}$$

53. Scent sprayer is based on Bernoulli's theorem.

54. Bernoulli's equation for steady, non-viscous, in-compressible flow expresses the conservation of energy.

55. Bernoulli's theorem can be seen in dynamic lift to aeroplane.

56. By equation of continuity

$A_1v_1 = A_2v_2$

$$v_2 = \left(\frac{A_1}{A_2}\right) v_1 = \left(\frac{r_1}{r_2}\right)^2 v_1 = \left(\frac{0.2}{0.1}\right)^2 \times 20$$

$= 80$ cm/s

57. Final velocity is terminal velocity which is independent from 'h'

$$v_T = \frac{2r^2}{9}\frac{(\rho-\sigma)}{\eta}g$$

58. $v_T \propto r^2 \therefore \frac{v_T'}{v_T} = \left(\frac{R}{r}\right)^2$ or $v_T' = \left(\frac{R}{r}\right)^2 v_T$ ...(1)

when 2 droplets of radius r coalesce the new radius R of the big drop will be given by

$$\frac{4}{3}\pi R^3 = 2\times\frac{4}{3}\pi r^3 \Rightarrow \frac{R}{r} = 2^{1/3} \quad ...(2)$$

from equation (1) and (2)

$v'_T = (2^{1/3})^2 \times 5 = 4^{1/3} \times 5$ cm/sec.

59. $$v_T = \frac{2r^2}{9}\frac{(\rho-\sigma)}{\eta}g$$

$v_T = 0.2$ m/s for gold

$\rho = 19.5$ kg/m^3 (gold)

$\sigma = 1.5$ kg /m^3

From above data we can find value of $\frac{r^2}{n}$

putting in 2nd case for silver

$\rho_2 = 10.5$ kg/m^3

$$\boxed{v_T = 0.1 \text{ m/s}}$$

60. $$v_T = \frac{2r^2}{9}\frac{(\rho-\sigma)}{\eta}g$$

$v_T \propto r^2 \quad \therefore \frac{v_{T_2}}{v_{T_1}} = \frac{r_2^2}{r_1^2}$

$$\therefore v_{T_2} = v_{T_1}\left(\frac{r_2}{r_1}\right)^2 = 20\times\left(\frac{1}{2}\right)^2 = 5 \text{ cm/s.}$$

61. The velocity of falling rain drop attain limited value because of viscous force exerted by air.

62. Poise is the unit of viscosity.

63. $$v_T = \frac{2r^2}{9}\frac{(\rho-\sigma)}{\eta}g$$

$v_T \propto r^2 \quad \Rightarrow \frac{v_{T_1}}{v_{T_2}} = \left(\frac{r_1}{r_2}\right)^2 = \left(\frac{1}{2}\right)^2 = 1:4$

64. $$v_T = \frac{2R^2}{9}\frac{(\rho-\sigma)}{\eta}g$$

$v_T \propto R^2$ but $M \propto R^3$

$\Rightarrow v_T \propto \frac{M}{R}$

65. $$v_T = \frac{2}{9}r^2\frac{(\rho_w - \rho_{air})g}{\eta}$$

$\because \rho_{air} \approx 0$

$$\Rightarrow v_T = \frac{2}{9}\times\frac{\left(1.50\times10^{-6}\right)^2\times10^3\times10}{2\times10^{-5}}$$

$= 2.5 \times 10^{-4}$ m/s

66. Elastic membrane is formed on the surface of water due to surface tension. This help's spider & insects to move and run on the surface of water.

67. $2T\ell = mg$ (soap film has two free surface)

$$T = \frac{mg}{2\ell} = \frac{1.5\times10^{-2}}{2\times0.3} = 0.025 \text{ N/m}$$

68. $\Delta E = 4\pi R^2 T\left(n^{1/3}-1\right)$

$= 4\pi\frac{D^2}{4} T\,[(27)^{1/3}-1] = 2\pi D^2 T$

69. Excess pressure in air bubble just below the water surface. $P_1 = \frac{2T}{r}$

Excess pressure inside a drop $P_2 = \frac{2T}{r}$

So $P_1 = P_2$

70. By conservation of volume

$$\frac{4}{3}\pi R^3 = n\left[\frac{4}{3}\pi r^3\right] \Rightarrow n = \left[\frac{R}{r}\right]^3 \Rightarrow r = \frac{R}{n^{1/3}}$$

$R = n^{1/3} r$

$$P_{ex.} \propto \frac{1}{r} \Rightarrow \frac{(P_{ex})_{big}}{(P_{ex})_{small}} = \frac{r}{R} = \frac{1}{n^{1/3}} = \frac{1}{2}$$

71. $h = \frac{2T\cos\theta}{r\rho g}$

If the radius of capillary is reduced to half, the rise of liquid column becomes two times because $h \propto \frac{1}{r}$

72. $h = \frac{2T\cos\theta}{r\rho g}$

$$h \propto \frac{1}{r}\ 0 \Rightarrow \frac{h_2}{h_1} = \frac{r_1}{r_2} \Rightarrow h_2 = \frac{h_1}{2}$$

Mass of water = V × ρ_{water}

$$\frac{M'}{M} = \frac{\pi(2r)^2 \times \frac{h}{2} \times \rho_w}{\pi r^2 \times h \times \rho_w} = 2$$

$\Rightarrow M' = 2M$

73. When water rise upto a height h then mass of liquid rise $m = (\pi r^2 h)\rho$

$\because$ total mass be located at centre of mass.

then potential energy $U = mg\frac{h}{2} = (\pi r^2 H\rho)g\frac{h}{2}$

by Zurin law r × h = constant

$\therefore$ U = constant

74. $\frac{2T}{r} = h\rho g$

$$h = \frac{2T}{r\rho g} = \frac{2\times 75}{\left(\frac{0.1}{2}\times 10^{-1}\right)\times 1\times 1000} = 30\text{ cm}$$

75. Inside a satellite, water will rise upto the top level but sill not overflow. Radius of curvature (R') increases in such a way that final height h' is reduced and given by $h' = \frac{hR}{R'}$. (It is in accordance with Zurin's law).

76. When two bubbles coalesce then total number of molecules of air will remain same and temperature will also remain constant

so $n_1 + n_2 = n \Rightarrow P_1V_1 + P_2V_2 = PV$

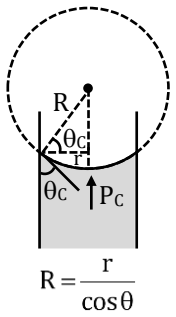
$$\Rightarrow \frac{4T}{r_1}\left(\frac{4}{3}\pi r_1^3\right) + \frac{4T}{r_2}\left(\frac{4}{3}\pi r_2^3\right) = \frac{4T}{r}\left(\frac{4}{3}\pi r^3\right)$$

$\Rightarrow r = \sqrt{r_1^2 + r_2^2}$

77. Spherical shape of rain-drop is due to surface tension which tries to minimize the surface area.

78. $h = \frac{2T}{r\rho g}$

$\Rightarrow h \propto \frac{1}{r}$



$R = \frac{r}{\cos\theta}$

79. $h = \frac{2T}{r\rho g}$

$\Rightarrow h \propto \frac{1}{g}$

$\Rightarrow$ at moon $g' = \frac{g}{6}$

80. For hemispherical

R = r

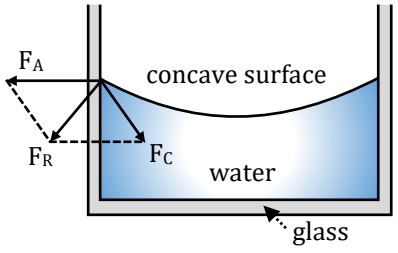
i.e. $\cos\theta = 1$

$\theta = 0°$

as $R\cos\theta = r$

in capillary tube

81.



Liquid rises up for acute angle of contact.

82. Splitting of big drop into smaller droplets.
In this process area increases, surface energy increases, internal energy decreases, temperature decreases, and energy is absorbed.
For reverse process energy is liberated.

83. $h = \frac{2T\cos\theta}{r\rho g}$

$h \propto \frac{T}{\rho} \Rightarrow \frac{h_1}{h_2} = \frac{T_1\rho_2}{T_2\rho_1} = \frac{60}{50} \times \frac{0.6}{0.8} = \frac{9}{10}$

84. Surface tension of liquid lead, which tries to minimize the surface area and gives a spherical shape.

85. S.E. = T × 2A
= 5 × 2 × 0.02
= 2 × 10^{-1} J

86. $h = \frac{2T\cos\theta}{r\rho g}$

$\Rightarrow r = \frac{2T\cos\theta}{h\rho g}$

$= \frac{2\times75\times10^{-3}\times1}{3\times10^{-2}\times10^{3}\times10}$

= 0.5 × 10^{-3} m
diameter d = 2r = 10^{-3} m = 1 mm

87. W = TΔA
W = TA
as ΔA = A

88. $P_1 - P_2 = \frac{4T}{r}$

$\Rightarrow \frac{4T}{r_1} - \frac{4T}{r_2} = \frac{4T}{r}$

$\Rightarrow r = \frac{r_1 r_2}{r_2 - r_1}$

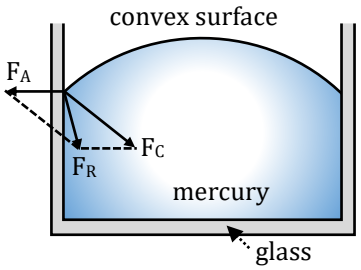
$r = \frac{r_1 r_2}{r_2 - r_1} = \frac{4\times5}{5-4}$ cm. = 20 cm.

89. Work done (surface energy) in blowing of a soap bubble of radius r :
W = T × ΔA or W = T × 2 × $4\pi r^2 = 8\pi r^2 T$
[∵ soap bubble has two surfaces]

$E = T \times 8\pi \left(r_2^2 - r_1^2\right)$

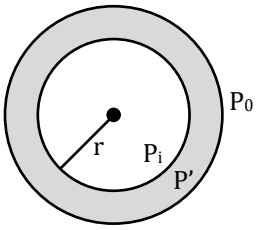
= T × 8π ($4r^2 - r^2$)
= $24\pi r^2 T$

90.



A liquid does not wet the sides of a solid, for obtuse (more than 90°) angle of contact.

91. Since a soap bubble has two surfaces, excess pressure will get doubled as compared to a drop



$P_i - P' = \frac{2T}{r}$, $P' - P_0 = \frac{2T}{r} \Rightarrow$ excess pressure

$= P_i - P_0 = \frac{4T}{r}$

92. Inside a freely falling elevator, water will rise upto the top level but still not overflow. Radius of curvature (R') increases in such a way that final height h' is reduced and given by $h' = \frac{hR}{R'}$.
(It is in accordance with Zurin's law).

93. On liquid force due to surface tension = $(2\pi R)T\cos\theta_C$
In equilibrium : force due to S.T = weight of rise liquid
$(2\pi R)T\cos\theta_C = mg$

$T = \frac{\text{Weight}}{2\pi R} = \frac{6.2\times10^{-4}}{2\times3.14\times2\times10^{-3}} = 5\times10^{-2}$ N/m

94. On liquid force due to surface tension = $(2\pi r)T\cos\theta_C$
In equilibrium : force due to S.T = weight of rise liquid
$(2\pi r)T\cos\theta_C = mg$
$2\pi r.T$ = weight

$r = \frac{\text{weight}}{2\pi T}$

$= \frac{6.28\times10^{-4}}{2\times3.14\times5\times10^{-2}} = 2\times10^{-3}$ m

95. On liquid force due to surface tension = $(2\pi r)T\cos\theta_C$

In equilibrium : force due to S.T = weight of rise liquid

$(2\pi r)T\cos\theta_C = mg$

$2\pi r.T$ = weight

$2\pi r = \frac{\text{weight}}{T}$

$= \frac{75\times10^{-4}}{6\times10^{-2}}$

$= 12.5 \times 10^{-2}$ m

96. By conservation of volume

$$\frac{4}{3}\pi R^3 = n\left[\frac{4}{3}\pi r^3\right]$$

$$\Rightarrow n = \left[\frac{R}{r}\right]^3$$

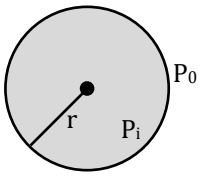
$$\Rightarrow r = \frac{R}{n^{1/3}}$$

Radius of big drop $R' = n^{1/3}.r = 2^{1/3}.R$

$$\frac{\text{Surface energy before change}}{\text{Surface energy after change}}$$

$$= \frac{n4\pi R^2T}{4\pi R'^2T} = \frac{2.R^2}{2^{2/3}.R^2} \quad = 2^{1/3}$$

97.

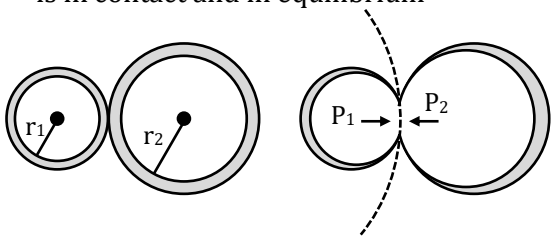


$\Rightarrow$ excess pressure = $P_i - P_0 = \frac{2T}{r}$

98. $W = 4\pi\left(\frac{D}{2}\right)^2\left(n^{\frac{1}{3}}-1\right)T = \pi D^2\left(n^{1/3}-1\right)T$

$= 3.14 \times (0.2 \times 10^{-2})^2(10 - 1) = 7.9 \times 10^{-6}$J

99. $\because r_1 < r_2 \quad \therefore P_1 > P_2$ Small portion of bubbles is in contact and in equilibrium



$$\Rightarrow P_1 - P_2 = \frac{4T}{r}$$

$$\Rightarrow \frac{4T}{r_1} - \frac{4T}{r_2} = \frac{4T}{r}$$

$$\Rightarrow r = \boxed{\frac{r_1r_2}{r_2 - r_1}}$$

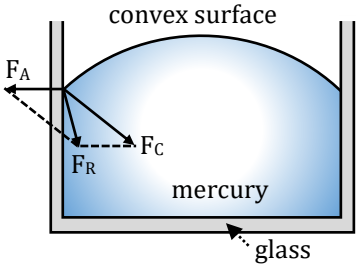
100. On liquid force due to surface tension = $(2\pi R)T\cos\theta_C$

In equilibrium : force due to S.T = weight of rise liquid

$(2\pi R)T\cos\theta_C = W$

$$\frac{W_2}{W_1} = \frac{4\pi(2R)^2T}{4\pi R^2T} = 4 \Rightarrow W_2 = 4W_1 = 4W$$

101.



A liquid does not wet the sides of a solid, for obtuse (more than 90°) angle of contact.

102. $h = \frac{2T\cos\theta}{r\rho g}$

$= \frac{2\times70\times1}{\frac{1}{28}\times1\times980} = 4$ cm

103. $W = T\times\Delta A$

$= 20 \times 2\ [120 - 60] = 2400$ erg

104. Work done = $8\pi TR^2$

$$= 8\pi\times60\times10^{-3}\times\left(\frac{2}{10}\right)^2 = 8\pi\times6\times10^{-2}\times\frac{4}{10^2}$$

$W = 192\ \pi \times 10^{-4}$ J

105. Detergent decreases the oil-water surface tension and helps in removing dirty greasy stains.

106. $(P_{ex})_1 = 2(P_{ex})_2$

$$\frac{4T}{r_1} = 2\left(\frac{4T}{r_2}\right) \Rightarrow \frac{r_1}{r_2} = \frac{1}{2}$$

$$\frac{V_1}{V_2} = \left(\frac{r_1}{r_2}\right)^3 = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

107. For soap film we have two free surface

$W = T \times 2\Delta A$

$= 3\times10^{-2}\times 2[4 \times 5 - 4 \times 4] \times 10^{-4}$

$= 24 \times 10^{-6}$ J

Exercise-II (Previous Year Questions)

1. $Y = \dfrac{\frac{F}{A}}{\frac{\Delta\ell}{\ell}} \Rightarrow \Delta\ell = \dfrac{F\ell}{AY}$

But $V = A\ell$ so $A = \dfrac{V}{\ell}$ (V = volume)

Therefore $\Delta\ell = \dfrac{F\ell^2}{VY} \propto \ell^2$

2. As surface area decreases so energy is released.

Released energy

$= 4\pi R^2T[n^{1/3}-1]$ where $R = n^{1/3}r$

$= 4\pi R^3T\left[\dfrac{1}{r}-\dfrac{1}{R}\right]$ $= 3VT\left[\dfrac{1}{r}-\dfrac{1}{R}\right]$

3. As we know

$B = \dfrac{P}{\frac{\Delta V}{V}}$

so $\dfrac{\Delta V}{V} = \dfrac{P}{B}$

Now $P = \rho gh$ & compressibility 'C' $= \dfrac{1}{B}$

so $\dfrac{\Delta V}{V} = \rho gh(C)$

$= 10^3 \times 9.8 \times 2700 \times 45.4 \times 10^{-11}$

$= 1.201 \times 10^{-2}$

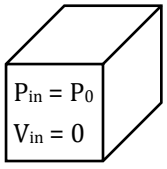
4. By Bernaulli's equation

$P + \dfrac{1}{2}\rho v^2 = P_0 + 0$

$P_0 - P = \dfrac{1}{2}\rho v^2$

$F = \dfrac{1}{2}\rho v^2 A$

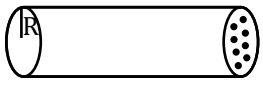
$F = 2.4 \times 10^5$ N upward



5. Av = constant

$\pi R^2 V = n\pi r^2 v_1 \Rightarrow$

$v_1 = \dfrac{VR^2}{nr^2}$



6. $Y = \dfrac{F\ell}{A\Delta\ell} \Rightarrow \Delta\ell = \dfrac{F\ell}{AY}$

$(\Delta\ell)_{Steel} = (\Delta\ell)_{Brass}$

$\Rightarrow \dfrac{W_S\ell}{AY_S} = \dfrac{W_B\ell}{AY_B}$

$\Rightarrow \dfrac{W_S}{W_B} = \dfrac{Y_S}{Y_B} = \dfrac{2}{1}$

7. water will rise upto the top level but will not overflow. Radius of curvature (R') increases in such a way that final height h' is reduced and given by $h' = \dfrac{hR}{R'}$ (It is in accordance with Zurin's law)

8. Pressure = 150 mm Hg

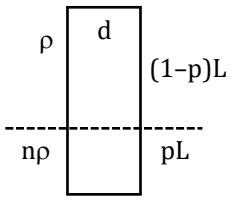
Pumping rate $= \dfrac{dV}{dt} = \dfrac{5\times10^{-3}}{60}$ m³/s

Power of heart = $P.\dfrac{dV}{dt} = \rho gh \times \dfrac{dV}{dt}$

$= (13.6 \times 10^3 \text{ kg/m}^3)\ (10) \times (0.15) \times \dfrac{5\times10^{-3}}{60}$

$= \dfrac{13.6\times5\times0.15}{6} = 1.70$ watt

9. Weight of cylinder = Upthurst by the two liquids



$L\ A\ d\ g = (pL)\ A\ (n\rho)g + (1-p)\ L\ A\ \rho\ g$

$\Rightarrow d = (1-p)\rho + pn\ \rho = [1 + (n-1)p]\rho$

10. $W = T(2\Delta A)$ $\{\Delta A = (20-8)\ cm^2\}$

$\Rightarrow T = \dfrac{W}{2\Delta A} = \dfrac{3\times10^{-4}}{2\times12\times10^{-4}} = 0.125\ Nm^{-1}$

11. $h = \dfrac{2T\cos\theta}{\rho gr}$

As r, h, T are same, $\dfrac{\cos\theta}{\rho} = \text{constant}$

$\Rightarrow \dfrac{\cos\theta_1}{\rho_1} = \dfrac{\cos\theta_2}{\rho_2} = \dfrac{\cos\theta_3}{\rho_3}$

As $\rho_1 > \rho_2 > \rho_3$

$\Rightarrow \cos\theta_1 > \cos\theta_2 > \cos\theta_3 \Rightarrow \theta_1 < \theta_2 < \theta_3$

As water rises so θ must be acute

So, $0 \le \theta_1 < \theta_2 < \theta_3 < \pi/2$

12. $B = \dfrac{\Delta P}{-\frac{\Delta V}{V}}$, $\dfrac{\Delta V}{V} = \dfrac{3\Delta R}{R}$

$B = \dfrac{\Delta P}{\frac{-3\Delta R}{R}} \Rightarrow -\dfrac{\Delta R}{R} = \dfrac{P}{3B}$ $(\Delta P = P)$

13. $\rho_0 g \times 140 \times 10^{-3} = \rho_w g \times 130 \times 10^{-3}$

$$\rho_0 = \frac{130}{140} \times 10^3 \approx 928\ kg/m^3$$

14. Rate of heat produced

$$\frac{dQ}{dt} = F_v \times v_T \quad \because v_T = \frac{2r^2}{9\eta}(\rho - \sigma)g \propto r^2$$

$$= 6\pi\eta r v_T \times v_T$$

$$\Rightarrow \frac{dQ}{dt} \propto r v_T^2 \propto r^5$$

15. $Y = \frac{F\ell}{A\Delta\ell} \quad \because V = A\ell \quad$ so $\ell = \frac{V}{A}$

$$\text{So } F = \frac{YA\Delta\ell}{\ell} = \frac{YA^2\Delta\ell}{V} \propto A^2$$

$$\frac{F_1}{F_2} = \left(\frac{A_1}{A_2}\right)^2 \Rightarrow \frac{F}{F_2} = \left(\frac{A}{3A}\right)^2 = \frac{1}{9} \Rightarrow F_2 = 9F$$

16. $P = P_0 + \rho g Z_0$ .....(i)

Also, $P = P_0 +$ .....(ii)

From (i) & (ii)

$$\rho g Z_0 = \frac{4T}{R}$$

$$\therefore \quad Z_0 = \frac{4T}{\rho g R} = \frac{4 \times 2.5 \times 10^{-2}}{10^3 \times 10 \times 10^{-3}}$$

$= 10^{-2}$ m = 1 cm

17. $U = \frac{1}{2}$ (force)(elongation)

$$= \frac{1}{2}(Mg)\ell = \frac{1}{2}Mg\ell$$

18. velocity of efflux $v = \sqrt{2gh}$

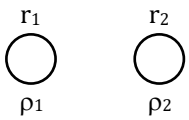
volume flow rate = $Av = A\sqrt{2gh}$

$= (2 \times 10^{-6})\ (2 \times 10 \times 2)^{1/2}$

$= 4\sqrt{10} \times 10^{-6}\ m^3/s$

$\cong 12.6 \times 10^{-6}\ m^3/s$

19.



$$v_T = \frac{2r^2(\sigma - \rho)g}{9\eta}$$

$$\frac{v_1}{v_2} = \left(\frac{r_1}{r_2}\right)^2 \frac{(\sigma_1 - \rho)}{(\sigma_2 - \rho)}$$

$$= \left(\frac{1}{2}\right)^2 \left(\frac{8\rho_2 - 0.1\rho_2}{\rho_2 - 0.1\rho_2}\right) = \frac{79}{36}$$

20. $\rho_{oil}\ h_{oil} = \rho_{water}\ h_{water}$

$$\Rightarrow \rho_{oil} = \frac{1000(15)}{20} = 750\ kg\ m^{-3}$$

22. $m \propto r \Rightarrow \frac{m_2}{m_1} = \frac{r_2}{r_1} \Rightarrow \frac{m_2}{5} = \frac{2r}{r} \Rightarrow m_2 = 10g$

23. $Y = \frac{FL}{A\Delta L} = \frac{MgL}{A(L_1 - L)}$

24. When angle of contact $\geq 90°$ then liquid doesn't wet solid.

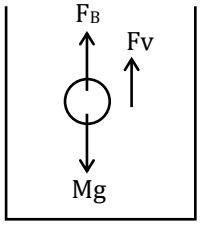
25. $76\ cm \times \rho_{Hg} \times g = h \times \rho_L \times g$

$$h = 76\,cm \times \frac{\rho_{Hg}}{\rho_L} = 76\ cm \times \frac{13600}{760} = 13.6\,m$$

26. Mass = M

Density of ball = d

Density of glycerine = $\frac{d}{2}$



$$F_B = V_s \rho_\ell g = V\frac{d}{2}g$$

$F_g = Mg = vdg$

for constant velocity, $F_{net} = 0$

$\therefore \quad F_B + F_v = M_g$

$$F_v = M_g - F_B = Vdg - \frac{Vdg}{2} = \frac{Vdg}{2} = \frac{Mg}{2}$$

31. $E = 2T(4\pi R^2)$

$= 2\ (0.03)\ (4)\ (3.14)\ (2 \times 10^{-2})^2$

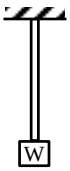
$= 3.01 \times 10^{-4} J$

32. Venturimeter works an Bernoulli's principle

33. Stress = $\frac{IRF}{A}$

Stress = $\frac{W}{A}$

(Here A Cross-sectional Area)



34. $F = 6\pi r\eta v$

$$= (6)(3.14)\left(\frac{1 \times 10^{-3}}{2}\right)(0.8 \times 10^{-1})(2)$$

$= 1.5 \times 10^{-3}$ N

35. Pressure is a scalar quantity.

36. $\frac{\text{E.P.E}}{\text{Volume}} = \frac{1}{2}(\text{stress})(\text{strain})$

$= \frac{1}{2}(Y)(\text{strain})^2$

$= \frac{1}{2}(Y)\left(\frac{\Delta L}{L}\right)^2$

$= \frac{1}{2}(2\times10^{11})\left(\frac{1\times10^{-3}}{100\times10^{-2}}\right)^2$

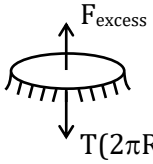
37. $Y = \frac{F\ell}{A\Delta\ell}$

$\Delta\ell = \frac{\left(\frac{F}{A}\right)\ell}{Y}$

$\Delta\ell = \frac{8\times10^8\times1}{2\times10^{11}}$

$\Delta\ell = 4\,\text{mm}$

38.



$F_{excess} = (T)\,(2\pi R)$

$= (0.07)\left[2\times\frac{22}{7}\times4.5\times10^{-2}\right]$

$= 44\times4.5\times10^{-4}$

$= 44\times4.5\times10^{-4}$

$= 19.8\times10^{-3}\,N$

$= 19.8\,mN$

39. $F = YA\alpha\Delta\theta$

$= 0.5 \times 10^{11} \times 10^{-3} \times 10^{-5} \times (100 - 0)$

$= 50 \times 10^3 N$

40. From continuity equation,

Speed at Y is less than speed at X

$\therefore K_2 < K_1$

Exercise-III (Analytical Questions)

1. I. In water adhesive force is greater than cohesive force. $\therefore$ It wets the capillary, whereas in mercury, cohesive force is greater.

II. $\Delta P = \frac{4T}{R} \Rightarrow \Delta P \propto \frac{1}{R}$

3. In case of ice and water, the volume of water displaced by ice is equal to the volume of water formed by melting the ice, therefore level remains unchanged.

5. Velocity of efflux is given by

$V = \sqrt{\frac{2gh}{1-\frac{a^2}{A^2}}}$

$a \rightarrow$ area of orifice (hole)

$A \rightarrow$ Area of tank

7. $\Delta\ell = \frac{F\ell}{Ay}$

for coper wire :

$(\Delta\ell)_1 = \frac{500\times8}{0.5\times10^{-4}\times10^{11}}$

$(\Delta\ell)_1 = 0.8$ mm

for steel wire :

$(\Delta\ell)_2 = \frac{500\times4}{0.5\times10^{-4}\times2\times10^{11}}$

$(\Delta\ell)_2 = 0.2$ mm

Total elongation :

$\Delta\ell = (\Delta\ell)_1 + (\Delta\ell)_2$

$= 1$ mm

8. $\because \quad P_{e_x} = \frac{4T}{r}$

$\therefore \quad P_{e_x} \propto \frac{1}{r}$

The air pressure is greater inside the smaller bubble.

Hence air flows from the smaller to the larger bubble.

9. In given condition

$\therefore$ Tension in a rod is = F

$\therefore$ Stress = F/A

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10. When a needle is placed carefully on the surface of water, it floats on the surface of water due to the surface tension of water, which does not allow the needle to sink. In case of a steel ball, the surface tension of the water is not sufficient to keep it floating, so it sinks down.

11. On spraying, surface area increases and hence surface energy; this increase in surface energy is on the expense of decrease in internal energy and hence temperature decreases.

12. Viscosity of liquids is greater than gases.
Insoluable impurities decreases the cohesive force acting between the molecule of liquid. Therefore surface tension decreases.

13. For a substance:

$B = \frac{1}{C}$

$BC = 1$

for ideal rigid body

strain = 0

$\therefore \ \eta = \frac{\text{shear stress}}{\text{shear strain}}$

$\eta = \infty$

14. For concave meniscus

$F_A > \frac{F_C}{\sqrt{2}} \Rightarrow \theta_C < 90°$

For convex meniscus

$F_A > \frac{F_C}{\sqrt{2}} \Rightarrow \theta_C > 90°$

For plane meniscus

$F_A = \frac{F_C}{\sqrt{2}} \Rightarrow \theta_C = 90°$

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Thermal Physics

SOLUTIONS

Exercise-I (Conceptual Questions)

1. $\frac{C-0°}{100°}=\frac{F-32°}{180°}$ [As C = F]

$\frac{C}{100°}=\frac{C-32°}{180°}$

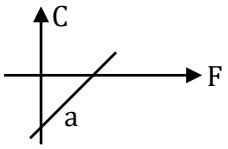
C = F = –40°

2. $C=\frac{5}{9}(F-32)$

$\Delta C=\frac{5}{9}\Delta F$

$\Delta F=\frac{9}{5}\times 25=45\,°F$

3. $C=\frac{5}{9}F-\frac{5}{9}\times 32$



y = mx – C

4. Slope of line AB

$=\frac{\Delta C}{\Delta F}=\frac{100-0}{212-32}=\frac{100}{180}=\frac{5}{9}$

5. $\frac{C}{5}=\frac{F-32}{9}\Rightarrow\frac{-183}{5}=\frac{F-32}{9}$

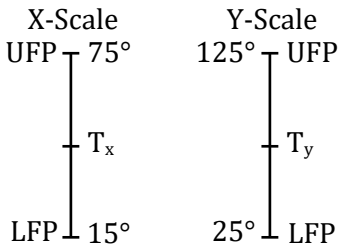
5[F–32] = –1647

F = – 329.4 + 32

F ≈ –297 °F

6. Pyrometer is used to measure a temperature of sun by using radiation

7.



$\frac{T_x-15}{75-15}=\frac{T_y-25}{125-25}$

Given $T_x = 60$; $T_y = ?$

$\frac{60-15}{75-15}=\frac{T_y-25}{100}$

$75 = T_y - 25$

$T_y = 100$ °Y

8. TFFT

9. $\alpha_{Rod} > \alpha_{Frame}$, then rod may touch floor.

10. $\gamma=\frac{\Delta V}{V}\times\frac{1}{\Delta T}$

$\gamma=\frac{0.15}{100}\times\frac{1}{24}\Rightarrow \qquad \alpha=\frac{\gamma}{3}$

$\alpha=\frac{0.15}{100}\times\frac{1}{24}\times\frac{1}{3}=2.0\times10^{-5}/°C$

11. Always increases (Imagine as a photographic enlargement)

12. $\Delta\ell=\ell\alpha\Delta T\Rightarrow\alpha=\frac{\Delta\ell}{\ell\Delta T}$

$\alpha_1=\frac{1}{1\times100}=10^{-2}$

$\alpha_2=\frac{2}{100}=2\times10^{-2}$

$\alpha_3=\frac{3}{1.5\times50}=4\times10^{-2}$

$\alpha_4=\frac{4\times5}{2.5\times20.5}=8\times10^{-2}$ (maximum)

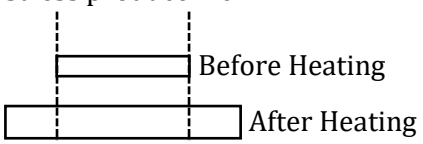
13. $Y=\frac{F\ell}{A\Delta\ell}=\frac{F\ell}{A\ell\Delta T\alpha}=\frac{F}{A\alpha\Delta T}\qquad(\Delta L=L\alpha\Delta L)$

$F = YA\,\alpha\,\Delta T$

$F=10^{11}\times10^{-3}\times10^{-6}\times100=10^{4}\,N$

14. We known that, when rod is freely expand,

stress produce = 0



stress α strain

so strain = 0

15. Heat = $\frac{mgh}{J}=\frac{5\times9.8\times30}{4.2}$ cal = 350 cal

16. $\frac{1}{2}\times\left(\frac{1}{2}mv^2\right)=msd\theta\Rightarrow d\theta=\frac{v^2}{4s}$

17. $Q = ms_{ice}\Delta T + mL_F + ms_{water}\Delta T + mL_V$

Q = (1)(0.5)(10) + (1)(80) + (1)(1)(100) + (1)(540)

Q = 5 + 80 + 100 + 540 = 725 cal.

or 725 × 4.18 = 3030 joule.

18. 2kg ice at – 20°C + 5 kg water at 20°C

$Q_{gain} = Q_{lost}$

$2\times\frac{1}{2}\times20+M\times80=5\times1\times20$

M = 1 kg

Water = 5 + 1 = 6 kg

19. 2 mgh = J msΔT

2 × 5 × 10 × 10 = 2 × 1 × 4.2 × 1000 × ΔT

ΔT = 0.12 °C

20. ms ΔT = m'L

2.5 × 500 × 0.1 = m × 80

m = 1.56 kg

21. at 1kg ice at –10 °C + 4.4 kg of water at 30 °C

Heat gain = Heat loss

$1000\times\frac{1}{2}\times10+1000\times80+1000\ (T-0)$

= 4.4 × 1000 × 1 × (30–T)

$\Rightarrow$ 5.4 T = 4.4 × 30 – 85

$\Rightarrow$ T = 8.7 °C

22. $Q = mL_f + ms\,\Delta\theta + mL_v$

= 1 × 80 + 1 × 1 × 100 + 1 × 536 = 716 cal

23. $536\ cal/g=\frac{536\times4.2J}{10^{-3}kg}$ = 2.25 × 10^6 J/kg

24. Heat given by water 20 °C → 0 °C

Q_1 = 10 × 1 × (20–0) = 200 cal

Heat required to melt 10 g ice

Q_2 = mL = 10 × 80 = 800 cal

$Q_1 < Q_2$ Hence complete ice will not melt

$\therefore$ equilibrium temperature = 0 °C

25. $\Delta Q = msd\theta$

420 J = 10 g × 4.2 × dθ

dθ = 10 °C

26. msΔT = dQ

$\frac{dT}{dQ}=\frac{1}{ms}=\frac{1}{\text{Heat capacity}}$ (For vapour)

27. Heat required for vapourisation = Rate × time

= 42 × (30–20) = mL = 5 × L

$\Rightarrow$ L = 84 kJ/kg

28. $mgh=\frac{mL}{5}\quad\Rightarrow\quad h=\frac{L}{5g}$

29. ΔQ = mL + msΔT + mL + WΔT

= 10 × 80 + 10 × 1 × 100 + 10 × 540 + 10 × 100

= 8200 cal

30. Q_1 = msΔT = 10 × 0.53 × 40 = 212 → (S)

Q_2 = mL = 10 × 80 → (P)

Q_3 = msΔT = 10 × 1 × 100 = 1000 → (Q)

Q_4 = mL = 10 × 540 = 5400 → (R)

31. Thermal capacity of a body is the quantity of heat required to raise its temperature by a unit degree.

32. Heat required is = 2 × 10^3 × 1 × 50 × 4.2 joule

Heat supplied per sec = 1000 – 160 = 840 J/s

So, Heat available for unit time (If total time taken is t)

$P=\frac{Q}{t}\Rightarrow\quad 840=\frac{2\times10^3\times1\times50\times4.2}{t}$

t = 500 sec = 8 min 20 sec

33. $P=\frac{Q}{t}\Rightarrow P=\frac{ms\Delta T}{t}$

P = 2100 J/s ; $\frac{m}{t}$ = 20 g/s

T_1 = 10 °C $\quad T_2$ = ?

$\left(\frac{2100}{4.2}\right)\frac{cal}{s}$ = 20 × 1 (T_2 – 10)

T_2 = 35 °C

34. As $R=\frac{L}{KA}\Rightarrow\frac{L_1}{K_1A}=\frac{L_2}{K_2A}\Rightarrow\frac{L_1}{L_2}=\frac{K_1}{K_2}=\frac{5}{3}$

35. $Q\propto\frac{r^2}{\ell}\Rightarrow\frac{Q_1}{Q_2}=\frac{r^2}{L}\times\frac{2L}{(2r)^2}\Rightarrow Q_2=2Q_1$

36. $\frac{Q}{t}=\frac{KA(\theta_1-\theta_2)}{L}$

$4000=\frac{400\times100\times10^{-4}(\theta_1-\theta_2)}{10\times10^{-2}}$

$\theta_1-\theta_2$ = 100 °C

37. $\frac{(9K)(A)(100-\theta)}{18}=\frac{K(A)(\theta-0)}{6}$

900 – 9θ = 3θ

$\theta=\frac{900}{12}$ = 75 °C

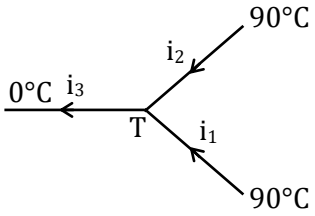
38. $\frac{\Delta Q}{\Delta t}$ = same

So $\frac{KA(20-10)}{\ell}=\frac{2KA(10-\theta)}{\ell}$

$\Rightarrow\quad 2\theta=10$

$\Rightarrow\theta$ = 5 °C

39.



$i_1 + i_2 = i_3$

$$\frac{KA}{\ell}(90-T)+\frac{KA}{\ell}(90-T)=\frac{KA}{\ell}(T-0)$$

$180 - 2T = T$

$3T = 180 \Rightarrow T = 60\ °C$

40 Thermal conductivity (K) depend's on material

41. Heat resistance $R=\frac{\ell}{kA}\Rightarrow R\propto\frac{\ell}{A}$

R is minimum for option IV

So conduction of heat will be more.

42. Gravitation force is required for convection

43. Convection (density difference)

44. $$R=\frac{dQ}{dt}\propto\frac{A}{\ell}\propto\frac{r^2}{\ell}\Rightarrow\frac{R_1}{R_2}=\frac{\left(r_1/r_2\right)^2}{\left(\ell_1/\ell_2\right)}\frac{4/9}{1/2}=\frac{8}{9}$$

45. $L_1 = L_2$, $(\theta_1 - \theta_2)$ is same

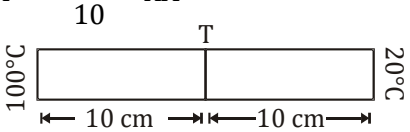
$$\frac{K_1A_1}{L_1}(\theta_1-\theta_2)=\frac{K_2A_2}{L_2}(\theta_1-\theta_2)$$

$K_1A_1 = K_2A_2$

46. $$\frac{(100-T)}{10}KA=\frac{(T-20)}{10}KA$$

$2T = 120$

$T = 60\ °C$



47. $R_{eq} = R_1 + R_2$

$$\frac{2L}{K_{eq}A}=\frac{L}{KA}+\frac{L}{2KA}\Rightarrow K_{eq}=\frac{4}{3}K$$

48. Under steady state, the temperature of a body does not change with time but is different at different point of the body.

49. $$\frac{dQ}{dt}=\frac{KA}{L}(\theta_1-\theta_2)=\frac{(0.2)(10)(40-20)}{2\times10^{-3}}$$

$= 20 \times 10^3 = 2 \times 10^4\ J$

50. Given $\frac{dQ}{Adt}=10$; $K = 0.5$

$$\frac{dQ}{dt}=KA\frac{dT}{dx}\Rightarrow\frac{dT}{dx}=\frac{1}{K}\times\frac{dQ}{Adt}$$

$$\Rightarrow\ \frac{dT}{dx}=\frac{10}{0.5}=20\ °C/cm$$

51. $$R=\frac{\ell}{KA}=R=\frac{\ell}{KA}=\frac{[L]}{\frac{\left[ML^2T^{-2}\times L\right]}{L^2T\theta}\left[L^2\right]}$$

$$R=\left[M^{-1}L^{-2}T^3\theta\right]$$

OR

$$R=\frac{\Delta T}{dQ/dT}$$

$$R=\frac{\theta'}{ML^2T^{-3}}$$

52. Low specific heat means low heat capacity and high conductivity means it will transfer heat to food quickly.

53. Convection (density difference)

54. Metal has high thermal conductivity

55. $$\frac{dQ}{dt}\propto\frac{A}{\ell}\propto\frac{r^2}{\ell},\frac{Q_1}{Q_2}=\left(\frac{2}{1}\right)^2\times\frac{4}{1}=16:1$$

56. $$\frac{KA}{1}(0-\theta)=\frac{(3K)(A)}{2}(\theta-100)$$

$-2\theta = 3\theta - 300 \Rightarrow \theta = 60\ °C$

57. Mud is a bad conductor of heat

58. $$\frac{K_1A(T_1-T)}{d_1}=\frac{K_2A(T-T_2)}{d_2}\ \text{As}\ \left(\frac{dQ_1}{dt}=\frac{dQ_2}{dt}\right)$$

$$T=\frac{K_1T_1d_2+K_2T_2d_1}{K_1d_2+K_2d_1}$$

59. There is a vaccume in the bulb.

60. Convection is due to density difference

61. Convection heat the air molecules which move upward, while high density molecule comes down for low density (pressure)

62. $Q = e_r\sigma AT^4t \Rightarrow \frac{Q}{t}=0.6\sigma AT^4$

63. $r_1 = 1\ m$; $T_1 = 4 \times 10^3\ K$

$r_2 = 4\ m$; $T_2 = 2 \times 10^3\ K$

$$\frac{E_1}{E_2}=\frac{A_1T_1^4}{A_2T_2^4}\ \Rightarrow\ \frac{E_1}{E_2}=\frac{r_1^2T_1^4}{r_2^2T_2^4}$$

$$\Rightarrow\frac{E_1}{E_2}=\frac{(1)^2(4\times10^3)^4}{(4)^2(2\times10^3)^4}=\frac{(1)^2(4)^4}{(4)^2(2)^4}$$

$$\Rightarrow\frac{E_1}{E_2}=\frac{1}{1}$$

64. $R_1 \propto (600)^4 - (200)^4$

$R_2 \propto (400)^4 - (200)^4$

$$\frac{R_2}{R_1} = \frac{(16+4)(16-4)}{(36+4)(36-4)} = \frac{20\times12}{40\times32}$$

$$R_2 = \frac{3}{16}R$$

65. $E = \sigma T^4$

$E' = \sigma (1.1\ T)^4$

$E' = (1.1)^4 \sigma T^4 = (1.1)^4 E$

$E' = 1.21 \times 1.21E = 1.4641E$

$$\%\Delta E = \frac{E'-E}{E'}\times100\% = 0.4641\times100\%$$

$= 46.41\%$

66. $$\frac{E}{E'} = \frac{\sigma(400)^4\times8\times4}{\sigma(600)^4\times4\times2}$$

$$\frac{E}{E'} = \left(\frac{2}{3}\right)^4\times4$$

$$E' = \frac{81}{64}E$$

67. $$\frac{E_1}{E_2} = \frac{E}{E_2} = \left(\frac{546}{273}\right)^4$$

$$E_2 = \frac{E}{16}$$

68. Use $\frac{\theta_1-\theta_2}{t} = K\left(\frac{\theta_1+\theta_2}{2}-\theta_0\right)$

$$\Rightarrow \frac{95-90}{30} = K\left(\frac{95+90}{2}-\theta_0\right) \quad ...(i)$$

$$\Rightarrow \frac{55-50}{70} = K\left(\frac{55+50}{2}-\theta_0\right) \quad ...(ii)$$

By dividing equation (i) by (ii),

$$\frac{7}{3} = \frac{185-2\theta_0}{105-2\theta_0}$$

$\Rightarrow 735 - 14\theta_0 = 555 - 6\theta_0$

$8\theta_0 = 180 \Rightarrow \theta_0 = \frac{180}{8} = 22.5\ °C$

69. $Q = C\Delta T \quad \Rightarrow \quad \frac{dQ}{dt} = C\frac{dT}{dt}$

$$C_1\left(\frac{dT}{dt}\right)_1 = C_2\left(\frac{dT}{dt}\right)_2$$

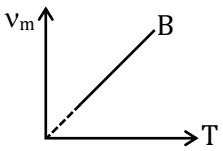
Here C = thermal capacity.

$$\frac{(dT/dt)_1}{(dT/dt)_2} = \frac{C_2}{C_1} = \frac{4}{1}$$

70. Newton's law of cooling is used to calculate the specific heat of liquid .

71. $\lambda_m \propto \frac{1}{T} \Rightarrow \lambda_m = \frac{b}{T}$

$$\lambda_m = \frac{c}{\nu_m}$$



$\nu_m \propto$ temperature

72. $\lambda_m \uparrow T \downarrow$

wavelength of blue colour is lesser than red so the temperature of blue star is more.

73. $\lambda_{m_1} T_1 = \lambda_{m_2} T_2$

$2597\ K \times 12000Å = 4800Å \times T_2$

$T_2 = 6492.5\ K \simeq 6219.5\ °C$

74. Kirchhoff's Law :- good emitter is a good absorber

75. $\lambda_m = \frac{b}{T}$ weins law

76. Black absorb more, so emit more.

77. $\frac{d\theta}{dt} \propto (\theta - \theta_0)$ (Newton's law)

78. $$\frac{E_1}{E_2} = \frac{\sigma T_1^4}{\sigma T_2^4} \quad \Rightarrow \quad \frac{E}{2E} = \left(\frac{1000}{T}\right)^4$$

$T = 2^{1/4} \times 1000 = \left(\sqrt{2}\right)^{1/2} \times 1000$

$T = 1.19 \times 1000 = 1190\ K$

79. $Q \propto r^2T^4 \Rightarrow \frac{Q_1}{Q_2} = \left(\frac{1}{4}\right)^2 \times \left(\frac{2}{1}\right)^4 = \frac{1}{1}$

80. Good emitter is a good absorber

81. $E \propto T^4 \Rightarrow \frac{E_1}{E_2} = \left(\frac{T_1}{T_2}\right)^4$

$$\Rightarrow \frac{10J}{E_2} = \left(\frac{300}{600}\right)^4 = \frac{1}{16}$$

$E_2 = 160\ J$

82. $$\frac{10^6}{16\times10^6} = \frac{\sigma(400)^4}{\sigma(T)^4} \qquad [E = \sigma T^4]$$

$T^4 = (2 \times 400)^4$

$\Rightarrow T = 800\ K = 527\ °C$

83. Solar constant S

$$S \propto \frac{1}{d^2} \Rightarrow \frac{S'}{S} = \frac{(d)^2}{(d')^2}$$

$$S' = \frac{S.d^2}{(0.4d)^2} \Rightarrow S' = \frac{2}{(0.4)^2} = 12.5\ cal/min\ cm^2$$

84. $T \propto \frac{1}{\lambda}$

$$\frac{Q_1}{Q_2} = \frac{A_1}{A_2}\left(\frac{T_1}{T_2}\right)^4 = \frac{A_1}{A_2}\left(\frac{\lambda_2}{\lambda_1}\right)^4 = \left(\frac{r_1}{r_2}\right)^2\left(\frac{\lambda_2}{\lambda_1}\right)^4$$

$$= \frac{6^2}{18^2} \times \frac{15^4}{5^4} = \frac{1}{3^2} \times 3^4 = 9$$

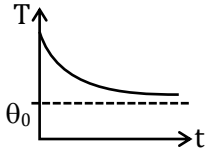
85. $\frac{T_1}{T_2} = \frac{\lambda_2}{\lambda_1} = \frac{560}{420} = \frac{28}{21} = \frac{4}{3}$

86. $\frac{dT}{dt} = -k(T - \theta_0) \Rightarrow \frac{dT}{T - \theta_0} = -kdt$

$\ln(T - \theta_0) = -kt + c$

$T - \theta_0 = e^{-kt + c}$

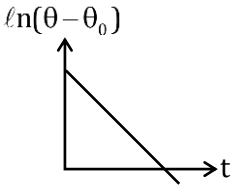
$T = \theta_0 + e^{-kt + c}$



87. $\frac{d\theta}{dt} = -k(\theta - \theta_0)$

$\frac{d\theta}{(\theta - \theta_0)} = -kdt$

$\ell n(\theta - \theta_0) = -kt + c$



88. $-\frac{d\theta}{dt} = k(\theta - \theta_0)$

As difference between body temperature and surrounding temperature decreases. Rate of cooling decreases with time.

89. $\lambda_m \propto \frac{1}{T}$

90. good absorber is good emitter

91. $\frac{60-50}{10} = K\left(\frac{60+50}{2} - 25\right)$

$\Rightarrow 1 = K(30)$ ...(i)

$\Rightarrow \frac{50-T}{10} = K\left(\frac{50+T}{2} - 25\right)$

$\Rightarrow \left(\frac{50-T}{10}\right) = K.\frac{T}{2}$ ...(ii)

Solving equation (i) and (ii), we find

T = 42.85 °C

92. good absorber is good emitter

93. Thermal radiation are infrared radiation

94. $\lambda_m \propto \frac{1}{T}$ according to weins law

95. $\frac{R_{C1}}{R_{C2}} = \frac{R_{F1}}{R_{F2}} = \frac{r_2}{r_1} = \frac{2r}{r} = 2:1$

96. $E \propto T^4$ and $E_{max} \propto T^5$ (wein's 5th power law)

97. Kirchhoff's law :-

$\frac{e_\lambda}{a_\lambda} = E_\lambda$

98. Case ...(1) $\frac{80-50}{5} = K\left[\frac{80+50}{2} - 20\right]$

Case ...(2) $\frac{60-30}{t} = K\left[\frac{60+30}{2} - 20\right]$

On solving t = 9 min.

99. $\frac{80-60}{60} = K\left(\frac{80+60}{2} - 30\right)$

$\Rightarrow \frac{1}{3} = K(40)$ ...(i)

$\Rightarrow \frac{60-50}{t} = K\left(\frac{60+50}{2} - 30\right)$

$\Rightarrow \frac{10}{t} = K(25)$ ...(ii)

Solving equation (i) and (ii), we find

t = 48 Sec.

100. Wein's Law $\lambda_m \propto \frac{1}{T}$

101. Amount of radiation emitted and absorbed by it will be equal.

102. Ideal black body, totally absorb heat radiation of all the wavelengths.

103. $\frac{E_1}{E_2} = \frac{r_1^2 T_1^4}{r_2^2 T_2^4} = \left(\frac{8}{2}\right)^2 \cdot \left(\frac{400}{800}\right)^4 \Rightarrow \frac{E_1}{E_2} = 1$

104. Wein's Law $\lambda_m \propto \frac{1}{T}$

105. $R_H \propto \text{Area}(A)$, so same for both

106. $R_F = \frac{d\theta}{dt} \propto \frac{1}{\rho r} \Rightarrow \frac{R_{F_1}}{R_{F_2}} = \frac{\rho_2 r_2}{\rho_1 r_1} = \frac{1 \times 2}{2 \times 1} = 1:1$

107. $\lambda_m T$ = constant

108. Spectral energy distribution curve is a continuous curve for ideal black body (IBB)

109. $Q = \sigma T^4$ At = $5.6 \times 10^{-8} \times 10^{12} \times 0.1 \times 60$

$Q = \frac{5.67 \times 10^4 \times 6}{4.2}$ = 81000 cal

110. $E = \sigma T^4 \Rightarrow \frac{E}{E'} = \frac{T}{(T/2)^4} \Rightarrow E' = \frac{E}{16}$

111. Definite absorption takes place in chromosphere

112. $E \propto T^4$

113. For ideal black body (IBB) : $\boxed{a = 1}$

114. $N = \frac{PV}{kT} = \frac{1.3\times10^5\times7\times10^{-3}}{1.38\times10^{-23}\times273}$

$N = 2.41 \times 10^{23}$

115. All real gas behave as an ideal gas at high temperature, low pressure and low density.

116. μ = constant, T = constant, $P_aV_a = P_bV_b$

117. At same temperature pressure and volume $(N_1=N_2=N_3)$ Avogadro law

118. All temperature and pressure.

119. $P = \frac{1}{3}\left(\frac{mN}{V}\right)v_{rms}^2$

$P \propto N$ (no of molecule)

$N \rightarrow$ double, so P become double.

120. $PV = \mu RT \Rightarrow P = \frac{\mu RT}{V}$

$PV^{2/3} = C \Rightarrow \frac{\mu RT}{V}\times V^{2/3} = C$

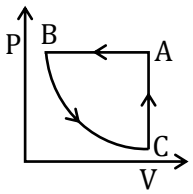
$TV^{-1/3} = C \Rightarrow T \propto V^{1/3}$

$T\uparrow \Rightarrow V\uparrow$

121. $A \rightarrow B \Rightarrow$ P = constant & T↓ So V↓ $\Rightarrow$ IBC

$B \rightarrow C \Rightarrow$ T = constant & P↓ So V↑ $\Rightarrow$ ITE

$C \rightarrow A \Rightarrow$ V = constant & P↑ So T↑ $\Rightarrow$ IC



122. $P = \frac{\mu RT}{V}$

$P^2V = C \Rightarrow \left(\frac{T}{V}\right)^2 \times V = C$

$\frac{T^2}{V} = C \Rightarrow \frac{T_2^2}{T_1^2} = \frac{V_2}{V_1}$ $(\because V_2 = 2V_1)$

$T_2 = \sqrt{2}T$

123. P = constant

$V \propto T \Rightarrow \frac{V_1}{V_2} = \frac{T_1}{T_2}$

$T_2 = T_1 \times \frac{V_2}{V_1} \Rightarrow T_2 = 300 \times \frac{500}{250}$

$T_2 = 600$ K = 327 °C

124. $\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2} \Rightarrow V_2 = \frac{P_1}{P_2}\times\frac{T_2}{T_1}\times V_1$

$V_2 = \frac{1}{0.5}\times\frac{270}{300} \times 500\ m^3 = 900\ m^3$

125. V = constant $\Rightarrow \frac{P_1}{M_1T_1} = \frac{P_2}{M_2T_2}$

Final mass $M_2 = \frac{P_2}{T_2}\times\frac{M_1T_1}{P_1}$

$M_2 = \frac{P/2}{300}\times\frac{6g\times400}{P} = -4\,g$

Leak out mass = $M_1 - M_2 = 6 - 4 = 2$ g

126. High temperature and low density.

Note : All real gas behave as an ideal gas at high temperature, low pressure and low density.

127. T = constant

$P_1V_1 = P_2V_2$

$P_2 = \frac{P_1V_1}{V_2}$

$P_2 = 0.8 \times \frac{5}{8}$

$P_2 = 0.5$ m

128. $PV = \mu RT$

$P = \frac{\rho}{M_w}RT \Rightarrow P \propto \rho$ (At const. temp.)

129. $PV = \mu RT \Rightarrow PV = \frac{M}{M_w}RT \Rightarrow P = \frac{M}{V}\frac{RT}{M_w}$

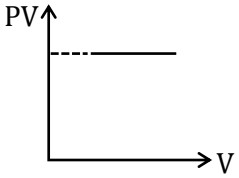
$P = \frac{\rho RT}{M_w} \Rightarrow \rho = \frac{PM_w}{RT} \Rightarrow \rho \propto \frac{P}{T}$

130. Closed container $\Rightarrow$ V = constant

$\frac{\Delta P}{P} = \frac{\Delta T}{T} \Rightarrow T = \frac{P}{\Delta P} = \Delta T$

$T = \frac{100}{0.4} \times 1\ K = 250\ K$

131. PV = constant $\Rightarrow$ T = constant



132. $N = \frac{PV}{kT} = \frac{13.8\,N/m^2 \times 1m^3}{1.38\times10^{-23}\,J/K\times300K}$

$N = 3.3 \times 10^{21}$ molecule

133. Closed contain (V = constant)

Initially,

$M_1 = 10$ kg

$P_1 = 10^7\ N/m^2$

let M′ kg gas leak

Finally $M_2 = (10 - M')$kg

$P_2 = 2.5 \times 10^6\ N/m^2$

given T = constant for a given gas

$PV = \frac{MRT}{M_w}$ V = constant

$P \propto M$ M_w = constant

So $\frac{P_2}{P_1} = \frac{M_2}{M_1}$ T = Constant

By solving. M′ = 7.5 kg

134. Dalton's law of partial pressure

or $P_{mix} = \frac{P_1V_1 + P_2V_2}{V_{mix}}$ (At const. temp.)

135. $PV = \mu RT \Rightarrow PV = \frac{M}{Mw}RT$

O_2 M = 5gm

$M_w = 32$ gm/mole

$PV = \frac{5}{32}RT$

136. Have same mass but negligible volume. (Point mass)

137. $\frac{P}{\rho} = \frac{RT}{M_w}$

$\rho = \frac{PM_w}{RT}$ $M_w = mN_A$

$\rho = \frac{Pm}{kT}$ $\frac{R}{N_A} = k$

138. $\left.\begin{aligned} N_1 &= \frac{P_1V_1}{kT_1} \\ N_2 &= \frac{P_2V_2}{kT_2} \end{aligned}\right\} \Rightarrow \frac{N_1}{N_2} = \frac{P_1V_1}{T_1}\times\frac{T_2}{P_2V_2}$

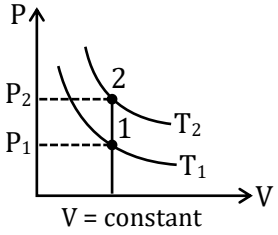
139. At constant temperature

$\therefore P_1V_1 = P_2V_2 \Rightarrow PV = P_2$.

$\frac{V}{4} \Rightarrow P_2 = 4P$

So, $P_2 = 4$ atm

140.



At point 1 and point 2

Let V = constant ($P \propto T$)

so $P_2 > P_1$

$T_2 > T_1$

141. Real gas equation

for μ = 1 mole gas

$\left(P + \frac{a}{V^2}\right)(V - b) = RT$

142. $v_{rms} = \sqrt{\frac{3RT}{M_w}}$

143. $v_{rms} \propto \sqrt{T} \Rightarrow \frac{v_1^2}{v_2^2} = \frac{T_1}{T_2}$

$\left(\frac{v}{v/2}\right)^2 = \frac{600}{T}$

$T = \frac{600}{4} = 150\ K = -123\ °C$

144. $v_{rms} \propto \sqrt{T}$ (Does not depend on pressure)

$\frac{v_{rms_1}}{v_{rms_2}} = \sqrt{\frac{T_1}{T_2}} \Rightarrow \frac{200}{v_{rms_2}} = \sqrt{\frac{300}{400}}$

$v_{rms_2} = \frac{400}{\sqrt{3}}$ m/s

145. average speed $< C > \propto \frac{1}{\sqrt{M_w}}$

$\frac{< C_H >}{< C_{He} >} = \sqrt{\frac{4}{1}}$

$< C_H > = 2 < C_{He} >$

146. $v_{rms} \propto \sqrt{\frac{T}{M_w}}$

$v' \propto \sqrt{\frac{(T/2)}{(2M_w)}} \propto \frac{1}{2}\sqrt{\frac{T}{M_w}}$

Hence v_{rms} become half

147. $v \propto \sqrt{\frac{T}{M_w}}$

$\frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2} \times \frac{M_{w_2}}{M_{w_1}}}$

$\frac{1930}{v_2} = \sqrt{\frac{300}{1200} \times \frac{32}{2}}$

$\frac{1930}{v_2} = \sqrt{4} \Rightarrow \frac{1930}{V_2} = 2$

$V_2 = \frac{1930}{2} = 965 \text{ m/s}$

148. $v_{r.m.s.} \propto \sqrt{\frac{1}{M_w}} \Rightarrow \frac{(v_{r.m.s.})_{O_2}}{(v_{r.m.s.})_{H_2}} = \sqrt{\frac{2}{32}} = \frac{1}{4}$

149. v_{rms} = same $\Rightarrow$ $T \propto M_w$

$T_{He} = \frac{4}{2} \times 273 = 546 \text{ K} = 273°C$

150. Any speed depends on temperature and molecular weight only so if temperature constant then mean square velocity will also constant.

151. Value of v_{rms} of the molecules of gas is more than escape velocity.

152. $v_{r.m.s} = \sqrt{\frac{v_1^2 + v_2^2 + v_3^2 + v_4^2 + v_5^2}{5}}$

$= \sqrt{\frac{2^2 + 3^2 + 4^2 + 5^2 + 6^2}{5}} = 4.24 \text{ m/s}$

153. Independent of its pressure but directly proportional to the square root of its kelvin temperature.

154. $v_{rms} \propto \frac{1}{\sqrt{M_w}}$

155. $P = \frac{1}{3}\rho v_{rms}^2 = \frac{1}{3} \times 8.99 \times 10^{-2} \times (3180)^2$

$= 3 \times 10^5 \text{ N/m}^2 = 3 \text{ atm}$

156. $P = \frac{1}{3}\left(\frac{M}{V}\right)v_{rms}^2$

P(pressure) $\propto v^2_{rms}$

pressure become four times

157. $v_{rms} = \sqrt{\frac{3RT}{M_w}} = \sqrt{\frac{3kT}{m}}$

$v_{rms} \propto \frac{1}{\sqrt{m}}$ $\quad m \rightarrow$ mass of molecule

158. $v_{rms} = \sqrt{\frac{3P}{\rho}} = \sqrt{3}\sqrt{\frac{P}{\rho}} = 1.71\sqrt{\frac{P}{\rho}}$

$v_{avg} = \sqrt{\frac{8}{\pi}\frac{P}{\rho}} = 1.59\sqrt{\frac{P}{\rho}}$

$v_{mp} = \sqrt{\frac{2P}{\rho}} = 1.41\sqrt{\frac{P}{\rho}}$

for a given gas P and ρ = constant

$v_{rms} > v_{avg} > v_{mp}$

159. $(v_{rms})_{H_2} = (v_{es})_{earth}$

$\Rightarrow \sqrt{\frac{3RT}{M_w}} = 11.2 \times 10^3 \text{ m/s}$

$\Rightarrow \frac{3 \times 8.314 \times T}{2 \times 10^{-3}} = 125 \times 10^6$

$\Rightarrow T \simeq 10^4 \text{ K}$

160. $P = \frac{1}{3}\left(\frac{mN}{V}\right)v_{rms}^2$

$m \rightarrow$ mass of molecule.

161. $v_{rms} \propto \sqrt{\frac{T}{M_w}}$

than $\frac{v'}{v} = \sqrt{\frac{T'}{M_w} \times \frac{M_w}{T}}$

$v' = \sqrt{\frac{2T}{M_w/2} \times \frac{M_w}{T}} \times v$

$v' = 2v$

162. v_{rms} = same $\Rightarrow$ $T \propto M_w$

$\frac{T_{O_2}}{T_{H_2}} = \frac{(M_w)_{O_2}}{(M_w)_{H_2}} \Rightarrow T_{H_2} = \frac{2}{32} \times 320 = 20 \text{ K}$

163. $v_{rms} > v_{mp}$ so different.

$v_{rms} = \sqrt{\frac{3P}{\rho}} = 1.73\sqrt{\frac{P}{\rho}}$

$v_{mp} = \sqrt{\frac{2P}{\rho}} = 1.41\sqrt{\frac{P}{\rho}}$

For given gas P, ρ are constant

164. $v_{rms} > v_{avg} > v_{mp}$

165. We know that total K.E = $\frac{f}{2}kT$

$f \rightarrow$ degree of freedom

for diatomic gas

$f = f_T + f_R$ $\quad f_T=3$

$\quad f_R=2$

$(K.E)_T = \frac{f_T}{2}kT = \frac{3}{2}kT$

$(K.E)_R = \frac{f_R}{2}kT = \frac{2}{2}kT$

$\frac{(K.E)_T}{(K.E)_R} = \frac{3}{2}$

166. gram specific heat difference is

$\left(c_p - c_v = r = \frac{R}{M_w}\right)$

for H_2 $\quad c_p - c_v = a = \frac{R}{2}\left(M_w(H_2) = 2\right)$

for O_2 $\quad c_p - c_v = b = \frac{R}{32}\left(M_w(O_2) = 32\right)$

$a = \frac{R}{2}$; $b = \frac{R}{32}$

$a = 16b$

167. $U = U_1 + U_2$

$= \mu_1 C_{v_1} T + \mu_2 C_{v_2} T$

$= 2\times\frac{5}{2}RT + 4\times\frac{3}{2}RT$

$= 5RT + 6RT = 11RT$

168. $E \propto T$

$\frac{E_1}{E_2} = \frac{T_1}{T_2} \Rightarrow \frac{6.21\times10^{-21}}{E_2} = \frac{300}{500}$

$E_2 = \frac{5}{3} \times 6.21 \times 10^{-21} = 10.35 \times 10^{-21}$ J

169. At same temperature

$v_{rms} = \sqrt{\frac{3kT}{m}}$ so $\frac{v_A}{v_B} = \sqrt{\frac{m_B}{m_A}}$

$\frac{P_A}{P_B} = \frac{m_A v_A}{m_B v_B} = \frac{m_A}{m_B}\times\sqrt{\frac{m_B}{m_A}}$

$\Rightarrow P_A = \left(\frac{m_A}{m_B}\right)^{1/2} P_B$

170. $P = \frac{2}{3}\frac{E_T}{V}$

$P = \frac{2}{3}\times\frac{1.52\times10^5}{200\times10^{-3}} = 5 \times 10^5$ N/m^2

171. $M_{mix} = \frac{\mu_1 M_1 + \mu_2 M_2}{\mu_1 + \mu_2} = \frac{5\times4 + 2\times2}{7} = \frac{24}{7}$ g

172. $C_{V_{mix}} = \frac{2\times\frac{5}{2}R + 5\times\frac{3}{2}R}{5+2}$

$C_{P_{mix}} = \frac{2\times\frac{7}{2}R + 5\times\frac{5}{2}R}{5+2}$

$\gamma_{mix} = \frac{C_{P_{mix}}}{C_{V_{mix}}} = \frac{39/7}{25/7} = 1.56$

OR

$\gamma_{mix} = \frac{\mu_1 C_{P_1} + \mu_2 C_{P_2}}{\mu_1 C_{V_1} + \mu_2 C_{V_2}}$

$\gamma_{mix} = \frac{2\times\frac{7}{2}R + 5\times\frac{5}{2}R}{2\times\frac{5}{2}R + 5\times\frac{3}{2}R}$

173. $E_{mix} = E_1 + E_2$

$\frac{(n_1+n_2)fRT_{mix}}{2} = \frac{n_1 fRT_1}{2} + \frac{n_2 fRT_2}{2}$

$T_{mix} = \frac{n_1 T_1 + n_2 T_2}{n_1 + n_2}$

174. Total K.E = $\frac{5}{2}\mu RT$

for 1 mole = $\frac{5}{2}RT$

$= \frac{5}{2} \times 2 \times 300 = 1500$ cal

175. Mean K.E Per gram molecule = Molar K.E

$E_{mole} = \frac{3}{2}RT$

176. $\gamma = \frac{C_P}{C_V} = \frac{C_V + R}{C_V} = \frac{\frac{1}{2}fR + R}{\frac{1}{2}fR} = \frac{f+2}{f}$

$\Rightarrow \gamma f - f = 2 \Rightarrow f = \frac{2}{\gamma - 1}$

177. $E_T = \frac{3}{2}PV$

$\frac{E_T}{V} = \frac{3}{2}P$ $\quad E = \frac{E_T}{V}$ energy density

$P = \frac{2}{3}E$

178. $C_{V_{mix}} = \frac{\mu_1 C_{V_1} + \mu_2 C_{V_2}}{\mu_1 + \mu_2}$

179. Absolute zero temperature is one at which motion of molecules freeze. Hence K.E. of a molecules become zero.

180. $\frac{R}{C_v} = 0.67$

$C_v = \frac{R}{0.67} = \frac{100}{67}R = \frac{3}{2}R$

so, f = 3 (Monoatomic)

181. Total K.E = $\frac{f}{2}NkT$

Total K.E $\propto$ N (no. of gas molecule)

182. KE $\propto$ T

$\frac{E_1}{E_2} = \frac{T_1}{T_2}$

$T_2 = 2 \times 293$

$T_2 = 586$ K = 313°C

183. Avg. Kinetic Energy = $\frac{3}{2}kT$

$\Rightarrow$ Avg. K.E $\propto$ T

184. K.E associated with per degree of freedom of a molecule is $\frac{1}{2}kT$

185. The average of distances travelled between two successive collisions is mean free path.

186. The molecules are transferring their momentum to the wall.

187. $\Delta U = \mu C_v \Delta T$ (always) = $\frac{1}{2}$ = $\mu R \Delta T$, ΔT = same

$\Rightarrow \frac{U_1}{U_2} = \frac{1}{1}$

188. specific heat of an ideal gas is independent on temperature

189. Can have any value between 0 to ∞ because it depends on temperature and process.

190. Use $T_{mix} = \frac{\mu_1 f_1 T_1 + \mu_2 f_2 T_2}{\mu_1 f_1 + \mu_2 f_2}$

191. E $\propto$ T

192. K.E $\propto$ T $\Rightarrow$ T = absolute temperature

193. $(K.E)_{avg} = \frac{f}{2}kT$ [for monoatomic gas f = 3]

$(K.E)_{avg} = \frac{3}{2}kT$

194. A diatomic molecule has 5 degree of freedom.

195. $\gamma_{mix} = \frac{C_{Pmix}}{C_{Vmix}} = \frac{\mu_1 C_{P_1} + \mu_2 C_{P_2}}{\mu_1 C_{V_1} + \mu_2 C_{V_2}}$

$\gamma_{mix} = 1.55$

196. 1st law of thermodynamics is based on law of conservation of energy.

197. Q = 500 cal = 500 × 4.18 = 2093 J

W = –100 J

$\Delta U = Q - W$

$\simeq$ 2193 J

198. Given Q = – 20 J, W = – 8 J

$\Delta U = U_f - U_i$ So, $Q = \Delta U + W$

$-20 = U_f - U_i - 8$

$-20 = U_f - 30 - 8 \Rightarrow U_f = 18$ joule

199. $Q_1 = \Delta U_1 + W_1$

$Q_1 = 16 - 20 = -4$ kJ

$Q_2 = \Delta U_2 + W_2$

$W_2 = Q_2 - \Delta U_2$ $\quad (\because \Delta U_1 = \Delta U_2)$

= 9 – (–4) = 13 kJ

200. $Q_{ABC} = 90$ J and $W_{ABC} = 30$ J

$\therefore$ $\Delta U = 90 - 30 = 60$ J

if $W_{ADC} = 20$ J

$Q_{ADC} = \Delta U_{ADC} + W_{ADC} = 60$ J + 20 J = 80 J

201. W = Area of ellipse

$W = \pi\left(\frac{P_2 - P_1}{2}\right)\left(\frac{V_2 - V_1}{2}\right)$

202. W = Area of LMNOL = PV

203. $\Delta U = \mu C_V \Delta T = 2 \times 4.96 \times 2 = 19.84$ cal

204. Work done always depends on path.

205. Change in internal energy is same and $W_A > W_B$ So, $Q_A > Q_B$

206. FLOT $Q = W + \Delta U$

Q = +110 J

ΔU = +40 J

110 = 40 + W

W = + 70 J

207. According to FLOT :-

$Q = W + \Delta U$ $\Delta U = mC_V\Delta T$

$Q = W + mC_v\Delta T$

208. $W = \frac{1}{2} \times 4P_0 \times 3V_0 = 6P_0V_0$

209. For cyclic process

$\Delta U = U_f - U_i = 0$

Zero in all the four cases.

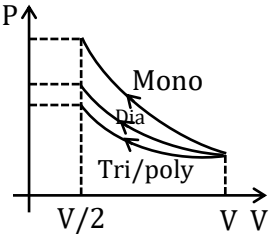
210. Heat capacity = $\frac{Q}{\Delta T}$

At constant volume W = 0

$Q = \Delta U = 80$ J

Heat capacity = $\frac{80}{20}$ = 4 J/K

211.



$P_2 = P_1\left(\frac{V_1}{V_2}\right)^{\gamma}$ $V_1 = V$ and $V_2 = V/2$

Hence $P_2 = P_1(2)^{\gamma}$

$\gamma\uparrow$ $P_2\uparrow$

As $\gamma_{Mono} > \gamma_{Dia} > \gamma_{Tri}$

then $P_{Mono} > P_{Dia} > P_{Tri}$

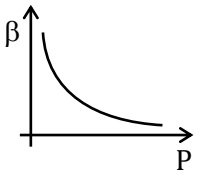
212. At constant temperature

PV = constant

On differentiation

$VdP + PdV = 0$

$\frac{dP}{dV} = -\frac{P}{V} \Rightarrow \frac{dV}{dP} = -\frac{V}{P}$



$\beta = \frac{-\left(\frac{dV}{dP}\right)}{V} = \frac{1}{P}$

$\beta = \frac{1}{P}$ Hence $\beta \propto \frac{1}{P}$

213. $\beta = \gamma P = \frac{7}{5} \times 10^5$ N/m^2 = 1.4 × 10^5 N/m^2

214. β = P (for isothermal)

215. $Slope_{adiabatic} \propto \gamma$

Slope of 1 < Slope of 2

$\gamma_1 < \gamma_2$ and $\gamma_{mono} > \gamma_{dia}$

So, 2 is monoatomic & 1 is diatomic

216. For adiabatic

$\frac{\Delta P}{\Delta V} = -\gamma\frac{P}{V} \Rightarrow \frac{\Delta P}{P} = -\gamma\frac{\Delta V}{V}$

217. $TV^{\gamma-1}$ = constant

$\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$

$\frac{T_2}{300} = \left(\frac{27}{8}\right)^{\frac{5}{3}-1} = \left(\frac{27}{8}\right)^{2/3} = \frac{3^2}{2^2} = \frac{9}{4}$

$T_2 = 300 \times \frac{9}{4} = 675$ K

$\Delta T = T_2 - T_1 = 675 - 300 = 375$ K

218. For adiabatic process

$P^{1-\gamma}T^{\gamma}$ = constant

$P \propto T^{\left(\frac{-\gamma}{1-\gamma}\right)} \propto T^{\left(\frac{\gamma}{\gamma-1}\right)}$

as per question $P \propto T^C$

on comparing

$C = \frac{\gamma}{\gamma-1} = \frac{5/3}{5/3-1} = \frac{5}{2}$

219. $\frac{W}{Q} = \frac{\mu R\Delta T}{\mu C_p\Delta T} = \frac{R}{C_p} = \frac{R}{\frac{5}{2}R} = \frac{2}{5} \times 100\% = 40\%$

220. $B \rightarrow$ Adiabatic $\rightarrow P \propto \frac{1}{V^{\gamma}}$ $V\uparrow P\downarrow$ (more steep)

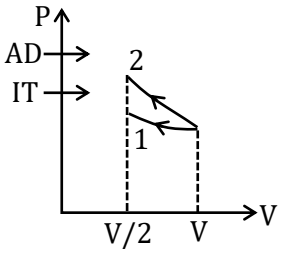
$A \rightarrow$ Isothermal $\rightarrow P \propto \frac{1}{V}$ $V\uparrow P\downarrow$ (less steep)

(C, D) $\rightarrow$ positive slope

221. Slope of adiabatic = γ (Slope of isothermal)

Isothermal for A and C while adiabatic for B and D

222.



In isothermal process T = constant, ΔT=0

But V↓ , P↑

In adiabatic process (P, V, T) are variable,

$V\downarrow$, $W = -ve$, $\Delta U = +ve$ $(Temp\uparrow)$

V↓, P↑

Option (3) is correct

223. $\frac{W}{Q}=\frac{\mu R\Delta T}{\mu C_P\Delta T}=\frac{R}{C_P}=\frac{R}{\frac{5}{2}R}=\frac{2}{5}$

$W=\frac{2}{5}Q$

224. For polytropic process

$W=\frac{\mu R\Delta T}{1-x}$

Here $x=-2$

$W=\frac{\mu R(T_2-T_1)}{1-(-2)}$

$W==\frac{1}{3}R(T_2-T_1)$

225. $\frac{dU}{\Delta Q_P}=\frac{\mu C_V dT}{\mu C_P dT}=\frac{C_V}{C_P}=\frac{1}{\gamma}=\frac{5}{7}$

$\left(\because\ \gamma=\frac{7}{5}\right)$ for diatomic gas

226. Adiabatic relation between pressure and temperature.

$P^{(1-\gamma)}T^{\gamma}$ = constant

227. PV^{γ} = constant $\Rightarrow$ $\frac{\mu RT}{V}V^{\gamma}$ = constant

$TV^{\gamma-1}$ = constant

If $V\uparrow\Rightarrow T\downarrow\ U\downarrow$ or $V\downarrow\Rightarrow T\uparrow U\uparrow$

May decrease and may also increase

228. $\Delta U=\mu C_V\Delta T=1\times\frac{3}{2}R\times 100$

$\Delta U=1\times\frac{3}{2}\times 8.31\times 100=12.48\times 10^2$ J

229.

V = constant	P = constant
W = 0	$W\neq 0$
$(Q)_V=(\Delta U)_V+0$	$(Q)_P=(\Delta U)_P+(W)_P$

So $(Q)_P>(Q)_V$

$\mu C_p\Delta T>\mu C_v\Delta T$

$C_P>C_V$

Work is done in the expansion of the gas at constant pressure.

So option (1) is correct

230. In adiabatic process $\Delta Q=0$

$\Delta U+\Delta W=0$

$\Delta U=-\Delta W$

When gas adiabaticaly compressed, then $V\downarrow$

ΔV = –ve, So ΔW = –ve

ΔU = +ve $\Rightarrow U\uparrow$

$\mu C_v(T_f-T_i)>0$

$T_f>T_i$ hence temperature $\uparrow$

231. $\frac{T_1^{\gamma}}{P_1^{\gamma-1}}=\frac{T_2^{\gamma}}{P_2^{\gamma-1}}$

$T_2=\left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}\times T_1$

$T_2=\left(\frac{1}{2}\right)^{0.5/1.5}\times 300=\frac{1}{2^{1/3}}\times 300$

T_2= 240 K = –33°C

232. $C=\frac{\Delta Q}{\mu\Delta T}$

for adiabatic $\Delta Q=0$

$C_{AD}=0$

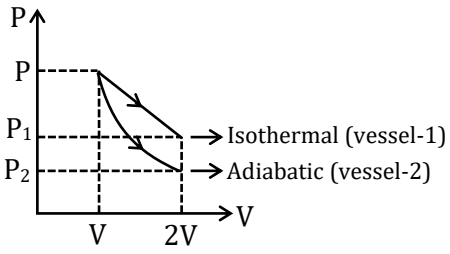
233. Gas suddenly compressed ($V\downarrow$)

so

W = – ve and Q = 0 (for adiabatic)

ΔU = +ve (temperature increase)

234.



$P_1>P_2$, final pressure in vessel (1) is more than vessel (2)

$W_1>W_2$ $\qquad$ $W\propto$ Area

so, In first vessel, both pressure and work done are more.

235. Heat is rejected by gas

Q = W (for isothermal process $\Delta U=0$)

when gas compressed $V\downarrow$, ΔV = –ve

W = –ve

Q = –ve (heat rejected)

236. $(\text{slope})_{AD}=-\gamma\frac{P}{V}$

$(\text{slope})_{AD}=\gamma\left(-\frac{P}{V}\right)$

$(\text{slope})_{AD}=\gamma(\text{slope})_{IT}$

237. Work done in cyclic process is area enclosed.

$\text{Area}=W=\frac{1}{2}BH$

$W_{ABCD}=\frac{1}{2}(3V_1-V_1)(4P_1-P_1)$

$W_{ABCD}=3P_1V_1$

238. $W = -\frac{\mu R(T_2 - T_1)}{\gamma - 1}$

$W = -\frac{(1)(8.3)(102-27)}{1.5-1} = -1245$ J

239. According to FLOT

$\delta Q = \delta W + dU$

Isometric V = Constant $\Delta V = 0$

$\delta Q = dU$

240. $W = P\Delta V = 10^3 \times 0.25$

$W = 250$ J

241. Q = W, when $\Delta U = 0$, T = constant (isothermal process)

242. $P_1V_1^{\gamma} = P_2V_2^{\gamma} \Rightarrow P_2 = \left(\frac{V_1}{V_2}\right)^{\gamma} \times P_1$

$P_2 = \left(\frac{V}{V/8}\right)^{4/3} \times P_0 = 16\,P_0$

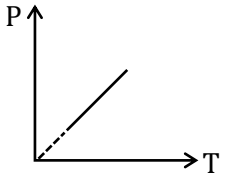
243. Work equals to area under the P-V graph or curve and V-axis.

244. In adiabatic process Q = 0

Total heat content of the system remain constant.

245. Isometric V = constant

$P \propto T$



246. Adiabatic equation in terms of P, V

PV^{γ} = constant

247. $\Delta U = Q - W = 300 - 0$; V = constant ; W = 0

Change in internal energy per gram

$= \Delta U/M = \frac{300}{50} = 6$ cal/g

248. $T_B > T_A = T_C > T_D \Rightarrow U = \mu C_V T$

$U_B > U_A = U_C > U_D \Rightarrow U \propto T$

249. Free expansion as $P_{ext} = 0 \Rightarrow W = 0$

250. Heat cannot be converted in 100% work, but reverse is true.

251. Heat can't flow from the body at lower temperature to body at higher temperature is a consequences of second law of thermodynamics.

252. $\eta = 1 - \frac{T_2}{T_1} = \frac{W}{Q}$

$1 - \frac{300}{900} = \frac{W}{3\times10^6}$

$\frac{2}{3} = \frac{W}{3\times10^6}$

$W = 2 \times 10^6$ cal $= 2 \times 4.2 \times 10^6$ J $= 8.4 \times 10^6$ J

253. $COP = \frac{T_C}{T_H - T_C}$

254. $(\text{slope})_{\text{adiabatic}} = \gamma\,(\text{slope})_{\text{isothermal}}$

BC, DA $\rightarrow$ adiabatic

AB, CD $\rightarrow$ isothermal

255. Exchange of heat take place in isothermal process

A $\rightarrow$ B	$\Delta U = 0$	(T=constant)
	W = +ve	(V$\uparrow$)
	Q = W =+ve	(Heat absorbed)
C $\rightarrow$ D	$\Delta U = 0$	(T = constant)
	W = –ve	(V$\downarrow$)
	Q = –ve	(Heat rejected)

256. $\eta = 1 - \frac{T_2}{T_1}$

257. $\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1}$

$\Rightarrow \frac{T_2}{T_1} = \frac{Q_2}{Q_1}$

$\Rightarrow \frac{T_2}{400} = \frac{80}{100}$

$\Rightarrow T_2 = 320$ K $= 47$ °C

258. $\frac{\text{Work done}}{\text{Heat supplied}} = \frac{W}{Q_1} = \frac{T_1 - T_2}{T_1}$

$\Rightarrow \frac{800}{Q_1} = \frac{600-300}{600}$

$\Rightarrow Q_1 = 1600$ J

259. $\frac{W}{Q_1} = \frac{T_1 - T_2}{T_1}$

$\frac{W}{6\times10^4} = \frac{500-400}{500} \Rightarrow W = 1.2 \times 10^4$ cal

260. $COP = \frac{T_2}{T_1 - T_2}$

$T_1 = 300$ K , $T_2 = 263$ K

$COP = \frac{263}{300-263} = 7.1$

Exercise-II (Previous Year Questions)

1. $m_1L_v + m_1s\Delta\theta = m_2s\Delta\theta$

$m \times 540 + m \times 1 \times 20 = 20 \times 1 \times 70$

$m_1 \times 560 = 1400$

$m_1 \approx 2.5$ g

$\therefore$ Total mass of water = 20 + 2.5 = 22.5 g

2. $\frac{T_1 - T_2}{t} = K\left(\frac{T_1 + T_2}{2} - T_0\right)$

$T_0 \rightarrow$ surrounding temperature.

3. Mean free path $\lambda_m = \frac{1}{\sqrt{2}\pi d^2 n}$

where d = diameter of molecule

n = molecular density

$\Rightarrow \quad \lambda_m \propto \frac{1}{r^2}$

4. In isothermal process : $P_1V_1 = P_2V_2$

$PV = P_2(2V) \quad \Rightarrow \quad P_2 = \frac{P}{2}$

In adiabatic process

$\left(\frac{P}{2}\right)(2V)^{\gamma} = P'(16V)^{\gamma}$

$P' = \frac{P}{2} \times \frac{1}{32} = \frac{P}{64}$

5. $W_{cyclic} = \frac{1}{2}[3P_0 + P_0] \times V_0 - 3P_0V_0$

$+\frac{1}{2}[3P_0 + P_0]V_0 - P_0V_0 = 0$

6. $\Delta U = \mu C_V \Delta T \quad \& \quad T = \frac{PV}{\mu R}$

so $\Delta T = T_2 - T_1 = \frac{P_2V_2 - P_1V_1}{\mu R}$

so $\Delta U = \frac{\mu R}{\gamma - 1}\left(\frac{P_2V_2 - P_1V_1}{\mu R}\right) = \frac{P_2V_2 - P_1V_1}{\gamma - 1}$

$\Rightarrow \quad \Delta U = \frac{-8 \times 10^3}{2/5} = -20$ kJ

7. From Wein's displacement law

$\lambda_m \propto \frac{1}{T}$

Now from sequence 'VIBGYOR'

$(\lambda_m)_P < (\lambda_m)_R < (\lambda_m)_Q$

So $T_P > T_R > T_Q$

8. For Engine & refrigerators operating between two same temperatures

$\eta = \frac{1}{1+\beta} \Rightarrow \frac{1}{10} = \frac{1}{1+\beta} \Rightarrow \beta = 9$

$\beta = \frac{Q_2}{W}$ (From the principle of refrigerator)

$9 = \frac{Q_2}{10} \Rightarrow Q_2 = 90$ J

9. $\gamma = 1 + \frac{2}{n}$ (Here degree of freedom $\rightarrow$ n)

10. In cyclic process ABCA,

$\Delta U_{cyclic} = 0$

$Q_{cyclic} = W_{cyclic}$

$Q_{AB} + Q_{BC} + Q_{CA}$ = closed loop area.

$400 + 100 + Q_{CA} = \frac{1}{2} \times (2 \times 10^{-3}) \times 4 \times 10^4$

$400 + 100 - Q_{AC} = 40$

$Q_{AC} = 460$ J

11. Rate of heat flow $\propto$ temperature difference between two ends

$\Rightarrow \quad \frac{dQ}{dt} \propto (T_1 - T_2) \qquad T_1 > T_2$

Here temperature difference in both case is same (i.e. 10 °C)

So, rate of heat flow will also be same

So, $\frac{dQ}{dt} = 4$ J/s

12. According to ideal gas equation

$P = \frac{\rho RT}{M_w} \Rightarrow M_w = \frac{\rho RT}{P}$

so $\frac{M_A}{M_B} = \frac{\rho_A}{\rho_B} \cdot \frac{T_A}{T_B} \cdot \frac{P_B}{P_A} = (1.5)\ (1)\left(\frac{1}{2}\right)$

$\Rightarrow \quad \frac{M_A}{M_B} = \frac{3}{4}$

13. Molecular mass $M_w = 4.0$ g

$v_{sound} = \sqrt{\frac{\gamma RT}{M_w}} \quad \Rightarrow \qquad \gamma = \frac{M_w v_s^2}{RT} = 1.6$

So, $C_p = \gamma C_v = 1.6 \times 5.0 = 8.0$ J K^{-1} mol^{-1}

14. Coefficient of performance of refrigerator

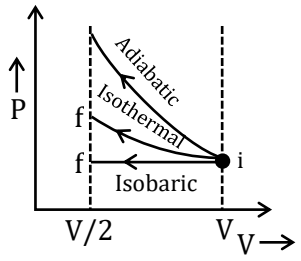
$COP = \frac{T_L}{T_H - T_L}$

Where $T_L \rightarrow$ lower Temperature

and $T_H \rightarrow$ Higher Temperature

So, $5 = \frac{253}{T_H - 253} \Rightarrow T_H = 303.6K \simeq 31°C$

15.



work done on the gas

$W_{isochoric} = 0$

and $W_{adiabatic} > W_{Isothermal} > W_{Isobaric}$

16. $d_f = \dfrac{d_i}{(1 + \gamma\Delta T)}$

fractional change

$= \dfrac{d_i - d_f}{d_i} = 1 - \dfrac{d_f}{d_i}$

$= 1 - (1 + \gamma\Delta T)^{-1}$

$= 1 - (1 - \gamma\Delta T)$

$\because (1+x)^n \; 1 + nx$

$= \gamma\Delta T$

$= 5 \times 10^{-4} \times 40$

$= 0.020$

17. $\beta = \dfrac{Q_2}{W} = \dfrac{T_2}{T_1 - T_2}$ (Where Q_2 is heat removed)

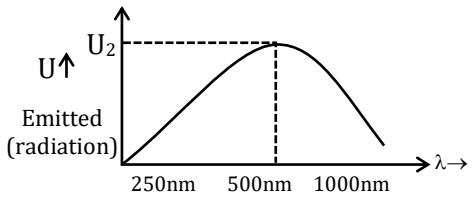
$\Rightarrow \quad \dfrac{600 \times 4.2}{W} = \dfrac{277}{303 - 277}$

$\Rightarrow$ W = 236.5 joule

$\Rightarrow$ Power $= \dfrac{W}{t} = \dfrac{236.5 \text{ joule}}{1 \text{ sec}}$ = 236.5 watt.

18. Maximum amount of emitted radiation corresponding to $\lambda_m = \dfrac{b}{T}$

$\lambda_m = \dfrac{2.88 \times 10^6 \text{ nmK}}{5760\text{K}} = 500$ nm



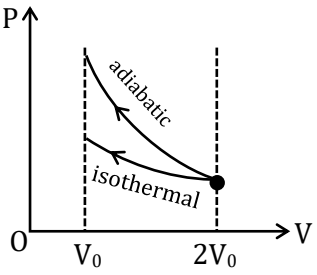
From the graph $U_1 < U_2 > U_3$

19. Change in length for both rods should be same

$\Delta\ell_1 = \Delta\ell_2 \Rightarrow \ell_1\alpha_1\Delta T = \ell_2\alpha_2\Delta T \Rightarrow \ell_1\alpha_1 = \ell_2\alpha_2$

20. $v \propto \sqrt{T} \Rightarrow \dfrac{v}{200} = \sqrt{\dfrac{400}{300}} \Rightarrow v = \dfrac{400}{\sqrt{3}}$ m/s

21.



W_{ext} = negative of area with volume-axis

W(adiabatic) > W(isothermal)

22. $\dfrac{mgh}{4} = mL \Rightarrow h = \dfrac{4L}{g} = \dfrac{4 \times 3.4 \times 10^5}{10} = 136$ km.

23. Let θ be the final common temperature. Further, let s_c and s_h be the average heat capacities of the cold and hot (initially) bodies respectively (where $s_c < s_h$ given)

From, principle of calorimetry,

heat lost = heat gained

$s_h(100°C - \theta) = s_c\theta$

$\therefore \quad \theta = \dfrac{s_h}{(s_h + s_c)} \times 100°C = \dfrac{100°C}{\left(1 + \dfrac{s_c}{s_h}\right)}$

$\because \quad s_c / s_h < 1 \qquad \therefore \; 1 + s_c / s_h < 2$

$\therefore \quad \theta > \dfrac{100°C}{2}$ or $\theta > 50°C$

OR

Body at 100°C has more heat capacity then body at 0°C so final temperature must be greater than 50°C.

24. Newton's laws of cooling

$\dfrac{T_1 - T_2}{t} = k\left(\dfrac{T_1 + T_2}{2} - T\right)$

$\dfrac{3T - 2T}{10} = k\left(\dfrac{5T - 2T}{2}\right) \Rightarrow \dfrac{T}{10} = k\left(\dfrac{3T}{2}\right)$ ...(i)

$\dfrac{2T - T'}{10} = k\left(\dfrac{2T + T'}{2} - T\right)$

$\Rightarrow \dfrac{2T - T'}{10} = k\left(\dfrac{T'}{2}\right)$ ...(ii)

By solving (i) and (ii) $T' = \dfrac{3}{2}T$

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25. PV^x = constant (Polytropic process)
Heat capacity in polytropic process is given by $\left[C = C_V + \frac{R}{1-x}\right]$
Given that PV^3 = constant $\Rightarrow x = 3$ ...(1)
also gas is monoatomic so $C_V = \frac{3}{2}R$ ...(2)
by formula
$$C = \frac{3}{2}R + \frac{R}{1-3} = \frac{3}{2}R - \frac{R}{2} = R$$

26. Heat delivered = Q_1
$$COP(\beta) = \frac{Q_2}{W} = \frac{Q_1 - W}{W} = \frac{Q_1}{W} - 1 = \frac{T_2}{T_1 - T_2}$$
$$\Rightarrow \quad \frac{Q_1}{W} = 1 + \frac{t_2 + 273}{t_1 - t_2} = \frac{t_1 + 273}{t_1 - t_2}$$

27. $\frac{P}{\rho} = \frac{RT}{M_w}$ (Ideal gas equation)
$$\Rightarrow \quad \rho = \frac{PM_w}{RT} = \frac{P \times (mN_A)}{kN_A T} = \frac{Pm}{kT}$$

28. In parallel $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$
$$\frac{K_{eq}(2A)}{\ell} = \frac{K_1 A}{\ell} + \frac{K_2 A}{\ell}$$
$$K_{eq} = \frac{K_1 + K_2}{2}$$

29. Process (1) $\rightarrow$ volume constant $\rightarrow$ Isochroic
Process (2) $\rightarrow V\uparrow \Rightarrow V\downarrow \rightarrow$ adiabatic
Process (3) $\rightarrow$ Temperature constant $\rightarrow$ Isothermal
Process (4) $\rightarrow$ Pressure constant $\rightarrow$ Isobaric

30. $P \propto r^2 T^4 \Rightarrow \frac{P_1}{P_2} = \left(\frac{r_1}{r_2}\right)^2 \left(\frac{T_1}{T_2}\right)^4$
P_2 = 1800 watt

31. $\eta = \frac{1}{1+\beta}$
$$\frac{1}{10} = \frac{1}{1+\beta} \Rightarrow \beta = 9$$
$$\beta = \frac{Q_2}{W} \Rightarrow 9 = \frac{Q_2}{10} \Rightarrow Q_2 = 90 \text{ J}$$

32. $U = \frac{f}{2}nRT$
$$U_{total} = \frac{5}{2}(2)RT + \frac{3}{2}(4)RT$$
U_{total} = 11RT

33. Q = 54 cal = 54 × 4.18 joule = 225.72 joule
$W = P[V_{steam} - V_{water}]$ [For water 0.1 gram=0.1 cc]
= $1.013 \times 10^5[167.1 \times 10^{-6} - 0.1 \times 10^{-6}]$ joule
= $1.013 \times 167 \times 10^{-1}$ = 16.917 joule
By FLOT
$\Rightarrow \quad \Delta U = Q - W = 225.72 - 16.917 = 208.8$ joule

34. $P = \sigma AT^4 \Rightarrow P \propto T^4$
According to Wein's law $T \propto \frac{1}{\lambda_m}$
$$\Rightarrow \quad P \propto \left(\frac{1}{\lambda_m}\right)^4 \Rightarrow \frac{P_2}{P_1} = \left(\frac{\lambda_{m_1}}{\lambda_{m_2}}\right)^4$$
$$\Rightarrow \quad \frac{P_2}{P_1} = \left(\frac{\lambda_0}{\frac{3}{4}\lambda_0}\right)^4 \Rightarrow \frac{nP}{P} = \frac{256}{81} \Rightarrow n = \frac{256}{81}$$

35. $V \propto T \Rightarrow W = P\Delta V = \mu R \Delta T$
$Q = \mu C_P \Delta T$
for mono atomic gas f = 3
$$C_P = \left(\frac{f}{2}R + R\right) = \frac{5}{2}R \text{ So } \frac{W}{Q} = \frac{\mu R \Delta T}{\mu C_P \Delta T} = \frac{2}{5}$$

36. $\eta = \left(\frac{T_1 - T_2}{T_1}\right) \times 100$
where T_1 = 373, T_2 = 273
$$\Rightarrow \quad \eta = \frac{100}{373} \times 100 = 26.8\%$$

37. $v_{es} = v_{rms} \quad \Rightarrow \quad 11.2 \times 10^3 = \sqrt{\frac{3kT}{m}}$
$$\Rightarrow T = \frac{\left(11.2 \times 10^3\right)^2 m}{3k}$$
Putting value of m and k
T = 8.360×10^4 K

38. Adiabatic process
$\Delta Q = 0$

39. KE $\propto$ Temperature
As temperature increases KE also increases

40. At any temperature
$(\Delta\ell)_{Cu} = (\Delta\ell)_{Al}$
$\ell_1\alpha_1\Delta T = \ell_2\alpha_2\Delta T$
$88 \times 1.7 \times 10^{-5} = \ell_2 \times 2.2 \times 10^{-5}$
ℓ_2 = 68 cm

41. $\frac{dQ}{dt} = -(K)A\frac{dT}{dx}$

$\frac{J}{s} = (K)m^2\frac{kelvin}{m}$

(K) = watt m^{-1} K^{-1}

42. $\frac{T_1 - T_2}{t} = K\left(\frac{T_1 + T_2}{2} - Ts\right)$

$\frac{80-70}{12} = K\left(\frac{80+70}{2} - 25\right)$ ...(1)

$\frac{70-60}{t} = K\left(\frac{70+60}{2} - 25\right)$ ...(2)

on solving : t = 15 min

43. $KA\frac{[0-(-26)]}{x}dt = A(dx)\rho L$

$\Rightarrow \frac{dx}{dt} = \frac{26K}{\rho L x}$

44. Hydrogen $\rightarrow$ 7/5 (diatomic),
Helium $\rightarrow$ 5/3 (monoatomic), X $\rightarrow$ 9/7

45. $Q = \Delta U + W \Rightarrow mL = \Delta U + P(V_2 - V_1)$
$\Rightarrow 1(2256) = \Delta U + 1 \times 10^5 (1670 \times 10^{-6})$
$\Rightarrow \Delta U = 2089$ J

46. Average thermal energy = $\frac{3}{2}K_B T$

where 3 is translational degree of freedom
For monoatomic gas total degree of freedom f = 3 (translational degree of freedom)

47. Mean free path for a gas sample

$\lambda_m = \frac{1}{\sqrt{2}\pi d^2 n}$

where d is diameter of a gas molecule and n is molecular density

48. Heat supplied $\Delta Q = MS\Delta T$
For same material 'S' same.

$\Delta Q \propto M$ and $M = \frac{4}{3}\pi r^3\rho \Rightarrow \Delta Q \propto r^3$

$\frac{\Delta Q_1}{\Delta Q_2} = \left(\frac{r_1}{r_2}\right)^3 = \left(\frac{1.5}{1}\right)^3 = \frac{27}{8}$

49. For an ideal gas sample

$\frac{P}{\rho} = \frac{RT}{M_w}$

$\rho = \frac{PM_w}{RT} = \frac{249\times10^3\times2\times10^{-3}}{8.314\times300} = 0.199$

$\rho \simeq 0.2$ kg/m^3

50. Free expansion i.e. expansion against vacuum is adiabatic in nature for all type of gases. It should be noted that final temperature is equal to initial temperature for ideal gases.

51. Efficiency of cannot engine

$\eta = 1 - \frac{T_2}{T_1}$

T_1 = temperature of source
T_2 = temperature of sink

52. $\lambda = \frac{1}{\sqrt{2}\pi d^2 n} \propto \frac{1}{d^2}$

λ = mean free path
d = effective diameter of molecule
n = number density of molecules

53. PV = nRT

$\Rightarrow P = \frac{1}{V}.\left(\frac{m}{M_0}\right)RT = \left(\frac{m}{V}\right)\left(\frac{RT}{M_0}\right)$

$\Rightarrow P = \frac{\rho RT}{M_0}$

$\rho = \frac{m}{v}$ = mass density

& M_0 = molar mass

54. P = constant $\Rightarrow$ Isobaric process

55. By using Wien's law $\lambda_m T$ = constant,

we have $T \propto \frac{1}{\lambda_m}$

As $\lambda_B < \lambda_Y < \lambda_R$ (VIBGYOR)
$\Rightarrow T_A > T_C > T_B$

56. According to Newton's law of cooling

$\frac{T_1 - T_2}{t} = K\left[\frac{T_1 + T_2}{2} - T_0\right]$

For 1st cup of coffee,

$\Rightarrow \frac{90-80}{t} = K\left[\frac{90+80}{2} - 20\right]$ ...(1)

For 2nd cup of coffee,

$\Rightarrow \frac{80-60}{t'} = K\left[\frac{80+60}{2} - 20\right]$ ...(2)

Divide (1) by (2)

$\frac{t'}{2t} = \frac{65}{50} \Rightarrow t' = \frac{13}{5}t$

57. Root mean square speed of gas molecules

$$\upsilon_{rms} = \sqrt{\frac{3RT}{M}}$$

Pressure exerted by ideal Gas

$$P = \frac{1}{3}\rho\upsilon_{rms}^2$$

$$P = \frac{1}{3}mn\upsilon^2$$

$\rho = mn$, $\upsilon_{rms}^2 = \vec{\upsilon}^2$

Average kinetic energy of a molecule

$$KE = \frac{3}{2}KT$$

Total internal energy of 1 mole of a diatomic gas

$$U = \frac{5}{2}\mu RT$$

or $U = \frac{5}{2}RT$ (For 1 mole diatomic gas)

63. $v_{rms} \propto \sqrt{T}$

$$\frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}}$$

= let initial speed is v

As speed is increased by 3 times so final speed become 4v

$$\Rightarrow \frac{v}{4v} = \sqrt{\frac{223}{T}}$$

T = 3568 K

So temp. in °C = 3568 – 273 = 3295°C

64. Efficiency of carnot engine

$$\%\eta = \left(1 - \frac{T_{sink}}{T_{source}}\right) \times 100$$

T_{source} = 327°C = 600 K

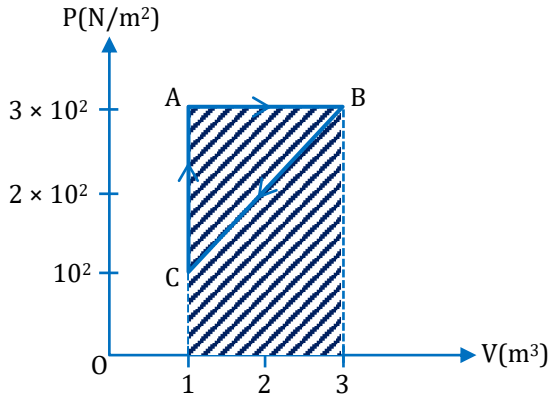
$$50 = \left(1 - \frac{T_{sink}}{600}\right) \times 100$$

$$\frac{1}{2} = 1 - \frac{T_{sink}}{600}$$

$$T_{Sink} = 300K$$

So temp. of sink is °C = 300 – 2763 = 27°C

65.



AB is isobaric process

$W_{AB} = P(V_2 - V_1)$

$W_{AB} = 3 \times 10^2 (3-1)$

$W_{AB} = 3 \times 100 \times 2$

W_{AB} = 600 J

66. $$P_{mix} = \frac{(\mu_1 + \mu_2)RT_{mix}}{V_{mix}}$$

$$= \frac{(0.2+0.3)\times 8.3\times 200}{200\times 10^{-6}}$$

$$= \frac{0.5\times 8.3\times 200}{200\times 10^{-6}}$$

$P_{mix} = 4.15 \times 10^6\ P_a$

67. For path bc volume is constant

so work done is zero

$\Rightarrow W = 0$

68. PV = nRT

$$\Rightarrow T = \frac{P}{nR}.V$$

comparing with y = mx

$$m = \text{slope} = \frac{P}{nR} \propto P$$

$\Rightarrow P_1 > P_2 > P_3$

69. Houses made of concrete roofs overlaid with foam keep the room colder during summer.

70. Water is a good coolant because of it's high specific heat capacity.

71. Equilibrium state of a thermodynamic system is described by thermodynamic variables such as pressure, temperature, volume etc.

72. Let vibrational degrees of freedom be f_{vib}
Translatory degrees of freedom = 3
Rotatory degrees of freedom = 3
Total degree of freedom = f + 6

$$\gamma = 1 + \frac{2}{f_{total}} \Rightarrow \gamma = 1 + \frac{2}{f_{vib} + 6}$$

$$\Rightarrow f_{vib} = \frac{2(4-3\gamma)}{\gamma - 1}$$

$$\Rightarrow \text{Vibrational modes} = \frac{(4-3\gamma)}{\gamma - 1}$$

Exercise-III (Analytical Questions)

1. Celsius scale ranges from 0°C to 100°C Fahrenheit scale ranges from 32°F to 212°F Kelvin scale ranges from 273.15 K to 373.15 K unknown scale ranges from x to y.

2. When a bimetallic strip is heated it bends in the from of an arc with material having greater temperature expansion co-efficient going on outer side. It is used in thermostate.

3. Since the coefficient of linear expansion for brass is greater than steel so system should be cooled to take out the disc from the plate.

4. When heat is supplied to the solid its temperature may increase (phase heating) or it may remains constant (state change).

5. A good cooking utensil should have low specific heat, low thermal capacity and high coefficient of thermal conductivity.

6. In BC temperature remains constant
ice $\rightarrow$ water
As density of ice is less than water so after melting volume decreases.

7. During melting temperature of substance remains constant.

8. Heat $\rightarrow$ Energy in transit
Internal energy $\rightarrow$ It is sum of energy due to attractive force and due to random motion.
Temperature $\rightarrow$ Degree of hotness or coldness of a substance.
Specific heat $\rightarrow$ Amount of heat required to raise temperature of unit mass by 1°C.

9. If t = 0
$\Rightarrow a(\uparrow), r\downarrow, e(\uparrow)$ {Here, a + r = 1 and $a \propto e$}
$\Rightarrow a(\downarrow), r\uparrow, e(\downarrow)$

11. $P = \frac{dQ}{dt} = \sigma A T^4 \propto r^2$

$$R = \frac{d\theta}{dt} = \frac{\sigma A T^4}{ms} \propto \frac{A}{v} \propto \frac{r^2}{r^3} \propto \frac{1}{r}$$

13. $\lambda_m \propto \frac{1}{T}$

(As temperature increases λ_m decreases)

14. $V_{RMS} \propto \sqrt{\frac{T}{M_w}}$

V_{RMS} also depends on nature of gas.

15. $V \propto \sqrt{\frac{T}{M_w}}$

$\Rightarrow T \rightarrow \text{const.} \Rightarrow V \propto \frac{1}{\sqrt{M_w}}$

$\Rightarrow (M_w)_{N_2} > (M_w)_{He}$

$\Rightarrow V_{He} > V_{N_2}$

and $KE = \frac{f}{2} kT$; $KE \propto f$

for N_2 gas ; f↑ KE↑

16. As $\frac{p}{\rho} = \frac{RT}{M_w} \Rightarrow p = \left(\frac{M}{V}\right)\left(\frac{RT}{M_w}\right)$

at constant temperature $\Rightarrow p \propto \frac{1}{V}$

V↑ and P↓

18. Remember

$$V_s = \sqrt{\frac{\gamma RT}{M_w}} ; V_{Avg} = \sqrt{\frac{8}{\pi}\frac{RT}{M_w}}$$

$$V_{RMS} = \sqrt{\frac{3RT}{M_w}} ; V_{mp} = \sqrt{\frac{2RT}{M_w}}$$

19. (A) Adiabatic bulk modulus

$$= -\left(\frac{dP}{dV}\right) \times V = -\left(-\gamma \frac{P}{V}\right) \times V = +\gamma P$$

(B) Slope of isothermal curve PV = K

$PdV + VdP = 0$

$$(\text{slope})_{IT} = \frac{dP}{dV} = -\frac{P}{V}$$

(C) $C_V = \frac{f}{2}R,\ C_P = \left(\frac{f}{2}+1\right)r,\ \gamma = \frac{C_P}{C_V}$

Therefore $\gamma = 1+\frac{2}{f}$

So, $f = \frac{2}{\gamma-1}$

(D) $C_P = \frac{\gamma R}{\gamma-1}$

$\frac{C_P}{R} = \frac{\gamma}{\gamma-1}$

20. $Q = W + \Delta U$

$Q = 0$

$\Delta U = -W$

21. Process ab $\rightarrow$ (Isothermal process)

$T = \text{constant} \Rightarrow U = \text{constant} \Rightarrow \Delta U = 0$

and ρ increasing

Process bc $\Rightarrow$ ρ = constant

then isochoric process

Process cd $\Rightarrow$ (Isothermal process)

$T = \text{constant} \Rightarrow U = \text{constant} \Rightarrow \Delta U = 0$

and ρ decreasing.

Process da $\Rightarrow$ (Isothermal process)

ρ = constant then isochoric process

22. (i) $dU = nC_v dT$ is valid for all types of process.

(ii) Temperature of the system can change in adiabatic process, polytropic process.

(iii) $dQ = dU + dW$ (First law) is valid for all types of process.

(iv) The process in which heat exchange does not occurs is adiabatic process and free expansion.

23. $C_P - C_V = R$

$C_P \Rightarrow Q = W + \Delta U$

$C_V \Rightarrow Q = \Delta U$

(in isobaric process)

24. In a cyclic process $dU = 0 \Rightarrow U \rightarrow$ constant

In an isothermal process $dU = 0$

$\Rightarrow U \rightarrow$ constant

If $\Delta Q = \Delta W$ means $dU = 0 \Rightarrow U \rightarrow$ constant

25. $Q = 0$ for adiabatic

(i) $nCdT = 0 \Rightarrow C = 0$

(ii) $C = \frac{\Delta Q}{mdT}$ for IT ; $dT = 0$

$c = \infty$

(iii) $C_V = \frac{f}{2}R;\quad C_P = \left(\frac{f+2}{2}\right)R$

f dependents on nature of gas

(iv) $Q = mS\Delta T$ ...(1)

$Q = nC\Delta T$ ...(2)

$n = \frac{m}{M_w}$

$mS = nC$

$mS = \frac{m}{M_w}C \Rightarrow C = M_wS$

26. (A) Adiabatic expansion – Temperature decrease so decrease in internal energy

(B) Isobaric expansion – Volume increases and hence temperature increases since P $\rightarrow$ constant ($V \propto T$)

So internal energy increases

(C) Isothermal expansion – T $\rightarrow$ constant

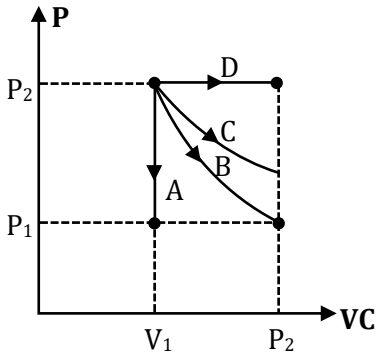
So, constant internal energy

(D) Isochoric process – V $\rightarrow$ constant

So, No work done

As $\omega = P\Delta V = 0$

27.



For (i) : $W = 0$ & $\Delta T < 0 \Rightarrow Q < 0$ & $\Delta U < 0$

For (ii) : $Q = 0, W > 0 \Rightarrow \Delta U < 0$

For (iii) : $\Delta T = 0, \Rightarrow \Delta U = 0 \Rightarrow Q = W > 0$

For (iv) : $V \propto T \Rightarrow W > 0$ & $\Delta U > 0 \Rightarrow Q > 0$

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Oscillation

SOLUTIONS

Exercise-I (Conceptual Questions)

1. Force constant
$K = m\omega^2$
$K = m(2\pi n)^2$
$K = m4\pi^2n^2$

2. $\frac{d^2x}{dt^2} = -\left(\frac{k}{m}\right)x$

3. $\frac{d^2x}{dt^2} = -kx$

compare this given equation with $\frac{d^2x}{dt^2} = -\omega^2 x$

$\omega^2 = k$ $\omega = \sqrt{k}$

$\frac{2\pi}{T} = \sqrt{k}$ $T = \frac{2\pi}{\sqrt{k}}$

4. Maximum value of tan is ∞

5. $\omega = 20$

$\frac{2\pi}{T} = 20$ $T = \frac{2\pi}{20} = \frac{\pi}{10}s$

6. At $x = +A$, $\theta = \frac{\pi}{2}$

7. $\omega = 3000 \Rightarrow n = \frac{\omega}{2\pi}$

8. acceleration is always towards equilibrium position (mean position) where as displacement is away from mean position.

9. Four times of amplitude

10. At extreme position displacement is maximum and phase $\theta = \frac{\pi}{2}$

11. $V_{max} = A\omega = 10(20) = 200$ m/s

12. Phase constant $= \frac{\pi}{3} = 60°$

13. t = 0 (x = 0) ——— (x = A) $t = \frac{T}{4}\left(\frac{\pi}{2}\text{Angle}\right)$

$\frac{\pi}{6}$ phase will develop when particle move from mean position towards amplitude position. It mean particle move in +x direction
$x = a\sin\omega t$
$x = a\sin\theta$

$x = a\sin\frac{\pi}{6} = a\sin30° = \frac{a}{2}$

14. (1) → ——— ← (2)
x = 0

Particle (1) start its motion from mean position so phase angle is 0.
Particle (2) move in –ve x direction at mean position the phase angle becomes π so phase difference between the two particle is $\pi - 0 = \pi$

15. $x = A\sin\omega t$

$\frac{\sqrt{3}}{2}A = A\sin\left(\frac{2\pi\times3}{T}\right)$

$\frac{\sqrt{3}}{2} = \sin\left(\frac{6\pi}{T}\right)$

$\frac{\pi}{3} = \frac{6\pi}{T}$, T = 18 seconds

16. Time required between mean to half of amplitude $\left(\frac{a}{2}\right)$ position in t_1 is $= \frac{T}{12}$

Time required between $\left(\frac{a}{2}\right)$ to a position is

$t_2 = \frac{T}{6}$

$\frac{t_1}{t_2} = \frac{\frac{T}{12}}{\frac{T}{6}} = \frac{1}{2} = 1:2$

17. t = 0 (Mean position) ——— $\frac{T}{4}$

Maximum displacement (amplitude position)

18. Required time from mean to half of amplitude position is $= \frac{T}{12} = \frac{6}{12} = \frac{1}{2}$

21. $v_0 = A\omega$, $x = \frac{A}{2}$

$v = \omega\sqrt{A^2 - x^2} \Rightarrow v = \omega\sqrt{A^2 - \frac{A^2}{4}}$

$v = \omega\sqrt{\frac{3}{4}A^2} \Rightarrow \frac{\sqrt{3}}{2}A\omega = \frac{\sqrt{3}}{2}v_0$

22. $v = \frac{\sqrt{3}}{2}$ (Velocity at mean position)

$v = \frac{\sqrt{3}}{2} a\omega$

$\Rightarrow \quad v = \omega\sqrt{a^2 - x^2}$

$\Rightarrow \quad \frac{\sqrt{3}}{2} a\omega = \omega\sqrt{a^2 - x^2}$

$\Rightarrow \quad x^2 = a^2 - \frac{3}{4}a^2$

$\Rightarrow \quad x^2 = \frac{a^2}{4} \Rightarrow x = \frac{a}{2}$

23. $a = \omega^2 x$ (magnitude)

$20 = \omega^2(5) \quad \Rightarrow \quad \omega^2 = 4 \Rightarrow \quad \omega = 2\,\frac{\text{Rad}}{\text{s}}$

24. $V = \omega\sqrt{A^2 - x^2}$

$= \frac{2\pi}{T}\sqrt{5^2 - 3^2} = \frac{2\pi}{\pi}\sqrt{16} = 8$ cm/sec.

OR

$x = A \sin \omega t \Rightarrow 3 = 5 \sin \omega t \Rightarrow \sin \omega t = \frac{3}{5}$

$\therefore \quad \cos \omega t = \frac{4}{5} \quad \therefore V = A\omega \cos \omega t$

so $\quad V = 5 \times \frac{2\pi}{\pi} \times \frac{4}{5}$

25. $V_{max} = A\omega = 100 \qquad ...(1)$

$a_{max} = A\omega^2 = 157 \qquad ...(2)$

Divided (1) by (2)

$\frac{1}{\omega} = \frac{100}{157}$

$\frac{T}{2\pi} = \frac{100}{157} \qquad T = \frac{2 \times 3.14 \times 100}{157} = 4$

26. $x = 1$ cm $\qquad a = 1$ cm/s^2

$V = 1$ cm/s

$a = \omega^2 x \Rightarrow \omega^2 = 1 \quad \Rightarrow \omega = 1 \qquad T = 2\pi$

27. Length of SHM path (2A) is 4cm

$2A = 4 \qquad$ So $\qquad$ amp $A = 2$ cm

$V = A\omega \quad \Rightarrow \quad 12 = 2(2\pi n)$

$n = \frac{12}{4\pi} \Rightarrow \frac{3}{\pi}$ Hz

28. The velocity at mean position is non zero.

29. $a = -\omega^2 x$,

So graph is straight line with negative slope

30. $V_{max} = A\omega \Rightarrow A\left(\frac{2\pi}{T}\right) = 0.40\left(\frac{2\pi}{0.05}\right)$

$\Rightarrow 16\pi$ m/s

31. $V = \omega\sqrt{A^2 - x^2} \qquad V_{max} = 100 = A\omega$

$50 = 10\sqrt{100 - x^2} \quad 100 = 10\omega$

$25 = 100 - x^2 \qquad \omega = 10$

$x^2 = 75 = 5$ cm

OR

$100 = 50 \cos \omega t$

$\cos \omega t = \frac{1}{2} \qquad \therefore \qquad \sin \omega t = \frac{\sqrt{3}}{2}$

so $\quad x = A \sin \omega t$

$x = 5\sqrt{3}$ cm

32. Time period T = 0.04 sec

So frequency $n = \frac{1}{T} = \frac{1}{0.04} = 25$ Hz

33. $V_{max} \propto A$

34. At extreme position V is zero. But acceleration (a) is maximum

At mean position V is maximum but acceleration (a) is zero.

35. The force and acceleration of particle is zero at mean position.

36. In SHM the velocity is maximum at central (mean) position.

37. $x = 5.0 \cos(2\pi t + \pi) = -5.0 \cos 2\pi t$

Velocity = $\frac{dx}{dt} = 10\pi \sin 2\pi t$

Acceleration = $\frac{d^2x}{dt^2} = 20\pi^2 \cos 2\pi t$

$\therefore \quad$ At t = 1.5 sec,

$x = -5.0 \cos 3\pi = -5.0\,(-1) = 5$

Velocity = $\frac{dx}{dt} = 10\pi(\text{zero}) = 0$

Acceleration = $\frac{d^2x}{dt^2} = 20\pi^2(-1) = -20\pi^2$

38. $Y = 3 \sin\,(100\,t + \frac{\pi}{6})$

$Y = A \sin\,(\omega\,t + \phi)$

$A = 3, \quad \omega = 100$

maximum velocity

$V_{max} = A\omega = 3 \times 100 = 300$

39. $v_{max} = A\omega \Rightarrow 4.4 = 7 \times 10^{-3}\omega \Rightarrow \omega = \frac{4400}{7}$

So $T = \frac{14\pi}{4400} = 0.01$ second

40. In one complete oscillation the displacement is zero, so average velocity is also zero.

41. If particle starts oscillation from extreme position then $x = A\cos\omega t$

$\frac{A}{2} = A\cos\omega t \Rightarrow \cos\omega t = \cos\frac{\pi}{3} \Rightarrow \omega t = \frac{\pi}{3}$

$\frac{2\pi}{T}t = \frac{\pi}{3} \Rightarrow t = \frac{T}{6}$

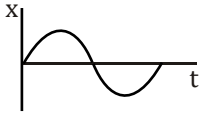
42. In one complete oscillation the displacement is zero so total work done is also zero.

43. $a = -\omega^2 x \Rightarrow \frac{\text{acceleration}}{\text{displacement}} = \text{constant}$

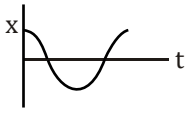
44. $Y = 2\cos(\pi t + \pi/6)$

here $\omega = \pi \Rightarrow \frac{2\pi}{T} = \pi \Rightarrow T = 2$ second

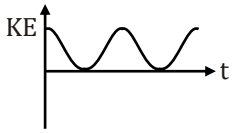
45. $x = A\sin\omega t$



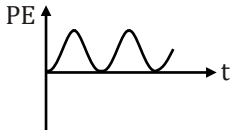
46. $V = \frac{dx}{dt} = A\omega\cos\omega t$



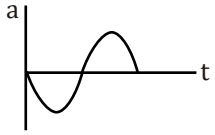
47. K.E. $= \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2A^2\cos^2\omega t$



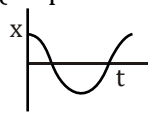
48. PE $= \frac{1}{2}Kx^2 = \frac{1}{2}KA^2\sin^2\omega t$



49. $a = -\omega^2 x \Rightarrow a = -\omega^2(A\sin\omega t)$

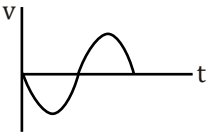


50. $x = A\cos\omega t$ (Displacement – time)



51. $\frac{dx}{dt} = -A\omega\sin\omega t$

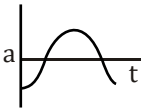
$V = -A\omega\sin\omega t$



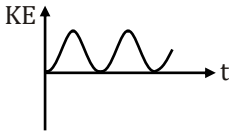
52. $V = -A\omega\sin\omega t$

$\frac{dv}{dt} = -A\omega^2\cos\omega t$

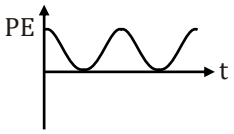
$a = -A\omega^2\cos\omega t$



53. $KE = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2A^2\sin^2\omega t$



54. $PE = \frac{1}{2}kx^2 = \frac{1}{2}kA^2\cos^2\omega t$



55. $E = \frac{1}{2}kA^2$

56. T.E. $= \frac{1}{2}m\omega^2A^2 = 2\pi^2m\frac{A^2}{T^2}$

57. $E = \frac{1}{2}m\omega^2A^2$

K.E. $= \frac{1}{2}m\omega^2(A^2 - x^2)$ $\qquad x = \frac{A}{2}$

$= \frac{1}{2}m\omega^2\left(A^2 - \frac{A^2}{4}\right)$

$(K.E.) = \left(\frac{1}{2}m\omega^2A^2\right) = \frac{3E}{4}$

OR

$\sin\omega t = \frac{1}{2} \quad \therefore \quad \cos\omega t = \frac{\sqrt{3}}{2}$

$KE = E_0\cos^2\omega t$

58. $E = \frac{1}{2}m\omega^2A^2$ $\qquad x = \frac{A}{2}$

P.E. $= \frac{1}{2}m\omega^2x^2 = \frac{1}{2}m\omega^2\frac{A^2}{4} = \frac{E}{4}$

OR

$\because \quad \sin\omega t = \frac{1}{2} \quad \therefore \quad U = E_0\sin^2\omega t$

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59. Length of SHM is 8 cm
So 2A = 8
Amp of SHM A = 4 cm
K.E. = P.E.

$\frac{1}{2}m^2(A^2 - x^2) = \frac{1}{2}m\omega^2x^2$

$A^2 - x^2 = x^2$

$2x^2 = A^2 \quad \Rightarrow \quad x = \frac{A}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$ cm

60. $(PE.)_{AV} = \frac{1}{2}(P.E.)_{max} = \frac{1}{2}\left(\frac{1}{2}Ka^2\right) = \frac{1}{4}Ka^2.$

61. T.E. $= \frac{1}{2}m\omega^2A^2.$

Does not depend on time & displacement.
Ans. is (1) option

62. $(KE)_{max} = \frac{1}{2}KA^2$

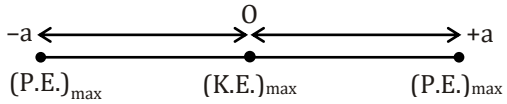
$5 = \frac{1}{2}K(10 \times 10^{-2})^2$

$K = \frac{10}{(10\times10^{-2})^2} = 1000$ N/m

63. $F = -\frac{dU}{dx} = \frac{-d}{dx}(8x^2) \quad \Rightarrow -16x$

(F) = −16(−2) = 32 Dyne.

64.



Mean position

displacement is ±a

65. $U = E_0\sin^2\omega t = \frac{E_0}{4}$

$\sin\omega t = \frac{1}{2}$

$x = a\sin\omega t = \frac{a}{2}$

66. $\frac{\frac{1}{2}KA^2}{\frac{1}{2}K(A^2 - x^2)} = \frac{A^2}{A^2 - \frac{A^2}{4}} = \frac{4}{3}$

Ans. is (3) option

67. $\frac{1}{2}Kx^2 = \frac{1}{2}K(A^2 - x^2)$

$\Rightarrow \quad x^2 = \frac{A^2}{2} \quad \Rightarrow \quad \frac{x}{A} = \frac{1}{\sqrt{2}}$

68. In this conservative field total energy remains constant.

69. <E> = <V>

70. $U = \frac{1}{2}Kx^2 \quad \Rightarrow \quad U \propto x^2$

or $\frac{U'}{U} = \left(\frac{x'}{x}\right)^2 = \left(\frac{3}{1}\right)^2 = 9 \quad \Rightarrow \quad U' = 9U$

71. $E = \frac{1}{2}Kx^2$

$\frac{E'}{E} = \frac{(x/2)}{x} = \frac{1}{4} \quad \Rightarrow \quad E' = \frac{E}{4}$

72. $f = \frac{1}{2\pi}\sqrt{\frac{K}{m}} \propto \sqrt{K}$

Both springs are in parallel.
So $f' \propto \sqrt{2K} \propto \sqrt{2}f$

73. $f \propto \sqrt{K'}$,
both springs are in parallel so $K' = K_1 + K_2$

74. $\upsilon = \frac{1}{2\pi}\sqrt{\frac{K}{m}}$

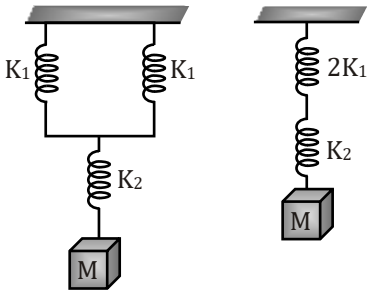
$\upsilon' = \frac{1}{2\pi}\sqrt{\frac{K}{4m}} = \frac{1}{2}\left(\frac{1}{2\pi}\sqrt{\frac{K}{m}}\right) = \frac{\upsilon}{2}$

75. In series combination

$\frac{1}{K_S} = \frac{1}{K_1} + \frac{1}{K_2} \Rightarrow K_S = \frac{K_1K_2}{K_1 + K_2}$

76. $T = 2\pi\sqrt{\frac{M}{K_1 + K_2}}$

77.



$\frac{1}{K_{eq}} = \frac{1}{2K_1} + \frac{1}{K_2}$

$K_{eq} = \left[\frac{1}{2K_1} + \frac{1}{K_2}\right]^{-1}$

78. $T = 2\pi\sqrt{\frac{m}{K}}$

$2 = 2\pi\sqrt{\frac{m}{K}}$

$2+1 = 2\pi\sqrt{\frac{m+2}{K}} \quad \Rightarrow \quad \frac{4}{9} = \frac{m}{m+2}$

$4m + 8 = 9m \qquad 5m = 8 \qquad m = 1.6$ kg

79. Time period of spring pendulum does not depends on 'g'.

80. Time period will remain same

81. In form of P.E.

82. Spring watch (independent of g)

83. $\nu \propto \sqrt{K}$

Now K' = 2K So $\frac{\nu'}{\nu} = \frac{\sqrt{2K}}{\sqrt{K}} = \sqrt{2} \Rightarrow \nu' = \sqrt{2}\nu$

84. $U_1 = U_2$

$\frac{1}{2}K_1x_1^2 = \frac{1}{2}K_2x_2^2 \quad \Rightarrow \quad \frac{F_1^2}{2K_1} = \frac{F_2^2}{2K_2}$

$\frac{F_1^2}{F_2^2} = \frac{K_1}{K_2}$ or $\frac{F_1}{F_2} = \sqrt{\frac{K_1}{K_2}}$

85. Length of one part is 3 times of other

$\ell_1 = 3\ell_2$

$\frac{K_1}{K} = \frac{\ell}{\ell_1} = \frac{\ell_1+\ell_2}{\ell_1} = \frac{4}{3}$

$K_1 = \frac{4}{3}K$

86. $K \propto \frac{1}{\ell}$

$\because \quad \ell' = \frac{\ell}{n} \qquad \therefore \quad K' = nK$

87. $T \propto \sqrt{m}$ for spring oscillation

$\frac{T'}{T} = \sqrt{\frac{m'}{m}} = \sqrt{\frac{40}{10}} = 2 \quad \Rightarrow \quad T' = 2T$

88. T of simple pendulum does not depend on mass of bob

89. $V_{max} = A\omega$

90. $T = 2\pi\sqrt{\frac{\ell}{g}} = 2\pi\sqrt{\frac{39.2}{9.8\times\pi^2}} = 2\pi\sqrt{\frac{4}{\pi^2}} = 2\pi\times\frac{2}{\pi}$

$\Rightarrow 4$

91. $T = 2\pi\sqrt{\frac{\ell}{g}} \propto \sqrt{\ell}$

$T' \propto \sqrt{4\ell} \propto 2\sqrt{\ell} \propto 2T$ (Double)

92. T = 2 sec

One extreme to other extreme is

$= \frac{T}{2} = \frac{2}{2} = 1$ sec

93. $T = 2\pi\sqrt{\frac{\ell}{g}}$, T = 2 seconds so $\ell \approx 1$m

94. In artificial satellite time period becomes ∞.

95. Due to frictional forces the mechanical energy of oscillator converts in heat energy and finally oscillation stops.

96. $T = 2\pi\sqrt{\frac{\ell}{g}\left(\frac{R_E}{\ell+R_E}\right)}$, if $\ell = R_e$ then T = 59.8

97. If ℓ is large $\ell + R_E \approx \ell$

then t = 84.6 min $\approx 1\frac{1}{2}$ hour

98. W = 0, as tension force is always perpendicular to displacement.

99. When lift is at rest T = $\sqrt{\frac{\ell}{g}}\,2\pi$

When lift is ascending

$T = 2\pi\sqrt{\frac{\ell}{g+g/3}} = 2\pi\sqrt{\frac{3\ell}{4g}} = \frac{\sqrt{3}}{2}\left(2\pi\sqrt{\frac{\ell}{g}}\right) = \frac{\sqrt{3}}{2}T$

100. $T = 2\pi\sqrt{\frac{\ell}{g}} \qquad \ell\downarrow \therefore T\downarrow$

101. If velocity is same then acceleration will be zero and there will be no variation in time period.

Exercise-II (Previous Year Questions)

1. Displacement, x = Acos(ωt)

Velocity, $v = \frac{dx}{dt} = -A\omega\sin(\omega t)$

Acceleration, $a = \frac{dv}{dt} = -A\omega^2\cos(\omega t)$

2. $Y = y_1 + y_2$

$= a\sin\omega t + b\cos\omega t$

$$= \sqrt{a^2+b^2}\left[\frac{a}{\sqrt{a^2+b^2}}\sin\omega t + \frac{b}{\sqrt{a^2+b^2}}\cos\omega t\right]$$

$$= \sqrt{a^2+b^2}\left[\cos\phi\sin\omega t + \sin\phi\cos\omega t\right]$$

$$\left[\text{Let} \quad \cos\phi = \frac{a}{\sqrt{a^2+b^2}}, \quad \sin\phi = \frac{b}{\sqrt{a^2+b^2}}\right]$$

$$Y = \sqrt{a^2+b^2}\sin(\omega t+\phi)$$

3. $v = \omega\sqrt{A^2-x^2}$

$v_1 = \omega\sqrt{A^2-x_1^2} \Rightarrow v_1^2 = \omega^2(A^2 - x_1^2)$ ...(1)

$v_2 = \omega\sqrt{A^2-x_2^2} \Rightarrow v_2^2 = \omega^2[A^2 - x_2^2]$

$v_1^2 - v_2^2 = \omega^2[A^2 - x_1^2 - A^2 + x_2^2]$

$$\omega^2 = \frac{v_1^2-v_2^2}{x_2^2-x_1^2}$$

$$\omega = \sqrt{\frac{v_1^2-v_2^2}{x_2^2-x_1^2}} \Rightarrow \frac{2\pi}{T} = \sqrt{\frac{v_1^2-v_2^2}{x_2^2-x_1^2}}$$

$$T = 2\pi\sqrt{\frac{x_2^2-x_1^2}{v_1^2-v_2^2}}$$

4. **Case-A :** When stretched by same amount

$$\frac{W_P}{W_Q} = \frac{\frac{1}{2}K_P.x^2}{\frac{1}{2}K_Q.x^2} = \frac{K_P}{K_Q} > 1$$

$\therefore$ $W_P > W_Q$

Case-B : When stretched by same force

$$\frac{W_P}{W_Q} = \frac{F^2/2K_P}{F^2/2K_Q} = \frac{K_Q}{K_P} < 1$$

$\therefore$ $W_P < W_Q$

5. $\alpha = \omega^2 A$

$\beta = \omega A$

$$\therefore \quad \frac{\alpha}{\beta} = \omega \Rightarrow \frac{\alpha}{\beta} = \frac{2\pi}{T} \Rightarrow T = \frac{2\pi\beta}{\alpha}$$

6. $T = 2\pi\sqrt{\frac{m}{k}}$

$3 = 2\pi\sqrt{\frac{m}{k}}$ ...(1)

$5 = 2\pi\sqrt{\frac{m+1}{k}}$ ...(2)

$$\frac{(1)^2}{(2)^2} \Rightarrow \frac{9}{25} = \frac{m}{m+1} \Rightarrow m = \frac{9}{16}$$

7. Length of the spring segments $= \frac{\ell}{6}, \frac{\ell}{3}, \frac{\ell}{2}$

As we know $K \propto \frac{1}{\ell}$

so spring constants for spring segments will be

$K_1 = 6K, K_2 = 3K, K_3 = 2K$

so in parallel combination

$K'' = K_1 + K_2 + K_3 = 11K$

in series combination

$K' = K$ (As it will become original spring)

so $K' : K'' = 1 : 11$

8. Amplitude A = 3 cm

When particle is at x = 2 cm,

its |velocity| = |acceleration|

i.e., $\omega\sqrt{A^2-x^2} = \omega^2 x \Rightarrow \omega = \frac{\sqrt{A^2-x^2}}{x}$

$$T = \frac{2\pi}{\omega} = 2\pi\left(\frac{2}{\sqrt{5}}\right) = \frac{4\pi}{\sqrt{5}}$$

9. $|a| = \omega^2 x$

$\Rightarrow 20 = \omega^2 5$

$\Rightarrow \omega^2 = 4$

$\Rightarrow \omega = 2 \quad \Rightarrow \quad \frac{2\pi}{T} = 2$

$\Rightarrow T = \pi$ s

10. $y = A_0 + A\sin\omega t + B\cos\omega t$

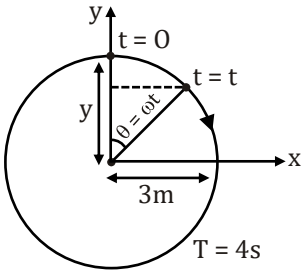
$y = A_0 + \sqrt{A^2+B^2}\sin(\omega t + \phi)$

A_0 is mean position, and $\sqrt{A^2+B^2}$ is amplitude

11. Displacement = zero in one complete oscillation

$\Rightarrow$ Average velocity $= \frac{\text{Displacement}}{T} = 0$

12.



$\omega = \frac{2\pi}{4} = \frac{\pi}{2}$

For y-projection,

$y = A\cos\omega t$

$\Rightarrow \; y = 3\cos\left(\frac{\pi t}{2}\right)$

13. Distance in one oscillation = 4A

14. Time of flight = $\sqrt{\frac{2h}{g}} \propto \frac{1}{\sqrt{g}}$

Time period of pendulum = $2\pi\sqrt{\frac{l}{g}} \propto \frac{1}{\sqrt{g}}$

Ratio of time of flight & time period of pendulum is independent of g. Hence t' = 2T'

15. Displacement (x) equation of SHM

$x = A\sin(\omega t + \phi)$ ...(1)

$\frac{dx}{dt} = A\omega\cos(\omega t + \phi)$

acceleration (a) = $\frac{d^2x}{dt^2}$

$a = -\omega^2 A\sin(\omega t + \phi)$

$a = \omega^2 A\sin(\omega t + \phi + \pi)$ ...(2)

from (1) & (2), phase difference between displacement and acceleration is π.

16. Option (3) is a combination of SHM of same ω and same axis so its resultant is also a SHM which is periodic.

17. Displacement equation of SHM of frequency 'n'

$x = A\sin(\omega t) = A\sin(2\pi nt)$

Now,

Potential energy

$U = \frac{1}{2}kx^2 = \frac{1}{2}KA^2\sin^2(2\pi nt)$

$U = \frac{1}{2}kA^2\left[\frac{1-\cos(2\pi(2n)t)}{2}\right]$

So frequency of potential energy = 2n

18. $F = kx$

$10 = k(5\times10^{-2})$

$k = \frac{10}{5\times10^{-2}} = 2\times10^2 = 200\ N/m$

Now $T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{2}{200}} = \frac{2\pi}{10}$

23. $x = A\sin(\omega t)$

$\frac{dx}{dt} = v = A\omega\cos(\omega t)$

$\frac{dv}{dt} = a = -\omega^2 A\sin(\omega t)$

$a = -\left(\frac{2\pi}{8}\right)^2 \times 1\sin\left(\frac{2\pi}{8}\times 2\right)$

$\Rightarrow a = -\frac{\pi^2}{16}\times\sin\left(\frac{\pi}{2}\right)$

$\therefore \boxed{a = \frac{-\pi^2}{16}\,m/s^2}$

24. $x = 5\sin(\pi t + \frac{\pi}{3})$

comparing with $x = A\sin(\omega t + \phi)$

we get A = 5 m and $\omega = \pi$

$\Rightarrow \; T = \frac{2\pi}{\omega} = \frac{2\pi}{\pi} = 2\,sec.$

25. $T = 2\pi\sqrt{\frac{\ell}{g}}$

$T' = 2\pi\sqrt{\frac{\ell'}{g}} = 2\pi\sqrt{\frac{\ell/2}{g}}$

$T' = \frac{T}{\sqrt{2}} = \frac{x}{2}T$

$\Rightarrow x = \sqrt{2}$

26. PE = KE

$\frac{1}{2}kx^2 = \frac{1}{2}k(A^2 - x^2) \qquad \Rightarrow x = \pm\frac{A}{\sqrt{2}}$

Exercise-III (Analytical Questions)

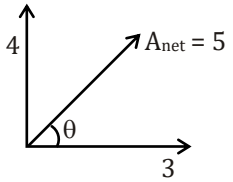
1. $x = 3\sin 4\pi t + 4\cos 4\pi t$

$\tan\theta = \frac{4}{3}$

$\theta = 53°$

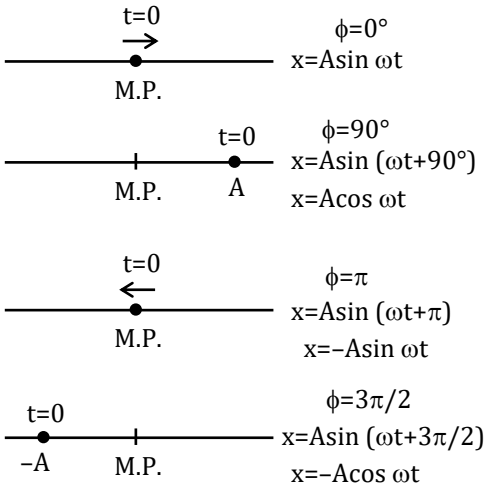
$x = 5\sin(4\pi t + 53°)$

(SHM)

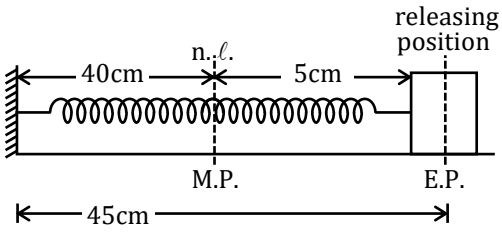


2. Theory

3.



4.



elongation in spring = 5 cm

amplitude = 5 cm

$v_{max} = A\omega = 5\sqrt{\frac{500}{1}} \quad \therefore \ \omega = \sqrt{\frac{k}{m}}$

$50\sqrt{5}$ cm/s

5. (a) Lift is at rest a = 0

(b) lift is moving downward with constant speed

$\Rightarrow$ constant velocity $\Rightarrow$ a = 0

in both cases a = 0 so $g_{eff} = g$

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

(c) In this case lift is moving with increasing speed so acceleration is downward.

$g_{eff} = g - a.$

6. Energy $E = \frac{1}{2}KA^2 = \frac{1}{2}m\omega^2A^2$

$v_{max} = A\omega = A\sqrt{\frac{K}{m}}$

$$T = 2\pi\sqrt{\frac{m}{K}}$$

$(P.E.)_{max} = \frac{1}{2}KA^2$

(i) Mass is doubled

$\Rightarrow$ Time period become $\sqrt{2}$ times.

(ii) Amplitude $\Rightarrow$ double

Energy $\Rightarrow$ 4 times

$v_{max} \Rightarrow$ 2 times

$(P.E.)_{max} \Rightarrow$ 4 times

(iii) K $\Rightarrow$ double

Energy = double

$v_{max} = \sqrt{2}$ times

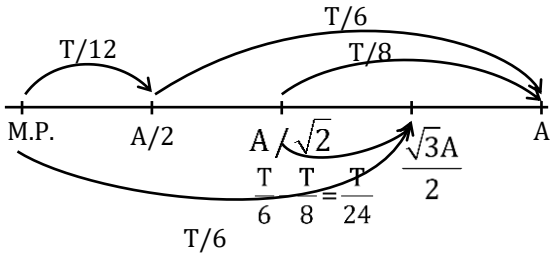
T $\Rightarrow \frac{1}{\sqrt{2}}$ times

$(P.E.)_{max}$ = 2 times

(iv) Angular frequency $\left(\omega = \sqrt{\frac{k}{m}}\right)$ is halfed

means k becomes 1/4 times so maximum speed is halfed.

7.

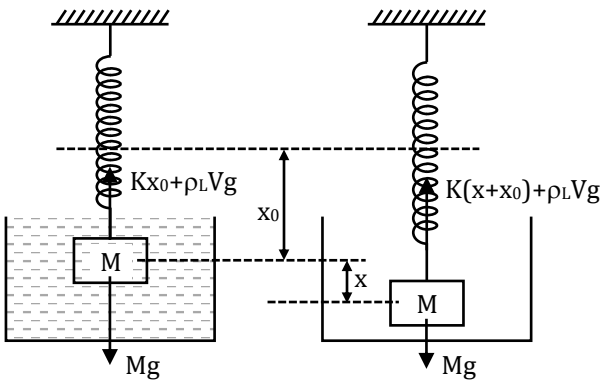


11. $a = -\omega^2x$

13. $A = A_0e^{-\frac{bt}{2m}}$

$$A = \frac{F_0}{m\sqrt{\left(\omega_d^2 - \omega^2\right)^2 + \left(\frac{b\omega_d}{m}\right)^2}}$$

14.



(1) $\rho_LVg + Kx_0 = Mg$

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(2) $F_R = -[K(x + x_0) + \rho_L Vg - Mg]$

$= -[Kx + Kx_0 + \rho_L Vg - Mg]$

$= -Kx$

(3) $a = \frac{F_R}{M} = -\frac{Kx}{M} \Rightarrow \omega = \sqrt{\frac{K}{M}} \Rightarrow T = 2\pi\sqrt{\frac{M}{K}}$

15. It is a periodic oscillatory and since released from small height it is SHM

16. $x = 2 + 3\,\frac{1-\cos 2\omega t}{2} = 2 + \frac{3}{2} - \frac{3}{2}\cos 2\omega t$

$= \frac{7}{2} - \frac{3}{2}\cos 2\omega t \Rightarrow \left(x - \frac{7}{2}\right) = \frac{3}{2}\cos 2\omega t$

17. $\frac{xy}{z} = \frac{A\omega.f}{A\omega^2} = \frac{f}{\omega} = \frac{f}{2\pi f} = \frac{1}{2\pi}$

π constant and dimension less.

18. At the mean position if $U_0 < 0$, then $K_{max} > E_{total}$

19. $T = 2\pi\sqrt{\frac{m}{k}}$

22. Here $T = \frac{2\pi}{\omega} = \frac{2\pi}{\pi} = 2\pi\sqrt{\frac{\ell}{g}} \Rightarrow \frac{\ell}{g} = \frac{1}{\pi^2}$ But g $= \pi^2$ therefore $\ell = 1$ m

23. It is due to resonance i.e. bridge may collapse.

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Wave Motion

SOLUTIONS

Exercise-I (Conceptual Questions)

3. distance = speed × time

6. Velocity of wave = $\frac{600}{2}$ = 300 m/s

Frequency = 500 Hz, Wave length $\lambda = \frac{3}{5}$ m

Number of wavelength = $\frac{600}{3/5}$ = 1000

7. Speed of sound wave is medium dependent term.

8. $y = 4\sin\frac{\pi}{2}\left(8t - \frac{x}{8}\right) = 4\sin\left(4\pi t - \frac{\pi}{16}x\right)$

Compare by $y = A\sin(\omega t - kx)$, $\omega = 4\pi$

$k = \frac{\pi}{16} \Rightarrow v = \frac{\omega}{k}$

v = 64 cm / sec. → +ve x-direction.

9. $y = 4\sin\left[\pi\left(\frac{t}{5} - \frac{x}{9}\right) + \frac{\pi}{6}\right]$ compare this equation by

$y = a\sin(\omega t - kx + \phi)$

$\omega = \frac{\pi}{5}$ $k = \frac{\pi}{9}$

a = 4 $\frac{T}{2} = 5$ $\frac{\lambda}{2} = 9$

T = 10 s, λ = 18 cm, v = nλ = 1.8 cm/s

10. Amplitude becomes double

Hence it becomes = 0.5

and frequency becomes half

hence ω becomes = π (half) and travelling in opposite direction with same speed so K will also become half

$\left\{v = \frac{\omega}{K} = \text{const.}\right\}$

$\therefore$ $y = 0.5\cos(\pi t + \pi x)$

11. $v_{max} = A\omega$

$v_{max} = 3 \times 10$

= 30 units

12. $v = \frac{\omega}{k} = \frac{6}{3} = 2$ units

13. Wave is a disturbance, which carries energy and momentum without transport of mass of matter.

15. $y_1 = a\sin\omega t$, $y_2 = a\cos\omega t = a\sin(\omega t + \frac{\pi}{2})$

16. Distance between two consecutive crests is 5 m which is equal to λ = 5m, freq. n = 2 Hz

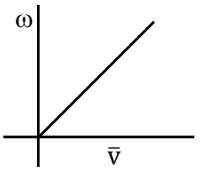
v = nλ = 10 m/s

17. $\omega = 2\pi n$

$\omega \propto n$ (2π = const)

$\omega \propto \frac{1}{\lambda}$ $\bar{\nu} = \frac{1}{\lambda}$

$\Rightarrow$ $\omega \propto \bar{\nu}$



19. v = 10

n = 100 $\therefore \lambda = \frac{10}{100}$ = 0.1 m = 10 cm

$\frac{\Delta\phi}{2\pi} = \frac{\Delta\lambda}{\lambda} \Rightarrow \frac{\Delta\phi}{2\pi} = \frac{2.5}{10} \Rightarrow \Delta\phi = \frac{\pi}{2}$

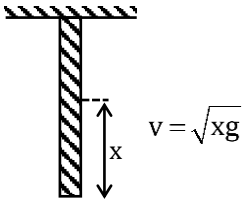
20. At the upper end –

$v = \sqrt{2.5\times10} = 5$ m/s

at a point 0.5 m distance from bottom -

$v = \sqrt{0.5\times10} = \sqrt{5}$

= 2.24 m/s



21. $v = \sqrt{\frac{T}{\mu}} \Rightarrow \frac{\omega}{K} = \sqrt{\frac{T}{\mu}} \Rightarrow \frac{\left(\frac{2\pi}{0.04}\right)}{\left(\frac{2\pi}{0.50}\right)} = \sqrt{\frac{T}{0.04}}$

$\Rightarrow$ $\frac{0.50}{0.04}\times\frac{0.50}{0.04} = \frac{T}{0.04}$

$\therefore$ $T = \frac{50}{4} \times 0.5 = 6.25$ N

22. $v_{wave} = \frac{\omega}{k}$

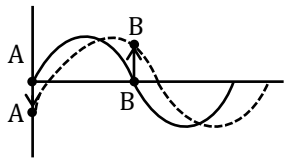
$v_{particle} = A\omega$

On comparing

$v_{w1} = 2$m/s, $v_{w2} = 3$m/s, $v_{w3} = \frac{11}{5}$m/s

$v_{p1} = 12$cm/s, $v_{p2} = 36$cm/s, $v_{p3} = 44$cm/s

23.



from fig. A is moving down & B is moving up and the phase at A is greater the phase at B.

24. $v = \frac{\omega}{k} = \frac{30}{1} = 30$ m/s

$d = 1.3 \times 10^{-4}$ kg/m, $v^2 d = T$

then $T = 1.17 \times 10^{-1}$N

26. $v_{O_2} = \sqrt{\frac{\gamma RT}{M_w}} = V$ $\quad v_{H_2+O_2} = \sqrt{\frac{\gamma RT}{(M_w)_{mix}}}$

$(M_w)_{mix} < M_w$ $\quad$ so $\quad v_{H_2+O_2} > V_{O_2}$

27. $v = \sqrt{\frac{E}{\rho}}$

28. $I = \frac{1}{2}\rho v a^2\omega^2 \Rightarrow I \propto n^2a^2$

29. frequency does not depends upon medium, velocity of sound decreases when it goes from water to air. $v = f\lambda$ $\quad (\because$ $f =$ const.)

so λ also decreases

30. Frequency doesn't change on changing the medium

32. $L = 10\log_{10}\frac{I}{I_0} \Rightarrow \log_{10}\frac{I}{I_0} = 3 \Rightarrow \frac{I}{I_0} = 10^3$

33. Velocity of sound

$$v = \sqrt{\frac{\gamma P}{\rho}}$$

as $\rho_d > \rho_m$

$\therefore \upsilon_m > \upsilon_d$

34. $V = v \Rightarrow \frac{\omega}{k} = \omega A \Rightarrow A = \frac{1}{k} = \frac{1}{(2\pi/\lambda)}$

$$\Rightarrow A = \frac{\lambda}{2\pi}$$

35. Energy same so $a_1^2n_1^2 = a_2^2n_2^2$

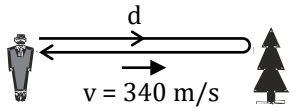
36. $\frac{V_1}{V_2} = \sqrt{\frac{\rho_2}{\rho_1}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$

37. $\frac{\Delta\phi}{2\pi} = \frac{\Delta t}{T} \Rightarrow \Delta\phi = \frac{1}{12} \times 2\pi = \frac{\pi}{6}$

38. $v \propto \frac{1}{\sqrt{\rho}}$

39. Let distance between cliff and mountain be d

$t = \frac{2d}{v} \Rightarrow 1 = \frac{2d}{340} \Rightarrow$ distance(d) $= \frac{340}{2} = 170$m



40. $A_R = \sqrt{a_1^2 + a_2^2 + 2a_1a_2\cos\phi}$

41. In superposition, only redistribution of energy occurs.

42. (1) Same phase $I_R = 4I$

(2) $\phi = 90°$, $I_R = I + I + 2\sqrt{I}\sqrt{I}\cos 90° = 2I$

43. $I_1 = I$ $\quad I_2 = I$ $\quad \phi = 0$

$I_R = I + I + 2\sqrt{I}\sqrt{I}\cos 0 = 4I$

44. $\frac{a_1}{a_2} = \frac{3}{4}$

$\frac{I_{max}}{I_{min}} = \left(\frac{a_1 + a_2}{a_1 - a_2}\right)^2 = \frac{(3+4)^2}{(3-4)^2} = \frac{49}{1}$

45. $I = I_1 + I_2 + 2\sqrt{I_1I_2}\cos\phi$

for $I_{max} \Rightarrow \phi = 0 \Rightarrow I_{max} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2$

46. $\frac{I_{max.}}{I_{min.}} = \frac{\left(\sqrt{I_1} + \sqrt{I_2}\right)^2}{\left(\sqrt{I_1} - \sqrt{I_2}\right)^2} = \frac{(3+1)^2}{(3-1)^2} = \frac{16}{4} = \frac{4}{1}$

47. A $\quad$ B

n $\quad$ n ± x

n + x $\quad$ or $\quad$ n - x

If frequency of B is n + x then on waxing beats decrease.

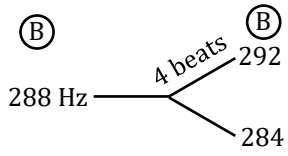
48. 250 ± 8 → 242 or 258 $\qquad$ 270 ± 12 → 282 or 258

common → 258 Hz

49. Number of beats $\Delta n = n_1 - n_2$

$2 = \frac{v}{\lambda_1} - \frac{v}{\lambda_2}$ $\quad \Rightarrow \quad 2 = \frac{v}{2} - \frac{v}{2.02}$

$2 = \frac{v}{202} \Rightarrow v = 404$

50. 

After loading wax, frequency of A decrease. So beats frequency will also decrease.

Ans. would be 292

52. for minima/destructive interferance

$\Delta\phi = (2n+1)\pi$

$\frac{2\pi}{\lambda}\ \Delta\lambda = (2n+1)\pi$

$\Delta\lambda = (2n+1)\ \frac{\lambda}{2}$

53. $I_{max} = 4I,\ \frac{I_{max}}{I} = 4$

54. $n_1 = 202$ Hz

$n_2 = 200$ Hz

$b = n_1 - n_2 = 2$

55. $a = \sqrt{a^2 + a^2 + 2a^2\cos\phi}$

$\Rightarrow \cos\phi = -\frac{1}{2} \qquad \Rightarrow \phi = \frac{2\pi}{3}$

56. $T = \frac{1}{|\nu_1 - \nu_2|}$

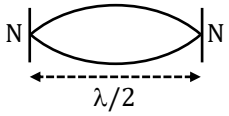
57. beats per second $= n_2 - n_1 = 253 - 250 = 3$

beats per minuts $= 3 \times 60 = 180$

59. Fundamental frequency

$n = \frac{1}{2L}\sqrt{\frac{T}{\mu}}\ \ [\because\ \mu = \frac{M}{L}\ ]\ \ \therefore n = \frac{1}{2}\sqrt{\frac{T}{ML}}$

60. v = 700 cm/sec = 7 m/sec, L = 2 m



$n = \frac{V}{2\ell} = \frac{7}{2\times 2} \Rightarrow n = \frac{7}{4}$ Hz

61. $y = a\cos(\omega t - kx)$

x = 0 is a node

So rigid end reflection

$y = a\cos(\omega t + kx + \pi)$

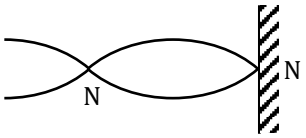
$y = -a\cos(\omega t + kx)$ $[\cos(180 + \theta) = -\cos\theta]$

62. Here $\lambda = 4$ cm

Now distance between two consecutive nodes

$= \frac{\lambda}{2} = 2$cm

63. n = 100 Hz., Given $\frac{\lambda}{2} = 10$ cm



$v = n\lambda = 100 \times 0.2 \qquad v = 20$ m/s

65. $n \propto \sqrt{T}$

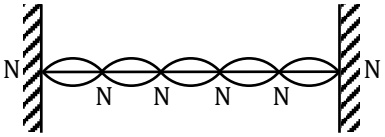
66. At earth $n = \frac{P}{2L}\sqrt{\frac{Mg}{\mu}} \Rightarrow n = \frac{P}{2L}\sqrt{\frac{g}{\mu}}$

(T = Mg M=1 Kg)

At moon $n' = \frac{P}{2L}\sqrt{\frac{M'g}{6\mu}}$

n' = n When M' = 6 Kg

67. p = 5, ℓ = 10 m



$v = \sqrt{\frac{T}{\mu}} = 20$ m/s, $n = \frac{P}{2\ell}\sqrt{\frac{T}{\mu}}$

68. $N\underset{\lambda/4}{\text{—}}A\underset{\lambda/4}{\text{—}}N\underset{\lambda/4}{\text{—}}A\underset{\lambda/4}{\text{—}}N$

$4\left(\frac{\lambda}{4}\right) = 1.21\text{Å} \quad \Rightarrow \quad \lambda = 1.21$ Å

69. $n \propto \sqrt{T} \Rightarrow \frac{n_1}{n_2} = \sqrt{\frac{T_1}{T_2}}$

$\Rightarrow \frac{n-10}{n} = \sqrt{\frac{129.6}{160}} = \frac{36}{40}$

$40n - 400 = 36\,n \Rightarrow n = 100$ Hz

70. $\nu \propto \sqrt{T}$

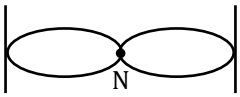
71. In stationary wave, energy does not propogate.

72. $n \propto \sqrt{T}$ $T\downarrow$ $n\downarrow$

73. Plucking distance

$$x = \frac{\ell}{2p} = \frac{\ell}{4}$$

touching distance = $\ell/2$



74. $n = \frac{1}{2\ell}\sqrt{\frac{T}{\pi r^2 d}} \Rightarrow \frac{n_2}{n_1} = \sqrt{\frac{T_2}{T_1}\times\frac{r_1^2}{r_2^2}\times\frac{d_1}{d_2}}$

$$= \sqrt{\frac{2}{1}\times\frac{1}{4}\times\frac{2}{1}} = 1$$

$\Rightarrow \quad n_2 = n_1 = n$

75. $n \propto \sqrt{T}$

76. $n = \frac{1}{2\ell}\sqrt{\frac{T}{\pi r^2 d}}$

$$n' = \frac{1}{2\times 3\ell}\sqrt{\frac{3T}{9\pi r^2 d}} = \frac{n}{3\sqrt{3}}$$

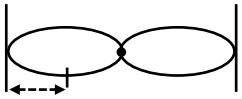
77. $n \propto \frac{1}{\ell}$

$$\frac{n_1}{n_2} = \frac{\ell_2}{\ell_1} \Rightarrow \frac{800}{1000} = \frac{\ell_2}{50}$$

$\ell_2 = 40$ cm

78. $n \propto \sqrt{T}$

79. $y = A \cos kx \cos \omega t$



80. 0.25m

$$\frac{\lambda}{4} = 0.25 \qquad \lambda = 1m$$

$$n = \frac{V}{\lambda} = \frac{1}{1}\sqrt{\frac{T}{m}} = \sqrt{\frac{20}{0.5\times 10^{-3}}} = 200 \text{ Hz}$$

81. When T↑ and velocity (v)↑ Then n↑

82. Length of air column decreases,

$\therefore$ frequency increases.

83. e = 1 cm $\qquad \ell_1 = 15$ cm

$$e = \frac{\ell_2 - 3\ell_1}{2}$$

$$1 = \frac{\ell_2 - 3\times 15}{2}$$

$2 = \ell_2 - 45$

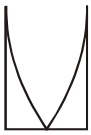
$\ell_2 = 47$ cm

84. Next 3 overtones are in ratio of 3 : 5 : 7

Only odd harmonic are present.

85. For closed pipe 

$$n = \frac{v}{4\ell} = 512$$

For open pipe 

$$n = \frac{v}{2\ell} = 1024$$

n = 1024 Hz.

86. $n = \frac{v}{4\ell}$



n = 264 Hz

v = 330 m/s

Then

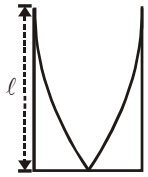
ℓ = 31.25 m

87. Fundamental frequency

$$n = \frac{v}{4\ell}$$

v = 320 m/s

ℓ = 1 m., n = 80 Hz



For closed pipe

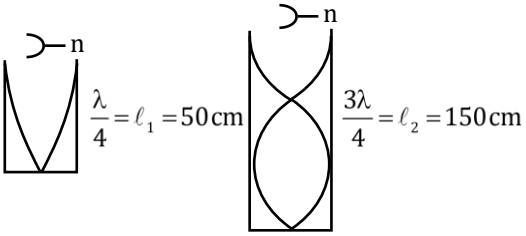
$n_1 : n_2 : n_3$ = ... = 1 : 3 : 5 : .....

frequency 80, 240, 400, ...

88. Fundamental frequency of open organ pipe

$$n = \frac{v}{2\ell} = \frac{330}{2\times 0.3} = 550 Hz$$

89.



length will be 3 times

90. $\frac{v}{4\ell} = 260, \quad \ell = 31.7$ cm

91. $\lambda = \frac{v}{n} = \frac{330}{330} = 1$ m

particle vibrate maximum at a distance

$\lambda/4$. So $\frac{\lambda}{4} = 0.25$m

92. $L_0 = 33$ cm

n = 1000 Hz $\quad 1000 = \frac{N \times 330}{2 \times .33} \Rightarrow N = 2$

i.e. 1st overtone

2nd harmonic

93. $n = (2m+1)\ \frac{V}{4L}$

$L = (2m+1)\ \frac{V}{4n}$

For L to be minima,

mode should be fundamental i.e. m = 0

$L = \frac{V}{4n}$

94. $n_1 = \frac{v}{4\ell_1}, \ n_2 = \frac{v}{4\ell_2}$

$\ell_1 = 0.75$ m $\quad \ell_2 = 0.770$ m

$n_1 - n_2 = 3 \Rightarrow \frac{v}{4}\left(\frac{1}{\ell_1} - \frac{1}{\ell_2}\right) = 3 \Rightarrow v = 346.5$ m/s

95. $n = \frac{Nv}{2L} \quad \Rightarrow \quad L = \frac{Nv}{2n}$

for L to be minm N = 1

$L = \frac{v}{2n} = \frac{350}{2 \times 350} = 50$ cm

96. Frequency remains unchanged on change medium

n = 60 kHz

$\lambda = \frac{v}{n} = \frac{330}{60 \times 10^3} = 5.5 \times 10^{-3}$ m = 5.5 mm

97. $\frac{v}{4\ell}$ = fundamental frequency = 1500 Hz

Normal person can listen frequency 20,000 Hz.

1500 × x = 20,000, x = 13 (No. of Harmonic)

Then overtones and frequency in case of closed organ pipe, ratio of frequency

1,	3,	5,	7,	9,	11,	13
I	II	III	IV	V	VI	

Then 6 overtones.

98. $n_N = (2n+1)\frac{v}{4L}$

$n_N = (2n+1)\frac{320}{4 \times 1}$

$n_N = (2n + 1) \times 80$ = odd multiple of 80

At 300 Hz resonance will not occur.

99. For open organ pipe

$n_1 = \frac{v}{2\ell}$



for COP $n_2 = \frac{v}{4(\ell/2)} = \frac{2v}{4\ell} \Rightarrow n_2 = \frac{v}{2\ell}$, $n_1 = n_2$

100. It occurs due to end correction.

Exercise-II (Previous Year Questions)

1. Total length of string $\ell = \ell_1 + \ell_2 + \ell_3$

But frequency $\propto \frac{1}{\text{length}}$

so $\frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3}$

2. Frequency COP, $n_N = (2N + 1)\ \frac{v}{4L}$

for N = 0, $n_0 = 100$ Hz

N = 1, $n_1 = 300$ Hz

N = 2, $n_2 = 500$ Hz

N = 3, $n_3 = 700$ Hz

N = 4, $n_4 = 900$ Hz

N = 5, $n_5 = 1100$ Hz Total = 6

Which are less than 1250 Hz.

3. $\frac{V}{4L_C} = \frac{3V}{2L_0}$

$L_0 = 4L_C \times \frac{3}{2}$

$L_0 = 6 \times 20 = 120$ cm

4. Two consecutive resonant frequencies for a string fixed at both ends will be

$\frac{nv}{2\ell}$ and $\frac{(n+1)v}{2\ell} \Rightarrow \frac{(n+1)v}{2\ell} - \frac{nv}{2\ell} = 420 - 315$

$\frac{v}{2\ell} = 105$ Hz

Which is the minimum resonant frequency

5. $T_1 = m_2g$

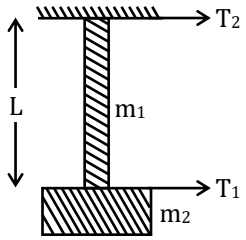
$T_2 = (m_1 + m_2)g$

Velocity $\propto \sqrt{T}$ (n=const.)

$\Rightarrow \lambda \propto \sqrt{T}$

$\Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{\sqrt{T_1}}{\sqrt{T_2}}$

$\Rightarrow \frac{\lambda_2}{\lambda_1} = \sqrt{\frac{m_1 + m_2}{m_2}}$



6. First minimum resonating length for closed organ pipe = $\frac{\lambda}{4} = 50\,cm$

$\therefore$ Next larger length of air column

$= \frac{3\lambda}{4} = 150\,cm$

7. For second overtone (3rd harmonic) in open organ pipe,

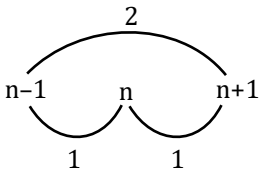
$\frac{3\lambda}{2} = \ell_0 \Rightarrow \lambda = \frac{2\ell_0}{3}$

for first overtone (3rd harmonic) in closed organ pipe,

$\frac{3\lambda}{4} = \ell_C \Rightarrow \lambda = \frac{4\ell_C}{3} = \frac{4L}{3}$

So, $\frac{2\ell_0}{3} = \frac{4L}{3} \Rightarrow \ell_0 = 2L$

8.



Now divide 1 second into 1, 1, 2 equal divisions

$\frac{1}{1}$

$\frac{1}{1}$

$\frac{1}{2}$ $\frac{2}{2}$

By eliminating common time instants, total maxima in one second is 2.

So, two beats per second will be heard.

9. Difference between any two consecutive frequencies of COP = $\frac{2v}{4\ell} = 260 - 220 = 40$ Hz

$\Rightarrow \frac{v}{4\ell} = 20\,Hz$

So fundamental frequency = 20 Hz

10. $V = 2f(\ell_2 - \ell_1)$ $\because \lambda = 2(\ell_2 - \ell_1)$

$= 2(320) \times (0.73 - 0.20)$ $V = f\lambda$

$= (640) \times (0.53) = 339.2$ m/s

11. $f_{oop} = 3f_{cop} \Rightarrow \frac{v}{2\ell_{oop}} = \frac{3v}{4\ell_{cop}}$

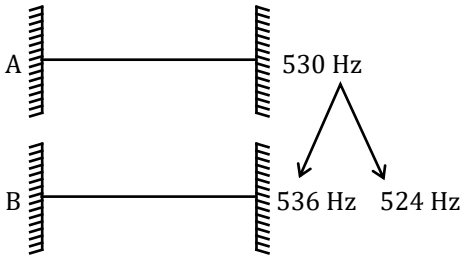
$\Rightarrow \ell_{oop} = \frac{2}{3} \times \ell_{cop} = \frac{2}{3} \times 20 = 13.3$ cm

12. Here $\lambda = 2(l_2 - l_1)$

where $l_1 = 9.75$ cm, $l_2 = 31.25$ cm

so $v = 2n(l_2 - l_1) = 2 \times 800\ (31.25 - 9.75) = 344$ m/s

13. Guitar string i.e. string is fixed from both ends



Frequency $\propto \sqrt{\text{Tension}}$

If tension in B slightly decreases then frequency of B decreases.

If B is 536 Hz, as the frequency decreases, beats with A also decreases.

If B is 524 Hz, as the frequency decreases, beats with A increases.

$\therefore$ Original frequency of B will be 524 Hz.

14. Frequency of stretched string

$$n = \frac{1}{2\ell}\sqrt{\frac{T}{\mu}}$$

Here T and μ are constant so $n \propto \frac{1}{\ell}$

Therefore $\frac{n'}{n} = \frac{\ell}{\ell'}$

$\Rightarrow \frac{180}{120} = \frac{90}{\ell} \Rightarrow \ell' = 60$ cm

17. $\frac{n_{oop}}{n_{cop}} = \frac{\frac{V}{2L}}{\frac{V}{4L}}$

$\Rightarrow \frac{n_{oop}}{n_{cop}} = \frac{2}{1}$

18. Interference is obtained by coherent waves superposition.
A and C has same frequency.

19. Coefficient of t in equation y = Asin(ωt–kx) is angular frequency.

So, comparing $\omega = 2\pi\frac{a}{\lambda}$

$2\pi f = 2\pi\frac{a}{\lambda}$

$f = \frac{a}{\lambda}$

Exercise-III (Analytical Questions)

1. Same length of organ pipe means wavelength in both the pipe are same but velocity of sound is different in both the pipes due to different gases inside it.

3. $v = \sqrt{\frac{\gamma RT}{M_W}} = \sqrt{\frac{\gamma P}{\rho}}$

$(M_W)_{dry\ air} > (M_W)_{moist\ air}$

$(\rho)_{dry\ air} > (\rho)_{moist\ air}$

4. $y = 0.02 \sin 2\pi (10t - 5x)$

compare the about equation with

$y = A \sin(\omega t - kx)$

$A = 0.02$ m

$\omega = 20\pi$ rod/s

$k = 10\pi$ m^{-1}

$v = \frac{\omega}{k} = \frac{20\pi}{10\pi} = 2$

$\omega = 20\pi = 2\pi f$

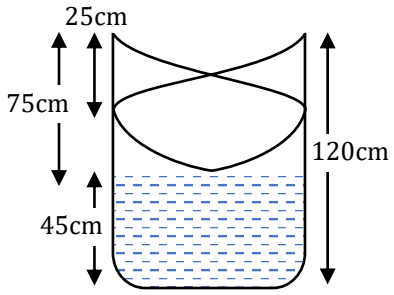
$f = 10$ Hz

$k = 10\pi = \frac{2\pi}{\lambda} \Rightarrow \lambda = 0.2$ m

$v_{max} = A\omega = 0.02 \times 20\pi = 0.4\pi$ m/s

5. As $V = f\lambda$

$\lambda = \frac{V}{f} = \frac{340}{340} = 1$m



first Resonance length

$R_1 = \frac{\lambda}{4} = \frac{1}{4}$m $= 25$cm

$\therefore R_2 = \frac{3\lambda}{4} = \frac{3}{4}$m $= 75$cm

$\therefore R_3 = \frac{5\lambda}{4} = \frac{5}{4}$m $= 125$cm

i.e. third resonance does not establish

$\therefore$ minimum length of H_2O column to have the resonance = 45 cm

$\therefore$ Distance between two successive nodes = $\frac{\lambda}{2}$

$= \frac{1}{2}$m $= 50$cm

and maximum length of H_2O column to create resonance i.e. 120 – 25 = 95 cm

OR

h = 120 cm

f = 340 Hz

v = 340 m/s

$\lambda = \frac{v}{f} = \frac{340}{340} = 100$ cm

(1) $\frac{\lambda}{4} = \frac{100}{4} = 25$cm (air) $\Rightarrow$ water level = 95 cm

(2) $\frac{3\lambda}{4} = 75$cm (air) $\Rightarrow$ water level = 45 cm

(3) $\frac{5\lambda}{4}$ = 125 cm (not possible)

So maximum water level = 95 cm

Minimum water level = 45 cm

Distance between two successive nodes

$= \frac{\lambda}{2} = \frac{100}{2}$ = 50 cm

6. From $\sqrt{\frac{\gamma RT}{M_W}} = \sqrt{\frac{\gamma P}{\rho}}$

(i) Temperature 2 times $\Rightarrow$ speed double

(ii) Temperature constant there

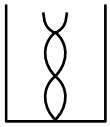
$P\uparrow \quad \rho\uparrow \quad v_{sound}$ = const

(iv) M_W is 4 times $\Rightarrow$ speed half.

7. For closed organ pipe fundament frequency

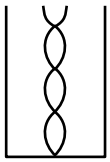
$f = \frac{v}{4\ell}$

Third overtone $f = \frac{7v}{4\ell}$ (7 time of fundamental frequency)



Second overtone $f = \frac{5v}{4\ell}$

No. of nodes = 3



Third overtone $f = \frac{7v}{4\ell}$

No. of A.N. = 4

8. Theory

9. $v = \sqrt{\frac{\gamma RT}{M\omega}} = \sqrt{\frac{\gamma\rho}{\rho}}$

$(M_\omega)_{dry\ air} > (M_\omega)_{moist\ air}$

$(\rho)_{dry\ air} > (\rho)_{moist\ air}$

10. First resonating length $\ell_1 = \frac{\lambda}{4}$

Second resonating length $\ell_2 = \frac{3\lambda}{4}$

11. Theory

13. Wavelength depends on length which is fixed. Therefore wavelength does not change.

Further $v = \sqrt{T/m}$ or $v \propto T^{1/2}$

$\therefore$ % change in $v = \frac{1}{2}$ (% change in T)

$= \frac{1}{2}(2) = 1\%$

i.e., Speed and hence frequency will change by 1%.

Change in frequency is 15 Hz which is 1% of 1500 Hz.

Therefore, original frequency should be 1500 Hz.

OR

$f = \frac{v}{2\ell} = \frac{1}{2\ell}\sqrt{\frac{T}{\mu}}$

$\frac{\Delta t}{f} = \frac{1}{2}\frac{\Delta T}{T}$

$\frac{15}{f} = \frac{1}{2}\frac{2}{100}$

f = 1500 Hz (original frequency)

$v = \sqrt{\frac{T}{\mu}} \Rightarrow \frac{\Delta v}{v} = \frac{1}{2}\frac{\Delta T}{T}$

$\frac{\Delta v}{v} \times 100 = \frac{1}{2} \times 2\% = 1\%$

$\because$ Length of string is fixed that means wavelength remain same.

14. For closed pipe,

$f = n\left(\frac{v}{4\ell}\right) m = 1,3,5,.....$

For n = 1, $f_1 = \left(\frac{v}{4\ell}\right)$ n = 1, 3, 5.....

For n = 1, $f_1 = \frac{v}{4\ell} = \frac{320}{4\times1} = 80Hz$

For n = 3, $f_3 = 3f_1$ = 240 Hz

For n = 5, $f_5 = 5f_1$ = 400 Hz

OR

L = 1m = 100 cm

v = 320 m/s

fundament freq. $f = \frac{v}{4\ell} = \frac{320}{4\times1}$ = 80 Hz

3rd harmonic = 3 × 80 = 240 Hz

5th harmonic = 5 × 80 = 400 Hz

15. $y = 8\sin^2\pi\left[\frac{t}{0.02} - \frac{x}{100}\right]$

$= 4\left[1 - \cos 2\pi\left(\frac{t}{0.02} - \frac{x}{100}\right)\right]$

$\Rightarrow y - 4 = -4\cos\left(\frac{2\pi t}{0.02} - \frac{2\pi x}{100}\right)$

For (A) : Amplitude = 4 cm

For (B) :

Frequency $f = \frac{\omega}{2\pi} = \left(\frac{2\pi}{0.02}\right)\frac{1}{2\pi} = 50\text{Hz}$

For (C) : Velocity of wave = $\frac{\omega}{k}$ = 5000cm/s

= 50ms^{-1}

For (D) : Maximum particle velocity

= $A\omega = (4 \times 10^{-2})\left(\frac{2\pi}{0.02}\right) = 4\pi\ \text{ms}^{-1}$

= $400\ \pi\ \text{cm-s}^{-1}$

16. The wave speed depends on properties of the medium, not on how you generate the wave. For a string. $v = \sqrt{T_s/\mu}$. Increasing the tension or decreasing the linear density (lighter string) will increase the wave speed.

17. $v_1 = \frac{\omega}{k} = \frac{4}{2} = 2\text{m/s}$,

$v_2 = \frac{10}{4} = 2.5\text{m/s}$,

$v_3 = \frac{12}{6} = 2\text{m/s}$,

$v_4 = \frac{16}{8} = 2\text{m/s}$,

$v_5 = \frac{20}{10} = 2\ \text{m/s}$

as $v = \sqrt{\frac{T}{\mu}}$ so T_2 is different.

18. At node $\cos(10\pi x) = 0$ &

at antinode $\cos(10\pi x) = 1$

& $\omega = 50\ \pi$, $k = 10\pi$ & $v = \frac{\omega}{k}, k = \frac{2\pi}{\lambda}$

AIM: FREE EDUCATION TO ALL APUL

Units, Dimensions and Measurement

SOLUTIONS

Exercise-I (Conceptual Questions)

1. SI system has seven fundamental units.

2. Density is a derived quantity

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

3. Kilogram is a unit

4. (a) 10^{-6} sec (b) 366 days (c) 27.3 days
(d) 3.08×10^{16} m (not the unit of time)

5. Energy = Work = (Force) (Displacement)
= N × m = Joule

6. Theory-Kelvin

7. Momentum = mv = $M^1L^1T^{-1}$ kg × m/sec.

8. $\text{Pressure} = \frac{\text{Force}}{\text{Area}} = \frac{N}{m^2}$

9. $F = Ma = Kg \times \frac{m}{sec^2}$

10. $\frac{N}{Kg} = \frac{\text{Force}}{\text{Mass}} = \text{Acceleration}$

11. Velocity = $[L^1T^{-1}]$
= $(10^{-6}\text{ m})\ (10^{-6}\text{ sec})^{-1}$ = m/sec.

12. $\text{Power} = \frac{\text{Eerngy}}{\text{time}}$ = J/sec. = Watt or Kilowatt

13. Kilowatt hour = Power × time = Energy

14. 1 nanometer = 10^{-9} m = 10^{-7} cm

15. Angstrom = 10^{-10}m, nanometre = 10^{-9}m
Fermi = 10^{-15}m, attometre = 10^{-18}m

16. 1 Fermi = 10^{-15} meter

17. Velocity = $[L^1T^{-1}] \rightarrow$ derived

18. Temperature being a fundamental quantity itself cannot be expressed in terms of any of these.

19. $\text{Velocity} = \frac{\text{displacement}}{\text{time}} \frac{(L^1)}{(T^1)}$

It can be derived from other two so it is a derived.

20. $d = 10^3$ kg/m^3

$$= \frac{10^3 \times 10^3 \text{ gram}}{10^{-6}\text{ cm}^3} = 1 \text{ gram/cm}^3$$

21. 1 dyne = 10^{-5} N

1 cm = 10^{-2} m

$$\text{Surface Tension} = \frac{70 \text{ dyne}}{\text{cm}} = \frac{70 \times 10^{-5}\text{ N}}{10^{-2}\text{ m}}$$

$$= 7 \times 10^{-2} \text{ N/m}$$

22. (a) $[\text{Time}] = \left[\frac{\text{Length}}{\text{Velocity}}\right] = \text{Unchanged}$

(b) $[\text{Mass}] = \left[\frac{\text{Force}}{\text{Length} \times \text{Time}^{-2}}\right] = \text{Unchanged}$

(c) $[\text{Momentum}] = \left[\text{Mass} \times \frac{\text{Length}}{\text{Time}}\right] \propto [\text{Length}]$

so double

(d) $[\text{Energy}] = \left[\frac{\text{Mass} \times \text{Length}^2}{\text{Time}^2}\right] \propto [\text{Length}^2]$

So becomes four times of initial

23. [Power] = [Force × Velocity]
so becomes one fourth

24. Angle = Dimensionless but unit = radian
relative density = Dimensionless and is unitless so dimensionless quantity may have a unit.

25. For unitless quantity dimensions are always zero i.e. they never have non zero dimension.

26. Coefficient of friction is unitless & dimenssionless.

27. $\text{Relative density} = \frac{\text{Density of substance}}{\text{Density of water}}$

$$= \frac{\text{kgm}^{-3}}{\text{kgm}^{-3}} = \text{dimensionless}$$

28. 1 Fermi = 10^{-15} m; 1 Micron = 10^{-6} m
1 angstrom = 10^{-10} m; radian $\rightarrow$ Angle

29. Force $\rightarrow [M^1L^1T^{-2}]$

Pressure $\rightarrow [M^1L^{-1}T^{-2}]$

Gravitational constant $\rightarrow [M^{-1}L^3T^{-2}]$

30. Energy = Work = (Force) (displacement)

$= [M^1L^1T^{-2}]$

31. Impulse = Force × Time

$$= \left(\frac{\text{Linear momentum}}{\text{time}}\right) \times \text{time}$$

= linear momentum

32. Torque= (Force) × (distance)

Energy = Work = (Force) × (displacement)

33. Dimensional analysis cannot be used to derive relations involving trigonometric, exponential functions and relations involving (+) & (–) sign

34. In e^x, x must be dimensionless so in $e^{-A/B}$, A & B must have same dimensions also A+B and A–B are not possible because quantities with different dimensions can't be added or subtracted.

35. Dimensionally correct equation may be correct or incorrect but dimensionally incorrect equation is always incorrect e.g. $p = \frac{mv}{t}$ is dimensionally as well as experimentally wrong.

36. Dimensionally correct equation may be correct or incorrect but dimensionally incorrect equation is always incorrect e.g. $p = \frac{mv}{t}$ is dimensionally as well as experimentally wrong.

37. $x = at + bt^2$

$[x] = [at] = [bt^2]$

$$b = \frac{x}{t^2} = m/hr^2$$

38. $F = at + bt^2$

dimensions of at and bt^2 are same as that of F

$at = F \Rightarrow a[T] = [M^1L^1T^{-2}]$

$a = [M^1L^1T^{-3}]$

$bt^2 = F \Rightarrow b[T]^2 = [M^1L^1T^{-2}]$

$b = [M^1L^1T^{-4}]$

39. $S = 1/3\ ft^3$

$$[S] = [ft^3] \Rightarrow [f] = \frac{[S]}{[t^3]} = [L^1T^{-3}]$$

40. $P = P_0e^{-\alpha t^2}$

αt^2 must be dimensionless

$$\text{Thus dimensions of } \alpha = \frac{1}{t^2} = [T^{-2}]$$

41. $F = Pt^{-1} + Q \Rightarrow [F] = [PT^{-1}]$

$$P = \frac{[M^1L^1T^{-2}]}{[T^{-1}]} = [M^1L^1T^{-1}] = \text{Momentum}$$

42. $$\eta = \frac{F}{6\pi av} = \frac{M^1L^1T^{-2}}{[L^1][L^1T^{-1}]} = M^1L^{-1}T^{-1}$$

43. θ in $\cos\theta$ and $\sin\theta$ is dimensionless

$Bx = M^0L^0L^0 \Rightarrow BL^1 = M^0L^0T^0 \Rightarrow B = M^0L^{-1}T^0$

$Dt = M^0L^0T^0 \Rightarrow DT^1 = M^0L^0T^0 \Rightarrow D = M^0L^0T^{-1}$

$$\frac{D}{B} = \frac{M^0L^0T^{-1}}{M^0L^{-1}T^0} = L^1T^{-1}$$

44. $x = at^2 + b$

$[x] = [at^2] = [b]$

$a = [L^1T^{-2}];\ b = [L^1]$

$$\sqrt{ab} = \sqrt{[L^2T^{-2}]} = [L^1T^{-1}]$$

45. $y = r \sin(\omega t - kx)$

$[\omega t - kx] = [M^0L^0T^0]$

$[\omega t = [kx] = [M^0L^0T^0]$

$$\left[\frac{\omega}{k}\right] = \left[\frac{x}{t}\right] = [L^1T^{-1}]$$

46. $y = a \sin[At - Bx + C]$

$[At - Bx + C] = [M^0L^0T^0]$

$[At] = [Bx] = [C] = [M^0L^0T^0]$

$[A] = [T^{-1}]; [B] = [L^{-1}]; [C] = [M^0L^0T^0]$

47. at + 3 = dimensionless

$\Rightarrow$ at = dimensionless $\Rightarrow aT^1 = M^0L^0T^0$

$a = M^0L^0T^{-1}$

48. $\frac{dv}{dt} = AT - Bv$

$$\left[\frac{dv}{dt}\right] = [AT] = [Bv]$$

$[L^1T^{-2}] = [AT]\ ; [L^1T^{-2}] = [B]\ [L^1T^{-1}]$

$[A] = [L^1T^{-3}] \qquad [B] = [T^{-1}]$

49. $\frac{d^2y}{dx^2}=\left[\frac{L^1}{T^2}\right]=[L_1T^{-2}]$

50. $M = E^aV^bF^c$

$[M^1] = [ML^2T^{-2}]^a\,[L^1T^{-1}]^b\,[M^1L^1T^{-2}]c$

$[M^1L^0T^0] = [M^{a+c}\,L^{2a+b+c}\,T^{-2a-b-2c}]$

$a + c = 1;\ 2a + b + c = 0;\ -2a - b - 2c = 0$

51. $F = D^aV^bA^c$

$[M^1L^1T^{-2}] = [M^1L^{-3}]^a\,[L^1T^{-1}]^b\,[L^2]^c$

$[M^1L^1T^{-2}] = [M^a\,L^{-3a+b+2c}\,T^{-b}]$

$a = 1;\ -3a + b + 2c = 1;\ -b = -2$

54. According to the rules of significant figures

$0.007\ m^2$ has one significant figures.

2.64×10^{24} kg has three significant figures.

$0.0006032\ m^2$ has four significant figures.

6.3200 J has five significant figures.

55. $P = 0.0030 \Rightarrow 2$ SF

$Q = 2.40 \Rightarrow 3$ SF

$R = 3000\ m \Rightarrow 4$ SF

56. $4621 = 4.621 \times 10^3$

$\approx 4.6 \times 10^3 = 4600$

57. L+B = 4.431 cm = 4.4 cm

(Final answer must be upto as many decimal places as in number with least decimal places)

58.
$$\begin{array}{r} 5.27 \\ -0.2 \\ \hline \end{array}$$

$5.07 \simeq 5.1$ J

59. $17.8 \times 3.1143 = 55.4354$

In product & division the number of S.F. in the answer would be equal to least S.F. figure of one of the individual qunaity.

Answer will have 3 S.F.

60. $1013 = 1.013 \times 10^3 \approx 10^3$ order of magnitude = 3

61. Sum of lengths of two rods = L_1+L_2 = (22.10 ± 0.10)cm

62. Thickness of cylinder can be written as

$t = t \pm \Delta t$

$t = r_2 - r_1 = 5.23 - 4.89 = 0.34$

$\Delta t = \Delta r_1 + \Delta r_2 = 0.015 + 0.005 = 0.02$

$t = (0.34 \pm 0.02)$ cm

63. $R = \frac{V}{I} \Rightarrow \frac{\Delta R}{R}=\frac{\Delta V}{V}+\frac{\Delta I}{I}$

$\Rightarrow \frac{\Delta R}{R}\times 100=\left(\frac{5}{100}\times 100+\frac{0.2}{10}\times 100\right)=7\%$

64. If % change in radius = x%

then % change in area = 2x%

and % change in volume = 3x%

65. Circumference (S) = $2\pi r$

$\frac{\Delta S}{S}\times 100=\frac{\Delta r}{r}\times 100=2\%$

66. $V=\frac{4}{3}\pi r^3=\frac{4}{3}\pi\left(\frac{d}{2}\right)^3=\frac{\pi d^3}{6}$

$\%V = 3\%d \Rightarrow 1.5 = 3\%d \Rightarrow \%d = 0.5\%$

67. Volume of cube V = $(\text{side})^3$

$\therefore V = a^3$. Take log and differentiate.

$\frac{\Delta V}{V}\times 100=3\frac{\Delta a}{a}\times 100 = 3(\pm 2\%) = \pm 6\%.$

68. $A=\ell^2 \Rightarrow \frac{\Delta A}{A}\times 100=2\frac{\Delta \ell}{\ell}\times 100=2(2\%)=4\%$

69. $f(x) = x^2$

Relative error = $2\left(\frac{\Delta x}{x}\right)$

70. $KE = \frac{1}{2}mv^2$

$\frac{\Delta KE}{KE}=\frac{\Delta m}{m}+2\frac{\Delta v}{v}$

$\frac{\Delta KE}{KE}\times 100 = 2 + 2 \times 3 = 8\%$

71. $H = I^2Rt$

$\Rightarrow \frac{\Delta H}{H}\times 100=\frac{2\Delta I}{I}\times 100+\frac{\Delta R}{R}\times 100+\frac{\Delta t}{t}\times 100$

$=2(2\%)+1\%+1\% = 6\%$

72. $I = 2/5\ MR^2$

$\frac{\Delta I}{I}=\frac{\Delta M}{M}+2\frac{\Delta R}{R}$

$\frac{\Delta I}{I}\times 100=1\%+2\times 2\%=5\%$

73. Resistance of a wire $R=\frac{\rho l}{\pi r^2}$

$\frac{\Delta R}{R}\times 100=\frac{\Delta \rho}{\rho}\times 100+\frac{\Delta \ell}{\ell}\times 100+2\frac{\Delta r}{r}\times 100$

$= 1\% + 2\% + 2(3\%) = 9\%$

74. $P=\frac{F}{A}=\frac{F}{L^2}$ For max. error

$$\frac{\Delta P}{P}\times 100=\frac{\Delta F}{F}\times 100+\frac{2\Delta L}{L}\times 100$$

$$= 4\% + 2(2\%) = 8\%$$

75. $T=2\pi\sqrt{\frac{\ell}{g}}\Rightarrow g=\frac{4\pi^2\ell}{T^2}$

$$\frac{\Delta g}{g}=\frac{\Delta \ell}{\ell}+\frac{2\Delta T}{T}=y+2x$$

76. $T=2\pi\sqrt{\frac{\ell}{g}}\Rightarrow g=\frac{4\pi^2\ell}{T^2}$

$$\frac{\Delta g}{g}\times 100=\frac{\Delta \ell}{\ell}\times 100+\frac{2\Delta T}{T}\times 100$$

$$=1\%+2(3\%)=7\%$$

77. $X = ab^2/c^3 \Rightarrow \frac{\Delta X}{X}=\frac{\Delta a}{a}\pm 2\frac{\Delta b}{b}\pm 3\frac{\Delta c}{c}$

$$\frac{\Delta X}{X}\times 100=\pm 1\%\pm 2\times 3\%\pm 3\times 2\%=\pm 13\%$$

78. Given : $X=\frac{2k^3I^2}{m\sqrt{n}}$

$$\therefore\ \frac{\Delta X}{X}=3\frac{\Delta k}{k}+2\frac{\Delta l}{l}+\frac{\Delta m}{m}+\frac{1}{2}\frac{\Delta n}{n}$$

Percentage error in X

$$\frac{\Delta X}{X}\times 100=\left(3\frac{\Delta k}{k}+2\frac{\Delta l}{l}+\frac{\Delta m}{m}+\frac{1}{2}\frac{\Delta n}{n}\right)\times 100$$

$$= 3 \times 1\ \% + 2 \times 1\% + 3\% + \frac{1}{2} \times 4\%$$

$$= 3\% + 2\% + 3\% + 2\% = 10\ \%$$

Hence, the value of X is uncertain by 12 %.

79. $\frac{\Delta X}{X}\times 100=a\frac{\Delta M}{M}\%+b\frac{\Delta L}{L}\%+c\frac{\Delta T}{T}\%$

$$= a\alpha\% + b\beta\% + c\gamma\%$$

$$\frac{\Delta X}{X}\times 100 = (a\alpha + b\beta + c\gamma)\%$$

80. $X = [M^{-1}L^3T^{-2}]$

$$\left(\frac{\Delta x}{x}\right)100\%=\left[\frac{1\Delta M}{M}+\frac{3\Delta L}{L}+\frac{2\Delta T}{T}\right]100\%$$

$$= 2\% + 3(3\%) + 2(4\%) = 19\%$$

81. $PV^{3/2}=K\Rightarrow P=\frac{K}{V^{3/2}}$

$$\%P = 1\%K - \frac{3}{2}\%V = 1(0) - \frac{3}{2}(-0.5) = 0.75\%$$

82. Zero error is due to erroneous manufacture of instrument so it is a instrumental error $\Rightarrow$ systematic error.

84. 40 VSD = 38 MSD

$$1\ \text{MSD} = \frac{40}{38}\ \text{VSD}$$

L.C = 1 MSD – 1 VSD

85. 10 MSD = 1 cm

1 MSD = 0.1 cm

20 VSD = 16 MSD

$$1\ \text{VSD} = \frac{16}{20}\text{MSD}$$

L.C. = 1 MSD – 1 VSD

86. 10 MSD = 1 cm

1 MSD = 0.1 cm

10 VSD = 9 MSD $\Rightarrow$ 1 VSD = 0.9 MSD

1 VSD = 0.09 cm

L.C. = 1 MSD – 1 VSD = 0.1 – 0.09

= 0.01 cm

T.R. = MSR + VSR = 2.8 cm + 7(0.01 cm)

= 2.87 cm

87. The correct value of diameter is upto 3 digits after the decimal as the least count is 0.001 cm

88. $LC=\frac{\text{Pitch}}{\text{Total circular divisions}}=\frac{5\text{mm}}{100}$

= 0.05 mm

89. In 5 rotation distance moved = 5 mm

In 1 rotation distance moved = 1 mm

= Pitch

$$LC=\frac{\text{Pitch}}{\text{Total circular divisions}}=\frac{1\text{mm}}{100}$$

= 0.01 mm = 0.001 cm

90. Total reading (T.R.) = MSD + CSR

= (3 mm) + 25 × (L.C.)

L.C. = 0.01 mm

91. $\text{L.C.} = \dfrac{\text{Pitch}}{\text{no. of division on circular scale}}$

$= \dfrac{1}{100} = 0.01 \text{ mm}$

T.R. = MSR + CSR

= 2 × 1 mm + 25 × L.C.

= 2 × 1 mm + 25 × 0.01 mm

True value (T.V.) = T.R. – error

= 2.25 – 5 × 0.01

= 2.20 mm

Exercise-II (Previous Year Questions)

1. $[\text{mass}] = \left[\dfrac{\text{Force}}{\text{Acceleration}}\right] = \left[\dfrac{\text{Force}}{\text{Velocity / time}}\right]$

$= [F\, V^{-1}\, T]$

2. Applying dimensional analysis

$S \propto E^a V^b T^c$

$[M^1L^0T^{-2}] = k\,[M^1L^2T^{-2}]^a\,[L^1T^{-1}]^b\,[T^1]^c$

$[M^1L^0T^{-2}] = k\,[M^aL^{2a}T^{-2a}]\,[L^bT^{-b}]\,[T^c]$

$[M^1L^0T^{-2}] = k[M]^a[L]^{2a+b}[T]^{-2a-b+c}$

Comparision

$a = 1$ $\quad 2a+b=0 \quad -2=-2a-b+c$

$b=-2 \quad -2=-2(1)+2+c$

$c = -2$

So the dimensional formula for surface tension will be $[E^1V^{-2}T^{-2}]$

Alternate solution :

$\text{Surface Tension} = \dfrac{\text{Surface energy}}{\text{Area}}$

$[\text{Surface tension}] = \dfrac{[E]}{[V\cdot T]^2} = [E\, V^{-2}\, T^{-2}]$

3. $\nu_c \propto [\eta^x \rho^y r^z]$

$[L^1T^{-1}] \propto [M^1\, L^{-1}\, T^{-1}]^x\,[M^1L^{-3}]^y\,[L^1]^z$

$[L^1T^{-1}] \propto [M^{x+y}]\,[L^{-x-3y+z}]\,[T^{-x}]$

taking comparision on both size

$x + y = 0, -x - 3y + z = 1, -x = -1$

$\Rightarrow \quad x = 1, y = -1, z = -1$

4. $\ell \propto h^x G^y c^z$

$M^0L^1T^0 = (ML^2T^{-1})^x\,(M^{-1}L^3T^{-2})^y\,(LT^{-1})^z$

$= M^{x-y}\, L^{2x+3y+z}\, T^{-x-2y-z}$

Equating :

$$\left.\begin{aligned} x-y&=0\\ 2x+3y+z&=1\\ -x-2y-z&=0\end{aligned}\right\} \Rightarrow x=\frac{1}{2}; y=\frac{1}{2}; z=-\frac{3}{2}$$

$\Rightarrow \ell \propto \dfrac{\sqrt{hG}}{c^{3/2}}$

5. $[L] = [c]^a\,[G]^b\left[\dfrac{e^2}{4\pi\varepsilon_0}\right]^c$

$[L] = [LT^{-1}]^a\,[M^{-1}L^3T^{-2}]^b\left[ML^3T^{-2}\right]^c$

$[L] = L^{a+3b+3c}\, M^{-b+c}\, T^{-a-2b-2c}$

$a + 3b + 3c = 1$

$-b + c = 0$

$a + 2b + 2c = 0$

On solving,

$a = -2, b = \dfrac{1}{2}, c = \dfrac{1}{2}$

$\therefore L = \dfrac{1}{c^2}\left[G.\dfrac{e^2}{4\pi\varepsilon_0}\right]^{\frac{1}{2}}$

6. Reading of screw gauge

= MSR + VSR × LC + zero error

= 0.5 cm + 25 × 0.001 cm + 0.004 cm

= 0.529 cm

7. $x = \dfrac{A^2B^{1/2}}{C^{1/3}D^3}$

$\dfrac{\Delta x}{x} = \dfrac{2\Delta A}{A} + \dfrac{1}{2}\dfrac{\Delta B}{B} + \dfrac{1}{3}\dfrac{\Delta C}{C} + 3\dfrac{\Delta D}{D}$

$\Rightarrow \dfrac{\Delta x}{x}\times 100 = 2(1\%) + \dfrac{1}{2}(2\%) + \dfrac{1}{3}(3\%) + 3(4\%) = 16\%$

8. $\dfrac{dQ}{dt} = -(K)A\dfrac{dT}{dx}$

$\dfrac{J}{s} = (K)m^2\dfrac{\text{kelvin}}{m}$

$(K) = \text{watt } m^{-1}\, K^{-1}$

9. $n(\text{VSD}) = (n-1)\text{MSD} \Rightarrow 1\text{ VSD} = \dfrac{(n-1)}{n}\text{ MSD}$

$\text{Least count} = 1\text{MSD} - 1\text{VSD} = \left[1 - \dfrac{(n-1)}{n}\right]\text{MSD}$

$\text{Least count} = \dfrac{1}{n}\text{MSD} = \dfrac{1}{n}\left(\dfrac{1}{n}\right)\text{cm} = \dfrac{1}{n^2}\text{cm}$

10. In subtraction the number of decimal places in the result should be equal to the number of decimal places of that term in the operation which contain lesser number of decimal places.

$$\begin{array}{r} 9.99 \\ -0.0099 \\ \hline 9.98 \end{array}$$

11. $[\text{stress}] = \left[\frac{\text{Force}}{\text{Area}}\right] = \left[\frac{M^1L^1T^{-2}}{L^2}\right] = M^1L^{-1}T^{-2}$

12. $\text{L.C.} = \frac{\text{Pitch}}{\text{Number of division on circular scale}}$

$\Rightarrow 0.01\text{mm} = \frac{\text{Pitch}}{50} \Rightarrow \text{Pitch} = 0.5 \text{ mm}$

13. $1' = \left(\frac{1}{60}\right)^{\circ} = \frac{1}{60} \times \frac{\pi}{180}$ radian

14. Mean of observation

$= \frac{1.25+1.24+1.27+1.21+1.28}{5} = 1.25 \text{ s}$

Mean absolute error

$= \frac{0+0.01+0.02+0.04+0.03}{5} = 0.02 \text{ s}$

% error = $\frac{0.02}{1.25} \times 100 = 1.6\%$

15. $E \propto F^a A^b T^c$

$[M^1L^2T^{-2}] = [M^1L^1T^{-2}]^a [LT^{-2}]^b [T]^c$

Solving,

$a = 1$ ...(i)

$a + b = 2 \Rightarrow b = 1$ ...(ii)

$-2a - 2b + c = -2 \Rightarrow c = 2$ ...(iii)

$\Rightarrow \quad a = 1 \quad b = 1 \quad c = 2$

$E = [F] [A] [T^2]$

16. Least count = $\frac{1\text{mm}}{100} = 0.01\text{mm}$

Diameter = main scale reading + circular scale reading

Diameter = $0 + 52 \times 0.01$ mm

= 0.52 mm = 0.052 cm

17. E = energy = $[ML^2T^{-2}]$

G = Gravitational constant = $[M^{-1}L^3T^{-2}]$

So $\frac{E}{G} = \frac{[E]}{[G]} = \frac{ML^2T^{-2}}{M^{-1}L^3T^{-2}} = [M^2L^{-1}T^0]$

18. Plane angle and solid angle are dimensionless but have units.

19. $[MLT^{-2}A^{-2}]$ = Magnetic permeability

20. $Y = \frac{\text{stress}}{\text{strain}}$ (Strain is dimensionless)

$\therefore$ [Y] = [stress] = [Pressure]

21. $\frac{F_e}{F_G} = \frac{\frac{Kq_1q_2}{r^2}}{\frac{Gm_1m_2}{r^2}}$

$2.4 \times 10^{39} = \frac{K}{G} \times \frac{(1.6 \times 10^{-19})^2}{(9.11 \times 10^{-31} \times 1.67 \times 10^{-27})}$

$\frac{K}{G} = \frac{2.4 \times 10^{39} \times 15.2137 \times 10^{-58}}{2.56 \times 10^{-38}}$

$= 14.26 \times 10^{19}$

$= 1.426 \times 10^{20}$

$\approx 10^{20}$

22. Error arise due to unpredictable fluctuation in temperature and voltage supply are $\rightarrow$ random errors.

23. $\rho = \frac{M}{V}$

$\rho = \frac{M}{\pi r^2 \ell}$

$\frac{\Delta\rho}{\rho} = \frac{\Delta M}{M} + \frac{2\Delta r}{r} + \frac{\Delta \ell}{\ell}$

$\frac{\Delta\rho}{\rho}\% = \left[\frac{0.002}{0.4} + \frac{2(0.001)}{(0.3)} + \frac{0.02}{5}\right] \times 100\%$

$= \frac{1}{2}\% + \frac{2}{3}\% + \frac{2}{5}\%$

$= 1.6\%$

24. Mean diameter = $\frac{d_1+d_2+d_3+d_4+d_5}{5}$

$= \frac{3.33+3.32+3.34+3.33+3.32}{5}$

$= 3.328 \approx 3.33$

25. Angular momentum = $[ML^2T^{-1}]$

Coeff of thermal conductivity = $[MLT^{-3}K^{-1}]$

Torque = $[ML^2T^{-2}]$

Gravitational constant = $[M^{-1}L^3T^{-2}]$

So, gravitational constant has power of (–1) of M.

26. Vernier Constant = MSD – VSD

$$= \text{MSD} - \frac{N}{N+1}\text{MSD}$$

$$= \frac{1}{N+1}(\text{MSD})$$

$$= \frac{1}{N+1}(0.01)\text{cm}$$

$$= \frac{1}{100(N+1)}$$

27. Solid angle (Ω) = $\frac{A}{r^2}$

It is dimensionless quantity

So from options

Option (1) Strain & Angle both are dimensionless

28. Dimensional Formula of

$[\alpha t^2] = [F]$

$[\alpha] = \left[\frac{MLT^{-2}}{T^2}\right] = [MLT^{-4}]$

and $[\beta t] = [F]$

$[\beta] = \frac{MLT^{-2}}{T} = [MLT^{-3}]$

Option (2) is satisfied as $\frac{\alpha t}{\beta} = \frac{[MLT^{-4}][T]}{[MLT^{-3}]}$

$= [M^0L^0T^0]$

29. $R = \frac{V}{I} = \frac{200}{20} = 10\ \Omega$

$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$

$\frac{\Delta R}{10} = \frac{4}{200} + \frac{0.2}{20}$

$\Delta R = \frac{6}{200}(10) = 0.3$

$\therefore (10 \pm 0.3)\Omega$

30. LC = $\frac{\text{pitch}}{\text{No.of divisions on circular scale}}$

= $\frac{1}{100}\text{mm} = 0.01\text{mm}$

Reading = MSR + LC × CSR

= 1 + 0.01 × 63

= 1.63 mm

= 0.163 cm

31. $V = \frac{AX^2}{\sqrt{x}+B}$

$[B] = [\sqrt{x}] \Rightarrow [B] = L^{1/2}$

$[V] = \left[\frac{AX^2}{\sqrt{x}+B}\right] \Rightarrow ML^2T^{-2} = \frac{[A]L^2}{\sqrt{L}}$

$ML^2T^{-2} = [A]\ L^{3/2} \Rightarrow [A] = ML^{1/2}T^{-2}$

$\frac{A^2}{B} = \frac{M^2 \perp T^{-4}}{L^{1/2}} = M^2L^{1/2}T^{-4}$

Exercise-III (Analytical Questions)

2. Reason is false because error in measurement can have any value.

eg. :- A student can read 23 as 25 from the device from which he is measuring a physical quantity.

3. Assertion is true because lower least count means lower error.

Since accuracy increases when least count decreases.

Reason is correct explanation because higher accuracy indicates error will be smaller.

4. Relative error is unitless and dimensionless. Absolute error has dimension.

5. Least count = $\frac{\text{Pitch}}{\text{No. of circular scale divisions}}$

$\therefore$ Statement I is false.

Statement II is true because smaller pitch means, smaller least count. $\therefore$ Greater accuracy.

6. [Stress] = [Modulus of Elasticity] = [Pressure] = [Energy density] = $[ML^{-1}T^{-2}]$

[Strain] = [Angle] = $[M^0L^0T^0]$

[Torque] = [Energy] = $[ML^2T^{-2}]$

7. Electrical resistance [Use $P = I^2R$]

$R = \frac{[P]}{[I^2]} = \frac{[ML^2T^{-3}]}{[A^2]} = [ML^2T^{-3}A^{-2}]$

Capacitance $\left[\text{use } U = \frac{1}{2}\frac{Q^2}{C}\right]$

$[C] = \frac{[Q^2]}{[U]} = \frac{[AT]^2}{[ML^2T^{-2}]} = [M^{-1}L^{-2}T^4A^2]$

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Magnetic field [use F = ILB]

$$[B]=\frac{[F]}{[I][L]}=\frac{[MLT^{-2}]}{[A][L]}=[MT^{-2}A^{-1}]$$

Inductance [use $U=\frac{1}{2}LI^2$]

$$[L]=\frac{[U]}{[I]^2}=\frac{[ML^2T^{-2}]}{[A^2]}=[ML^2T^{-2}A^{-2}]$$

8. Boltzmann constant (use $E=\frac{3}{2}KT$]

$[K] = ML^2T^{-2}K^{-1}$

Coefficient of viscosity $\left(\text{use } \eta=\frac{F}{A}\left(\frac{\Delta x}{\Delta v}\right)\right)$

$[\eta] = [ML^{-1}T^{-1}]$

Planck's constant (use $E = h\nu$)

$[h] = [ML^2T^{-1}]$

Thermal conductivity $\left(\text{use } K=\frac{i_t L}{A\Delta\theta}\right)$

$[K] = [MLT^{-3}K^{-1}]$

9. [Energy density] = [Young's modulus] = $[ML^{-1}T^{-2}]$

10. [h] = [Angular impulse] = [Angular momentum]
= $[ML^2T^{-1}]$